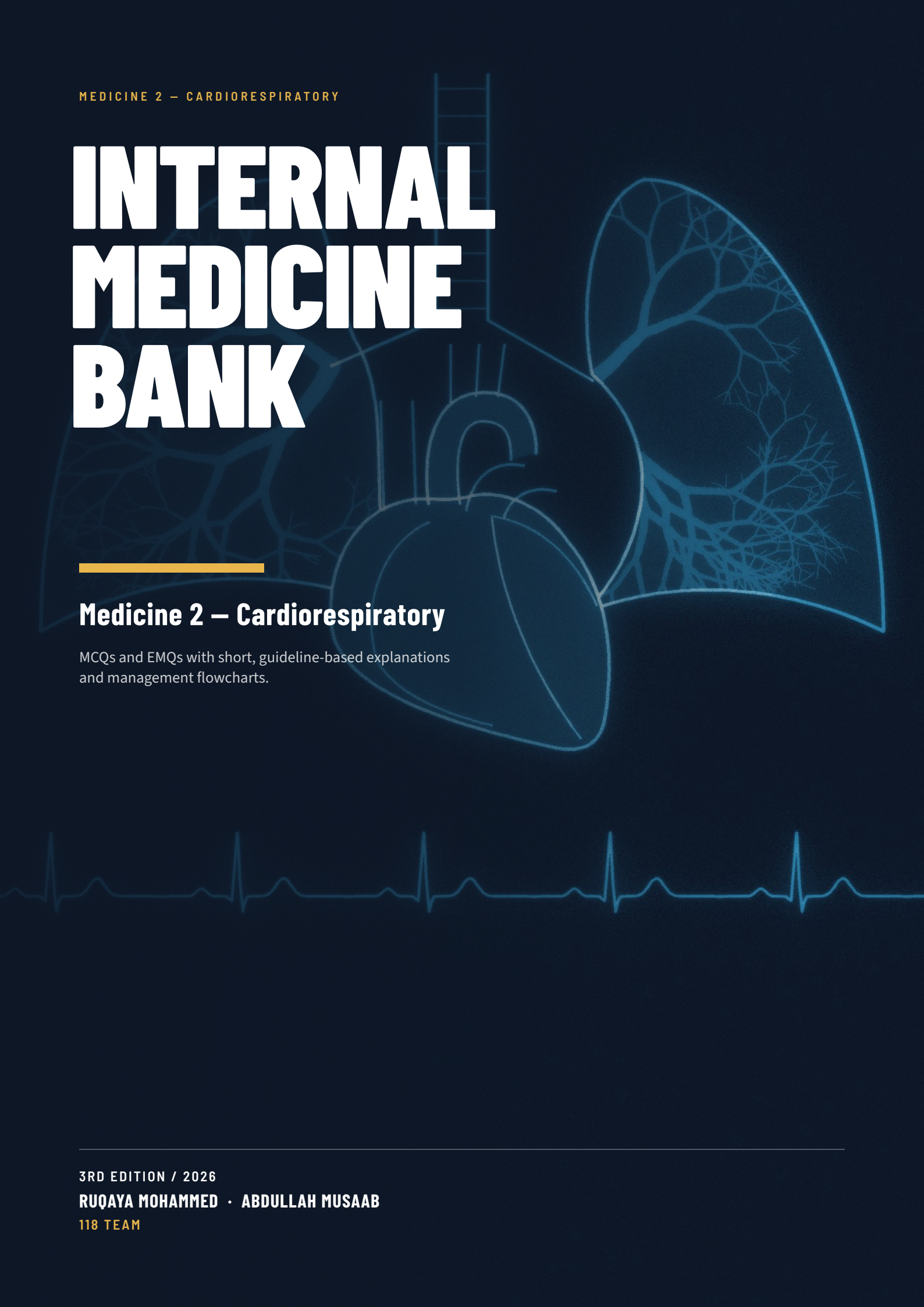


MEDICINE 2 – CARDIORESPIRATORY

INTERNAL MEDICINE BANK



Medicine 2 – Cardiorespiratory

MCQs and EMQs with short, guideline-based explanations and management flowcharts.

3RD EDITION / 2026

RUQAYA MOHAMMED · ABDULLAH MUSAAB

118 TEAM

3RD EDITION / 2026

INTERNAL MEDICINE BANK

Medicine 2 — Cardiorespiratory

**Ruqaya Mohammed
Abdullah Musaab**

3rd Edition / 2026

118 Team · t.me/AR_med118

COLOPHON

THIS EDITION

The 3rd Edition of the Internal Medicine Bank for Medicine 2 (cardiovascular and respiratory medicine). 1,512 questions — 1000 MCQs and 512 EMQ items in 19 chapters — with answers and explanations.

WHAT CHANGED IN THE 3RD EDITION

Every key was checked against current guidance (ESC, NICE, BTS, GINA, GOLD, WHO); 297 questions were corrected, and every explanation was rewritten and then cut to two or three lines that say why the answer is right. 220 questions whose idea had already been asked were removed: the question kept is the one from the earliest source in the chapter, and a highlighted note under it says how many times the idea was asked and in which exams or books. Numbering now restarts for each source: the local exams of a chapter (39th batch end-block and final papers, end-block, formative & previous years, log book) share one sequence, and each international book has its own; EMQs are numbered by theme. All changes are logged in the corrections file that accompanies this book.

MANAGEMENT FLOWCHARTS

The old summary tables were removed. Each chapter now opens with step-by-step flowcharts that combine the summaries of the source books with current guidelines, so that the next step in management can be read at a glance.

ARTWORK

Chapter and cover artwork was drawn for this edition; each illustration shows the subject of its chapter.

IMPORTANT

This book is a revision aid. Guidelines change: always check current local and national guidance before applying anything to patient care.

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CARDIOVASCULAR SYSTEM

01	Coronary & Vascular Disease Stable and unstable coronary disease, myocardial infarction and its complications, peripheral arterial and venous disease, and diseases of the aorta.	134 Q	10
02	Heart Failure & Cardiomyopathy Acute and chronic heart failure across the ejection-fraction spectrum, the drugs that change prognosis, and the cardiomyopathies.	86 Q	38
03	Congenital Heart Disease Shunts, obstructive lesions and cyanotic disease in children and adults: murmurs, ECG and X-ray clues, and when to intervene.	21 Q	58
04	Arrhythmias & Conduction Disease Atrial fibrillation, supraventricular and ventricular tachycardias, heart block and cardiac arrest: ECG recognition, rate and rhythm control, and anticoagulation.	90 Q	66
05	Valvular Heart Disease Aortic and mitral stenosis and regurgitation, right-sided valve disease and prosthetic valves: bedside signs, echo criteria and timing of intervention.	39 Q	88
06	Pericardial & Inflammatory Heart Disease Pericarditis, tamponade and constriction; infective endocarditis; myocarditis; and rheumatic fever — the inflammatory diseases of the heart and their look-alikes.	66 Q	100
07	Hypertension Diagnosis and grading, investigation for target-organ damage and secondary causes, drug choice, and hypertensive emergencies.	28 Q	116
08	Cardiovascular Pharmacology Mechanisms, indications, interactions and adverse effects of antiplatelets, anticoagulants, antianginals, antiarrhythmics and lipid-lowering drugs.	15 Q	126
09	Mixed Cardiology Integrated cardiology questions across the whole system — examination, investigation and management — for final revision.	81 Q	134
10	Cardiology EMQs Extended matching questions on the cardiovascular system: chest pain, murmurs, pulses, ECGs, arrhythmias and cardiovascular drugs.	298 Q	152

RESPIRATORY SYSTEM

11	Obstructive Airway Disease Asthma and COPD side by side: diagnosis and spirometry, acute severe attacks and exacerbations, oxygen, ventilation and long-term management.	80 Q	190
12	Respiratory Infection & Inflammation Pneumonia, tuberculosis, bronchiectasis, lung abscess and granulomatous disease: organisms, severity assessment and treatment.	79 Q	210
13	Respiratory Failure & ARDS Type 1 and type 2 respiratory failure, blood gases, oxygen and non-invasive ventilation, and the acute respiratory distress syndrome.	22 Q	228

14	Pleural Disease Pleural effusion, empyema, pneumothorax and haemothorax: Light criteria, aspiration and drainage, and mesothelioma.	42 Q	236
15	Interstitial Lung Disease Idiopathic pulmonary fibrosis, sarcoidosis, hypersensitivity pneumonitis, occupational and drug-induced lung disease.	60 Q	250
16	Pulmonary Vascular Disease Pulmonary embolism — probability, imaging and treatment — and pulmonary hypertension.	14 Q	266
17	Lung Cancer Presentation, paraneoplastic syndromes, diagnosis, staging and treatment of lung cancer and other thoracic tumours.	53 Q	274
18	Mixed Respiratory Medicine Integrated respiratory questions across the whole system for final revision.	90 Q	288
19	Respiratory EMQs Extended matching questions on the respiratory system: breathlessness, haemoptysis, clubbing, chest X-rays, blood gases and pleural fluid.	214 Q	308
BACK MATTER			
—	Index		335

HOW TO USE THIS BOOK

FLOWCHARTS FIRST

Each chapter opens with management flowcharts updated to current guidelines: follow the arrows from the presenting problem, answer the ? decision boxes, and read the next step.

MCQS AND EMQS

MCQs are numbered 1, 2, 3 ... within their source. EMQs are numbered **Theme 1, Theme 2 ...**; each theme lists its options A, B, C ... and its scenarios are numbered 1, 2, 3

REPEATED QUESTIONS

When an idea was asked more than once, one question is kept — the one from the earliest source — and a highlighted note under it says how many times and where else it was asked. Repeats are a strong signal of a high-yield topic.

REPEATED 3* · ALSO ASKED IN

Previous-years exam · Harrison

LOCAL EXAMS FIRST, THEN THE BOOKS

Questions come grouped by source. **Local Exams** — the 39th-batch end-block and final papers, end-block exams 2025, formative and previous-years exams and the log book — come first and share one numbering. Each international book (Davidson, Harrison, Crash Course, PreTest, Passmedicine, PasTest, Irfan, Get Ahead, other sources) follows with its own numbering from 1.

ANSWERS AFTER EVERY SET

Answers follow each set of up to 25 questions, each with a two- or three-line explanation. Tap a question number to jump to its answer and the answer number to jump back.

UPDATED ANSWERS AND INDEX

Answers marked **UPDATED** were corrected in this edition. The index lists diseases, drugs, tests and signs with the **page numbers** of the questions that test them.

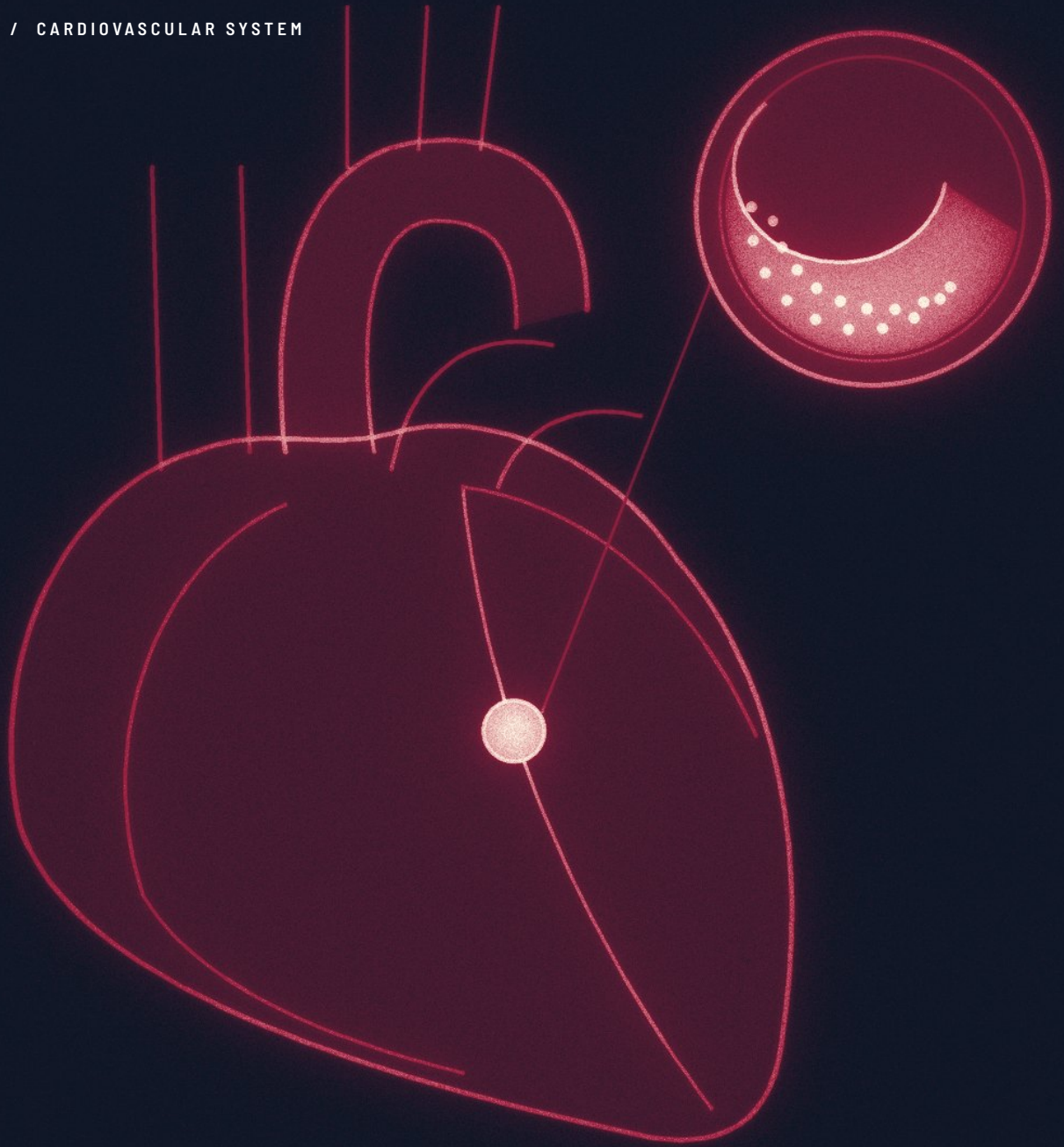
AT A GLANCE

CHAPTER	MCQS	EMQ SCENARIOS	REPEATED IDEAS
01 Coronary & Vascular Disease	134	0	19
02 Heart Failure & Cardiomyopathy	86	0	13
03 Congenital Heart Disease	21	0	2
04 Arrhythmias & Conduction Disease	90	0	14
05 Valvular Heart Disease	39	0	7
06 Pericardial & Inflammatory Heart Disease	66	0	15
07 Hypertension	28	0	2
08 Cardiovascular Pharmacology	15	0	0
09 Mixed Cardiology	81	0	3
10 Cardiology EMQs	0	298	5
11 Obstructive Airway Disease	80	0	15
12 Respiratory Infection & Inflammation	79	0	17
13 Respiratory Failure & ARDS	22	0	1
14 Pleural Disease	42	0	5
15 Interstitial Lung Disease	60	0	8
16 Pulmonary Vascular Disease	14	0	3
17 Lung Cancer	53	0	3
18 Mixed Respiratory Medicine	90	0	4
19 Respiratory EMQs	0	214	6
Total	1000	512	142

QUESTION SOURCES

In every chapter the banks appear in this order. “Merged” counts questions removed because their idea was already asked in an earlier or better question; each is credited in the repeat caption of the question that was kept.

SOURCE	QUESTIONS	MERGED
39 Blocks Questions End-block examinations of the 39th batch (2024–2025), recalled and verified	253	14
39th Batch Final Papers Final exam Paper 1 & Paper 2 of the 39th batch, recalled	63	1
End-Block Exams End-block & induction exams 2025	13	19
Formative & Previous Years College formative & previous-years exam questions	128	47
Log Book Log-book questions 2025	40	19
Davidson After Davidson's Principles & Practice of Medicine self-assessment	171	9
Harrison After Harrison's Self-Assessment & Board Review	244	12
Crash Course After Crash Course SBAs & EMQs	175	36
PreTest After PreTest Medicine	54	9
Passmedicine After Passmedicine	80	12
PasTest After PasTest	42	7
Irfan Irfan EMQ collection	56	8
Get Ahead After Get ahead! Medicine EMQs	154	—
Other Sources Other colleges & sources	39	27
Total	1512	220



01

CORONARY & VASCULAR DISEASE

Coronary & Vascular Disease

Stable and unstable coronary disease, myocardial infarction and its complications, peripheral arterial and venous disease, and diseases of the aorta.

134

MCQs

0

EMQ SCENARIOS

7

SOURCES

19

REPEATED IDEAS

IN THIS CHAPTER

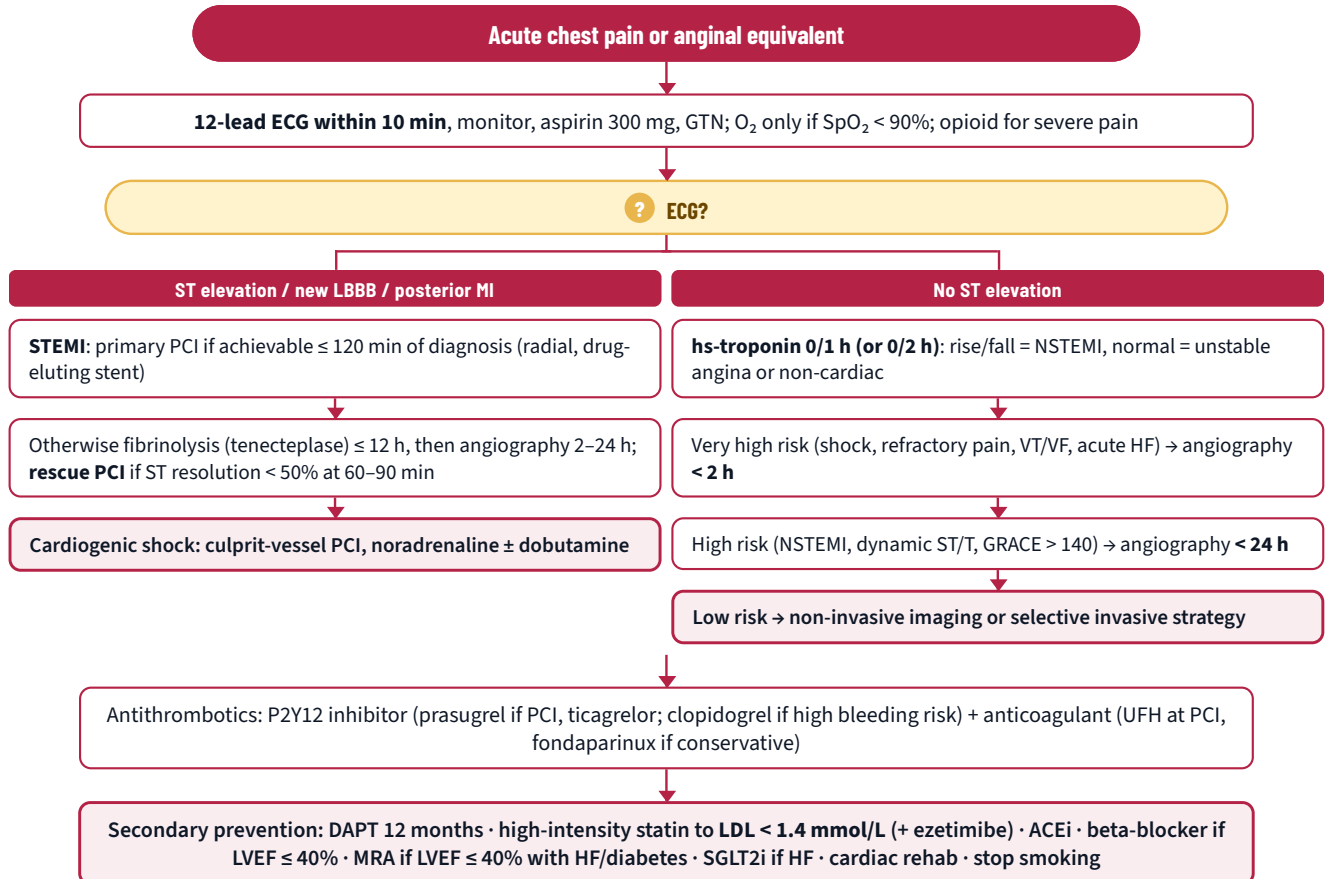
01	Management flowcharts	3 charts	page 12
02	Local Exams MCQs	44 Q	page 14
03	Davidson MCQs	25 Q	page 20
04	Harrison MCQs	31 Q	page 24
05	Crash Course MCQs	12 Q	page 31
06	PreTest MCQs	10 Q	page 33
07	Passmedicine MCQs	6 Q	page 35
08	Other Sources MCQs	6 Q	page 36

01

Step by step

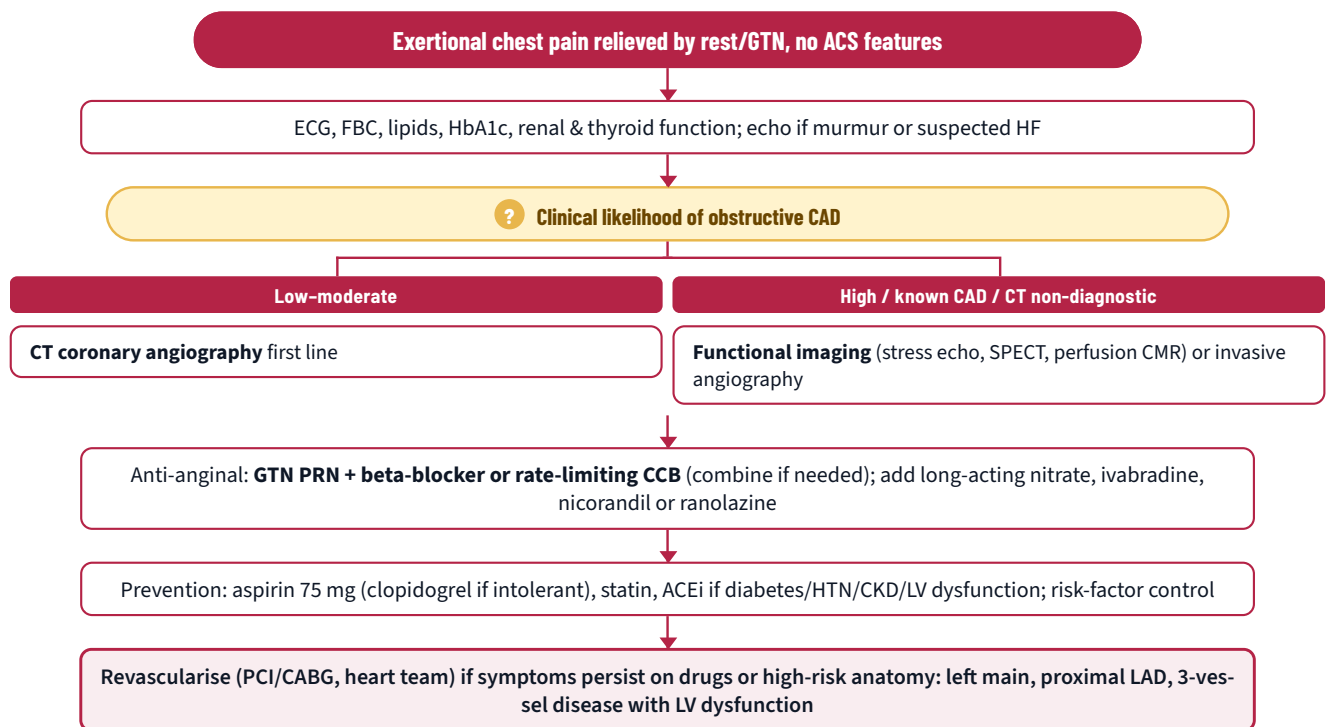
Acute chest pain: suspected acute coronary syndrome

ESC 2023 ACS • NICE NG185



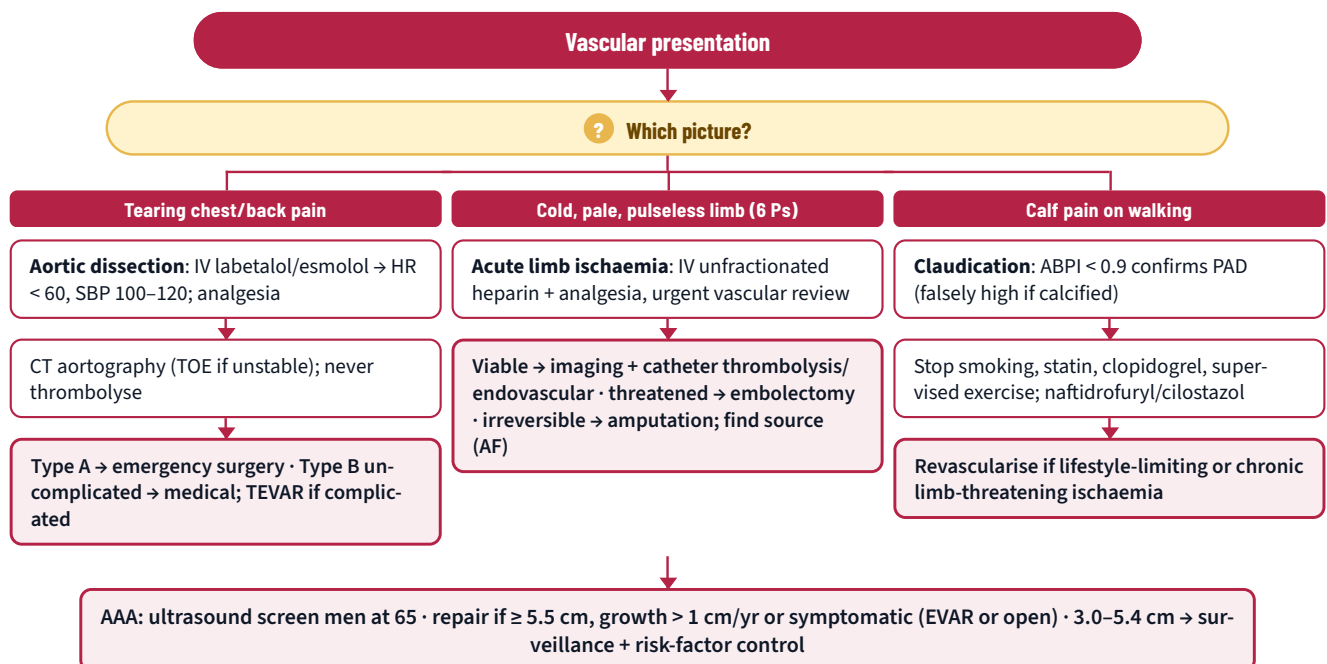
Stable chest pain: chronic coronary syndrome

ESC 2024 CCS · NICE CG95



Aorta and limb ischaemia

ESC 2024 AORTIC & PAD · NICE NG156



Local Exams MCQs

SET 1 OF 2

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

CASE • QUESTIONS 1–5

A 58-year-old man with hypertension has had recurrent chest discomfort and breathlessness for 2 hours. ECG shows ST elevation in V1–V4. He takes ramipril 5 mg twice daily and metformin 1 g twice daily; random blood glucose is 165 mg/dL and HbA1c 6.3%.

- 1 What is the next investigation?
 - A Stress echocardiography
 - B Coronary CT angiography
 - C Cardiac enzymes (troponin)
 - D Treadmill exercise test
 - E Chest radiography
 - 2 Which drug should be added to his treatment?
 - A Amlodipine
 - B Aspirin
 - C Spironolactone
 - D Sitagliptin
 - E Glimepiride
- REPEATED 3* • ALSO ASKED IN**
Log book 2025 • Crash Course
- 3 What is the most common cause of death in such a patient?
 - A Cardiac arrhythmias
 - B Acute kidney injury
 - C Thromboembolic complications
 - D Uncontrolled blood glucose
 - E Uncontrolled hypertension
- REPEATED 2* • ALSO ASKED IN**
Previous-years exam
- 4 Two days later he develops new sharp chest pain with a low-grade fever. What is the most likely cause?
 - A Pulmonary embolism
 - B Pulmonary oedema
 - C New acute MI
 - D Acute pericarditis
 - E Infective endocarditis
 - 5 Which drug has NO prognostic benefit in this patient?
 - A Metoprolol
 - B Rosuvastatin
 - C Aspirin
 - D Ramipril
 - E Amlodipine

- 6 A 62-year-old man presents 60 minutes after the onset of severe substernal chest pain radiating to the left arm, with sweating and nausea. BP 98/60 mmHg, HR 54 bpm, RR 20, SpO₂ 96% on air. ECG shows 3 mm ST elevation in II, III and aVF with reciprocal ST depression in I and aVL. Bedside echo shows inferior wall hypokinesia with a normal-sized right ventricle. He has had chewable aspirin and is being prepared for primary PCI. Which next step is most appropriate before PCI?
 - A IV metoprolol 5 mg to reduce infarct size
 - B Sublingual glyceryl trinitrate
 - C 1 L normal saline bolus to improve preload
 - D Loading dose of ticagrelor
 - E IV diltiazem to treat the bradycardia
- 7 A 68-year-old man with diabetes and chronic kidney disease (eGFR 32 mL/min) has had chest discomfort for 4 hours. BP 142/86 mmHg, HR 96 bpm. ECG shows ST depression in V4–V6 and high-sensitivity troponin is elevated and rising. He is pain-free after aspirin, oxygen and IV nitrates. Creatinine is 2.3 mg/dL (baseline 1.9). He is haemodynamically stable with no heart failure. Which management is most appropriate?
 - A Immediate coronary angiography (within 2 hours)
 - B IV fibrinolytic therapy
 - C Early invasive strategy within 24 hours
 - D Delay angiography until creatinine returns to baseline
 - E Switch aspirin to clopidogrel monotherapy
- 8 A 40-year-old man has severe central chest pain and a blood pressure of 200/120 mmHg. Aortic dissection is suspected. Which drug is first-line to control his blood pressure?
 - A IV labetalol
 - B IV sodium nitroprusside
 - C IV furosemide
 - D IV hydralazine

CASE · QUESTIONS 9-10

A 74-year-old woman with hypertension has had sudden pain in the left leg for 1 hour. Pulse 88/min, irregularly irregular; BP 170/96 mmHg. The left leg is pale and cold with reduced sensation, and the dorsalis pedis and posterior tibial pulses are absent. She takes aspirin and candesartan.

9 What is the most likely diagnosis?

- A Deep vein thrombosis
- B Cellulitis
- C Acute arterial occlusion
- D Ruptured Baker cyst
- E Venous insufficiency

REPEATED 2* · ALSO ASKED IN
Davidson

10 What is the initial treatment?

- A Clopidogrel
- B Warfarin
- C Alteplase
- D Intravenous unfractionated heparin

11 A 45-year-old heavy smoker is admitted with severe central chest pain on minimal exertion that began 14 hours ago. Which statement about management is NOT correct?

- A LMW heparin is appropriate
- B IV thrombolytic therapy is life-saving
- C Chewed aspirin is beneficial
- D Oral metoprolol is indicated

REPEATED 4* · ALSO ASKED IN
End-block exam 2025 · Previous-years exam · Davidson

12 A 58-year-old man has burning chest pain on walking relieved by rest; his ECG shows ST depression and T-wave inversion in II, III and aVF. Which of the following is NOT a risk factor for his disease?

- A Hypertension
- B Diabetes mellitus
- C High triglycerides
- D High HDL cholesterol

REPEATED 2* · ALSO ASKED IN
39th batch end-block (cardiology)

13 A 65-year-old man develops chest pain; the ECG shows ST depression and T-wave inversion in V1-V4. Which of the following is an unmodifiable risk factor for his disease?

- A Smoking
- B Genetics (family history)
- C Diabetes mellitus
- D Hypertension

14 A patient with a past myocardial infarction has a paradoxical (dyskinetic) impulse on the chest wall and persistent ST elevation on the ECG. What is the diagnosis?

- A Heart failure
- B Left ventricular aneurysm
- C New MI
- D Dressler's syndrome

REPEATED 3* · ALSO ASKED IN
39th batch end-block (cardiology) · Passmedicine

15 A 73-year-old smoker has leg pain on walking that is relieved by rest. Which of the following is NOT a feature of this disease?

- A Paralysis of the limb
- B Redness of the limb
- C Absent posterior tibial pulse
- D Cold limb

16 Which of the following is true about stable angina?

- A Pain is relieved by NSAIDs
- B Pain lasts more than 30 minutes
- C Pain is relieved by nitrates
- D Pain occurs at rest

REPEATED 2* · ALSO ASKED IN
Previous-years exam

17 At what diameter should elective repair of an abdominal aortic aneurysm be considered in the absence of contraindications?

- A 1 cm
- B 3 cm
- C 5.5 cm
- D 10 cm

18 Which feature is least consistent with a diagnosis of unstable angina?

- A Chest pain on minimal exertion
- B Chest pain not relieved by sublingual nitrate
- C Persistent ST-segment elevation on ECG
- D More than two episodes of rest pain in the last 24 hours
- E T-wave inversion on ECG

REPEATED 3* · ALSO ASKED IN
Formative exam · Previous-years exam

19 A patient with acute myocardial infarction 2 days ago develops sudden severe breathlessness. Pulse 112/min, small volume with pulsus paradoxus; BP 90/55 mmHg; the JVP is raised and the heart sounds are soft. What is the most likely diagnosis?

- A Ventricular aneurysm
- B Cardiac tamponade
- C Ventricular septal defect
- D Dressler's syndrome

REPEATED 2* · ALSO ASKED IN
Formative exam

- 20** A 67-year-old man with hypertension and hyperlipidaemia is about to be discharged 6 days after an acute MI treated with primary PCI. He has no symptoms of heart failure, but the pre-discharge echo shows LVEF 38%. He takes aspirin, metformin, ticagrelor, atorvastatin, lisinopril and metoprolol. Which medication is most appropriate to add?
- A Digoxin
 - B Spironolactone (a mineralocorticoid receptor antagonist)
 - C Switch metoprolol to carvedilol
 - D Oral furosemide
 - E Stop lisinopril and start candesartan

CASE · QUESTIONS 21–22

A 48-year-old man with diabetes has had acute retrosternal chest pain for 3 hours. Pulse 102/min, small volume; BP 110/65 mmHg. ECG shows ST elevation in I, aVL and V1–V4.

- 21** What is the most appropriate next investigation?
- A Chest X-ray
 - B Cardiac enzymes (troponin)
 - C Echocardiography
 - D CT pulmonary angiography
- 22** What is the most appropriate management?
- A Unfractionated heparin
 - B Low-molecular-weight heparin
 - C Oral aspirin 100 mg
 - D Urgent percutaneous coronary intervention

REPEATED 2× · ALSO ASKED IN
Davidson

- 23** A 50-year-old man with hypertension has breathlessness, difficulty swallowing and central chest pain radiating to the back. The chest X-ray shows a widened mediastinum. What is the most likely diagnosis?
- A Thoracic aortic aneurysm
 - B Acute coronary syndrome
 - C Pulmonary embolism
 - D Acute pericarditis
- 24** A 50-year-old man with hypertension has a thoracic aortic aneurysm causing chest pain, dysphagia and a widened mediastinum. What is the most likely serious complication if it is left untreated?
- A Cardiac tamponade
 - B Aortic dissection
 - C Ischaemic heart disease
 - D Haemothorax

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

- 25** A 71-year-old male smoker with hypertension and type 2 diabetes has noticed a pulsating abdominal mass and has had dull abdominal pain radiating to the back for 2 months. What is the most likely diagnosis?
- A Diverticulitis
 - B Ureteric colic
 - C Abdominal aortic aneurysm
 - D Chronic pancreatitis
 - E Mesenteric ischaemia

Answers 1–25**LOCAL EXAMS MCQS**

- 1** **C Cardiac enzymes (troponin)** — Troponin confirms myocardial necrosis and adds prognosis, but with ST elevation primary PCI is arranged on the ECG alone. Stress tests and CT angiography are for stable pain.
- 2** **B Aspirin** — Every ACS patient gets aspirin (300 mg load, then 75 mg daily) plus a P2Y12 inhibitor. Amlodipine has no prognostic benefit and HbA1c 6.3% needs no extra drug.
- 3** **A Cardiac arrhythmias** — Most MI deaths are early (often pre-hospital) from ventricular fibrillation; hence monitoring and rapid defibrillation. Cardiogenic shock dominates later in-hospital deaths.
- 4** **D Acute pericarditis** — Sharp positional pain with low-grade fever 1–3 days after a transmural MI is early post-infarct pericarditis; treat with aspirin. Dressler syndrome comes weeks later.
- 5** **E Amlodipine** — Prognosis after MI improves with DAPT, high-intensity statin, ACE inhibitor and beta-blocker (± MRA/SGLT2i if LVEF low). Amlodipine only relieves angina or lowers BP.
- 6** **D Loading dose of ticagrelor** — Load a P2Y12 inhibitor (ticagrelor/prasugrel) with aspirin before primary PCI. In inferior or STEMI with hypotension, beta-blockers, diltiazem and nitrates can cause shock.
- 7** **C Early invasive strategy within 24 hours** — Rising troponin with dynamic ST depression is NSTEMI (high risk): angiography within 24 h. Immediate (< 2 h) only if unstable; CKD needs hydration, not delay.
- 8** **A IV labetalol** — In dissection give an IV beta-blocker (labetalol/esmolol) first: target HR < 60/min and SBP 100–120 mmHg. Vasodilators alone cause reflex tachycardia.
- 9** **C Acute arterial occlusion** — Sudden pale, cold, pulseless leg with sensory loss in AF is acute arterial (embolic) occlusion. DVT gives a warm swollen leg with pulses present.
- 10** **D Intravenous unfractionated heparin** — **UPDATED** Acute limb ischaemia: IV unfractionated heparin, analgesia and urgent vascular referral for embolectomy or thrombolysis. Long-term anticoagulation follows for AF.
- 11** **B IV thrombolytic therapy is life-saving** — Thrombolysis is harmful without ST elevation and is not given > 12 h after onset even for STEMI. Aspirin, P2Y12 inhibitor, fondaparinux and beta-blocker are all appropriate.
- 12** **D High HDL cholesterol** — High HDL cholesterol is protective (reverse cholesterol transport). Hypertension, diabetes and high triglycerides are risk factors for coronary disease.

- 13** **B Genetics (family history)** — Unmodifiable risk factors: age, male sex and family history of premature CAD. Smoking, diabetes, hypertension, lipids and obesity are modifiable.
- 14** **B Left ventricular aneurysm** — Persistent ST elevation weeks after MI with a dyskinetic apical impulse is LV aneurysm; echo confirms. Risks: heart failure, VT and mural thrombus.
- 15** **A Paralysis of the limb** — **UPDATED** Claudication (chronic PAD) causes absent pulses, cool hairless skin, pallor on elevation and rubor. Paralysis belongs to acute limb ischaemia (the 6 Ps).
- 16** **C Pain is relieved by nitrates** — Stable angina: exertional tightness lasting minutes, relieved by rest or GTN within ~5 min. Rest pain > 20 min suggests ACS.
- 17** **C 5.5 cm** — **UPDATED** Repair an AAA at ≥ 5.5 cm, growth > 1 cm/year or if symptomatic (NICE NG156). Aneurysms of 3.0–5.4 cm are kept under ultrasound surveillance.
- 18** **C Persistent ST-segment elevation on ECG** — Unstable angina has no necrosis; ECG is normal or shows ST depression/T inversion. Persistent ST elevation means STEMI and needs immediate reperfusion.
- 19** **B Cardiac tamponade** — Hypotension, raised JVP, soft heart sounds and pulsus paradoxus days after MI is tamponade from free-wall rupture. A VSD would give a loud pansystolic murmur.
- 20** **B Spironolactone (a mineralocorticoid receptor antagonist)** — After MI with LVEF $\leq 40\%$ plus heart failure or diabetes, add an MRA (eplerenone/spironolactone) to ACE inhibitor and beta-blocker; it reduces mortality.
- 21** **B Cardiac enzymes (troponin)** — Anterolateral STEMI: troponin confirms necrosis among the tests listed, but reperfusion never waits for it. CTPA is for suspected PE.
- 22** **D Urgent percutaneous coronary intervention** — STEMI < 12 h needs immediate reperfusion: primary PCI if possible within 120 min of diagnosis, otherwise fibrinolysis. Antithrombotics alone do not open the artery.
- 23** **A Thoracic aortic aneurysm** — **UPDATED** Widened mediastinum with dysphagia and breathlessness from compression in a hypertensive man suggests thoracic aortic aneurysm; confirm with CT angiography.
- 24** **B Aortic dissection** — Untreated thoracic aneurysms dissect or rupture; risk rises steeply above ~5.5 cm (the repair threshold, lower in Marfan). Tamponade follows dissection.
- 25** **C Abdominal aortic aneurysm** — An expansile (pulsatile) mass with back pain in an older male smoker is an abdominal aortic aneurysm; new or persistent pain suggests expansion and needs urgent imaging.

SET 2 OF 2

CASE · QUESTIONS 26–28

A 71-year-old male smoker with hypertension and type 2 diabetes has noticed a pulsating abdominal mass and has had dull abdominal pain radiating to the back for 2 months.

- 26** What is the gold-standard diagnostic test?
- A Doppler ultrasound
 - B Abdominal X-ray
 - C CT angiography
 - D MRI
 - E ECG
- 27** He suddenly develops pain in his right leg, and the dorsalis pedis pulse is absent. What is the most likely diagnosis?
- A Chronic limb ischaemia
 - B Cellulitis
 - C Acute limb ischaemia
 - D Deep vein thrombosis
 - E Compartment syndrome
- 28** What is the first-line management for this complication?
- A Antiplatelet therapy
 - B Anticoagulation
 - C Antibiotics
 - D Leg elevation
 - E Fasciotomy

CASE · QUESTIONS 29–33

A 25-year-old man has exertional chest pain. His father died of a myocardial infarction at the age of 44. Fasting lipids: total cholesterol 550 mg/dL, LDL cholesterol 280 mg/dL, HDL cholesterol 35 mg/dL, triglycerides 120 mg/dL.

- 29** What is the most likely diagnosis?
- A Familial hypercholesterolaemia
 - B Familial hypertriglyceridaemia
 - C Familial combined hyperlipidaemia
 - D Familial chylomicronaemia
 - E Raised lipoprotein(a)
- 30** If untreated, what is the most likely complication?
- A Premature coronary artery disease and myocardial infarction
 - B Childhood stroke
 - C Acute pancreatitis
 - D Hypothyroidism
 - E Gallstones
- 31** What is the best initial treatment?
- A Lifestyle modification only
 - B Statin therapy
 - C Fibrate therapy
 - D Niacin
 - E PCSK9 inhibitor monotherapy

32 Despite adherence to a high-intensity statin for 3 months, his LDL cholesterol remains above target. What is the next step?

- A Add ezetimibe
- B Switch to a fibrate
- C Add niacin
- D Add a fish-oil supplement
- E Return to lifestyle modification only

33 What is the mode of inheritance of this disorder?

- A Autosomal dominant
- B Autosomal recessive
- C X-linked recessive
- D Mitochondrial
- E Sporadic mutation

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

34 You review a 62-year-old man who has recently been discharged from hospital in Hungary following a myocardial infarction. He brings a copy of an echocardiogram report which shows his left ventricular ejection fraction is 48%. On examination his pulse is 78/min and regular, blood pressure is 124 / 72 mmHg and his chest is clear. His current medications include aspirin, simvastatin and lisinopril. What is the most appropriate next step in terms of his medication?

- A Make no changes
- B Add bisoprolol
- C Add atenolol
- D Add furosemide

35 A 60-year-old businessman has central crushing chest pain radiating to both arms after running to catch a bus. The pain is relieved by rest and his ECG 1 hour later is normal. What is the most likely diagnosis?

- A Aortic dissection
- B Unstable angina
- C Stable angina
- D Pericarditis

36 Which one of the following conditions is most associated with aortic dissection?

- A Bicuspid aortic valve
- B Acromegaly
- C Adult polycystic kidney disease
- D Sarcoidosis

37 A 63-year-old man with hypertension presents to the emergency department with sudden-onset tearing chest pain radiating to the back. What is the most likely diagnosis?

- A Oesophageal spasm
- B Pneumothorax
- C Aortic dissection
- D Angina

REPEATED 4* · ALSO ASKED IN
Log book 2025 · Other Sources ×2

Log-book questions

38 A 78-year-old woman has crushing central chest pain. Which investigation should be performed first to diagnose an ST-elevation myocardial infarction?

- A Chest X-ray
- B 12-lead ECG
- C Echocardiography
- D Coronary CT angiography
- E Stress echocardiogram

REPEATED 2* · ALSO ASKED IN
Harrison

39 Which risk stratification score is most commonly used in patients with acute myocardial infarction?

- A CHADS2 score
- B APACHE II score
- C TIMI risk score
- D GRACE score
- E HAS-BLED score

40 In a patient with STEMI presenting to a centre without percutaneous coronary intervention (PCI) capability, what is the preferred reperfusion strategy?

- A Immediate coronary artery bypass grafting
- B Intravenous thrombolytic therapy
- C Conservative management with high-dose statins
- D Urgent PCI referral despite delay
- E Use of oral beta-blockers only

41 A patient has chronic heart failure due to ischaemic heart disease. What definitive treatment should be considered if indicated?

- A Long-term diuretic therapy only
- B Coronary revascularization
- C Beta-blocker monotherapy
- D Lifestyle modification only
- E Implantable cardioverter-defibrillator

42 Which investigation is most appropriate to confirm the diagnosis of aortic dissection?

- A Transthoracic echocardiogram
- B CT angiography of the aorta
- C Chest X-ray
- D MRI of the brain
- E D-dimer assay

43 What is the best treatment strategy for a patient with aortic dissection (Type A) presenting in this manner?

- A Conservative management with oral antihypertensives
- B Urgent surgical repair after blood pressure control
- C Elective percutaneous stenting
- D Thrombolytic therapy
- E High-dose corticosteroids

REPEATED 2* · ALSO ASKED IN
Davidson

- 44** Which of the following is a common complication of aortic dissection?
- A** Cardiac tamponade
 - B** Mitral valve prolapse
 - C** Atrial fibrillation
 - D** Ventricular septal defect
 - E** Pericardial friction rub

Answers 26–44

LOCAL EXAMS MCQS

- 26** **C CT angiography** — Ultrasound is for screening and surveillance; CT angiography is the reference test, measuring the aneurysm, its extent and any leak, and planning EVAR or open repair.
- 27** **C Acute limb ischaemia** — Sudden pain with a lost pulse is acute limb ischaemia, here from an embolus of mural thrombus in the aneurysm. Look for the 6 Ps (pain, pallor, pulselessness, perishing cold, paraesthesia, paralysis).
- 28** **B Anticoagulation** — Give IV unfractionated heparin at once to stop clot propagation, then revascularise (embolectomy or thrombolysis). Elevation worsens ischaemia.
- 29** **A Familial hypercholesterolaemia** — Very high LDL, normal triglycerides and premature coronary disease in a first-degree relative is familial hypercholesterolaemia (LDL-receptor defect).
- 30** **A Premature coronary artery disease and myocardial infarction** — Lifelong high LDL causes premature atherosclerosis and MI decades early. Pancreatitis complicates severe hypertriglyceridaemia, not high LDL.
- 31** **B Statin therapy** — Start a high-intensity statin (atorvastatin 40–80 mg) with lifestyle advice; diet alone cannot correct a genetic LDL excess. Fibrates target triglycerides.
- 32** **A Add ezetimibe** — Add ezetimibe (blocks intestinal cholesterol absorption) to the statin; if LDL is still above target, add a PCSK9 inhibitor. Niacin and fish oil do not reduce events here.
- 33** **A Autosomal dominant** — FH is autosomal dominant (LDLR, APOB or PCSK9 variants), so half of first-degree relatives are affected and cascade testing of the family is recommended.
- 34** **B Add bisoprolol** — Every post-MI patient should take a beta-blocker unless contraindicated; bisoprolol is preferred if EF is reduced. Furosemide is only for congestion.
- 35** **C Stable angina** — Predictable exertional pain relieved by rest with a normal ECG is stable angina: GTN, antianginal drug and CT coronary angiography. Unstable angina occurs at rest.
- 36** **A Bicuspid aortic valve** — Bicuspid aortic valve has an associated aortopathy and raises dissection risk. Others: hypertension (commonest), Marfan, Loeys–Dietz, coarctation, Turner, pregnancy, cocaine.
- 37** **C Aortic dissection** — Sudden tearing pain radiating to the back in a hypertensive patient is aortic dissection until proved otherwise; confirm with CT aortography and control BP and heart rate.
- 38** **B 12-lead ECG** — **UPDATED** STEMI is diagnosed on a 12-lead ECG within 10 min: ST elevation in 2 contiguous leads or new LBBB. Troponin is not needed to start reperfusion.
- 39** **D GRACE score** — **UPDATED** GRACE (age, HR, SBP, creatinine, Killip, arrest, ST deviation, troponin) is the recommended ACS risk score. CHA2DS2-VASc and HAS-BLED are for AF.
- 40** **B Intravenous thrombolytic therapy** — If primary PCI cannot be done within 120 min, give fibrinolysis (tenecteplase) within 12 h, then transfer for angiography in 2–24 h (pharmaco-invasive).
- 41** **B Coronary revascularization** — In ischaemic cardiomyopathy with angina or viable myocardium, CABG improves survival (STICH) and treats the cause. Diuretics and ICDs do not.
- 42** **B CT angiography of the aorta** — CT angiography of the whole aorta confirms dissection in a stable patient; TOE if unstable. Chest X-ray and transthoracic echo cannot exclude it.
- 43** **B Urgent surgical repair after blood pressure control** — Type A dissection (ascending aorta) kills 1–2%/hour untreated: emergency surgery after IV beta-blocker BP/HR control. Type B is usually medical.
- 44** **A Cardiac tamponade** — Proximal dissection can rupture into the pericardium causing tamponade, a leading cause of death; also acute AR, coronary occlusion and branch ischaemia.

Davidson MCQs

- 1 A 74 year old woman presents with breathlessness. She is found to have an elevated jugular venous pressure (JVP). Which of the following conditions is most likely to explain this physical finding?
 - A Aortic stenosis
 - B Dehydration
 - C Exacerbation of asthma
 - D Increased left atrial pressure
 - E Recurrent pulmonary embolism
 - 2 You assess a 62 year old woman 2 days after treatment for anterior myocardial infarction. On examination she is tachycardic and tachypnoeic, and has a harsh systolic murmur radiating to the right side of the chest. There are fine inspiratory crepitations audible at the lung bases. What is the most likely explanation for these findings?
 - A Acute aortic incompetence
 - B Left ventricular free wall rupture
 - C Papillary muscle rupture and mitral incompetence
 - D Post-infarction pericarditis with pericardial rub
 - E Rupture of the interventricular septum
- REPEATED 2* • ALSO ASKED IN**
Other Sources
- 3 A 55 year old man with type 2 diabetes presents with a 1-hour history of severe central chest pain. Which of the following statements is true?
 - A A normal baseline troponin and elevated 6-hour troponin level is suspicious of myocardial infarction
 - B A normal ECG excludes myocardial infarction
 - C A normal initial troponin level excludes myocardial infarction
 - D Failure of chest pain to resolve with nitrates confirms myocardial infarction
 - E T-wave inversion on the ECG confirms myocardial infarction
 - 4 Which of the following patients is a suitable candidate for an implantable cardiac defibrillator?
 - A A 26 year old man with polymorphic ventricular tachycardia (torsades de pointes) occurring after cocaine use
 - B A 48 year old man who presents with acute inferior myocardial infarction complicated within the first 6 hours by ventricular fibrillation
 - C A 55 year old woman with syncope; ECG monitoring shows sinus rhythm with third-degree atrioventricular block
 - D A 75 year old man with syncope; ambulatory ECG shows sinus bradycardia and daytime sinus pauses of up to 5 seconds
 - E An 80 year old man with a history of anterior myocardial infarction 6 months previously; he is fit, has never experienced arrhythmia, and a cardiac magnetic resonance scan shows poor left ventricular function (left ventricular ejection fraction 28%)
 - 5 A 54 year old security guard who is obese and enjoys drinking alcohol and cigarette smoking with his friends has a diet high in saturated fats. He has an acute myocardial infarction. Which lifestyle risk factor has the strongest association with myocardial infarction?
 - A Excess alcohol
 - B High-saturated fat diet
 - C Obesity
 - D Sedentary activity
 - E Smoking
 - 6 A 36 year old smoker has sudden onset of chest pain whilst out walking in a remote island of Scotland. He attends the local hospital and is found to have ST segment elevation myocardial infarction. Which treatment has the strongest time-dependent benefit (i.e. the quicker received, the better the outcome) for ST segment elevation myocardial infarction?
 - A Aspirin
 - B B-blocker
 - C Heparin
 - D Percutaneous coronary intervention
 - E Tissue plasminogen activator
 - 7 Which of the following biomarkers is a structural protein rather than a cardiac enzyme?
 - A Aspartate aminotransferase
 - B Creatine kinase
 - C Creatine kinase MB
 - D Lactate dehydrogenase
 - E Troponin I

- 8** A patient has a stent placed in his right coronary artery. On return to the ward he develops severe chest pain and becomes very unwell, and stent thrombosis is suspected. Which ECG feature would suggest that the stent has occluded?
- A Anterior T-wave inversion
 - B Atrial fibrillation
 - C Atrioventricular block
 - D ST elevation in I, aVL and V6
 - E ST elevation in V2–V5
- 9** A 72 year old woman has had 'indigestion' for 4 days with vomiting and sweating. She presents to the emergency department where a delayed presentation inferior ST segment elevation myocardial infarction is diagnosed. She has already developed Q waves in leads II, III and aVF. One day after admission to hospital, she suddenly deteriorates with severe breathlessness, low blood pressure and sudden onset of pulmonary oedema. What is the most likely cause?
- A Acute papillary muscle rupture
 - B Acute pericarditis
 - C Atrial septal defect
 - D Free wall rupture
 - E Mural thrombus
- 10** A patient admitted to the emergency department with severe chest pain and ST segment deviation suddenly collapses and is found not to be breathing or have a pulse. A cardiac arrest call is made. What is the most likely cause of his collapse?
- A Asystole
 - B Complete heart block
 - C Free wall rupture
 - D Pulseless electrical activity
 - E Ventricular fibrillation
- 11** A 39 year old heavy smoker presents with calf pain on walking and is referred to a vascular surgeon for assessment. Which clinical feature would be most reassuring?
- A Capillary refill < 2 seconds
 - B Cold temperature
 - C Hair loss
 - D Pallor
 - E Pulselessness
- 12** Limb ischaemia can take many forms and has varied causes. This may result in sudden acute vessel occlusion from arterial spasm or thrombosis, or more chronic processes. What is the most likely underlying cause of severe limb ischaemia in an otherwise well, thin 30 year old heavy smoker?
- A Atherosclerosis
 - B Atrial fibrillation
 - C Buerger's disease
 - D Diabetes mellitus
 - E Raynaud's disease
- 13** An 81 year old non-smoker presents with chest pain and an ECG with ST segment depression. His troponin concentration is 456 ng/L (reference range < 34 ng/L). He is treated with an angioplasty and stent 2 days later. At the same time, a 60 year old smoker with diabetes has a large anterior ST segment elevation myocardial infarction, has ventricular fibrillation in the ambulance and has immediate defibrillation. He undergoes immediate percutaneous coronary intervention on arrival at hospital and has a troponin concentration of > 50 000 ng/L. A medical student asks who has the better prognosis. What is the biggest predictor of mortality following acute myocardial infarction?
- A Age
 - B Cardiac arrest
 - C ECG changes
 - D Smoking
 - E Troponin concentration
- 14** A 43 year old man has an extensive anterior myocardial infarction and has received antiplatelet, anticoagulant and statin therapy. He is referred for an echocardiogram. What will transthoracic echocardiography most usefully assess in this setting?
- A Cardiac arrhythmia
 - B Future prognosis
 - C Left ventricular function and the presence of mural thrombus
 - D Myocardial scar formation
 - E Thrombus in the left atrium
- 15** A 43-year-old man with an extensive anterior myocardial infarction undergoes coronary angiography. Which angiographic features predict the greatest benefit from coronary artery bypass graft surgery?
- A Diabetes mellitus and diffuse three-vessel coronary heart disease
 - B Left main stem stenosis and significant left ventricular systolic dysfunction
 - C Severe proximal disease of the left anterior descending coronary artery
 - D Three-vessel coronary heart disease with good left ventricular function
 - E Two-vessel coronary heart disease
- 16** A man with an extensive anterior myocardial infarction, left main stem and triple-vessel disease is referred for coronary artery bypass surgery. The surgeon wants to know how much of the anterior wall is scar rather than viable myocardium. Which imaging modality is best?
- A Computed tomography
 - B Coronary angiography
 - C Echocardiography
 - D MRI
 - E Stress echocardiography

- 17** An anaesthetist is seeking advice regarding a patient with coronary heart disease, diabetes mellitus and a murmur. Which of the following is NOT a significant risk factor for perioperative myocardial infarction during non-cardiac surgery?
- A Aortic stenosis with a peak gradient of 25 mmHg
 - B Diabetes mellitus treated with insulin and associated with renal failure
 - C Recent (within 4 weeks) stenting of a severe proximal stenosis in the left anterior descending coronary artery
 - D Recent acute coronary syndrome
 - E Severe left ventricular dysfunction
- 18** A 67 year old woman presents with predictable exertional angina pectoris when climbing steep inclines. She has been commenced on aspirin, statin and a B-blocker. She attends your clinic for assessment. Which of the following suggests the patient is at low risk of future events?
- A Poor exercise tolerance
 - B Poor left ventricular function
 - C Post-infarct angina
 - D Recent onset of symptoms
 - E ST segment depression during stage 3 of the Bruce Protocol
- 19** You review a 50 year old smoker 2 months after successful treatment for a myocardial infarction. Which intervention has the greatest benefit to prevent a recurrence of myocardial infarction?
- A ACE inhibitor therapy
 - B Aspirin
 - C Regular and frequent aerobic exercise
 - D Smoking cessation
 - E Statin therapy
- 20** A 55 year old man with a history of poorly controlled hypertension presents with a history of sudden-onset central chest pain. There are no diagnostic electrocardiogram (ECG) abnormalities, and an interval troponin concentration is not diagnostic of myocardial infarction. What diagnosis should be confirmed or excluded next?
- A Anxiety
 - B Aortic dissection
 - C Myocarditis
 - D Pericarditis
 - E Pneumothorax
- 21** Which of the following pathologies can be associated with an early diastolic murmur?
- A Long QT syndrome type 1
 - B Marfan's syndrome
 - C Mitral valve prolapse
 - D Myotonic dystrophy
 - E Wolff-Parkinson-White syndrome
- 22** A 75 year old man is incidentally found to have a pulsatile swelling in his abdomen on a routine health check. He is sent for an abdominal ultrasound scan, which confirms the presence of an abdominal aortic aneurysm. Which risk factor is protective against the formation and expansion of an abdominal aortic aneurysm?
- A Diabetes mellitus
 - B Family history of aneurysm disease
 - C Hypercholesterolaemia
 - D Hypertension
 - E Smoking
- 23** A 65 year old smoker with hypertension is found to have an abdominal aortic aneurysm on population screening with ultrasound. Which intervention will most reduce his future risk of aortic aneurysm rupture?
- A Angiotensin-converting enzyme (ACE) inhibitor
 - B B-blocker
 - C Smoking cessation
 - D Statin
 - E Warfarin
- 24** A short young woman presents with severe chest pain, vomiting and a sinus tachycardia. She is in the last trimester of pregnancy and has had normal blood pressure and observations at antenatal care. She is admitted for observation but is later found collapsed and in cardiac arrest. Despite attempts at resuscitation, mother and child die. Postmortem reveals an aortic dissection. What is the most likely underlying cause for the dissection?
- A Coarctation of the aorta
 - B Intramural haematoma
 - C Marfan's syndrome
 - D Pregnancy
 - E Undiagnosed hypertension
- 25** A 23 year old woman is referred for evaluation because she is very tall, has problems with her vision and has a heart murmur. Her family physician is concerned that she may have Marfan's syndrome. What structural gene is associated with an abnormality in Marfan's syndrome?
- A Fibrillin
 - B Myosin heavy chain
 - C Myosin-binding protein
 - D Titan
 - E Troponin

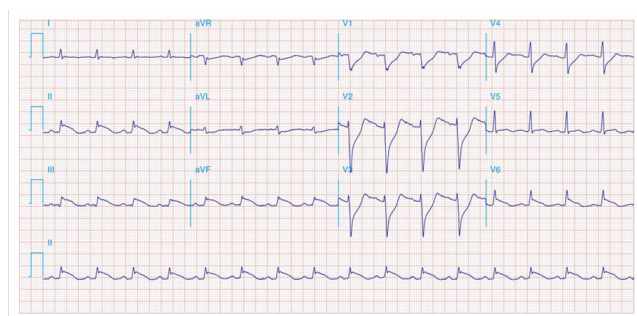
Answers 1-25

- 1 **E Recurrent pulmonary embolism** — JVP reflects right atrial pressure; recurrent PE causes pulmonary hypertension and RV strain, raising it. Dehydration lowers the JVP.
- 2 **E Rupture of the interventricular septum** — New harsh pansystolic murmur at the left sternal edge with collapse 2–5 days after anterior MI is septal rupture (VSD). Papillary rupture gives an apical MR murmur.
- 3 **A A normal baseline troponin and elevated 6-hour troponin level is suspicious of myocardial infarction** — Troponin may be normal early: a rise above the 99th centile with dynamic change on repeat (1–3 h hs-assay) supports MI. One normal test does not exclude it.
- 4 **E An 80 year old man with a history of anterior myocardial infarction 6 months previously; he is fit, has never experienced arrhythmia, and a cardiac magnetic resonance scan shows poor left ventricular function (left ventricular ejection fraction 28%)** — Primary-prevention ICD: LVEF $\leq 35\%$ ≥ 40 days after MI on optimal therapy with good life expectancy. VF within 48 h of MI is reversible; heart block needs a pacemaker.
- 5 **E Smoking** — Smoking is the lifestyle factor most strongly linked to MI (roughly doubles risk, more in the young). Stopping halves the excess risk within a few years.
- 6 **E Tissue plasminogen activator** — Fibrinolysis benefit falls steeply with time (best within 2–3 h, small after 12 h). PCI benefit is less time-dependent; aspirin and heparin are not.
- 7 **E Troponin I** — Troponin I/T are structural myofibrillar proteins, highly cardiac-specific and raised for 7–14 days. CK, CK-MB, AST and LDH are less specific enzymes.
- 8 **C Atrioventricular block** — RCA occlusion gives inferior STEMI and, as it supplies the SA and AV nodes, bradycardia and AV block. ST elevation in I, aVL, V6 is circumflex territory.
- 9 **A Acute papillary muscle rupture** — Sudden pulmonary oedema and hypotension after inferior MI is posteromedial papillary muscle rupture (single blood supply) with acute MR; echo, IABP, surgery.
- 10 **E Ventricular fibrillation** — Sudden pulseless collapse during acute ischaemia is most often VF; immediate defibrillation restores circulation. Free-wall rupture occurs days later.
- 11 **A Capillary refill < 2 seconds** — Capillary refill < 2 s shows good distal perfusion. Coldness, pallor, hair loss and absent pulses mean advanced ischaemia; ABPI < 0.9 confirms PAD.
- 12 **C Buerger's disease** — Buerger disease (thromboangiitis obliterans): segmental arteritis in young heavy smokers with digital ischaemia and thrombophlebitis; only stopping smoking halts it.
- 13 **A Age** — Age is the strongest predictor of death after MI and weighs most in the GRACE score; prompt PCI after resuscitated VF carries a better outlook.
- 14 **C Left ventricular function and the presence of mural thrombus** — After large anterior MI, echo assesses LV function (guides ACEi, MRA, ICD) and detects apical thrombus needing anticoagulation. MRI best shows scar.
- 15 **B Left main stem stenosis and significant left ventricular systolic dysfunction** — **UPDATED** CABG gives the greatest survival benefit in left main disease and multivessel disease with LV dysfunction. One- or two-vessel disease with good LV gains symptoms only.
- 16 **D MRI** — **UPDATED** Late gadolinium-enhanced MRI shows scar and its transmural extent: > 50% transmural scar is unlikely to recover after revascularisation.
- 17 **A Aortic stenosis with a peak gradient of 25 mmHg** — A peak aortic gradient of 25 mmHg is only mild AS. Recent ACS, recent stenting, severe LV dysfunction, insulin-treated diabetes and CKD raise perioperative risk.
- 18 **E ST segment depression during stage 3 of the Bruce Protocol** — Reaching Bruce stage 3 before ischaemia shows a high ischaemic threshold and low risk. Early ST depression or poor LV function means high risk.
- 19 **D Smoking cessation** — Stopping smoking after MI cuts death and reinfarction by a third to a half, more than any single drug; aspirin, statin and ACE inhibitor add further benefit.
- 20 **B Aortic dissection** — Sudden severe chest pain in a hypertensive with no MI on ECG/troponin: exclude aortic dissection before antithrombotics or thrombolysis, which could be fatal.
- 21 **B Marfan's syndrome** — An early diastolic murmur is aortic regurgitation; in Marfan syndrome aortic root dilatation stretches the valve ring. MVP gives a click and late systolic murmur.
- 22 **A Diabetes mellitus** — Diabetes is consistently linked to fewer and slower-growing AAAs. Smoking is the strongest risk factor; also male sex, age, family history and hypertension.
- 23 **C Smoking cessation** — Stopping smoking slows AAA growth and lowers rupture risk; drugs (ACEi, beta-blocker, statin) have not been shown to slow growth.
- 24 **D Pregnancy** — Pregnancy (third trimester/peripartum) predisposes to dissection through medial hormonal change and high cardiac output; this woman had no Marfan features.
- 25 **A Fibrillin** — Marfan syndrome is autosomal dominant, due to FBN1 (fibrillin-1) mutations: tall, ectopia lentis, aortic root dilatation and MVP. Sarcomere genes cause HCM.

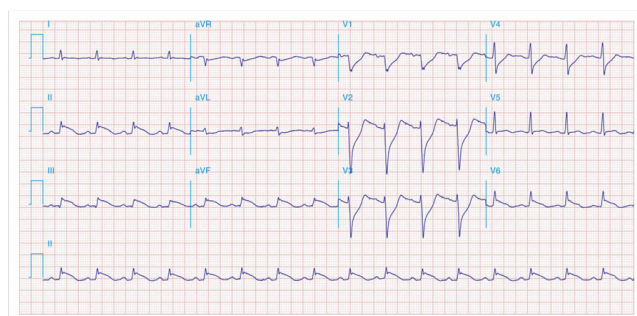
Harrison MCQs

SET 1 OF 2

- 1** A 56-year-old construction worker with hypertension and a history of smoking has had 30 minutes of nausea, breathlessness and chest pressure. His initial ECG is shown. Which of the following is NOT present on this ECG?

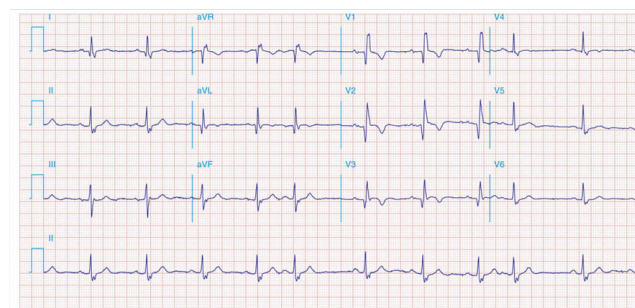


- A** Inferior myocardial ischaemia
B P waves
C Posterior myocardial ischaemia
D Sinus tachycardia
E Ventricular tachycardia
- 2** The ECG shown is from a 56-year-old man with 30 minutes of chest pressure and breathlessness. Which coronary artery is most likely occluded?



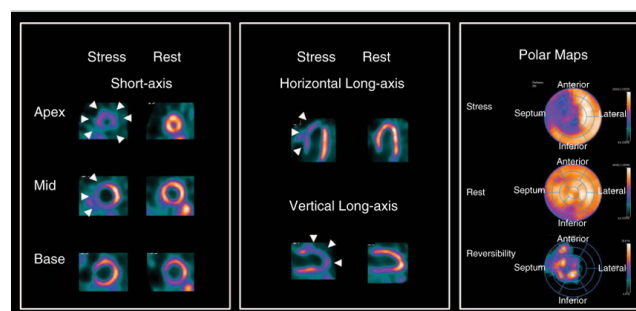
- A** First septal perforator
B Proximal left anterior descending artery
C Proximal left circumflex artery
D Proximal left anterior descending and left circumflex arteries
E Right coronary artery

- 3** A 48-year-old woman attends for a first visit. She reports 'some heart trouble' in the past and takes tablets for cholesterol and blood pressure. Her ECG is shown. Which statement about this ECG is true?



- A** It is likely she has suffered a prior myocardial infarction
B She is in normal sinus rhythm
C The presence of a left bundle branch block on this ECG indicates dyssynchronous mechanical contraction
D The presence of premature ventricular contractions and tachycardia is concerning for electrolyte imbalance
E The presence of anterior T-wave inversions is concerning in this ECG for acute myocardial ischaemia

- 4** A 45-year-old woman who smokes, with previous breast cancer treated by mastectomy and radiotherapy, has had 6 months of breathlessness, nausea and sweating when walking up a hill. Resting ECG and examination are normal, and an exercise SPECT (technetium-99m sestamibi) perfusion scan (shown) demonstrates reversible ischaemia. Which arterial territory is most likely involved?



- A** Left anterior descending artery
B Left circumflex artery
C Left main coronary artery
D Posterior descending artery
E Right coronary artery

- 5** A 62-year-old man with hypertension, hyperlipidaemia and a history of smoking had a large anterior MI 4 days ago treated with PCI to the LAD. His EF is 25%, he has had no ventricular arrhythmias, and he has bibasilar crackles and a raised JVP. He takes aspirin, clopidogrel and atorvastatin. All of the following will reduce his mortality over the next 40 days EXCEPT:
- A** Automated implantable defibrillator
 - B** Eplerenone
 - C** Lisinopril
 - D** Metoprolol
 - E** All of the above will reduce Mr. Wittstine's mortality
- 6** Ms. Wilson is a 66-year-old postmenopausal U.S. postal employee. While walking her mail route over the past 9 months, she has routinely experienced chest pressure and dyspnoea while climbing a certain steep hill. The pressure resolves when she rests for about 3 minutes. She has not missed any work due to these symptoms. You suspect she is experiencing angina. What term and Canadian Cardiovascular Society (CCS) Functional Class of angina is appropriate to describe her symptoms?
- A** Stable angina – CCS class I
 - B** Stable angina – CCS class II
 - C** Stable angina – CCS class III
 - D** Stable angina – CCS class IV
 - E** Unstable angina
- 7** A 50-year-old accountant with a history of cigarette use completed an exercise treadmill ECG stress test. During his test, he developed upsloping ST depressions in leads II, III, and aVF. His peak heart rate achieved was 130 bpm. His blood pressure increased from 120/80 to 155/95 mmHg. What is the appropriate interpretation of this stress test?
- A** Negative stress test; inappropriate hypertensive response to exercise
 - B** Negative stress test; nonspecific ST changes
 - C** Nondiagnostic stress test; normal blood pressure response
 - D** Positive stress test; inappropriate hypertensive response to exercise
 - E** Positive stress test; normal blood pressure response
- 8** A 70-year-old woman presents to the emergency department with 3 hours of acute shortness of breath and crushing substernal chest pressure. Her ECG shows 1-mm ST depression in leads II, III, and aVF without any ST elevation. Initial serum troponin level is mildly elevated. You suspect that she is having a non-ST-segment elevation myocardial infarction (NSTEMI). All of the following are causes of NSTEMI EXCEPT:
- A** Coronary vasospasm
 - B** Increased myocardial oxygen demand produced by systemic conditions
 - C** Microvascular endothelial dysfunction
 - D** Partially occluding thrombus forming on disrupted atherothrombotic coronary plaque
 - E** Progressive coronary atherosclerosis leading to severe mechanical obstruction
- 9** A 57-year-old man with hypertension, hyperlipidaemia and a history of smoking has had exertional chest pressure for 3 weeks, occurring with less exertion over the past week and tonight at rest, lasting about 10 minutes and radiating to the jaw. His ECG shows new T-wave inversion in I, II, aVL, V5 and V6. Which diagnosis is appropriate?
- A** Non-ST-segment elevation myocardial infarction
 - B** ST-segment elevation myocardial infarction
 - C** Stable angina
 - D** Unstable angina
 - E** More information is required before making a diagnosis
- 10** A 42-year-old man with anuric diabetic end-stage renal disease missed his last dialysis and is breathless. HR 105/min, BP 185/100 mmHg, bibasilar crackles and a raised JVP; the ECG shows sinus tachycardia without ST deviation, and troponin I is 0.14 ng/mL (normal < 0.06). What are the most appropriate next steps?
- A** Administer 80 mg of IV furosemide. Follow serial troponin I values and ECGs
 - B** Administer 80 mg of aspirin. Refer for pharmacologic nuclear stress testing
 - C** Initiate dual antiplatelet therapy (aspirin and clopidogrel) and heparin. Follow serial troponin I values and ECGs
 - D** Initiate dual antiplatelet therapy (aspirin and clopidogrel) and heparin. Refer for urgent coronary angiography
 - E** Initiate urgent hemodialysis with ultrafiltration for fluid removal. Follow serial troponin I values and ECGs

- 11** A 67-year-old fry chef at the local waffle house presents to the emergency department with the sudden onset of crushing substernal chest pain. He has a history of hypertension, hyperlipidaemia, and erectile dysfunction for which he takes daily amlodipine, simvastatin, and tadalafil. On your evaluation, his blood pressure is 145/85 mmHg, and heart rate is 90 bpm. His lungs are clear, and jugular venous pulse is not elevated. There are no murmurs or gallops on cardiac auscultation. He continues to complain of chest discomfort, rating it an 8/10. Initial troponin I result is elevated at 0.52 ng/dL (normal <0.06 ng/dL), and ECG reveals 1-mm ST depression in leads II, III, aVF, V5, and V6. All of the following pharmacologic therapies are reasonable to administer EXCEPT:

- A Aspirin orally
- B Clopidogrel orally
- C Heparin intravenously
- D Metoprolol intravenously
- E Nitroglycerin sublingually

- 12** You are evaluating a patient with chest pain in the emergency department. The ECG shows no Q waves or ST-segment changes. You wish to know whether the patient is suffering a myocardial infarction. Which of the following biomarkers is the preferred biochemical marker of myocardial infarction?

- A Creatine kinase
- B CK-MB band
- C Erythrocyte sedimentation rate
- D Lactate dehydrogenase
- E Troponin I

REPEATED 2* • ALSO ASKED IN
Crash Course

- 13** During coronary angiography a catheter-tip thrombus embolises and completely occludes the left anterior descending artery. If it were not removed, when would CK-MB first be expected to rise in a standard serum assay?

- A 5–10 minutes
- B 1–2 hours
- C 4–8 hours
- D 12–24 hours
- E 24–48 hours

- 14** The greatest delay between symptom onset and definitive therapy for patients with ST-segment elevation myocardial infarction usually occurs in what time interval?

- A Between onset of pain and patient's decision to call for help
- B Between patient call and emergency medical team arrival
- C Between emergency medical team arrival and arrival at the hospital
- D Between arrival at the hospital and decision to initiate reperfusion therapy
- E Between decision to initiate reperfusion therapy and the actual initiation of this therapy

- 15** A 61-year-old man presented to the emergency department 2 hours ago with anginal chest pain and ST-segment elevations in I, aVL, and V1–V3. At the time, the interventional cardiologist on call was involved in a complex case and was unable to take the patient for prompt primary percutaneous coronary intervention; thus tissue plasminogen activator (tPA) was administered. However, 120 minutes after administration of tPA, the patient continues to complain of chest pain, and his ECG shows no resolution of ST-segment elevations. What is the next most appropriate step?

- A Administer a second dose of tPA
- B Administer streptokinase 1.5 million units over 1 hour
- C Proceed with urgent coronary artery bypass grafting
- D Proceed with urgent percutaneous coronary intervention
- E Wait for 60 more minutes before pursuing any other therapeutic options

- 16** A 65-year-old woman with hypertension has exertional breathlessness and chest pressure after 2–3 blocks. Angiography shows high-grade focal stenoses in the proximal LAD, circumflex and right coronary arteries. When counselling her about PCI versus CABG, which statement is true?

- A CABG has a lower mortality rate at 1 year than PCI for patients like Ms. Constance
- B At 1 year, patients like Ms. Constance undergoing CABG have a higher risk of needing other procedures for revascularization than patients undergoing PCI
- C PCI carries a higher risk of stroke than CABG
- D The characteristics of coronary anatomy and the cardiologist's anticipation of success with PCI should factor into the decision
- E PCI has a lower rate of myocardial infarction at 1 year than CABG for patients like Ms. Constance

- 17** A man with exertional chest pressure has infero-apico-posterior ischaemia on a perfusion scan. Angiography shows a left-dominant system with a chronic total occlusion (CTO) of the circumflex artery, filled by collaterals from the LAD; the operator cannot cross it with an antegrade wire. Which statement about treating the CTO is true?

- A If the CTO cannot be crossed with any wire in the antegrade approach, the operator should always abandon percutaneous repair and consider referral for coronary bypass surgery
- B Attempting to access the CTO via a retrograde approach through a collateral vessel is an option when attempting percutaneous revascularization
- C Although the presence of a CTO is associated with a higher mortality rate, successful PCI of a CTO has never been shown to lower this mortality rate
- D Coronary artery bypass surgery of a CTO is rarely successful due to the atretic nature of the vessel distal to the chronically occluded segment
- E In studies, successful revascularization of chronic total coronary occlusions is associated with symptom relief but no improvement in left ventricular function

18 A 67-year-old man had CABG 12 years ago (LIMA to LAD and three saphenous vein grafts). He now has a month of crescendo angina, and angiography shows a significant stenosis in the body of the vein graft to the first obtuse marginal branch. Which statement is true?

- A** Graft failure of an SVG is almost always from thrombus embolized from the aortic anastomosis site
- B** PCI to this saphenous vein graft will require deployment of a distal protective device
- C** PCI on a saphenous vein graft is too risky to warrant proceeding. Aggressive medical management with antianginal therapy should be pursued
- D** The risk of PCI to an SVG is identical to native-vessel PCI
- E** Venous and arterial graft conduits (for example, SVG and LIMA grafts) have identical patency rate over the first 5 years following bypass grafting

19 A 75-year-old man has diagnostic coronary angiography via the femoral artery. A week later he has abdominal pain, nausea and reduced urine output, a temperature of 38.3°C and livedo reticularis on both legs. Creatinine is 2.7 mg/dL (previously 1.1), eosinophils 21% and ESR 92 mm/h. What is the most likely diagnosis?

- A** Acute interstitial nephritis
- B** Atheroembolic renal disease
- C** Churg-Strauss syndrome
- D** Contrast-induced nephropathy
- E** Hypereosinophilic syndrome

REPEATED 2x • ALSO ASKED IN
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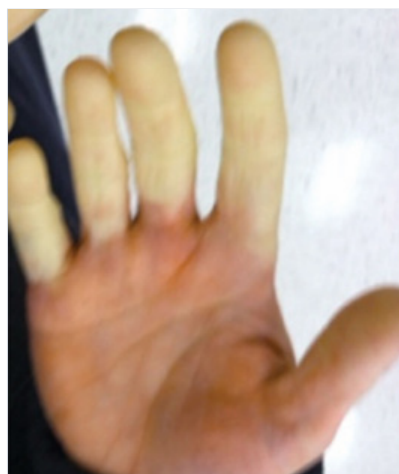
20 A 75-year-old smoker with diabetes and hypertension has burning calf pain after walking two blocks, relieved by rest, and nocturnal calf aching relieved by sitting up. Which statement is most likely true?

- A** He has critical aorto-iliac stenosis
- B** His pulse-volume recording will be sharp and spiked
- C** The most likely cause is fibromuscular dysplasia of the femoral arteries
- D** The next investigation should be MR angiography
- E** His ankle-brachial pressure index is likely to be below 0.9

21 A 75-year-old smoker with diabetes and hypertension has symptomatic peripheral arterial disease with an ankle-brachial index of 0.82 on both sides. Which statement about his prognosis and treatment is true?

- A** Anticoagulation with warfarin is superior to antiplatelet agents in preventing adverse cardiovascular events in patients with PAD
- B** B-Adrenergic blockers have been shown to worsen limb ischaemia and should not be used in PAD
- C** He should exercise daily, walking to the point of maximal claudication before stopping to rest and allowing his symptoms to resolve
- D** His greatest risk of morbidity and mortality over the ensuing 5 years is progression to critical limb ischaemia
- E** Vasodilators are first-line therapy for symptomatic PAD

22 A 32-year-old construction foreman presents with exertional pain in his bilateral forearms and hands. He smokes one pack per day but otherwise has no significant past medical history. He has an easily palpable brachial pulse but very faint radial and ulnar pulses. A picture of his hand is shown. Angiography of his upper extremity reveals smooth tapering segmental lesions in the small distal arterial vessels. Which of the following treatments has the greatest chance of success?



- A** Cilostazol
- B** Enoxaparin
- C** Prednisone
- D** Smoking cessation
- E** Warfarin

23 Which of the following statements regarding chronic venous disease is true?

- A** More men than women suffer from varicose veins
- B** More men than women suffer from chronic venous insufficiency with oedema
- C** More than 50% of patients over 70 years old have chronic venous insufficiency
- D** Most patients with chronic venous insufficiency will develop venous ulcers
- E** All of the above are true

24 Which of the following is the longest vein in the body?

- A** Azygous vein
- B** Greater saphenous vein
- C** Inferior vena cava
- D** Left common iliac vein
- E** Right common femoral vein

- 25** You are evaluating a 77-year-old woman from Ohio with a history of heart failure with preserved ejection fraction, diabetes mellitus, and a prior left lower extremity deep venous thrombosis treated with anticoagulation. For the past several months, she has complained of a cramping, burning sensation in her legs. Today she presents for evaluation of a skin ulcer, as shown. On examination, there is ulceration over the medial malleolus with nonpitting woody oedema and skin hyperpigmentation. This ulcer is most likely due to which of the following?



- A Arachnid envenomation
- B Bacillus anthracis infection
- C Chronic venous insufficiency
- D Diabetic foot ulcer
- E Peripheral arterial disease

Answers 1-25

HARRISON MCQs

- 1** **E Ventricular tachycardia** — The ECG shows sinus tachycardia with anterior ST elevation and inferior reciprocal depression (anterior STEMI). Narrow QRS with P waves excludes VT.
- 2** **B Proximal left anterior descending artery** — **UPDATED** ST elevation in V1–V4 with reciprocal inferior depression means proximal LAD occlusion (anterior wall and septum). RCA gives inferior, circumflex lateral/posterior changes.
- 3** **A It is likely she has suffered a prior myocardial infarction** — Pathological Q waves (> 40 ms or > ¼ of the R wave) in contiguous leads mean an old transmural infarct; echo can confirm a regional wall-motion defect.
- 4** **A Left anterior descending artery** — A reversible defect in the apex, anterior wall and septum is LAD territory. Mediastinal radiotherapy accelerates coronary disease, especially of the proximal LAD.
- 5** **A Automated implantable defibrillator** — ACE inhibitors, beta-blockers and eplerenone cut early post-MI mortality; an ICD in the first 40 days does not (DINAMIT, IRIS). Reassess EF after 40 days.
- 6** **B Stable angina – CCS class II** — Predictable exertional angina limiting ordinary activity slightly (e.g. walking uphill, stairs quickly) is CCS class II. Class III = one flight at normal pace.
- 7** **C Nondiagnostic stress test; normal blood pressure response** — The test needs ≥ 85% of age-predicted maximum HR (~145/min at 50); a peak of 130 is non-diagnostic. A BP rise to 155/95 on exercise is normal.
- 8** **C Microvascular endothelial dysfunction** — NSTEMI causes: non-occlusive plaque thrombus (commonest), spasm, progressive narrowing, inflammation and supply–demand mismatch. Microvascular dysfunction causes microvascular angina.
- 9** **E More information is required before making a diagnosis** — Crescendo then rest angina with new T inversion is NSTEMI-ACS; only troponin separates NSTEMI from unstable angina, so the diagnosis awaits it (treatment starts now).
- 10** **E Initiate urgent hemodialysis with ultrafiltration for fluid removal. Follow serial troponin I values and ECGs** — A small troponin rise with fluid overload in an anuric dialysis patient is demand injury: urgent dialysis with ultrafiltration and serial troponin/ECG. Furosemide fails in anuria.
- 11** **E Nitroglycerin sublingually** — Nitrates are contraindicated within 48 h of tadalafil (24 h for sildenafil): profound hypotension. Aspirin, P2Y12 inhibitor, anticoagulant and beta-blocker are fine.

- 12 E Troponin I** — High-sensitivity troponin I/T is the preferred MI marker: a rise/fall with one value > 99th centile plus ischaemic evidence. CK-MB and LDH are less specific.
- 13 C 4–8 hours** — CK-MB rises 4–8 h after infarction, peaks ~24 h and normalises by 48–72 h (useful for reinfarction). Myoglobin rises earliest but is non-specific.
- 14 A Between onset of pain and patient's decision to call for help** — The longest delay in STEMI care is the patient's delay in calling for help, longest in the elderly, women and diabetics; system delays are now short.
- 15 D Proceed with urgent percutaneous coronary intervention** — ST elevation resolving < 50% at 60–90 min after fibrinolysis with ongoing pain is failed reperfusion: immediate rescue PCI. Repeat lysis is ineffective.
- 16 D The characteristics of coronary anatomy and the cardiologist's anticipation of success with PCI should factor into the decision** — PCI vs CABG in multivessel disease depends on anatomy (SYNTAX), diabetes, LV function and chance of complete revascularisation: a heart-team decision.
- 17 B Attempting to access the CTO via a retrograde approach through a collateral vessel is an option when attempting percutaneous revascularization** — When antegrade wiring of a CTO fails, a retrograde approach through collaterals is an option. CTO-PCI relieves angina; no proven mortality benefit.
- 18 B PCI to this saphenous vein graft will require deployment of a distal protective device** — PCI to a degenerated vein graft risks distal embolisation and no-reflow, so use an embolic protection device. LIMA grafts outlast vein grafts (> 90% vs ~50% at 10 years).
- 19 B Atheroembolic renal disease** — Days–weeks after catheterisation: livedo, blue toes, AKI, eosinophilia and high ESR = cholesterol atheroembolism. Contrast nephropathy starts within 48–72 h without eosinophilia.
- 20 E His ankle–brachial pressure index is likely to be below 0.9** — **UPDATED** Claudication with night pain eased by dependency is PAD; ABPI < 0.9 confirms it (< 0.4 critical). Calcified vessels in diabetes can falsely raise the ABPI.
- 21 C He should exercise daily, walking to the point of maximal claudication before stopping to rest and allowing his symptoms to resolve** — **UPDATED** Supervised exercise to near-maximal claudication is first line (NICE CG147). His main risk is MI or stroke; add clopidogrel and a statin. Few progress to limb loss.
- 22 D Smoking cessation** — Distal limb ischaemia with tapering small-artery occlusions in a young smoker is Buerger disease; complete tobacco abstinence is the only effective treatment.
- 23 B More men than women suffer from chronic venous insufficiency with oedema** — Varicose veins are commoner in women, but chronic venous insufficiency with oedema is slightly commoner in men. Venous ulcers affect only 1–2% of adults.
- 24 B Greater saphenous vein** — The great saphenous vein, from the medial foot to the saphenofemoral junction, is the longest vein; it is the usual CABG conduit and the commonest site of varicosities.
- 25 C Chronic venous insufficiency** — A medial malleolar ulcer with haemosiderin pigment, lipodermatosclerosis and oedema after DVT is venous; treat with compression once ABPI is adequate.

SET 2 OF 2

- 26** An 84-year-old woman with diabetes and hyperlipidaemia had an anterior STEMI 3 weeks ago, treated with primary PCI and a drug-eluting stent to the left anterior descending artery. She has been well since on metoprolol, aspirin, clopidogrel, rosuvastatin, lisinopril and insulin. Her ophthalmologist has scheduled elective cataract surgery for next week but considers the bleeding risk too high while she takes both antiplatelet drugs, and asks your advice. What should you recommend?
- A** Stop aspirin, continue clopidogrel and proceed with surgery
- B** Stop clopidogrel, continue aspirin and proceed with surgery
- C** Stop both antiplatelet drugs and proceed with surgery
- D** Postpone surgery for at least 6 months, preferably 12, when clopidogrel can be stopped
- E** Proceed with surgery on both drugs, as she will never be able to stop either
- 27** Four days after an open aortobifemoral bypass, a CT scan shows a pseudoaneurysm just proximal to the aortic origin of the graft. Which of the following best defines a pseudoaneurysm?
- A** A focal dilation in a vessel in which the intimal and medial layers are disrupted and the dilated segment is lined by adventitia only
- B** A focal dilation of a vessel only involving one portion of the circumference
- C** Apparent dilation of a vessel due to intrinsic narrowing proximal and distal to the point of apparent narrowing
- D** Dilation of a vessel, though not to the size necessary to be diagnosed as a true aneurysm
- E** The appearance of an aneurysmal dilation of a vessel on imaging due to the angulation of the artery and imaging technique
- 28** All of the following factors are associated with elevated risk for the development of an aortic aneurysm EXCEPT:
- A** Aging
- B** Cigarette smoking
- C** Female sex
- D** Hypercholesterolemia
- E** Hypertension

- 29** You are seeing Mr. Walker in clinic for follow-up. He is a 22-year-old man with a family history of sudden death and arterial aneurysms. He is quite tall with widely spaced eyes and a broad uvula. On his first visit, you order a transthoracic echocardiogram to investigate a diastolic murmur and note an aortic root aneurysm of 4.9 cm. He most likely has a mutation in which of the following genes?
- A** Fibrillin-1
 - B** SMAD3
 - C** Smooth muscle-specific α -actinin
 - D** TGF- β receptor
 - E** Type III procollagen
- 30** Which of the following patients with aortic dissection or hematoma is best managed without surgical therapy?
- A** A 45-year-old female with a dissection involving the aorta distal to the great vessel origin but cephalad to the renal arteries
 - B** A 74-year-old male with a dissection involving the root of the aorta
 - C** A 58-year-old male with aortic dissection involving the distal aorta and the bilateral renal arteries
 - D** A 69-year-old male with an intramural hematoma within the aortic root
 - E** All of the above patients require surgical management of their aortic disease
- 31** An incidental CT finding in a 32-year-old woman is dilatation of the ascending aorta to 4.3 cm. All of the following are recognised causes of this finding EXCEPT:
- A** Giant cell arteritis
 - B** Rheumatoid arthritis
 - C** Syphilis
 - D** Systemic lupus erythematosus
 - E** Takayasu arteritis

Answers 26–31

HARRISON MCQS

- 26** **D Postpone surgery for at least 6 months, preferably 12, when clopidogrel can be stopped** — After ACS with a drug-eluting stent, DAPT continues ~12 months; stopping clopidogrel at 3 weeks risks fatal stent thrombosis. Defer elective surgery (≥ 6 , ideally 12 months).
- 27** **A A focal dilation in a vessel in which the intimal and medial layers are disrupted and the dilated segment is lined by adventitia only** — A pseudoaneurysm is a contained rupture held only by adventitia or surrounding tissue (after puncture, surgery, trauma). A true aneurysm involves all three layers.
- 28** **C Female sex** — Degenerative aortic aneurysms are several times commoner in men, so female sex is not a risk factor. Age, smoking, hypertension and family history are.
- 29** **D TGF- β receptor** — Aortic root aneurysm with hypertelorism, bifid uvula and familial sudden death is Loeys–Dietz (TGF- β R1/2); aneurysms dissect smaller, so repair earlier than Marfan.
- 30** **A A 45-year-old female with a dissection involving the aorta distal to the great vessel origin but cephalad to the renal arteries** — Uncomplicated type B dissection (beyond the left subclavian) is treated medically with IV beta-blockade. Malperfusion, rupture or expansion need repair; any ascending involvement needs surgery.
- 31** **D Systemic lupus erythematosus** — Ascending aortic dilatation: GCA, Takayasu, syphilis, spondyloarthritis, Marfan/Loeys–Dietz/EDS, bicuspid valve. SLE is not a recognised cause.

12 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1 A 48-year-old woman has a routine health check: total cholesterol 5.8 mmol/L, triglycerides 2.7 mmol/L, BMI 29 kg/m², non-smoker, 10 units of alcohol a week, no personal or family history, and a QRISK3 score of 12%. Which step in her initial management is LEAST needed?
 - A Low carbohydrate diet
 - B High consumption of monounsaturated fats
 - C Regular aerobic exercise
 - D Start medical therapy with statins
 - E Decrease body mass index to <25
- 2 A 50-year-old lawyer presents to the cardiology outpatient clinic with recurrent episodes of dull central chest pain radiating up to her jaw and down her left shoulder, associated with sweating and nausea following exertion and alleviated by rest. She is a heavy smoker and her father died of a heart attack in his 40s. What is the most definitive diagnostic test in this case?
 - A Echocardiography
 - B Stress testing with echocardiography
 - C Stress testing with myocardial perfusion scintigraphy
 - D Coronary angiography
 - E Electrocardiography during an attack
- 3 A 55-year-old man presents with acute central chest pain, radiating to his neck and left shoulder, and associated with one episode of vomiting. His medical history includes hypertension and hyperlipidaemia. ECG and troponin show evidence of an acute myocardial infarction (MI). Primary percutaneous coronary intervention is unavailable within close distance; therefore a decision is made to perform thrombolytic therapy. This therapy is most beneficial when done within which of the following time limits from the onset of pain?
 - A Within 2 days
 - B Within 18 hours
 - C Within 12 hours
 - D Within 24 hours
 - E Within 36 hours
- 4 You are asked to review an 89-year-old arteriopath who complains of severe pain at rest in both feet, the toes of which are cold and purple. He has had a recent course of heparin and aspirin for suspected unstable angina. The major arterial pulses are present in the limbs, but there are areas of lacelike purplish discoloration on the skin over both knees. What is the most likely explanation for these findings?
 - A Allergic reaction to the heparin
 - B Cholesterol embolism
 - C Haemorrhage from the anticoagulants
 - D Heparin-induced thrombocytopenia
 - E Sepsis
- 5 A 53-year-old man comes to see his GP complaining of chest pain. The GP performs an ECG which demonstrates ischaemic changes. The following are all risk factors for ischaemic heart disease EXCEPT?
 - A Hyperlipidaemia
 - B Smoking
 - C Hypertension
 - D High homocysteine levels
 - E High dietary folic acid intake
- 6 A 56-year-old woman presents to AE complaining of sudden-onset central chest pain. She comments that it came on at rest. She is referred for an exercise test and coronary angiogram. The exercise test is stopped early due to chest pain and ST depression in the lateral leads. However, she has normal coronary arteries on the angiogram. What is the most likely diagnosis?
 - A Decubitus angina
 - B Prinzmetal angina
 - C Cardiac syndrome X
 - D Unstable angina
 - E Stable angina

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Passmedicine

- 7 You are on call in AE when a patient with sudden-onset chest pain is admitted. You request an ECG which shows evidence of T-wave inversion in leads V2–V6 but no evidence of ST elevation. You prescribe aspirin, enoxaparin and clopidogrel. What is the most appropriate dose of enoxaparin?
 - A 40 mg once a day
 - B 20 mg once a day
 - C 1 mg/kg once a day
 - D 1.5 mg/kg twice a day
 - E 1 mg/kg twice a day

- 8** You are a house officer on call when you are asked to see a patient with chest pain. The patient tells you that he suffers from angina and this pain is similar to previous episodes of chest pain. An ECG is performed which rules out an MI. What is the next most appropriate step in management?
- A Atenolol
 - B Glyceryl trinitrate (GTN)
 - C Diltiazem
 - D Nicorandil
 - E Aspirin
- 9** An ECG is performed on a 45-year-old man as part of a routine medical check-up. His past medical history includes a previous MI some years ago. The ECG shows presence of Q waves. Q waves in inferior MI appear in which ECG leads?
- A I and II
 - B II, III and aVF
 - C V1, V2 and V3
 - D I, aVF and V6
 - E V4, V5 and V6
- 10** A 56-year-old man presents with central chest pain at rest. He comments that he feels short of breath and nauseous. An ECG demonstrates tall R waves, ST segment depression and tall T waves in leads V1 and V2. What is the most likely diagnosis?
- A Anterior myocardial infarct
 - B Lateral myocardial infarct
 - C Inferior myocardial infarct
 - D Antero-lateral myocardial infarct
 - E Posterior myocardial infarct
- 11** Which ECG change would lead you to initiate thrombolytic therapy?
- A Right bundle-branch block
 - B ST segment elevation in left ventricular leads
 - C Widespread ST segment depression
 - D Q waves in septal leads
 - E Prolongation of the QT interval
- 12** A 43-year-old man has an acute MI treated with immediate angioplasty. All of the following are LATE complications of myocardial infarction (weeks to months) EXCEPT:
- A Dressler syndrome
 - B Left ventricular aneurysm
 - C Mural thrombus with systemic embolism
 - D Ventricular septal rupture
 - E Chronic heart failure

Answers 1-12

CRASH COURSE MCQS

- 1** **D Start medical therapy with statins** — NICE: lifestyle advice comes first for raised CV risk; atorvastatin 20 mg is then offered at QRISK3 $\geq 10\%$ as a shared decision, so it is the least urgent step.
- 2** **D Coronary angiography** — Invasive coronary angiography is the reference standard and allows PCI. NICE uses CT coronary angiography first, then functional imaging if non-diagnostic.
- 3** **C Within 12 hours** — Fibrinolysis is indicated within 12 h of STEMI onset when PCI is not possible within 120 min; it works best in the first 3 h.
- 4** **B Cholesterol embolism** — Painful blue toes with livedo but palpable pulses after anticoagulation or instrumentation is cholesterol embolism; it may also cause AKI and eosinophilia.
- 5** **E High dietary folic acid intake** — Lipids, smoking, hypertension and high homocysteine are risk factors. Folic acid lowers homocysteine and is not a risk factor.
- 6** **C Cardiac syndrome X** — Typical angina with exercise ST depression but normal angiogram is cardiac syndrome X (microvascular angina), commoner in women. Prinzmetal gives ST elevation at rest.
- 7** **E 1 mg/kg twice a day** — ACS dose of enoxaparin is 1 mg/kg SC 12-hourly (once daily if CrCl < 30). 40 mg daily is prophylaxis; fondaparinux 2.5 mg is preferred if no early angiography.
- 8** **B Glyceryl trinitrate (GTN)** — Treat an angina attack with sublingual GTN; call an ambulance if pain persists 5 min after the second dose. Beta-blockers or CCBs prevent attacks.
- 9** **B II, III and aVF** — The inferior wall is seen by II, III and aVF (usually RCA). Septal = V1–V2, anterior = V3–V4, lateral = I, aVL, V5–V6.
- 10** **E Posterior myocardial infarct** — Posterior MI mirrors in V1–V2: tall R, ST depression and upright T; V7–V9 show ST elevation. Treat as STEMI (circumflex or RCA).
- 11** **B ST segment elevation in left ventricular leads** — Reperfusion is indicated for ischaemic pain with ST elevation in 2 contiguous leads, new LBBB or true posterior MI. ST depression alone is not an indication.
- 12** **D Ventricular septal rupture** — **UPDATED** Septal, free-wall and papillary rupture are early complications (2–7 days). Dressler syndrome, LV aneurysm, mural thrombus and heart failure come weeks later.

10 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1** A 60-year-old male patient is receiving aspirin, an angiotensin-converting enzyme inhibitor, nitrates, and a beta-blocker for chronic stable angina. He presents to the ER with an episode of more severe and long-lasting anginal chest pain each day over the past 3 days. His ECG and cardiac enzymes are normal. Which of the following is the best course of action?

A Admit the patient and add intravenous digoxin
 B Admit the patient and begin low-molecular-weight heparin
 C Admit the patient for thrombolytic therapy
 D Admit the patient for observation with no change in medication
 E Increase the doses of current medications and follow closely as an outpatient

REPEATED 2* • ALSO ASKED IN
Other Sources

- 2** A 42-year-old man who smokes has sudden crushing central chest pain, nausea, sweating and breathlessness; the ECG shows anteroseptal ST elevation but cardiac enzymes stay normal. Emergency angiography shows only a 25% LAD stenosis and no intervention is performed. Which treatment would most reduce the risk of further episodes?

A Placement of a percutaneous drug-eluting coronary artery stent
 B Placement of a percutaneous non-drug-eluting coronary artery stent
 C Beginning therapy with an ACE inhibitor
 D Beginning therapy with a beta-blocker
 E Beginning therapy with a calcium-channel blocker

REPEATED 2* • ALSO ASKED IN
Other Sources

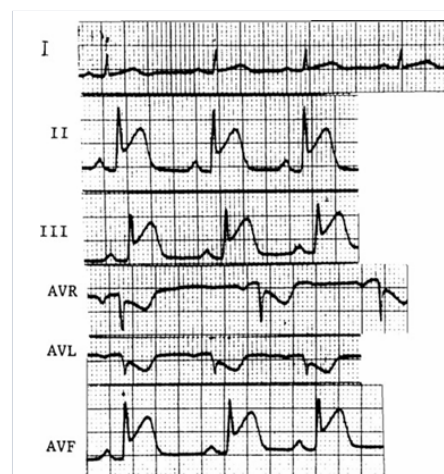
- 3** A 55-year-old man admitted with chest pain has normal cardiac enzymes, BP 140/90–150/100 mmHg, a soft early systolic murmur at the upper left sternal edge, and a resting ECG showing left ventricular hypertrophy with secondary ST-T changes. Why is an exercise ECG not an appropriate test for him?

A Anticipated difficulty with the patient's anxiety (ie, he might falsely claim chest pain during the test)
 B Increased risk associated with high blood pressure readings
 C Concern about the heart murmur, a relative contraindication to stress testing
 D The presence of LVH with ST-T changes on baseline ECG
 E Concern that this represents the onset of unstable angina with unacceptable risk of MI with stress testing

- 4** A 70-year-old male with a history of coronary artery disease presents to the emergency department with 2 hours of substernal chest pressure, diaphoresis, and nausea. He reports difficulty "catching his breath." An electrocardiogram shows septal T-wave inversion. The patient is given 325-mg aspirin and sublingual nitroglycerin while awaiting the results of his blood work. His troponin I is 0.65 ng/mL (normal < 0.04 ng/mL). The physician in the emergency department starts the patient on low-molecular-weight heparin. His pain is 3/10. Blood pressure is currently 154/78 and heart rate is 72. You are asked to assume care of this patient. What is the best next step in management?

A Arrange emergency cardiac catheterisation
 B Begin intravenous thrombolytic therapy
 C Admit to a monitored bed and repeat enzymes and ECG in 6 hours without further treatment
 D Begin intravenous furosemide
 E Load a P2Y12 inhibitor (clopidogrel 300–600 mg or ticagrelor 180 mg)

- 5** A 55-year-old obese woman develops pressure-like substernal chest pain lasting 1 hour. She works as a housekeeper; lifting and exertion have precipitated similar pain in the recent past. There is a positive family history of gallstones (mother and sister). Her ECG is shown below. Which of the following is the most likely diagnosis?

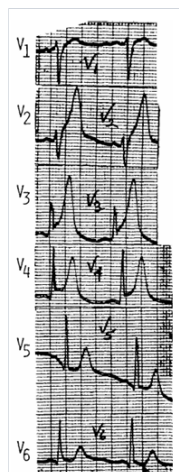


A Costochondritis
 B Acute anterior myocardial infarction
 C Acute inferior myocardial infarction
 D Pericarditis
 E Gastroesophageal reflux

6 A 68-year-old male complains of pain in his calves while walking. He notes bilateral foot pain, which awakens him at night. His blood pressure is 117/68. Physical examination reveals diminished bilateral lower extremity pulses. An ankle:brachial index measures 0.6. The patient's current medications include aspirin and hydrochlorothiazide. Which of the following is the best initial management plan for this patient's complaint?

- A** Smoking cessation therapy, warfarin
- B** Smoking cessation therapy, graduated exercise regimen, cilostazol
- C** Smoking cessation therapy, schedule an arteriogram
- D** Smoking cessation therapy, warfarin, peripherally acting calcium-channel blocker
- E** Smoking cessation therapy, consultation with a vascular surgeon

7 A 48-year-old male with a history of hypercholesterolemia presents to the ER after 1 hour of substernal chest pain, nausea, and sweating. His ECG is shown below. There is no history of hypertension, stroke, or any other serious illness. Which of the following is most appropriate at this time?



- A** Aspirin, calcium-channel blocker, morphine and primary PCI
- B** Aspirin, beta-blocker and morphine
- C** Aspirin, ticagrelor, heparin and primary PCI
- D** Aspirin and IV thrombolysis after 3 hours of observation
- E** Aspirin and beta-blocker

8 A 67-year-old male presents to your office after community ultrasound screening revealed an aortic aneurysm measuring 3.0×3.5 cm. Physical examination confirms a palpable, pulsatile, nontender abdominal mass just above the umbilicus. The patient's medical conditions include hypertension, hyperlipidaemia, and tobacco use. What is the best recommendation for the patient to consider?

- A** Watchful waiting is the best course until the first onset of abdominal pain
- B** Surgery is indicated except for the excess operative risk represented by the patient's risk factors
- C** Serial follow-up with ultrasound, CT, or MRI is indicated, with the major determinant for surgery being aneurysmal size greater than 5 to 6 cm
- D** Serial follow-up with ultrasound, CT, or MRI is indicated, with the major determinant for surgery being involvement of a renal artery
- E** Unlike stents in coronary artery disease, endovascular stent grafts have proven unsuccessful in the management of AAAs

9 A 70-year-old female has been healthy except for hypertension treated with a thiazide diuretic. She presents with sudden onset of a severe, tearing chest pain, which radiates to the back and is associated with dyspnoea and diaphoresis. Blood pressure is 210/94. Lung auscultation reveals bilateral basilar rales. A faint murmur of aortic insufficiency is heard. The BNP level is elevated at 550 pg/mL (Normal < 100). ECG shows nonspecific ST-T changes. Chest x-ray suggests a widened mediastinum. Which of the following choices represents the best initial management?

- A** IV furosemide plus IV loading dose of digoxin
- B** Percutaneous coronary intervention with consideration of angioplasty and/or stenting
- C** Blood cultures and rapid initiation of vancomycin plus gentamicin, followed by echocardiography
- D** IV beta-blocker to control heart rate, IV nitroprusside to control blood pressure, transesophageal echocardiogram
- E** IV heparin followed by CT pulmonary angiography

10 A 67-year-old male presents to your clinic to establish primary care; he is asymptomatic. He has a history of hypertension for which he takes hydrochlorothiazide. His father had a myocardial infarction at age 62. The patient smoked until 5 years ago, but has been abstinent from tobacco since then. His blood pressure in the office today is 132/78. Aside from being overweight, the remainder of the physical examination is unremarkable. Which of the following preventive health interventions would be most appropriately offered to him today?

- A** Carotid ultrasound to evaluate for carotid artery stenosis
- B** Abdominal ultrasound to evaluate for aortic aneurysm
- C** Lipoprotein(a) assay to evaluate coronary heart disease risk
- D** Exercise (treadmill) stress testing to evaluate for coronary artery disease
- E** Homocysteine level to evaluate coronary heart disease risk

Answers 1–10

PRETEST MCQs

- 1 **B** Admit the patient and begin low-molecular-weight heparin — Crescendo rest angina with normal ECG and troponin is unstable angina: admit and add anticoagulation (LMWH/fondaparinux). Thrombolysis is only for ST elevation.
- 2 **E** Beginning therapy with a calcium-channel blocker — Transient ST elevation with near-normal coronaries is vasospasm: a CCB (diltiazem/nifedipine) with nitrates and stopping smoking. Non-selective beta-blockers may worsen spasm.
- 3 **D** The presence of LVH with ST-T changes on baseline ECG — Resting ST-T changes (LVH strain, LBBB, digoxin, WPW, pacing) make exercise ECG uninterpretable; use functional imaging or CT coronary angiography.
- 4 **E** Load a P2Y12 inhibitor (clopidogrel 300–600 mg or ticagrelor 180 mg) — UPDATED Stable NSTEMI already on aspirin, nitrate and LMWH: load a P2Y12 inhibitor to complete DAPT, then angiography within 24 h. Lysis is harmful without ST elevation.
- 5 **C** Acute inferior myocardial infarction — ST elevation in II, III, aVF with reciprocal depression in aVL is acute inferior STEMI. Pericarditis gives widespread concave ST elevation with PR depression.
- 6 **B** Smoking cessation therapy, graduated exercise regimen, cilostazol — Claudication (ABPI 0.6): stop smoking, statin, antiplatelet and supervised exercise; add cilostazol or naftidrofuryl if needed. Revascularise only if lifestyle-limiting or critical.
- 7 **C** Aspirin, ticagrelor, heparin and primary PCI — UPDATED Anterior STEMI within 1 h: aspirin, potent P2Y12 inhibitor (ticagrelor/prasugrel), anticoagulation and immediate primary PCI.
- 8 **C** Serial follow-up with ultrasound, CT, or MRI is indicated, with the major determinant for surgery being aneurysmal size greater than 5 to 6 cm — A 3.0–3.5 cm AAA is monitored with ultrasound while treating risk factors; repair at ≥ 5.5 cm, growth > 1 cm/year or symptoms. EVAR is an alternative to open repair.
- 9 **D** IV beta-blocker to control heart rate, IV nitroprusside to control blood pressure, transesophageal echocardiogram — Tearing back pain, severe hypertension, new AR and widened mediastinum is proximal dissection: IV beta-blocker, then nitroprusside, urgent TOE/CT and surgery.
- 10 **B** Abdominal ultrasound to evaluate for aortic aneurysm — One-off ultrasound AAA screening is offered to men 65–75 who have ever smoked (all men at 65 in the UK); it reduces aneurysm deaths.

6 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 64-year-old man with a history of type 2 diabetes mellitus is admitted with chest pain to the Emergency Department. An ECG shows ST elevation in the anterior leads and he is thrombolysed and transferred to the Coronary Care Unit (CCU). His usual medication includes simvastatin, gliclazide and metformin. How should his diabetes be managed whilst in CCU?
 - A Stop metformin. Continue gliclazide at a higher dose
 - B Stop metformin & gliclazide. Start subcutaneous insulin (basal-bolus regime)
 - C Continue metformin & gliclazide at same dose
 - D Stop metformin & gliclazide. Start intravenous insulin infusion
 - E Stop metformin & gliclazide. Start subcutaneous insulin (biphasic insulin regime)
- 2 A 72-year-old man is admitted to the Emergency Department with chest pain. On initial assessment he is noted to be pale, have a heart rate of 40/min and a blood pressure of 90/60 mmHg. Which one of the coronary arteries is most likely to be affected?
 - A Posterior descending
 - B Left anterior descending
 - C Right coronary
 - D Anterior interventricular
 - E Left circumflex
- 3 A 62-year-old patient presents to the Emergency Department with a 25 minute history of crushing central chest pain. ECG shows ST elevation in leads I and aVL. Which coronary territory is likely to be affected?
 - A Lateral
 - B Posterior
 - C Anteroseptal
 - D Anterolateral
 - E Inferior

- 4** A 71-year-old man with a four-month history of exertional chest pain is reviewed. The pain typically comes on when he is walking up a hill, is centrally located and radiates to the left arm. It then settles with rest after about 2-3 minutes. Clinical examination and a resting 12 lead ECG are normal. Following NICE guidelines, what is the most appropriate diagnostic strategy?
- A CT coronary angiography
 - B Manage as angina, no further diagnostic tests required
 - C Exercise tolerance test
 - D MPS with SPECT
 - E Invasive coronary angiography
- 5** A 62-year-old man is admitted to hospital following a myocardial infarction. Four days after admission he develops a further episode of central crushing chest pain. Which is the best cardiac marker to investigate his chest pain?
- A LDH
 - B Troponin I
 - C Troponin T
 - D CK-MB
 - E AST
- 6** A 38-year-old lady presents to the emergency department with increased shortness of breath for the last 5 days. On full history taking, you find she has also recently had some episodes of chest pain on exertion. She is normally fit and well but does report generalized aching and a high temperature approximately 2 weeks ago. On examination you find that she is haemodynamically stable with a blood pressure on 100/65 mmHg in the right arm and a heart rate of 95 bpm. The radial pulse on the right side is absent. Heart sounds are normal and the apex beat is non-displaced. Her oxygen saturations are 95% on air and her respiratory rate at rest is 20 breaths per minute. Given the history and examination findings given, what is the most likely diagnosis?
- A Type B aortic dissection
 - B Acute myocarditis
 - C Community acquired pneumonia
 - D Takayasu's arteritis
 - E Peripheral arterial embolus

Answers 1-6

PASSMEDICINE MCQs

- 1** **D Stop metformin & gliclazide. Start intravenous insulin infusion** — In ACS, glucose > 11 mmol/L is treated with a dose-adjusted IV insulin infusion (NICE CG130). Stop metformin around angiography and sulfonylureas acutely.
- 2** **C Right coronary** — Chest pain with bradycardia and hypotension suggests inferior/RV MI; the RCA supplies the AV node and RV. Nodal block here is usually transient and atropine-responsive.
- 3** **A Lateral** — ST elevation in I and aVL is a high lateral infarct (circumflex or first diagonal). Anteroseptal = V1–V4; inferior = II, III, aVF.
- 4** **A CT coronary angiography** — NICE: CT coronary angiography is first line for suspected angina; functional imaging if non-diagnostic. Exercise ECG is no longer used for diagnosis.
- 5** **D CK-MB** — Troponin stays raised 7–14 days, so for reinfarction at day 4 CK-MB (normal by 2–3 days) is more useful; a new hs-troponin rise > 20% is also used.
- 6** **D Takayasu's arteritis** — A young woman with prior systemic illness, chest pain and an absent radial pulse has Takayasu arteritis (large-vessel vasculitis); diagnose by CT/MR angiography, treat with steroids.

6 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

- 1** A 65-year-old man presents with central crushing chest pain for the first time. He is transferred immediately to the closest cardiac unit to undergo a primary percutaneous coronary intervention. There is thrombosis of the left circumflex artery only. Angioplasty is carried out and a drug-eluting stent is inserted. What are the most likely changes to have occurred on ECG during admission?
- A ST depression in leads V1–4
 - B ST elevation in leads V1–6
 - C ST depression in leads II, III and AVF
 - D ST elevation in leads V5–6
 - E ST elevation in leads II, III and AVF
- 2** A 55-year-old man has just arrived in accident and emergency complaining of 20 minutes of central crushing chest pain. Which feature is most indicative of myocardial infarction at this moment in time?
- A Inverted T waves
 - B ST depression
 - C ST elevation
 - D Q waves
 - E Raised troponin

- 3** A 49-year-old man is rushed to accident and emergency complaining of a 20-minute history of severe, crushing chest pain. After giving the patient glyceryl trinitrate (GTN) spray, he is able to tell you he suffers from hypertension and type 2 diabetes and is allergic to aspirin. The most appropriate management is:
- A Aspirin
 - B Morphine
 - C Dipyridamole
 - D Clopidogrel
 - E Warfarin
- 4** A 44-year-old woman attends her local accident and emergency department with a history of at least six months of frequent central chest pain in the early morning or during the night. She had no chest pain on exertion. This had been a particularly severe attack, lasting over 2 hours. Her pulse rate is 84/minute in sinus rhythm, and blood pressure is 134/86 mmHg. The ECG shows anterior ST segment elevation, but troponin levels do not rise. Subsequent coronary angiography is normal. What is the most likely diagnosis?
- A MI
 - B Stable angina
 - C Unstable angina
 - D Anxiety
 - E Variant angina
- 5** A 56-year-old man presents to the accident and emergency department with a 2-hour history of central chest pain radiating to the left arm. He is anxious, nauseated and sweaty. His pulse rate is 120/minute in sinus rhythm and the ECG reveals ST elevation in leads II, III and aVF. The troponin level is significantly raised. This is certainly acute MI. Which is the most likely coronary vessel to be occluded?
- A Circumflex artery
 - B Left anterior descending artery
 - C Right coronary artery
 - D Left main coronary artery
 - E Posterior descending artery
- 6** A 64-year-old man is brought to the emergency department 30 minutes after the sudden onset of substernal chest pain and nausea while playing tennis. He has hypertension and hyperlipidaemia. Current medications include hydrochlorothiazide and atorvastatin. He does not smoke cigarettes or drink alcohol. He is diaphoretic and distressed. Pulse is 110/min, respirations are 31/min, and blood pressure is 85/55 mm Hg. Examination shows jugular venous distention. Crackles are heard over the lower lung fields bilaterally. Capillary refill time is delayed; the extremities are cool to touch. A 12-lead ECG shows ST elevations in leads II, III, and aVF. Which of the following is the most appropriate next step in management?
- A Perform a percutaneous transluminal coronary angioplasty
 - B Administer sublingual nitroglycerin
 - C Start single antiplatelet therapy
 - D Start intravenous metoprolol therapy
 - E Perform a coronary artery bypass graft surgery

Answers 1-6

- 1** **D ST elevation in leads V5-6** — Circumflex occlusion gives lateral ST elevation (I, aVL, V5-V6), often with posterior changes, and is the infarct most often 'silent' on the ECG.
- 2** **C ST elevation** — At 20 min the earliest diagnostic sign is ST elevation (after hyperacute T waves). Troponin may still be normal and Q waves take hours.
- 3** **D Clopidogrel** — **UPDATED** With true aspirin allergy, use clopidogrel (a P2Y₁₂ inhibitor) as the antiplatelet, plus anticoagulation and reperfusion. Warfarin and dipyridamole have no role.
- 4** **E Variant angina** — Early-morning rest pain with transient ST elevation, no troponin rise and normal coronaries is vasospastic (Prinzmetal) angina: treat with CCBs and nitrates.
- 5** **C Right coronary artery** — Inferior STEMI (II, III, aVF) is due to RCA occlusion in ~80%. Check V4R for RV infarction: give fluids and avoid nitrates.
- 6** **A Perform a percutaneous transluminal coronary angioplasty** — Inferior STEMI with cardiogenic shock: emergency PCI of the culprit artery is the only intervention that improves survival (SHOCK). Nitrates and beta-blockers worsen hypotension.



02

HEART FAILURE & CARDIOMYOPATHY

Heart Failure & Cardiomyopathy

Acute and chronic heart failure across the ejection-fraction spectrum, the drugs that change prognosis, and the cardiomyopathies.

86

MCQs

0

EMQ SCENARIOS

7

SOURCES

13

REPEATED IDEAS

IN THIS CHAPTER

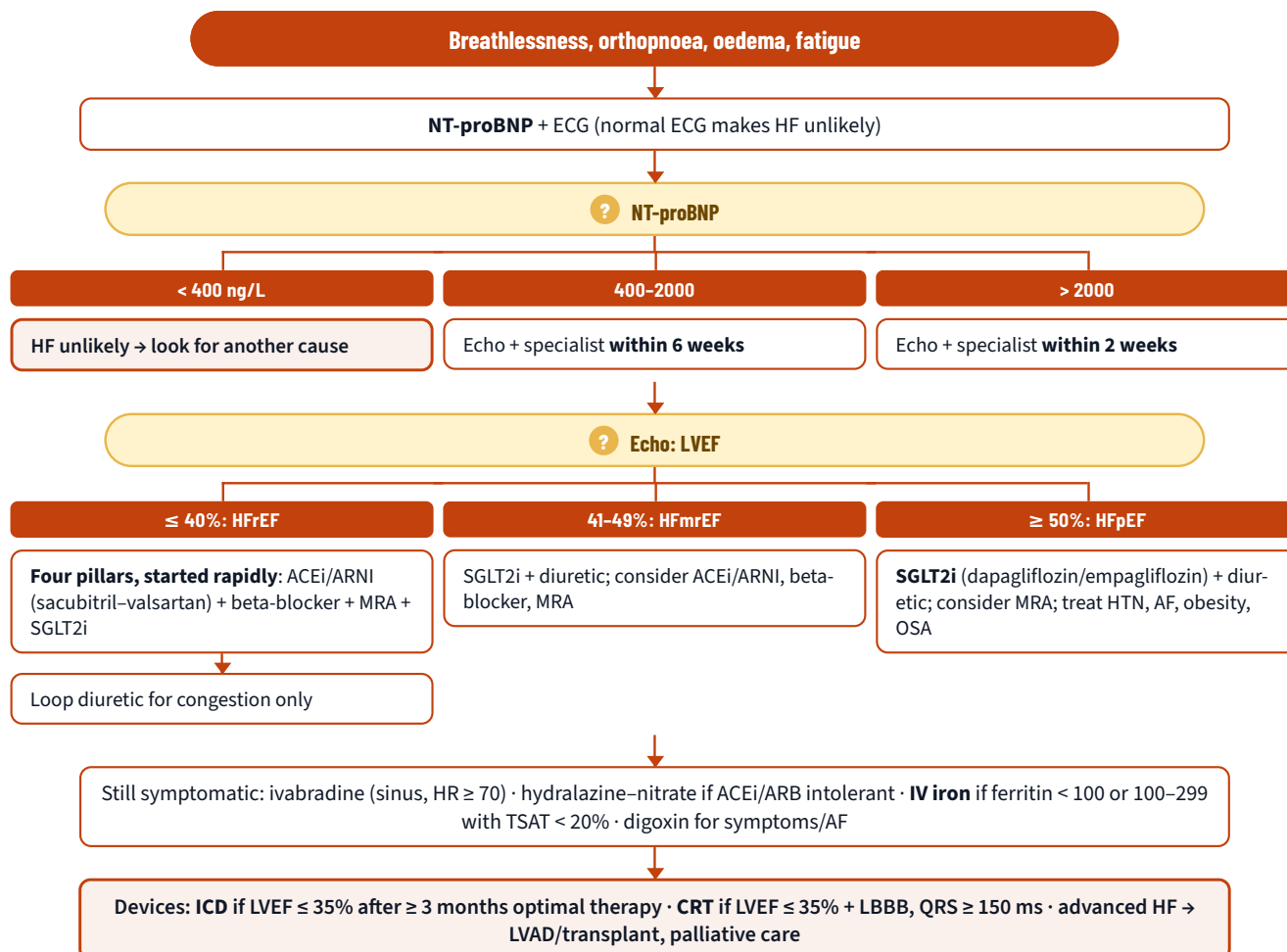
01	Management flowcharts	3 charts	page 40
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02

Step by step

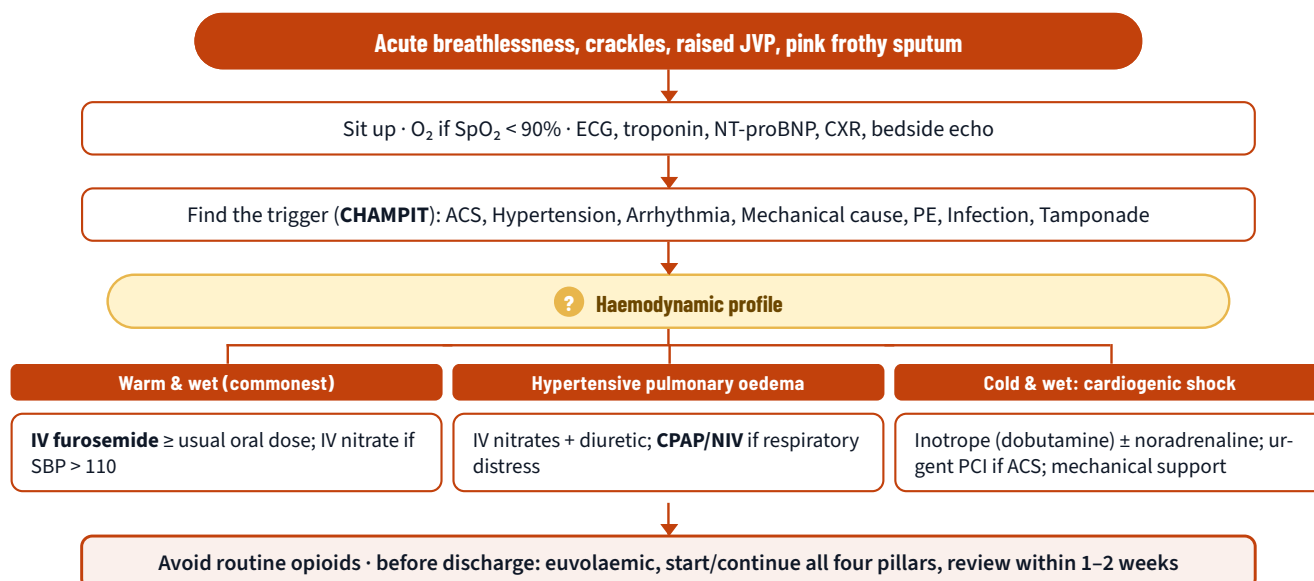
Chronic heart failure: diagnosis to treatment

ESC 2021 + 2023 UPDATE • NICE NG106



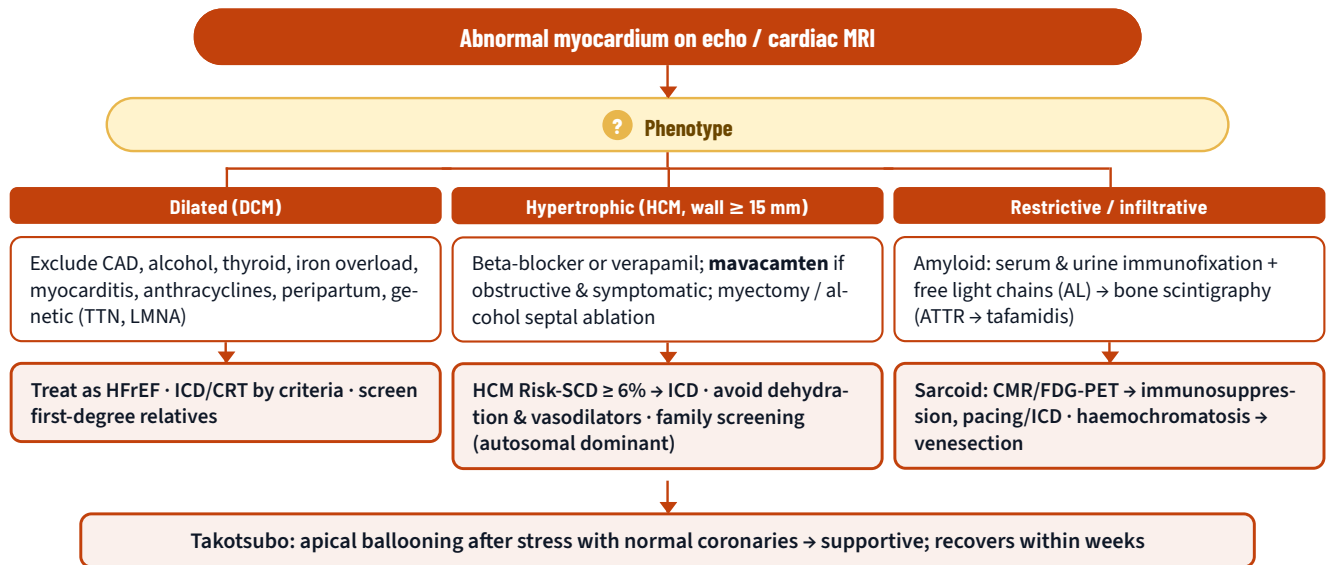
Acute heart failure

ESC 2021 • NICE CG187



Cardiomyopathies

ESC 2023 CARDIOMYOPATHY



Local Exams MCQs

SET 1 OF 2

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

CASE • QUESTIONS 1–5

A 65-year-old man presents to the emergency department with breathlessness that has gradually increased over two months, difficulty sleeping and the need for extra pillows; he has coughed up frothy pink sputum. He has poorly controlled hypertension and coronary artery disease and takes lisinopril, aspirin, atorvastatin and bisoprolol; he smokes (12 pack-years). He is tachypnoeic with bilateral fine crackles, a raised JVP and leg oedema. Chest X-ray shows cardiomegaly.

1 What is the most likely diagnosis?

- A Acute exacerbation of COPD
- B Pulmonary embolism
- C Acute respiratory distress syndrome
- D Acute decompensated heart failure
- E Acute asthma attack

REPEATED 2* • ALSO ASKED IN
Previous-years exam

2 Which drug should be started immediately?

- A Valsartan
- B Heparin
- C Furosemide
- D Diltiazem
- E Clopidogrel

REPEATED 3* • ALSO ASKED IN
Crash Course • Other Sources

3 Which of the following is NOT a complication of this condition?

- A Cardiac arrhythmias
- B Hyponatraemia
- C Hyperkalaemia
- D Hyperlipidaemia
- E Raised blood urea

4 At this stage, which is the most important investigation?

- A ECG
- B Arterial blood gases
- C Echocardiography
- D Coronary angiography
- E Serum electrolytes

REPEATED 4* • ALSO ASKED IN
Log book 2025 ×2 • Crash Course

5 Which drug can improve this patient's symptoms?

- A Spironolactone
- B Propranolol
- C Flecainide
- D Meloxicam
- E Bretylium

6 A 61-year-old man with long-standing hypertension and ischaemic cardiomyopathy has worsening breathlessness and orthopnoea. BP 124/70 mmHg, HR 92 bpm, SpO₂ 96%; bibasal crackles and a raised JVP. He takes bisoprolol 5 mg, furosemide 40 mg and atorvastatin 40 mg daily. Creatinine 2.6 mg/dL (baseline 1.8), eGFR 25 mL/min, potassium 5.4 mmol/L, BNP 2100 pg/mL. Echo: LVEF 28%, mild MR, normal RV. Ramipril was stopped 2 days ago because of worsening renal function and rising potassium. Which medication gives a mortality benefit and is appropriate now?

- A Start sacubitril/valsartan
- B Add spironolactone
- C Add ivabradine
- D Start digoxin
- E Start hydralazine plus isosorbide dinitrate

7 A 69-year-old man with long-standing hypertension, untreated obstructive sleep apnoea and obesity (BMI 36) has had progressive exertional breathlessness for 2 months. BP 162/88 mmHg, HR 78, SpO₂ 95%; raised JVP, S4, mild crackles and ankle oedema. BNP 620 pg/mL. Echo: EF 55%, grade II diastolic dysfunction, increased LV mass, PASP 48 mmHg. Creatinine 1.0 mg/dL, potassium 4.2 mmol/L, HbA1c 6.1%. He takes losartan, amlodipine and chlortalidone and is eu-volaemic after diuresis. Which therapy gives the best evidence-based reduction in heart-failure hospitalisation?

- A Switch losartan to sacubitril/valsartan
- B Add spironolactone
- C Start dapagliflozin
- D Start isosorbide mononitrate
- E Start metoprolol succinate

8 A 30-year-old woman at 34 weeks of pregnancy has progressive breathlessness, orthopnoea and leg oedema. ECG shows sinus tachycardia; echo shows a dilated left ventricle with LVEF 25% and no significant valve disease. BP 98/60 mmHg, HR 112 bpm. Which is the most appropriate management during pregnancy?

- A ACE inhibitor and spironolactone
- B Beta-blocker and loop diuretic
- C Low-molecular-weight heparin only
- D Ivabradine for rate control
- E Sacubitril/valsartan

9 All of the following drugs improve prognosis in heart failure EXCEPT:

- A Lisinopril
- B Spironolactone
- C Digoxin
- D Bisoprolol

REPEATED 4* · ALSO ASKED IN

39th batch end-block (cardiology) · End-block exam 2025 · Previous-years exam

10 Which of the following is true regarding cardiomyopathy?

- A Cardiomyopathy in Duchenne or Becker muscular dystrophy presents mainly with diastolic dysfunction
- B Ischaemia should be excluded in older patients with dilated cardiomyopathy
- C Constrictive cardiomyopathy is the leading cause of sudden death in athletes
- D Restrictive cardiomyopathy is the easiest type to diagnose
- E Cardiac resynchronisation therapy is best indicated in hypertrophic cardiomyopathy

11 A 68-year-old woman attends her GP for annual review of her heart failure. She is breathless walking up hills, her BP is 165/90 mmHg, and she takes furosemide and aspirin. Which medication is most appropriate to add?

- A Hydrochlorothiazide
- B Isosorbide mononitrate
- C Spironolactone
- D Enalapril

REPEATED 3* · ALSO ASKED IN

Crash Course · Passmedicine

END-BLOCK EXAMS 2025

End-block & induction exams

12 A 30-year-old woman presents with a 2-month history of progressive effort intolerance and an episode of paroxysmal nocturnal dyspnoea three weeks ago. Her past medical history is negative for hypertension or diabetes. On examination, her jugular venous pressure is elevated up to her earlobes, she has a soft, tender hepatomegaly, bilateral pitting oedema up to the knees, bibasal crepitations, and an audible S3 on cardiac auscultation. A chest X-ray reveals cardiomegaly with interstitial infiltrates, and echocardiography shows global left ventricular hypokinesia with an ejection fraction of 25–30%. What is the most likely diagnosis?

- A Dilated cardiomyopathy leading to congestive heart failure
- B Restrictive cardiomyopathy
- C Hypertrophic cardiomyopathy
- D Constrictive pericarditis
- E Valvular heart disease

REPEATED 2* · ALSO ASKED IN

Crash Course

13 A 30-year-old woman has dilated cardiomyopathy with biventricular failure and an ejection fraction of 25–30%. Which of the following is a cornerstone of the medical treatment of her systolic heart failure?

- A ACE inhibitors
- B Calcium channel blockers
- C Antiarrhythmic agents
- D Nitrates
- E Antibiotics

REPEATED 3* · ALSO ASKED IN

Log book 2025 ×2

14 In the same woman with dilated cardiomyopathy and an ejection fraction of 25–30%, which of the following is another key medication?

- A Beta-blockers
- B Statins
- C Anticoagulants
- D Nitrates
- E Corticosteroids

15 In the same woman with dilated cardiomyopathy, oedema and pulmonary crackles, which of the following is also an important part of her medical therapy?

- A Diuretics
- B Antiplatelet agents
- C Antibiotics
- D Calcium-channel blockers
- E Proton pump inhibitors

16 Which complication is commonly associated with dilated cardiomyopathy with reduced ejection fraction?

- A Arrhythmias
- B Pulmonary fibrosis
- C Aortic dissection
- D Deep venous thrombosis
- E Pericardial effusion

17 Which of the following is another potential complication of dilated cardiomyopathy?

- A Thromboembolism
- B Oesophageal varices
- C Chronic obstructive pulmonary disease
- D Myocardial infarction
- E Liver cirrhosis

18 In a patient with dilated cardiomyopathy and heart failure, an audible third heart sound most likely indicates:

- A Volume overload of the left ventricle
- B Mitral stenosis
- C Pericardial constriction
- D Right ventricular hypertrophy
- E Atrial septal defect

FORMATIVE & PREVIOUS YEARS

- 19** Which one of the following is least likely to cause dilated cardiomyopathy?
- A Alcohol
 - B Wilson's disease
 - C Hypertension
 - D Haemochromatosis
- 20** The following are Causes of dilated cardiomyopathy EXCEPT:
- A Beta-myosin heavy chain mutations
 - B Alcohol
 - C Pregnancy
 - D Long standing tachycardia
- 21** A 67-year-old man with a history of chronic obstructive pulmonary disease and ischaemic heart disease is taken to the Emergency Department with dyspnoea. On examination his respiratory rate is 24/min JVP is not elevated and crackles are heard in both lung bases. Which other finding would most strongly indicate that his dyspnoea is secondary to isolated left ventricular failure?
- A Gallop rhythm
 - B Tachycardia
 - C Pulsus alternans
 - D Peripheral oedema

REPEATED 2* · ALSO ASKED IN
Davidson

LOG BOOK 2025

- 22** A 65-year-old man with known heart failure is evaluated and found to have a reduced ejection fraction. How is his heart failure best classified?
- A HFpEF (heart failure with preserved ejection fraction)
 - B HFrEF (heart failure with reduced ejection fraction)
 - C High-output heart failure
 - D Right-sided heart failure only
 - E Diastolic dysfunction only
- 23** In a patient with HFrEF who experiences symptoms on exertion and has evidence of structural heart disease, which stage of heart failure is most likely?
- A Stage A
 - B Stage B
 - C Stage C
 - D Stage D
- 24** In a heart failure patient with left bundle branch block (LBBB), which intervention may improve left ventricular function?
- A Increased dose of diuretics
 - B Cardiac resynchronization therapy
 - C High-dose beta-blockers
 - D Coronary angioplasty
 - E Implantable defibrillator only
- 25** A 56-year-old man with long-standing heart failure and hypertension is expected to show which electrocardiographic abnormality?
- A Right ventricular hypertrophy
 - B Left ventricular hypertrophy
 - C Atrial fibrillation
 - D Complete heart block
 - E Prolonged QT interval

Answers 1-25

LOCAL EXAMS MCQs

- 1** **D Acute decompensated heart failure** — Orthopnoea, pink frothy sputum, crackles, raised JVP, oedema and cardiomegaly is acute decompensated heart failure. COPD/asthma give wheeze without a raised JVP.
- 2** **C Furosemide** — Congestion in acute heart failure is treated with an IV loop diuretic (furosemide), sitting up, oxygen if hypoxic and nitrates if BP allows. Diltiazem is negatively inotropic.
- 3** **D Hyperlipidaemia** — Heart failure causes arrhythmias, hyponatraemia, hyperkalaemia (ACEi/MRA) and raised urea (cardiorenal). Hyperlipidaemia is a cause of the underlying CAD, not a complication.
- 4** **C Echocardiography** — Echocardiography is key: EF (HFrEF/HFmrEF/HFpEF), valves, wall motion, pericardium and filling pressures. Angiography comes later if ischaemia is suspected.
- 5** **A Spironolactone** — Spironolactone (MRA) improves symptoms and survival in HFrEF (RALES). Flecainide raises mortality in structural heart disease; NSAIDs worsen fluid retention.
- 6** **E Start hydralazine plus isosorbide dinitrate** — If ACEi/ARB/ARNI cannot be used (renal impairment, hyperkalaemia), hydralazine plus isosorbide dinitrate reduces HFrEF mortality and does not raise potassium.
- 7** **C Start dapagliflozin** — In HFpEF, SGLT2 inhibitors (dapagliflozin, empagliflozin) reduce heart-failure hospitalisation and CV death; diuretics for congestion and treat comorbidities.
- 8** **B Beta-blocker and loop diuretic** — Peripartum cardiomyopathy during pregnancy: beta-blocker and loop diuretic (hydralazine/nitrate if needed). ACEi, ARB, ARNI and MRA are unsafe before delivery.
- 9** **C Digoxin** — ACEi/ARB/ARNI, beta-blockers, MRAs and SGLT2 inhibitors reduce HFrEF mortality. Digoxin reduced admissions but not mortality (DIG trial).
- 10** **B Ischaemia should be excluded in older patients with dilated cardiomyopathy** — In older patients with dilated cardiomyopathy, exclude coronary disease before calling it non-ischaemic, as revascularisation may help.

- 11 **D Enalapril** — A heart-failure patient on only furosemide and aspirin lacks prognostic therapy: start an ACE inhibitor (also treats her BP), then beta-blocker, MRA and SGLT2i.
- 12 **A Dilated cardiomyopathy leading to congestive heart failure** — Biventricular failure with cardiomegaly and global hypokinesia (EF 25–30%) in a young woman is dilated cardiomyopathy. Restrictive disease keeps systolic function.
- 13 **A ACE inhibitors** — ACE inhibitors (or ARB/ARNI) reduce HFrEF mortality and remodelling and are a cornerstone with beta-blockers, MRAs and SGLT2i. Non-DHP CCBs worsen HFrEF.
- 14 **A Beta-blockers** — Evidence-based beta-blockers (bisoprolol, carvedilol, metoprolol succinate) cut HFrEF mortality by ~a third; start low when euvoalaemic and titrate slowly.
- 15 **A Diuretics** — Loop diuretics relieve congestion and symptoms but do not improve survival; use the lowest dose that keeps the patient euvoalaemic.
- 16 **A Arrhythmias** — DCM predisposes to AF and ventricular arrhythmias, and sudden death is common; consider an ICD if EF stays $\leq 35\%$ despite optimal therapy.
- 17 **A Thromboembolism** — Stasis in a dilated, poorly contracting ventricle (and atria in AF) causes mural thrombus and embolism; anticoagulate for AF or documented LV thrombus.
- 18 **A Volume overload of the left ventricle** — S3 reflects rapid filling of a dilated, volume-overloaded ventricle with high filling pressure (gallop rhythm). Opening snap = MS; pericardial knock = constriction.
- 19 **C Hypertension** — DCM causes: alcohol, haemochromatosis, myocarditis, genetic, peripartum, anthracyclines, tachycardia. Hypertension causes concentric LVH and HFpEF, so is least likely.
- 20 **A Beta-myosin heavy chain mutations** — Beta-myosin heavy chain (MYH7) mutations classically cause HCM. Alcohol, pregnancy and persistent tachycardia cause DCM (the last two may reverse).
- 21 **A Gallop rhythm** — With COPD and IHD together, a gallop rhythm (S3) is the most specific sign that breathlessness is from LV failure with high filling pressure.
- 22 **B HFrEF (heart failure with reduced ejection fraction)** — By LVEF: HFrEF $\leq 40\%$, HFmrEF 41–49%, HFpEF $\geq 50\%$ with raised filling pressures. High-output failure is a separate category.
- 23 **C Stage C** — ACC/AHA stages: A at risk, B structural disease without symptoms, C structural disease with symptoms, D refractory. NYHA classes grade symptoms separately.
- 24 **B Cardiac resynchronization therapy** — CRT (biventricular pacing) for HFrEF with EF $\leq 35\%$, LBBB and QRS ≥ 150 ms despite optimal therapy restores synchrony and improves EF and survival.
- 25 **B Left ventricular hypertrophy** — Long-standing hypertension causes LVH: tall voltages (SV1 + RV5/6 > 35 mm) with lateral ST-T 'strain'. RVH suggests pulmonary hypertension.

SET 2 OF 2

- 26 A 16-year-old boy presents with early fatigue during sports, two syncopal episodes, and a family history of sudden cardiac death. What is the most likely diagnosis?
- A Arrhythmogenic right ventricular cardiomyopathy
B Hypertrophic cardiomyopathy
C Long QT syndrome
D Aortic stenosis
E Mitral valve prolapse
- REPEATED 3* · ALSO ASKED IN**
Crash Course x2
- 27 In hypertrophic cardiomyopathy, what is the primary cause of the systolic murmur heard on examination?
- A Mitral regurgitation due to valve prolapse
B Fixed obstruction at the aortic valve
C Dynamic left ventricular outflow tract obstruction
D Atrial septal defect flow
E Ventricular septal defect
- 28 During the Valsalva manoeuvre in a patient with hypertrophic cardiomyopathy, what happens to the intensity of the murmur?
- A Decreases
B No change
C Increases
D The murmur disappears
E It becomes a diastolic murmur
- REPEATED 3* · ALSO ASKED IN**
Harrison · Other Sources
- 29 What is the best next diagnostic test to confirm the suspected diagnosis of hypertrophic cardiomyopathy?
- A Cardiac catheterization
B Exercise stress test
C Echocardiography
D Chest X-ray
E Holter monitoring
- 30 The syncope experienced by patients with hypertrophic cardiomyopathy is most commonly due to:
- A Bradycardia
B Ventricular arrhythmias and dynamic outflow obstruction
C Hypotension from dehydration
D Conduction block
E Pulmonary embolism

Answers 26–30

LOCAL EXAMS MCQS

- 26** **B Hypertrophic cardiomyopathy** — Exertional fatigue and syncope with familial sudden death in a teenager is HCM, the commonest cause of sudden death in young athletes; ECG and echo first.
- 27** **C Dynamic left ventricular outflow tract obstruction** — The HCM murmur comes from dynamic LVOT obstruction (septal hypertrophy plus systolic anterior motion of the mitral valve); it does not radiate to the carotids.
- 28** **C Increases** — Valsalva (like standing) reduces preload, the LV cavity shrinks and outflow obstruction worsens, so the HCM murmur gets louder; squatting and handgrip soften it.
- 29** **C Echocardiography** — Echo confirms HCM: unexplained wall thickness ≥ 15 mm (often septal), SAM and an outflow gradient. MRI and Holter refine risk.
- 30** **B Ventricular arrhythmias and dynamic outflow obstruction** — Syncope in HCM is usually from ventricular arrhythmia or a surge in outflow obstruction; unexplained syncope is a major sudden-death risk factor favouring an ICD.

17 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1** The term 'orthopnoea' refers to breathlessness (dyspnoea) in a particular situation. Which answer below describes that situation?
- A After several hours of sleep
 - B Due to asthma
 - C Immediately on lying flat
 - D On exertion
 - E On sitting upright
- 2** A 65 year old female presents with chest pain, and the 12-lead ECG shows evidence of acute inferior myocardial infarction complicated by hypotension. An echocardiogram is performed and shows markedly reduced movement of the right ventricular walls, indicating that right ventricular infarction has occurred. Left ventricular function is only mildly impaired. Which of the following physical signs would be expected in this situation?
- A Tachycardia, a late systolic murmur and ascites
 - B Tachycardia, and absent jugular venous pulse because of inability to develop right heart pressure
 - C Tachycardia, acute development of peripheral oedema and acute ascites
 - D Tachycardia, basal crepitations and a third heart sound
 - E Tachycardia, elevated jugular venous pulse due to failure of right ventricular pump function, and hepatomegaly
- 3** What relationship does Starling's Law of the heart describe?
- A Between blood pressure and cardiac output
 - B Between cardiac filling and blood pressure
 - C Between cardiac filling and cardiac output
 - D Between heart rate and blood pressure
 - E Between heart rate and cardiac output
- 4** What underlying pathophysiological changes is chronic cardiac failure associated with?
- A Activation of the renin–angiotensin aldosterone system (RAAS)
 - B Inhibition of the RAAS
 - C Inhibition of the sympathetic nervous system
 - D Reduced production of brain natriuretic peptide (BNP)
 - E Systemic vasodilatation
- 5** Which of the following best describes dilated cardiomyopathy?
- A A disease of the myocardium characterised by chamber enlargement and thinning of the left and right ventricular walls
 - B A disease of the myocardium characterised by disproportionate thickening of the interventricular septum
 - C A disease of the myocardium characterised by infiltration of myocardial tissue resulting in restricted contraction and relaxation
 - D Isolated dilatation of the atria, causing atrial fibrillation
 - E Isolated dilatation of the right ventricle, causing ventricular tachycardia
- 6** Which of the following is a cause of dilated cardiomyopathy?
- A A high-cholesterol diet
 - B Heavy alcohol consumption
 - C Mutation in cardiac sodium channel gene
 - D Obesity
 - E Recreational cannabis use

- 7** By which of the following features is hypertrophic cardiomyopathy usually characterised?
- A** Asymmetric left ventricular hypertrophy with marked thickening of the interventricular septum
 - B** Asymmetric left ventricular hypertrophy with marked thickening of the anterior left ventricular wall
 - C** Hypertrophy of both atria and both ventricles
 - D** Hypertrophy of the left ventricle and atrophy of the right ventricle
 - E** Symmetrical left ventricular hypertrophy
- 8** Cardiac transplantation is considered in which group of patients with cardiomyopathy?
- A** Asymptomatic patients
 - B** Frail elderly patients with end-stage heart failure
 - C** Patients who do not wish to take life-long medication
 - D** Patients who have poor quality of life despite optimal drug therapy
 - E** Patients who have symptoms but good quality of life on optimal drug therapy
- 9** A 48 year old woman with no significant previous medical history collapses while running a marathon. Despite attempts at resuscitation, she does not survive. Postmortem examination reveals asymmetric left ventricular hypertrophy with disproportionate thickening of the interventricular septum. A postmortem diagnosis of hypertrophic cardiomyopathy is made. What is the most likely cause of this patient's sudden collapse?
- A** Atrial fibrillation
 - B** Left ventricular failure
 - C** Pulmonary embolism
 - D** Right ventricular failure
 - E** Ventricular arrhythmia
- 10** A 30 year old woman has recently been diagnosed with dilated cardiomyopathy. Her diagnosis was made with echocardiography, which showed moderate left ventricular dilatation and impairment. She has noticed herself becoming increasingly fatigued on moderate exertion. Her younger sister died suddenly the previous year and she is very worried about the risk of sudden death. Which of the following treatments is known to reduce her risk of sudden death?
- A** Aspirin
 - B** B-blocker (e.g. metoprolol)
 - C** Calcium channel blocker (e.g. verapamil)
 - D** Loop diuretic (e.g. furosemide)
 - E** Percutaneous coronary intervention (PCI)
- 11** A 55 year old woman presents with a history of acute, severe, constricting central chest pain associated with anterior ST segment elevation on the 12-lead ECG. She immediately undergoes coronary angiography, which shows no evidence of coronary artery disease and no coronary occlusion. An echocardiogram shows left ventricular apical dilatation, with normal left ventricular basal contraction. Which of the following factors is most likely to have precipitated this illness?
- A** Acute emotional stress
 - B** Cigarette smoking
 - C** Excessive alcohol consumption
 - D** Genetic factors
 - E** Viral infection
- 12** The following medical treatments are all associated with improved symptoms in patients with heart failure due to left ventricular systolic dysfunction. However, which of the treatments has NOT been shown to also improve survival?
- A** Bisoprolol
 - B** Enalapril
 - C** Furosemide
 - D** Sacubitril-valsartan
 - E** Spironolactone
- 13** An 80 year old woman presents with shortness of breath and swollen ankles. Her ECG showed some poor R-wave progression. She was referred for an echocardiogram and was found to have a high ejection fraction. Which of these conditions is the most likely cause of her presentation?
- A** Acute myocarditis
 - B** Aortic stenosis
 - C** Dilated cardiomyopathy
 - D** Ischaemic cardiomyopathy with extensive infarction
 - E** Restrictive cardiomyopathy
- 14** Neuroendocrine system activation is a feature of heart failure. Abnormalities of which hormone can cause heart failure rather than result from heart failure?
- A** Aldosterone
 - B** Angiotensin II
 - C** Catecholamines
 - D** Thyroxine
 - E** Vasopressin (antidiuretic hormone, ADH)
- 15** An army recruit is referred for assessment because there is a family history of sudden cardiac death and an uncle was diagnosed with hypertrophic cardiomyopathy. In what manner is hypertrophic cardiomyopathy commonly inherited?
- A** Autosomal dominant
 - B** Autosomal recessive
 - C** Never inherited
 - D** X-linked dominant
 - E** X-linked recessive

- 16** An 83 year old woman presents with acute pulmonary oedema, BP of 180/100 mmHg and a SaO₂ of 85%. Which treatment is **UNLIKELY** to be helpful in this setting?
- A Furosemide
 - B Intravenous dobutamine
 - C Intravenous nitrates
 - D Non-invasive ventilation
 - E Supplementary oxygen therapy
- 17** A 43 year old woman with a past history of breast cancer is referred with a gradual onset of breathlessness. An echocardiogram demonstrates a dilated poorly contracting left ventricle. You wish to investigate potential causes of her dilated cardiomyopathy. Which of the following would be an irreversible cause of her dilated cardiomyopathy?
- A Alcohol excess
 - B Anthracycline chemotherapy
 - C Haemochromatosis
 - D Hypothyroidism
 - E Thyrotoxicosis

Answers 1-17

DAVIDSON MCQS

- 1** **C Immediately on lying flat** — Orthopnoea is breathlessness soon after lying flat (venous redistribution), relieved by sitting up. PND wakes the patient after hours of sleep.
- 2** **E Tachycardia, elevated jugular venous pulse due to failure of right ventricular pump function, and hepatomegaly** — RV infarction with inferior MI: hypotension, raised JVP, hepatomegaly and clear lungs. Give IV fluids and avoid nitrates and diuretics.
- 3** **C Between cardiac filling and cardiac output** — Frank-Starling: stroke volume and output rise with ventricular filling (preload) up to a plateau; in heart failure the curve is flat, so extra filling causes congestion.
- 4** **A Activation of the renin-angiotensin aldosterone system (RAAS)** — Low output activates RAAS and the sympathetic system: salt/water retention, vasoconstriction and remodelling — the targets of ACEi/ARNI, MRAs and beta-blockers.
- 5** **A A disease of the myocardium characterised by chamber enlargement and thinning of the left and right ventricular walls** — DCM is dilatation (often biventricular) with thin walls and systolic dysfunction not explained by loading or coronary disease. Septal thickening = HCM.
- 6** **B Heavy alcohol consumption** — Heavy alcohol (> 80 g/day for years) is a common, reversible cause of DCM. SCN5A mutations cause Brugada and long QT3.
- 7** **A Asymmetric left ventricular hypertrophy with marked thickening of the interventricular septum** — HCM: asymmetric LVH with disproportionate septal thickening (septum/posterior wall > 1.3), myocyte disarray and dynamic outflow obstruction in ~two-thirds.
- 8** **D Patients who have poor quality of life despite optimal drug therapy** — Transplantation is for stage D heart failure with poor quality of life despite optimal drug and device therapy and no contraindications; LVAD is an alternative.
- 9** **E Ventricular arrhythmia** — Sudden death in HCM is from VT/VF arising in fibrotic myocardium, often on exertion; risk is scored (ESC HCM Risk-SCD) and high-risk patients get an ICD.
- 10** **B B-blocker (e.g. metoprolol)** — Beta-blockers reduce total and sudden death in HFrEF (as do MRAs and ARNI); consider an ICD if EF stays ≤ 35%. Verapamil is harmful; screen the family.
- 11** **A Acute emotional stress** — ST elevation, normal coronaries and apical ballooning in a postmenopausal woman after emotional stress is takotsubo cardiomyopathy; LV function usually recovers.
- 12** **C Furosemide** — Loop diuretics relieve symptoms but do not improve survival. Bisoprolol, enalapril, sacubitril/valsartan, spironolactone and SGLT2i reduce mortality.
- 13** **E Restrictive cardiomyopathy** — Heart failure with high EF and a pseudo-infarct ECG (low voltages) in an elderly woman suggests restrictive (amyloid) cardiomyopathy.
- 14** **D Thyroxine** — Thyroid disease causes heart failure: thyrotoxicosis gives high-output failure and AF; hypothyroidism a dilated heart. Check TFTs in all new heart failure.
- 15** **A Autosomal dominant** — HCM is autosomal dominant with variable penetrance (MYH7, MYBPC3); screen first-degree relatives with ECG and echo ± genetic testing.
- 16** **B Intravenous dobutamine** — Hypertensive pulmonary oedema: oxygen/CPAP, IV nitrates and IV furosemide. Dobutamine is for cardiogenic shock and only raises oxygen demand here.
- 17** **B Anthracycline chemotherapy** — Anthracycline (doxorubicin) cardiotoxicity is cumulative, dose-dependent and largely irreversible. Alcohol, haemochromatosis and thyroid DCM can improve.

17 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

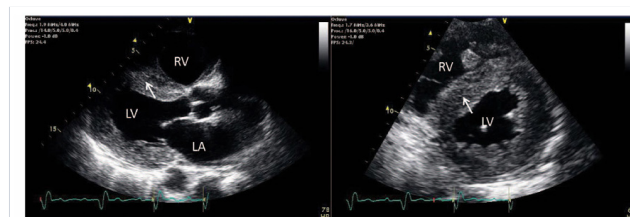
Harrison MCQs

- 1** Regarding the epidemiology and prognosis of heart failure, which of the following statements is true?
- A** Among patients with heart failure with reduced ejection fraction, coronary artery disease is the most common cause
 - B** Anaemia is a frequent cause of heart failure among patients with a previously structurally normal heart
 - C** Due to advances in medical and device therapy, the prognosis of all patients with heart failure is now excellent, with >90% surviving more than 1 year after diagnosis
 - D** Familial, or genetic, cardiomyopathy is quite rare, although it should be screened for aggressively to aid diagnosis and early treatment among presymptomatic affected family members
 - E** Heart failure with preserved ejection fraction makes up a minority of the total population of patients with heart failure

REPEATED 4* • ALSO ASKED IN
Crash Course • Other Sources ×2

- 2** A 75-year-old man presents to your emergency department appearing quite ill. His family says he has not had his normal energy for the last 6 months, and they noted he was confused and lethargic for the last day or two. As you take a history from the family, you palpate the patient's radial pulse and notice a regular beat-to-beat variability of pulse amplitude, although his rhythm is regular. Indeed, as you later take his blood pressure, you note that only every other phase I (systolic) Korotkoff sound is audible as the cuff pressure is slowly lowered and that this is independent of the respiratory cycle. Based on this, you suspect this patient has which of the following?
- A** Atrial fibrillation
 - B** Cardiac tamponade
 - C** Constrictive pericarditis
 - D** Pulmonary embolism
 - E** Severe left ventricular dysfunction
- 3** A 78-year-old man is admitted to the intensive care unit with decompensated heart failure. He has long-standing ischaemic cardiomyopathy. ECG shows atrial fibrillation and left bundle branch block. Chest radiograph shows cardiomegaly and bilateral alveolar infiltrates with Kerley B lines. Which of the following is least likely to be present on physical examination?
- A** Fourth heart sound
 - B** Irregular heart rate
 - C** Kussmaul sign
 - D** Reversed splitting of the second heart sound
 - E** Third heart sound

- 4** A 66-year-old smoker who has recently travelled in South America has progressive exertional breathlessness and fatigue. Pulse 104/min irregularly irregular, BP 96/76 mmHg, with a raised JVP and marked leg oedema. Echocardiography shows an LVEF of 55%; images are shown. To establish the cause of his heart failure, which test is most appropriate next?



- A** Bone marrow biopsy
 - B** Genetic testing for transthyretin mutations
 - C** Positron emission tomography cardiac stress test
 - D** Serologies for *Trypanosoma cruzi*
 - E** Serum and urine protein electrophoresis and light chain assay
- 5** A 71-year-old woman with ischaemic cardiomyopathy and a left ventricular ejection fraction of 38% has been hospitalized for the past week with acute decompensated heart failure. After diuresis and medication optimization, she is feeling immensely better. She is on the maximally tolerated doses of an angiotensin-converting enzyme (ACE) inhibitor, B-blocker, and appropriate diuretic dose. You are planning for discharge today. On rounds, the nurse notes that the patient had several short (5-10 beats) runs of nonsustained ventricular tachycardia (NSVT) and multiple premature ventricular contractions (PVCs) overnight, although she remained asymptomatic. A medical student on the team asks if the NSVT carries any prognostic significance and if any intervention is needed. What is the most appropriate response?
- A** "NSVT is common in patients with cardiomyopathy and carries no significance."
 - B** "NSVT is concerning in this patient. We should suppress the PVCs and NSVT with amiodarone."
 - C** "NSVT is associated with an increased mortality in patients with heart failure. We will refer her for an automated implantable defibrillator."
 - D** "NSVT is associated with an increased mortality in patients with heart failure. We should suppress the NSVT with flecainide."
 - E** "NSVT is associated with an increased mortality in patients with heart failure. However, suppression of PVCs and NSVT with antiarrhythmic drugs does not change this prognosis."

- 6** A 68-year-old man with a history of myocardial infarction and congestive heart failure is comfortable at rest. However, when walking to his car, he develops dyspnoea, fatigue, and sometimes palpitations. He must rest for several minutes before these symptoms resolve. Which of the following is his New York Heart Association classification?
- A Class I
 - B Class II
 - C Class III
 - D Class IV
- 7** A 47-year-old postmenopausal woman is seen for onset of severe dyspnoea over the last few weeks. She reports no preceding chest pain, cough, sputum, or fever, although she does report leg swelling. Physical examination is notable for a blood pressure of 145/78 mmHg and heart rate of 123 bpm. Exophthalmos is present, as well as bilateral inspiratory crackles occupying approximately one-third of the lower chest with neck vein distention. She has a third heart sound with no murmur. Bilateral lower extremity oedema is also present, with warm extremities and a fine hand tremor. Which of the following is the most likely pathophysiologic explanation for her heart failure?
- A Anaemia with high-output state
 - B Chronic systemic hypertension with resultant left ventricular hypertrophy and nonsystolic heart failure
 - C Hemochromatosis with subsequent restrictive cardiomyopathy
 - D Myocardial infarction with depressed left ventricular systolic function
 - E Thyrotoxicosis with high-output state
- 8** From a pathophysiologic perspective, all of the following are upregulated in heart failure with reduced ejection fraction EXCEPT:
- A Angiotensin II
 - B B-type natriuretic peptide (BNP)
 - C Calcium uptake into the sarcoplasmic reticulum
 - D Norepinephrine
 - E Tumour necrosis factor
- 9** A 67-year-old man presents to the emergency department with shortness of breath. He is found to have an elevated serum BNP. In addition to heart failure, all of the following may also cause an elevation in BNP EXCEPT:
- A Advanced age
 - B Female sex
 - C Obesity
 - D Pulmonary embolism with right heart strain
 - E Renal insufficiency
- 10** A ventilated patient with cor pulmonale has a tidal volume of 500 mL, FiO_2 0.4, PEEP 20 cmH₂O and a rate of 20/min; pH 7.40, PCO_2 40 mmHg and SaO_2 86%. All of the following would increase his right ventricular afterload EXCEPT:
- A Increasing FiO_2 to raise SaO_2 to 95%
 - B Increasing PEEP to 35 cmH₂O
 - C Increasing tidal volume to 750 mL
 - D Reducing the respiratory rate to 10/min
- 11** Mr. George is a 52-year-old man with long standing hypertension and poorly controlled diabetes. He presents complaining of several months of breathlessness with exertion, acute episodes of shortness of breath when recumbent, and lower extremity oedema. On examination, he has an elevated jugular venous pulse and an S4 on auscultation. Echocardiography shows a left ventricular ejection fraction of 55% with a large left atrium. You suspect this patient has the syndrome of heart failure with preserved ejection fraction (HFpEF). Drugs targeting which of the following have convincingly demonstrated mortality reduction for patients with HFpEF?
- A Angiotensin-converting enzyme
 - B Angiotensin receptor
 - C Phosphodiesterase-5
 - D Sodium-potassium-ATPase
 - E None of the above
- 12** You are evaluating a 64-year-old woman with a history of nonischaemic cardiomyopathy. She presents to the emergency department for shortness of breath. You note that she has gained 11 kg since her last cardiology appointment 2 months ago. Physical examination confirms findings associated with acute decompensated heart failure, including pulmonary rales, elevated jugular venous pulse, abdominal ascites, lower extremity oedema, and a square wave blood pressure Valsalva response. Her extremities are warm, and blood pressure is 110/78 mmHg with a heart rate of 75 bpm. Her laboratory studies return with a sodium of 128 mEq/L and creatinine of 2.5 mg/dL (which is increased from her prior level of 1.2 mg/dL). Chest x-ray shows a diffuse alveolar filling pattern consistent with pulmonary oedema. What is the next most appropriate step?
- A Digoxin 250 micrograms IV
 - B Furosemide 40 mg IV
 - C Large-bore central line and ultrafiltration
 - D Pulmonary artery catheter for haemodynamic monitoring
 - E Dobutamine 5 micrograms/kg/min titrated to urine output

- 13** Which of the following statements regarding nesiritide is true in patients with acute decompensated heart failure requiring invasive haemodynamic monitoring?
- A Nesiritide has no significant effect on blood pressure
 - B Nesiritide improves renal perfusion in states of acute decompensated heart failure
 - C Nesiritide is associated with a decreased rate of rehospitalization, but not death, from acute decompensated heart failure
 - D Nesiritide reduces pulmonary capillary wedge more rapidly than IV nitrates
 - E None of the statements are true
- 14** A 21-year-old man is lethargic and breathless a week after a viral upper respiratory infection. BP 88/72 mmHg, HR 115, SpO₂ 90%, with crackles, a raised JVP, an S3 and cool extremities; creatinine 2.3 mg/dL, raised BNP and lactate. Echo shows an LVEF of 15% with global hypokinesia. Before dobutamine takes effect, which haemodynamic parameter is most likely increased?
- A Cardiac output
 - B Left ventricular stroke work index
 - C Mixed venous oxygen saturation
 - D Stroke volume
 - E Systemic vascular resistance
- 15** Which of the following statements regarding the initiation of B-blockers versus ACE inhibitors in HFrEF is most correct?
- A ACE inhibitors should always be started before B-blockers
 - B B-blockers should always be started before ACE inhibitors
 - C It does not matter which is initiated first; both are essential and should be optimally titrated
 - D ACE inhibitors are more important than B-blockers in reducing mortality
 - E B-blockers have a stronger dose effect on survival than ACE inhibitors
- 16** A 22-year-old woman with a history of type 1 diabetes mellitus is on optimal doses of the three medications forming the foundation of neurohormonal therapy for HFrEF. Which additional therapy is most appropriate at this time?
- A Digoxin
 - B Addition of valsartan
 - C Aliskiren
 - D Ivabradine
 - E None – she is already on optimal therapy
- 17** A 21-year-old man has HFrEF. Which statement about prophylactic therapy to prevent sudden cardiac death is most appropriate?
- A An ICD is indicated if the LVEF remains 35% or less despite at least 3 months of optimal medical therapy
 - B An ICD is indicated only in NYHA class IV
 - C An ICD is not recommended in ischaemic cardiomyopathy
 - D An ICD should be delayed until after 5 years of optimal therapy
 - E No device therapy is ever indicated

Answers 1-17

- 1** **A Among patients with heart failure with reduced ejection fraction, coronary artery disease is the most common cause** — Coronary disease is the commonest cause of HFrEF; about half of heart failure is HFpEF, and 1-year mortality is still ~10–30%.
- 2** **E Severe left ventricular dysfunction** — Pulsus alternans (alternating strong and weak beats in regular rhythm, independent of respiration) indicates severe LV systolic dysfunction.
- 3** **A Fourth heart sound** — S4 needs atrial contraction into a stiff ventricle, so it cannot occur in AF. S3, reversed splitting (LBBB) and Kussmaul's sign all fit.
- 4** **E Serum and urine protein electrophoresis and light chain assay** — Thick 'sparkling' myocardium with biatrial enlargement and preserved EF suggests amyloidosis: first exclude AL with serum/urine electrophoresis and free light chains.
- 5** **E "NSVT is associated with an increased mortality in patients with heart failure. However, suppression of PVCs and NSVT with antiarrhythmic drugs does not change this prognosis."** — NSVT in heart failure predicts higher mortality, but antiarrhythmic suppression does not help (class I harmful, CAST); optimise HF therapy and electrolytes.
- 6** **C Class III** — Comfortable at rest but symptoms on less than ordinary activity (e.g. a short walk) is NYHA class III.
- 7** **E Thyrotoxicosis with high-output state** — Heart failure with warm peripheries, tachycardia, exophthalmos and tremor is thyrotoxic high-output failure: beta-blocker, diuretic and treat the thyroid.
- 8** **C Calcium uptake into the sarcoplasmic reticulum** — In HFrEF angiotensin II, noradrenaline, natriuretic peptides and TNF rise, but SR calcium uptake falls (SERCA2a down-regulated), impairing relaxation.
- 9** **C Obesity** — BNP rises with age, female sex, CKD, AF, PE, sepsis and ACS. Obesity lowers BNP, so a 'normal' value does not exclude heart failure in obese patients.
- 10** **A Increasing FiO₂ to raise SaO₂ to 95%** — Correcting hypoxaemia relieves hypoxic pulmonary vasoconstriction and lowers RV afterload. High PEEP, large tidal volumes and hypercapnia raise it.

- 11** **E None of the above** — No drug has convincingly reduced mortality in HFpEF; SGLT2i reduce the combined endpoint mainly via fewer admissions. Treat congestion and comorbidities.
- 12** **B Furosemide 40 mg IV** — 'Warm and wet': start an IV loop diuretic (at least the oral dose). Inotropes are for 'cold' low-output states; ultrafiltration for diuretic resistance.
- 13** **D Nesiritide reduces pulmonary capillary wedge more rapidly than IV nitrates** — Nesiritide (recombinant BNP) lowers wedge pressure faster than nitrates but causes hypotension and did not improve outcomes (ASCEND-HF).
- 14** **E Systemic vascular resistance** — In cardiogenic shock output falls and sympathetic vasoconstriction raises systemic vascular resistance (cool peripheries); mixed venous saturation falls.
- 15** **C It does not matter which is initiated first; both are essential and should be optimally titrated** — Beta-blocker or ACEi first gives similar outcomes (CIBIS III); start all four pillars early in low doses and titrate to target.
- 16** **E None – she is already on optimal therapy** — Adding an ARB or aliskiren to ACEi + MRA raises hyperkalaemia and renal risk without benefit; SGLT2i need caution in type 1 diabetes (ketoacidosis), so no change.
- 17** **A An ICD is indicated if the LVEF remains 35% or less despite at least 3 months of optimal medical therapy** — Primary-prevention ICD: LVEF $\leq$ 35% despite $\geq$ 3 months of optimal therapy, NYHA II–III and reasonable survival. Recheck EF after myocarditis or peripartum disease.

7 QUESTIONS • AFTER CRASH COURSE SBAS & EMQS

Crash Course MCQs

- 1** A 65-year-old man presents to the heart failure outpatient clinic with increased shortness of breath and swollen ankles. On examination his pulse was 100 beats/min, blood pressure 100/60 mmHg and jugular venous pressure (JVP) +10 cm water. The patient currently takes furosemide 40 mg BD, spironolactone 12.5 mg, bisoprolol 2.5 mg OD and ramipril 2.5 mg BD. Which of the following is true?
- A** Diuretics reduce the degree of neurohormonal activation in heart failure
 - B** Diuretics alone rarely provide rapid symptomatic relief
 - C** Beta-blockers are contraindicated
 - D** Treatment with spironolactone reduces mortality
 - E** Angiotensin-converting enzyme (ACE) inhibitors are unlikely to induce hypotension if an adequate diuresis has been achieved with diuretics
- 2** A 65-year-old man has NYHA class III chronic heart failure. Despite conventional therapy with appropriate dosages of a diuretic, an ACE inhibitor and a beta-adrenergic blocker, his left ventricular ejection fraction hovers around 35%, and he continues to have shortness of breath on exertion. His latest urea and electrolytes test results were normal and you consider adding digoxin to his treatment regimen. Which one of the following is true regarding digoxin therapy in this situation?
- A** A reasonable dosage is 0.50 mg/day orally
 - B** Serial drug levels are generally not necessary
 - C** A loading dose will be necessary
 - D** It is not likely to improve the ejection fractions
 - E** It is the treatment of choice if the patient's ECG shows AV block
- 3** A 46-year-old woman presents with shortness of breath. She comments that she feels particularly breathless when lying down and uses five pillows to sleep at night. On examination her pulse rate is regular at 122 beats/min and her blood pressure is stable at 122/65 mmHg. You note bibasal crackles when listening to her chest. You suspect heart failure and order a CXR and echocardiogram. The following may all be features on her CXR EXCEPT?
- A** Kerley B lines
 - B** Hilar haziness
 - C** Fluid in the left horizontal interlobar fissure
 - D** Upper lobe venous engorgement
 - E** Cardiomegaly
- 4** A 65-year-old man presents with shortness of breath at rest. Physical examination reveals evidence of pitting oedema over the sacrum and leg oedema to the mid thigh. An echocardiogram is requested which demonstrates an ejection fraction of 0.4. What is the next most appropriate step in management?
- A** ACE inhibitor
 - B** Angiotensin II receptor antagonist
 - C** Beta-blocker
 - D** Calcium channel antagonist
 - E** Diuretic
- 5** A 72-year-old man with known cardiac failure comes to see his GP. He is currently on a diuretic. His GP decides to commence an ACE inhibitor. The following are all side effects of ACE inhibitors EXCEPT?
- A** Renal failure
 - B** Hypokalaemia
 - C** Rash
 - D** Angioedema
 - E** Cough

- 6** A 21-year-old fit university student with no past medical history has sudden loss of consciousness 1 hour into a gym session. CPR is administered by bystanders. On arrival of the paramedics, he has regained consciousness. What is the most effective means of preventing sudden death in high-risk patients with asymptomatic HCM?
- A Amiodarone
 - B An implantable cardioverter-defibrillator (ICD)
 - C Chronic dual-chamber pacing
 - D Metoprolol
 - E Verapamil
- 7** An elderly woman complains of abdominal distension, due to ascites, upper abdominal pain from an enlarged tender liver and flatulence. She is mildly jaundiced and has marked pitting of the legs. Her arterial pulse is rapid and irregular, and she is sitting propped up in bed where it is evident that her face is cyanosed, and her jugular veins are engorged and pulsating in systole. She is passing little urine and the urine contains 2+ of protein. What is the most likely diagnosis?
- A Cirrhosis of the liver
 - B Congestive heart failure
 - C Malignant ascites
 - D Nephrotic syndrome
 - E Superior mediastinal obstruction

Answers 1-7

- 1** **D Treatment with spironolactone reduces mortality** — Spironolactone cut HFrEF mortality by ~30% (RALES), eplerenone in milder HF (EMPHASIS-HF). Diuretics relieve congestion but activate RAAS.
- 2** **B Serial drug levels are generally not necessary** — In chronic HF digoxin is given 62.5–250 µg daily without loading; levels only if toxicity, renal impairment or interactions. It reduces admissions, not mortality.
- 3** **C Fluid in the left horizontal interlobar fissure** — Heart-failure CXR: cardiomegaly, upper-lobe diversion, bat-wing oedema, Kerley B lines, effusions, fissure fluid. The left lung has no horizontal fissure.
- 4** **E Diuretic** — Breathless at rest with gross oedema: relieve congestion first with a loop diuretic; start/titrate ACEi and beta-blocker once less congested.
- 5** **B Hypokalaemia** — ACE inhibitors cause hyperkalaemia (less aldosterone), not hypokalaemia; also cough, angioedema, first-dose hypotension and AKI. Check U&E before and after dose changes.
- 6** **B An implantable cardioverter-defibrillator (ICD)** — An ICD is the only proven way to prevent sudden death in HCM: after aborted arrest (as here) or when the ESC 5-year risk is ≥ 6%.
- 7** **B Congestive heart failure** — Ascites, tender liver, jaundice, AF, oedema, orthopnoea and systolic JVP waves (TR) mean congestive (biventricular) failure. Cirrhosis does not raise the JVP.

5 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 15-year-old male presents to your office on the advice of his football coach. The patient started playing football this year and suffered a syncopal episode at practice yesterday. He reports that he was sprinting with the rest of the team and became lightheaded. He lost consciousness and fell to the ground, regaining consciousness within one or two minutes. He suffered no trauma during the event. He has no prior history of head injury or recent illness. He has had no prior episodes of syncope. The patient is adopted and family history unavailable. Physical examination is unremarkable. What is the best course of action regarding this patient's syncopal episode?
 - A Perform an ECG and echocardiogram. The patient may not return to competitive sports until results are available
 - B Perform an ECG. The patient may not return to competitive sports until results are available
 - C Perform an ECG. The patient may return to competitive sports pending the results
 - D Reassurance. The patient may return to competitive sports provided he increases his water consumption during practice times
 - E Reassurance. The patient may return to competitive sports with no restrictions
- 2 A 55-year-old patient presents to you after a 3-day hospital stay for gradually increasing shortness of breath and leg swelling while away on a business trip. He was told that he had congestive heart failure, but is asymptomatic now, with normal vital signs and physical examination. An echocardiogram shows an estimated ejection fraction of 38%. The patient likes to keep medications to a minimum. He is currently on aspirin and simvastatin. Which would be the most appropriate additional treatment?
 - A Begin an ACE inhibitor and then add a beta-blocker on a scheduled basis
 - B Begin digoxin plus furosemide on a scheduled basis
 - C Begin spironolactone on a scheduled basis
 - D Begin furosemide plus nitroglycerin
 - E Given his preferences, no other medication is needed unless shortness of breath and swelling recur
- 3 A 34-year-old woman is referred by an OB- GYN colleague for the onset of fatigue and dyspnoea on exertion 1 month after her second vaginal delivery. Physical examination reveals a laterally displaced PMI, elevated jugular venous pressure and 2+ pitting lower extremity oedema. Echocardiogram shows systolic dysfunction with an ejection fraction of 30%. Which statement most accurately describes her condition?
 - A This disease may occur unexpectedly years after pregnancy and delivery
 - B About half of similar patients will recover completely
 - C The condition is idiosyncratic; the risk of recurrence with a future pregnancy is no greater than average
 - D This condition will require a different therapeutic approach than the typical dilated cardiomyopathy
 - E This condition will require endomyocardial biopsy for diagnosis
- 4 A 72-year-old Caucasian male presents with shortness of breath that awakens him at night. At baseline he is able to walk less than a block before stopping to catch his breath. Physical examination findings include bilateral basilar rales and neck vein distention. The patient has a known history of congestive heart failure, and his last echocardiogram revealed an ejection fraction of 25%. The patient is compliant with a medication regimen including an ACE inhibitor, beta-blocker, and loop diuretic. Blood pressure is well controlled. What additional treatment should you begin next?
 - A Spironolactone
 - B Aspirin
 - C Amlodipine
 - D Warfarin
 - E Hydralazine and isosorbide dinitrate

REPEATED 3* • ALSO ASKED IN
Other Sources ×2

- 5** A 32-year-old male presents to your office with concern about progressive fatigue and lower extremity oedema. He has experienced decreased exercise tolerance over the past few months, and occasionally awakens coughing at night. Past medical history is significant for sickle cell anaemia and diabetes mellitus. He has had multiple admissions to the hospital secondary to vasoocclusive crises since the age of three. Physical examination reveals a displaced PMI, but is otherwise unremarkable. ECG shows a first degree AV block and low voltage. Chest x-ray shows an enlarged cardiac silhouette with clear lung fields. Which of the following would be the best initial diagnostic approach?
- A** Order serum iron, iron-binding capacity, and ferritin level
 - B** Order brain-natriuretic peptide (BNP)
 - C** Order CT scan of the chest
 - D** Arrange for placement of a 24-hour ambulatory cardiac monitor
 - E** Arrange for cardiac catheterization

Answers 1-5

PRETEST MCQs

- 1** **A Perform an ECG and echocardiogram. The patient may not return to competitive sports until results are available** — Syncope during exertion is a red flag (HCM, anomalous coronaries, long QT, ARVC, WPW): ECG and echo, and no competitive sport until cleared.
- 2** **A Begin an ACE inhibitor and then add a beta-blocker on a scheduled basis** — EF 38% after heart failure (stage C) needs prognostic therapy: ACEi (or ARB/ARNI) plus beta-blocker, then MRA and SGLT2i, even if now asymptomatic.
- 3** **B About half of similar patients will recover completely** — Peripartum cardiomyopathy (EF < 45% late pregnancy to 5 months postpartum): about half recover fully; recurrence in later pregnancies is common.
- 4** **A Spironolactone** — HFrEF still NYHA III on ACEi, beta-blocker and diuretic: add an MRA (spironolactone) with K⁺/renal monitoring; SGLT2i and ARNI follow.
- 5** **A Order serum iron, iron-binding capacity, and ferritin level** — Multiply transfused patients develop iron-overload cardiomyopathy: check ferritin and transferrin saturation, then cardiac MRI T2*; chelation can reverse it.

4 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1** A 71-year-old woman is admitted with acute dyspnoea to the Emergency Department. Oxygen saturations are 94% on 28% supplementary oxygen and her respiratory rate is 30/min. A rapid B-type natriuretic peptide (BNP) assay is reported as follows: BNP 62 pg/ml What is the best interpretation of this result?
- A** No conclusion can be drawn from this result
 - B** Pulmonary embolism is the most likely cause of her symptoms
 - C** If a further BNP level is above 50 pg/ml after one hour then this is diagnostic of heart failure
 - D** Heart failure is unlikely to be the cause of her dyspnoea
 - E** Heart failure is highly likely to be the cause of her dyspnoea
- 2** A 79-year-old man is admitted with congestive cardiac failure. Bloods on admission show: BNP 354 pg/ml Which one of the following would result from elevated BNP levels?
- A** Decreased sodium diuresis
 - B** Vasoconstriction of the coronary arteries
 - C** Inhibition of the renin-angiotensin-aldosterone system
 - D** Vasoconstriction of the pulmonary vessels
 - E** Increased sympathetic tone

- 3** Where is B-type natriuretic peptide mainly secreted from?
- A Atrial myocardium
 - B Juxtaglomerular cells
 - C Zona glomerulosa
 - D Ventricular myocardium
 - E Hypothalamus
- 4** You have ordered a B-type natriuretic peptide (BNP) test on a patient with suspected heart failure. It has come back as being slightly elevated. Which one of the following factors may account for a falsely elevated BNP?
- A ACE inhibitor therapy
 - B Beta-blocker therapy
 - C Furosemide therapy
 - D Obesity
 - E COPD

Answers 1-4

PASSMEDICINE MCQS

- 1** **D Heart failure is unlikely to be the cause of her dyspnoea** — BNP < 100 pg/mL (NT-proBNP < 300 acutely) has a high negative predictive value: acute heart failure is unlikely, so look for another cause.
- 2** **C Inhibition of the renin-angiotensin-aldosterone system** — BNP causes natriuresis, vasodilatation and inhibits RAAS and sympathetic activity. Sacubitril raises natriuretic peptides by blocking neprilysin.
- 3** **D Ventricular myocardium** — BNP/NT-proBNP is secreted mainly by ventricular myocytes in response to stretch; ANP comes from the atria.
- 4** **E COPD** — BNP can be raised without LV failure in COPD (RV strain), PE, AF, sepsis, CKD, cirrhosis and age > 70. Obesity and HF drugs lower it.

6 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

- 1** A 78-year-old woman is admitted with heart failure secondary to aortic stenosis. Which sign is most likely to be present?
- A Pleural effusion on chest x-ray
 - B Raised jugular venous pressure (JVP)
 - C Bilateral pedal oedema
 - D Bibasal crepitations
 - E Atrial fibrillation
- 2** A 78-year-old woman with a 3-day history of shortness of breath and white frothy sputum is suspected of having congestive cardiac failure. Which chest x-ray sign is NOT typically seen in congestive cardiac failure?
- A Lower lobe diversion
 - B Cardiomegaly
 - C Pleural effusions
 - D Alveolar oedema
 - E Kerley B lines
- 3** A 71-year-old man with congestive heart failure complains of nausea, anorexia, and yellow rings around lights. Which medication is most likely responsible?
- A Lisinopril
 - B Spironolactone
 - C Digoxin
 - D Furosemide
 - E Bisoprolol
- 4** A 55-year-old woman with hypertension and a 30-pack-year smoking history presents with epigastric pain, sweating, and breathlessness for 45 minutes. ABG shows pH 7.49, pCO₂ 32 mm Hg, and pO₂ 57 mm Hg. Which abnormality best explains her condition?
- A Decreased transpulmonary pressure
 - B Decreased minute ventilation
 - C Increase pulmonary capillary pressure
 - D Increased pulmonary capillary permeability
 - E Decreased total body haemoglobin
- 5** A 66-year-old man with hypertension, hyperlipidaemia, and diabetes presents with a 3-week history of exertional dyspnoea. Cardiac exam reveals an S3 gallop. What further evaluation is most likely to show?
- A Elevated serum brain natriuretic peptide levels
 - B Calcification of the aortic valve
 - C Bounding pulses of peripheral arteries
 - D Earlier onset of abnormal cardiac sound with standing
 - E Ruptured papillary muscle
- 6** A 58-year-old man, a long-term smoker, has a 5-day history of worsening dyspnoea and fatigue; his chest X-ray shows a perihilar 'bat's-wing' pattern. What is the most likely cause of his findings?
- A Left ventricular failure
 - B Tricuspid regurgitation
 - C Mitral stenosis
 - D Pericarditis
 - E Cardiomyopathy

REPEATED 2* • ALSO ASKED IN
Other Sources

Answers 1-6

OTHER SOURCES MCQS

- 1 **D Bibasal crepitations** — Decompensated aortic stenosis first raises LA and pulmonary venous pressure, giving bibasal crackles; raised JVP and oedema follow only with right failure.
- 2 **A Lower lobe diversion** — Venous hypertension causes upper-lobe diversion; 'lower-lobe diversion' is not a sign. Cardiomegaly, effusions, alveolar oedema and Kerley B lines are typical.
- 3 **C Digoxin** — Nausea, anorexia and yellow vision/halos are digoxin toxicity, promoted by hypokalaemia, CKD and interacting drugs; give Digifab for life-threatening toxicity.
- 4 **C Increase pulmonary capillary pressure** — Acute LV failure (likely MI) raises pulmonary capillary pressure, causing hydrostatic oedema, hypoxaemia and hyperventilation (respiratory alkalosis). Permeability = ARDS.
- 5 **A Elevated serum brain natriuretic peptide levels** — Exertional breathlessness with an S3 suggests heart failure; a raised BNP/NT-proBNP supports it and NICE then arranges echo (within 2 or 6 weeks).
- 6 **A Left ventricular failure** — **UPDATED** A perihilar bat-wing pattern is alveolar oedema; in a breathless smoker the likeliest cause is LV failure. TR causes systemic, not pulmonary, congestion.



03

CONGENITAL HEART DISEASE

Congenital Heart Disease

Shunts, obstructive lesions and cyanotic disease in children and adults: murmurs, ECG and X-ray clues, and when to intervene.

21

MCQs

0

EMQ SCENARIOS

6

SOURCES

2

REPEATED IDEAS

IN THIS CHAPTER

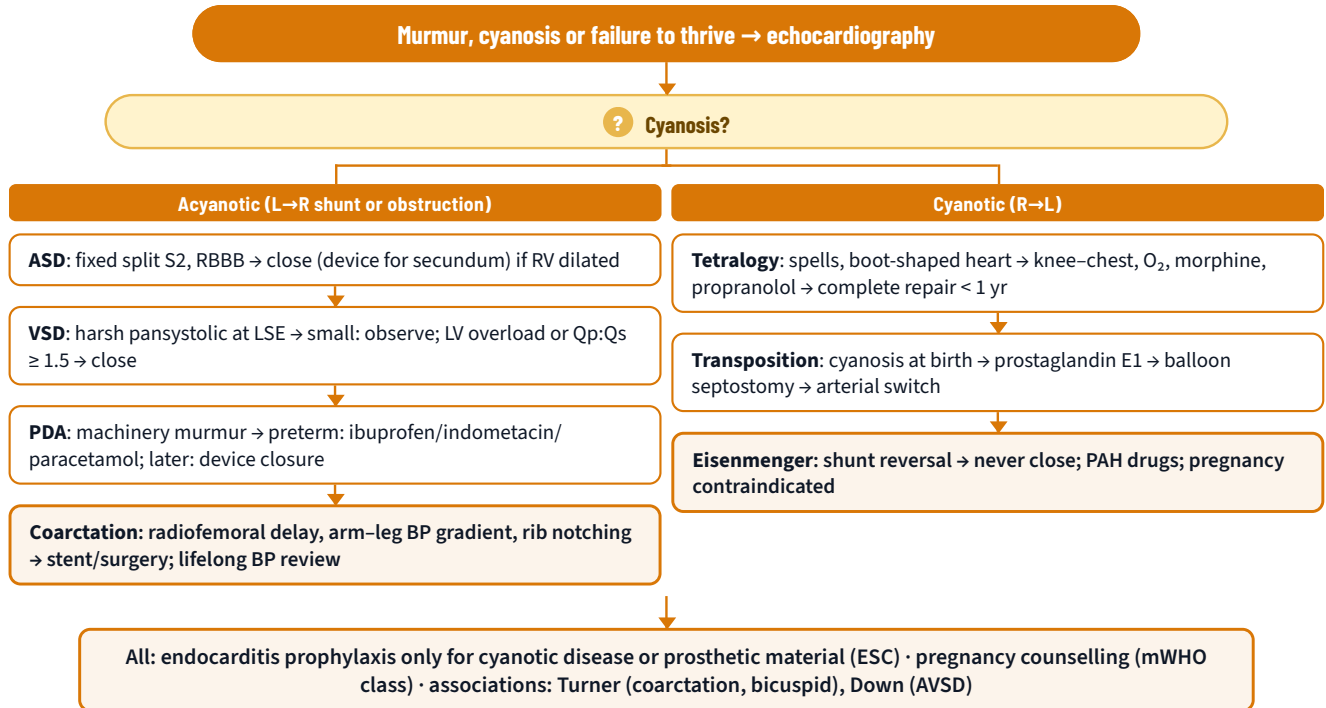
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03

Step by step

Congenital heart disease: from clue to action

ESC 2020 ACHD



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 32-year-old woman is referred with an incidental systolic murmur in the pulmonary area and fixed splitting of the second heart sound. The chest X-ray shows pulmonary plethora with a prominent main pulmonary artery, and the ECG shows incomplete right bundle branch block. What is the most likely diagnosis?

A Pulmonary stenosis
B Persistent ductus arteriosus
C Ventricular septal defect
D Atrial septal defect

REPEATED 2× • ALSO ASKED IN
End-block exam 2025

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 2** Which of the following congenital heart defects is most associated with a bicuspid aortic valve?

A Coarctation of the aorta
B Atrial septal defect
C Ventricular septal defect
D Tetralogy of Fallot

LOG BOOK 2025

Log-book questions

- 3** A 18-year-old man with a history of poorly controlled hypertension is noted to have a blood pressure of 160/90 mmHg in the right arm and 120/80 mmHg in the left arm. What is the most likely underlying cause of his high blood pressure?

A Essential hypertension
B Renal artery stenosis
C Coarctation of the aorta
D Pheochromocytoma
E Aortic regurgitation

REPEATED 2× • ALSO ASKED IN
Log book 2025

- 4** Which chest X-ray finding is most typically associated with coarctation of the aorta?

A Cardiomegaly
B Rib notching
C Pulmonary oedema
D Hyperlucent lung fields
E Pleural effusion

- 5** What is the definitive treatment for significant coarctation of the aorta (with hypertension or a large gradient)?

A Lifelong medical management with beta-blockers
B Repair of the coarctation (endovascular stenting or surgery)
C Angioplasty with a drug-eluting balloon
D Observation only

- 6** A 24-year-old asymptomatic woman with a small secundum atrial septal defect (ASD) is planning pregnancy. What is the expected risk during pregnancy?

A High risk of maternal mortality
B Significant risk of heart failure
C Minimal risk with appropriate follow-up
D Immediate need for surgical repair
E High risk of developing Eisenmenger's syndrome

- 7** A 24-year-old woman has a small, unrepaired secundum atrial septal defect. Is antibiotic prophylaxis against infective endocarditis indicated for a routine dental extraction?

A High risk requiring prophylactic antibiotics
B Moderate risk requiring lifelong prophylaxis
C Minimal risk, so prophylaxis is not indicated
D Risk only if the defect is large
E Risk only with prior endocarditis history

- 8** Which of the following best describes the second heart sound (S2) in patients with an ASD?

A Paradoxical splitting of S2
B Wide splitting of S2 that varies with respiration
C Fixed splitting of S2
D Single S2 without splitting
E Loud S2 with a prominent P2

- 9** What is the most common electrocardiographic finding in patients with a secundum ASD?

A Left axis deviation
B Left bundle branch block
C Incomplete right bundle branch block
D Atrial fibrillation
E Ventricular pre-excitation

- 10** What is the risk of developing Eisenmenger's syndrome in a patient with a small secundum ASD?

A Very high risk
B Moderate risk
C Low risk
D No risk at all
E Risk only if associated with ventricular septal defect

Answers 1–10

LOCAL EXAMS MCQs

- 1 **D Atrial septal defect** — Pulmonary flow murmur, fixed wide S2 splitting and incomplete RBBB (RSR' in V1) with plethoric lungs is a secundum ASD.
- 2 **A Coarctation of the aorta** — About half of patients with coarctation have a bicuspid aortic valve; both are common in Turner syndrome.
- 3 **C Coarctation of the aorta** — Hypertension in a young man with an arm BP difference suggests coarctation at or near the left subclavian origin (left arm lower). Check radiofemoral delay.
- 4 **B Rib notching** — Inferior rib notching (ribs 3–8) from dilated intercostal collaterals is classic of coarctation, rarely before age 8; the '3 sign' is another clue.
- 5 **B Repair of the coarctation (endovascular stenting or surgery)** — **UPDATED** Significant coarctation (gradient ≥ 20 mmHg or hypertension) is repaired: stenting in suitable adults, surgery in infants. Lifelong follow-up for recoarctation.
- 6 **C Minimal risk with appropriate follow-up** — A small secundum ASD without pulmonary hypertension is well tolerated in pregnancy (mWHO II); routine follow-up is enough.
- 7 **C Minimal risk, so prophylaxis is not indicated** — Isolated secundum ASD has very low endocarditis risk; prophylaxis is not indicated (NICE never routinely; ESC only for high-risk lesions).
- 8 **C Fixed splitting of S2** — In ASD the shunt keeps RV volume high throughout respiration, so S2 is widely split with a fixed interval. Paradoxical splitting = LBBB or severe AS.
- 9 **C Incomplete right bundle branch block** — Secundum ASD typically shows RSR' in V1 (incomplete RBBB) with right axis deviation; left axis deviation suggests a primum defect.
- 10 **C Low risk** — Eisenmenger develops in a minority of large unrepaired shunts (< 10% of adult ASDs); a small secundum ASD carries low risk.

3 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 A 17 year old male presents to the emergency department with an episode of collapse. Witnesses report he became extremely blue at the time of collapse, which occurred on walking. The patient tells you he has a history of congenital heart disease. On examination you note he is centrally cyanosed. Which of the following congenital conditions is the most likely explanation for this presentation?
 - A Coarctation of the aorta
 - B Congenital heart block
 - C Patent foramen ovale
 - D Tetralogy of Fallot
 - E Wolff–Parkinson–White syndrome
- 2 Which of the following is true of Eisenmenger's syndrome?
 - A Breathlessness and fatigue are uncommon symptoms
 - B It occurs in patients with patent foramen ovale
 - C Left to right shunting occurs because of pulmonary hypertension
 - D Life expectancy is markedly reduced
 - E Patients are peripherally but not centrally cyanosed
- 3 A 48 year old woman registers with a new family physician. She tells the doctor she had a small hole in her heart from birth but that it did not require any treatment. On examination, pulse is 70 beats/min and regular; BP 122/76 mmHg. You detect a loud, high-pitched systolic murmur at the left sternal border, accompanied by a thrill. Which of the following conditions would explain the history and physical findings?
 - A Anterior mitral leaflet prolapse
 - B Atrial septal defect
 - C Patent foramen ovale
 - D Persistent ductus arteriosus
 - E Ventricular septal defect

Answers 1–3

DAVIDSON MCQs

- 1 **D Tetralogy of Fallot** — Exertional collapse with deep cyanosis in congenital heart disease is a tetralogy of Fallot spell (RVOT spasm increases right-to-left shunt).
- 2 **D Life expectancy is markedly reduced** — Eisenmenger syndrome: shunt reversal causes cyanosis, clubbing and erythrocytosis; life expectancy is markedly reduced and pregnancy is very high-risk.
- 3 **E Ventricular septal defect** — A small restrictive VSD gives a loud harsh pansystolic murmur with thrill at the left sternal edge despite a trivial shunt (maladie de Roger).

3 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 You are working in a rural health clinic in Northern India. You evaluate an 8-year-old boy who has never seen a physician. His mother tells you that he is unable to keep up with his peers in terms of physical activity. On your initial examination of his skin, you notice clubbing and cyanosis in his feet, but his hands appear normal. Without any further examination, you suspect that he has which of the following congenital abnormalities?
 - A Atrial septal defect
 - B Dextro-transposition of the great arteries (TGA)
 - C Patent ductus arteriosus with secondary pulmonary hypertension
 - D Tetralogy of Fallot
 - E Ventricular septal defect
- 2 Mrs. Hellwig is a 25-year-old woman with systemic lupus erythematosus (SLE) complicated by nephropathy, hemolytic anaemia, and pleuritis. Her disease is well controlled on therapy. She recently discovered that she is pregnant, and presents today for prenatal counseling. She specifically is concerned about the effect her autoimmune disease may have on the infant. You tell her that the most common cardiac complication in children born to mothers with SLE is:
 - A AV block
 - B Coronary artery disease
 - C Dilated cardiomyopathy
 - D Pulmonary hypertension with right ventricular (RV) failure
 - E Sterile, Libman-Sacks endocarditis
- 3 A 37-year-old woman with no significant past medical history except for a childhood murmur is evaluated for sudden-onset severe pain in her right lower extremity. Examination is notable for pallor distal to the right knee, a cold leg, and absent dorsalis pedis pulse on the right, with the left leg being normal. Which of the following studies is likely to diagnose the underlying reason for the patient's presentation?
 - A Angiography of right lower extremity
 - B Blood cultures
 - C Echocardiogram with bubble study
 - D Serum cytoplasmic antineutrophil cytoplasmic antibodies (c-ANCA)
 - E Venous ultrasound of right upper extremity

Answers 1-3

HARRISON MCQs

- 1 **C Patent ductus arteriosus with secondary pulmonary hypertension** — Clubbing and cyanosis of toes but not fingers (differential cyanosis) is PDA with pulmonary hypertension and shunt reversal.
- 2 **A AV block** — Maternal anti-Ro/La antibodies damage the fetal conduction system, causing congenital complete heart block; monitor the fetal heart from 16 weeks.
- 3 **C Echocardiogram with bubble study** — Limb ischaemia in a young woman with a childhood murmur suggests paradoxical embolism through an ASD/PFO; echo with bubble study shows the shunt.

1 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 You are helping with school sports physicals and see a 16-year-old boy who has had trouble keeping up with his peers. Which of the following auscultatory findings suggests a previously undiagnosed ventricular septal defect?
- A A systolic crescendo-decrescendo murmur heard best at the upper right sternal border with radiation to the carotids; the murmur is augmented with exercise
 - B A systolic murmur at the pulmonic area and a diastolic rumble along the left sternal border
 - C A holosystolic murmur at the mid-left sternal border
 - D A diastolic decrescendo murmur at the mid-left sternal border
 - E A continuous murmur through systole and diastole at the upper left sternal border

Answer 1

PRETEST MCQS

- 1 **C** A holosystolic murmur at the mid-left sternal border — VSD gives a harsh holosystolic murmur at the mid/lower left

sternal border. ASD = pulmonary flow murmur; PDA = continuous machinery murmur.

2 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 3-day-old neonate born prematurely at 34 weeks' gestation has been slow to wean off the ventilator since birth. On examination, she has been found to have a continuous heart murmur. An echocardiogram has detected a patent ductus arteriosus. No other structural heart abnormalities have been found. A chest x-ray shows cardiomegaly and mildly congested lung fields. What initial treatment should be started to manage this condition?
- A Emergency surgery
 - B Indomethacin
 - C Beta-blockers
 - D Intravenous fluids
 - E Prostaglandins
- 2 Which one of the following complications is least associated with ventricular septal defects?
- A Right heart failure
 - B Aortic regurgitation
 - C Eisenmenger's complex
 - D Infective endocarditis
 - E Atrial fibrillation

Answers 1-2

PASSMEDICINE MCQS

- 1 **B** Indomethacin — Significant PDA in a preterm infant is closed medically with a COX inhibitor (ibuprofen/indomethacin) or paracetamol; prostaglandin E1 keeps it open.
- 2 **E** Atrial fibrillation — VSD complications: heart failure, Eisenmenger, aortic regurgitation, endocarditis. AF reflects atrial overload (ASD, mitral disease), not VSD.

2 QUESTIONS • OTHER COLLEGES & SOURCES

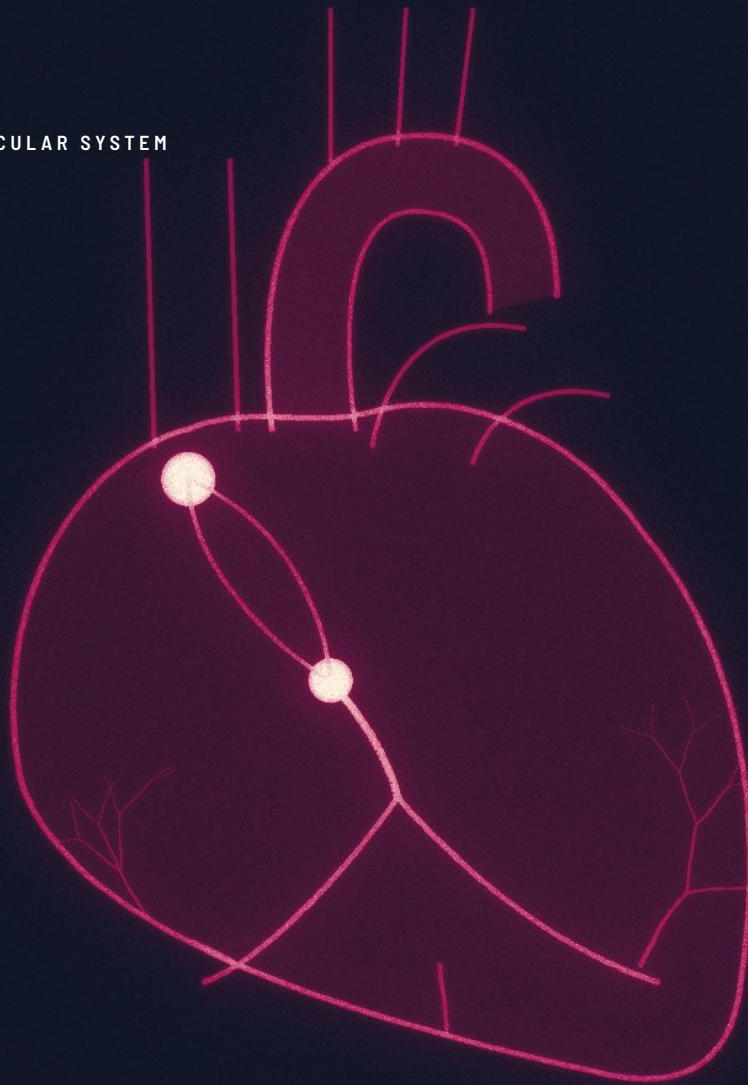
Other Sources MCQs

- 1** You are called to see a 21-year-old man in the ER with new onset slurred speech and left hemiparesis. Auscultation reveals a systolic murmur at the pulmonic region with a diastolic rumble along the left sternal border and a fixed split second heart sound. What is the likely cause of his symptoms?
- A Ventricular septal defect
 - B Atrial septal defect
 - C Patent ductus arteriosus
 - D Aortic insufficiency
 - E Coarctation of the aorta
- 2** A 16-year-old boy is diagnosed with a small VSD on screening due to family history of HOCM. He is asymptomatic with a small left-to-right shunt. What is the most likely significant complication?
- A Pulmonary hypertension
 - B Heart failure
 - C Dysrhythmias
 - D Endocarditis
 - E Shunt reversal (right to left flow)

Answers 1-2

OTHER SOURCES MCQS

- 1** **B Atrial septal defect** — Fixed S2 splitting with flow murmurs and a stroke in a young adult suggests paradoxical embolism across an ASD; confirm with bubble echo and consider closure.
- 2** **D Endocarditis** — A small VSD causes no overload or pulmonary hypertension; its main long-term risk is infective endocarditis. Many small muscular VSDs close spontaneously.



04

ARRHYTHMIAS & CONDUCTION DISEASE

Arrhythmias & Conduction Disease

Atrial fibrillation, supraventricular and ventricular tachycardias, heart block and cardiac arrest: ECG recognition, rate and rhythm control, and anticoagulation.

90

MCQs

0

EMQ SCENARIOS

7

SOURCES

14

REPEATED IDEAS

IN THIS CHAPTER

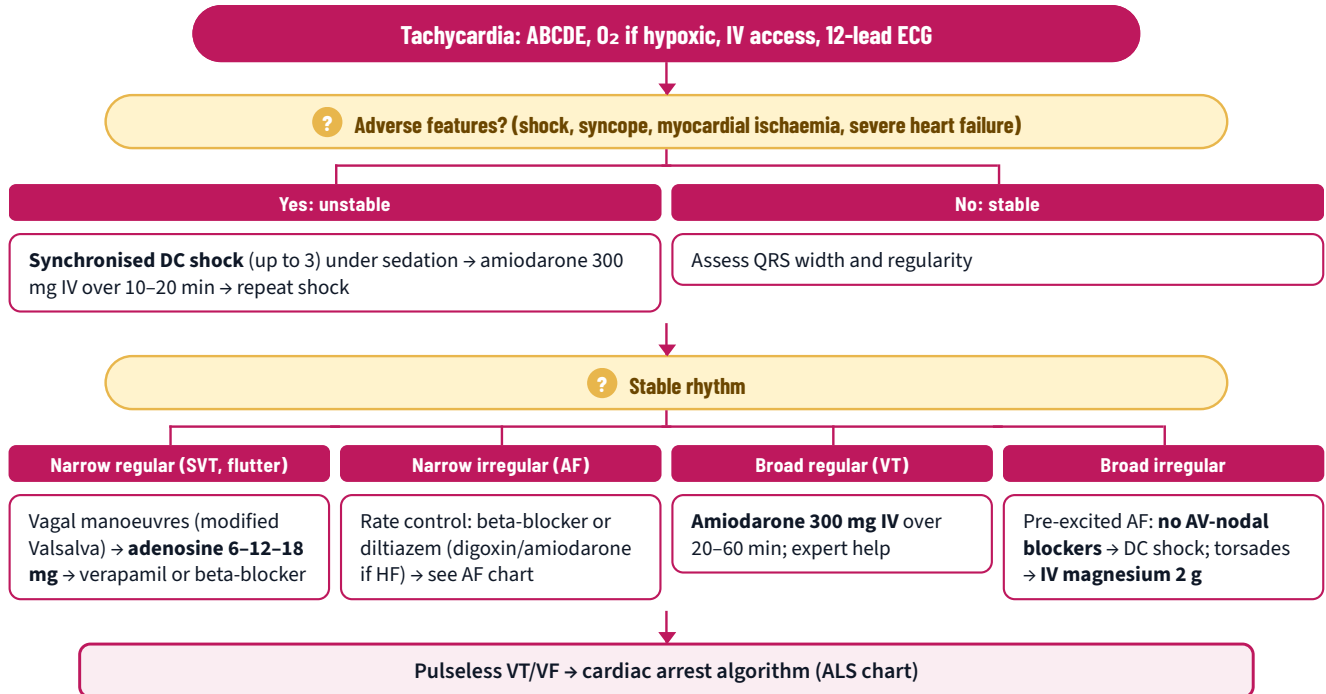
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03	Davidson MCQs	7 Q	page 72
04	Harrison MCQs	28 Q	page 74
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08	Other Sources MCQs	5 Q	page 86

04

Step by step

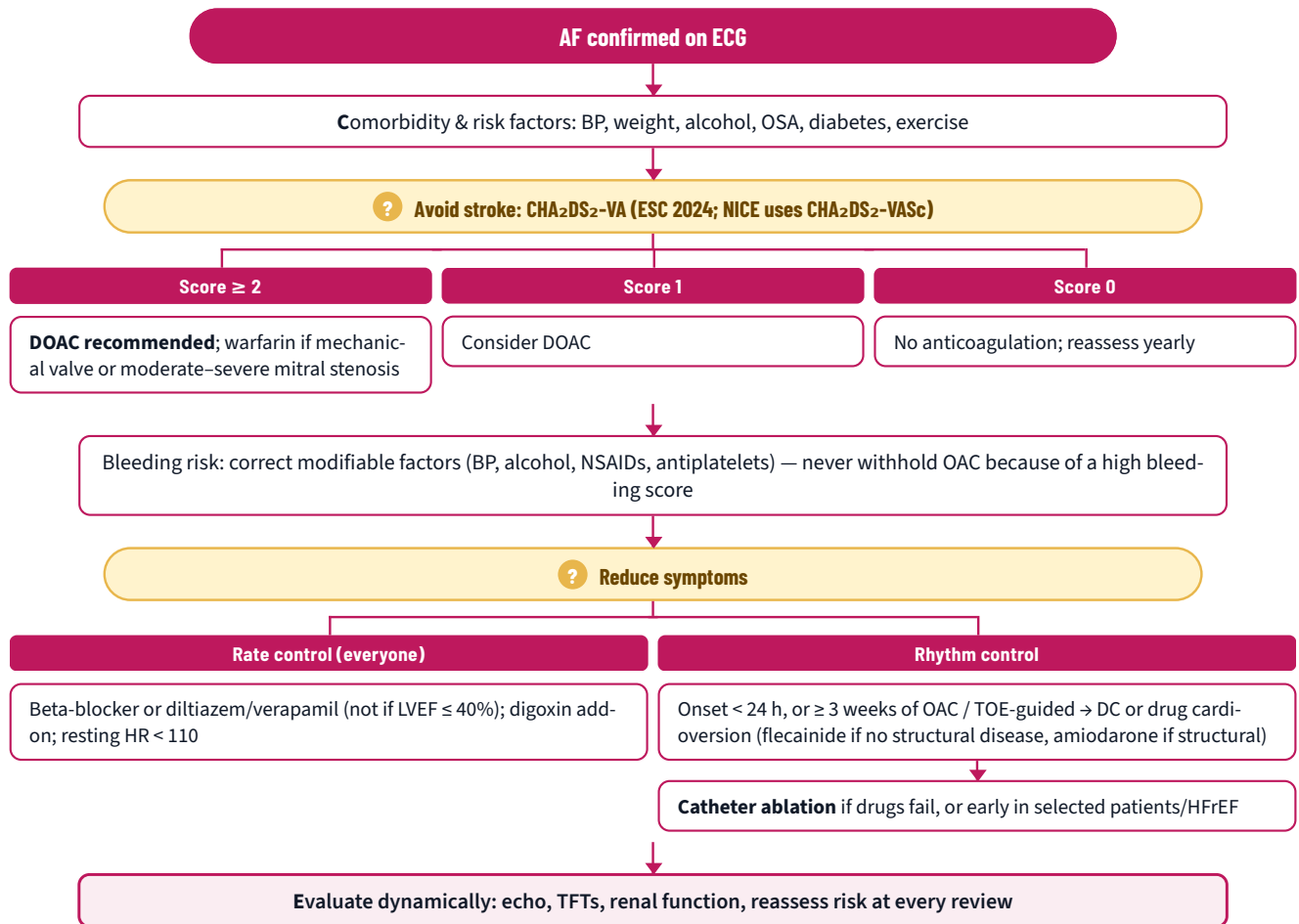
Tachycardia with a pulse

RESUSCITATION COUNCIL UK 2021 • ERC 2025



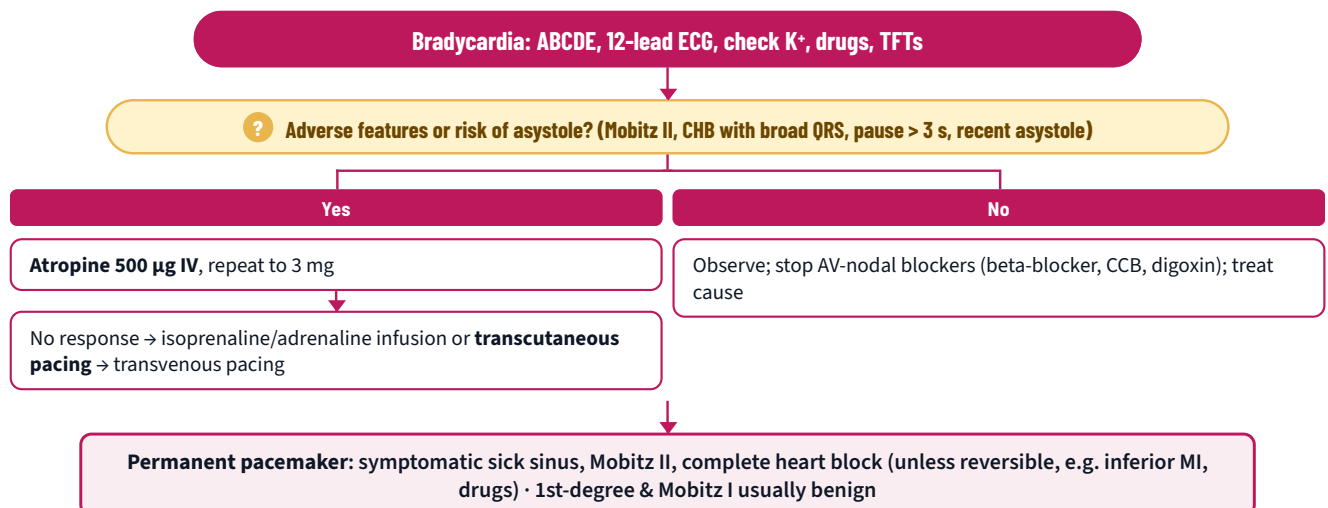
Atrial fibrillation: AF-CARE

ESC 2024 AF · NICE NG196



Bradycardia and heart block

RESUSCITATION COUNCIL UK 2021



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 59-year-old man with hyperthyroidism has palpitations and sweating. HR 150 bpm, BP 130/70 mmHg, SpO₂ 98%. ECG shows atrial fibrillation with a rapid ventricular response. TSH <0.01 mU/L with markedly raised free T₄; echo shows normal LV function (EF 60%) and no structural heart disease. What is the best initial rate-control therapy?

A IV diltiazem
B IV amiodarone
C Propranolol
D Digoxin
E Electrical cardioversion

- 2** A 68-year-old woman with long-standing heart failure and paroxysmal AF presents with vomiting, dizziness and palpitations after recently starting clarithromycin for pneumonia. She takes digoxin, furosemide and metoprolol. HR 40 bpm, BP 92/58 mmHg; ECG shows junctional bradycardia with occasional ventricular ectopics. Potassium 5.9 mmol/L, creatinine 1.8 mg/dL (baseline 1.0). Which ECG finding is most characteristic of digoxin effect?

A Short QT interval
B Delta waves
C Down-sloping ("scooped") ST depression
D Tall peaked T waves
E Broad QRS with LBBB pattern

- 3** The most common cause of atrial fibrillation is:

A Hypertension
B Valvular heart disease
C Lone atrial fibrillation
D Ischaemic heart disease

REPEATED 3* • ALSO ASKED IN

End-block exam 2025 • Previous-years exam

- 4** Which statement is true of Mobitz type I (Wenckebach) second-degree heart block?

A It usually indicates underlying ischaemia
B It is usually symptomatic
C The ECG shows progressive prolongation of the PR interval
D It is treated with a permanent pacemaker

REPEATED 3* • ALSO ASKED IN

End-block exam 2025 • Crash Course

- 5** All of the following are therapeutic options for atrial fibrillation EXCEPT:

A Digoxin
B Verapamil
C Radiofrequency pulmonary vein isolation
D Implantable cardioverter-defibrillator

REPEATED 2* • ALSO ASKED IN

Previous-years exam

- 6** A 66-year-old man with hypertension, ischaemic heart disease and atrial fibrillation has BP 160/100 mmHg and an irregularly irregular pulse of 120/min at routine follow-up. He takes captopril, furosemide, aspirin and spironolactone. What is the best add-on therapy?

A Amlodipine
B Metoprolol
C Digoxin
D Clopidogrel

- 7** A patient with acute MI suddenly starts snoring and collapses. The monitor shows a broad-complex rhythm at 168/min with fusion beats. What is the most likely diagnosis?

A Atrial fibrillation
B Ventricular fibrillation
C Ventricular tachycardia
D Supraventricular tachycardia

- 8** A 48-year-old man with diabetes has had acute retrosternal chest pain for 3 hours. Pulse 102/min, small volume; BP 110/65 mmHg. ECG shows ST elevation in I, aVL and V1–V4. Later he develops dizziness and palpitations; pulse 36/min and the ECG shows AV dissociation. Which complication has developed?

A First-degree heart block
B Mobitz type I block
C Mobitz type II block
D Complete heart block

REPEATED 2* • ALSO ASKED IN

Crash Course

CASE • QUESTIONS 9–11

A 77-year-old woman with diabetes, hypertension and heart failure presents with irregular palpitations. Heart rate 135/min, irregularly irregular; BP 70/50 mmHg. She takes metformin and lisinopril.

9 What is the first-line treatment?

- A Digoxin
- B Amiodarone
- C Synchronised DC cardioversion
- D Metoprolol

REPEATED 2* • ALSO ASKED IN
Other Sources

10 After she is stabilised, which drug is given to reduce her risk of thromboembolism?

- A Aspirin
- B Clopidogrel
- C Beta-blocker
- D Dual antiplatelet therapy
- E Rivaroxaban (DOAC)

11 If she remains symptomatic despite drug treatment, which measure gives long-term relief?

- A Pacemaker implantation and AV-node ablation
- B Aspirin
- C Bisoprolol
- D Canagliflozin

12 Which of the following drugs is NOT used to control the heart rate in atrial fibrillation?

- A Metoprolol
- B Diltiazem
- C Digoxin
- D Amiodarone

13 In a patient with atrial fibrillation, which of the following is NOT associated with an increased risk of stroke (thromboembolism)?

- A Coronary artery disease
- B Heart failure
- C Diabetes mellitus
- D Obesity
- E Previous stroke

14 All of the following are recognised causes of a prolonged PR interval EXCEPT:

- A Digoxin toxicity
- B Hypocalcaemia
- C Hypokalaemia
- D Rheumatic fever

REPEATED 3* • ALSO ASKED IN
End-block exam 2025 • Previous-years exam

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

15 The most important complication of atrial fibrillation is

- A Chest pain
- B Stroke due to thrombo-embolism
- C Dyspnoea
- D Headache

16 A 45-year-old man is found collapsed and an ECG on arrival shows torsades de pointes. Which of his medications is most likely to have contributed?

- A Bisoprolol
- B Risperidone
- C Cimetidine
- D Phenytoin

17 A 76-year-old woman is admitted with palpitations. During the cardiovascular examination you notice irregular cannon a' waves. Which one of the following would account for this finding?

- A Complete heart block
- B Atrial fibrillation with tricuspid stenosis
- C Atrio-ventricular nodal re-entry tachycardia
- D Tricuspid regurgitation

18 A patient with known Wolff–Parkinson–White syndrome has a regular narrow-complex tachycardia at 190/min (orthodromic AVRT). He is haemodynamically stable and vagal manoeuvres have failed. Which treatment is appropriate?

- A IV digoxin
- B IV verapamil
- C IV adenosine
- D IV lidocaine

Answers 1–18

1 **C Propranolol** — Thyrotoxic AF: a non-selective beta-blocker (propranolol) controls rate and adrenergic symptoms; antithyroid treatment restores sinus rhythm. Amiodarone worsens thyroid disease.

2 **C Down-sloping ("scooped") ST depression** — Digoxin effect is down-sloping ('reverse tick') ST depression with T flattening and short QT; toxicity shows as brady/junctional rhythms, ectopics or atrial tachycardia with block.

3 **A Hypertension** — **UPDATED** Hypertension is the commonest condition underlying AF (atrial pressure load); others: IHD,

heart failure, valve disease, thyroid, alcohol, obesity, sleep apnoea.

4 **C The ECG shows progressive prolongation of the PR interval** — Mobitz I: progressive PR prolongation until a dropped beat; AV-nodal, often benign, rarely needs pacing. Mobitz II (infranodal) progresses and needs pacing.

5 **D Implantable cardioverter-defibrillator** — AF is treated with rate control, rhythm control (cardioversion, drugs, ablation) and anticoagulation. An ICD treats ventricular arrhythmias only.

- 6 B Metoprolol** — AF at 120/min with hypertension and IHD: a beta-blocker (metoprolol) controls rate, BP and angina. He also needs a DOAC rather than aspirin.
- 7 C Ventricular tachycardia** — Fusion beats prove AV dissociation and are diagnostic of VT; collapse means pulseless VT, treated like VF with immediate defibrillation.
- 8 D Complete heart block** — A rate of 36/min with P waves independent of QRS is complete heart block; after anterior MI it is infranodal (septal necrosis) and needs pacing.
- 9 C Synchronised DC cardioversion** — AF with shock (BP 70/50) needs immediate synchronised DC cardioversion under sedation, with heparin, whatever the AF duration.
- 10 E Rivaroxaban (DOAC)** — CHA2DS2-VASc = 6 (age $\geq 75 \times 2$, female, HTN, diabetes, HF): long-term anticoagulation, preferably a DOAC (rivaroxaban/apixaban). Antiplatelets are not adequate.
- 11 A Pacemaker implantation and AV-node ablation** — Symptomatic AF despite drugs in an older HF patient: 'pace and ablate' (pacemaker then AV-node ablation) gives permanent rate control; PV isolation is the rhythm option.
- 12 D Amiodarone** — Rate control uses beta-blockers, diltiazem/verapamil or digoxin. Amiodarone is mainly for rhythm control and too toxic for long-term rate control.
- 13 D Obesity** — CHA2DS2-VASc: heart failure, hypertension, age, diabetes, stroke/TIA, vascular disease, female sex. Obesity raises AF risk but is not in the score.
- 14 B Hypocalcaemia** — PR > 200 ms: AV-nodal drugs, hypokalaemia, rheumatic fever, Lyme disease, inferior MI, fibrosis. Hypocalcaemia prolongs the QT, not the PR.
- 15 B Stroke due to thrombo-embolism** — The key complication of AF is ischaemic stroke from left atrial appendage thrombus (risk $\times 5$); hence CHA2DS2-VASc and anticoagulation.
- 16 B Risperidone** — Antipsychotics (risperidone, haloperidol) block hERG channels, prolong QT and cause torsades. Treat with IV magnesium and stop the drug.
- 17 A Complete heart block** — Irregular cannon a waves mean AV dissociation (complete heart block, VT); regular ones occur in junctional rhythms. AF abolishes a waves.
- 18 C IV adenosine** — **UPDATED** Orthodromic AVRT is narrow and AV-node-dependent, so adenosine terminates it. In pre-excited AF (irregular, broad) AV-nodal blockers are dangerous.

7 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

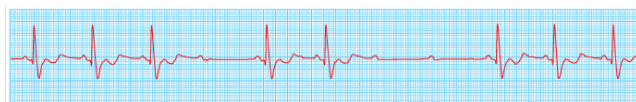
Davidson MCQs

- 1** A 75 year old woman presents to her family physician with a 24-hour history of rapid, irregular palpitations accompanied by fatigue. In an elderly patient, what is the most likely cause of palpitations?

A Atrial ectopic (premature) beats
 B Atrial fibrillation
 C Supraventricular tachycardia
 D Ventricular ectopic (premature) beats
 E Ventricular tachycardia

REPEATED 2* • ALSO ASKED IN
 Davidson

- 2** An 85 year old man presents with a 6-month history of sudden episodes of lightheadedness, which last up to 15 seconds. He is admitted to hospital with an episode of syncope resulting in facial injury. Examine the rhythm strip below. Which conduction abnormality does this show?



A Complete (third-degree) AV block
 B Left bundle branch block
 C Mobitz type II second-degree AV block
 D Sinus bradycardia
 E Wenckebach (Mobitz type I) second-degree AV block

- 3** Which of the following rhythms is NOT commonly associated with sick sinus syndrome (sinoatrial disease)?

A Atrial fibrillation
 B Atrial tachycardia
 C Sinus bradycardia
 D Sinus pauses
 E Ventricular tachycardia

- 4** A 75 year old woman has a history of hypertension and diabetes. She presents with atrial fibrillation. What is her CHA2DS2-VASc score?

A 2
 B 3
 C 4
 D 5
 E 6

- 5** A patient has sudden-onset rapid palpitations; the ECG shows a regular narrow-complex tachycardia at 180/min and he is haemodynamically stable. What should be used first to try to terminate the rhythm?

A Direct current cardioversion
 B Intravenous adenosine
 C Intravenous B-blocker
 D Oral B-blocker
 E Vagal manoeuvres, e.g. Valsalva manoeuvre

REPEATED 2* • ALSO ASKED IN
 Harrison

- 6** For which of the following scenarios would a permanent pacemaker be an appropriate treatment?
- A** Paroxysmal atrial fibrillation
 - B** Prevention of sudden death due to ventricular fibrillation
 - C** Sick sinus syndrome associated with syncope
 - D** Sinus bradycardia in an athlete
 - E** Supraventricular tachycardia
- 7** Which of the following is associated with excessive alcohol consumption?
- A** Atrial fibrillation
 - B** Diverticulitis
 - C** Hypertrophic cardiomyopathy
 - D** Hypotension
 - E** Supraventricular tachycardia

Answers 1-7

DAVIDSON MCQS

- 1** **B Atrial fibrillation** — Rapid irregular palpitations in an older person are most likely AF (prevalence ~10% over 80); confirm with an ECG during symptoms.
- 2** **C Mobitz type II second-degree AV block** — Constant PR with sudden dropped P waves is Mobitz II: infranodal, progresses to complete block and causes Stokes–Adams attacks — permanent pacemaker.
- 3** **E Ventricular tachycardia** — Sick sinus syndrome: sinus bradycardia, pauses, SA block, often with AF/atrial tachycardia (tachy–brady). VT is not a feature.
- 4** **D 5** — Age ≥ 75 (2) + female (1) + hypertension (1) + diabetes (1) = 5: high stroke risk; offer a DOAC.
- 5** **E Vagal manoeuvres, e.g. Valsalva manoeuvre** — Stable regular narrow-complex tachycardia: vagal manoeuvres first (modified Valsalva), then adenosine 6–12–18 mg rapid bolus. DC shock if unstable.
- 6** **C Sick sinus syndrome associated with syncope** — Permanent pacing for symptomatic bradycardia (sick sinus with syncope), Mobitz II and complete heart block. Athletes' bradycardia is physiological.
- 7** **A Atrial fibrillation** — Alcohol triggers AF ('holiday heart') and heavy intake raises AF incidence; abstinence reduces recurrence. It also causes dilated cardiomyopathy.

28 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

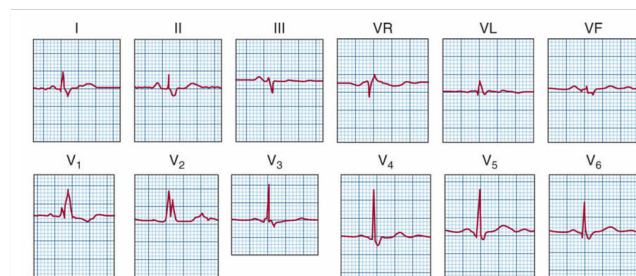
SET 1 OF 2

- 1** A 55-year-old woman with no past medical history presents with syncope; her ECG (A) shows advanced AV block, and she has painful red nodules on her shins (B). What is the most likely cause of both the cardiac and skin abnormalities?



- A Carney syndrome
- B Hypothyroidism
- C Sarcoidosis
- D Systemic lupus erythematosus
- E Tertiary syphilis

- 2** Which abnormality is shown on this 12-lead ECG?



- A First-degree heart block
- B Left bundle branch block
- C P-pulmonale
- D Right bundle branch block
- E S1Q3T3 (McGinn-White pattern) indicative of right ventricular strain

- 3** A 21-year-old competitive cyclist, asymptomatic and without syncope, has a resting heart rate of 45/min with occasional pauses up to 2.2 seconds. BP 108/72 mmHg; examination is otherwise normal, and the ECG shows sinus rhythm with PR 128 ms and QRS 80 ms. What advice is appropriate?

- A Perform electrophysiologic evaluation for consideration of a pacemaker
- B He should not compete in the upcoming race due to concern for bradycardia and will need a tilt table test
- C No further follow-up is needed. Good luck with the race!
- D Obtain a 48-hour Holter monitor tracing
- E He should undergo a treadmill stress ECG to determine the presence of chronotropic competence

- 4** A 60-year-old man is undergoing an electrophysiology study for evaluation of syncope. After careful venous cannulation and placement of conductance and pacing catheters and after administration of 0.2 mg/kg of propranolol and 0.04 mg/kg of atropine, his heart rate is 65 bpm. After stopping the drugs and allowing adequate time for washout, his superior/lateral right atrium is paced 140 bpm. On cessation of this overdrive pacing, his next sinus beat occurs 1800 msec later. Based on these observations, this patient can be diagnosed with which of the following?

- A Amyloid cardiomyopathy
- B AV nodal disease
- C Paroxysmal atrial fibrillation
- D SA nodal disease
- E Tachy-brady syndrome

- 5** All of the following are reversible causes of SA node dysfunction EXCEPT:
- A Hypothermia
 - B Hypothyroidism
 - C Increased intracranial pressure
 - D Lithium toxicity
 - E Radiation therapy
- 6** Which of the following is a risk factor for development of thromboembolism in patients with the tachycardia-bradycardia variant of sick sinus syndrome?
- A Age >50 years
 - B Atrial enlargement
 - C Diabetes mellitus
 - D Prothrombin 20210 mutation
 - E None of the above; there is no increased risk of thromboembolism with the tachycardia-bradycardia variant of sick sinus syndrome
- 7** Normal cells within the AV node exhibit a property known as decremental conduction. If you wanted to demonstrate this property during an electrophysiology study, what maneuver could you perform?
- A Pace the right atrium at serially decreasing cycle length and measure the pace-to-His conduction time
 - B Pace the ventricle and record right atrial potentials
 - C Administer atropine 0.04 mg/kg and record AV nodal conduction time by measuring the time it takes an atrial pacing beat to reach the His bundle
 - D Administer metoprolol 10 mg intravenously and record AV nodal conduction time by measuring the time it takes an atrial pacing beat to reach the His bundle
 - E Administer adenosine 12 mg intravenously and record the AV nodal recovery time
- 8** An 87-year-old man with a history of well-treated hypertension and aortic stenosis has become symptomatic from his aortic stenosis over the last 2 months. Yesterday, he underwent surgical aortic valve replacement with a bioprosthetic 25-mm valve with excellent intraoperative results. He was rapidly weaned from cardiopulmonary bypass and extubated within 24 hours. Per surgical protocol, he had temporary epicardial pacing wires placed on the ventricular surface and has been pacing at 90 bpm. On rounds this morning, you briefly pause his ventricular pacing to check his underlying rhythm. You note an atrial rate of 80 bpm, but a ventricular rate of 32 bpm with a wide QRS complex. There is no relationship between the P waves and QRS complexes. This patient's ventricular bradycardia is most likely due to which of the following?
- A Development of a systemic disease such as sarcoidosis or Lyme disease causing AV node dysfunction
 - B Endocarditis of the new aortic bioprosthesis causing AV node block
 - C SA node disease
 - D Slowed AV nodal recovery after overdrive pac ing
 - E Surgical injury to the AV node
- 9** An 82-year-old man has had two episodes of sudden syncope without warning in the past week, one causing facial bruising. During a further 'spell' in clinic, the rhythm strip shown is recorded. What type of block is present, matched to the appropriate management?
- 
- A Complete SA nodal block - Permanent pacemaker implantation
 - B First-degree AV block - Administer atropine
 - C Second-degree Mobitz type II SA nodal block - Exercise treadmill study
 - D Second-degree Mobitz type I AV block - No intervention necessary
 - E Second-degree Mobitz type II AV node block - Permanent pacemaker implantation
- 10** A 19-year-old long-distance runner, who finished in the top 10 of the local marathon last year, presents for cardiac evaluation after his primary care physician ordered a Holter monitor for screening purposes. On his Holter report, several episodes of second-degree, Mobitz I (Wenckebach) AV block were noted, all occurring during sleep. The patient reports no symptoms but thinks he may have a grandfather who had a pacemaker implanted at an advanced age. What is the most appropriate next step?
- A Exercise treadmill stress ECG
 - B Invasive electrophysiology study
 - C Reassurance
 - D Refer for pacemaker implantation
 - E Serologic testing including thyroid-stimulating hormone levels
- 11** A 47-year-old woman with a history of tobacco abuse and ulcerative colitis is evaluated for intermittent palpitations. She reports that for the last 6 months, every 2-4 days she notes a sensation of her heart "flip-flopping" in her chest for approximately 5 minutes. She has not noted any precipitating factors and has not felt light headed or had chest pains with these episodes. Her physical examination is normal. A resting ECG reveals sinus rhythm and no abnormalities. Aside from checking serum electrolytes, which of the following is the most appropriate testing?
- A Abdominal computed tomography (CT) with oral and intravenous (IV) contrast
 - B Event monitor
 - C Holter monitor
 - D Reassurance with no further testing needed
 - E Referral for electrophysiology study

- 12** A 47-year-old woman with a normal examination and ECG has brief 'flip-flopping' palpitations every few days. Event monitoring shows that the symptoms correspond to atrial premature contractions only. Which statement is true?

- A** Atrial premature contractions are less common than ventricular premature contractions on extended ECG monitoring
- B** Echocardiography is indicated to determine if structural heart disease is present
- C** Metoprolol should be initiated for symptom control
- D** The patient should be reassured that this is not a dangerous condition and does not require further evaluation
- E** The patient should undergo stress test to determine if ischaemia is present

REPEATED 2* • ALSO ASKED IN PreTest

- 13** An 18-year-old volleyball player with rare palpitations and light-headedness has a normal examination; her ECG shows a PR interval of 60 ms, QRS duration of 140 ms and a delta wave (WPW pattern). Which finding is reassuring that she is at low risk and does not need catheter ablation?

- A** Ability to increase the heart rate to 185 bpm on an exercise treadmill test
- B** Electrophysiology study demonstrating that the accessory pathway has both antegrade and retrograde conduction properties
- C** Electrophysiology study demonstrating that the accessory pathway is located in the posteroseptal region
- D** Exercise treadmill study demonstrating disappearance of the delta wave and wide QRS at a heart rate of 120 bpm
- E** Holter monitoring demonstrating occasional runs of atrial fibrillation

- 14** An 85-year-old woman with no prior cardiac history presents to the emergency department with 2 hours of palpitations. Blood pressure, oxygen saturation, and heart rate are normal, although you note an irregularly irregular rhythm on examination. ECG shows an irregularly irregular, narrow QRS without discernable P waves at a rate of 75 bpm. Echocardiogram reveals no structural heart disease. Despite the normal heart rate, the patient is quite symptomatic with her atrial fibrillation and wants to pursue achievement of sinus rhythm. All of the following interventions may be beneficial EXCEPT:

- A** Adenosine intravenously
- B** Amiodarone intravenously
- C** Direct current cardioversion
- D** Dofetilide orally
- E** Flecainide orally

- 15** A 79-year-old man with ischaemic cardiomyopathy (LVEF 30%) and hypertension has no new symptoms; BP 108/56 mmHg, HR 88/min. His rhythm strip is shown. Based on this, he now has a definite (class I) indication for which therapy?

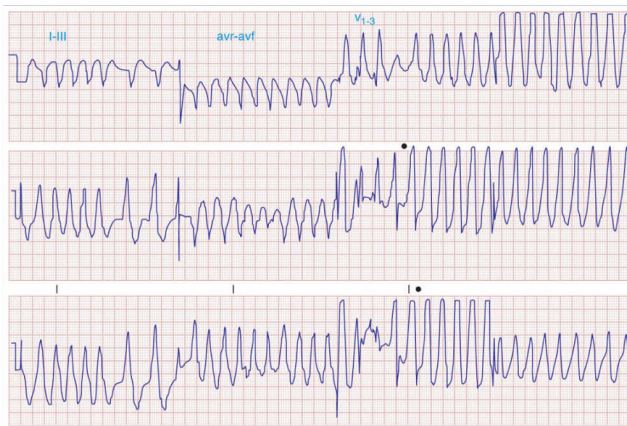


- A** Amiodarone 400 mg daily
- B** Aspirin 325 mg daily
- C** Flecainide 600 mg PRN palpitations
- D** Systemic anticoagulation with warfarin or a novel oral anticoagulant
- E** Transesophageal echocardiography followed by direct current (DC) cardioversion

- 16** Which of the following statements regarding restoration of sinus rhythm after atrial fibrillation is true?

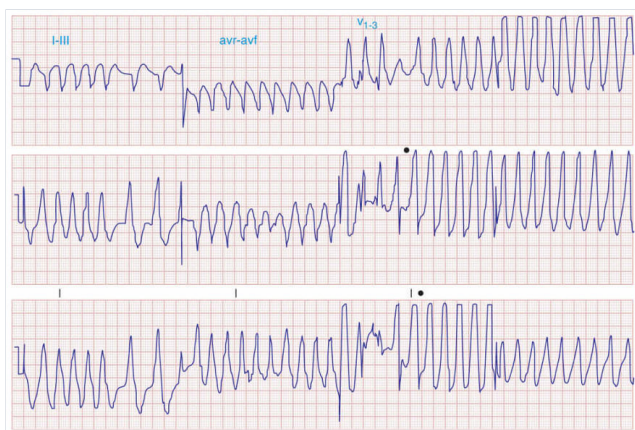
- A** Dofetilide may be safely started as an outpatient
- B** In patients who are treated with pharmacotherapy and are found to be in sinus rhythm, a prolonged Holter monitor should be worn to determine if anticoagulation can be safely stopped
- C** Patients who have pharmacologically maintained sinus rhythm after atrial fibrillation have improved survival compared with patients who are treated with rate control and anticoagulation
- D** Recurrence of atrial fibrillation is uncommon when pharmacotherapy is used to maintain sinus rhythm

- 17** A young woman collapses while jogging and is unresponsive, with BP 50 mmHg (palpable) and a heart rate of about 280/min. Her 12-lead ECG is shown. What is the next most appropriate step?



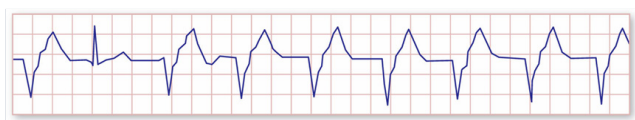
- A** Amiodarone 150 mg IV stat
- B** Lidocaine 1 mg IV stat
- C** Metoprolol 10 mg IV
- D** DC defibrillation
- E** Emergent CT scan with contrast to evaluate for pulmonary embolism

- 18** A young woman collapses while jogging (BP 50 mmHg, heart rate about 280/min). Her 12-lead ECG is shown. What is the underlying rhythm?



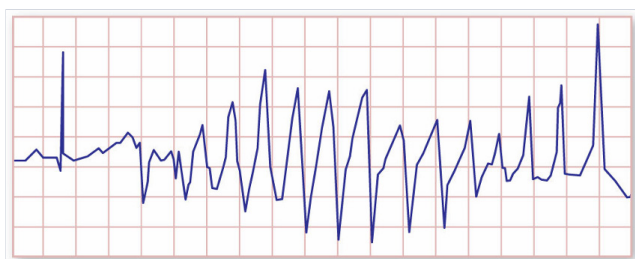
- A Atrial fibrillation with antegrade conduction through an accessory tract
- B Atrial flutter with 1:1 conduction
- C AV reentry tachycardia
- D AV nodal reentry tachycardia
- E Ventricular tachycardia

- 19** A 67-year-old man was given thrombolysis 3 hours ago for an anterolateral STEMI. His monitor now shows the rhythm strip below. What rhythm does this represent?



- A Atrial fibrillation
- B Complete heart block with a junctional escape
- C Idioventricular rhythm
- D Normal sinus rhythm with a left bundle branch block
- E Ventricular tachycardia

- 20** A 47-year-old woman on methadone takes leftover erythromycin for a respiratory infection and develops palpitations and light-headedness. The rhythm strip shows recurrent non-sustained runs of the arrhythmia shown. IV magnesium sulfate 2 g does not stop the episodes, and her potassium is normal. What is the next most appropriate step?



- A Metoprolol 5 mg IV repeatedly until her resting heart rate is 60 bpm
- B Amiodarone 150 mg IV
- C Emergent referral for implantable defibrillator
- D IV isoproterenol infusion titrated to a rate of 100-120 bpm
- E Sedation and defibrillation

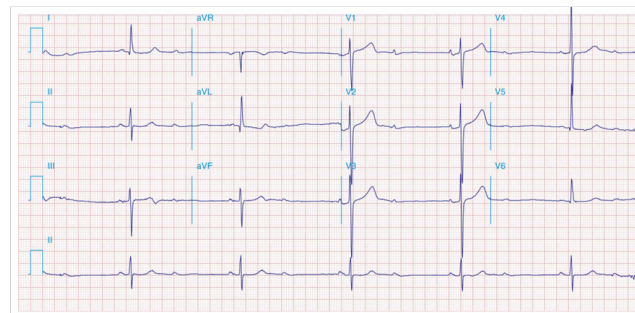
- 21** In an ECG with wide complex tachycardia, which of the following clues most strongly supports the diagnosis of ventricular tachycardia?

- A Atrial-ventricular dissociation
- B Classic right bundle branch block pattern
- C Irregularly irregular rhythm with changing QRS complexes
- D QRS duration >120 msec
- E Slowing of rate with carotid sinus massage

- 22** An 18-year-old man with no symptoms has an ECG at a pre-university medical showing an irregular rhythm: every QRS is preceded by an identical normal P wave with a constant PR interval, and the rate speeds up with inspiration and slows with expiration. What is the most appropriate next step?

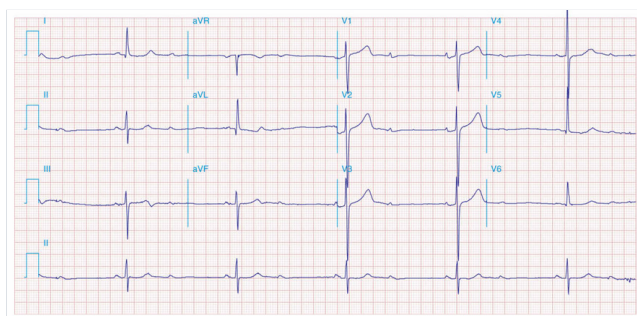
- A Echocardiography
- B Electrophysiology study and planned ablation of an ectopic atrial focus
- C Exercise ECG stress test
- D Holter monitor to define arrhythmia burden
- E Reassurance

- 23** A 22-year-old triathlete with type 1 diabetes is recovering from diabetic ketoacidosis. While she is asleep, the monitor shows bradycardia; the ECG is shown (2:1 AV block with narrow QRS complexes). Where is the block most likely to be?



- A AV node
- B Bachmann bundle
- C His bundle
- D Left bundle branch
- E SA node

- 24** A young athlete has 2:1 AV block with narrow QRS complexes while asleep (ECG shown), with the block located in the AV node. What would be expected if IV atropine were given?



- A** Increase sinus rate and maintain 2:1 AV block
- B** Increase sinus rate and improve heart block to 1:1 conduction
- C** Increase sinus rate and worsen heart block to >2:1
- D** No change in sinus rate and worsen heart block to >2:1
- E** No change

- 25** An 87-year-old man with Parkinson disease takes levodopa. After running out of his medication and failing to obtain a refill, he presents to the emergency department with a parkinsonian exacerbation. On initial evaluation, he has the ECG shown in Figure V-52. Which of the following is the next most appropriate step?



- A** Anticoagulation with IV heparin
- B** Check serum thyroid-stimulating hormone (TSH) and obtain targeted respiratory history
- C** DC cardioversion
- D** Metoprolol 5 mg IV
- E** No further testing or intervention

Answers 1-25

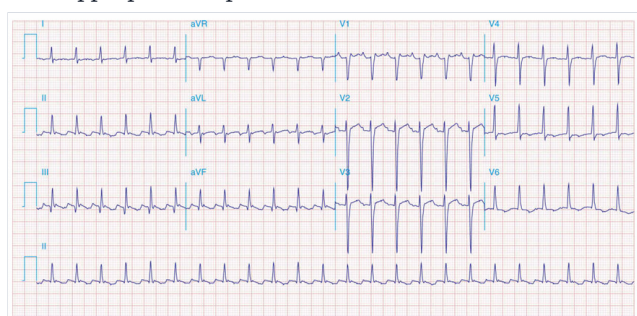
HARRISON MCQs

- 1** **C Sarcoidosis** — Heart block plus erythema nodosum suggests sarcoidosis: granulomas in the septum/conduction system; confirm with cardiac MRI or FDG-PET.
- 2** **D Right bundle branch block** — Broad QRS with rSR' ('M') in V1–V2 and slurred S in I, aVL, V5–V6 is RBBB. LBBB gives a broad QS in V1 and notched R in V6.
- 3** **C No further follow-up is needed. Good luck with the race!** — Sinus bradycardia, first-degree and Wenckebach block are physiological in asymptomatic athletes; no tests or restriction needed.
- 4** **D SA nodal disease** — A prolonged sinus node recovery time after overdrive pacing (1800 ms) with low intrinsic rate means intrinsic SA node disease.
- 5** **E Radiation therapy** — Reversible SA node dysfunction: hypothermia, hypothyroidism, raised ICP, drugs, hyperkalaemia, vagal tone. Radiation causes permanent fibrosis.
- 6** **B Atrial enlargement** — Tachy-brady syndrome includes AF/flutter; atrial enlargement (and age, prior stroke, LV dysfunction) adds embolic risk. A pacemaker does not remove it.
- 7** **A Pace the right atrium at serially decreasing cycle length and measure the pace-to-His conduction time** — Decremental conduction (slower conduction at faster rates) is shown by atrial pacing at shorter cycle lengths with a lengthening AH interval until Wenckebach.
- 8** **E Surgical injury to the AV node** — The AV node lies near the aortic annulus, so AVR/TAVI can cause complete heart block; pace permanently if it persists beyond ~5–7 days.
- 9** **E Second-degree Mobitz type II AV node block - Permanent pacemaker implantation** — Constant PR then sudden dropped
- 10** **C Reassurance** — Nocturnal Mobitz I in an asymptomatic athlete is high vagal tone; it resolves with exercise. Reassure.
- 11** **B Event monitor** — Monitoring depends on symptom frequency: Holter for daily symptoms, event recorder for every few days, implantable loop recorder for rare episodes.
- 12** **D The patient should be reassured that this is not a dangerous condition and does not require further evaluation** — **UPDATED** Atrial premature complexes are common and benign in a normal heart; reassure and reduce caffeine/alcohol. Beta-blocker only if troublesome.
- 13** **D Exercise treadmill study demonstrating disappearance of the delta wave and wide QRS at a heart rate of 120 bpm** — Sudden loss of the delta wave on exercise shows a pathway with a long refractory period, so low risk of pre-excited AF degenerating to VF.
- 14** **A Adenosine intravenously** — Recent-onset AF converts with DC shock, flecainide or amiodarone. Adenosine only transiently blocks the AV node and does not convert AF.
- 15** **D Systemic anticoagulation with warfarin or a novel oral anticoagulant** — AF with heart failure, hypertension, vascular disease and age ≥ 75 (CHA2DS2-VASc ≥ 5): oral anticoagulation is class I. Flecainide is contraindicated.
- 16** **B In patients who are treated with pharmacotherapy and are found to be in sinus rhythm, a prolonged Holter monitor should be worn to determine if anticoagulation can be safely stopped** — Rhythm control did not improve survival (AFFIRM) and asymptomatic recurrence is common, so anticoagulation follows CHA2DS2-VASc, not apparent sinus rhythm.

- 17 **D DC defibrillation** — Irregular very fast broad complexes of varying shape are pre-excited AF; with collapse give immediate DC shock. AV-nodal blockers can precipitate VF.
- 18 **A Atrial fibrillation with antegrade conduction through an accessory tract** — **UPDATED** Irregularly irregular broad complexes with varying width and RR < 250 ms is AF conducting down an accessory pathway. VT and AVRT are regular.
- 19 **C Idioventricular rhythm** — A regular broad ventricular rhythm at 60–110/min soon after thrombolysis is accelerated idioventricular rhythm, a benign reperfusion marker.
- 20 **D IV isoproterenol infusion titrated to a rate of 100–120 bpm** — Torsades from methadone + erythromycin persisting after magnesium: raise the rate with isoprenaline or overdrive pacing to shorten QT. Amiodarone worsens it.
- 21 **A Atrial-ventricular dissociation** — AV dissociation (independent P waves, fusion or capture beats) is diagnostic of VT.

SET 2 OF 2

- 26 A 68-year-old man with non-ischaemic cardiomyopathy, hypertension and an unexplained TIA 6 months ago has felt unwell for 5 days; BP 110/78 mmHg, HR 150/min. His ECG is shown. Two doses of IV metoprolol 5 mg slow the rate to 75/min and he feels better, and ablation is planned. What is the next most appropriate step?

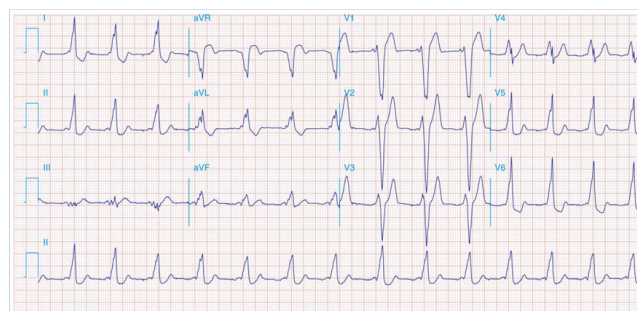


- A Amiodarone
B Aspirin
C Digoxin
D Furosemide
E Systemic anticoagulation

Prior MI, QRS > 160 ms and chest-lead concordance also favour VT.

- 22 **E Reassurance** — **UPDATED** Respiratory sinus arrhythmia (rate rises on inspiration, falls on expiration) is normal, especially in the young; all beats are sinus, so reassure.
- 23 **A AV node** — 2:1 block with narrow QRS in a sleeping young athlete is almost certainly vagally mediated AV-nodal block; infranodal block usually has broad QRS.
- 24 **B Increase sinus rate and improve heart block to 1:1 conduction** — **UPDATED** Atropine removes vagal tone at SA and AV nodes, raising the rate and restoring 1:1 conduction in nodal block; in infranodal block it may worsen the ratio.
- 25 **E No further testing or intervention** — Parkinsonian tremor artefact (4–6 Hz) can mimic flutter, AF or VT; regular QRS complexes seen through it confirm sinus rhythm, so no treatment.

- 27 A 22-year-old woman presents with palpitations. Her ECG (shown) has a short PR interval and delta waves. Where is her accessory pathway likely to be?



- A Anterior septum
B Anterolateral left ventricle
C Inferior left ventricle
D Inferior septum
E Right ventricle

- 28 Mr. Cooper underwent tPA administration for acute ST-segment elevation myocardial infarction 1 hour previously. He had an excellent response with resolution of his ST-segment elevation and chest pain. However, the nurse pages you to evaluate a change in his rhythm. You note a wide complex ventricular rhythm with a rate of 75 bpm. Mr. Cooper continues to feel well without new symptoms, and his blood pressure is 120/80 mmHg. The next most appropriate therapeutic maneuver is which of the following?
- A Administer amiodarone 150 mg IV over 10 minutes
B Administer flecainide 400 mg orally
C Administer metoprolol 5 mg IV every 5 minutes × 3
D Carotid sinus massage
E Continue observation

Answers 26–28

HARRISON MCQs

- 26** **E Systemic anticoagulation** — Saw-tooth waves in II, III, aVF with 2:1 conduction is typical flutter; its stroke risk equals AF, and CHA2DS2-VASc is high, so anticoagulate before ablation.
- 27** **E Right ventricle** — A negative delta wave and LBBB-like QRS in V1 indicate a right-sided accessory pathway; a positive delta in V1 indicates a left-sided pathway.

- 28** **E Continue observation** — A wide ventricular rhythm at 75/min soon after successful lysis in a well patient is accelerated idioventricular rhythm: observe. Suppression is harmful.

13 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1** A 73-year-old woman has chest discomfort and a slow pulse of 53/min with occasional pauses. Her ECG shows sinus bradycardia with intermittent pauses of 3 seconds in which no P wave appears. What is the most likely diagnosis?
- A Sick sinus syndrome
B First-degree atrioventricular (AV) block
C Mobitz type 1 block
D Mobitz type 2 block
E Complete heart block
- 2** A 28-year-old man with no past medical history and not on medications presents to the emergency department with palpitations for several hours and was found to have supraventricular tachycardia. Carotid massage was attempted without success. What is the treatment of choice to stop the attack?
- A Intravenous (IV) lignocaine
B IV digoxin
C IV amiodarone
D IV adenosine
E IV quinidine

REPEATED 4* • ALSO ASKED IN
PreTest • Other Sources ×2

- 3** A 75-year-old smoker with ischaemic heart disease and previous heart failure presents with a 6-hour history of palpitations and increasing breathlessness. He is alert, BP 128/76 mmHg, SpO₂ 94% on air, with no chest pain, and his chest is clear. The ECG shows a regular narrow-complex tachycardia at 160/min, and vagal manoeuvres have had no effect. What is the most appropriate next step?
- A Slow the heart rate with IV atenolol to help identify the rhythm
B IV lidocaine 50 mg bolus followed by an infusion of 4 mg/min
C IV adenosine to terminate the rhythm or reveal the underlying atrial activity
D Immediate DC cardioversion, because the rhythm may be ventricular tachycardia
E Withhold oxygen because he is a smoker

- 4** A 57-year-old man sees his general practitioner because of a strong family history of heart disease. An ECG shows sinus rhythm with a constant PR interval of 300 ms (0.3 s), and every P wave is followed by a QRS complex. What is the most likely diagnosis?
- A Sick sinus syndrome
B First-degree AV block
C Mobitz type 1 block
D Mobitz type 2 block
E Complete heart block

REPEATED 2* • ALSO ASKED IN
Other Sources

- 5** You are the junior doctor on call and a nurse asks you to review an ECG. It shows regular P waves with a constant PR interval in the conducted beats, but some P waves are not followed by a QRS complex; there is no progressive PR prolongation before the dropped beats. What is the most likely diagnosis?
- A Sick sinus syndrome
B First-degree AV block
C Mobitz type 1 block
D Mobitz type 2 block
E Complete heart block

- 6** You are a house officer on call when you are asked to see a 48-year-old man who is found to be unresponsive by the nursing staff. You cannot obtain a palpable carotid pulse and the cardiac monitor shows ventricular fibrillation. What is the first appropriate measure?
- A Sodium bicarbonate 50 mmol IV
B Immediate defibrillation (biphasic 150–200 J)
C Calcium chloride 10 mL IV
D Adrenaline 1 mg IV
E Carotid sinus massage

REPEATED 3* • ALSO ASKED IN
PreTest • Other Sources

- 7** At an ECG teaching session, a consultant describes a patient admitted to the coronary care unit with a regular atrial rate of 300/min and a regular ventricular rate of 150/min. What is the most likely diagnosis?
- A Atrial flutter
 - B First-degree AV block
 - C Mobitz type 1 block
 - D Mobitz type 2 block
 - E Complete heart block
- 8** A patient on the coronary care unit develops new atrial flutter with an atrial rate of 300/min and a ventricular rate of 150/min; it began about 6 hours ago. He is haemodynamically stable, and a rhythm-control strategy is chosen. What is the most effective first-line treatment to restore sinus rhythm?
- A Sotalol
 - B Radiofrequency catheter ablation
 - C Procainamide
 - D Lidocaine
 - E Synchronised DC cardioversion
- 9** A previously well 36-year-old woman presents to her GP with a 2-month history of attacks of palpitations and sweating. The attacks come on suddenly and are unrelated to food, posture or exercise. She does not smoke or drink alcohol. Which of the following investigations is the LEAST useful in evaluating her?
- A Thyroid function tests
 - B Urinary collection for catecholamines
 - C Holter monitor
 - D Echocardiography
 - E Upper gastrointestinal study
- 10** A 57-year-old Bangladeshi patient is admitted to accident and emergency (AE) with palpitations. He speaks very little English but tells you he has a 'heart problem'. On examination you note his pulse is irregular at 145 beats/min. An ECG is performed which demonstrates no P waves. Which management plan would you instigate first?
- A Bisoprolol
 - B Flecainide
 - C Radiofrequency catheter ablation
 - D Procainamide
 - E Lidocaine
- 11** A 37-year-old woman develops palpitations after recent surgery. Her pulse is regular at 175/min and BP is 125/70 mmHg; she is otherwise well. She has asthma. The ECG shows a regular narrow-complex tachycardia with no visible P waves. What is the next most appropriate step in management?
- A Adenosine
 - B Verapamil
 - C Carotid sinus massage
 - D Direct current (DC) cardioversion
 - E Flecainide
- 12** A 16-year-old boy is brought to the emergency department after collapsing during football training. He is now alert and orientated, and his vital signs and examination are normal. He remembers feeling dizzy before collapsing but recalls little else. He is normally healthy and takes no regular medication. Blood tests have been sent. What is the most appropriate next step?
- A Admit him for cardiac monitoring
 - B Request a neurology review
 - C Order an ECG
 - D Order electroencephalography (EEG)
 - E Schedule a tilt-table test
- 13** An otherwise healthy 50-year-old man presents to his GP with palpitations and is found to have an irregularly irregular pulse. The episode lasts less than 2 hours in total and resolves without treatment. Full blood count, biochemistry, thyroid function, ECG and echocardiogram are all normal. Which one of the following is the most appropriate treatment?
- A Aspirin
 - B Clopidogrel
 - C Do nothing
 - D Dipyridamole
 - E Warfarin

REPEATED 2* · ALSO ASKED IN
Passmedicine

Answers 1-13

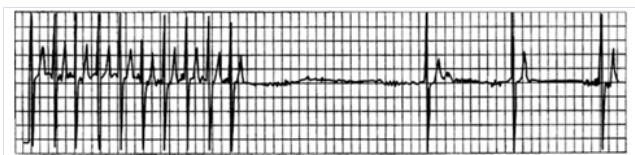
- 1** **A Sick sinus syndrome** — **UPDATED** Sinus pauses with absent P waves plus bradycardia in an older person is sick sinus syndrome. In AV block P waves are present but not conducted.
- 2** **D IV adenosine** — If vagal manoeuvres fail, adenosine (6 mg rapid bolus, then 12 and 18 mg) terminates AV-node-dependent SVT; verapamil if asthmatic.
- 3** **C IV adenosine to terminate the rhythm or reveal the underlying atrial activity** — **UPDATED** Stable regular narrow-complex tachycardia: after vagal manoeuvres give adenosine, which ends AVNRT/AVRT or unmasks flutter waves. Cardiovert only if adverse features.
- 4** **B First-degree AV block** — PR > 200 ms with every P conducted is first-degree AV block (ageing, vagal tone, AV-nodal drugs, inferior MI); no treatment needed alone.

- 5 **D Mobitz type 2 block** — Dropped P waves with a fixed PR in conducted beats is Mobitz II: infranodal, can progress abruptly to complete block, so pace permanently.
- 6 **B Immediate defibrillation (biphasic 150–200 J)** — UPDATED Pulseless VF is a shockable rhythm: defibrillate immediately, then 2 min CPR. Adrenaline is given after the third shock; bicarbonate and calcium are not routine.
- 7 **A Atrial flutter** — Atrial rate 300/min with ventricular rate 150/min is atrial flutter with 2:1 block; suspect it in any regular narrow tachycardia near 150/min.
- 8 **E Synchronised DC cardioversion** — UPDATED Synchronised DC cardioversion restores sinus rhythm in > 90% of flutter (drugs work poorly); anticoagulate as for AF. CTI ablation prevents recurrence.
- 9 **E Upper gastrointestinal study** — Paroxysmal palpitations with sweating: check TFTs, metanephrines (phaeochromocytoma), ambulatory ECG and echo. An upper GI study has no role.
- 10 **A Bisoprolol** — UPDATED Irregular tachycardia without P waves is AF; first-line rate control in a stable patient is a beta-blocker (bisoprolol). Flecainide is for rhythm control.
- 11 **C Carotid sinus massage** — Stable regular narrow-complex tachycardia: vagal manoeuvres first (carotid massage if young, no bruit); then adenosine, or verapamil if asthmatic.
- 12 **C Order an ECG** — Syncope during exercise is a cardiac red flag (HCM, long QT, WPW, Brugada, ARVC); a 12-lead ECG is the essential first test.
- 13 **C Do nothing** — Brief self-terminating AF with CHA2DS2-VASc 0 and a normal heart needs no anticoagulation; antiplatelets are never used for AF stroke prevention.

8 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 An 82-year-old woman is admitted for observation after presenting with dizziness, and the rhythm strip shown is recorded on the monitor. She has no history of cardiac disease, diabetes or hypertension. On questioning she reports several previous episodes of transient dizziness and one brief syncope. Examination is unremarkable. Which of the following is the best plan of care?



- A Reassurance; this is a benign condition and needs no treatment
- B Reassurance and advice not to drive until symptom-free; no other treatment
- C Nuclear cardiac stress testing, with treatment depending on the result
- D Start aspirin
- E Permanent pacemaker implantation

- 2 An active 78-year-old woman with hypertension presents with a new left hemiparesis; CT excludes haemorrhage, and cardiac monitoring shows atrial fibrillation. She was in sinus rhythm at a check-up 3 months earlier. Optimal management at discharge includes review of her antihypertensive therapy, a ventricular rate-control agent and which of the following?

- A An implantable cardioverter-defibrillator/permanent pacemaker to avoid the need for anticoagulation
- B Immediate direct-current cardioversion
- C Antiplatelet therapy such as aspirin, without anticoagulation
- D Antiplatelet therapy plus warfarin with a target INR of 1.5
- E Long-term oral anticoagulation (a DOAC, or warfarin with a target INR of 2.0–3.0)

- 3 A 72-year-old man with poorly controlled hypertension develops a viral upper respiratory infection. On the second day he has palpitations and attends the emergency department. His BP is 118/78 mmHg and he is otherwise well. The rhythm strip (lead V1) is shown. What is the best next step in management?



- A Administration of intravenous metoprolol
- B Administration of intravenous adenosine
- C Administration of intravenous amiodarone
- D Emergent electrical cardioversion
- E Initiation of chest compressions and preparation for semielective intubation

- 4 An asymptomatic 65-year-old man with diabetes attends the emergency department with a sports-related right shoulder injury. His pulse is irregular, and the rhythm strip shown is recorded; BP and the rest of the examination are normal. What is the best immediate management?



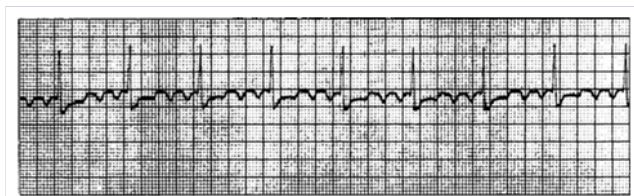
- A Atropine
- B Isoprenaline
- C Permanent pacemaker implantation
- D Electrical cardioversion
- E Observation

- 5 A patient has been on the coronary care unit for 24 hours with an acute anterior myocardial infarction. He develops the rhythm shown, although his BP remains stable at 110/68 mmHg. What is the best next step in management?



- A Cardioversion
- B Pacing (temporary transvenous, then assessment for a permanent pacemaker)
- C Digoxin
- D Propranolol
- E Lidocaine

- 6 A 62-year-old man with COPD develops a viral upper respiratory infection and starts an over-the-counter decongestant. Shortly afterwards he has palpitations and attends the emergency department, where the rhythm strip shown is recorded. What is the most likely diagnosis?



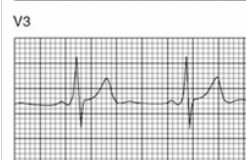
- A Normal sinus rhythm
- B Junctional rhythm
- C Atrial flutter with 4:1 atrioventricular block
- D Atrial tachycardia with 2:1 atrioventricular block
- E Atrial fibrillation

- 7 An 80-year-old woman is admitted with dizziness. Monitoring shows atrial fibrillation with a rapid ventricular response, which is controlled with a beta-blocker. Echocardiography shows an enlarged left atrium, an ejection fraction of 50% and no diastolic dysfunction. She is now asymptomatic, with BP 130/80 mmHg and an irregularly irregular pulse of about 80/min. What is the best management strategy for her arrhythmia?

- A Electrical cardioversion plus prolonged anticoagulation
- B Electrical cardioversion without anticoagulation
- C Chemical cardioversion plus prolonged anticoagulation
- D Chemical cardioversion without anticoagulation
- E Continued rate control plus prolonged anticoagulation

REPEATED 2* · ALSO ASKED IN
Crash Course

- 8 During medical screening of new military recruits, an 18-year-old man reports several episodes of palpitations and syncope over the past few years. Examination is unremarkable. Excerpts from his ECG are shown. What is the most likely diagnosis?



- A Previous myocardial infarction secondary to coronary artery disease
- B Congenital long QT syndrome
- C Hypertrophic obstructive cardiomyopathy
- D Pre-excitation (Wolff-Parkinson-White) syndrome
- E Rheumatic mitral stenosis

Answers 1-8

PRETEST MCQs

- 1 **E Permanent pacemaker implantation** — Rapid SVT ending in a long pause with dizziness/syncope is tachy-brady sick sinus syndrome: permanent pacemaker, then rate-control drugs safely.
- 2 **E Long-term oral anticoagulation (a DOAC, or warfarin with a target INR of 2.0–3.0)** — UPDATED Ischaemic stroke with AF (CHA2DS2-VASc ≥ 6): lifelong anticoagulation, DOAC preferred (or warfarin INR 2–3), started within days depending on infarct size.
- 3 **A Administration of intravenous metoprolol** — Flutter waves ~250/min with 2:1 block in a stable patient: rate control first with IV metoprolol, then anticoagulation and elective cardioversion/ablation.
- 4 **E Observation** — Progressive PR lengthening until a dropped beat is Mobitz I (AV-nodal, benign): observe and review AV-nodal drugs if asymptomatic.
- 5 **B Pacing (temporary transvenous, then assessment for a permanent pacemaker)** — Complete heart block after anterior MI is infranodal with an unreliable escape rhythm: pace (temporary, then assess permanent) even if BP is stable.
- 6 **C Atrial flutter with 4:1 atrioventricular block** — UPDATED Regular saw-tooth waves ~300/min with a QRS after every fourth is atrial flutter with 4:1 block; decongestants and lung disease are triggers.
- 7 **E Continued rate control plus prolonged anticoagulation** — Older patient with dilated LA, now rate-controlled and asymptomatic: continue rate control plus long-term anticoagulation (CHA2DS2-VASc ≥ 3).
- 8 **D Pre-excitation (Wolff–Parkinson–White) syndrome** — Short PR, delta wave and wide QRS is pre-excitation (WPW); palpitations or syncope warrant an EP study and pathway ablation.

11 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 38-year-old man with no past medical history is referred after an abnormal preoperative ECG, and long QT syndrome is confirmed on repeat ECGs. His father had a heart condition diagnosed in his 20s, and he himself has had episodes of syncope after exertion. He is started on a beta-blocker and an implantable cardioverter-defibrillator is planned. Loss of function of which of the following is most commonly associated with his condition?
 - A AV node
 - B Calcium channels
 - C Potassium channels
 - D SA node
 - E Sodium channels
- 2 A 55-year-old man with ischaemic heart disease presents with palpitations for the past 10 days. His pulse is 130/min and irregularly irregular. He had one previous episode of atrial fibrillation 3 months ago, which was terminated by elective cardioversion after anticoagulation with warfarin. Which term best describes his arrhythmia?
 - A Paroxysmal atrial fibrillation
 - B Atrial flutter
 - C Permanent atrial fibrillation
 - D Persistent atrial fibrillation
 - E Secondary atrial fibrillation
- 3 A 72-year-old man with hypertension and ischaemic heart disease presents with light-headedness. He takes bisoprolol, amlodipine and aspirin. Heart rate is 38/min, BP 96/54 mmHg, respiratory rate 18/min, SpO₂ 97% on air and temperature 37.1 °C. The ECG shows regular P–P and R–R intervals with no relationship between the P waves and the QRS complexes. What would you expect to hear on auscultation of the heart?
 - A Loud S1
 - B Soft S1
 - C Variable intensity of S1
 - D Loud S2
 - E Soft S2
- 4 A 56-year-old man is brought in by paramedics after fainting this morning; he is now conscious but pale and light-headed, with no injuries. His heart rate is 37/min, respiratory rate 16/min and BP 105/70 mmHg. The ECG shows sinus bradycardia with a prolonged PR interval. What is the initial management?
 - A IV atropine
 - B IV adenosine
 - C External pacing
 - D IM adrenaline
 - E Oral atropine

- 5** A 75-year-old woman is admitted to the acute medical unit with pneumonia. Her only past history is a transient ischaemic attack 2 months ago. On admission the ECG shows atrial fibrillation with a ventricular rate of 103/min. She is treated with IV fluids and antibiotics and improves; two further ECGs overnight show sinus rhythm, and next day she is medically fit for discharge. What is the single most appropriate management of her atrial fibrillation?
- A No follow-up required
 - B Repeat ECG in two weeks
 - C Oral anticoagulation
 - D Aspirin
 - E 24-hour tape and consider oral anticoagulation
- 6** A 30-year-old woman is brought to the emergency department with sudden-onset palpitations. BP is 124/84 mmHg and pulse 150/min. The ECG shows a regular narrow-complex tachycardia. Vagal manoeuvres fail, IV access is obtained and adenosine 6 mg is given with no effect. What is the most appropriate next step?
- A Intravenous adenosine 12 mg
 - B Intravenous adenosine 6 mg
 - C Intravenous verapamil 2.5 mg
 - D Radio-frequency ablation
 - E Electrical cardioversion
- 7** Each of the following may cause left bundle branch block EXCEPT:
- A Cardiomyopathy
 - B Atrial septal defect (ostium secundum)
 - C Hypertension
 - D Idiopathic fibrosis
 - E Ischaemic heart disease
- 8** A 35-year-old man presents to the emergency department after an episode of syncope. The ECG shows sinus rhythm at 85/min, QRS duration 110 ms, PR interval 180 ms and corrected QT interval 500 ms. Which of the following is the most likely cause of the ECG abnormality?
- A Hyperkalaemia
 - B Hypercalcaemia
 - C Hypokalaemia
 - D Pyrexia
 - E Digoxin therapy
- 9** A 34-year-old woman is admitted after a collapse, and the ECG shows polymorphic ventricular tachycardia. Which one of the following is NOT associated with an increased risk of torsades de pointes?
- A Tricyclic antidepressants
 - B Subarachnoid haemorrhage
 - C Hypercalcaemia
 - D Romano-Ward syndrome
 - E Hypothermia
- 10** A 68-year-old man presents with a 4-day history of palpitations and increasing breathlessness on exertion. The ECG shows atrial fibrillation at 118/min. He has brittle asthma, hypertension and heart failure, and a recent echocardiogram showed a left ventricular ejection fraction of 32%. What is the most appropriate drug to control his heart rate?
- A Verapamil
 - B Sotalol
 - C Diltiazem
 - D Digoxin
 - E Amiodarone
- 11** Which one of the following is NOT an indication for insertion of a temporary pacemaker?
- A Complete heart block after an inferior MI, with normal blood pressure
 - B Complete heart block after an anterior MI, with normal blood pressure
 - C Bradycardia-dependent torsades de pointes recurring despite IV magnesium
 - D Mobitz type II heart block after an anterior MI, with normal blood pressure
 - E Symptomatic bradycardia not responding to drug treatment

Answers 1-11

- 1** **C Potassium channels** — Congenital long QT is most often loss of potassium-channel function (KCNQ1 LQT1, hERG LQT2); treat with nadolol/propranolol and avoid QT drugs.
- 2** **D Persistent atrial fibrillation** — AF lasting > 7 days (or needing cardioversion after 7 days) is persistent; paroxysmal ends within 7 days, permanent means rhythm control abandoned.
- 3** **C Variable intensity of S1** — In complete heart block the varying AV interval makes S1 intensity vary (with intermittent cannon a waves). Stop bisoprolol and pace.
- 4** **A IV atropine** — **UPDATED** Bradycardia with an adverse feature (syncope): IV atropine 500 µg, repeat to 3 mg; then transcutaneous pacing or isoprenaline/adrenaline infusion.
- 5** **C Oral anticoagulation** — CHA2DS2-VASc 5 (age 75 = 2, female, prior TIA = 2): anticoagulate (DOAC); AF during acute illness often recurs. Aspirin is not used.
- 6** **A Intravenous adenosine 12 mg** — Adenosine is given 6 mg, then 12 mg, then 18 mg by rapid bolus with flush; repeating 6 mg rarely works. Verapamil if adenosine fails.

- 7** **B Atrial septal defect (ostium secundum)** — LBBB occurs with IHD, hypertension/LVH, AS, cardiomyopathy and conduction fibrosis. Secundum ASD causes RBBB with right axis deviation.
- 8** **C Hypokalaemia** — **UPDATED** QTc 500 ms is prolonged (high torsades risk); hypokalaemia (flat T, U waves) is a common cause. Hypercalcaemia and digoxin shorten the QT.
- 9** **C Hypercalcaemia** — Torsades needs a long QT: TCAs, SAH, hypothermia, congenital LQTS, low K/Mg/Ca. Hypercalcaemia shortens the QT. Treat with IV magnesium 2 g.
- 10** **D Digoxin** — AF with HFrEF and brittle asthma: rate-limiting CCBs are negatively inotropic and beta-blockers are precluded, so digoxin is used.
- 11** **A Complete heart block after an inferior MI, with normal blood pressure** — **UPDATED** CHB after inferior MI is nodal (narrow escape, atropine-responsive, usually resolves), so pacing is not needed if stable; after anterior MI it is.

5 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

- 1** A 79-year-old woman is admitted to the coronary care unit with unstable angina and started on appropriate antiplatelet and anti-anginal therapy. She has hypertension; fasting glucose is normal and total cholesterol is 5.2 mmol/L. She is found to be in atrial fibrillation. What is the most appropriate treatment to reduce her risk of stroke?
- A** Aspirin and clopidogrel
B Digoxin
C Cardioversion
D Aspirin alone
E Oral anticoagulation (a DOAC or warfarin)
- 2** A 29-year-old woman sees her GP with fatigue, palpitations and weight loss without dieting. Her pulse is about 120/min and irregularly irregular, BP 142/89 mmHg and BMI 19 kg/m²; there are no added heart sounds. An ECG confirms atrial fibrillation. What is the most useful next investigation?
- A** Thyroid function tests (TSH, free T4)
B ECG
C Chest x-ray
D Full blood count
E Fasting blood sugar
- 3** A 62-year-old man with a previous myocardial infarction presents with palpitations that began 3 hours ago; this is his first known episode. The ECG shows atrial fibrillation at about 130/min without acute ischaemic changes. He has mild central chest discomfort, but BP is 128/80 mmHg and he is not distressed. A rhythm-control strategy is preferred. Which of the following is the most appropriate first-line treatment?
- A** Heparin and digoxin at the standard daily dose
B Heparin followed by synchronised DC cardioversion
C Bisoprolol plus verapamil, and warfarin
D IV flecainide
E Wait for spontaneous reversion to sinus rhythm
- 4** A 69-year-old woman with paroxysmal atrial fibrillation and asthma is evaluated. Episodes are infrequent, and echocardiography shows a structurally normal heart. Which statement regarding her management is correct?
- A** Digoxin effectively prevents recurrences of the arrhythmia
B Anticoagulation is not necessary
C Sotalol is a suitable choice to prevent recurrences
D Amiodarone is contraindicated
E Oral flecainide taken as needed may be an effective way to terminate an attack
- 5** A 67-year-old man presents with a 3-day history of shortness of breath. His pulse is irregular at 140/min, and the ECG shows atrial fibrillation. What would you expect to see in the jugular venous pulse?
- A** Raised JVP with normal waveform
B Large 'v waves'
C Cannon 'a waves'
D Absent 'a waves'
E Large 'a waves'

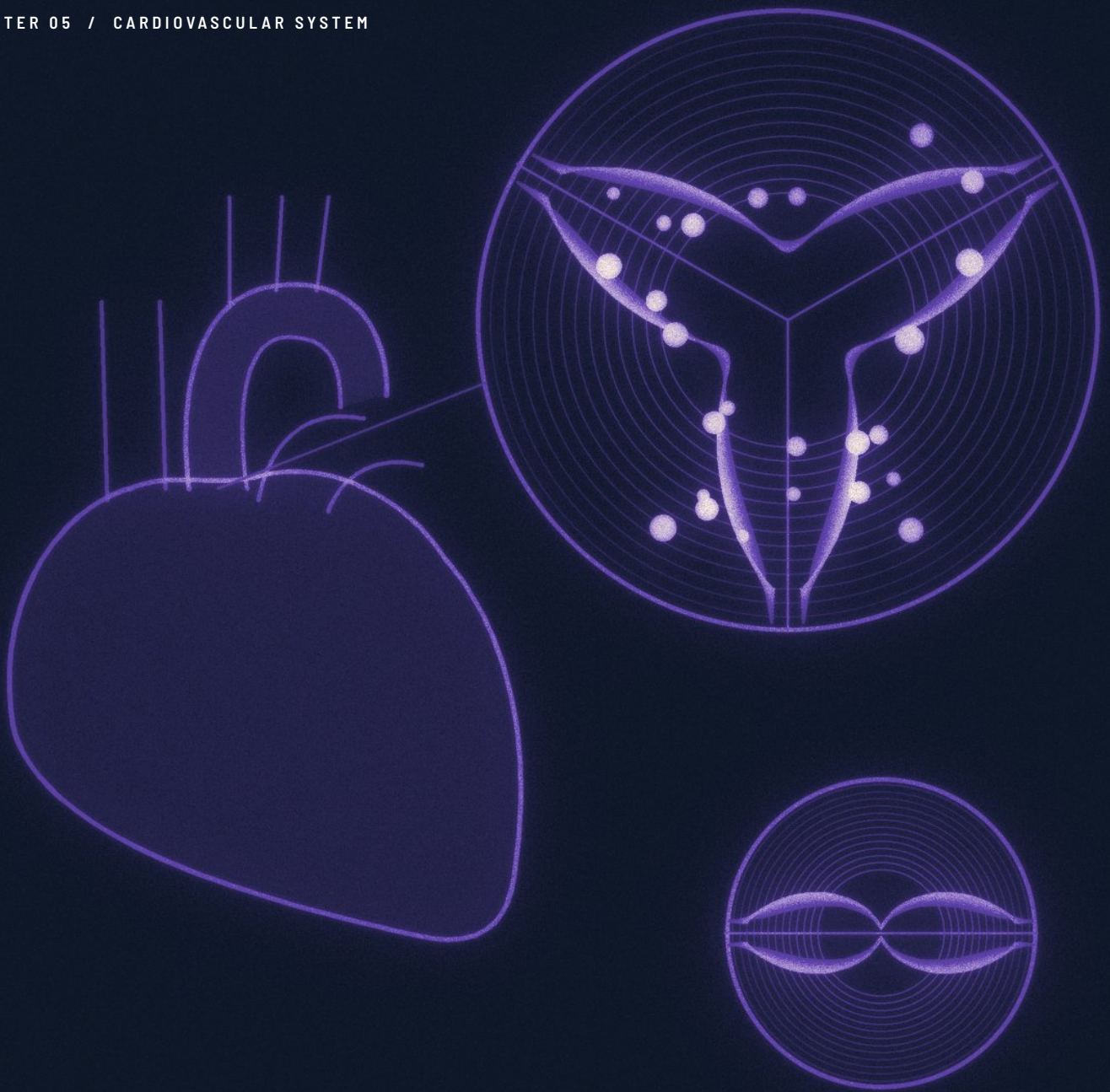
Answers 1-5

OTHER SOURCES MCQs

- 1** **E Oral anticoagulation (a DOAC or warfarin)** — **UPDATED** CHA₂DS₂-VASc 5: oral anticoagulation. After ACS with AF: brief

triple therapy (≤ 1 week), then DOAC + clopidogrel to 12 months, then DOAC alone.

- 2 **A** **Thyroid function tests (TSH, free T4)** — Weight loss, fatigue and new AF in a young woman suggest hyperthyroidism: check TSH and free T4 (recommended in all new AF).
- 3 **B** **Heparin followed by synchronised DC cardioversion** — **UPDATED** AF < 48 h can be cardioverted once heparin is started. Prior MI contraindicates flecainide, so use DC cardioversion (or amiodarone).
- 4 **E** **Oral flecainide taken as needed may be an effective way to terminate an attack** — **UPDATED** Infrequent paroxysmal AF in a normal heart: 'pill-in-the-pocket' flecainide with an AV-nodal blocker. Sotalol is contraindicated in asthma.
- 5 **D** **Absent 'a waves'** — The a wave is atrial contraction, so it is absent in AF. Cannon a waves = AV dissociation; large v waves = tricuspid regurgitation.



05

VALVULAR HEART DISEASE

Valvular Heart Disease

Aortic and mitral stenosis and regurgitation, right-sided valve disease and prosthetic valves: bedside signs, echo criteria and timing of intervention.

39

MCQs

0

EMQ SCENARIOS

7

SOURCES

7

REPEATED IDEAS

IN THIS CHAPTER

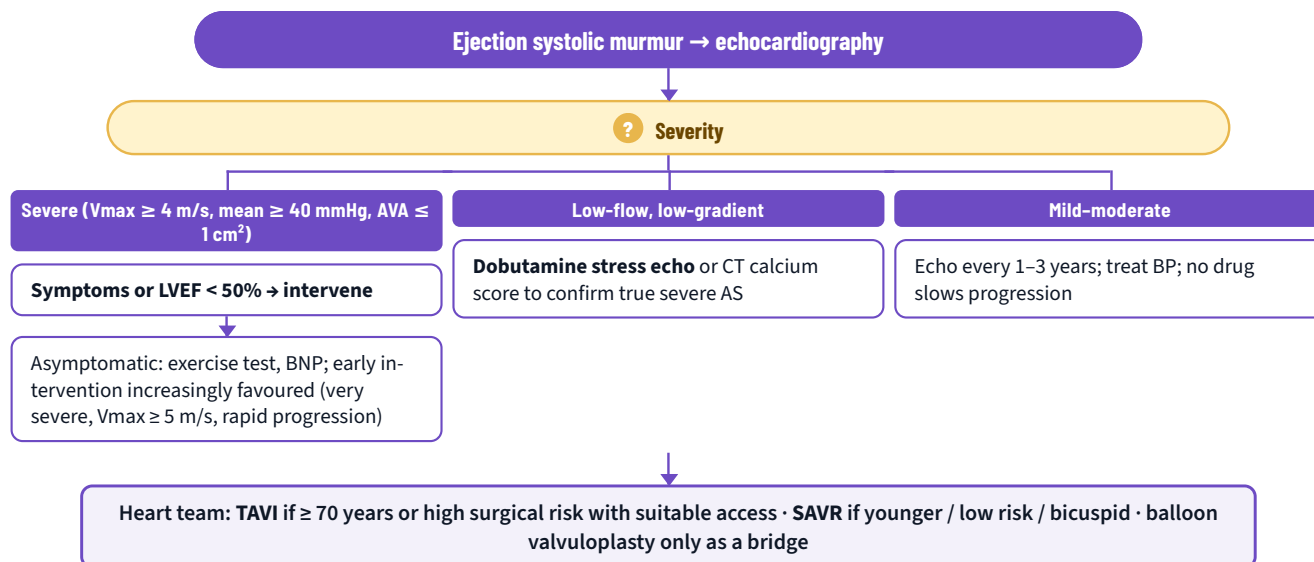
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02	Local Exams MCQs	13 Q	page 92
03	Davidson MCQs	4 Q	page 94
04	Harrison MCQs	1 Q	page 95
05	Crash Course MCQs	6 Q	page 95
06	PreTest MCQs	3 Q	page 96
07	Passmedicine MCQs	7 Q	page 97
08	Other Sources MCQs	5 Q	page 98

05

Step by step

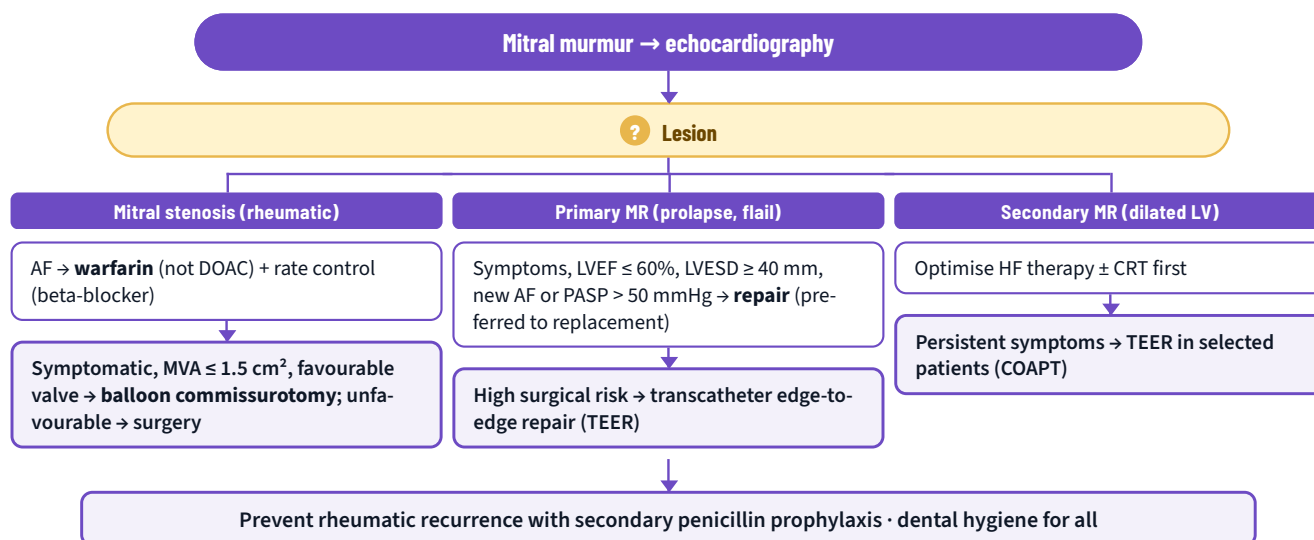
Aortic stenosis

ESC/EACTS 2025 VALVULAR HEART DISEASE



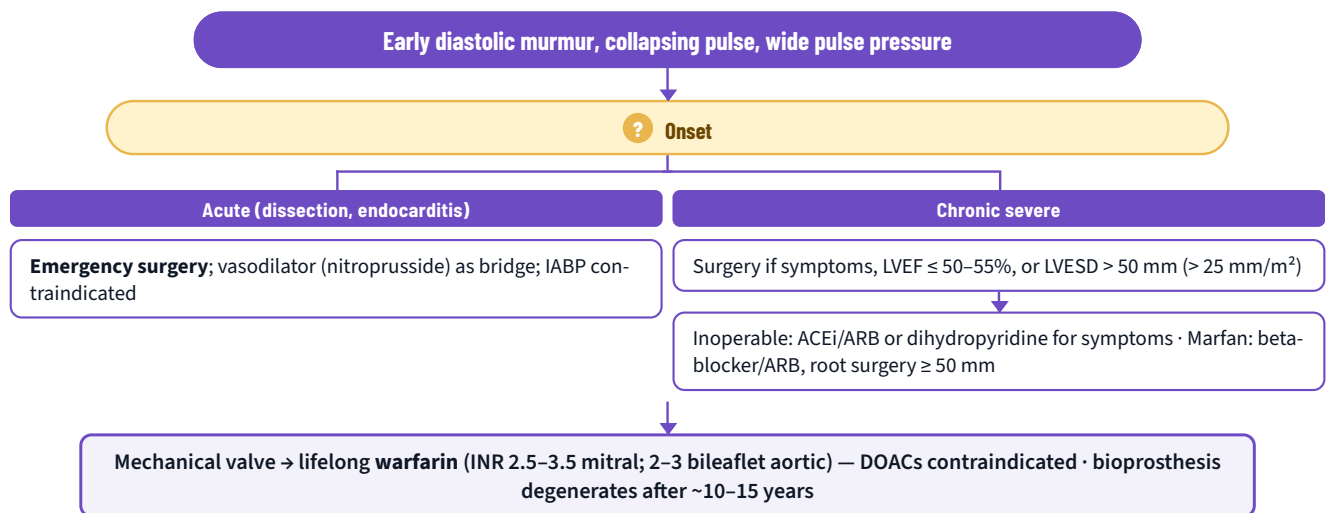
Mitral valve disease

ESC/EACTS 2025 VALVULAR HEART DISEASE



Aortic regurgitation and prosthetic valves

ESC/EACTS 2025 VALVULAR HEART DISEASE



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 74-year-old woman has progressive exertional breathlessness, angina and presyncope on climbing stairs. BP 138/76 mmHg, HR 78/min. There is a harsh late-peaking ejection systolic murmur at the right upper sternal edge, a soft S2 and a slow-rising (parvus et tardus) carotid pulse. Echocardiography: aortic valve area 0.8 cm², peak velocity 4.4 m/s, mean gradient 48 mmHg, LVEF 42% and moderate concentric LV hypertrophy. What is the best next step?

- A High-dose beta-blocker
- B Elective surgical or transcatheter aortic valve replacement
- C Stress echocardiography to confirm severity
- D Medical treatment and repeat echocardiography in 6–12 months
- E ACE inhibitor to lower afterload

- 2** A 36-year-old woman has exertional breathlessness and palpitations. There is a loud S1 and a low-pitched rumbling mid-diastolic murmur at the apex, best heard with the bell in the left lateral position and accentuated by brief exercise. Which additional finding most strongly supports mitral stenosis?

- A An opening snap after S2
- B Soft S1 with an apical pansystolic murmur
- C Wide fixed splitting of S2
- D Early diastolic decrescendo murmur at the left sternal edge
- E Continuous murmur that increases with inspiration

- 3** A young woman has breathlessness and palpitations, and an early diastolic murmur is heard at the left sternal edge. Aortic regurgitation is suspected. Which finding argues AGAINST this diagnosis?

- A Collapsing (water-hammer) pulse
- B Blood pressure 120/95 mmHg
- C To-and-fro femoral bruit on compression (Duroziez sign)
- D Visible nail-bed capillary pulsation (Quincke sign)
- E Soft mid-diastolic murmur at the apex (Austin Flint murmur)

REPEATED 3* • ALSO ASKED IN
39th batch end-block (cardiology) ×2

- 4** A 35-year-old man has exertional and paroxysmal nocturnal dyspnoea. Pulse 82/min regular, BP 135/85 mmHg; there are bibasal fine crackles and a pansystolic murmur at the apex radiating to the left axilla. What is the most likely diagnosis?

- A Aortic stenosis
- B Mitral regurgitation
- C Hypertrophic cardiomyopathy
- D Tricuspid regurgitation

REPEATED 3* • ALSO ASKED IN
Crash Course • Other Sources

- 5** An elderly man with hypertension treated with lisinopril has exertional breathlessness. BP is 165/55 mmHg. There is a grade 2/6 early diastolic murmur at the left sternal edge and a low-pitched mid-diastolic rumble at the apex; S1 is normal and there is no opening snap. What is the diagnosis?

- A Aortic regurgitation
- B Aortic stenosis
- C Mitral stenosis
- D Mitral regurgitation

REPEATED 4* • ALSO ASKED IN
Previous-years exam • Crash Course • Other Sources

- 6** A 32-year-old woman with rheumatic heart disease for 20 years has breathlessness and fatigue. There is a soft S1, a loud S3 and an apical pansystolic murmur radiating to the axilla. Which investigation will NOT help in the diagnosis of her valve lesion?

- A Cardiac enzymes
- B Echocardiography
- C Cardiac catheterisation
- D Left ventriculography

- 7** A patient has breathlessness, fatigue and leg oedema. There is a raised JVP with large v waves, a left parasternal (right ventricular) heave, a loud P2 and a pansystolic murmur at the lower left sternal edge. Echocardiography shows right ventricular hypertrophy with structurally normal tricuspid leaflets and functional tricuspid regurgitation. What is the most likely underlying diagnosis?

- A Primary tricuspid valve disease
- B Pulmonary hypertension
- C Pulmonary regurgitation
- D Dilated cardiomyopathy

FORMATIVE & PREVIOUS YEARS

- 8 A 71-year-old man with rheumatic heart disease for 20 years presents with breathlessness on exertion. BP is 128/78 mmHg. On examination the apex beat is displaced, the first heart sound is soft and the second heart sound is normal. Which valvular lesion is most likely?
- A Mitral stenosis
 - B Mitral regurgitation
 - C Aortic stenosis
 - D Aortic regurgitation
 - E Tricuspid regurgitation
- 9 A 42-year-old woman presents with breathlessness and chest discomfort. There is a loud first heart sound, an opening snap after S2 and a low-pitched mid-diastolic murmur at the apex. Which valve lesion is most likely?
- A Mitral stenosis
 - B Aortic stenosis
 - C Tricuspid stenosis
 - D Pulmonary regurgitation
 - E Ventricular septal defect

REPEATED 2* · ALSO ASKED IN
Crash Course

LOG BOOK 2025

- 10 A 62-year-old woman with known rheumatic mitral stenosis presents with progressive breathlessness, ankle swelling and abdominal distension. She is in sinus rhythm. Examination shows a raised JVP, a left parasternal heave, tender hepatomegaly and pitting oedema to the knees. What is the most likely cause of her new symptoms?
- A Acute myocardial infarction
 - B Progression to pulmonary hypertension with right heart failure
 - C Pericardial effusion
 - D Atrial fibrillation
 - E Mitral regurgitation
- 11 What is the typical character of the apex beat in a patient with mitral stenosis?
- A Displaced and forceful
 - B Hyperdynamic and bouncing
 - C Soft and not displaced
 - D Absent
 - E Tapping only
- 12 Which auscultatory finding most suggests pulmonary hypertension in a patient with mitral stenosis?
- A Diminished S2
 - B Loud P2 component of S2
 - C Split S1
 - D Ejection click
 - E Opening snap
- 13 In which patient with mitral stenosis is percutaneous balloon mitral commissurotomy indicated?
- A Asymptomatic with severe calcification
 - B Symptomatic patients with favorable valve morphology and minimal mitral regurgitation
 - C All patients with any degree of stenosis
 - D Only in patients with associated atrial fibrillation
 - E Only in elderly patients

Answers 1-13

- 1 **B Elective surgical or transcatheter aortic valve replacement** — **UPDATED** Symptomatic severe AS (area < 1 cm², mean gradient ≥ 40 mmHg) with EF < 50% needs AVR — surgical or TAVI by heart team. No drug improves outlook.
- 2 **A An opening snap after S2** — **UPDATED** An opening snap after S2, with a loud S1 and apical mid-diastolic rumble, is mitral stenosis; the shorter the S2–snap interval, the more severe the stenosis.
- 3 **B Blood pressure 120/95 mmHg** — **UPDATED** Significant chronic AR gives a wide pulse pressure (high systolic, low diastolic), so 120/95 mmHg argues against it. Acute AR may have a near-normal pulse pressure.
- 4 **B Mitral regurgitation** — Exertional dyspnoea and PND with an apical pansystolic murmur radiating to the axilla is mitral regurgitation (rheumatic or prolapse in the young); echo confirms.
- 5 **A Aortic regurgitation** — **UPDATED** Early diastolic murmur with wide pulse pressure is AR; the apical rumble is an Austin Flint murmur, told from MS by a normal S1 and no opening snap.
- 6 **A Cardiac enzymes** — Echo (TTE/TOE) and catheterisation define a chronic valve lesion; cardiac enzymes detect myocardial necrosis and say nothing about valve disease.
- 7 **B Pulmonary hypertension** — **UPDATED** Loud P2, RV heave and RVH with functional TR (normal leaflets) means pulmonary hypertension with right heart failure; then find its cause.
- 8 **B Mitral regurgitation** — **UPDATED** A displaced apex with a soft S1 in rheumatic disease is mitral regurgitation (LV volume overload). MS gives a loud S1 and tapping undisplaced apex.
- 9 **A Mitral stenosis** — **UPDATED** Loud S1, opening snap and an apical mid-diastolic rumble are mitral stenosis, usually rheum-

atic. Tricuspid stenosis is at the lower left sternal edge; aortic stenosis is systolic.

- 10 **B** **Progression to pulmonary hypertension with right heart failure** — **UPDATED** Long-standing MS causes pulmonary hypertension and RV failure (raised JVP, RV heave, hepatomegaly, oedema); this indicates valvotomy or surgery.
- 11 **E** **Tapping only** — In MS the apex is undisplaced and tapping (a palpable loud S1) because the LV is protected from overload.

- 12 **B** **Loud P2 component of S2** — Pulmonary hypertension causes a loud (even palpable) P2, often with an RV heave. The opening snap and loud S1 come from the mitral valve itself.
- 13 **B** **Symptomatic patients with favorable valve morphology and minimal mitral regurgitation** — Balloon mitral commissurotomy: symptomatic MS (area ≤ 1.5 cm²) with pliable non-calcified leaflets, $\leq$ mild MR and no LA thrombus on TOE.

4 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 A 34-year-old woman with a past history of rheumatic fever presents with breathlessness on exertion and a malar flush. She is in sinus rhythm, and echocardiography confirms mitral stenosis. Which physical sign is she likely to have?
- A Ejection systolic murmur
B Mid-systolic click
C Pre-systolic accentuation
D Quiet second heart sound
E Thrusting apex beat
- 2 An 80-year-old man is found to have an incidental ejection systolic murmur, and his family physician feels a systolic thrill at the base of the heart. What is the most likely cause of the thrill?
- A Aortic stenosis
B Large atrial septal defect
C Mitral stenosis
D Pulmonary hypertension
E Right ventricular hypertrophy
- 3 A 43-year-old man has a routine health check at work. He has a murmur, isolated systolic hypertension (180/60 mmHg) and left ventricular hypertrophy on his ECG. Echocardiography shows significant regurgitant flow across the aortic valve. Which clinical sign is he likely to have?
- A Opening snap
B Tapping, undisplaced apex beat
C Head nodding in time with the heartbeat (de Musset sign)
D Pulsus paradoxus
E Slow-rising pulse
- 4 A 58-year-old man with known mitral valve prolapse presents with 4 weeks of fever, malaise and weight loss after a dental extraction. Transthoracic echocardiography shows a vegetation on his native mitral valve with no abscess, and he has no embolic complications. Blood cultures grow viridans streptococci fully sensitive to penicillin, and his renal function is normal. What is the most appropriate antibiotic regimen to start?
- A Intravenous meropenem
B Intravenous benzylpenicillin and gentamicin
C Intravenous flucloxacillin
D Intravenous vancomycin and gentamicin
E Oral phenoxymethylpenicillin

Answers 1-4

DAVIDSON MCQs

- 1 **C** **Pre-systolic accentuation** — In MS with sinus rhythm, atrial contraction causes pre-systolic accentuation of the rumble; it disappears when AF develops.
- 2 **A** **Aortic stenosis** — **UPDATED** A basal systolic thrill with an ejection murmur in an elderly man is most likely calcific aortic stenosis; echo grades severity.
- 3 **C** **Head nodding in time with the heartbeat (de Musset sign)** — **UPDATED** Pulse pressure 120 mmHg with significant AR gives hyperdynamic signs: collapsing pulse, Corrigan, de Musset (head nodding), Quincke and Duroziez.
- 4 **B** **Intravenous benzylpenicillin and gentamicin** — **UPDATED** Penicillin-sensitive streptococcal native-valve IE: IV benzylpenicillin 4 weeks (or 2 weeks with gentamicin if uncomplicated). Vancomycin is for allergy.

1 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1** An 87-year-old woman with severe COPD (FEV₁ 0.76 L), poorly controlled diabetes and stage 3 chronic kidney disease presents with exertional chest pressure, leg swelling, breathlessness and rapidly worsening fatigue. There is a harsh late-peaking systolic murmur at the right upper sternal edge radiating to the carotids, an inaudible S2 and slow-rising, delayed carotid pulses. Echocardiography confirms aortic stenosis with a valve area of 0.59 cm². The cardiothoracic surgeon declines surgical aortic valve replacement because her predicted in-hospital mortality exceeds 15%. Which statement to her is reasonable?
- A** For this condition, medications have been shown to reduce the risk of death
 - B** It does not matter that you are not a surgical candidate: your condition carries a low 1- and 5-year mortality, and medication can control your symptoms
 - C** Percutaneous balloon aortic valvotomy is a good non-surgical option with excellent short- and long-term results
 - D** As you are not a surgical candidate, no other procedure can replace your valve, so we should consider hospice care
 - E** We should refer you for assessment for transcatheter aortic valve implantation (TAVI)

Answer 1

HARRISON MCQs

- 1** **E** We should refer you for assessment for transcatheter aortic valve implantation (TAVI) — Symptomatic severe AS has

~50% 2-year mortality without replacement; at prohibitive surgical risk, refer for TAVI (heart-team assessment).

6 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1** A 62-year-old man presents with angina and breathlessness. On examination there is an ejection systolic murmur at the upper right sternal edge radiating to the neck. What is the most likely diagnosis?
- A** Mitral stenosis
 - B** Mitral regurgitation
 - C** Aortic stenosis
 - D** Aortic regurgitation
 - E** Tricuspid stenosis
- 2** A 42-year-old man is reviewed in the cardiology clinic after an echocardiogram showing right ventricular dysfunction. He has a raised JVP and a pansystolic murmur at the lower left sternal edge. What is the most likely diagnosis?
- A** Mitral stenosis
 - B** Mitral regurgitation
 - C** Tricuspid regurgitation
 - D** Aortic stenosis
 - E** Aortic regurgitation

- 3** A 62-year-old man presents with angina and breathlessness. There is an ejection systolic murmur at the upper right sternal edge radiating to the neck. Which investigation is most likely to establish the diagnosis?

A Chest X-ray (CXR)
 B Echocardiogram
 C ECG
 D Coronary angiography
 E 24-hour ECG

REPEATED 2* · ALSO ASKED IN
 Other Sources

- 4** A 42-year-old man presents with shortness of breath and haemoptysis. There is a loud first heart sound and a rumbling mid-diastolic murmur at the apex. An echocardiogram and chest X-ray are requested. All of the following are chest X-ray features of this condition EXCEPT:

A Large right atrium
 B Large left atrium
 C Kerley B lines
 D Pulmonary venous hypertension
 E Narrowed carina

- 5** A 20-year-old student sees his GP with blurred vision; a year ago he had a dislocated lens. On examination he has pectus excavatum, a high-arched palate and joint hypermobility. Which cardiac disorder is often associated with his condition?

A Aortic valve stenosis
 B Coarctation of the aorta
 C Mitral valve prolapse
 D Ventricular septal defect
 E Ebstein anomaly

- 6** A 65-year-old man has 3 months of exertional central chest pain and breathlessness that settle with rest, and lately dizziness on exertion. There is an ejection systolic murmur radiating to the carotids. ECG shows tall R waves in I, aVL and V4–V6 with deep S waves in V1–V3. What is the most likely diagnosis?

A Myocardial infarction
 B Unstable angina
 C Stable angina
 D Aortic stenosis
 E Hypertrophic cardiomyopathy

REPEATED 5* · ALSO ASKED IN
 PreTest · Other Sources ×3

Answers 1–6

CRASH COURSE MCQs

- 1** **C Aortic stenosis** — Ejection systolic murmur at the upper right sternal edge radiating to the carotids is aortic stenosis; angina, dyspnoea and syncope signal severe disease.
- 2** **C Tricuspid regurgitation** — Pansystolic murmur at the lower left sternal edge louder on inspiration, with a raised JVP and large v waves, is tricuspid regurgitation.
- 3** **B Echocardiogram** — Transthoracic echo with Doppler confirms AS and grades it (velocity, gradient, area) with LV function; angiography is done before surgery.

- 4** **E Narrowed carina** — In MS the enlarged LA straightens the left heart border, gives a double right border and splays (widens) the carina; upper-lobe diversion and Kerley B lines.
- 5** **C Mitral valve prolapse** — Ectopia lentis, pectus, high palate and hypermobility suggest Marfan syndrome, commonly with MVP; aortic root dilatation is the dangerous lesion.
- 6** **D Aortic stenosis** — **UPDATED** Exertional angina, dyspnoea and dizziness/syncope with an ejection murmur and LVH on ECG is severe aortic stenosis. HCM is younger, and its murmur does not radiate to the carotids.

3 QUESTIONS · AFTER PRETEST MEDICINE

PreTest MCQs

- 1** A 72-year-old man has intermittent breathlessness on exertion, palpitations and a cough occasionally productive of blood. On auscultation, a low-pitched diastolic rumbling murmur is faintly heard at the apex. What is the most likely cause of the murmur?

A Rheumatic fever as a youth
 B Long-standing hypertension
 C A silent MI within the past year
 D A congenital anomaly
 E Anaemia from chronic blood loss

- 2** A 68-year-old man is intubated in the emergency department because of pulmonary oedema. His BP is 150/90 mmHg. Urgent echocardiography shows an ejection fraction of 45% and severe mitral regurgitation. Despite aggressive diuresis with furosemide, he still requires mechanical ventilation for pulmonary oedema. What is the best next step?

A Arrange mitral valve replacement surgery
 B Insert an intra-aortic balloon pump
 C Start metoprolol
 D Add a second loop diuretic
 E Start IV afterload reduction (e.g. nitroprusside or IV enalaprilat)

- 3** A nurse at a community clinic has listed patients scheduled for a dental extraction who may need antibiotic prophylaxis. Which of these patients is most likely to benefit from antibiotic prophylaxis to prevent infective endocarditis?
- A** 17-year-old boy with coarctation of the aorta
 - B** 26-year-old woman with a ventricular septal defect repaired in childhood
 - C** 42-year-old woman with mitral valve prolapse
 - D** 65-year-old man with a prosthetic aortic valve
 - E** 72-year-old woman with aortic stenosis

REPEATED 2* • ALSO ASKED IN
Other Sources

Answers 1-3

PRETEST MCQS

- 1** **A Rheumatic fever as a youth** — Dyspnoea, AF and haemoptysis with an apical diastolic rumble is mitral stenosis, usually a late consequence of childhood rheumatic fever.
- 2** **E Start IV afterload reduction (e.g. nitroprusside or IV enalaprilat)** — **UPDATED** Severe MR with pulmonary oedema and adequate BP: IV afterload reduction (nitroprusside/nitrates) increases forward flow; IABP if hypotensive; then surgery.
- 3** **D 65-year-old man with a prosthetic aortic valve** — Dental prophylaxis (AHA/ESC) only for highest risk: prosthetic valves, previous IE, unrepaired cyanotic CHD. NICE does not recommend it routinely.

7 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1** A 62-year-old man who had a mechanical mitral valve replacement 4 years ago is reviewed. Which long-term antithrombotic therapy is he likely to be taking?
- A** Nothing
 - B** Direct acting oral anticoagulants
 - C** Aspirin
 - D** Aspirin + clopidogrel for the first 12 months
 - E** Warfarin
- 2** A 34-year-old man presents to his GP with fatigue and exertional breathlessness. He has no significant past history, takes no medication, and does not smoke or drink alcohol. He is very tall (2.08 m) and thin, with a high-arched palate. Auscultation reveals a pansystolic murmur that is loudest in inspiration. What is the most likely explanation for his symptoms?
- A** Mitral regurgitation
 - B** Mitral stenosis
 - C** Pulmonary regurgitation
 - D** Tricuspid regurgitation
 - E** Ventricular septal defect
- 3** A 45-year-old woman who injects drugs presents with breathlessness and bilateral leg swelling. Her pulse is regular, there is a pansystolic murmur heard loudest at the lower left sternal edge, and there is shifting dullness in the abdomen. What is the most likely finding in her jugular venous pulse?
- A** Absent a waves
 - B** Cannon a waves
 - C** Prominent a waves
 - D** Prominent v waves
 - E** Prominent x descent
- 4** Which one of the following conditions is most associated with a bisferiens pulse?
- A** Cardiac tamponade
 - B** Severe left ventricular failure
 - C** Aortic stenosis
 - D** Patent ductus arteriosus
 - E** Mixed aortic valve disease
- 5** Which one of the following is a cause of a soft second heart sound?
- A** Aortic stenosis
 - B** Atrial septal defect
 - C** Mitral stenosis
 - D** Mitral regurgitation
 - E** Pulmonary hypertension

- 6 Which of the following conditions is NOT associated with the development of aortic regurgitation?
- A Rheumatic fever
 - B Ankylosing spondylitis
 - C Marfan's syndrome
 - D Syphilis
 - E Dilated cardiomyopathy

- 7 A 2-year-old boy is seen by his GP because his parents are concerned that he is struggling to gain weight and is excessively breathless on exertion. He has known congenital pulmonary stenosis, managed conservatively so far, and the parents ask whether intervention may be needed. What murmur is likely to be heard on examination?
- A Ejection systolic, louder on expiration
 - B Ejection systolic, louder on inspiration
 - C Holo-systolic, louder on expiration
 - D Holo-systolic, louder on inspiration
 - E Late systolic

Answers 1-7

PASSMEDICINE MCQS

- 1 **E Warfarin** — Mechanical valves need lifelong warfarin (INR ~2.5–3.5 mitral, 2.0–3.0 bileaflet aortic). DOACs are contraindicated (RE-ALIGN).
- 2 **D Tricuspid regurgitation** — A pansystolic murmur louder on inspiration (Carvallo) is tricuspid regurgitation; in Marfan it may follow tricuspid prolapse.
- 3 **D Prominent v waves** — **UPDATED** TR (e.g. right-sided IE in drug users) gives giant v (cv) waves in the JVP and a pulsatile liver. Cannon a waves mean AV dissociation.
- 4 **E Mixed aortic valve disease** — A bisferiens (double-peaked) pulse occurs in mixed aortic valve disease with predominant AR, severe AR and HOCM. Pure AS gives a slow-rising pulse.
- 5 **A Aortic stenosis** — **UPDATED** Severe calcific AS gives a soft or absent A2 (and reversed splitting). Pulmonary hypertension gives a loud P2; ASD gives fixed splitting.
- 6 **E Dilated cardiomyopathy** — AR arises from leaflet disease (rheumatic, IE, bicuspid) or root dilatation (Marfan, ankylosing spondylitis, syphilis). DCM causes functional MR/TR.
- 7 **B Ejection systolic, louder on inspiration** — Pulmonary stenosis: ejection systolic murmur at the upper left sternal edge, louder on inspiration, with click and soft P2; balloon valvuloplasty if severe.

5 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

- 1 A patient admitted with pneumonia is found to have a heart murmur. Which finding points to mitral regurgitation?
- A Murmur louder on inspiration
 - B Murmur louder with patient in left lateral position
 - C Murmur louder over the right 2nd intercostal space midclavicular line
 - D Corrigan's sign
 - E Narrow pulse pressure
- 2 A 59-year-old man attends for a routine check-up. He is asymptomatic, and examination reveals a mid-systolic click followed by a late systolic murmur at the apex. What is the most likely diagnosis?
- A Barlow syndrome (mitral valve prolapse)
 - B Austin Flint murmur
 - C Patent ductus arteriosus
 - D Graham Steell murmur
 - E Carey Coombs murmur
- 3 A 57-year-old woman with a history of rheumatic fever and essential hypertension presents with worsening breathlessness and reduced exercise tolerance. Which of the following is NOT a clinical sign of mitral stenosis?
- A Malar flush
 - B Atrial fibrillation
 - C Pan-systolic murmur which radiates to axilla
 - D Tapping, undisplaced apex beat
 - E Right ventricular heave
- 4 An asymptomatic 31-year-old woman is told she has mitral valve prolapse after an echocardiogram. Which statement is correct?
- A Beta-blocker therapy is indicated
 - B ACE inhibitor therapy is indicated
 - C One or both leaflets of the mitral valve are pushed back into the left atrium during systole
 - D Significant mitral regurgitation will eventually develop
 - E Exercise should be restricted

- 5 A 78-year-old man presents with a 3-week history of productive cough, leg swelling and fatigue; he has been afebrile throughout. He had a porcine (bioprosthetic) mitral valve replacement for severe mitral regurgitation 12 years ago. Examination shows bibasal crackles, a laterally displaced apex beat and a grade 3/6 low-pitched diastolic murmur at the apex. What is the most likely cause of his symptoms?
- A Infective endocarditis
 - B Chronic obstructive pulmonary disease
 - C Structural degeneration of the bioprosthetic valve
 - D Pneumonia
 - E Pulmonary embolism

Answers 1-5

OTHER SOURCES MCQS

- 1 **B Murmur louder with patient in left lateral position** — Mitral murmurs are loudest at the apex with the patient in the left lateral position; MR is pansystolic, radiates to the axilla, louder in expiration.
- 2 **A Barlow syndrome (mitral valve prolapse)** — Mid-systolic click then late systolic murmur is mitral valve prolapse (Barlow); standing/Valsalva bring the click earlier, squatting delays it.
- 3 **C Pan-systolic murmur which radiates to axilla** — MS signs: malar flush, AF, tapping apex, loud S1, opening snap, RV heave. A pansystolic murmur to the axilla is mitral regurgitation.
- 4 **C One or both leaflets of the mitral valve are pushed back into the left atrium during systole** — **UPDATED** MVP is systolic displacement of a leaflet > 2 mm into the LA; most are asymptomatic and need no drugs, but echo follow-up is advised.
- 5 **C Structural degeneration of the bioprosthetic valve** — **UPDATED** Bioprosthetic valves degenerate after ~10 years (earlier in mitral position); a new apical diastolic murmur means prosthetic MS. Exclude endocarditis; redo or valve-in-valve.



06

PERICARDIAL & INFLAMMATORY HEART DISEASE

Pericardial & Inflammatory Heart Disease

Pericarditis, tamponade and constriction; infective endocarditis; myocarditis; and rheumatic fever — the inflammatory diseases of the heart and their look-alikes.

66

MCQs

0

EMQ SCENARIOS

7

SOURCES

15

REPEATED IDEAS

IN THIS CHAPTER

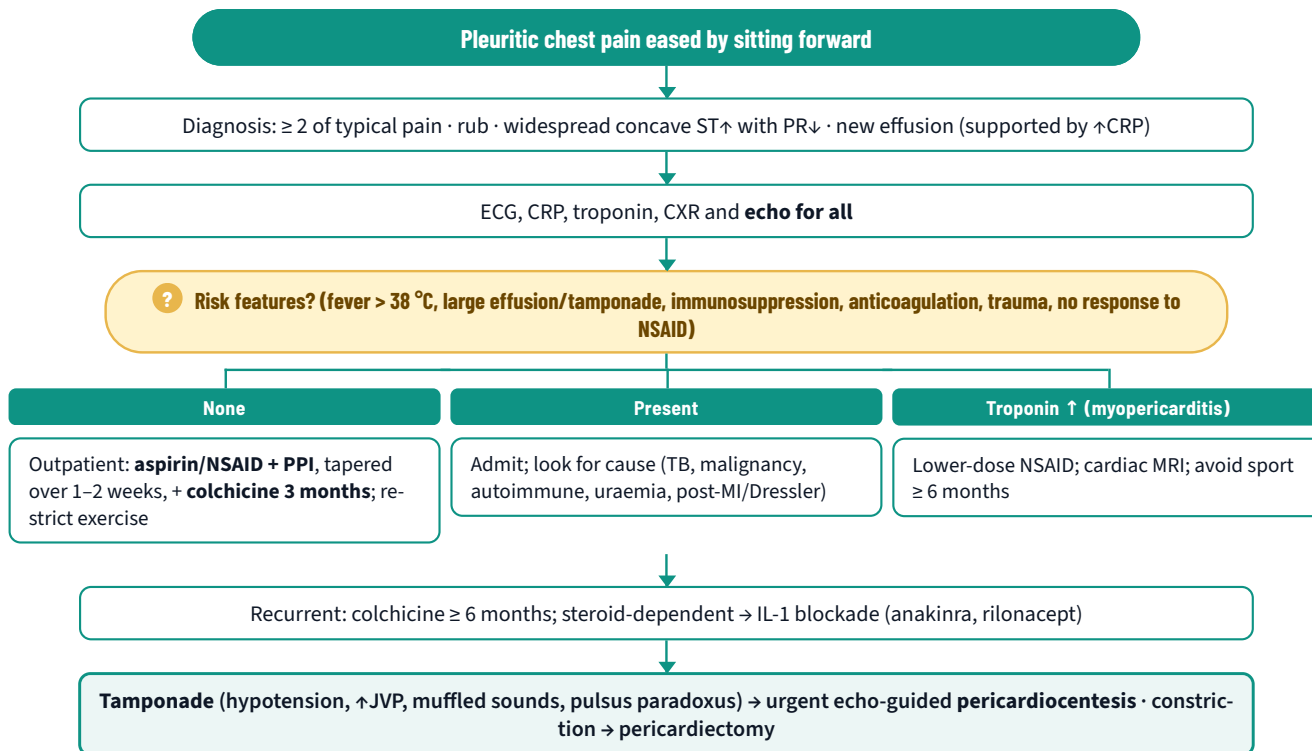
01	Management flowcharts	3 charts	page 102
02	Local Exams MCQs	30 Q	page 104
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06

Step by step

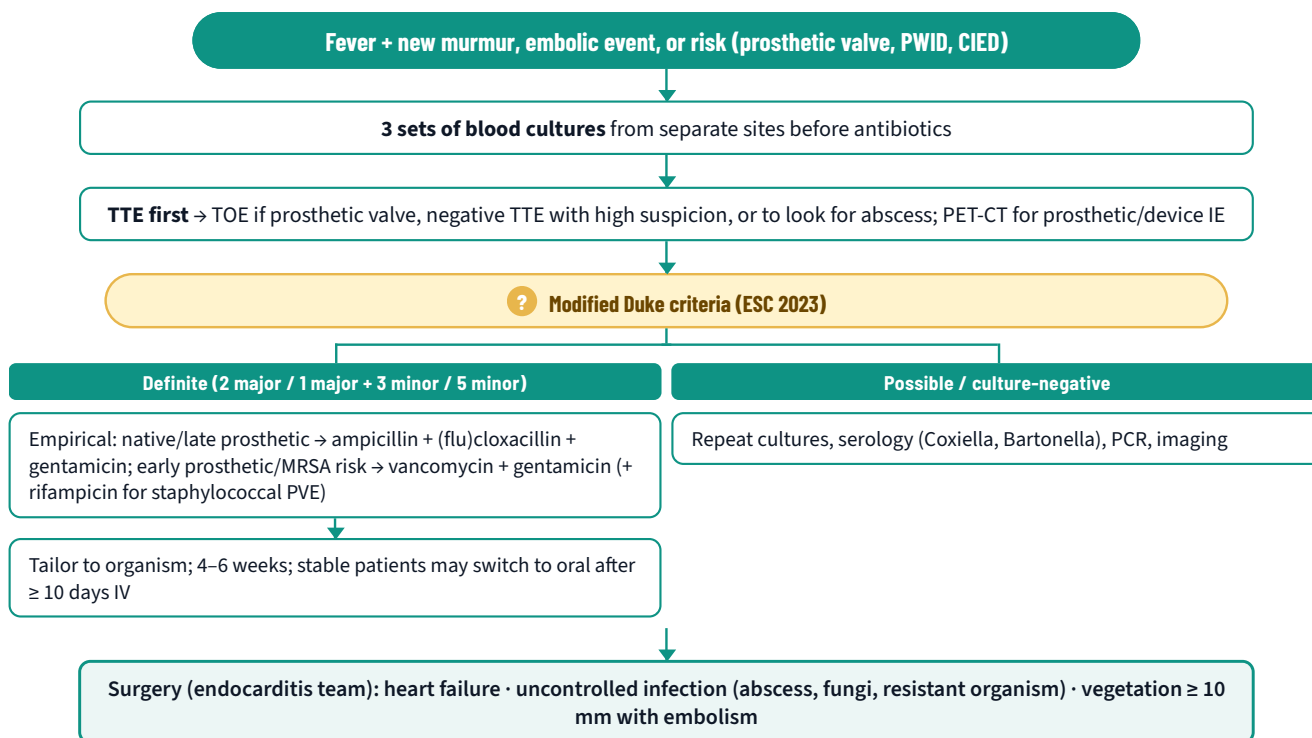
Acute pericarditis and tamponade

ESC 2025 MYOCARDITIS & PERICARDITIS



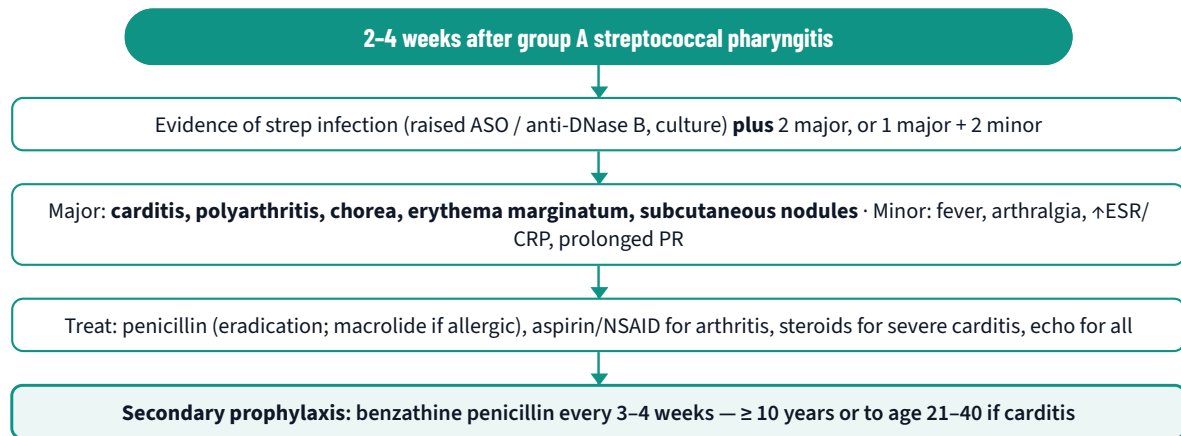
Infective endocarditis

ESC 2023 ENDOCARDITIS



Acute rheumatic fever

AHA REVISED JONES 2015 · WHO



Local Exams MCQs

SET 1 OF 2

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 48-year-old woman with breast cancer on recent chemotherapy has 4 days of progressive breathlessness, chest discomfort and fatigue. BP 92/64 mmHg, HR 118, RR 24. JVP is raised, heart sounds are muffled and pulsus paradoxus is 14 mmHg. ECG shows low-voltage QRS complexes; echo shows a large circumferential pericardial effusion with right atrial systolic collapse. What is the most appropriate next step?

A IV furosemide
 B Urgent pericardiocentesis
 C High-dose NSAID and colchicine
 D CT chest
 E IV fluids and observe for 24 hours

- 2** A 27-year-old man has had sharp, stabbing chest pain radiating to the shoulder and left jaw for 3 days with a low-grade fever. A to-and-fro scratching sound is heard over the precordium and the ECG shows ST elevation in I, II, aVF and V2–V5. What is the most likely diagnosis?

A Inferior MI
 B Lateral MI
 C Acute myocarditis
 D Unstable angina
 E Acute pericarditis

REPEATED 2* • ALSO ASKED IN
 Crash Course

- 3** A 27-year-old has had sharp, stabbing chest pain radiating to the shoulder and left jaw for 3 days with a low-grade fever. A to-and-fro scratching sound is heard over the precordium and the ECG shows ST elevation in I, II, aVF and V2–V5. Four days later he develops retrosternal oppression; the JVP is raised, pulsus paradoxus is present and BP is 80/45 mmHg. Which complication has developed?

A Heart failure
 B Cardiac tamponade
 C Dilated cardiomyopathy
 D Restrictive cardiomyopathy
 E Obstructive cardiomyopathy

REPEATED 2* • ALSO ASKED IN
 Harrison

- 4** A patient with acute pericarditis develops hypotension, a raised JVP and pulsus paradoxus. Which investigation best confirms this complication?

A ECG
 B Echocardiography
 C Cardiac enzymes
 D Chest X-ray
 E Treadmill test

- 5** A patient with acute pericarditis develops cardiac tamponade (BP 80/45 mmHg, raised JVP, pulsus paradoxus). What is the treatment of choice now?

A Digoxin
 B Diuretics
 C Pericardiocentesis
 D Nitrates
 E ACE inhibitor

REPEATED 2* • ALSO ASKED IN
 Log book 2025

- 6** Which of the following is a bad prognostic sign in acute rheumatic fever?

A Young age at first attack
 B Older age at first attack
 C No valvular lesion
 D Normal inflammatory markers

REPEATED 3* • ALSO ASKED IN
 End-block exam 2025 • Previous-years exam

- 7** Which of the following is a major criterion for the diagnosis of infective endocarditis?

A Predisposing valve lesion
 B Three or more positive blood cultures taken over more than one hour
 C IV drug use
 D Embolic phenomena

- 8** Which of the following is NOT a feature of cardiac tamponade?

A Hypotension
 B Collapsed JVP
 C Pulsus paradoxus
 D Muffled heart sounds
 E Low-voltage ECG

REPEATED 2* • ALSO ASKED IN
 Formative exam

- 9 A patient who had a mitral valve replacement 3 weeks ago develops infective endocarditis. Which organism is most likely responsible?

A Streptococcus sanguinis
 B Enterococcus faecium
 C Streptococcus bovis
 D Streptococcus milleri
 E Coagulase-negative staphylococcus

REPEATED 2* · ALSO ASKED IN
 Formative exam

- 10 A middle-aged woman has fatigue, abdominal distension and leg swelling. Her pulse is 112/min, regular and of small volume, there is an early diastolic sound (pericardial knock), and the chest X-ray shows calcification around the heart shadow. Chronic constrictive pericarditis is suspected. Which finding is AGAINST this diagnosis?

A Bilateral pitting leg oedema
 B Raised JVP
 C Prominent breathlessness
 D Massive ascites

- 11 All of the following are features of myocarditis EXCEPT:

A Acute heart failure
 B Shock
 C Pericardial rub
 D Chest pain

- 12 The age group most commonly affected by acute rheumatic fever is:

A Older adults
 B Neonates
 C Childhood (5–15 years)
 D Young adults

- 13 A 16-year-old girl has central chest pain and fever, and a pericardial rub is heard. Which ECG abnormality is NOT typical of acute pericarditis?

A PR depression
 B Diffuse concave ST elevation
 C T-wave inversion after the ST segments normalise
 D Hyperacute T waves

CASE · QUESTIONS 14–16

A 22-year-old man has fever, night sweats and weight loss. There are linear (splinter) haemorrhages under the nails and a new pansystolic murmur radiating to the left axilla.

- 14 What is the most likely underlying diagnosis?

A Mitral regurgitation
 B Ventricular septal defect
 C Infective endocarditis
 D Rheumatic heart disease
 E Tricuspid regurgitation

REPEATED 2* · ALSO ASKED IN
 Log book 2025

- 15 Which investigation should be performed first, before antibiotics are started?

A Blood cultures
 B Full blood count
 C General urine examination
 D Echocardiography
 E Chest X-ray

REPEATED 4* · ALSO ASKED IN
 Log book 2025 · Crash Course · Other Sources

- 16 Three months later he develops right-sided weakness. What is the most likely cause?

A Embolic stroke
 B Extradural haematoma
 C Subarachnoid haemorrhage
 D Ruptured cerebral aneurysm

- 17 In infective endocarditis, which of the following is NOT an indication for surgery?

A Persistent fever for more than 7 days despite antibiotics
 B Perivalvular abscess on echo
 C Young age
 D Development of heart failure

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

CASE · QUESTIONS 18–21

A 51-year-old man with mitral valve prolapse and type 2 diabetes has had intermittent fever, night sweats and fatigue for 10 days, with mild exertional breathlessness and occasional palpitations. He recently had a dental extraction. T 38.9 °C, pulse 122/min, BP 105/65 mmHg, RR 22/min. There is a pansystolic murmur at the apex radiating to the axilla and tender erythematous nodules on his fingertips.

- 18 What is the most likely causative organism?

A Staphylococcus aureus
 B Viridans streptococci
 C Enterococcus faecalis
 D Staphylococcus epidermidis
 E Streptococcus pyogenes

REPEATED 3* · ALSO ASKED IN
 Crash Course ×2

- 19 Which of the following findings is most specific to this disease?

A Fever and fatigue
 B Shortness of breath
 C Splinter haemorrhages
 D Splenomegaly
 E Tachycardia

- 20 The initial step of management should include:

A Antipyretics
 B Empirical intravenous antibiotics
 C Immediate surgical valve replacement
 D Anticoagulation
 E Antiarrhythmic drugs

- 21** Which of the following is the most serious potential complication of this disease?
- A Acute heart failure
 - B Pulmonary embolism
 - C Viral myocarditis
 - D Chronic liver disease
 - E Hypertension

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 22** A middle-aged woman presents with fatigue, abdominal distension and light-headedness. Constrictive pericarditis is suspected. Which of the following findings would be LEAST typical of this diagnosis?
- A Elevated jugular venous pressure
 - B Kussmaul's sign
 - C Pericardial knock
 - D Pulsus paradoxus
 - E Ascites

- 23** A 55-year-old man has progressive shortness of breath and ankle oedema. He is originally from Africa and probably had tuberculosis many years ago. His BP is 105/55 mmHg, the JVP is raised 5 cm with a rapid y descent, and there is marked ankle oedema. What is the most likely diagnosis?

- A Cardiac tamponade
- B Aortic regurgitation
- C Pericardial constriction
- D Restrictive cardiomyopathy

- 24** Dressler syndrome results from which of the following?

- A Autoimmune response
- B Viral infection
- C Protozoal infection
- D Bacterial infection

- 25** Which manifestation of acute rheumatic fever is responsible for its long-term organ damage?

- A Sydenham chorea
- B Arthritis
- C Erythema marginatum
- D Carditis

Answers 1-25

LOCAL EXAMS MCQs

- 1** **B Urgent pericardiocentesis** — Hypotension, raised JVP, muffled sounds and pulsus paradoxus with RA collapse is tamponade (likely malignant): urgent echo-guided pericardiocentesis. Diuretics are harmful.
- 2** **E Acute pericarditis** — Positional pain, friction rub and concave ST elevation across several territories is acute pericarditis; treat with aspirin/NSAID plus colchicine.
- 3** **B Cardiac tamponade** — A new raised JVP, pulsus paradoxus and hypotension days after pericarditis means tamponade; confirm with echo and drain urgently.
- 4** **B Echocardiography** — Echo shows the effusion, RA/RV diastolic collapse and a plethoric IVC, and guides pericardiocentesis. ECG and CXR are insensitive.
- 5** **C Pericardiocentesis** — Tamponade with hypotension needs urgent pericardiocentesis; diuretics, nitrates and ACEi worsen filling. Fluids only as a bridge.
- 6** **A Young age at first attack** — **UPDATED** In rheumatic fever, carditis and recurrence are commonest when the first attack is in young children, so young age is a bad sign; penicillin prophylaxis prevents recurrence.
- 7** **B Three or more positive blood cultures taken over more than one hour** — Duke major criteria: persistently positive blood cultures (e.g. 3 sets ≥ 1 h apart) or imaging evidence of endocardial involvement. IVDU and emboli are minor.
- 8** **B Collapsed JVP** — Tamponade raises all diastolic pressures, so the JVP is high (blunted y descent); a collapsed JVP means hypovolaemia. Beck's triad: hypotension, raised JVP, quiet heart sounds.
- 9** **E Coagulase-negative staphylococcus** — Early prosthetic valve endocarditis (within weeks of surgery) is mostly coagu-

lase-negative staphylococci or *S. aureus* acquired peri-operatively.

- 10** **C Prominent breathlessness** — Constriction causes systemic congestion (raised JVP, Kussmaul sign, early ascites, hepatomegaly) with clear lungs, so prominent breathlessness argues against it.
- 11** **C Pericardial rub** — Myocarditis gives chest pain, heart failure, arrhythmia and raised troponin; a rub means associated pericarditis. Cardiac MRI is the key test.
- 12** **C Childhood (5–15 years)** — Acute rheumatic fever follows group A streptococcal pharyngitis and peaks at 5–15 years; it is rare under 3 and uncommon in adults.
- 13** **D Hyperacute T waves** — Pericarditis: diffuse concave ST elevation with PR depression, then T inversion. Hyperacute T waves are the earliest sign of STEMI.
- 14** **C Infective endocarditis** — Fever, weight loss, a new regurgitant murmur and splinter haemorrhages is infective endocarditis: 3 sets of blood cultures before antibiotics, then echo.
- 15** **A Blood cultures** — **UPDATED** Take ≥ 3 sets of blood cultures from separate sites before antibiotics: they identify the organism and are a major Duke criterion; echo follows promptly.
- 16** **A Embolic stroke** — Embolisation of vegetations, most often to the brain, makes new focal weakness an embolic stroke; image urgently and consider early surgery.
- 17** **C Young age** — Surgery in IE: heart failure from valve dysfunction, uncontrolled infection (abscess, persistent bacteraemia, fungi) or large vegetations with emboli. Age is not an indication.

- 18 **B Viridans streptococci** — Subacute endocarditis on a prolapsing mitral valve after dental extraction is caused by oral viridans streptococci. *S. aureus* is acute and fulminant; *S. epidermidis* infects prosthetic valves.
- 19 **C Splinter haemorrhages** — Only the vascular signs (splinter haemorrhages, like his Osler nodes) point specifically to endocarditis; fever, dyspnoea, tachycardia and splenomegaly are non-specific.
- 20 **B Empirical intravenous antibiotics** — Take three sets of blood cultures, then start empirical IV antibiotics and tailor to the organism. Surgery is for heart failure, uncontrolled infection or large emboli.
- 21 **A Acute heart failure** — Leaflet or chordal destruction causes acute regurgitation and heart failure, the commonest cause of death in IE and the main reason for urgent surgery. Pulmonary emboli come from tricuspid IE.
- 22 **D Pulsus paradoxus** — **UPDATED** Constriction gives a raised JVP with rapid y descent, Kussmaul sign, pericardial knock and ascites. Pulsus paradoxus is the hallmark of tamponade.
- 23 **C Pericardial constriction** — Right-sided congestion with a steep y descent after previous TB is constrictive pericarditis; CT/MRI shows thickening or calcification.
- 24 **A Autoimmune response** — Dressler syndrome is an autoimmune pericarditis weeks after MI (antimyocardial antibodies); treat with aspirin/NSAID plus colchicine.
- 25 **D Carditis** — **UPDATED** Carditis is the only rheumatic fever manifestation causing permanent damage (chronic rheumatic valve disease); arthritis, rash and chorea resolve.

SET 2 OF 2

- 26 Which of the following conditions carries the LOWEST risk of infective endocarditis?
- A Rheumatic heart disease with mitral regurgitation
B Mitral valve prolapse without regurgitation
C Previous infective endocarditis
D Prosthetic heart valve
- 27 A 62-year-old man is admitted with pyrexia and found to have infective endocarditis. Which one of the following is most associated with a good prognosis?
- A *Staphylococcus aureus* infection
B Prosthetic valve endocarditis
C Culture negative endocarditis
D *Streptococcus viridans* infection
- 29 Which of the following clinical signs is characteristically associated with infective endocarditis?
- A Hepatomegaly
B Janeway lesions
C Xanthelasma
D Koilonychia
E Periorbital oedema
- 30 A 32-year-old woman has fever, chills, night sweats and a new apical pansystolic murmur radiating to the axilla. Infective endocarditis is suspected and three sets of blood cultures have been taken. She is unwell but haemodynamically stable. What is the most appropriate initial treatment?
- A High-dose oral antibiotics
B Empirical intravenous antibiotic therapy tailored after culture results
C Urgent valve replacement
D Anticoagulation therapy
E Observation and symptomatic treatment

LOG BOOK 2025

Log-book questions

- 28 A 30-year-old woman presents with chest pain following a week of fever, cough, and shortness of breath. Her cardiac examination reveals distant heart sounds and her chest X-ray shows an increased cardiac silhouette. What is the most likely cause of the increased cardiac silhouette?
- A Left ventricular hypertrophy
B Dilated cardiomyopathy
C Pericardial effusion
D Aortic aneurysm
E Pulmonary oedema

REPEATED 2x • ALSO ASKED IN
Log book 2025

Answers 26–30

- 26 **B Mitral valve prolapse without regurgitation** — **UPDATED** MVP without regurgitation has near-population endocarditis risk. Prosthetic valves and previous IE carry the highest risk.
- 27 **D Streptococcus viridans infection** — Viridans streptococcal native-valve IE is usually penicillin-sensitive with > 90% cure. *S. aureus*, prosthetic and culture-negative IE do worse.
- 28 **C Pericardial effusion** — Distant heart sounds with a newly enlarged globular silhouette after a viral illness is pericardial effusion (> 200–250 mL); echo confirms and checks for tamponade.
- 29 **B Janeway lesions** — **UPDATED** Janeway lesions (painless palmar/plantar haemorrhagic macules from septic emboli) are characteristic of endocarditis; a minor Duke criterion.

30 **B** Empirical intravenous antibiotic therapy tailored after culture results — After blood cultures, start empirical high-

dose IV bactericidal antibiotics, then tailor to the organism for 4–6 weeks. Surgery only for specific indications.

11 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 A 21-year-old man has a recent influenza-like illness with fever, myalgia and headache. He then develops pleuritic-type chest discomfort and breathlessness. His pulse is 105/min and regular and BP is 105/60 mmHg; the JVP is not raised. The first and second heart sounds are normal, and a loud, harsh to-and-fro sound is heard in both systole and diastole. What is the most likely diagnosis?
 - A Acute viral pericarditis
 - B Aortic valve endocarditis
 - C Mitral valve endocarditis
 - D Persistent ductus arteriosus
 - E Pulmonary embolism
 - 2 What is the appropriate initial treatment for the symptoms of acute pericarditis?
 - A Intravenous glucocorticoids
 - B Intravenous morphine
 - C Oral amiodarone
 - D Oral aspirin
 - E Rectal diclofenac
 - 3 Which of the following conditions may result in chronic pericardial constriction?
 - A Acute myocardial infarction
 - B Dilated cardiomyopathy
 - C Excessive alcohol consumption
 - D Osteoarthritis
 - E Tuberculosis
 - 4 A 75-year-old male smoker has a 6-week history of progressive exertional breathlessness and fatigue, and recently his ankles swell in the afternoon. His pulse is 100/min and regular and BP is 92/60 mmHg. The JVP is raised and rises on inspiration, the heart sounds are quiet with no added sounds, and there is bilateral pitting oedema to the knees. The chest X-ray shows apparent cardiomegaly with a globular cardiac silhouette. A pericardial effusion is suspected. Which of the following statements is true?
 - A A large effusion can be a sign of malignancy
 - B A pericardial rub is always heard if the effusion is large
 - C An ECG is the best investigation to confirm the diagnosis
 - D High-dose diuretic therapy will resolve the pericardial effusion
 - E In symptomatic patients, cardiac surgery is required to remove the pericardial fluid
 - 5 An 18-year-old man has sudden-onset sharp chest pain that is worse on deep inspiration or lying flat and relieved by sitting forward and taking shallow breaths. Acute pericarditis is suspected. What is the most specific ECG change in acute pericarditis?
 - A PR interval prolongation
 - B PR segment depression
 - C ST depression
 - D ST elevation
 - E T-wave inversion
- REPEATED 2* • ALSO ASKED IN**
Passmedicine
- 6 In patients with a pericardial effusion, what is the most important clinical sign to determine whether there is cardiac tamponade?
 - A Cyanosis
 - B Haematuria
 - C Peripheral oedema
 - D Pulsus paradoxus
 - E Raised JVP
 - 7 A 50-year-old man has 3 weeks of fever and influenza-like symptoms. He has a heart rate of 105/min and a wide pulse pressure (BP 140/45 mmHg). A murmur was suspected but is not heard on repeat examination. There is no evidence of chest or urinary infection. What are these findings most compatible with?
 - A Acute myocarditis
 - B Acute viral pericarditis
 - C Infective endocarditis affecting the aortic valve
 - D Infective endocarditis affecting the tricuspid valve
 - E Influenza
 - 8 An 18 year old woman presents with a sore throat and suspected acute rheumatic fever. Which of the following is a minor manifestation of acute rheumatic fever?
 - A Carditis
 - B Chorea
 - C Erythema marginatum
 - D Raised C-reactive protein
 - E Subcutaneous nodules

- 9** An 18 year old woman has a raised C-reactive protein, a rash consistent with erythema marginatum and pyrexia. However, the clinician remains uncertain about the diagnosis of acute rheumatic fever. Rapid response to which treatment will help to confirm the diagnosis?
- A Aspirin
 - B Bed rest
 - C Diuretics
 - D Glucocorticoids
 - E High-dose antibiotics
- 10** A 65-year-old man has a 4-week history of general malaise and lethargy. Two courses of antibiotics have temporarily improved his symptoms, but overall he continues to deteriorate, and his GP notes that he has become anaemic. He is admitted with fever; blood cultures are taken and an echocardiogram shows a mass on the mitral valve. Which organism is most likely to grow from his blood cultures?
- A Staphylococcus aureus
 - B Staphylococcus epidermidis
 - C Streptococcus faecalis
 - D Streptococcus gallolyticus
 - E Viridans streptococci
- 11** A 65-year-old man with a 4-week illness of malaise, anaemia and fever has a mitral valve mass on echocardiography. Before the blood culture results are known, a junior doctor examines him for signs of endocarditis. Which of the following is the commonest finding?
- A Haematuria
 - B Osler's nodes
 - C Roth's spots
 - D Splinter haemorrhages
 - E Subconjunctival haemorrhages

Answers 1-11

- 1** **A Acute viral pericarditis** — A to-and-fro scratchy sound in systole and diastole after a viral illness is a pericardial rub (acute viral pericarditis); treat with NSAID plus colchicine.
- 2** **D Oral aspirin** — Acute pericarditis: high-dose aspirin or ibuprofen tapered over 1–2 weeks plus colchicine for 3 months (halves recurrence). Steroids are second line.
- 3** **E Tuberculosis** — TB is the commonest cause of constriction worldwide; others are cardiac surgery, radiotherapy, viral/idiopathic pericarditis and connective tissue disease.
- 4** **A A large effusion can be a sign of malignancy** — A large effusion in an older smoker suggests malignancy; a rub is often absent, echo confirms, and diuretics do not help. Drain and send cytology.
- 5** **B PR segment depression** — PR depression (with PR elevation in aVR) is the most specific ECG sign of pericarditis; ST elevation is diffuse, concave and without reciprocal change.
- 6** **D Pulsus paradoxus** — Pulsus paradoxus (inspiratory SBP fall > 10 mmHg) is the key bedside sign that an effusion is causing tamponade; echo confirms.
- 7** **C Infective endocarditis affecting the aortic valve** — Prolonged fever with a new wide pulse pressure suggests aortic valve endocarditis with acute AR; its soft short murmur is easily missed.
- 8** **D Raised C-reactive protein** — Jones minor criteria: fever, arthralgia, raised ESR/CRP, prolonged PR. Carditis, polyarthritides, chorea, erythema marginatum and nodules are major.
- 9** **A Aspirin** — Rheumatic arthritis responds dramatically to aspirin within 24–48 h; a rapid response supports the diagnosis.
- 10** **E Viridans streptococci** — An indolent illness with anaemia and partial response to antibiotics is subacute native-valve IE, classically viridans streptococci.
- 11** **A Haematuria** — Microscopic haematuria (glomerulonephritis or renal emboli) is commoner in IE than splinters, Osler nodes or Roth spots; dipstick the urine.

2 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

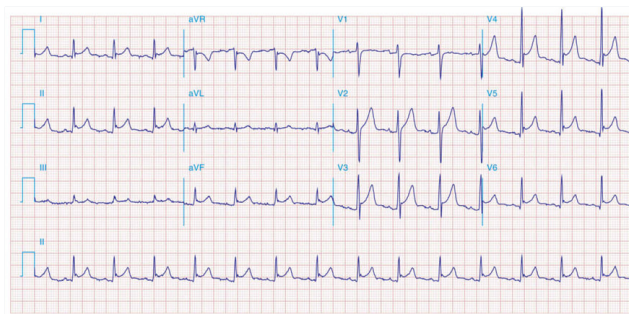
Harrison MCQs

- 1** A 75-year-old woman with widely metastatic non-small-cell lung cancer is admitted to intensive care with BP 73/52 mmHg after 3–5 days of fatigue and worsening breathlessness. Her neck veins are distended, and the chest X-ray shows a massive, water-bottle-shaped heart shadow without new pulmonary infiltrates. Which additional finding is most likely on examination?

- A Fall in systolic blood pressure >10 mmHg with inspiration
- B Lack of fall of the jugular venous pressure with inspiration
- C Late diastolic murmur with opening snap
- D Pulsus parvus et tardus
- E Rapid y-descent of jugular venous pressure tracing

REPEATED 3* • ALSO ASKED IN
Crash Course • PreTest

- 2** A 27-year-old doctor has had 1 week of cough, coryza and low-grade fever. Today he develops rapidly escalating chest pain that is worse on deep inspiration. His 12-lead ECG is shown. His BP is normal, he is now afebrile, and the JVP is not raised, but he is uncomfortable from the pain. What is the most appropriate next step?



- A Administer aspirin, intravenous heparin, sublingual nitroglycerin, and clopidogrel
- B Emergently obtain transthoracic echocardiogram with possible pericardiocentesis
- C Emergently perform coronary angiography to evaluate for acute myocardial infarction
- D Prescribe ibuprofen and colchicine
- E Refer for treadmill stress test

Answers 1-2

HARRISON MCQs

- 1** **A** Fall in systolic blood pressure >10 mmHg with inspiration — Hypotension, distended neck veins and a water-bottle heart in lung cancer is malignant tamponade, so pulsus paradoxus is expected. Kussmaul sign suggests constriction.
- 2** **D** Prescribe ibuprofen and colchicine — Widespread concave ST elevation with PR depression after a viral illness, no high-risk features: outpatient ibuprofen plus colchicine.

7 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1** A 42-year-old woman presents with chest pain. She comments that the pain is worse on inspiration but is relieved by leaning forward. Her past medical history includes breast carcinoma. What is the most likely diagnosis?
- A Acute pericarditis
 - B Pericardial effusion
 - C Pericardial tamponade
 - D Constrictive pericarditis
 - E Myocarditis
- 2** A 43-year-old woman with known breast cancer has increasing breathlessness. Heart sounds are soft and the apex beat cannot be felt. Which investigation is most likely to make the diagnosis?
- A CXR
 - B Echocardiogram
 - C ECG
 - D Arterial blood gas
 - E Coronary angiography

REPEATED 3* • ALSO ASKED IN
Log book 2025 • PreTest

- 3** A 45-year-old man attends the emergency department with worsening breathlessness and fevers. The medical team suspects a diagnosis of infective endocarditis. Which of the following is the most suggestive feature of infective endocarditis?
- A New or changing murmur
 - B Erythema marginatum
 - C Juxta-articular nodules
 - D Atrial fibrillation
 - E Lymphadenopathy
- 4** A 35-year-old man who injects drugs has night sweats and weight loss. He is febrile (38.5 °C) and infective endocarditis is suspected. Which organism is most likely responsible?
- A Streptococcus mutans
 - B Streptococcus sanguis
 - C Enterococcus faecalis
 - D Staphylococcus aureus
 - E Coxiella burnetii
- 5** A 40-year-old man with an underlying cardiac condition is about to have root canal treatment and a new crown fitted. For which of the following conditions is antibiotic prophylaxis against infective endocarditis recommended?
- A Atrial septal defect
 - B Ventricular septal defect
 - C Previous coronary artery bypass grafting involving the use of the internal mammary arteries
 - D Presence of a dual chamber pacemaker
 - E None of the above
- 6** Some clinical features of infective endocarditis result from immune-complex deposition. Which of the following is NOT an immune-complex phenomenon?
- A Splenic abscess
 - B Osler's nodes
 - C Roth spots
 - D Glomerulonephritis
 - E Positive rheumatoid factor
- 7** A previously well 45-year-old man has fevers and malaise. He has multiple splinter haemorrhages and a systolic murmur, and infective endocarditis is suspected. Regarding the modified Duke criteria, which statement is INCORRECT?
- A The presence of Osler's nodes is a minor criterion
 - B A positive rheumatoid factor is a minor criterion
 - C Echocardiographic evidence of new partial dehiscence of a prosthetic valve is a major criterion
 - D A single positive blood culture for Coxiella burnetii or a phase I IgG titre > 1:800 is a major criterion
 - E A definite diagnosis can be made with three minor criteria

REPEATED 2* · ALSO ASKED IN
Other Sources

Answers 1-7

- 1** **A Acute pericarditis** — Pleuritic chest pain relieved by sitting forward is acute pericarditis; with breast cancer consider malignant involvement and check echo.
- 2** **B Echocardiogram** — **UPDATED** Soft heart sounds and an impalpable apex in a patient with cancer suggest a malignant pericardial effusion; echocardiography confirms it and shows tamponade physiology.
- 3** **A New or changing murmur** — A new or changing regurgitant murmur with fever is the most suggestive feature of IE; new regurgitation on echo is a major Duke criterion.
- 4** **D Staphylococcus aureus** — In people who inject drugs, S. aureus is the commonest cause of IE, usually tricuspid with septic pulmonary emboli; cover MRSA where prevalent.
- 5** **E None of the above** — Prophylaxis only for highest risk (prosthetic valves, previous IE, unrepaired cyanotic CHD). ASD, VSD, CABG and pacemakers do not qualify.
- 6** **A Splenic abscess** — **UPDATED** Immune-complex phenomena: Osler nodes, Roth spots, glomerulonephritis, rheumatoid factor. Splenic abscess is embolic (vascular), like Janeway lesions.
- 7** **E A definite diagnosis can be made with three minor criteria** — **UPDATED** Definite IE (modified Duke): 2 major, 1 major + 3 minor, or 5 minor. Three minor criteria give only 'possible' endocarditis.

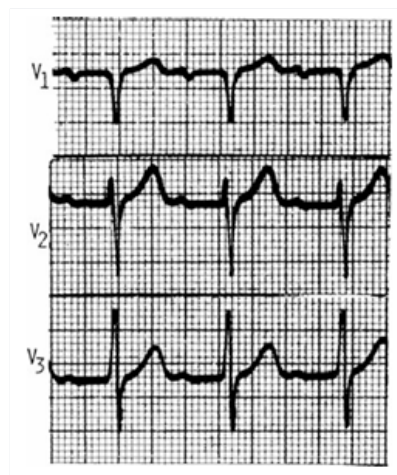
2 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

1 Two weeks after hospital discharge for documented myocardial infarction, a 65-year-old returns to your office concerned about low-grade fever and pleuritic chest pain. There is no associated shortness of breath. Lungs are clear to auscultation and the heart is free of murmur, gallop, or rub. ECG is unchanged from the last one in the hospital. Which therapy is most likely to be effective?

- A Antibiotics
- B Anticoagulation with warfarin (Coumadin)
- C An anti-inflammatory agent
- D An increase in antianginal medication
- E An anxiolytic agent

2 An 18-year-old man has fever and flitting pain in both knees and elbows; the right knee was red and swollen for 1 day during the previous week. He has a low-grade fever, a grade III/VI high-pitched apical systolic murmur radiating to the axilla and a soft mid-diastolic murmur at the base. A tender nodule is felt over an extensor tendon of the hand, and there are pink erythematous lesions on the abdomen, some with central clearing. Haematocrit 0.42, WBC $12 \times 10^9/L$ (80% neutrophils, 20% lymphocytes), ESR 60 mm/h. His ECG is shown. Which test is most critical to the diagnosis?



- A Blood cultures
- B Antistreptolysin O antibody
- C Echocardiogram
- D Antinuclear antibodies
- E Creatine kinase

REPEATED 2x • ALSO ASKED IN
Other Sources

Answers 1-2

PRETEST MCQS

1 **C An anti-inflammatory agent** — Fever and pleuritic pain 2 weeks after MI with no new ECG change is Dressler syndrome: aspirin/NSAID plus colchicine. Anticoagulation risks haemopericardium.

2 **B Antistreptolysin O antibody** — Migratory arthritis, carditis, nodules and erythema marginatum meet Jones criteria, which also need evidence of recent strep infection: a raised ASO (or anti-DNase B) titre.

7 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1** A 77-year-old man is under review on the ward after having a pacemaker inserted earlier that day. He initially presented to the emergency department two days ago with recurrent episodes of collapse, leading to a diagnosis of sick sinus syndrome. In the past hour, he has reported increasing chest discomfort and developed hypotension. On examination, he is cool and clammy peripherally, his jugular venous pressure (JVP) is at 5 cm above the sternal angle, and his heart sounds are indistinct. His current heart rate is 72 bpm with a blood pressure of 88/56 mmHg. What is the most likely diagnosis?

 - A Haemopericardium
 - B Iatrogenic mediastinitis
 - C Pacemaker lead migration
 - D Pericarditis
 - E Ventricular aneurysm
- 2** A 65-year-old man with type 2 diabetes attends the emergency department with increasing breathlessness over 7 days, much worse in the last 24 hours, and light-headedness. He has left shoulder pain and mild ongoing chest pain that improves on sitting up. Five weeks ago he had an anterior STEMI treated with a stent to the left anterior descending artery. The JVP is raised, peripheral pulses are weak and the heart sounds are muffled; the chest is clear with good air entry. Pulse 101/min regular, BP 89/50 mmHg (equal in both arms), temperature 36.0 °C, SpO₂ 95% on air, respiratory rate 20/min. The ECG shows Q waves in V1–V3 without ST change and small-amplitude complexes. What is the most likely diagnosis?

 - A Aortic dissection
 - B Cardiac tamponade
 - C Pneumothorax
 - D Pulmonary embolism
 - E Stent re-stenosis
- 3** A 30-year-old woman attends the emergency department with a 1-day history of severe, non-radiating central chest pain that eases on expiration. Cardiorespiratory examination is normal apart from a heart rate of 96/min. The ECG shows widespread ST elevation in the anterior, inferior and lateral leads. FBC and U&E are normal; troponin I is 0.8 µg/L (reference < 0.2 µg/L). What is the most likely diagnosis?

 - A Pulmonary embolism
 - B Acute coronary syndrome
 - C Hypertrophic obstructive cardiomyopathy
 - D Acute pericarditis
 - E Arrhythmogenic right ventricular cardiomyopathy
- 4** Which one of the following features is not part of the modified Duke criteria used in the diagnosis of infective endocarditis?

 - A Prolonged PR interval
 - B Positive serology for *Coxiella burnetii*
 - C Fever > 38°C
 - D Roth spots
 - E Positive microbiology from embolic fragments
- 5** A 72-year-old woman is due to have a tooth extracted next week. She had an aortic valve replacement 2 years ago. According to current ESC guidance, what antibiotic prophylaxis is recommended before this procedure?

 - A Chlorhexidine mouthwash
 - B General oral hygiene advice
 - C Intravenous co-amoxiclav
 - D Oral amoxicillin
 - E Oral phenoxymethylpenicillin
- 6** A 75-year-old man presents with a 6-week history of fevers, night sweats and lethargy. On examination, you note several Janeway lesions on his hands. There is a collapsing pulse and you note an early diastolic murmur. You take multiple sets of blood cultures and commence the patient on broad-spectrum IV antibiotics for presumed infective endocarditis. A trans-thoracic echo identifies large vegetation on the aortic valve with associated aortic regurgitation. You note the patient has a background of metastatic colorectal cancer and is currently receiving palliative chemotherapy. In light of the patient's past medical history, which of the following organisms are most likely to be responsible?

 - A *Streptococcus gallolyticus*
 - B *Brucella*
 - C *Coxiella burnetii*
 - D *Staphylococcus epidermidis*
 - E *Streptococcus viridans*
- 7** A 60-year-old man who had a mitral valve replacement 3 years ago presents acutely with high fever, rigors and fatigue 1 week after a large boil (furuncle) on his thigh. Echocardiography shows a new vegetation on the prosthetic valve, and three sets of blood cultures are taken. Which organism is most likely to be responsible?

 - A *Staphylococcus aureus*
 - B *Streptococcus gallolyticus*
 - C Coagulase-negative staphylococci
 - D *Enterococcus faecalis*
 - E *Candida albicans*

Answers 1-7

- 1 **A Haemopericardium** — Chest discomfort, hypotension, raised JVP and quiet heart sounds hours after pacing suggest lead perforation with haemopericardium/tamponade.
- 2 **B Cardiac tamponade** — Hypotension, raised JVP, muffled sounds and low voltages weeks after MI is tamponade from a post-MI effusion: urgent echo and pericardiocentesis.
- 3 **D Acute pericarditis** — Multi-territory ST elevation with pleuritic pain in a young woman is pericarditis; a small troponin rise means myopericarditis, not ACS.
- 4 **A Prolonged PR interval** — A prolonged PR is a Jones (rheumatic fever) criterion, not a Duke criterion; in IE new conduction block may signal an aortic root abscess.
- 5 **D Oral amoxicillin** — **UPDATED** A prosthetic valve is highest risk: ESC/AHA advise single-dose oral amoxicillin 2 g 30–60 min before extraction. NICE advises against routine prophylaxis.
- 6 **A Streptococcus gallolyticus** — Streptococcus gallolyticus (S. bovis) IE is strongly linked to colorectal neoplasia, so isolating it should prompt colonoscopy.
- 7 **A Staphylococcus aureus** — **UPDATED** Late prosthetic valve IE (> 1 year) resembles native IE; an acute illness after a furuncle points to S. aureus, which usually needs rifampicin and early surgery.

7 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

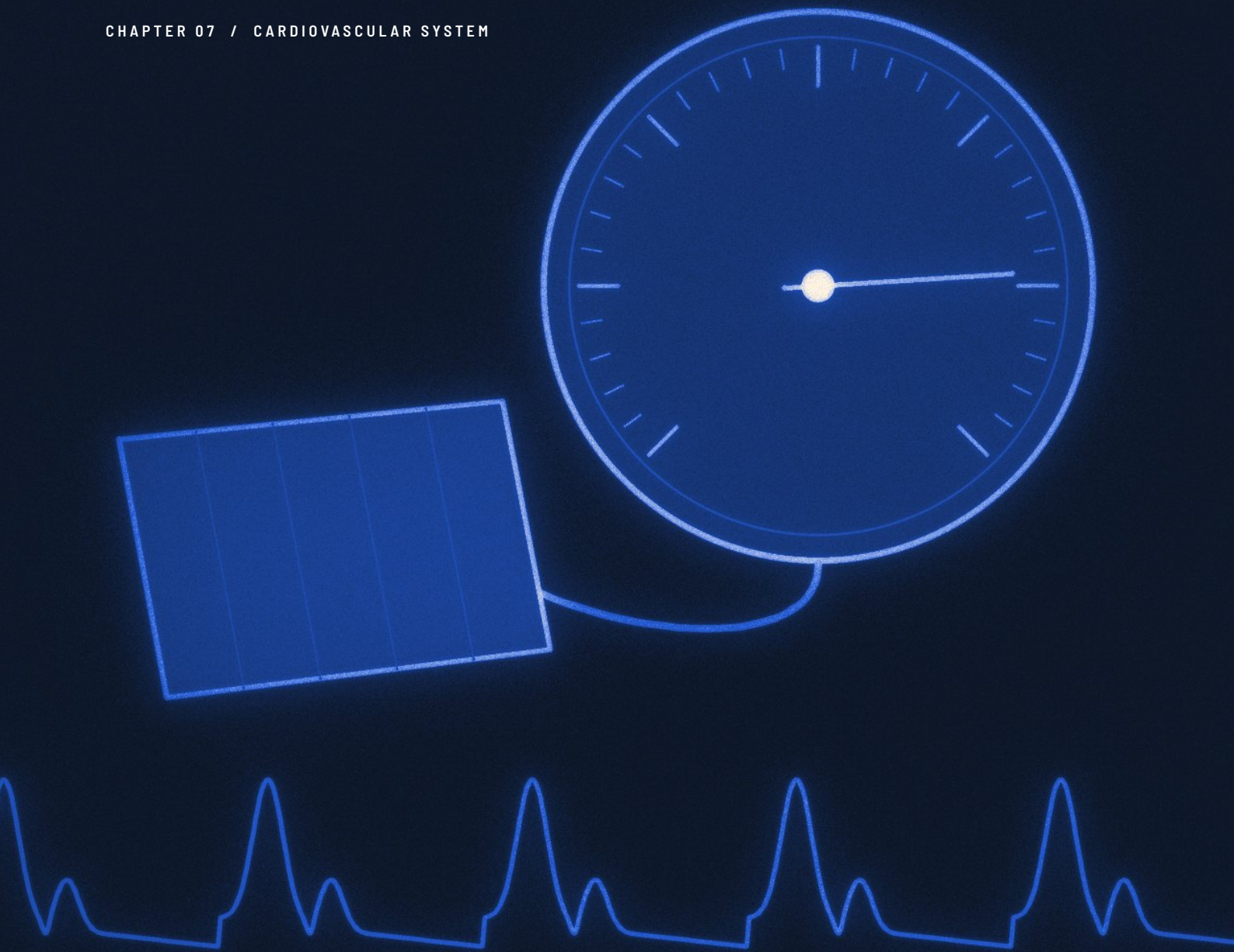
- 1 A 55-year-old man attends the emergency department with a dry cough and severe chest pain that began this morning. Two months ago he had an inferior myocardial infarction treated with a right coronary artery stent. He has hypertension and hypercholesterolaemia and takes aspirin, clopidogrel, atorvastatin and enalapril. Temperature 38.5 °C, pulse 92/min, respiratory rate 22/min, BP 130/80 mmHg. A high-pitched scratching sound is heard best with him leaning forward in expiration; the rest of the examination is normal. The ECG shows diffuse ST elevation and troponin I is 0.2 µg/L (0.2 ng/mL). What is the most likely cause of his symptoms?
 - A Constrictive pericarditis
 - B Early infarct-associated pericarditis
 - C Reinfarction
 - D Cardiac tamponade
 - E Dressler syndrome
- 2 A 12-year-old girl has high fever and swelling of the left ankle and knee 3 weeks after a sore throat. Two years ago she developed breathlessness and a generalised rash when given dicloxacillin for a skin infection. Temperature 38.8 °C, pulse 87/min, BP 98/62 mmHg. The left ankle and knee are swollen and tender with limited movement, and a grade 3/6 holosystolic murmur is heard best at the apex. Which antibiotic is most appropriate?
 - A Vancomycin
 - B Amoxicillin
 - C Clarithromycin
 - D Doxycycline
 - E Ciprofloxacin
- 3 A 43-year-old man with HIV who injects drugs presents with fever, night sweats and a holosystolic murmur that increases on inspiration. Which complication of his condition is most likely?
 - A Septic pulmonary emboli
 - B Splinter haemorrhages under the fingernails
 - C Painful nodules on the finger pads
 - D Painless haemorrhagic macules on the palms and soles
 - E Retinal haemorrhages
- 4 A 34-year-old man with a bicuspid aortic valve has fatigue, exertional breathlessness, low-grade fevers and splinter haemorrhages. He was started on empirical IV vancomycin, and blood cultures now grow *Cardiobacterium hominis*. What is the next step in management?
 - A Switch to IV gentamicin
 - B Continue IV vancomycin
 - C Switch to IV ceftriaxone
 - D Switch to IV ampicillin
 - E Switch to IV cefazolin
- 5 A 64-year-old man admitted 7 days ago with pancreatitis after abdominal trauma develops fever, breathlessness and a new holosystolic murmur. There is purulent discharge from his central venous catheter site. Which organism is most likely responsible?
 - A Pseudomonas aeruginosa
 - B Streptococcus gallolyticus
 - C Klebsiella pneumoniae
 - D Haemophilus aphrophilus
 - E Staphylococcus aureus

- 6** At autopsy, multiple small sterile nodules composed of immune complexes, fibrin and thrombus are found on both surfaces of the mitral valve, including the ventricular side. In which patient would these findings most likely be found?
- A** A 71-year-old man with acute high fever and nail-bed haemorrhages
 - B** A 54-year-old man after dental surgery
 - C** A 41-year-old woman with a facial rash and non-erosive arthritis
 - D** A 28-year-old HIV-positive woman who injects drugs
 - E** A 6-year-old girl with subcutaneous nodules and erythema marginatum
- 7** Three weeks after a transurethral resection of the prostate, a 70-year-old man has fever, malaise and limb pain. He has subungual petechiae, tender red papules on the fingers and a new holosystolic murmur. Blood cultures grow a non-haemolytic, PYR-positive Gram-positive coccus. Which organism is most likely responsible?
- A** *Staphylococcus aureus*
 - B** *Streptococcus gallolyticus*
 - C** *Staphylococcus epidermidis*
 - D** *Haemophilus aphrophilus*
 - E** *Enterococcus faecalis*

Answers 1-7

OTHER SOURCES MCQS

- 1** **E Dressler syndrome** — Fever, pleuritic pain, rub and diffuse ST elevation weeks after MI is Dressler syndrome; early infarct pericarditis occurs within 1–3 days.
- 2** **C Clarithromycin** — UPDATED Rheumatic fever: eradicate group A strep with penicillin/amoxicillin; with penicillin anaphylaxis use a macrolide (clarithromycin/azithromycin).
- 3** **A Septic pulmonary emboli** — UPDATED IE in people who inject drugs is usually tricuspid (murmur louder on inspiration) and embolises to the lungs as septic pulmonary emboli.
- 4** **C Switch to IV ceftriaxone** — *Cardiobacterium hominis* is a HACEK organism, often beta-lactamase-producing: switch to ceftriaxone for 4 weeks (6 for prosthetic valves).
- 5** **E *Staphylococcus aureus*** — Line-associated healthcare IE is most often *S. aureus*; remove the line, take cultures and give vancomycin until MRSA is excluded.
- 6** **C A 41-year-old woman with a facial rash and non-erosive arthritis** — Sterile vegetations on both sides of the mitral leaflets are Libman–Sacks endocarditis of SLE (malar rash, non-erosive arthritis), especially with antiphospholipid antibodies.
- 7** **E *Enterococcus faecalis*** — IE after a urological procedure with a non-haemolytic, PYR-positive Gram-positive coccus is *Enterococcus faecalis*: amoxicillin plus ceftriaxone or gentamicin ~6 weeks.



07

HYPERTENSION

Hypertension

Diagnosis and grading, investigation for target-organ damage and secondary causes, drug choice, and hypertensive emergencies.

28

MCQs

0

EMQ SCENARIOS

6

SOURCES

2

REPEATED IDEAS

IN THIS CHAPTER

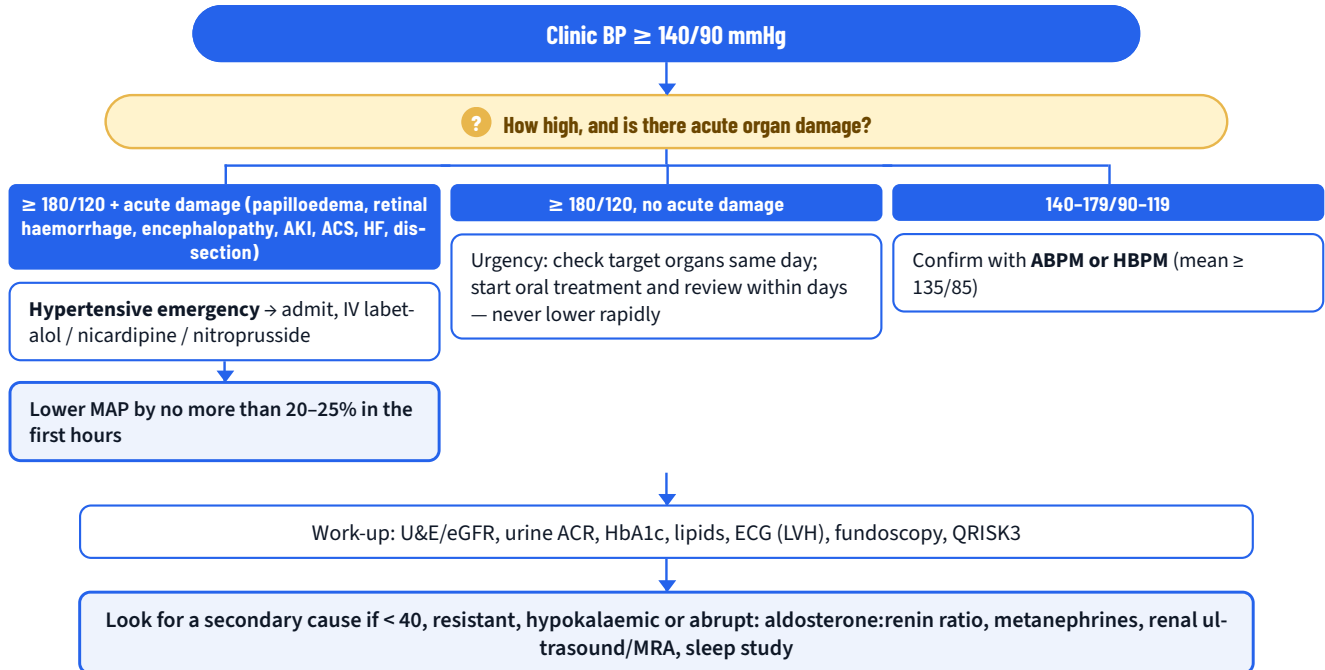
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02	Local Exams MCQs	7 Q	page 120
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07

Step by step

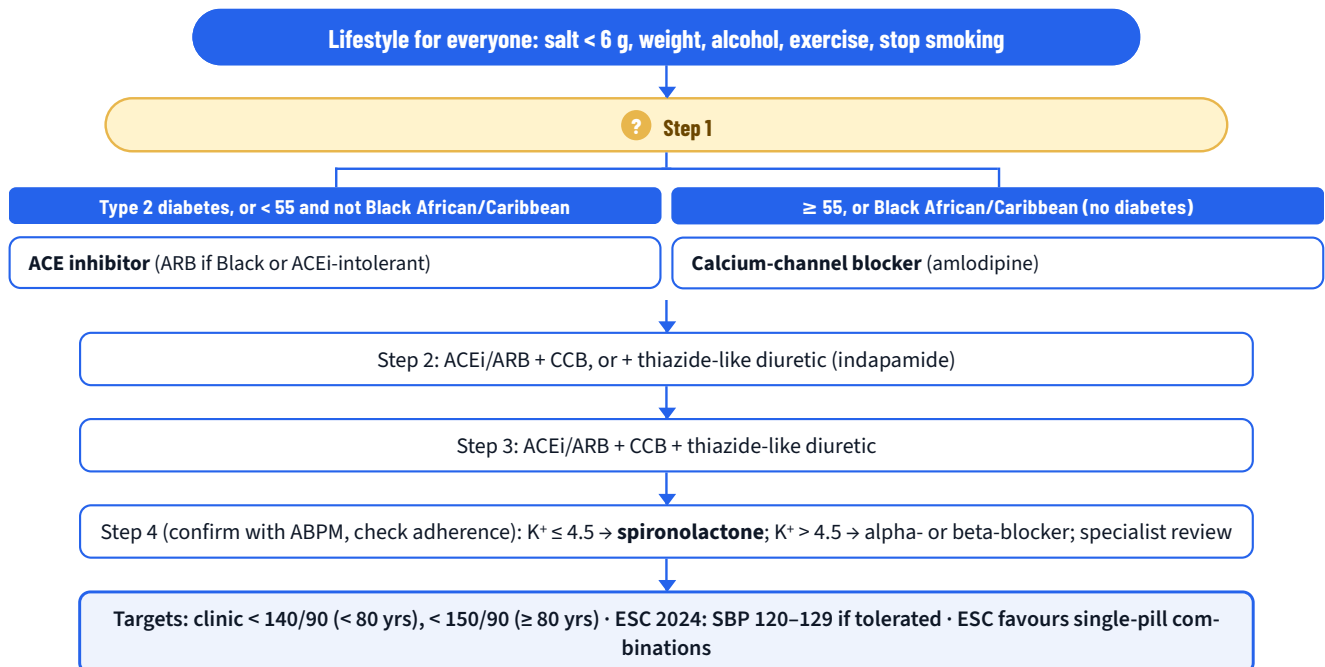
Raised blood pressure: diagnosis and urgency

NICE NG136 • ESC 2024 HYPERTENSION



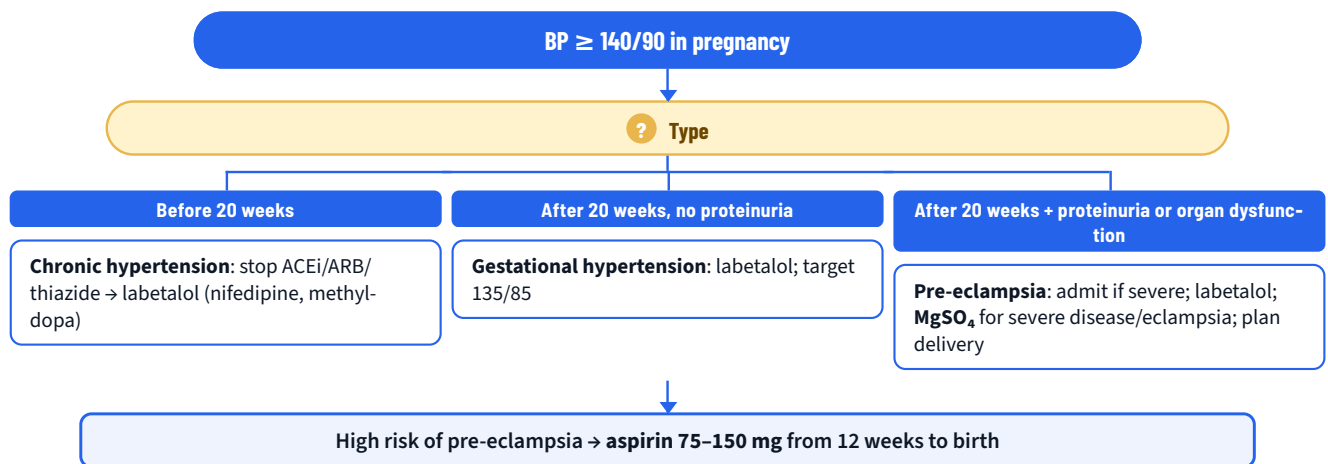
Drug treatment steps

NICE NG136 (2023) • ESC 2024



Hypertension in pregnancy

NICE NG133



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1 A 45-year-old man with newly diagnosed hypertension is referred for baseline investigations. Which test is NOT routinely indicated?
 - A Urea and electrolytes
 - B ECG
 - C CT aortography
 - D Lipid profile
 - 2 On routine examination a patient with hypertension has a blood pressure of 165/105 mmHg. According to the European (ESH/ESC) and British Hypertension Society classification, this is:
 - A High-normal blood pressure
 - B Grade 1 (mild) hypertension
 - C Grade 2 (moderate) hypertension
 - D Grade 3 (severe) hypertension
 - 3 Which of the following drug classes is NOT first-line antihypertensive therapy?
 - A ACE inhibitors
 - B Calcium-channel blockers
 - C Beta-blockers
 - D Angiotensin receptor blockers
 - E Thiazide-like diuretics
- REPEATED 3* • ALSO ASKED IN**
39th batch end-block (cardiology) • Formative exam
- 4 All of the following endocrine diseases are recognised causes of secondary hypertension EXCEPT:
 - A Acromegaly
 - B Congenital adrenal hyperplasia
 - C Conn's syndrome
 - D Addison's disease

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 5 A 45-year-old woman with type 2 diabetes mellitus is reviewed in the diabetes clinic. Her blood pressure was 145/95 mmHg and lisinopril was started. Baseline blood tests showed potassium 4.5 mmol/L, creatinine 159 μ mol/L (1.8 mg/dL) and eGFR 47 mL/min/1.73 m². Three months later, potassium is 4.9 mmol/L, creatinine 177 μ mol/L (2.0 mg/dL) and eGFR 44 mL/min/1.73 m². What is the most appropriate course of action?
 - A No action
 - B Switch to diuretics
 - C Reduce dose of lisinopril
 - D Stop lisinopril and arrange investigations to exclude renal artery stenosis

LOG BOOK 2025

Log-book questions

- 6 A 42-year-old man with known hypertension who ran out of his medication presents with a blood pressure of 230/140 mmHg, severe headache, dizziness and blurred optic disc margins (papilloedema) on fundoscopy. What is the most likely diagnosis?
 - A Hypertensive urgency
 - B White coat hypertension
 - C Isolated systolic hypertension
 - D Secondary hypertension
 - E Hypertensive emergency
- 7 A 42-year-old man with known hypertension who ran out of his medication presents with a blood pressure of 230/140 mmHg, severe headache, dizziness and papilloedema on fundoscopy. Which of the following is an appropriate first-line antihypertensive in this situation?
 - A Oral amlodipine
 - B Intravenous sodium nitroprusside
 - C Sublingual captopril
 - D Oral lisinopril
 - E Oral bendroflumethiazide

Answers 1-7

LOCAL EXAMS MCQS

- 1 **C CT aortography** — New hypertension work-up: U&E/eGFR, urine ACR and dipstick, HbA1c, lipids, ECG and fundoscopy. CT aortography is only for a specific clue.
- 2 **C Grade 2 (moderate) hypertension** — ESH/ESC grades: 1 = 140–159/90–99, 2 = 160–179/100–109, 3 = $\geq 180/\geq 110$ mmHg; 165/105 is grade 2. Confirm with ABPM/HBPM.
- 3 **C Beta-blockers** — First-line: ACEi/ARB, dihydropyridine CCB or thiazide-like diuretic. Beta-blockers are used only for compelling indications (angina, MI, HFrEF, AF).
- 4 **D Addison's disease** — Addison's disease causes salt loss and hypotension. Conn's, acromegaly and 11 β /17 α -hydroxylase CAH cause hypertension.
- 5 **A No action** — A creatinine rise < 30% (eGFR fall < 25%) with K⁺ < 6 after an ACE inhibitor is expected; continue the drug and monitor.
- 6 **E Hypertensive emergency** — **UPDATED** Severe hypertension with papilloedema (acute organ damage) is a hypertensive emergency (malignant hypertension): admit for controlled IV treatment.

- 7 **B Intravenous sodium nitroprusside** — UPDATED Hypertensive emergency needs a titratable IV drug (labetalol, nicardipine or nitroprusside); lower MAP by $\leq 20\text{--}25\%$ over the first hours.

3 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 A 56-year-old man presents with headache. On auscultation he has a loud second heart sound. Which of the following could explain this finding?
- A Aortic incompetence
 - B Essential hypertension
 - C Mechanical mitral valve replacement
 - D Mitral incompetence
 - E Postural hypotension
- 2 A 35-year-old executive has a reproducible blood pressure of 180/100 mmHg and is referred to clinic to look for an underlying cause. What is the commonest cause of secondary hypertension?
- A Congenital adrenal hyperplasia
 - B Cushing's syndrome
 - C Pheochromocytoma
 - D Renal disease
 - E Thyrotoxicosis
- 3 A 60-year-old man is referred by his family doctor because his blood pressure remains uncontrolled despite four drugs, and a secondary cause is suspected. What is the commonest cause of poorly controlled hypertension?
- A Conn's syndrome
 - B Glucocorticoid-suppressible hyperaldosteronism
 - C Hyper-reninaemia
 - D Poor adherence with medication
 - E Renal failure

Answers 1-3

DAVIDSON MCQS

- 1 **B Essential hypertension** — Systemic hypertension makes the aortic valve close forcefully, giving a loud A2. Aortic regurgitation softens A2.
- 2 **D Renal disease** — UPDATED Renal disease (parenchymal and renovascular) is the commonest secondary cause; primary aldosteronism is the commonest endocrine cause.
- 3 **D Poor adherence with medication** — Apparent resistant hypertension is most often non-adherence (pseudo-resistance); exclude it, white-coat effect and NSAIDs before seeking secondary causes.

7 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 All of the following statements regarding the epidemiology of hypertension are true EXCEPT:
- A Among people aged 60 years or older, average systolic blood pressure is higher in women than in men
 - B Diastolic blood pressure increases steadily with age throughout life
 - C Systolic blood pressure continues to rise with age throughout adult life
 - D In industrialised nations, blood pressure rises steadily during the first two decades of life
 - E Worldwide, around one-third of adults aged 30–79 years have hypertension
- 2 Disease of which organ is the most common cause of mortality in hypertensive patients?
- A Brain
 - B Heart
 - C Kidney
 - D Liver
 - E Lungs

- 3** A 28-year-old woman has had hypertension that is difficult to control since the age of 26 years, requiring increasing doses of medication. Her current regimen is labetalol 1000 mg twice daily, lisinopril 40 mg once daily, clonidine 0.1 mg twice daily and amlodipine 5 mg once daily. She appears well; blood pressure is 168/100 mmHg and heart rate 84 bpm. Cardiac examination is unremarkable, peripheral pulses are good and there is no oedema. There is no hirsutism, abnormal fat distribution or genital abnormality. Serum potassium is 2.8 mmol/L, bicarbonate 32 mmol/L and fasting glucose 6.3 mmol/L. What is the most likely diagnosis?
- A Congenital adrenal hyperplasia
 - B Fibromuscular dysplasia
 - C Cushing's syndrome
 - D Conn's syndrome
 - E Pheochromocytoma
- 4** A 28-year-old woman has resistant hypertension despite four antihypertensive drugs. She has no cushingoid features. Serum potassium is 2.8 mmol/L and bicarbonate 32 mmol/L. What is the best initial test to establish the diagnosis?
- A Renal vein renin levels
 - B 24-hour urine collection for metanephrines
 - C MRI of the renal arteries
 - D 24-hour urine collection for cortisol
 - E Plasma aldosterone/renin ratio
- 5** A 65-year-old engineer who never sees a doctor and takes no medications is brought to the emergency department with a headache that has worsened over the past day and confusion that began an hour ago. Blood pressure is 230/140 mmHg, heart rate 90 bpm and oxygen saturation 95%. He moves all limbs equally and has no cranial nerve deficit but is clearly delirious. The apex beat is forceful and displaced with an S4 gallop; the lungs are clear. Creatinine is 212 µmol/L (2.4 mg/dL), urine shows 2+ protein and haematuria, haematocrit is 0.32 with a normal platelet count, and the blood film shows schistocytes. Urgent brain MRI shows old microvascular changes but no acute infarct or haemorrhage. Which is the most reasonable approach to his blood pressure?
- A Clonidine 0.1 mg orally
 - B Lisinopril 20 mg orally
 - C Intravenous labetalol 20 mg followed by a continuous infusion, aiming for a mean arterial pressure of about 125 mmHg in the first hour
 - D Immediate-release nifedipine 90 mg orally
 - E Emergency plasma exchange
- 6** Which of the following statements regarding the physiology of renal perfusion is true?
- A Because of tight precapillary regulation, the glomerular capillary endothelium is relatively shielded from pressure injury
 - B Renal cortical blood flow is less than renal medullary blood flow, and together with the high metabolic needs of the cortex this leaves the cortex bordering on hypoxia
 - C Renal perfusion is tightly matched to the metabolic needs of the kidney
 - D Urinary albumin excretion predicts systemic atherosclerotic events and can appear years before clinical atherosclerotic disease
 - E Venous blood returning from the renal cortex has a lower oxygen content than venous blood returning from the renal medulla
- 7** Which of the following statements regarding renal artery stenosis is true?
- A An abnormally reduced renal artery flow velocity by Doppler ultrasound is predictive of haemodynamically significant stenosis
 - B Early in the course of renal artery stenosis, elevated systemic renin levels are typical
 - C In the general population, the presence of fibromuscular dysplasia is rare (<1% prevalence)
 - D Levels of renin activity observed in renal artery stenosis are predictive of response to medical therapy
 - E Usually the first clinical presentation of renal stenosis due to fibromuscular dysplasia is reduced renal function

Answers 1-7

HARRISON MCQS

- 1** **B** Diastolic blood pressure increases steadily with age throughout life — **UPDATED** Systolic BP rises throughout life; diastolic plateaus or falls after 50–60 as arteries stiffen, hence isolated systolic hypertension in the elderly.
- 2** **B** Heart — Heart disease (CHD, LVH, heart failure) is the commonest cause of death in hypertension; stroke is most tightly linked to BP level.

- 3 **D Conn's syndrome** — Resistant hypertension with hypokalaemia and alkalosis despite an ACE inhibitor suggests primary aldosteronism (Conn's); check aldosterone:renin ratio.
- 4 **E Plasma aldosterone/renin ratio** — Screen for primary aldosteronism with the aldosterone:renin ratio (correct K⁺ first); confirm by suppression test, then adrenal CT and venous sampling.
- 5 **C Intravenous labetalol 20 mg followed by a continuous infusion, aiming for a mean arterial pressure of about 125 mmHg in the first hour** — Malignant hypertension with encephalopathy and AKI: IV labetalol infusion, lowering MAP by no more than 20–25% in the first hour.
- 6 **D Urinary albumin excretion predicts systemic atherosclerotic events and can appear years before clinical atherosclerotic disease** — Albuminuria reflects generalised endothelial dysfunction and predicts CV events years ahead; hence urine ACR in the hypertension work-up.
- 7 **B Early in the course of renal artery stenosis, elevated systemic renin levels are typical** — Early renal artery stenosis raises renin (renin-dependent hypertension); a significant stenosis increases peak Doppler velocity.

6 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1 A man has been diagnosed with essential hypertension by his GP and is about to start treatment. Which of the following statements about the treatment of essential hypertension is true?
- A Alpha-blockers such as prazosin rarely cause postural hypotension
 - B Patients of African-Caribbean origin respond well to ACE inhibitor monotherapy
 - C ACE inhibitors are safe in bilateral renal artery stenosis
 - D Treatment with bendroflumethiazide may cause hypokalaemia
 - E ACE inhibitors commonly cause ankle swelling
- 2 A 50-year-old man has a persistently raised blood pressure of about 185/110 mmHg. He has had headaches for a few weeks but is otherwise well. His ECG shows increased QRS voltages with ST depression in the left ventricular leads, and fundoscopy shows arteriovenous nipping without haemorrhages or papilloedema. What is the most appropriate management?
- A Advise weight loss and see again in 2 months
 - B Arrange an exercise stress test
 - C Hospitalize and give urgent IV antihypertensive medication
 - D Begin oral antihypertensives and see in 2 months
 - E Begin oral antihypertensives and see in 3 days
- 3 A 50-year-old white man sees his GP for a blood pressure check, having recently been diagnosed with essential hypertension. His only other condition is type 2 diabetes mellitus without end-organ damage, well controlled on oral hypoglycaemic agents. His blood pressure has remained about 145/90 mmHg despite 3 months of diet and exercise. Which is the most appropriate first-line antihypertensive?
- A Start doxazosin
 - B Start bisoprolol
 - C Start a thiazide diuretic
 - D Start nifedipine
 - E Start enalapril
- 4 A 50-year-old man presents to the emergency department with a mild headache. He is known to have poorly controlled hypertension and is on multiple antihypertensive medications. On examination his blood pressure is 205/110 mmHg and there is no evidence of end-organ involvement. His headache is still present but there is no focal neurology identified. Which of the following is the least appropriate in the management of this patient's blood pressure?
- A Rapidly lowering blood pressure in the emergency department
 - B Adjust treatment and follow-up within 24 to 48 hours of presentation in the community
 - C Initiate a maintenance dose of an oral medication before discharge
 - D Consider a short observation period before discharge
 - E Discharge the patient, emphasizing the importance of close follow-up and compliance with medications
- 5 A 55-year-old man has been recently diagnosed with hypertension and commenced on treatment. He attends his GP to find out more about the condition. Which of the following is the commonest cause of hypertension?
- A Polycystic kidney disease
 - B Renin-producing tumour
 - C Undetermined cause
 - D Oral contraceptive pills
 - E Coarctation of aorta
- 6 A 65-year-old hypertensive patient is reviewed in the outpatient clinic complaining of visual disturbances. On examination you note flame-shaped haemorrhages. What is the most likely diagnosis?
- A Grade 1 retinal disease
 - B Grade 2 retinal disease
 - C Grade 3 retinal disease
 - D Grade 4 retinal disease
 - E Grade 5 retinal disease

REPEATED 2* • ALSO ASKED IN
Crash Course

Answers 1-6

- 1 **D Treatment with bendroflumethiazide may cause hypokalaemia** — **UPDATED** Thiazides cause hypokalaemia, hyponatraemia, gout and hyperglycaemia. Ankle swelling is from dihydropyridine CCBs, not ACE inhibitors.
- 2 **E Begin oral antihypertensives and see in 3 days** — Grade 3 hypertension with LVH and grade 2 retinopathy but no acute features: start oral treatment now and review within days.
- 3 **E Start enalapril** — **UPDATED** In type 2 diabetes NICE starts an ACE inhibitor at any age (ARB if Black or ACEi-intolerant); add CCB or thiazide-like diuretic next.
- 4 **A Rapidly lowering blood pressure in the emergency department** — Hypertensive urgency (no acute organ damage) is lowered gradually with oral drugs; rapid ED reduction risks stroke and ischaemia.
- 5 **C Undetermined cause** — About 90–95% of hypertension is primary (essential) with no single cause; secondary causes make up only 5–10%.
- 6 **C Grade 3 retinal disease** — Keith–Wagener: 1 narrowing/silver wiring, 2 AV nipping, 3 flame haemorrhages and cotton-wool spots, 4 papilloedema.

3 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 55-year-old woman presents to the emergency department with lethargy and a blood pressure of 250/150 mmHg. Her family say she complained of severe headache and visual disturbance earlier in the day. She has a history of asthma but no known kidney disease. Fundoscopy shows retinal haemorrhages. Which of the following is the best approach?
 - A Intravenous labetalol
 - B Continuous-infusion sodium nitroprusside
 - C Oral clonidine
 - D Sublingual nifedipine
 - E Intravenous loop diuretic
- 2 A 50-year-old construction worker continues to have a blood pressure of 160/95 mmHg even after a third agent is added to his antihypertensive regimen. Physical examination is normal, electrolytes are normal and he takes no over-the-counter medications. Which of the following is the best next step?
 - A Check pill count
 - B Evaluate for Cushing syndrome
 - C Check chest x-ray for coarctation of the aorta
 - D Obtain a renal angiogram
 - E Obtain an adrenal CT scan
- 3 You see a 45-year-old woman with diabetes mellitus for the first time in clinic. Her systolic blood pressure was above 140 mmHg at each of her last two clinic appointments, home blood pressure monitoring over the past week averaged 150/90 mmHg, and her clinic blood pressure today is 164/92 mmHg. What is the best next step in her blood pressure management?
 - A Ask her to keep a written record of her home blood pressure and bring it to a return appointment
 - B Advise a heart-healthy, low-sodium diet and refer her to a dietitian
 - C Prescribe an ACE inhibitor in addition to a heart-healthy diet
 - D Prescribe a dihydropyridine calcium-channel blocker in addition to a heart-healthy diet
 - E Arrange an echocardiogram to assess for end-organ damage

Answers 1-3

- 1 **B Continuous-infusion sodium nitroprusside** — Hypertensive emergency with asthma: avoid labetalol; use a nitroprusside (or nicardipine) infusion, lowering MAP $\leq 25\%$ in the first hour.
- 2 **A Check pill count** — Before calling hypertension resistant, exclude non-adherence (pill count, drug levels) and confirm with ABPM; then seek secondary causes.
- 3 **C Prescribe an ACE inhibitor in addition to a heart-healthy diet** — **UPDATED** In diabetes with confirmed hypertension, start an ACE inhibitor alongside lifestyle advice (reduces albuminuria and nephropathy).

2 QUESTIONS • AFTER PASSMEDICINE

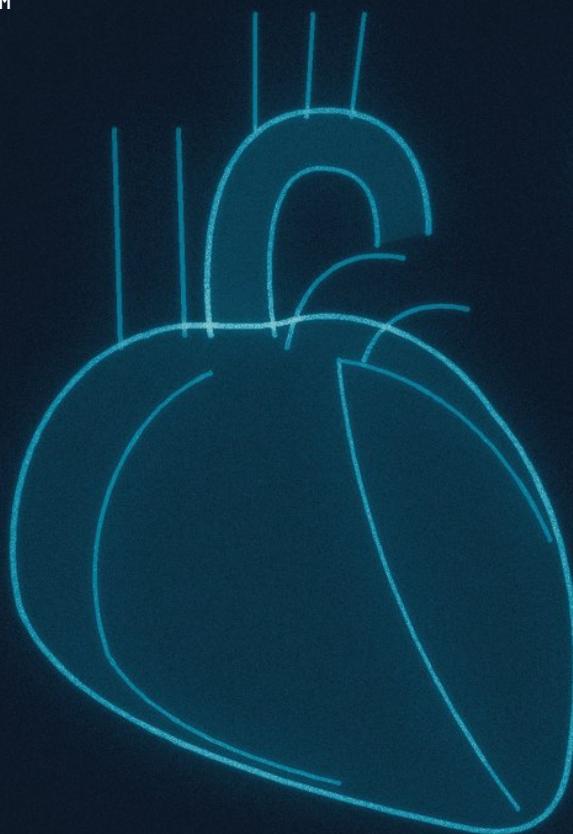
Passmedicine MCQs

- 1 A 45-year-old man with hypertension attends his GP for review. His last clinic reading was 156/80 mmHg, and ambulatory blood pressure monitoring shows a daytime average of 138/85 mmHg. He is already taking the maximum dose of ramipril. Which agent would be most appropriate to add?
- A Bisoprolol
 - B Doxazosin
 - C Indapamide
 - D Spironolactone
 - E Verapamil
- 2 Which one of the following statements regarding hypertension in pregnancy is NOT correct?
- A A rise above booking readings of more than 30 mmHg systolic or 15 mmHg diastolic suggests hypertension
 - B Pre-eclampsia occurs in around 5% of pregnancies
 - C A urine dipstick showing protein + is consistent with gestational hypertension
 - D A rise in blood pressure before 20 weeks suggests pre-existing hypertension
 - E In gestational hypertension the blood pressure rises in the second half of pregnancy

Answers 1-2

PASSMEDICINE MCQS

- 1 **C Indapamide** — Daytime ABPM 138/85 is above the < 135/85 target on maximum ACEi; step 2 adds a CCB or thiazide-like diuretic (indapamide).
- 2 **C A urine dipstick showing protein + is consistent with gestational hypertension** — Gestational hypertension is new BP $\geq$ 140/90 after 20 weeks without proteinuria; proteinuria makes it pre-eclampsia. Labetalol is first line.



08

CARDIOVASCULAR PHARMACOLOGY



Cardiovascular Pharmacology

Mechanisms, indications, interactions and adverse effects of antiplatelets, anticoagulants, antianginals, antiarrhythmics and lipid-lowering drugs.

15

MCQs

0

EMQ SCENARIOS

6

SOURCES

0

REPEATED IDEAS

IN THIS CHAPTER

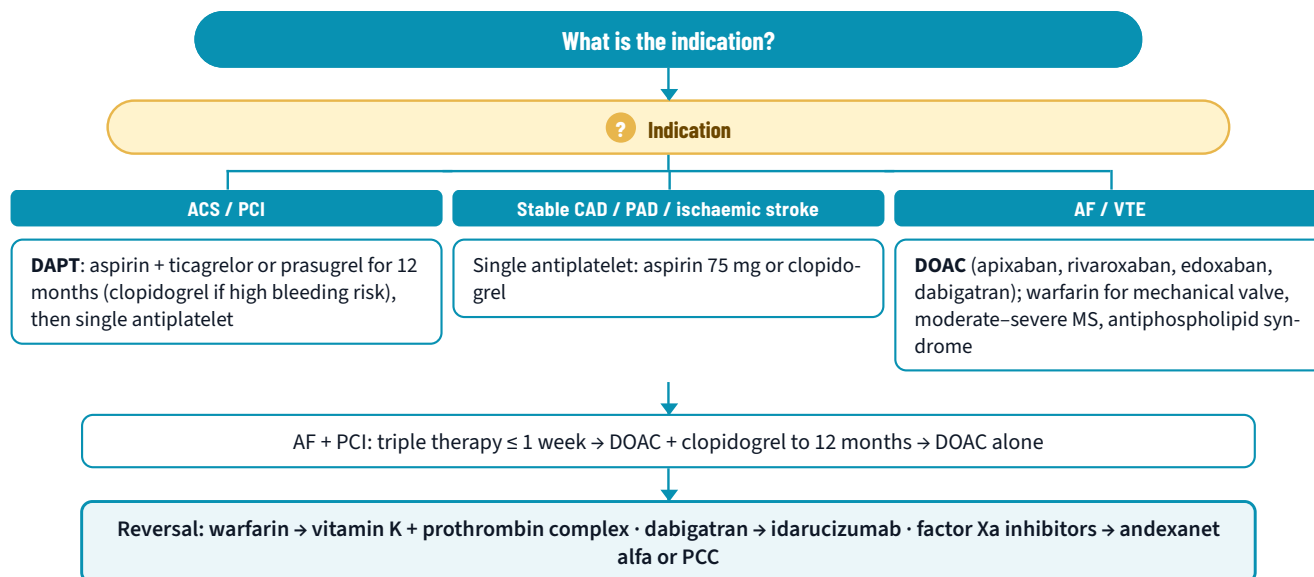
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08

Step by step

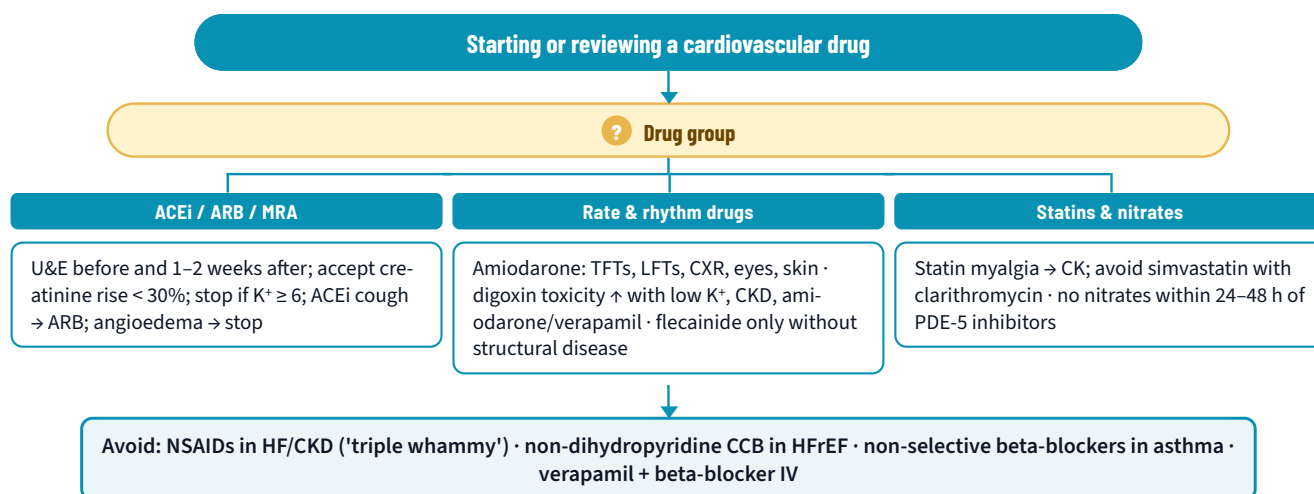
Choosing an antithrombotic

ESC 2023 ACS • ESC 2024 AF • NICE NG158



Prescribing cardiovascular drugs safely

BNF • NICE



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1 All of the following are antiplatelet drugs EXCEPT:
- A Warfarin
 - B Aspirin
 - C Clopidogrel
 - D Dipyridamole

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 2 Digoxin toxicity may be precipitated by all of the following EXCEPT:
- A Hypokalaemia
 - B Old age
 - C Renal failure
 - D Hepatic impairment
- 3 A 47-year-old man is admitted with an acute coronary syndrome. He has a history of peptic ulcer disease, and his cardiologist decides to use clopidogrel. What is the mechanism of action of clopidogrel?
- A Glycoprotein IIb/IIIa inhibitor
 - B Phosphodiesterase-5 inhibitor
 - C Non-selective phosphodiesterase inhibitor
 - D Blocks ADP binding to its platelet P2Y₁₂ receptor

Answers 1–3

LOCAL EXAMS MCQS

- 1 **A Warfarin** — Warfarin is an anticoagulant (vitamin K epoxide reductase; factors II, VII, IX, X). Aspirin, clopidogrel and dipyridamole are antiplatelets.
- 2 **D Hepatic impairment** — Digoxin is renally excreted, so hepatic impairment matters little. Renal impairment, hypokalaemia, hypomagnesaemia, hypercalcaemia and amiodarone/verapamil precipitate toxicity.
- 3 **D Blocks ADP binding to its platelet P2Y₁₂ receptor** — Clopidogrel is a prodrug (CYP2C19) that irreversibly blocks the platelet P2Y₁₂ ADP receptor. Omeprazole may reduce its activation; prefer lansoprazole.

Davidson MCQs

- 1 Loop diuretics such as furosemide and bumetanide have which of the following effects?
- A Diuresis due to inhibition of potassium and water reabsorption
 - B Diuresis due to inhibition of sodium and water reabsorption
 - C Diuresis due to inhibition of water reabsorption only
 - D Increased serum potassium levels due to enhanced distal tubule function
 - E Osmotic diuresis
- 2 In which of the following situations are beta-adrenoceptor antagonists (beta-blockers) indicated?
- A Acute left ventricular failure
 - B Cardiac failure associated with bradycardia
 - C Cardiogenic shock
 - D Chronic left ventricular systolic dysfunction
 - E High-output cardiac failure
- 3 A 71-year-old woman with hypertension presents with fatigue and rapid, irregular palpitations. She normally takes enalapril. Her pulse is irregularly irregular at 125 beats/min and her blood pressure is 128/86 mmHg; cardiovascular examination is otherwise normal. A 12-lead ECG shows atrial fibrillation with a poorly controlled ventricular rate and no other abnormality. Which of the following drugs is most suitable to control her heart rate?
- A Adenosine
 - B Amiodarone
 - C Beta-blocker
 - D Flecainide
 - E Lidocaine

- 4 Which of the following drugs is known to be effective in preventing stroke in patients with atrial fibrillation?
- A Amiodarone
 - B Apixaban
 - C Aspirin
 - D Beta-blocker
 - E Clopidogrel

- 5 Which of the following antiplatelet drugs is a phosphodiesterase inhibitor?
- A Cangrelor
 - B Clopidogrel
 - C Dipyridamole
 - D Prasugrel
 - E Ticagrelor

Answers 1-5

DAVIDSON MCQS

- 1 **B Diuresis due to inhibition of sodium and water reabsorption** — Loop diuretics block the $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ co-transporter in the thick ascending limb, the most potent diuresis; they cause hypokalaemia and alkalosis.
- 2 **D Chronic left ventricular systolic dysfunction** — Evidence-based beta-blockers cut mortality by about a third in chronic HFrEF (start low when euvolaemic). They are harmful in acute LVF and cardiogenic shock.
- 3 **C Beta-blocker** — AF rate control without heart failure: a beta-blocker or rate-limiting CCB first (target < 110/min). Her CHA₂DS₂-VASc is 3, so anticoagulate too.
- 4 **B Apixaban** — Oral anticoagulants cut AF stroke by ~two-thirds; apixaban beat warfarin with less bleeding (ARISTOTLE). Aspirin is not recommended.
- 5 **C Dipyridamole** — Dipyridamole inhibits platelet phosphodiesterase and adenosine reuptake (raises cAMP). Clopidogrel, prasugrel and ticagrelor block P2Y₁₂.

2 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 Which of the following beta-blockers has been shown to reduce mortality in heart failure with reduced ejection fraction (HFrEF)?
- A Bucindolol
 - B Nebivolol
 - C Metoprolol succinate
 - D Metoprolol tartrate
 - E All of the above
- 2 All of the following drugs used in the acute management of ST-elevation myocardial infarction are correctly matched to their mechanism of benefit EXCEPT:
- A Aspirin – reduction in thromboxane A₂
 - B Abciximab – inhibition of the glycoprotein IIb/IIIa receptor
 - C Beta-adrenergic antagonists – reduced myocardial oxygen consumption
 - D Clopidogrel – inhibition of the platelet adenosine diphosphate (ADP) receptor
 - E Nitroglycerin – reduced cardiac afterload

Answers 1-2

HARRISON MCQS

- 1 **C Metoprolol succinate** — **UPDATED** Mortality benefit in HFrEF is proven for bisoprolol, carvedilol and metoprolol succinate (MERIT-HF); short-acting metoprolol tartrate was inferior.
- 2 **E Nitroglycerin – reduced cardiac afterload** — GTN is mainly a venodilator: its benefit is preload (not afterload) reduction. Avoid in hypotension, RV infarction and after PDE-5 inhibitors.

1 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1 A 64-year-old female comes to your surgery with a past medical history of hypertension. She is on an ACE inhibitor as well as on a range of other medications. Which one of the following would be most likely to blunt the antihypertensive effects of an ACE inhibitor?
- A Chlorpromazine
 - B Allopurinol
 - C Ibuprofen
 - D Spironolactone
 - E Paracetamol

Answer 1

CRASH COURSE MCQs

- 1 **C Ibuprofen** — NSAIDs block renal prostaglandins, causing salt retention that blunts ACEi; with a diuretic they form the 'triple whammy' for AKI.

3 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 66-year-old man is diagnosed with a transient ischaemic attack and commenced on clopidogrel. He already takes a statin. After a week of treatment, he presents to his GP complaining of severe diarrhoea. His GP suspects that clopidogrel is the cause and decides to switch him to combination antiplatelet treatment. This includes modified-release dipyridamole. What is the mechanism of action of this drug?
- A Glycoprotein IIb/IIIa inhibitor
 - B Glycoprotein VI inhibitor
 - C Phosphodiesterase inhibitor
 - D Glycoprotein 1b inhibitor
 - E P2Y₁₂ inhibitor
- 2 A 60-year-old man is admitted following an acute coronary syndrome. He receives aspirin, clopidogrel, nitrates and morphine. His 6-month risk score is high and percutaneous coronary intervention is planned. He is therefore given intravenous tirofiban. What is the mechanism of action of this drug?
- A Inhibits the production of thromboxane A₂
 - B Activates antithrombin III
 - C Coronary vasodilator
 - D Glycoprotein IIb/IIIa receptor antagonist
 - E Reversible direct thrombin inhibitor
- 3 A 78-year-old man presents with a hot, red, swollen leg. On examination, the left leg is tender and hot to touch, with evidence of a recent superficial laceration he sustained whilst gardening. His blood tests show raised inflammatory markers and a negative D-dimer. He is treated for cellulitis with clarithromycin due to being penicillin-allergic. Which of his regular medications should be concurrently suspended?
- A Amitriptyline
 - B Aspirin
 - C Clopidogrel
 - D Simvastatin
 - E Tramadol

Answers 1-3

PASSMEDICINE MCQS

- 1 **C Phosphodiesterase inhibitor** — Dipyridamole is a phosphodiesterase inhibitor that also blocks adenosine reuptake; MR dipyridamole + aspirin is used after TIA/stroke if clopidogrel is not tolerated.
- 2 **D Glycoprotein IIb/IIIa receptor antagonist** — Tirofiban (like eptifibatide, abciximab) blocks platelet GP IIb/IIIa, the final common pathway of aggregation; now mainly bail-out during PCI.
- 3 **D Simvastatin** — Clarithromycin strongly inhibits CYP3A4, raising simvastatin levels and risking rhabdomyolysis; withhold the statin during the course.

1 QUESTIONS • OTHER COLLEGES & SOURCES

Other Sources MCQs

- 1 A 48-year-old woman who has been treated for essential hypertension for 3 months presents with a persistent dry cough that has not responded to linctus or antibiotics. Which antihypertensive is most likely responsible?
- A Amlodipine
 - B Lisinopril
 - C Bendroflumethiazide
 - D Furosemide
 - E Atenolol

Answer 1

OTHER SOURCES MCQS

- 1 **B Lisinopril** — ACE inhibitors cause dry cough in 10–15% (bradykinin accumulation), often weeks after starting; switch to an ARB.



09

MIXED CARDIOLOGY

Mixed Cardiology

Integrated cardiology questions across the whole system — examination, investigation and management — for final revision.

81

MCQs

0

EMQ SCENARIOS

6

SOURCES

3

REPEATED IDEAS

IN THIS CHAPTER

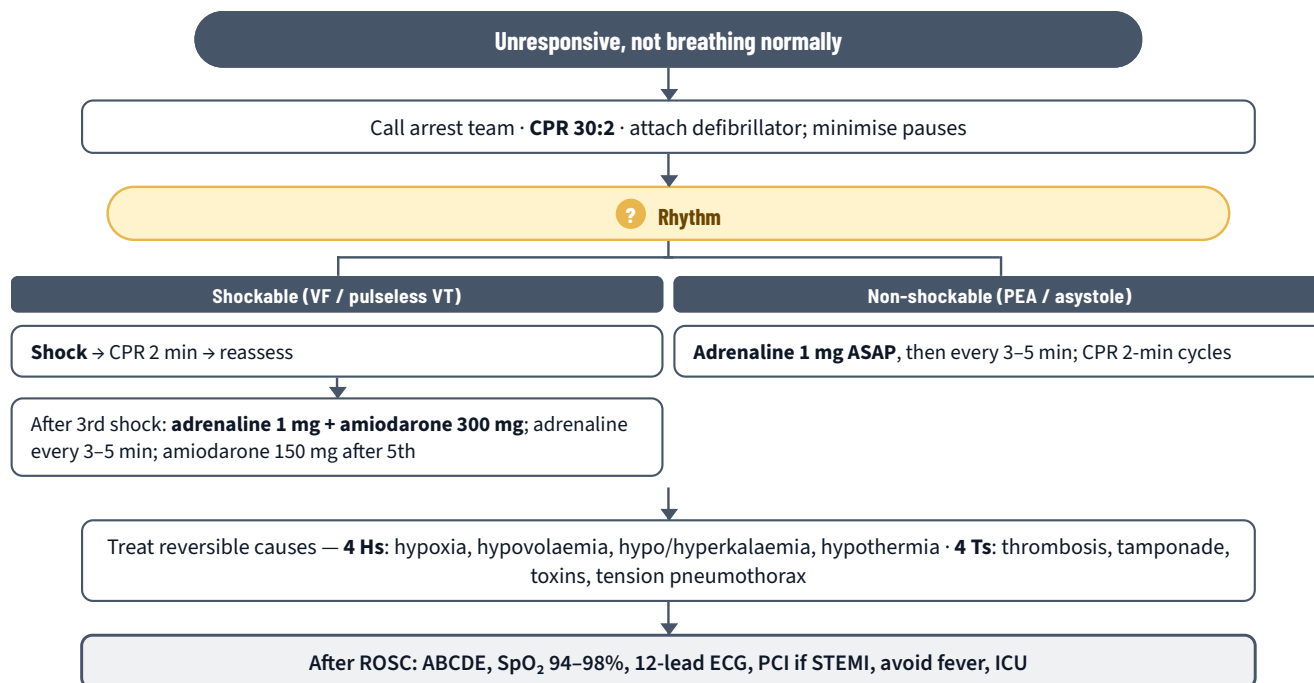
01	Management flowcharts	2 charts	page 136
02	Local Exams MCQs	1 Q	page 137
03	Davidson MCQs	4 Q	page 137
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09

Step by step

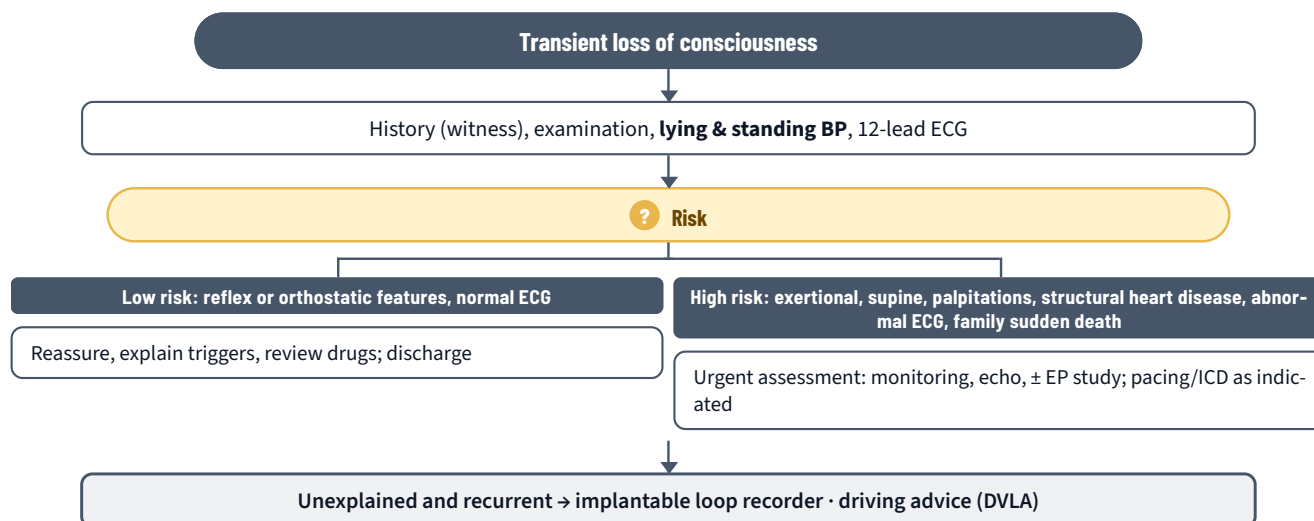
Cardiac arrest: advanced life support

RESUSCITATION COUNCIL UK 2021 • ERC 2025



Syncope

ESC 2018 SYNCOPES • NICE CG109



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1 A 24-year-old man with no previous medical problems faints during exercise. He recovers quickly with no neurological deficit, and the ECG shows sinus rhythm. Which is the most useful diagnostic test?
- A Carotid sinus massage
 - B Echocardiogram
 - C CT scan of the head
 - D Blood glucose

REPEATED 2* • ALSO ASKED IN
End-block exam 2025

Answer 1

LOCAL EXAMS MCQS

- 1 **B** Echocardiogram — Exertional syncope in a young person suggests structural disease (HCM, AS, anomalous coronaries); echocardiography is the key test.

4 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 In the management of cardiac arrest, which of the following most accurately describes basic life support (BLS)?
- A Administration of intravenous drugs and external defibrillation (the two 'D's)
 - B External cardiac massage only
 - C Support of airway, breathing and circulation (ABC)
 - D Support of airway, breathing and circulation, and assessment of disability and exposure (ABCDE)
 - E Support of airway, breathing and circulation, and assessment of disability and exposure, treatment of fibrillation (ABCDEF)
- 2 Which of the following statements is true of a pulseless electrical activity (PEA) cardiac arrest?
- A Cardiopulmonary resuscitation (CPR) should be carried out for 1 minute before the rhythm is reassessed
 - B Intravenous amiodarone will restore cardiac output
 - C It is initially managed with immediate defibrillation
 - D Reversible causes include hyperthyroidism and hypercalcaemia
 - E Reversible causes include hypothermia and hypoxia
- 3 Atrial myxoma is the most common primary cardiac tumour. Which of the following is true of atrial myxoma?
- A Atrial myxomas are usually malignant
 - B It occurs more commonly in the right atrium than in the left atrium
 - C Surgery is not indicated because atrial myxomas are benign
 - D Surgery is usually indicated to prevent embolic complications such as stroke
 - E The tumour commonly obstructs the aortic valve
- 4 A patient presents with an incidental finding of a mass in the left atrium whilst undergoing an echocardiogram for hypertension. What is the most likely cardiac tumour in this situation?
- A Angiosarcoma
 - B Atrial myxoma
 - C Fibroelastoma
 - D Fibroma
 - E Lipoma

Answers 1–4

DAVIDSON MCQS

- 1 **C** Support of airway, breathing and circulation (ABC) — **UPDATED** BLS is support of airway, breathing and circulation

without equipment: call for help, CPR 30:2 and early AED. Drugs belong to ALS.

- 2 **E** Reversible causes include hypothermia and hypoxia — PEA is non-shockable: CPR, adrenaline 1 mg ASAP then every 3–5 min, and treat the 4 Hs and 4 Ts (e.g. hypoxia, hypothermia).
- 3 **D** Surgery is usually indicated to prevent embolic complications such as stroke — Atrial myxoma (75% left atrial, from the

septum) can obstruct the mitral valve, embolise and cause fever/high ESR (IL-6); excise surgically.

- 4 **B** Atrial myxoma — The commonest primary cardiac tumour in adults is myxoma, a pedunculated LA mass on the septum; metastases are commoner overall.

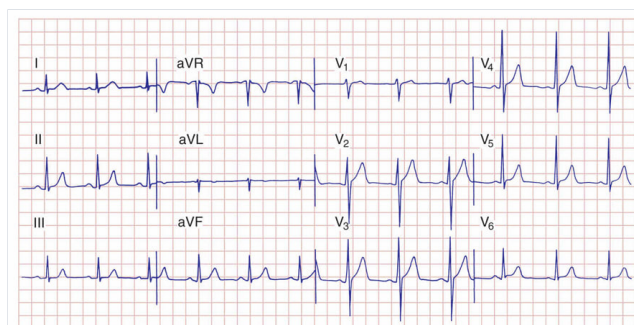
71 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

SET 1 OF 3

- 1 A country with a population and environment very similar to China is moving from the first stage of the epidemiological transition ('pestilence and famine') to the second ('receding pandemics'). Regarding the expected pattern of cardiovascular disease (CVD) in this country, which of the following is true?
- A As this nation enters the stage of receding pandemics, you would expect the majority of CVD morbidity and mortality to be due to cardiomyopathies secondary to infectious agents
 - B Each nation or geographical location progresses through the five stages of epidemiologic transition identically in regard to CVD risk
 - C In this nation, when CVD mortality peaks, you would expect stroke mortality to be greater than coronary heart disease mortality
 - D One would expect a very homogeneous epidemiologic pattern of CVD risk throughout this nation over time
 - E You anticipate that CVD mortality will inexorably climb as this nation progresses through the five stages of the epidemiologic transition
- 2 A 67-year-old man complains of exertional breathlessness, leg oedema and waking at night acutely short of breath. You assess his volume status by examining the jugular venous pulse (JVP). Which of the following statements about JVP measurement is true?
- A If done properly, the angle of inclination matters little to the measurement of JVP
 - B In normal patients, the JVP rises with inspiration due to the augmented volume loading of the right heart
 - C Measurement of the elevation of the top of the JVP and the sternal inflection point (angle of Louis) will provide a highly accurate measurement of central venous pressure
 - D The external jugular is preferred over the internal jugular vein due to its ease of visibility
 - E Venous pulsations above the clavicle in the sitting position are abnormal
- 3 Which of the following statements regarding blood pressure measurements is true?
- A Systolic pressure increases and diastolic pressure decreases when measured in more distal arteries
 - B Systolic leg blood pressures are usually as much as 20 mmHg lower than arm blood pressures
 - C The concept of 'white coat hypertension' (pressures measured in clinical settings significantly higher than elsewhere) has been shown to be a myth
 - D An inter-arm systolic difference of 25 mmHg is within normal limits
 - E Using a blood pressure cuff that is too small results in a marked underestimation of the true blood pressure
- 4 Which of the following statements regarding normal depolarization patterns of the heart is true?
- A Each normal sinus beat is initiated by spontaneous depolarization in the atrioventricular (AV) node
 - B The normal order of depolarization is as follows: sinoatrial (SA) node - atrial myocardium - AV node - His bundle - Purkinje fibers - ventricular myocardium
 - C The right bundle branch bifurcates into an anterior and posterior fascicle
 - D The SA node is unique in its ability to spontaneously depolarize, a quality known as automaticity
 - E Within the ventricular myocardium, depolarization sweeps from epicardium to endocardium
- 5 All of the following ECG waveforms are matched correctly to the cardiac cycle that they represent EXCEPT:
- A P wave – atrial repolarisation
 - B PR interval – conduction from the atria through the AV node and His–Purkinje system
 - C QRS complex – ventricular depolarisation
 - D T wave – ventricular repolarisation
 - E U wave – late ventricular repolarisation
- 6 Which of the following mean QRS vectors in the frontal electrocardiographic plane is matched appropriately to its designation?
- A -20 degrees - normal axis
 - B -35 degrees - right axis deviation
 - C -110 degrees - left axis deviation
 - D -80 degrees - extreme axis deviation
 - E All of the above are incorrect

7 What is the primary finding on the ECG shown?



- A Left ventricular hypertrophy
- B Normal ECG
- C Peaked T waves, possibly hyperkalaemia
- D Sinus bradycardia
- E ST elevation in the anterior precordial leads; suspect anterior myocardial ischaemia

8 The SA node serves as the dominant pacemaker of the heart in normal sinus rhythm. What property of SA nodal cells allows it to act as the primary pacemaker?

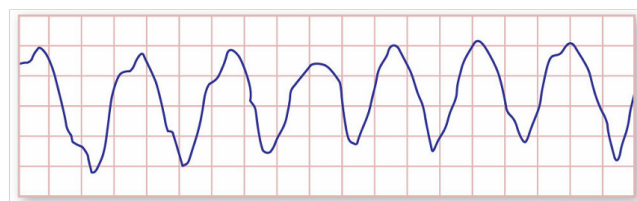
- A Location near the superior lateral area of the right atrium
- B More numerous intercalated discs of any other myocardial tissue
- C Most rapid phase 0 depolarization
- D Only cells with the ability to spontaneously depolarize
- E Spontaneous depolarization during phase 4 of the action potential at a more rapid rate than any other myocardial cells

9 A 76-year-old woman with hypertension, chronic obstructive pulmonary disease (COPD), diabetes mellitus and osteoporosis is brought to the emergency department after a fall at home, followed immediately by intense left hip pain. She was found by a neighbour after several hours and cannot remember whether she lost consciousness. She is exquisitely tender over the left hip, and the leg is shortened and externally rotated. Her mucous membranes are dry and skin turgor is reduced. Blood pressure is 170/80 mmHg and heart rate 130 beats/min. Her rhythm strip is shown (arrows mark P waves). What is the most appropriate first step for her tachycardia?



- A Adenosine 6 mg IV
- B DC cardioversion
- C Digoxin 250 micrograms IV
- D Metoprolol 5 mg IV
- E Pain control and IV fluids

10 After a snowstorm has kept people indoors for 4 days, a middle-aged woman is brought to the emergency department obtunded. Her neighbours had not seen her since the snow started, and the police found her unconscious on the floor. A radial pulse is barely palpable and blood pressure is 60 mmHg systolic. She has an arteriovenous dialysis fistula in the left arm. Her ECG is shown. What electrolyte abnormality do you expect?



- A Hypercalcaemia
- B Hyperkalaemia
- C Hypokalaemia
- D Hypomagnesaemia
- E Hypophosphatemia

REPEATED 2x · ALSO ASKED IN Harrison

11 An 87-year-old man with an orthotopic heart transplant presents with a mild resting tachycardia. Which of the following best explains his heart rate response?

- A The donor heart is hyper-innervated
- B The denervated donor heart lacks resting vagal tone and responds only to circulating catecholamines
- C Acute rejection of the graft
- D The transplant procedure permanently damages the SA node
- E None of the above

12 A 28-year-old man presents to the emergency department with breathlessness on exertion. He had an orthotopic heart transplant for non-ischaemic cardiomyopathy 5 years ago and has done well apart from one cytomegalovirus reactivation in the first year. For the past 3 months, he has become limited by breathlessness on progressively less exertion. He is adamant that he has no chest pain or pressure. He takes tacrolimus, mycophenolate mofetil and low-dose prednisolone and is fully adherent. Echocardiography shows normal LV function and wall thickness, and his resting ECG shows sinus rhythm at 80 beats/min. What is the most likely cause of his symptoms?

- A Antibody-mediated rejection
- B Cellular rejection
- C Coronary artery disease
- D Endocarditis
- E Medication side effect

- 13** A 19-year-old woman attending a travel clinic complains of a painless swelling of one leg. The skin is thickened with a woody texture, and a bedside ultrasound shows patent veins without thrombus. You suspect lymphoedema due to the most common cause of secondary lymphoedema worldwide. What cause do you suspect?
- Cancer involving the inguinal lymph nodes
 - Lymphogranuloma venereum
 - Lymphatic filariasis
 - Recurrent bacterial lymphangitis
 - Tuberculosis
- 14** A woman has chronic lymphoedema of the leg following recurrent streptococcal lymphangitis in childhood. She finds the swelling unsightly and asks about treatment. All of the following are reasonable treatments for chronic lymphoedema EXCEPT:
- Decongestive physiotherapy
 - Diuretic therapy
 - Frequent leg elevation
 - Intermittent pneumatic compression devices
 - Liposuction
- 15** A 72-year-old woman with ischaemic cardiomyopathy and a left ventricular ejection fraction of 35% has NYHA class II symptoms. She takes lisinopril 40 mg daily, carvedilol 25 mg twice daily and spironolactone 25 mg daily. Blood pressure is 102/82 mmHg and she is in sinus rhythm at 60 beats/min. Her eGFR is 55 mL/min/1.73 m² and potassium 4.8 mmol/L. Which additional medication will further reduce her mortality?
- Aliskiren
 - Digoxin
 - Ivabradine
 - Valsartan
 - Dapagliflozin
- 16** A 57-year-old man had a large anterior myocardial infarction 8 months ago, treated with primary PCI with an excellent result despite a late presentation. He is now asymptomatic on optimal medical therapy, with heart rate 65 beats/min and blood pressure 104/82 mmHg. His ECG shows sinus rhythm with right bundle branch block and a QRS duration of 135 ms. Echocardiography shows an ejection fraction of 25% with a large anterior dyskinetic area consistent with aneurysm. Which therapy is indicated to reduce his mortality?
- Implantable cardioverter defibrillator
 - Implantable cardioverter defibrillator with cardiac resynchronization capability (coronary sinus lead)
 - Ivabradine
 - Surgical ventricular remodeling
 - None of the above
- 17** A 25-year-old man had an orthotopic heart transplant for non-ischaemic cardiomyopathy 18 months ago. His resting heart rate has been consistently around 90 beats/min, and he feels well. What is the most appropriate next step regarding his resting heart rate?
- Exercise treadmill ECG testing to evaluate for chronotropic competence
 - Initiate diltiazem 120 mg twice daily
 - Initiate metoprolol tartrate 25 mg twice daily
 - Reassurance
 - Refer for electrophysiology study to evaluate for ectopic atrial tachycardia
- 18** A government committee on organ transplantation is considering a nationwide sharing strategy for donor hearts in a large country. Why is such a strategy not beneficial?
- Blood groups are distributed non-randomly across the country, preventing true nationwide sharing
 - Regions use highly variable criteria for grading severity of illness
 - The financial structure of reimbursement would not allow it
 - The tolerable cold ischaemic time of a harvested heart is too short to allow nationwide sharing
 - None of the above
- REPEATED 2x · ALSO ASKED IN**
Harrison
- 19** A 68-year-old woman with chemotherapy-related cardiomyopathy (EF 15%) has frequent admissions with acute decompensated heart failure despite optimal therapy. She is not a candidate for heart transplantation. Which of the following statements about other options to prolong her life is correct?
- Continuous infusion of milrinone will improve cardiac function and survival
 - A left ventricular assist device can improve quality of life but provides no survival benefit over optimal medical care
 - A left ventricular assist device can improve survival over the initial 2–5 years more than optimal medical therapy
 - Stem cell implantation has been shown to prolong life in patients with end-stage cardiomyopathy
 - There are no other options for prolonging life, and we should consider palliative measures in the future
- 20** Which of the following statements regarding the epidemiology of congenital heart disease (CHD) in high-income countries is true?
- CHD is extremely rare, complicating fewer than 1 in 1000 live births
 - With advances in surgery and pre- and postnatal care, survival of neonates with CHD now approaches 90%
 - Because fewer women with CHD become pregnant, the incidence of CHD in neonates is declining
 - The population of adults with CHD is declining because of improved prenatal screening
 - Women with CHD have no increased risk of complications during pregnancy compared with the general population

- 21** A 62-year-old man is being assessed for mitral valve surgery for severe mitral regurgitation. Transoesophageal echocardiography shows a small jet of right-to-left flow across the atrial septum, occurring when a small flap of tissue swings open (less than 1 mm). Which of the following explains this finding?
- Ostium primum atrial septal defect
 - Ostium secundum atrial septal defect
 - Partial anomalous pulmonary venous return
 - Patent foramen ovale
 - Sinus venosus atrial septal defect
- 22** A 21-year-old man has a newly diagnosed ventricular septal defect (VSD). Echocardiography confirms a 5-mm VSD in the muscular interventricular septum. Right heart catheterisation shows a mean pulmonary artery pressure of 20 mmHg, pulmonary vascular resistance of 2 Wood units and systemic vascular resistance of 6 Wood units, with a Qp:Qs of 1.25. What do you recommend?
- Cardiopulmonary exercise testing to define peak oxygen consumption
 - Closure with a percutaneously deployed septal occluder device
 - Consideration of heart and lung transplant
 - No intervention now; annual cardiology assessments and reimaging if any new symptoms emerge
 - Surgical correction via open heart surgery and patch repair
- 23** A 52-year-old patient presents with chest pain and raised cardiac enzymes. A superficial continuous murmur, not previously documented, is heard at the mid-sternal level. Echocardiography shows normal valves. Right heart catheterisation gives these oxygen saturations: superior vena cava 58%, right atrium 60%, mid-coronary sinus 91%, right ventricle 70%, pulmonary artery 70%. What abnormality accounts for these findings?
- Anomalous left coronary artery off the main pulmonary artery
 - Atrial septal defect
 - Coronary arteriovenous fistula to the coronary sinus
 - Patent ductus arteriosus
 - Ventricular septal defect
- 24** A 40-year-old man has a congenital bicuspid aortic valve. Which other structure is important to follow on his regular echocardiograms?
- Aortic root size
 - Left atrial size
 - Pulmonary artery pressures
 - Pulmonic valve function
 - Tricuspid valve regurgitation
- 25** All of the following are classic defining features of tetralogy of Fallot EXCEPT:
- Obstruction to RV outflow
 - Overriding aorta
 - RV hypertrophy
 - Tricuspid atresia
 - Ventricular septal defect

Answers 1-25

- 1** **C** In this nation, when CVD mortality peaks, you would expect stroke mortality to be greater than coronary heart disease mortality — In China-like populations, as CVD peaks, stroke mortality exceeds coronary mortality (high hypertension and salt intake).
- 2** **E** Venous pulsations above the clavicle in the sitting position are abnormal — The RA lies ~5 cm below the sternal angle, so venous pulsation above the clavicle when sitting upright means a raised JVP.
- 3** **A** Systolic pressure increases and diastolic pressure decreases when measured in more distal arteries — **UPDATED** Pulse amplification makes systolic pressure higher and diastolic lower in distal arteries (leg SBP > arm, basis of the ABPI).
- 4** **B** The normal order of depolarization is as follows: sinoatrial (SA) node - atrial myocardium - AV node - His bundle - Purkinje fibers - ventricular myocardium — Normal conduction: SA node → atria → AV node → His bundle → bundle branches → Purkinje fibres → ventricles (endocardium to epicardium).
- 5** **A** P wave – atrial repolarisation — **UPDATED** The P wave is atrial depolarisation; atrial repolarisation is hidden in the PR segment/QRS. T = ventricular repolarisation.
- 6** **A** -20 degrees - normal axis — Normal QRS axis is -30° to +90°, so -20° is normal; -30° to -90° is left axis deviation.
- 7** **B** Normal ECG — Sinus rhythm with normal axis, intervals, R-wave progression and no ST shift or tall T waves: a normal ECG.
- 8** **E** Spontaneous depolarization during phase 4 of the action potential at a more rapid rate than any other myocardial cells — SA node cells depolarise spontaneously in phase 4 (funny current) faster than any other tissue, so they reach threshold first and dominate.
- 9** **E** Pain control and IV fluids — Multifocal/sinus tachycardia driven by pain, dehydration and hypoxia after a hip fracture: treat the cause (analgesia, IV fluids, oxygen).
- 10** **B** Hyperkalaemia — Missed dialysis with a sine-wave ECG is pre-terminal hyperkalaemia: IV calcium, insulin-glucose and salbutamol, then urgent dialysis.
- 11** **B** The denervated donor heart lacks resting vagal tone and responds only to circulating catecholamines — **UPDATED** A denervated transplanted heart lacks vagal tone, so it runs at its intrinsic rate (~90–110/min) and responds only to circulating catecholamines.

- 12** **C** Coronary artery disease — Late after transplant, breathlessness without angina suggests cardiac allograft vasculopathy (denervated heart); diagnose by angiography.
- 13** **C** Lymphatic filariasis — Lymphatic filariasis (*Wuchereria bancrofti*) is the commonest cause of secondary lymphoedema worldwide; cancer treatment in high-income countries.
- 14** **B** Diuretic therapy — Lymphoedema is protein-rich fluid, so diuretics do not help. Use decongestive therapy, compression, skin care and exercise.
- 15** **E** Dapagliflozin — **UPDATED** HFREF on ACEi, beta-blocker and MRA: add an SGLT2 inhibitor (dapagliflozin), which cuts CV death and admissions with or without diabetes.
- 16** **A** Implantable cardioverter defibrillator — EF $\leq 30\%$ > 40 days after MI: a primary-prevention ICD reduces mortality (MADIT-II). CRT needs LBBB with QRS ≥ 150 ms.
- 17** **D** Reassurance — A resting rate of ~ 90 /min in a well heart-transplant recipient is expected (denervation); reassure.
- 18** **D** The tolerable cold ischaemic time of a harvested heart is too short to allow nationwide sharing — A donor heart tolerates only about 4 h of cold ischaemia, so it must be allocated within a limited radius of the donor.
- 19** **C** A left ventricular assist device can improve survival over the initial 2–5 years more than optimal medical therapy — In transplant-ineligible stage D heart failure, a durable LVAD (destination therapy) improves survival; inotropes do not.
- 20** **B** With advances in surgery and pre- and postnatal care, survival of neonates with CHD now approaches 90% — **UPDATED** CHD affects $\sim 1\%$ of births and $\sim 90\%$ of neonates now survive, so adults with CHD now outnumber children.
- 21** **D** Patent foramen ovale — A flap that opens intermittently between atria is a PFO ($\sim 25\%$ of adults), allowing paradoxical embolism; no treatment if incidental.
- 22** **D** No intervention now; annual cardiology assessments and reimaging if any new symptoms emerge — A small restrictive VSD with Qp:Qs < 1.5 and normal pulmonary pressures needs no closure; review periodically and maintain dental hygiene.
- 23** **C** Coronary arteriovenous fistula to the coronary sinus — **UPDATED** A coronary sinus saturation of 91% (normally ~ 30 – 40%) means arterial blood is entering it: a coronary AV fistula, causing a continuous murmur and steal.
- 24** **A** Aortic root size — Bicuspid aortic valve carries an aortopathy, so measure the aortic root/ascending aorta on every echo; repair at 55 mm (50 mm with risk factors).
- 25** **D** Tricuspid atresia — Tetralogy of Fallot: VSD, overriding aorta, RVOT obstruction and RV hypertrophy. Tricuspid atresia is a separate cyanotic lesion.

SET 2 OF 3

- 26** A 78-year-old man with a history of tobacco abuse, obesity, and diabetes mellitus is evaluated for dyspnoea on exertion. Echocardiography confirms severe aortic stenosis. Which of the following most likely contributed to the development of his lesion?
- Congenital bicuspid aortic valve
 - Diabetes mellitus
 - Occult rheumatic heart disease
 - Underlying connective tissue disease
 - None of the above
- 27** A 63-year-old man presents with new exertional syncope and is found to have severe aortic stenosis. What is the average life expectancy of untreated patients with this presentation?
- 5 years
 - 4 years
 - 3 years
 - 2 years
 - 1 year
- 28** An 82-year-old man with severe calcific aortic stenosis, who previously deferred surgery, now presents with hypotension, tachycardia, and signs of multi-organ dysfunction. Which of the following would be a reasonable therapeutic option to improve his short-term perfusion to permit aortic valve replacement?
- Atorvastatin 80 mg
 - Digoxin 250 μ g IV once and 125 μ g orally daily thereafter
 - Metoprolol 5 mg IV serial dosing targeting a resting heart rate of 60–70 bpm
 - Percutaneous aortic balloon valvuloplasty
 - Phenylephrine continuous IV infusion titrated to a mean arterial pressure of > 65
- 29** An 85-year-old man presents with increasing breathlessness and leg oedema. Echocardiography shows an ejection fraction of 25% with global hypokinesis, a calculated aortic valve area of 0.8 cm^2 and a mean gradient of 25 mmHg. What is the most appropriate next step to determine whether he would benefit from aortic valve replacement?
- Cardiac magnetic resonance imaging (MRI) to evaluate ventricular scar and aortic valve morphology
 - Cardiac positron emission tomography (PET) to determine ventricular viability
 - Coronary angiography to evaluate for the presence of obstructive coronary disease
 - Dobutamine stress echocardiography
 - Right heart catheterization to document cardiac output and filling pressures

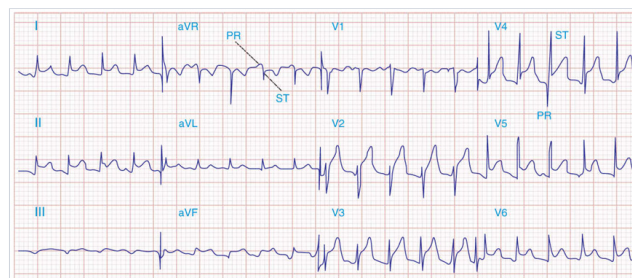
- 30** Which of the following parameters is typically reduced in chronic severe aortic regurgitation?
- A Diastolic blood pressure
 - B Left ventricular afterload
 - C Left ventricular diameter
 - D Left ventricular preload
 - E Total left ventricular stroke volume
- 31** A 21-year-old woman with a connective tissue disease presents with acute breathlessness and pulmonary oedema. Echocardiography shows severe aortic regurgitation with an avulsed right coronary cusp. While awaiting emergency surgery, which intervention is most likely to help maintain her end-organ perfusion?
- A Esmolol continuous IV infusion titrated to a heart rate of 60–70 bpm
 - B Intra-aortic balloon pump
 - C IV nitroprusside
 - D IV norepinephrine
 - E IV vasopressin
- 32** What is the most common cause of obstruction to left ventricular inflow worldwide?
- A Congenital mitral valve disease
 - B Cor triatriatum
 - C Infective endocarditis with large mitral valve vegetations
 - D Mitral annular calcification
 - E Rheumatic mitral disease
- 33** In cases of severe mitral stenosis, which of the following parameters is typically increased?
- A Cardiac output
 - B Left atrial pressure
 - C Left ventricular diameter
 - D Left ventricular end-diastolic pressure
 - E Pulmonary vascular compliance
- 34** In a patient with mitral stenosis, all of the following will raise left atrial pressure and may worsen pulmonary congestion EXCEPT:
- A Anaemia
 - B Isoprenaline
 - C Metoprolol
 - D Pregnancy
 - E Running on a treadmill
- 35** An 84-year-old woman with severe mitral stenosis develops new-onset atrial fibrillation. Which of the following interventions will NOT help lower her left atrial pressure?
- A IV furosemide
 - B Percutaneous balloon mitral valvotomy
 - C Synchronized DC cardioversion
 - D Transvenous pacemaker placement and pacing at a heart rate of 90 bpm
 - E All of the above will help lower left atrial pressure
- 36** A 34-year-old man with rheumatic mitral stenosis is asymptomatic, with a mitral valve area of 1.7 cm². On exercise his pulmonary artery systolic pressure reaches 40 mmHg. What management do you recommend?
- A Metoprolol 25 mg orally twice daily
 - B Percutaneous mitral balloon valvotomy
 - C Periodic cardiology assessments and echocardiographic monitoring
 - D Sildenafil 20 mg twice daily
 - E Surgical mitral valve replacement
- 37** A 65-year-old man with a 15-year history of non-ischaemic dilated cardiomyopathy (LVEF 30%) has severe functional (secondary) mitral regurgitation. He remains in NYHA class III despite optimal guideline-directed medical therapy and cardiac resynchronisation. He asks whether fixing his valve would improve his survival. What is the most accurate response?
- A Valve repair is only needed if you develop pulmonary hypertension or atrial fibrillation
 - B Neither surgical nor percutaneous valve repair has ever been shown to improve survival in patients like you
 - C Surgery should be considered only if repair is possible, because replacement would not improve survival
 - D Surgical repair has not been shown to improve survival, but in selected patients like you transcatheter edge-to-edge repair has been shown to reduce mortality
 - E Yes, your valve should have been fixed years ago
- 38** During right and left heart catheterisation of a patient with stenosis of two valves, the following pressures are recorded: right atrium (mean) 15 mmHg; right ventricle 25/6 mmHg; pulmonary artery 25/12 mmHg; pulmonary artery wedge pressure 12 mmHg; left ventricle 105/6 mmHg; aorta 105/75 mmHg. Which two valves are stenosed?
- A Aortic and mitral
 - B Aortic and tricuspid
 - C Mitral and pulmonic
 - D Mitral and tricuspid
 - E Pulmonic and tricuspid
- 39** You are evaluating a 50-year-old woman with idiopathic pulmonary arterial hypertension. She has severe tricuspid regurgitation, a dilated, hypokinetic right ventricle, and signs of right heart failure. What is the best treatment for her severe tricuspid regurgitation?
- A Diuretics and salt restriction accompanied by medical therapy targeting her elevated pulmonary artery pressures
 - B Percutaneous balloon valvotomy
 - C Percutaneous tricuspid valve repair
 - D Surgical mitral valve replacement
 - E Surgical tricuspid valve repair

- 40** A 21-year-old woman who has never seen a doctor is found to have an ejection systolic murmur in the left second intercostal space, preceded by an ejection click. Echocardiography confirms pulmonary stenosis with a peak gradient of 70 mmHg and a doming pulmonary valve without regurgitation. What is the best treatment?
- Diuretics and salt restriction
 - No treatment needed
 - Percutaneous balloon valvotomy
 - Percutaneous pulmonary valve replacement
 - Surgical pulmonary valve replacement
- 41** A 77-year-old patient with severe aortic stenosis develops signs of acute pulmonary oedema and is found to have a new soft, short apical systolic murmur consistent with acute severe mitral regurgitation due to ruptured chordae. Which parameter will likely increase as a result of the new severe mitral regurgitation?
- Aortic valve gradient
 - Calculated aortic valve area
 - Effective stroke volume
 - Ejection fraction
 - Left ventricular afterload
- 42** A man with rheumatic heart disease and moderate mitral stenosis presents with atrial fibrillation at a ventricular rate of 60 beats/min. Which of the following is the appropriate thromboembolic prophylaxis?
- Apixaban
 - Dabigatran
 - Rivaroxaban
 - Warfarin
 - More information is needed prior to initiating thromboembolic prophylaxis
- 43** All of the following are risk factors for the development of peripartum cardiomyopathy EXCEPT:
- Advanced maternal age
 - Malnutrition
 - Primiparity
 - Twin pregnancy
 - Use of tocolytics
- 44** A 25-year-old man with a diagnosis of heart failure has used a wheelchair for many years and has severe scoliosis. His ECG shows tall R waves in V1 and V2 with deep Q waves in V5 and V6, and echocardiography shows severe global left ventricular dysfunction. What is the most likely diagnosis?
- Amyotrophic lateral sclerosis
 - Atrial septal defect
 - Chronic thromboembolic disease
 - Duchenne muscular dystrophy
 - Ischaemic cardiomyopathy
- 45** Which of the following is the most common inheritance pattern for familial cardiomyopathies?
- Autosomal dominant driven by exon duplication
 - Autosomal dominant driven by missense mutations
 - Autosomal recessive driven by exon deletion
 - Autosomal recessive driven by exon duplication
 - X-linked driven by exon deletion
- 46** A 45-year-old man with obesity presents with several months of breathlessness, fatigue and a nocturnal cough. Echocardiography shows a dilated left ventricle with a globally reduced ejection fraction of 25%. Which blood test is routinely recommended in all patients with newly diagnosed heart failure to identify a reversible cause?
- Serum angiotensin-converting enzyme
 - Anti-nuclear antibodies
 - Erythrocyte sedimentation rate
 - Coxsackie virus serology
 - Thyroid-stimulating hormone
- 47** A 22-year-old student presents with lethargy and fatigue 3 days after a viral upper respiratory illness. He is obtunded, tachycardic and hypotensive, with cool peripheries and a raised jugular venous pressure. Echocardiography shows an ejection fraction of 30% with mild mitral regurgitation, and endomyocardial biopsy shows lymphocytic myocarditis. Which of the following statements about his prognosis and treatment is true?
- His chance of survival is <10% without cardiac transplantation. Emergent transplant listing is warranted
 - His chance of survival is >50%, with many similar patients having a full recovery in left ventricular function over the ensuing weeks to months. Aggressive pharmacologic and mechanical haemodynamic support is warranted
 - Immunosuppression with high-dose systemic steroids will increase his chance of survival
 - The presence and titer of anti-heart antibodies can help provide prognostic information for this patient
- 48** A 42-year-old woman with autoimmune (Hashimoto) thyroiditis is brought in after collapsing at home. She reports several days of worsening lethargy and chest discomfort. Echocardiography shows a left ventricular ejection fraction of 15%, and endomyocardial biopsy shows diffuse inflammatory infiltrates with multinucleated giant cells and myocyte necrosis. Which of the following statements about her diagnosis is true?
- Most patients with this form of cardiomyopathy recover with supportive care
 - Steroids alone are highly effective in treating this form of cardiomyopathy
 - The course is often rapidly progressive, with deterioration requiring urgent transplantation
 - Ventricular tachyarrhythmias are rare in this disease
 - None of the above is true

49 All of the following are associated with an elevated risk for development of cardiomyopathy during anthracycline chemotherapy EXCEPT:

- A Concomitant chest radiation
- B High total anthracycline dose
- C Preexisting cardiac disease
- D Trastuzumab administration
- E All of the above are associated with elevated risk

50 A 32-year-old man presents with chest pain after a week of coryza, cough, fever and muscle aches. The pain is constant, radiates to his left shoulder, and is worse lying flat and on deep inspiration. There is a scratchy extracardiac sound with three components per heartbeat. His ECG is shown, and troponin I is undetectable. What is the most appropriate treatment?



- A Aspirin 75 mg daily, metoprolol 25 mg twice daily and atorvastatin 80 mg daily
- B High-dose aspirin (750 mg–1 g every 8 hours) plus colchicine, with proton pump inhibitor cover
- C Heparin, aspirin and clopidogrel with immediate coronary angiography
- D Prednisolone 40 mg daily for 2 weeks, then tapered over 2 months
- E Thrombolysis with alteplase

Answers 26–50

- 26** **B** Diabetes mellitus — Calcific AS of a trileaflet valve at 78 shares atherosclerotic risk factors: age, diabetes, hypertension, smoking, lipids, CKD.
- 27** **C** 3 years — Untreated severe AS: average survival ~5 years after angina, 3 after syncope and 2 after heart failure; any symptom means valve replacement.
- 28** **D** Percutaneous aortic balloon valvuloplasty — Cardiogenic shock from critical AS: balloon aortic valvuloplasty temporarily relieves obstruction as a bridge to AVR/TAVI (restenosis within months).
- 29** **D** Dobutamine stress echocardiography — Low-flow low-gradient AS with poor EF: dobutamine stress echo separates true severe AS (gradient rises, area stays $\leq 1 \text{ cm}^2$) from pseudo-severe and shows contractile reserve.
- 30** **A** Diastolic blood pressure — In chronic severe AR diastolic pressure falls and pulse pressure widens; LV end-diastolic volume, stroke volume and afterload all increase.
- 31** **C** IV nitroprusside — Acute severe AR: nitroprusside ($\pm$ dobutamine) increases forward flow as a bridge to surgery. IABP is contraindicated; beta-blockers lengthen diastole.
- 32** **E** Rheumatic mitral disease — Rheumatic heart disease is by far the commonest cause of mitral stenosis/LV inflow obstruction worldwide; annular calcification in the elderly.
- 33** **B** Left atrial pressure — In MS left atrial pressure rises to drive flow across the valve; LV end-diastolic pressure is normal or low and output falls.
- 34** **C** Metoprolol — Anything raising heart rate or output raises LA pressure in MS. Beta-blockers (metoprolol) lengthen diastole and lower LA pressure.
- 35** **D** Transvenous pacemaker placement and pacing at a heart rate of 90 bpm — In MS, LA pressure falls with diuretics, valvotomy, cardioversion or rate control. Pacing at 90/min shortens diastole and raises it.
- 36** **C** Periodic cardiology assessments and echocardiographic monitoring — MVA 1.7 cm^2 is moderate MS; asymptomatic with exercise PASP $< 60 \text{ mmHg}$, so review and serial echo every 1–2 years.
- 37** **D** Surgical repair has not been shown to improve survival, but in selected patients like you transcatheter edge-to-edge repair has been shown to reduce mortality — **UPDATED** In secondary (functional) MR surgery does not improve survival, but transcatheter edge-to-edge repair reduced mortality in selected patients (COAPT).
- 38** **D** Mitral and tricuspid — **UPDATED** Diastolic gradients $\text{RA} > \text{RVDP}$ and $\text{wedge} > \text{LVEDP}$ mean tricuspid and mitral stenosis; no systolic gradients exclude pulmonary and aortic stenosis.
- 39** **A** Diuretics and salt restriction accompanied by medical therapy targeting her elevated pulmonary artery pressures — Functional TR from pulmonary hypertension is treated by diuretics, salt restriction and pulmonary vasodilator therapy; valve surgery is high-risk.
- 40** **C** Percutaneous balloon valvotomy — **UPDATED** Doming pulmonary valve with peak gradient $> 64 \text{ mmHg}$ is congenital pulmonary stenosis: percutaneous balloon valvotomy.

- 41** **D** Ejection fraction — New severe MR gives the LV a low-pressure outlet, so EF rises while forward flow and the aortic gradient fall.
- 42** **D** Warfarin — **UPDATED** AF with moderate–severe rheumatic MS needs warfarin (INR 2–3) regardless of CHA2DS2-VASc; rivaroxaban was inferior (INVICTUS).
- 43** **C** Primiparity — Peripartum cardiomyopathy risk factors: older age, multiparity, twins, pre-eclampsia/hypertension, African ancestry. Primiparity is not one.
- 44** **D** Duchenne muscular dystrophy — A wheelchair-bound young man with scoliosis and DCM has Duchenne muscular dystrophy (X-linked dystrophin); tall R in V1–V2 is typical.
- 45** **B** Autosomal dominant driven by missense mutations — Most familial cardiomyopathies are autosomal dominant missense mutations (MYH7, MYBPC3, TTN, LMNA); offer cascade screening.
- 46** **E** Thyroid-stimulating hormone — **UPDATED** TSH is part of routine tests in new heart failure, as hypo- and hyperthyroidism are reversible causes.
- 47** **B** His chance of survival is >50%, with many similar patients having a full recovery in left ventricular function over the ensuing weeks to months. Aggressive pharmacologic and mechanical haemodynamic support is warranted — Fulminant lymphocytic myocarditis has a good outcome if supported (inotropes, ECMO/VAD); LV function often recovers fully.
- 48** **C** The course is often rapidly progressive, with deterioration requiring urgent transplantation — Giant cell myocarditis (with autoimmune disease) progresses rapidly with heart block and VT; immunosuppression helps but many need urgent transplant.
- 49** **E** All of the above are associated with elevated risk — Anthracycline cardiotoxicity rises with cumulative dose (> 400–450 mg/m²), chest radiotherapy, heart disease, age extremes and trastuzumab.
- 50** **B** High-dose aspirin (750 mg–1 g every 8 hours) plus colchicine, with proton pump inhibitor cover — **UPDATED** Viral pericarditis (rub, diffuse concave ST elevation, PR depression, normal troponin): high-dose aspirin or NSAID with PPI plus colchicine.

SET 3 OF 3

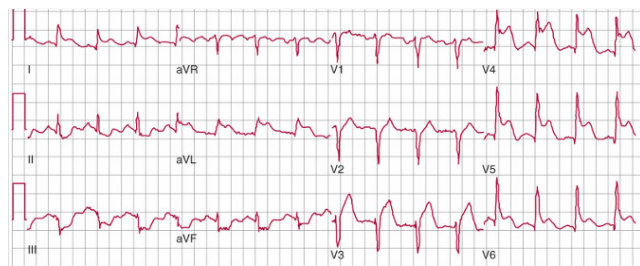
- 51** In a patient with cardiac tamponade, which of the following is likely to be increased during inspiration compared to expiration?
- A Left ventricular end-diastolic diameter
 - B Left ventricular stroke volume
 - C Mitral inflow velocity
 - D Systolic blood pressure
 - E Tricuspid inflow velocity
- 52** All of the following ECG characteristics aid in differentiating acute pericarditis from acute myocardial infarction EXCEPT:
- A Absence of the development of Q waves
 - B Concave shape to ST elevations
 - C PR depression
 - D ST elevation in V2
 - E T-wave inversions after return of the ST segments to baseline
- 53** A 35-year-old woman with a history of Hodgkin disease treated with chemotherapy and mediastinal irradiation presents with signs of constrictive pericarditis. Echocardiogram shows pericardial thickening and an abrupt cessation of ventricular filling in early diastole. What is the best approach for treatment?
- A Aggressive diuresis only
 - B Cardiac transplantation
 - C Mitral valve replacement
 - D Pericardial resection
 - E Pericardiocentesis
- 54** A 45-year-old man with a 3-month history of fevers, symmetrical arthralgia and fatigue develops sudden left hand weakness that resolves within 1 hour. Echocardiography shows a solitary 2-cm pedunculated mass in the left atrium arising from the interatrial septum near the fossa ovalis. What is the most appropriate next step?
- A Cardiothoracic surgical removal of the mass
 - B Catheter-based biopsy of the mass
 - C Repeat blood cultures and evaluation for HACEK organisms
 - D Serologies for ANAs, anti-DNA, and anticardiolipin antibodies
 - E Whole-body PET scan to evaluate for malignancy
- 55** Which of the following malignancies carries the highest relative risk of metastasis to the heart?
- A Glioblastoma
 - B Hepatocellular carcinoma
 - C Malignant melanoma
 - D Pancreatic adenocarcinoma
 - E Small-cell lung cancer
- 56** At a children's baseball game, a 12-year-old catcher is struck in the chest by a foul ball and immediately collapses in cardiac arrest. Which intervention is most critical for his survival?
- A Emergent transfer for percutaneous coronary intervention
 - B Emergent transfer for repair of an acute aortic dissection
 - C No intervention can impact survival
 - D Prompt cardiac surgery to repair an avulsed aortic valve
 - E Prompt defibrillation

- 57** A 47-year-old woman with a BMI of 37 kg/m² and type 2 diabetes asks about her future risk of heart attack and stroke. What is the most common cause of death in adults with type 2 diabetes mellitus?
- Coronary artery disease
 - Infection
 - Neuropathy
 - Renal failure
 - Stroke
- 58** A 49-year-old man with a history of heavy alcohol use presents with high-output heart failure. After treatment, his symptoms and left ventricular size improve rapidly. Which treatment most likely produced this improvement?
- Folate
 - Penicillamine
 - Thiamine
 - Thyroxine
 - Vitamin B12
- 59** A 65-year-old woman presents with fatigue, constipation, cold intolerance, brittle hair and leg swelling. Her TSH is 79.4 mU/L. Which cardiovascular measurement is NOT expected to decrease?
- Cardiac output
 - Heart rate
 - Pulse pressure
 - QT interval
 - Systolic blood pressure
- 60** In the current model describing the steps of atherosclerosis initiation and evolution, which of the following constitutes the first step?
- Arterial endothelial cells overexpress adhesion receptors for leukocytes
 - Lipoprotein particles undergo oxidative modifications
 - Low-density lipoproteins accumulate within the arterial intima
 - Macrophages and other leukocytes are recruited to the arterial intima
 - Macrophages become lipid-laden foam cells via endocytosis
- 61** Which of the following statements regarding the pathophysiologic consequences of an atheroma is true?
- An atheroma resulting in total vascular occlusion invariably causes infarction
 - An enlarging atheroma does not generally encroach on the arterial lumen until the plaque exceeds 40% of the internal elastic lamina
 - Most atheromas will eventually produce symptoms in the patient
 - Upon initial formation, the atheromatous plaque usually grows inward toward the vessel lumen
 - Vessels affected by atherogenesis tend to contract and get smaller in diameter
- 62** Under the 2013 ACC/AHA cholesterol guideline (whose statin benefit groups were retained in 2018), all of the following are statin benefit groups EXCEPT:
- All individuals with clinical atherosclerotic cardiovascular disease (ASCVD)
 - Individuals with a raised high-sensitivity C-reactive protein (CRP) regardless of LDL cholesterol or ASCVD risk
 - Individuals with LDL cholesterol of 4.9 mmol/L (190 mg/dL) or more without a secondary cause
 - Individuals aged 40–75 years without ASCVD or diabetes, with LDL 1.8–4.9 mmol/L (70–189 mg/dL) and a 10-year ASCVD risk of 7.5% or more
 - Individuals aged 40–75 years with diabetes but no ASCVD, with LDL 1.8–4.9 mmol/L (70–189 mg/dL)
- 63** A 50-year-old man with obesity attends his annual check-up. He has no known vascular disease and a 10-year ASCVD risk of 6%. His lipid profile shows total cholesterol 5.7 mmol/L (220 mg/dL), triglycerides 3.2 mmol/L (283 mg/dL), HDL cholesterol 0.75 mmol/L (29 mg/dL) and LDL cholesterol 3.4 mmol/L (132 mg/dL). He is concerned about his low HDL. Which intervention aimed at his low HDL has been shown to lower his cardiovascular risk?
- A peroxisome proliferator-activated receptor- δ agonist
 - A cholesteryl ester transfer protein inhibitor
 - Fenofibrate
 - Nicotinic acid
 - Weight loss and regular physical activity
- 64** All of the following are major determinants of myocardial oxygen demand EXCEPT:
- Heart rate
 - Heart rhythm
 - Myocardial contractility
 - Ventricular wall stress
 - All of the above are major determinants of myocardial oxygen demand
- 65** During an exercise stress test, how does the normal heart meet its increased oxygen demand?
- Increased myocardial oxygen extraction during near-constant coronary blood flow
 - Increased oxygen extraction from the ventricular chambers
 - Relative expansion of the diastolic time period allowing increased coronary flow
 - Vasodilation of epicardial coronary arteries leading to decreased resistance and increased flow
 - Vasodilation of intramyocardial coronary vessels leading to increased coronary flow
- 66** A man develops angina during an exercise stress test. Which of the following biochemical changes would NOT be expected in his ischaemic myocardium?
- An increase in cytosolic calcium
 - An increase in intracellular pH
 - A shift from aerobic to anaerobic metabolism
 - Depletion of myocardial stores of high-energy phosphates
 - Leakage of potassium from myocytes

67 Which of the following is an unusual site for radiation of anginal pain?

- A Back
- B Interscapular region
- C Root of the neck
- D Teeth
- E Trapezius muscle

68 A 62-year-old man with an acute anterior ST-elevation myocardial infarction (ECG shown) develops cardiogenic shock while being prepared for primary PCI: BP 78/50 mmHg, cool peripheries, clear lung fields and a lactate of 4.5 mmol/L. Which agent is most appropriate first to support his blood pressure?



- A Dopamine 5 µg/kg/min IV infusion
- B Noradrenaline (norepinephrine) IV infusion
- C IV glyceryl trinitrate 10 µg/min
- D IV vasopressin 0.05 units/min
- E IV furosemide 80 mg

69 A 72-year-old man with known ischaemic cardiomyopathy (EF 20%) presents in cardiogenic shock with pulmonary oedema: BP 82/50 mmHg, cool extremities, SpO₂ 84% despite high-flow oxygen and a respiratory rate of 34/min. All of the following are appropriate EXCEPT:

- A Dobutamine 5 µg/kg/min IV
- B Furosemide 60 mg IV once perfusion pressure is supported
- C Mechanical ventilation
- D Morphine 2 mg IV
- E Noradrenaline (norepinephrine) 5 µg/min IV

70 A 52-year-old man develops acute pulmonary oedema and hypotension 4 days after an inferior myocardial infarction, with a new apical systolic murmur; acute severe mitral regurgitation from papillary muscle rupture is suspected. What would a pulmonary artery catheter most likely show?

- A Low RA pressure, normal wedge, high CO
- B Normal RA pressure, low wedge, high SVR
- C Elevated wedge with prominent v waves
- D Equalization of pressures in all chambers
- E Low CO, low wedge, high RA pressure

71 All of the following statements about sudden cardiac death (SCD) are true EXCEPT:

- A Coronary artery disease is the commonest underlying substrate in adults
- B In about two-thirds of cases SCD is the first manifestation of heart disease or occurs in patients previously thought to be at low risk
- C Asystole and pulseless electrical activity are commonly encountered initial rhythms in SCD arrests
- D A parental history of SCD increases the risk in offspring
- E Most victims have a previously documented left ventricular ejection fraction of 35% or less

Answers 51-71

HARRISON MCQs

51 **E** Tricuspid inflow velocity — In tamponade inspiration increases RV filling (tricuspid inflow) while LV filling and SBP fall (pulsus paradoxus).

52 **D** ST elevation in V2 — Pericarditis: diffuse concave ST elevation, PR depression, no reciprocal change or Q waves. ST elevation in V2 occurs in both, so it does not help.

53 **D** Pericardial resection — Post-radiotherapy constriction (knock, dip-and-plateau) is treated definitively by pericardiectomy; diuretics give only temporary relief.

54 **A** Cardiothoracic surgical removal of the mass — A pedunculated LA mass at the fossa ovalis with fever, arthralgia and TIA is myxoma: surgical excision without delay.

55 **C** Malignant melanoma — Melanoma has the highest propensity for cardiac metastasis; lung and breast cancer are commonest by number.

56 **E** Prompt defibrillation — Commotio cordis: a precordial blow on the T-wave upslope triggers VF in a normal heart; immediate CPR and defibrillation save life.

57 **A** Coronary artery disease — Coronary artery disease is the leading cause of death in type 2 diabetes (2–4× risk, often silent ischaemia).

58 **C** Thiamine — High-output heart failure in a heavy drinker that responds rapidly is wet beriberi from thiamine deficiency; give thiamine before glucose.

59 **D** QT interval — Hypothyroidism lowers heart rate, stroke volume and output, raises SVR and prolongs (not shortens) the QT.

60 **C** Low-density lipoproteins accumulate within the arterial intima — Atherogenesis begins with LDL retention in the intima; oxidation, endothelial activation, monocyte recruitment and foam cells follow.

61 **B** An enlarging atheroma does not generally encroach on the arterial lumen until the plaque exceeds 40% of the internal elastic lamina — Arteries remodel outward, so a plaque does not narrow the lumen until it exceeds ~40% of the area within the internal elastic lamina (Glagov).

- 62** **B** Individuals with a raised high-sensitivity C-reactive protein (CRP) regardless of LDL cholesterol or ASCVD risk — Statin benefit groups: clinical ASCVD, LDL $\geq$ 4.9 mmol/L, diabetes aged 40–75, 10-year risk $\geq$ 7.5%. Raised hs-CRP alone is not one.
- 63** **E** Weight loss and regular physical activity — **UPDATED** HDL-raising drugs (CETP inhibitors, niacin, fibrates) did not cut events; weight loss and exercise improve HDL and overall risk.
- 64** **B** Heart rhythm — Myocardial oxygen demand depends on heart rate, contractility and wall stress (preload and afterload); rhythm itself is not a primary determinant.
- 65** **E** Vasodilation of intramyocardial coronary vessels leading to increased coronary flow — The heart already extracts ~70–75% of oxygen at rest, so extra demand is met by arteriolar dilatation and increased coronary flow.
- 66** **B** An increase in intracellular pH — Ischaemia shifts to anaerobic glycolysis: lactate and H^+ accumulate (pH falls), ATP depletes, calcium rises and potassium leaks out.
- 67** **E** Trapezius muscle — Angina radiates to arms, neck, jaw, epigastrium and back; radiation to the trapezius ridge is characteristic of pericarditis.
- 68** **B** Noradrenaline (norepinephrine) IV infusion — **UPDATED** Cardiogenic shock after MI: urgent PCI, with noradrenaline as first-line vasopressor (SOAP II), $\pm$ dobutamine. Routine IABP is not recommended.
- 69** **D** Morphine 2 mg IV — Morphine is not recommended in acute heart failure, especially in shock (hypotension, respiratory depression, worse outcomes).
- 70** **C** Elevated wedge with prominent v waves — **UPDATED** Acute MR (papillary rupture) into a small non-compliant LA gives a high wedge pressure with giant v waves. A saturation step-up in the RV suggests VSD.
- 71** **E** Most victims have a previously documented left ventricular ejection fraction of 35% or less — **UPDATED** Most sudden cardiac deaths occur in people without known heart disease or with EF > 35%; CAD underlies ~80% of adult cases.

2 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1** While on call, you are bleeped to see a man who has had a massive rectal bleed. He is drowsy, with a blood pressure of 65/40 mmHg and a pulse of 130 beats/min. While IV access is obtained, he arrests and CPR is started. The monitor shows organised QRS complexes, but there is no pulse. CPR continues and adrenaline 1 mg IV has just been given. What is the most important next step in management?
- A Rapid IV infusion of blood products via the major haemorrhage protocol
B DC shock 200 J
C Amiodarone 300 mg IV
D Repeat adrenaline 1 mg IV immediately
E Atropine 3 mg IV
- 2** A 44-year-old woman presents with chest pain of several days' duration. The pain is sharp and stabbing, located at the left sternal border, and worsened by coughing and by palpation. She has no family history of heart disease and no diabetes, hypertension, smoking or hyperlipidaemia. Physical examination, ECG and chest X-ray are all normal. Which further investigation is indicated?
- A Treadmill exercise ECG
B Stress echocardiography
C Cardiac catheterisation
D Stress myocardial perfusion imaging
E No further tests

Answers 1–2

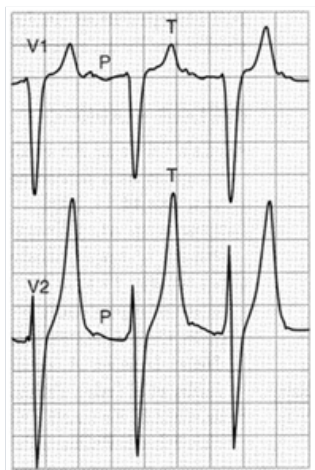
CRASH COURSE MCQs

- 1** **A** Rapid IV infusion of blood products via the major haemorrhage protocol — **UPDATED** PEA from massive haemorrhage will not reverse without volume: give blood products rapidly via the major haemorrhage protocol during CPR.
- 2** **E** No further tests — Sharp localised pain reproduced by palpation with normal ECG and CXR is musculoskeletal (costochondritis); no further tests, reassure and analgesia.

2 QUESTIONS • AFTER PRETEST MEDICINE

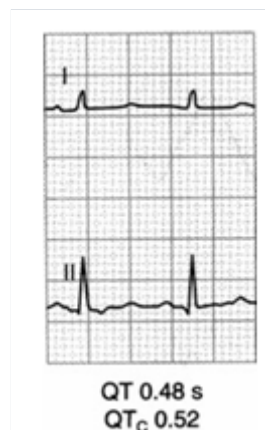
PreTest MCQs

- 1** A 70-year-old man with mild chronic kidney disease, diabetes mellitus and heart failure is admitted with reduced urine output, weakness and breathlessness. He takes several medications but cannot remember their names. Blood results are pending, and his ECG is shown. Based on the information available, what is the best initial step in management?



- A Administration of intravenous insulin
- B Administration of intravenous sodium bicarbonate
- C Administration of intravenous 3% hypertonic saline
- D Administration of oral sodium polystyrene sulfonate
- E Administration of intravenous calcium gluconate

- 2** A surgical colleague asks you to review a 54-year-old woman with ECG abnormalities the day after a sub-total thyroidectomy for toxic multinodular goitre. Her only medication is fentanyl for postoperative pain. She has no history of syncope and no family history of sudden cardiac death. Examination is unremarkable apart from a clean incision at the base of the neck. Her ECG is shown. What is the best next step?



- A Administration of intravenous magnesium sulfate
- B Measurement of serum ionized calcium
- C Stat noncontrast CT scan of the brain
- D Formal auditory testing
- E Reassure the patient that her ECG is normal for a woman her age

Answers 1-2

PRETEST MCQS

- 1** **E** Administration of intravenous calcium gluconate — Peaked T waves with flat P waves are hyperkalaemia: give IV calcium gluconate first, then insulin–glucose and salbutamol.
- 2** **B** Measurement of serum ionized calcium — Long QT from ST-segment lengthening the day after thyroidectomy suggests hypocalcaemia (parathyroid injury): measure ionised calcium.

1 QUESTIONS • AFTER PASSMEDICINE

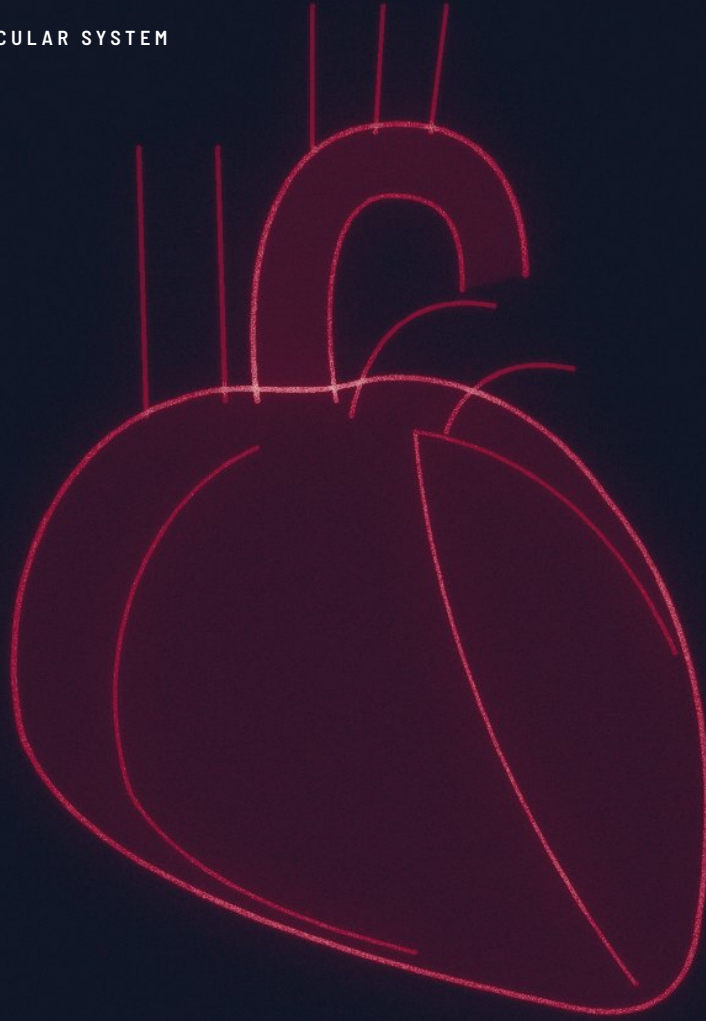
Passmedicine MCQs

- 1 A 56-year-old man attends his GP surgery for a routine blood pressure check and examination. He has no immediate concerns and his past medical history includes hypertension and asthma. He feels well in himself. On examination, he appears well and observations are normal except for a raised blood pressure of 155/87 mmHg. Upon auscultation of the heart, the GP notices an S4 heart sound. The GP then requests an ECG recording. Which part of the ECG does this sound coincide with?
- A P wave
 - B QRS complex
 - C ST segment
 - D T wave
 - E U wave

Answer 1

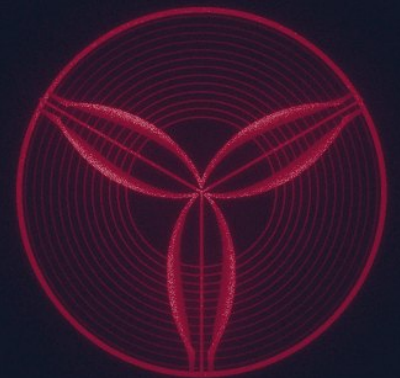
PASSMEDICINE MCQS

- 1 **A** P wave — S4 is atrial contraction into a stiff ventricle, so it falls just after the P wave; S1 coincides with QRS. No S4 in AF.



10

CARDIOLOGY EMQS



Cardiology EMQs

Extended matching questions on the cardiovascular system: chest pain, murmurs, pulses, ECGs, arrhythmias and cardiovascular drugs.

0

MCQS

298

EMQ SCENARIOS

5

SOURCES

5

REPEATED IDEAS

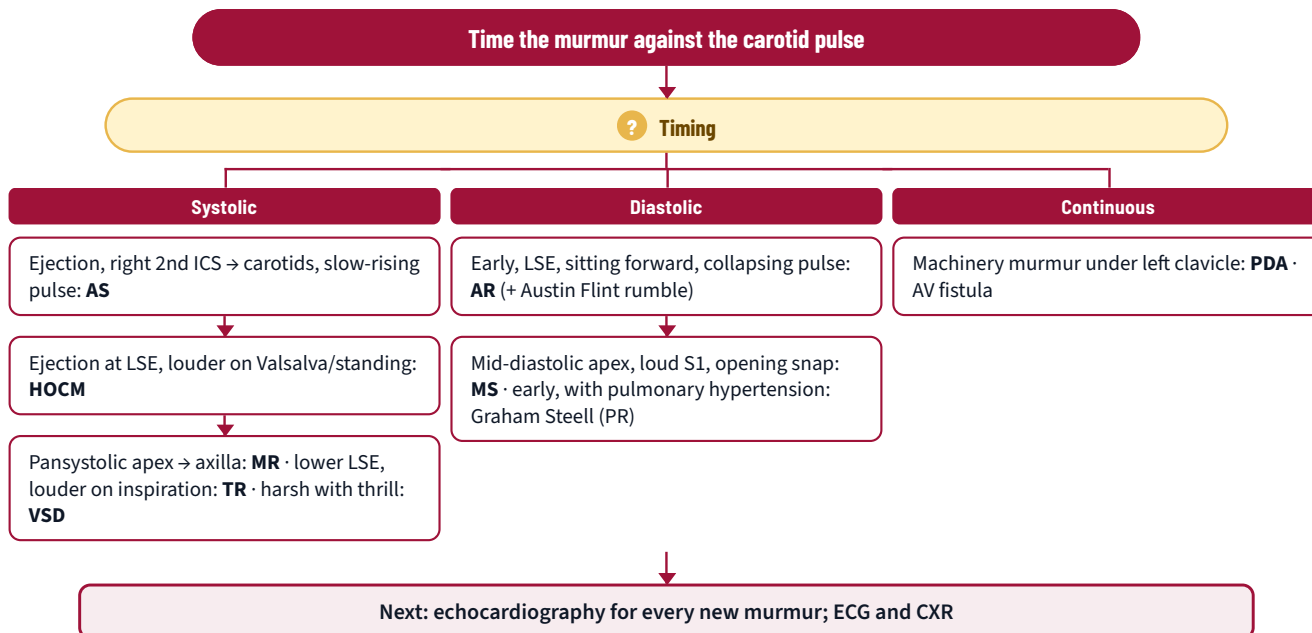
IN THIS CHAPTER

01	Management flowcharts	2 charts	page 154
02	Local Exams EMQs	123 Q	page 155
03	Crash Course EMQs	45 Q	page 168
04	Irfan EMQs	30 Q	page 172
05	Get Ahead EMQs	94 Q	page 175
06	Other Sources EMQs	6 Q	page 188

Step by step

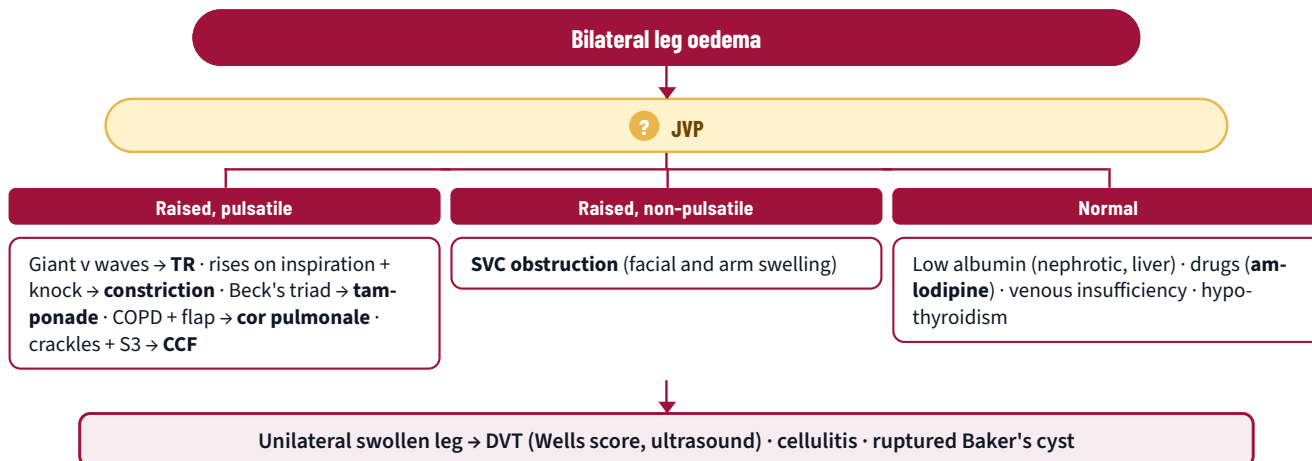
Murmur decoder

CLINICAL EXAMINATION



Leg oedema and the JVP

CLINICAL EXAMINATION



Local Exams EMQs

SET 1 OF 6

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

THEME 1 Leg oedema

- A Cor pulmonale
- B Amlodipine-induced oedema
- C Congestive heart failure
- D Tricuspid regurgitation
- E Constrictive pericarditis
- F Mitral regurgitation
- G Mitral stenosis
- H Chronic obstructive pulmonary disease
- I Idiopathic pulmonary fibrosis
- J Restrictive cardiomyopathy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 60-year-old smoker with diabetes has bilateral pitting leg oedema, a raised JVP, a pulsatile liver and a pansystolic murmur at the left lower sternal edge.
- 2 A 55-year-old woman with well-controlled hypertension on regular treatment and no diabetes has bilateral pitting leg oedema; the JVP is normal and all investigations are normal.
- 3 A 65-year-old heavy smoker with a long history of cough, sputum and severe breathlessness has a raised JVP, bilateral pitting oedema and a flapping tremor; there is a prolonged expiratory phase with coarse crackles and scattered wheeze.
- 4 A middle-aged woman has fatigue, abdominal distension and leg swelling. Pulse 112/min, regular, small volume with pulsus paradoxus; bilateral pitting oedema, raised JVP and an early third heart sound; hepatomegaly and ascites.
- 5 A 32-year-old woman with rheumatic heart disease for 20 years has breathlessness, fatigue, oedema and ascites; there is a soft S1, a loud S3 and an apical pansystolic murmur radiating to the axilla.

THEME 2 Chest pain (cardiorespiratory)

- A Pneumothorax
- B Pleural effusion
- C Empyema
- D Bronchial carcinoma
- E Acute pericarditis
- F Hypertrophic obstructive cardiomyopathy
- G Acute lobar pneumonia
- H Chronic obstructive pulmonary disease
- I Pulmonary infarction

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 25-year-old man has sudden sharp right-sided chest pain worse on deep breathing and coughing, a short painful dry cough, fever, rigors, vomiting and malaise; there is dullness, bronchial breathing and fine crackles over the right lower zone.
- 2 A 28-year-old tall smoker has sudden sharp right-sided chest pain and breathlessness; chest expansion is reduced on the right with a resonant percussion note and diminished breath sounds.
- 3 A 54-year-old woman with diabetes has high fever, rigors, sweating, sharp left-sided chest pain, breathlessness and purulent sputum despite 10 days of amoxicillin; there is finger clubbing, reduced expansion, dullness and markedly diminished breath sounds over the left lower and middle zones.
- 4 A 25-year-old woman has severe stabbing central chest pain radiating to the left jaw and shoulder, with fever and breathlessness; a friction rub is heard over the precordium.
- 5 An 18-year-old man has breathlessness, exertional chest pain and palpitations; there is a double apical impulse and an ejection systolic murmur.

THEME 3 Cardiovascular presentations

- A Hypertension
- B Hypertrophic obstructive cardiomyopathy
- C Dilated cardiomyopathy
- D Coarctation of the aorta
- E Fulminant myocarditis
- F Chronic constrictive pericarditis
- G Infective endocarditis
- H Cardiac tamponade
- I Acute myocardial infarction
- J Restrictive cardiomyopathy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 20-year-old student has headaches, weakness and leg cramps. BP is 165/95 mmHg in the arm and 120/75 mmHg in the leg; there is a systolic murmur and an ejection click in the aortic area.
- 2 A 26-year-old man has chest pain and breathlessness on exercise; there is a jerky pulse, a double apical impulse and a mid-systolic murmur at the base.
- 3 A 50-year-old man with lung cancer becomes breathless; pulse 112/min, regular and small-volume with pulsus paradoxus, BP 85/50 mmHg, a grossly raised JVP and soft heart sounds.

- 4** An 18-year-old woman who had a flu-like illness a week ago develops acute heart failure; BP 70/40 mmHg, non-specific ECG changes and a markedly raised troponin.
- 5** A patient has fatigue, a rapid low-volume pulse with pulsus paradoxus, a raised JVP, a loud early third heart sound, hepatomegaly, ascites and peripheral oedema.

THEME 4 Leg oedema (second paper)

- A** Cor pulmonale
- B** Deep vein thrombosis
- C** Congestive heart failure
- D** Tricuspid regurgitation
- E** Ruptured Baker cyst
- F** Mitral regurgitation
- G** Cellulitis
- H** Lymphoedema
- I** Restrictive cardiomyopathy
- J** Acute myocarditis
- K** Constrictive pericarditis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 35-year-old woman has right leg swelling with mild pain. She has taken the oral contraceptive pill for 8 months and recently travelled a long distance.
- 2** A 65-year-old heavy smoker with a long history of cough, sputum and severe breathlessness has a raised JVP, bilateral pitting oedema and a flapping tremor; there is a prolonged expiratory phase with coarse crackles and scattered wheeze.
- 3** A 62-year-old man with hypertension, diabetes and ischaemic heart disease has a painful, swollen, red and tender left leg with pitting oedema; the white cell count is raised and Doppler ultrasound is normal.
- 4** A middle-aged woman has fatigue, abdominal distension and leg swelling; pulse 112/min, regular and small-volume with pulsus paradoxus, a raised JVP, an early third heart sound, hepatomegaly and ascites.

- 5** A 66-year-old man with ischaemic heart disease on irregular treatment has progressive breathlessness for 6 months with a dry cough; there is a raised JVP, leg oedema and bibasal fine inspiratory crackles.

REPEATED 3* · ALSO ASKED IN
Formative exam x2

THEME 5 Hypertension

- A** Cushing's syndrome
- B** Systemic sclerosis
- C** Coarctation of the aorta
- D** Conn's syndrome
- E** Pregnancy
- F** Polycystic kidneys
- G** Malignant hypertension
- H** Hyperparathyroidism
- I** Renal artery stenosis
- J** Portal hypertension
- K** Pheochromocytoma

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 30-year-old woman with hypertension is found to have hypokalaemia and a mild metabolic alkalosis.
- 2** An anxious 26-year-old woman has episodes of chest pain and palpitations precipitated by stress and smoking. Her 24-hour urine shows raised catecholamines.
- 3** A 45-year-old woman has weight gain, muscle weakness and hirsutism; she is hypertensive with pedal oedema.
- 4** A 40-year-old man is brought to the emergency department with a severe headache; he has papilloedema and fundal haemorrhages, and BP 220/145 mmHg.
- 5** A 75-year-old heavy smoker with widespread peripheral vascular disease has hypertension.

REPEATED 2* · ALSO ASKED IN
Irfan

Answers · Themes 1-5

LOCAL EXAMS EMQS

Theme 1 · Leg oedema

- 1** **D Tricuspid regurgitation** — Raised JVP with giant v waves, pulsatile liver and a pansystolic murmur at the lower left sternal edge (louder on inspiration) is tricuspid regurgitation.
- 2** **B Amlodipine-induced oedema** — Bilateral ankle oedema with normal JVP and investigations in a treated hypertensive is amlodipine oedema (arteriolar dilatation); it responds poorly to diuretics.
- 3** **A Cor pulmonale** — COPD with hypercapnia (flap), raised JVP and oedema is cor pulmonale: right heart failure from hypoxic pulmonary hypertension.

- 4** **E Constrictive pericarditis** — Raised JVP, pericardial knock (early S3), hepatomegaly and ascites out of proportion to oedema with a small pulse is constrictive pericarditis.
- 5** **F Mitral regurgitation** — Rheumatic disease with an apical pansystolic murmur to the axilla, soft S1 and loud S3 is severe MR, with secondary right heart failure.

Theme 2 · Chest pain (cardiorespiratory)

- 1** **G Acute lobar pneumonia** — Fever, rigors and pleuritic pain with dullness, bronchial breathing and crackles over one zone is lobar pneumonia; assess with CURB-65 and CXR.

- 2 **A Pneumothorax** — Sudden pleuritic pain in a tall young smoker with reduced expansion, hyper-resonance and reduced breath sounds is a primary spontaneous pneumothorax.
- 3 **C Empyema** — Swinging fever despite antibiotics with dullness and absent breath sounds at the base is empyema; aspirate (pH < 7.2, pus) and insert a chest drain.
- 4 **E Acute pericarditis** — Sharp central pain to the shoulder, fever and a friction rub is acute pericarditis (saddle ST elevation, PR depression); NSAID plus colchicine.
- 5 **F Hypertrophic obstructive cardiomyopathy** — Exertional pain in a young man with a double apical impulse and ejection systolic murmur is HOCM; louder with Valsalva, confirm by echo.

Theme 3 • Cardiovascular presentations

- 1 **D Coarctation of the aorta** — Arm hypertension with lower leg pressures and an ejection click (bicuspid valve) in a young adult is coarctation; confirm by CT/MR angiography.
- 2 **B Hypertrophic obstructive cardiomyopathy** — Exertional symptoms with a jerky pulse, double apical impulse and ejection murmur are HOCM; aortic stenosis gives a slow-rising pulse.
- 3 **H Cardiac tamponade** — Hypotension, pulsus paradoxus, grossly raised JVP and soft heart sounds in lung cancer is malignant tamponade: urgent pericardiocentesis.
- 4 **E Fulminant myocarditis** — Cardiogenic shock with a markedly raised troponin a week after a viral illness is fulminant myocarditis; support, and recovery is often good.
- 5 **F Chronic constrictive pericarditis** — Raised JVP, pericardial knock, hepatomegaly, ascites and oedema is chronic constrictive pericarditis; tamponade is acute with soft heart sounds.

SET 2 OF 6

THEME 6 Chest pain

- A Pulmonary embolus
- B Pneumothorax
- C Lobar pneumonia
- D Costochondritis
- E Oesophageal spasm
- F Atrial fibrillation
- G Infective endocarditis
- H Aortic dissection
- I Cardiac tamponade
- J Herpes zoster infection
- K Tension pneumothorax
- L Pericarditis
- M Angina

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 63-year-old man with high blood pressure presents with sudden-onset tearing chest pain radiating to the back.

Theme 4 • Leg oedema (second paper)

- 1 **B Deep vein thrombosis** — Unilateral painful leg swelling on the pill after long-haul travel is DVT: Wells score, D-dimer/ultrasound, then a DOAC.
- 2 **A Cor pulmonale** — Long-standing COPD with a flap, raised JVP and oedema is cor pulmonale; LTOT improves survival in eligible patients.
- 3 **G Cellulitis** — Hot, red, tender swollen leg with raised WCC and normal Doppler is cellulitis; treat with flucloxacillin (doxycycline/clarithromycin if allergic).
- 4 **K Constrictive pericarditis** — Ascites and hepatomegaly out of proportion to oedema, raised JVP and pericardial knock indicate constrictive pericarditis; pericardiectomy is definitive.
- 5 **C Congestive heart failure** — Progressive breathlessness with raised JVP, oedema and bibasal crackles in poorly treated IHD is congestive heart failure; NT-proBNP and echo.

Theme 5 • Hypertension

- 1 **D Conn's syndrome** — Hypertension with hypokalaemia and alkalosis in a young woman suggests Conn's syndrome; screen with the aldosterone:renin ratio.
- 2 **K Pheochromocytoma** — Paroxysmal palpitations and chest pain with raised urinary catecholamines is pheochromocytoma; alpha-block before any beta-blocker.
- 3 **A Cushing's syndrome** — Weight gain, proximal weakness, hirsutism, oedema and hypertension suggest Cushing's syndrome; screen with overnight dexamethasone suppression.
- 4 **G Malignant hypertension** — BP 220/145 with papilloedema and retinal haemorrhages is malignant hypertension; lower by ≤ 25% in the first hours with IV labetalol or nitroprusside.
- 5 **I Renal artery stenosis** — New hypertension in an elderly smoker with widespread atherosclerosis suggests atheromatous renal artery stenosis; confirm with duplex or CT/MR angiography.

- 2 A 40-year-old woman develops sudden breathlessness at rest after hip replacement surgery; she is tachycardic and the ECG shows right axis deviation.
- 3 A 60-year-old businessman has central crushing chest pain radiating to both arms after running for a bus; it is relieved by rest and his ECG 1 hour later is normal.
- 4 A 21-year-old high-jumper has acute breathlessness and right-sided pleuritic chest pain; there is increased resonance and reduced expansion on the right.
- 5 A 23-year-old woman has localised left-sided chest pain that is worse on coughing and very tender to light pressure; it is relieved by aspirin and the ECG is normal.

REPEATED 2* • ALSO ASKED IN
Irfan

THEME 7 Treatment of arrhythmias

- A Intravenous adenosine
- B Oral warfarin
- C Oral lidocaine
- D Digoxin plus warfarin for a month
- E Intravenous magnesium plus ventricular pacing
- F Intravenous amiodarone
- G Low-molecular-weight heparin
- H Oral bisoprolol plus aspirin
- I DC shock plus heparin
- J Oral sotalol plus warfarin
- K None of the above

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man with heart failure and a reduced ejection fraction has had atrial fibrillation for more than 48 hours, with a ventricular rate of 120/min; he is haemodynamically stable. Choose his treatment before elective DC cardioversion.
- 2 Initial therapy for a 60-year-old woman who is severely compromised by acute atrial fibrillation.
- 3 A 55-year-old man with an acute myocardial infarction has recurrent runs of ventricular tachycardia despite a beta-blocker and correction of potassium and magnesium; he needs prophylaxis against recurrence.
- 4 A drug to aid diagnosis in a 50-year-old man with an unidentified regular narrow-complex tachycardia.
- 5 Prophylaxis of ventricular tachycardia with a varying QRS axis in a patient with a prolonged QT interval.

REPEATED 3* · ALSO ASKED IN
Formative exam · Irfan

THEME 8 Murmurs

- A Mitral stenosis
- B Aortic regurgitation
- C Ventricular septal defect
- D Atrial septal defect
- E Pulmonary hypertension
- F Aortic stenosis
- G Coarctation of the aorta
- H Hypertrophic obstructive cardiomyopathy
- I Mitral regurgitation
- J Pulmonary stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient with a tapping apex beat and a mid-diastolic rumbling murmur.
- 2 A patient with head nodding, a large-volume pulse, BP 130/50 mmHg and a diastolic murmur.
- 3 A young patient with a pansystolic murmur at the left parasternal area.

- 4 A young patient with fixed splitting of the second heart sound and a systolic murmur in the pulmonary area.
- 5 A patient with right upper abdominal discomfort, a distended JVP, a loud second heart sound and a pansystolic murmur of tricuspid regurgitation.
- 6 A patient with a slow-rising, small-volume pulse, a heaving apex and an ejection systolic murmur radiating to the neck.
- 7 A patient with a mid-systolic murmur best heard over the back between the scapulae and a left-arm blood pressure of 170/110 mmHg.

THEME 9 Vascular and structural heart disease

- A Chronic venous insufficiency
- B Acute limb ischaemia
- C Mitral regurgitation
- D Tetralogy of Fallot
- E Atrial septal defect
- F Deep vein thrombosis
- G Peripheral arterial disease (intermittent claudication)
- H Ventricular septal defect
- I Lymphoedema
- J Aortic stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A barber has aching pain in both legs at the end of a day of standing, and has started to develop skin pigmentation around the ankles.
- 2 A 32-year-old teacher has severe left leg pain of 1 hour and an irregular pulse.
- 3 A 40-year-old farmer with known rheumatic heart disease has breathlessness and an apical pansystolic murmur radiating to the axilla.
- 4 A 2-year-old child has episodes of cyanosis when crying, squats after play, has SpO₂ 89% and a harsh ejection systolic murmur at the left sternal edge.
- 5 A 20-year-old has fixed splitting of the second heart sound and a systolic murmur.

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

THEME 10 Lipid treatment

- A Atorvastatin
- B Ezetimibe
- C Fenofibrate
- D Nicotinic acid (niacin)
- E PCSK9 inhibitor (evolocumab)
- F Omega-3 fatty acids
- G Lifestyle modification

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 45-year-old man with coronary artery disease still has a high LDL cholesterol despite taking a statin. What is the next treatment?

- 2 A 30-year-old man has isolated hypertriglyceridaemia. Which drug acts by stimulating lipoprotein lipase?
- 3 A patient with hyperlipidaemia is started on a drug that inhibits HMG-CoA reductase and can lower LDL cholesterol by up to 60%.

Answers · Themes 6–10

LOCAL EXAMS EMQS

Theme 6 · Chest pain

- 1 **H Aortic dissection** — Sudden tearing pain to the back in a hypertensive man is aortic dissection; confirm with CT angiography and never thrombolysed.
- 2 **A Pulmonary embolus** — Sudden breathlessness and tachycardia after hip replacement with right axis deviation (RV strain) is PE; CTPA and anticoagulate.
- 3 **M Angina** — Exertional crushing pain relieved by rest with a normal ECG afterwards is stable angina; CT coronary angiography is the first test.
- 4 **B Pneumothorax** — Pleuritic pain and breathlessness in a tall young athlete with hyper-resonance on one side is spontaneous pneumothorax.
- 5 **D Costochondritis** — Localised chest-wall pain reproduced by light pressure and relieved by NSAIDs, with a normal ECG, is costochondritis.

Theme 7 · Treatment of arrhythmias

- 1 **D Digoxin plus warfarin for a month** — Before elective cardioversion of AF > 48 h: ≥ 3 weeks of anticoagulation (DOAC now preferred) with rate control (digoxin in HFrEF), continued ≥ 4 weeks after.
- 2 **I DC shock plus heparin** — AF with haemodynamic compromise (shock, ischaemia, pulmonary oedema) needs immediate synchronised DC cardioversion. Start heparin as soon as possible without delaying the shock.
- 3 **F Intravenous amiodarone** — Recurrent VT in acute MI despite beta-blocker and electrolyte correction: IV amiodarone; lidocaine is a second-line IV alternative.
- 4 **A Intravenous adenosine** — IV adenosine transiently blocks the AV node: it terminates AVNRT/AVRT and unmasks flutter, so it is diagnostic and therapeutic. Avoid in severe asthma.
- 5 **E Intravenous magnesium plus ventricular pacing** — Torsades de pointes: IV magnesium 2 g, correct K⁺, stop QT drugs, and override pacing or isoprenaline to prevent recurrence. Amiodarone is contraindicated.

Theme 8 · Murmurs

- 1 **A Mitral stenosis** — A tapping apex with a low-pitched apical mid-diastolic rumble is mitral stenosis, almost always rheumatic.
- 2 **B Aortic regurgitation** — Head nodding (de Musset's sign), a large-volume collapsing pulse and a wide pulse pressure

(130/50 mmHg) with an early diastolic murmur are chronic aortic regurgitation.

- 3 **C Ventricular septal defect** — A harsh pansystolic murmur at the left parasternal area (lower left sternal edge) in a young patient is a ventricular septal defect. Small defects are loudest and often close spontaneously.
- 4 **D Atrial septal defect** — Fixed wide S2 splitting with a pulmonary flow murmur is ASD; the ECG shows RBBB.
- 5 **E Pulmonary hypertension** — Loud P2 with raised JVP, hepatic congestion and functional TR indicates pulmonary hypertension with right heart failure.
- 6 **F Aortic stenosis** — Slow-rising small-volume pulse, heaving apex and ejection murmur to the carotids is aortic stenosis; symptoms mean valve replacement.
- 7 **G Coarctation of the aorta** — Arm hypertension with a systolic murmur between the scapulae is coarctation; check radiofemoral delay and rib notching.

Theme 9 · Vascular and structural heart disease

- 1 **A Chronic venous insufficiency** — Aching legs after standing with haemosiderin pigmentation at the ankles is chronic venous insufficiency; compression once ABPI is normal.
- 2 **B Acute limb ischaemia** — Sudden severe limb pain with an irregular pulse (AF) is embolic acute limb ischaemia: IV heparin and urgent embolectomy.
- 3 **C Mitral regurgitation** — An apical pansystolic murmur radiating to the axilla with breathlessness in rheumatic heart disease is mitral regurgitation.
- 4 **D Tetralogy of Fallot** — Cyanotic spells, squatting and an ejection systolic murmur in a toddler is tetralogy of Fallot.
- 5 **E Atrial septal defect** — Fixed S2 splitting with a pulmonary flow murmur in a young adult is ASD; echo confirms and significant defects are closed by device.

Theme 10 · Lipid treatment

- 1 **B Ezetimibe** — With LDL above target on a maximally tolerated statin, add ezetimibe first; a PCSK9 inhibitor follows if LDL remains high.
- 2 **C Fenofibrate** — Fibrates activate PPAR- α , increasing lipoprotein lipase activity and lowering triglycerides by up to 50%.
- 3 **A Atorvastatin** — High-intensity statins (atorvastatin 40–80 mg) inhibit HMG-CoA reductase and cut LDL by about 50–60%.

SET 3 OF 6

THEME 11 Cyanosis

- A Tetralogy of Fallot
- B Transposition of the great arteries
- C Bronchopneumonia
- D COPD
- E ARDS
- F Atrial septal defect
- G Ventricular septal defect
- H Patent ductus arteriosus
- I Methaemoglobinemia
- J Cold exposure
- K Pulmonary hypertension
- L Total anomalous pulmonary venous return

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 4-month-old infant has cyanosis and breathlessness. The chest X-ray shows a "boot-shaped" heart.
- 2 A 44-year-old patient has progressive breathlessness and central cyanosis. The ECG shows right ventricular hypertrophy and the chest X-ray shows enlarged pulmonary arteries.
- 3 A 60-year-old smoker has chronic cough, wheeze and cyanosis. The chest X-ray shows hyperinflated lungs.
- 4 A 30-year-old patient has a sudden fever, productive cough and cyanosis. The chest X-ray shows bilateral patchy infiltrates.
- 5 A 6-year-old child has cyanosis. The chest X-ray shows a "snowman" (figure-of-eight) heart.

THEME 12 Limb ischaemia

- A Chronic atherosclerotic ischaemia
- B Acute limb ischaemia
- C Reperfusion injury
- D Leriche syndrome
- E Buerger disease
- F Aortic dissection

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man with a mechanical mitral valve, receiving IV antibiotics for infective endocarditis, suddenly develops severe pain in his right leg while watching television. The leg is cold, pale and pulseless with loss of sensation below the knee; motor function is intact.
- 2 A man has bilateral buttock and thigh claudication, absent femoral pulses and erectile dysfunction.
- 3 After an emergency embolectomy the patient develops increasing pain, swelling and a tense calf.

- 4 A 30-year-old male smoker develops ischaemia of the distal digits. He has previously had superficial thrombophlebitis.

THEME 13 Cardiac emergencies

- A Aspirin 300 mg
- B Primary PCI
- C IV adenosine
- D Oxygen
- E IV low-molecular-weight heparin
- F Thrombolysis
- G Synchronised DC cardioversion
- H Beta-blocker
- I Calcium-channel blocker
- J 100% oxygen
- K IV morphine

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient with an acute STEMI in a rural area can be taken for PCI within 90 minutes.
- 2 A patient with an acute STEMI presents to you; the nearest hospital with PCI is 3 hours away. What is the best treatment?
- 3 A patient has palpitations; the ECG shows a regular narrow-complex tachycardia (SVT) and he is haemodynamically stable.
- 4 A patient has an irregular tachycardia at 146/min with hypotension and chest pain.

THEME 14 Chest pain

- A ST-elevation myocardial infarction
- B Stable angina
- C Acute pericarditis
- D Tietze syndrome
- E Pneumothorax
- F Pulmonary embolism
- G Herpes zoster
- H Prinzmetal (vasospastic) angina
- I Gastro-oesophageal reflux disease
- J Aortic dissection
- K Boerhaave syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient has typical central chest pain at rest radiating to the left arm.
- 2 A patient on long-term steroids has chest pain after meals and on lying down that is partly relieved by a proton-pump inhibitor.
- 3 A patient has chest pain relieved by sitting forward; the ECG shows ST elevation with PR depression.

- 4 A 50-year-old woman has central chest pain at rest with ST elevation on the ECG that resolves spontaneously.
- 5 A patient has tearing chest pain radiating to the back.

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

THEME 15 Pulse examination

- A Coarctation of the aorta
 B Acute CO₂ retention
 C Aortic regurgitation
 D Aortic stenosis
 E Cardiac tamponade
 F Constrictive pericarditis
 G Hypertrophic obstructive cardiomyopathy
 H Thyrotoxicosis
 I Complete heart block

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Radio-femoral delay in a young man with hypertension.
- 2 A bounding pulse, warm peripheries, a flapping tremor and drowsiness in a breathless patient with COPD.
- 3 A collapsing pulse with a wide pulse pressure and a 'pistol-shot' sound heard over the femoral arteries.
- 4 A slow-rising pulse with a narrow pulse pressure, a heaving apex beat and a fourth heart sound.
- 5 Pulsus paradoxus with hypotension, a raised JVP with an absent y descent and muffled heart sounds.

REPEATED 2* · ALSO ASKED IN
Irfan

Answers · Themes 11–15**Theme 11 · Cyanosis**

- 1 **A Tetralogy of Fallot** — A boot-shaped heart (upturned RV apex, concave pulmonary bay) with cyanosis in infancy is tetralogy of Fallot.
- 2 **K Pulmonary hypertension** — Progressive dyspnoea with RV hypertrophy and enlarged central pulmonary arteries is pulmonary hypertension; echo then right heart catheter.
- 3 **D COPD** — Chronic cough and wheeze in a smoker with hyperinflation is COPD; confirm with post-bronchodilator FEV₁/FVC < 0.7.
- 4 **C Bronchopneumonia** — Acute fever and cough with bilateral patchy shadowing is bronchopneumonia; ARDS needs a precipitant and refractory hypoxaemia.
- 5 **L Total anomalous pulmonary venous return** — The snowman heart is supracardiac total anomalous pulmonary venous return; TGA gives an "egg on a string".

Theme 12 · Limb ischaemia

- 1 **B Acute limb ischaemia** — Sudden cold, pale, pulseless leg from an embolus (vegetation or valve thrombus) is acute limb ischaemia; heparin and urgent embolectomy.
- 2 **D Leriche syndrome** — Aorto-iliac occlusion gives the Leriche triad: claudication, absent femoral pulses and impotence.
- 3 **C Reperfusion injury** — Reperfusion causes oedema and compartment syndrome; measure compartment pressure and perform fasciotomy.
- 4 **E Buerger disease** — Distal ischaemia with migratory thrombophlebitis in a young male smoker is thromboangiitis obliterans; stop smoking completely.

Theme 13 · Cardiac emergencies

- 1 **B Primary PCI** — Primary PCI is the reperfusion of choice when it can be delivered within 120 min of diagnosis (ideally <90 min).

- 2 **F Thrombolysis** — When PCI cannot be achieved within 120 min, give fibrinolysis (e.g. tenecteplase) within 10 min, then transfer for angiography.
- 3 **C IV adenosine** — Stable SVT: vagal manoeuvres first, then IV adenosine 6 mg, then 12 mg, as a rapid bolus.
- 4 **G Synchronised DC cardioversion** — An unstable tachyarrhythmia (shock, ischaemia, syncope, heart failure) needs synchronised DC cardioversion under sedation.

Theme 14 · Chest pain

- 1 **A ST-elevation myocardial infarction** — Ischaemic pain at rest radiating to the arm is an acute coronary syndrome; of these options it is STEMI, so get an ECG within 10 min.
- 2 **I Gastro-oesophageal reflux disease** — Postprandial, positional burning pain relieved by a PPI is gastro-oesophageal reflux, aggravated by steroids.
- 3 **C Acute pericarditis** — Pleuritic pain eased by leaning forward with saddle ST elevation and PR depression is acute pericarditis: NSAID plus colchicine.
- 4 **H Prinzmetal (vasospastic) angina** — Transient ST elevation with rest pain resolving spontaneously is coronary vasospasm (Prinzmetal); calcium-channel blockers and nitrates.
- 5 **J Aortic dissection** — Tearing pain radiating to the back is aortic dissection; check both arms' BP and arrange urgent CT aortography.

Theme 15 · Pulse examination

- 1 **A Coarctation of the aorta** — Radiofemoral delay with weak femorals and arm hypertension is the hallmark of coarctation; rib notching on CXR.
- 2 **B Acute CO₂ retention** — Bounding pulse, warm peripheries, flap and drowsiness in COPD reflect acute CO₂ retention: ABG, target SpO₂ 88–92%, NIV if acidotic.

- 3 C Aortic regurgitation** — Collapsing pulse, wide pulse pressure and pistol-shot femorals (Traube) indicate severe aortic regurgitation.
- 4 D Aortic stenosis** — A slow-rising pulse with a narrow pulse pressure, a heaving (pressure-loaded) apex and an S4 indicate

severe aortic stenosis. Hypertrophic cardiomyopathy gives a jerky pulse instead.

- 5 E Cardiac tamponade** — Pulsus paradoxus, hypotension, raised JVP with absent y descent and muffled sounds is tamponade; Kussmaul sign and knock suggest constriction.

SET 4 OF 6

THEME 16 Findings on cardiac auscultation

- A** Mitral regurgitation
B Bicuspid aortic valve
C Atrial septal defect
D Congenital pulmonary stenosis
E Coarctation of the aorta
F Ventricular septal defect
G Hypertrophic obstructive cardiomyopathy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 45-year-old woman has a pansystolic murmur best heard at the apex, radiating to the axilla.
- 2** A 29-year-old man has an ejection click and an ejection systolic murmur in the right 2nd intercostal space, with a soft second heart sound.
- 3** A 12-year-old boy has a soft ejection systolic murmur in the left 2nd intercostal space with fixed splitting of the second heart sound.
- 4** An 11-year-old boy has a harsh ejection systolic murmur heard best posteriorly between the scapulae; his upper-limb blood pressure is raised.
- 5** A 7-year-old boy has a harsh ejection systolic murmur with a thrill in the left 2nd intercostal space and a soft second heart sound.

THEME 17 Chest X-ray interpretation

- A** Heart failure
B Right upper lobe collapse
C Idiopathic pulmonary fibrosis
D Arteriovenous malformation
E Lung abscess
F Left atrial enlargement
G Pneumonia
H Pleural plaque
I Haemothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** The chest X-ray shows a double shadow at the right heart border, a prominent left atrial appendage and elevation of the left main bronchus.
- 2** CXR shows Kerley B lines, bat wing shadowing, prominent upper lobe vessels and cardiomegaly.
- 3** The chest X-ray shows the trachea deviated to the right, with the horizontal fissure and right hilum displaced upwards.

- 4** The chest X-ray shows bilateral reticular shadowing with small lung volumes, predominantly in the subpleural areas of both lower zones.

- 5** The chest X-ray of a man with fever and foul-smelling sputum after an alcoholic binge shows a round cavity with an air–fluid level.

THEME 18 Investigations in cardiology

- A** Cardiac enzymes (troponin)
B Transthoracic echocardiography
C Transoesophageal echocardiography
D Treadmill exercise test
E CT pulmonary angiography
F Blood cultures
G Doppler ultrasound

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 50-year-old man has had 90 minutes of central chest pain. His ECG shows no ST elevation, only non-specific T-wave flattening.
- 2** A 70-year-old woman has fever, sharp left-sided chest pain, a pericardial friction rub and diffuse ST elevation on her ECG.
- 3** A 30-year-old man has had two episodes of syncope during exercise. He has a jerky pulse and an ejection systolic murmur at the left sternal edge that becomes louder on standing and during the Valsalva manoeuvre. His brother died suddenly aged 25.
- 4** A 45-year-old smoker with exertional retrosternal chest pain recovering with rest.
- 5** A 15-year-old with congenital heart disease, prolonged fever, weight loss, diastolic murmur and other systemic signs.
- 6** A patient with a thyroid goitre is found to have an ejection systolic murmur at the left upper sternal edge with wide, fixed splitting of the second heart sound. Transthoracic echocardiography is reported as normal.
- 7** A 30-year-old woman develops sudden chest tightness, breathlessness and hypotension 3 days after an operation. Her ECG shows only sinus tachycardia.

THEME 19 ECG changes: ST elevation

- A Inferior ST-elevation myocardial infarction
- B Anterior ST-elevation myocardial infarction
- C Lateral ST-elevation myocardial infarction
- D Acute pericarditis
- E Vasospastic (Prinzmetal) angina
- F Left ventricular aneurysm

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 62-year-old man has had 2 hours of crushing central chest pain. His ECG shows 3 mm ST elevation in leads II, III and aVF with ST depression in I and aVL.
- 2 A 58-year-old smoker has had 1 hour of severe central chest pain with sweating. His ECG shows 4 mm ST elevation in V1–V4 with early Q waves.
- 3 A 66-year-old woman has 90 minutes of chest pain. Her ECG shows ST elevation in I, aVL, V5 and V6 with reciprocal ST depression in III and aVF.

- 4 A 28-year-old man has sharp central chest pain for 2 days, worse on lying flat and relieved by sitting forward, 1 week after a flu-like illness. His ECG shows concave ST elevation in I, II, aVL, aVF and V2–V6 with PR-segment depression; aVR shows ST depression.
- 5 A 45-year-old woman who smokes has recurrent episodes of chest pain at rest in the early morning. During an episode her ECG shows ST elevation in V1–V4 that resolves completely within minutes of sublingual GTN. Troponin is normal and coronary angiography shows no significant stenosis.
- 6 A 65-year-old man had an anterior myocardial infarction 8 weeks ago. He now has increasing breathlessness but no chest pain. His ECG still shows Q waves with 2 mm ST elevation in V1–V4, unchanged over several weeks, and troponin is not rising.

Answers • Themes 16–19

LOCAL EXAMS EMQS

Theme 16 • Findings on cardiac auscultation

- 1 **A Mitral regurgitation** — A pansystolic murmur at the apex radiating to the axilla is mitral regurgitation. A VSD murmur is at the left sternal edge and does not radiate to the axilla.
- 2 **B Bicuspid aortic valve** — Ejection click with an aortic-area ejection murmur in a young adult suggests bicuspid aortic valve; a soft A2 means significant stenosis.
- 3 **C Atrial septal defect** — Soft pulmonary flow murmur with fixed S2 splitting is ASD; the murmur is from increased flow across the pulmonary valve.
- 4 **E Coarctation of the aorta** — Systolic murmur loudest between the scapulae with arm hypertension is coarctation; check radiofemoral delay.
- 5 **D Congenital pulmonary stenosis** — Harsh ejection murmur with thrill in the left 2nd space and a soft delayed P2 is congenital pulmonary stenosis; balloon valvuloplasty.

Theme 17 • Chest X-ray interpretation

- 1 **F Left atrial enlargement** — Double right heart border, prominent LA appendage and elevated left main bronchus mean left atrial enlargement, classically mitral stenosis.
- 2 **A Heart failure** — Upper-lobe diversion, Kerley B lines, batwing shadowing and cardiomegaly are the CXR signs of left heart failure.
- 3 **B Right upper lobe collapse** — Trachea pulled right with elevated horizontal fissure and right hilum is right upper lobe collapse; bronchoscopy to exclude carcinoma.
- 4 **C Idiopathic pulmonary fibrosis** — Bilateral subpleural lower-zone reticulation with small lungs is IPF (UIP pattern); HRCT shows honeycombing.

- 5 **E Lung abscess** — A cavity with an air–fluid level after an alcohol binge is a lung abscess from aspiration (anaerobes, *S. aureus*).

Theme 18 • Investigations in cardiology

- 1 **A Cardiac enzymes (troponin)** — Ischaemic pain with a non-diagnostic ECG: high-sensitivity troponin at presentation and at 1–3 h to diagnose or exclude NSTEMI.
- 2 **B Transthoracic echocardiography** — Fever, pleuritic chest pain, a pericardial rub and diffuse ST elevation are acute pericarditis; transthoracic echocardiography is recommended in all cases to detect an effusion or tamponade.
- 3 **B Transthoracic echocardiography** — Exertional syncope, jerky pulse, murmur louder on Valsalva and familial sudden death suggest HCM; transthoracic echo is the key test.
- 4 **D Treadmill exercise test** — Exertional angina in a smoker: exercise testing is the best listed option, though NICE/ESC now prefer CT coronary angiography first.
- 5 **F Blood cultures** — Prolonged fever, weight loss and a murmur in congenital heart disease is IE: ≥ 3 blood culture sets before antibiotics.
- 6 **C Transoesophageal echocardiography** — Fixed S2 splitting with a flow murmur but normal TTE suggests an ASD missed by TTE (e.g. sinus venosus); TOE or cardiac MRI.
- 7 **E CT pulmonary angiography** — Sudden chest tightness and hypotension after surgery is PE until proven otherwise: CTPA (bedside echo if too unstable).

Theme 19 • ECG changes: ST elevation

- 1 **A Inferior ST-elevation myocardial infarction** — ST elevation in II, III, aVF with reciprocal I/aVL depression is inferior STEMI (RCA); record V4R and avoid nitrates if RV involved.

- 2 B Anterior ST-elevation myocardial infarction** — ST elevation V1–V4 with evolving Q waves is anterior STEMI (LAD), the worst-prognosis territory: primary PCI.
- 3 C Lateral ST-elevation myocardial infarction** — ST elevation in I, aVL, V5–V6 with reciprocal inferior depression is lateral STEMI (circumflex or diagonal).
- 4 D Acute pericarditis** — Widespread concave ST elevation with PR depression after a viral illness is acute pericarditis: NSAID plus colchicine.
- 5 E Vasospastic (Prinzmetal) angina** — Early-morning rest pain with transient ST elevation, normal troponin and normal coronaries is Prinzmetal angina: CCB and nitrates.
- 6 F Left ventricular aneurysm** — Persistent ST elevation with Q waves 8 weeks after anterior MI with heart failure suggests LV aneurysm; echo confirms.

SET 5 OF 6

THEME 20 Diagnosis of cardiac disease

- A** Peripartum cardiomyopathy
B Mitral stenosis with atrial fibrillation
C Patent ductus arteriosus
D Cardiac tamponade
E Atrial septal defect
F Aortic regurgitation
G Aortic stenosis
H Mitral regurgitation
I Pulmonary embolism
J Acute ST-elevation myocardial infarction

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 1-year-old child with repeated chest infections, a continuous harsh murmur in the left infraclavicular area and bounding pulses.
- 2** A 55-year-old hypertensive woman has had 3 hours of central chest pain. Her ECG shows ST elevation in V1–V4, troponin is raised, and echocardiography shows anterior wall akinesia with an ejection fraction of 35%.
- 3** A 50-year-old woman who had rheumatic fever in childhood has exertional breathlessness and orthopnoea. Her pulse is irregularly irregular, the apex beat is tapping and undisplaced, and there is a low-pitched mid-diastolic murmur at the apex. Chest X-ray shows upper-lobe venous diversion, but echocardiography shows normal left ventricular function.
- 4** A woman who is 8 months pregnant, with no previous heart disease, has acute breathlessness, orthopnoea, pink frothy sputum and a third heart sound.
- 5** A 70-year-old man with recurrent syncope, slow-rising pulse, narrow pulse pressure, and an ejection systolic murmur in the aortic area.

THEME 21 Treatment of cardiac emergencies

- A** Heparin followed by warfarin for 3–6 months
B Synchronised DC cardioversion
C Intravenous adenosine
D Intravenous dobutamine
E Temporary transvenous pacemaker
F Urgent pericardiocentesis
G Reassurance only

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

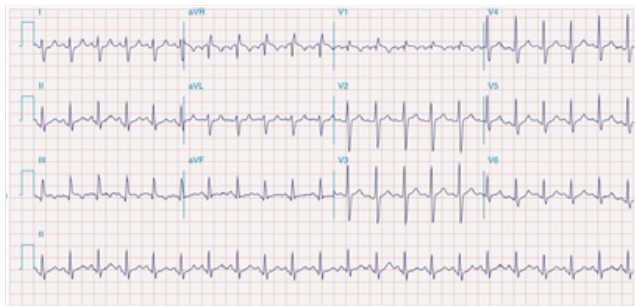
- 1** A 42-year-old man with diabetes and hypertension, 2 days after an inferior myocardial infarction, has repeated episodes of third-degree AV block with a ventricular rate of 30 beats/min and hypotension that do not respond to atropine.
- 2** A 32-year-old man on the coronary care unit becomes hypotensive with a raised JVP, pulsus paradoxus, muffled heart sounds and electrical alternans on the ECG.
- 3** A 46-year-old woman has sudden-onset regular palpitations at 180 beats/min. The ECG shows a regular narrow-complex tachycardia with no visible P waves. Her blood pressure is 118/76 mmHg, and vagal manoeuvres have failed.
- 4** A 70-year-old man has atrial fibrillation of a few hours' duration at 140 beats/min with acute pulmonary oedema and a blood pressure of 80/50 mmHg.
- 5** A 31-year-old man with no symptoms is found to have frequent unifocal ventricular ectopics (4% of beats on Holter monitoring). Echocardiography and electrolytes are normal.

THEME 22 Arrhythmia treatment (ECG cases)

- A** Intravenous adenosine
B Oral warfarin
C Intravenous lidocaine
D Correction of electrolyte disturbance
E Immediate defibrillation
F Beta-blocker
G Reassurance only
H Permanent pacemaker
I Oral sotalol plus anticoagulation
J Intravenous magnesium sulfate
K Cardiac resynchronisation therapy
L Oxygen, low-molecular-weight heparin and intravenous fluids

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

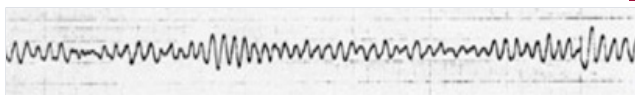
- 1** A 26-year-old woman taking the combined oral contraceptive pill presents with sudden breathlessness and pleuritic chest pain. SpO₂ is 90% on air and blood pressure 112/70 mmHg. Her ECG (shown) shows sinus tachycardia. CT pulmonary angiography confirms pulmonary embolism.



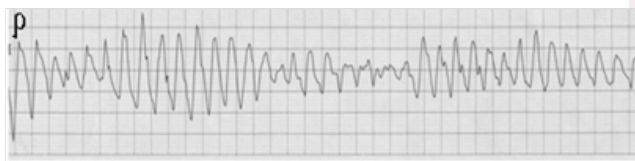
- 2** A 24-year-old man has muscle weakness and cramps after a week of profuse diarrhoea. Serum potassium is 2.4 mmol/L. His ECG (shown) shows ST depression, flattened T waves and prominent U waves.



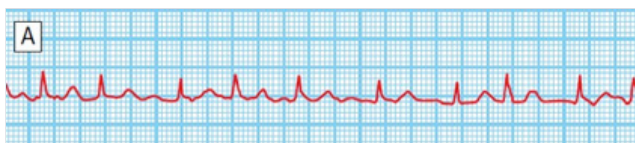
- 3** A 34-year-old man collapses in the emergency department waiting room. He is unresponsive, not breathing normally and has no pulse. The monitor rhythm is shown.



- 4** A 26-year-old woman taking erythromycin and citalopram has recurrent self-terminating episodes of palpitations and near-syncope. Her resting QTc is 560 ms, and the monitor during an episode is shown (polymorphic ventricular tachycardia with a QRS axis that twists around the baseline).



- 5** A 68-year-old woman with hypertension, diabetes and treated hypothyroidism (now euthyroid) has recurrent symptomatic paroxysms of the rhythm shown. Her CHA₂DS₂-VASc score is 4, and a rhythm-control strategy together with stroke prevention is planned.



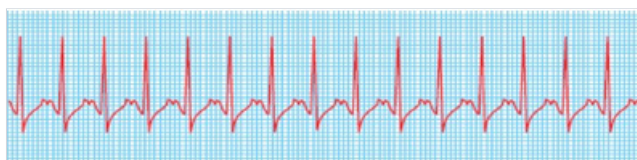
- 6** A 74-year-old man has recurrent syncope. He takes no rate-limiting drugs, and thyroid function and electrolytes are normal. Ambulatory monitoring captures the rhythm shown, a prolonged sinus pause with no P waves, at the time of a syncopal episode.



- A 40-year-old man complains of troublesome 'missed beats'. Echocardiography, electrolytes and thyroid function are normal. His symptoms have continued despite reassurance and stopping caffeine. His rhythm strip is shown.



- 8** A 30-year-old woman has sudden-onset regular palpitations. She is haemodynamically stable, and a modified Valsalva manoeuvre has failed. Her ECG is shown.



THEME 23 Cardiac murmurs

- A** Mitral stenosis
- B** Patent ductus arteriosus
- C** Pulmonary stenosis
- D** Aortic stenosis
- E** Mitral valve prolapse
- F** Tricuspid regurgitation
- G** Aortic regurgitation
- H** Tetralogy of Fallot
- I** Atrial septal defect
- J** Hypertrophic cardiomyopathy
- K** Ventricular septal defect
- L** Mitral regurgitation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A pansystolic murmur at the left fifth intercostal space, mid-clavicular line, radiating to the axilla, with a third heart sound.

- 2** Late systolic murmur with a mid-systolic click at the left 5th intercostal space.

Mid-diastolic rumbling murmur with a loud S1.

A patient with a collapsing pulse and an early diastolic murmur at the left sternal edge also has a mid-diastolic rumble at the apex, with a normal first heart sound and no opening snap.

- 5 A pansystolic murmur at the left lower sternal edge that increases with inspiration.

Answers · Themes 20–23

LOCAL EXAMS EMQs

Theme 20 · Diagnosis of cardiac disease

- 1 **C Patent ductus arteriosus** — Continuous machinery murmur below the left clavicle with bounding pulses and chest infections is PDA; device or surgical closure.
- 2 **J Acute ST-elevation myocardial infarction** — Chest pain, ST elevation, raised troponin and new anterior akinesia is acute STEMI: primary PCI, then ACEi and beta-blocker.
- 3 **B Mitral stenosis with atrial fibrillation** — Rheumatic history with congestion, preserved LV function, AF, tapping apex and diastolic rumble is MS with AF: warfarin.
- 4 **A Peripartum cardiomyopathy** — New heart failure at 8 months' pregnancy without another cause is peripartum cardiomyopathy (EF < 45%).
- 5 **G Aortic stenosis** — Recurrent syncope with a slow-rising pulse, narrow pulse pressure and aortic ejection murmur is severe AS: valve replacement.

Theme 21 · Treatment of cardiac emergencies

- 1 **E Temporary transvenous pacemaker** — Recurrent symptomatic complete heart block after inferior MI unresponsive to atropine needs temporary transvenous pacing.
- 2 **F Urgent pericardiocentesis** — Hypotension, pulsus paradoxus, raised JVP and electrical alternans is tamponade: urgent echo-guided pericardiocentesis.
- 3 **C Intravenous adenosine** — Stable regular narrow-complex tachycardia after failed vagal manoeuvres: IV adenosine 6, 12, then 18 mg.
- 4 **B Synchronised DC cardioversion** — Recent AF with pulmonary oedema and hypotension is unstable: synchronised DC cardioversion under sedation.
- 5 **G Reassurance only** — Frequent unifocal ectopics in an asymptomatic man with a normal heart are benign: reassure and cut caffeine and alcohol.

Theme 22 · Arrhythmia treatment (ECG cases)

- 1 **L Oxygen, low-molecular-weight heparin and intravenous fluids** — Non-massive PE (hypoxaemic, normotensive) on the pill: oxygen and therapeutic LMWH; thrombolysis only if haemodynamically compromised.

- 2 **D Correction of electrolyte disturbance** — Weakness after diarrhoea with ST depression, flat T and U waves is hypokalaemia: replace K⁺ (and Mg) with monitoring.
- 3 **E Immediate defibrillation** — Chaotic waveforms without QRS in a pulseless patient is VF: immediate defibrillation and CPR.
- 4 **J Intravenous magnesium sulfate** — Polymorphic VT with twisting axis on QT-prolonging drugs is torsades: IV magnesium 2 g and stop the drugs.
- 5 **I Oral sotalol plus anticoagulation** — Paroxysmal AF needing rhythm control with CHA2DS2-VASc 4: an antiarrhythmic (sotalol) plus anticoagulation (DOAC preferred).
- 6 **H Permanent pacemaker** — A long sinus pause coinciding with syncope and no reversible cause is sinus node dysfunction: permanent pacemaker.
- 7 **F Beta-blocker** — Symptomatic benign unifocal ventricular ectopics despite reassurance and no caffeine: a beta-blocker is first line.
- 8 **A Intravenous adenosine** — Stable regular narrow-complex SVT after failed modified Valsalva: IV adenosine with ECG recording.

Theme 23 · Cardiac murmurs

- 1 **L Mitral regurgitation** — Apical pansystolic murmur to the axilla with S3 is significant mitral regurgitation.
- 2 **E Mitral valve prolapse** — Mid-systolic click then late systolic murmur at the apex is mitral valve prolapse.
- 3 **A Mitral stenosis** — A low-pitched rumbling mid-diastolic murmur at the apex with a loud S1 (and often an opening snap) is mitral stenosis, heard best with the bell in the left lateral position in expiration.
- 4 **G Aortic regurgitation** — An apical mid-diastolic rumble with a collapsing pulse and early diastolic murmur is the Austin Flint murmur of severe AR.
- 5 **F Tricuspid regurgitation** — Pansystolic murmur at the lower left sternal edge louder on inspiration (Carvallo) is tricuspid regurgitation.

SET 6 OF 6

THEME 24 ECG interpretation

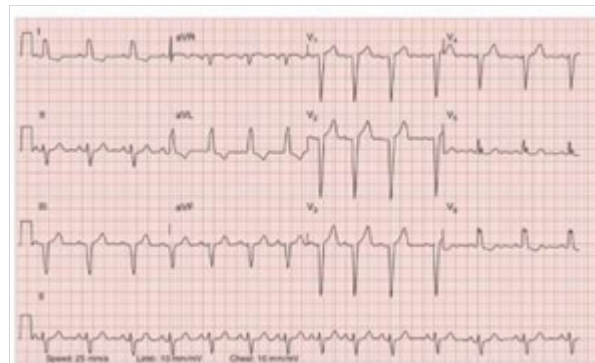
- A Subendocardial infarction
- B Pulmonary embolism
- C Right bundle-branch block
- D Mobitz type II second-degree heart block
- E Hyperkalaemia
- F Inferior myocardial infarction
- G Anterolateral myocardial infarction
- H Left bundle-branch block
- I Pericarditis
- J Atrial fibrillation
- K Mobitz type I second-degree heart block (Wenckebach)
- L Mitral stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

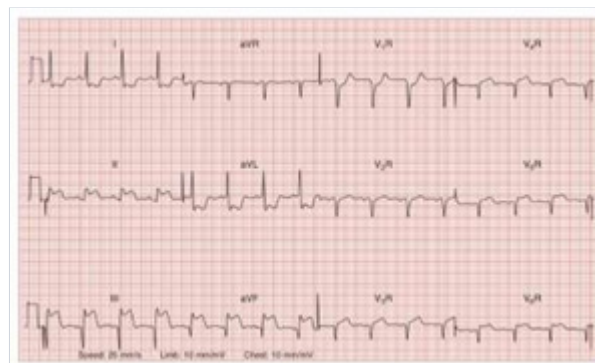
- 1 A 72-year-old woman presents with palpitations. Her rhythm strip (leads V1, II and V5) is shown: the QRS complexes are narrow and irregularly irregular, with no discernible P waves.



- 2 A 70-year-old man with no chest pain has a routine pre-operative ECG (shown), unchanged from 2 years ago. The QRS duration is 140 ms, with deep QS complexes and discordant ST elevation in V1–V3, left axis deviation and broad upright complexes in I and aVL.



- 3 A 60-year-old man has had 1 hour of central chest pain. He is hypotensive with clear lung fields and a raised JVP. His ECG, including right-sided chest leads (V1R–V6R), is shown: there is ST elevation in II, III and aVF (greater in III than II) with ST depression in I and aVL, and ST elevation in V4R.



- 4 A 25-year-old athlete is found to have an irregular pulse at a medical check. His ECG shows sinus P waves at a regular rate; the PR interval lengthens progressively over 3–4 beats until one P wave is not followed by a QRS complex, after which the cycle repeats.

Answers • Theme 24

LOCAL EXAMS EMQS

Theme 24 • ECG interpretation

- 1 **J Atrial fibrillation** — Irregularly irregular narrow complexes without P waves is AF; assess CHA2DS2-VASc and choose rate or rhythm control.
- 2 **H Left bundle-branch block** — Broad QRS with QS in V1–V3, discordant ST changes and left axis is old LBBB; echo for structural disease (new LBBB with pain = STEMI equivalent).

- 3 **F Inferior myocardial infarction** — Inferior STEMI with ST elevation in V4R, hypotension and clear lungs is RV infarction: fluids, no nitrates, primary PCI.
- 4 **K Mobitz type I second-degree heart block (Wenckebach)** — Progressive PR lengthening until a dropped beat with grouped beating is Mobitz I (Wenckebach), benign in athletes.

9 THEMES • 45 SCENARIOS • AFTER CRASH COURSE SBAS & EMQS

Crash Course EMQs

SET 1 OF 2

THEME 1 Cardiac murmurs

- A Aortic regurgitation
- B Aortic stenosis
- C Bicuspid aortic valve
- D Mitral regurgitation
- E Mitral stenosis
- F Patent ductus arteriosus
- G Transposition of the great arteries
- H Tricuspid regurgitation
- I Ventricular septal defect
- J Pulmonary stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 60-year-old man presents with heart failure. On examination he has a collapsing pulse, an early diastolic murmur and a displaced apex beat.
- 2 A 56-year-old man gives a history of rheumatic fever. He has flushed cheeks, an irregularly irregular pulse and a mid-diastolic murmur.
- 3 A 75-year-old woman presents to the emergency department following a sudden collapse. Her ECG shows left ventricular hypertrophy. On examination, a harsh ejection systolic murmur is heard at the right upper sternal edge radiating to both carotid arteries.
- 4 Three days after a myocardial infarction, a 64-year-old man suddenly deteriorates with acute left ventricular failure. He has a new pansystolic murmur at the apex radiating to the axilla.
- 5 A baby is cyanosed at birth and has a loud, long ejection systolic murmur at the left upper sternal edge. Tetralogy of Fallot is suspected. What lesion produces the murmur?

THEME 2 Chest pain

- A Angina pectoris
- B Pericarditis
- C Myocardial infarction
- D Aortic dissection
- E Reflux oesophagitis
- F Pulmonary infarct
- G Pneumonia
- H Costochondritis
- I Pneumothorax
- J Pulmonary embolus

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A middle-aged man has central crushing chest pain on exertion that is relieved by rest within a few minutes.
- 2 A 60-year-old man has central crushing chest pain at rest for the last hour, radiating to his jaw and left arm, with sweating.
- 3 A 65-year-old hypertensive man has sudden severe tearing chest pain radiating to his back.
- 4 A 30-year-old woman has sharp central chest discomfort with a localised tender area over the costal cartilages on palpation.
- 5 A 25-year-old man has sharp central chest pain that is worse on deep inspiration and lying flat, and relieved by sitting forward.

THEME 3 Cardiovascular drugs

- A Propranolol
- B Nicorandil
- C Aspirin
- D Nifedipine
- E Captopril
- F Digoxin
- G Bisoprolol
- H Bendroflumethiazide
- I Clopidogrel
- J Simvastatin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Used in the treatment of cardiac failure. Known to cause a persistent cough.
- 2 Contraindicated in renal artery stenosis.
- 3 Associated with headache, flushing, reflex tachycardia and ankle swelling.
- 4 First-line management in acute coronary syndrome.
- 5 Specifically contraindicated in asthmatics and individuals with peripheral vascular disease.

THEME 4 Clinical features of cardiac disease

- A Aortic stenosis
- B Aortic regurgitation
- C Mitral stenosis
- D Mitral regurgitation
- E Infective endocarditis
- F Rheumatic fever
- G Tricuspid stenosis
- H Tricuspid regurgitation
- I Pulmonary hypertension
- J Pulmonary embolism

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A pansystolic murmur auscultated at the apex and radiating into the axilla.
- 2 Associated with erythematous macules on the palms and haemorrhages under the nails.
- 3 Associated with pink rings with pale centres on the trunk and subcutaneous nodules over the extensor surfaces of the joints.
- 4 Associated with a slow rising carotid pulse and ejection systolic murmur at the right upper sternal border.
- 5 Known to cause a left parasternal heave and a loud pulmonary component of the second heart sound.

THEME 5 Murmurs

- A Aortic regurgitation
- B Aortic stenosis
- C Austin-Flint murmur
- D Flow murmur
- E Graham Steell murmur
- F Mitral regurgitation
- G Mitral stenosis
- H Tricuspid regurgitation
- I Tricuspid stenosis
- J Ventricular septal defect

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 60-year-old Egyptian man with a pansystolic murmur loudest at the right sternal edge, a raised JVP and a pulsatile liver.
- 2 A 70-year-old woman in atrial fibrillation, an apex beat palpable in the mid-axillary line and a loud pansystolic murmur which is heard all over the heart.
- 3 A 14-year-old girl with a flat occiput, single palmar crease and a pansystolic murmur. She has a chromosomal abnormality affecting chromosome 21.
- 4 An elderly gentleman with a collapsing pulse and an early diastolic murmur which is loudest when sitting forward.
- 5 A 50-year-old woman with severe rheumatic mitral stenosis and pulmonary hypertension has a high-pitched early diastolic murmur at the left upper sternal edge that is louder on inspiration.

Answers · Themes 1-5

Theme 1 · Cardiac murmurs

- 1 **A Aortic regurgitation** — Collapsing pulse, early diastolic murmur and displaced apex is chronic aortic regurgitation; operate once symptoms or LV dilatation appear.
- 2 **E Mitral stenosis** — Rheumatic history, malar flush, AF and a mid-diastolic rumble is mitral stenosis; anticoagulate with warfarin.
- 3 **B Aortic stenosis** — Syncope, LVH and a harsh ejection murmur to both carotids is severe aortic stenosis; valve replacement (surgical or TAVI).
- 4 **D Mitral regurgitation** — Acute LVF with a new apical pansystolic murmur to the axilla 3 days after MI is acute MR from papillary muscle rupture.
- 5 **J Pulmonary stenosis** — In tetralogy the loud ejection murmur comes from RV outflow (pulmonary) stenosis; the large VSD is silent.

Theme 2 · Chest pain

- 1 **A Angina pectoris** — Exertional crushing pain relieved by rest within minutes is stable angina; it also responds to GTN.

- 2 **C Myocardial infarction** — Crushing rest pain > 20 min radiating to jaw/arm with sweating is acute MI until proven otherwise: ECG within 10 min and troponin.
- 3 **D Aortic dissection** — Sudden tearing pain through to the back in a hypertensive is aortic dissection; check arm pulses, CT aortography, no thrombolysis.
- 4 **H Costochondritis** — Localised anterior pain reproduced by pressure over the costal cartilages is costochondritis; NSAIDs.
- 5 **B Pericarditis** — Pain worse on inspiration and lying flat, eased sitting forward, is acute pericarditis.

Theme 3 · Cardiovascular drugs

- 1 **E Captopril** — ACE inhibitors (captopril) improve HFrEF survival but cause dry cough in ~10%; switch to an ARB.
- 2 **E Captopril** — In bilateral renal artery stenosis GFR depends on angiotensin II, so ACE inhibitors can cause AKI.
- 3 **D Nifedipine** — Dihydropyridine CCBs (nifedipine) cause headache, flushing, reflex tachycardia and ankle oedema.
- 4 **C Aspirin** — Aspirin 300 mg chewed is given as soon as ACS is suspected; a P2Y₁₂ inhibitor is added.

- 5 **A Propranolol** — Non-selective propranolol causes bronchospasm in asthma and worsens peripheral vascular disease.

Theme 4 · Clinical features of cardiac disease

- 1 **D Mitral regurgitation** — An apical pansystolic murmur radiating to the axilla is mitral regurgitation.
- 2 **E Infective endocarditis** — Janeway lesions and splinter haemorrhages are vascular signs of infective endocarditis.
- 3 **F Rheumatic fever** — Erythema marginatum and subcutaneous nodules are major Jones criteria for acute rheumatic fever.
- 4 **A Aortic stenosis** — Slow-rising carotid pulse with an ejection murmur at the right upper sternal edge is aortic stenosis.
- 5 **I Pulmonary hypertension** — A left parasternal heave and loud P2 are signs of pulmonary hypertension; confirm with right heart catheter (mPAP > 20 mmHg).

SET 2 OF 2

THEME 6 Symptoms and signs of heart failure

- A Ankle swelling
- B Cough
- C Hepatomegaly
- D Orthopnoea
- E Paroxysmal nocturnal dyspnoea
- F Pink frothy sputum
- G Pulmonary fibrosis
- H Pulmonary oedema
- I Shortness of breath

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A patient has been on a cardiology ward for 3 weeks with a diagnosis of pulmonary oedema and is receiving large doses of furosemide. His systolic blood pressure is now 74 mmHg and rises when he is tilted head-down. He has finger clubbing, and fine end-inspiratory 'velcro' crackles are still heard throughout both lung fields.
- 2 A patient complains of waking in the middle of the night and racing out of bed to the window, gasping for breath.
- 3 A patient sleeps on four pillows at night so he can catch his breath.
- 4 A patient with a diagnosis of heart failure complains of this symptom several weeks after starting an ACE inhibitor.
- 5 Congestion due to right heart failure can give the liver a 'nutmeg' appearance and cause this sign.

Theme 5 · Murmurs

- 1 **H Tricuspid regurgitation** — Pansystolic murmur at the lower sternal edge with giant v waves and a pulsatile liver is TR (schistosomal pulmonary hypertension in Egypt).
- 2 **F Mitral regurgitation** — AF with an apex in the mid-axillary line and a loud pansystolic murmur is chronic severe MR.
- 3 **J Ventricular septal defect** — Down syndrome is associated with AV septal and ventricular septal defects; a pansystolic murmur fits a VSD.
- 4 **A Aortic regurgitation** — Collapsing pulse with an early diastolic murmur loudest sitting forward is aortic regurgitation.
- 5 **E Graham Steell murmur** — Graham Steell murmur: early diastolic pulmonary regurgitation from pulmonary hypertension (e.g. severe MS), louder on inspiration.

THEME 7 ECG findings

- A Anterior myocardial infarction
- B Atrial fibrillation
- C Atrial flutter with 2:1 block
- D Digitalis effect
- E Inferior myocardial infarction
- F Pericarditis
- G Posterior myocardial infarction
- H Sick sinus syndrome
- I Third-degree heart block
- J Ventricular tachycardia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 ST depression in V1–3 and tall R wave in V1 and V2.
- 2 Concave (saddle-shaped) ST elevation in most leads except aVR, with PR-segment depression.
- 3 ST elevation of 3 mm in leads II, III and aVF.
- 4 Broad QRS complexes at a regular rate of 45 beats/min; P waves are present at a faster rate but bear no relationship to the QRS complexes.
- 5 No P waves, narrow complex QRS rate of 150 beats/min with sawtooth baseline.

THEME 8 Heart disease

- A Acute myocardial infarction
- B Aortic stenosis
- C Atrial myxoma
- D Complete heart block
- E Cor pulmonale
- F Dilated cardiomyopathy
- G Fibrinous pericarditis
- H Infective endocarditis
- I Mitral valve stenosis
- J Tricuspid incompetence

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 58-year-old woman with known asymptomatic aortic stenosis presents with fever and feeling generally unwell. She has just had three fillings performed by her dentist.
- 2 A 49-year-old man presents to his GP with shortness of breath on walking up hills. He also experiences central chest pain which is relieved by resting, as well as occasional dizziness. He has a loud systolic murmur.
- 3 On the second day post-MI, a 71-year-old man develops pleuritic sounding chest pain. This is relieved by ibuprofen. The ECG demonstrates saddle-shaped ST changes.
- 4 A 63-year-old man presents to the emergency department with shortness of breath. He has a pansystolic murmur at the apex and a displaced apex beat. A chest X-ray shows a globular enlarged heart with a cardiothoracic ratio >50%. He admits to many years of heavy alcohol intake.
- 5 A 44-year-old woman with known COPD and a history of heavy smoking attends her respiratory clinic appointment. She now has ankle swelling and her exercise tolerance has fallen to about 15 metres.

THEME 9 Chest pain

- A Myocardial infarct
- B Pulmonary embolus
- C Pericarditis
- D Left lower lobe pneumonia
- E Rib fracture
- F Oesophageal reflux disease
- G Aortic dissection
- H Costochondritis
- I Angina pectoris
- J Herpes zoster

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 48-year-old woman is admitted with left-sided sharp chest discomfort that is worse on coughing. She has no sputum. She takes the combined oral contraceptive pill and propranolol. Her father and uncle died in their 60s from 'heart attacks'. She smokes 30 cigarettes a day. ECG: sinus tachycardia and right bundle-branch block. D-dimer 0.71 mg/L (reference <0.5); troponin I normal. Chest X-ray: normal.
- 2 A 58-year-old man with insulin-treated diabetes is admitted with intermittent central burning chest pain for the past month. It is worse after large meals and on lying flat and is eased by antacids. He has had atrial fibrillation and hypertension for 6 years and takes digoxin, atenolol, diclofenac and simvastatin. BMI 31 kg/m²; no chest wall tenderness; ejection systolic murmur in the aortic area. HbA1c 91 mmol/mol (10.5%), digoxin level 1.3 µg/L (therapeutic). ECG: rate-controlled AF with 1-mm sagging ST depression in V1–V6. An exercise stress test is negative. Chest X-ray: normal.
- 3 A 45-year-old man from Morocco attends the emergency department with a 1-week history of sharp central chest pain ('like a knife') that eases when he sits forward. There is no recent trauma. Examination is unremarkable. ECG: widespread concave ST elevation. Chest X-ray: scarring of the right apex and calcified hilar nodes. He confirms he was treated for tuberculosis in Morocco.
- 4 A 36-year-old man is brought to the emergency department with right-sided chest pain and breathlessness 2 days after falling while drunk. There is no fever. He is thin and smells of tobacco and alcohol. There is bruising and marked point tenderness over the right lateral chest wall with crepitus, and breath sounds are reduced on the right. Chest X-ray shows a small peripheral area devoid of lung markings on the right.
- 5 A 56-year-old man with insulin-treated diabetes is admitted at the request of a relative. He has had nagging chest pain for 2 days with a feeling of breathlessness. He is afebrile with no chest wall tenderness, clear lungs and normal heart sounds. Chest X-ray: normal. ECG: ST elevation of 3 mm in V4–V6 with T-wave inversion. Troponin I markedly raised.

Answers • Themes 6–9

Theme 6 • Symptoms and signs of heart failure

- 1 **G Pulmonary fibrosis** — Persistent velcro crackles with clubbing in a now volume-depleted patient is pulmonary fibrosis misdiagnosed as oedema; HRCT, reduce diuretic.

- 2 **E Paroxysmal nocturnal dyspnoea** — Waking at night gasping and rushing to the window is paroxysmal nocturnal dyspnoea, fairly specific for LV failure.
- 3 **D Orthopnoea** — Needing several pillows to breathe at night is orthopnoea; pillow count tracks severity.

- 4 **B Cough** — A dry cough weeks after starting an ACE inhibitor is from bradykinin; switch to an ARB after excluding pulmonary oedema.
- 5 **C Hepatomegaly** — Chronic congestion in right heart failure gives a 'nutmeg' liver and tender hepatomegaly.

Theme 7 • ECG findings

- 1 **G Posterior myocardial infarction** — ST depression with tall R waves in V1–V3 is a posterior MI (mirror image); V7–V9 show elevation.
- 2 **F Pericarditis** — Widespread saddle ST elevation (except aVR) with PR depression is pericarditis.
- 3 **E Inferior myocardial infarction** — ST elevation in II, III and aVF is inferior MI (RCA); check V4R and watch for AV block.
- 4 **I Third-degree heart block** — P waves unrelated to a slow broad escape rhythm (45/min) is third-degree heart block: pace.
- 5 **C Atrial flutter with 2:1 block** — Sawtooth flutter waves with a regular rate of 150/min is atrial flutter with 2:1 block.

Theme 8 • Heart disease

- 1 **H Infective endocarditis** — Fever after dental work with a known stenotic aortic valve is infective endocarditis until proven otherwise.
- 2 **B Aortic stenosis** — Exertional dyspnoea, angina and presyncope with a loud systolic murmur at 49 is symptomatic AS (likely bicuspid): valve replacement.

- 3 **G Fibrinous pericarditis** — Pleuritic pain with saddle ST changes on day 2 after MI is early fibrinous pericarditis; Dressler syndrome comes weeks later.
- 4 **F Dilated cardiomyopathy** — Heavy alcohol use with heart failure, functional MR and a globular heart is alcoholic DCM; abstinence may restore function.
- 5 **E Cor pulmonale** — Ankle oedema and severe limitation in COPD is cor pulmonale; LTOT and diuretics.

Theme 9 • Chest pain

- 1 **B Pulmonary embolus** — Pleuritic pain on the pill with tachycardia, new RBBB and raised D-dimer but normal CXR is PE; CTPA and anticoagulate meanwhile.
- 2 **F Oesophageal reflux disease** — Meal- and posture-related burning pain relieved by antacids on an NSAID is GORD; the sagging ST depression is digoxin effect. Stop NSAID, start PPI.
- 3 **C Pericarditis** — Positional pain with saddle ST elevation and previous TB with calcified hilar nodes suggests tuberculous pericarditis; echo and fluid ADA/culture.
- 4 **E Rib fracture** — Point tenderness and crepitus after a fall are rib fractures, here complicated by a small traumatic pneumothorax; analgesia and assess size.
- 5 **A Myocardial infarct** — ST elevation in V4–V6 with markedly raised troponin is an anterolateral STEMI; diabetics often have atypical pain. Reperfuse if < 12 h.

6 THEMES • 30 SCENARIOS • IRFAN EMQ COLLECTION

Irfan EMQs

SET 1 OF 2

THEME 1 Cardiac murmurs

- A Mitral stenosis
- B Aortic regurgitation
- C Atrial flutter
- D Tetralogy of Fallot
- E Atrial fibrillation
- F Atrial septal defect
- G Aortic stenosis
- H Pulmonary stenosis
- I Friction rub
- J Mitral regurgitation
- K Tricuspid regurgitation
- L Ventricular septal defect

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Tapping apex beat, loud S1, mid-diastolic murmur loudest at the apex in expiration lying on the left side.
- 2 Heaving undisplaced apex beat, absent A2 with ejection systolic murmur radiating to the carotids.
- 3 Pansystolic murmur heard best at lower left sternal edge during inspiration in a patient with pulsatile hepatomegaly.

- 4 Displaced, volume-overloaded apex. Soft S1, pansystolic murmur at apex radiating to axilla.
- 5 A 25-year-old man has a left parasternal heave and a harsh pansystolic murmur with a thrill at the lower left sternal edge, also audible at the apex. The murmur does not vary with respiration and the JVP is normal.

THEME 2 Jugular venous pressure

- A Tricuspid stenosis
- B Complete heart block
- C Ventricular fibrillation
- D Left heart failure
- E Tricuspid regurgitation
- F Mitral stenosis
- G Constrictive pericarditis
- H Aortic stenosis
- I Aortic regurgitation
- J Superior vena cava obstruction
- K Atrial fibrillation
- L Normal JVP

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Elevated JVP with absent pulsation.

- 2 Giant systolic 'v' waves.
- 3 Large 'a' waves and slow 'y' descent in JVP. Patient has ascites.
- 4 Cannon 'a' waves.
- 5 Raised JVP that rises on inspiration.

THEME 3 ECG abnormalities (Set 1)

- A Subendocardial infarction
- B Pulmonary embolus
- C Right bundle-branch block
- D Mobitz type II second-degree heart block
- E Hyperkalaemia
- F Inferior myocardial infarction
- G Anterolateral myocardial infarction
- H Left bundle-branch block
- I Hypokalaemia
- J Pericarditis
- K Atrial fibrillation
- L Pulmonary hypertension
- M Mitral stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 26-year-old woman presents acutely unwell with shortness of breath. Her ECG shows sinus tachycardia, deep S waves in I, inverted T waves in III and Q waves in III.
- 2 An ECG shows a dominant R wave with an RSR' pattern in V1, inverted T waves in V1–V3, deep wide S waves in I and V6, and a QRS duration of 140 ms.
- 3 Prolonged P–R interval, depressed ST, flattened T waves, prominent U waves.
- 4 An ECG shows sinus rhythm with broad (0.14 s), notched, bifid P waves best seen in lead II and a biphasic P wave with a deep negative terminal component in V1.
- 5 A 65-year-old man presents with chest pain radiating to the jaw. The ECG shows ST segment elevation in II, III and aVF, with T-wave inversion in V5 and V6.

THEME 4 ECG abnormalities (Set 2)

- A Hyperkalaemia
- B Atrial fibrillation
- C Mitral regurgitation
- D Hypocalcaemia
- E Left bundle-branch block
- F Atrial flutter
- G Normal ECG
- H Ventricular tachycardia
- I First-degree heart block
- J Right bundle-branch block
- K Cardiac tamponade
- L Hypokalaemia
- M Hypercalcaemia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 The ECG of a 55-year-old man being treated for hypertension with ramipril and spironolactone shows tall tented T waves.
- 2 A 34-year-old man is brought to the emergency department after a road traffic accident in which his chest struck the steering wheel. His neck veins are distended, the trachea is central and breath sounds are equal on both sides. He then loses his pulse while the monitor still shows organised electrical activity (pulseless electrical activity).
- 3 An 85-year-old man with pneumonia complains of palpitations. The ECG shows an irregularly irregular narrow-complex rhythm with absent P waves.
- 4 ECG of a 45-year-old man with sarcoidosis shows an 'M' pattern in V5 and inverted T waves in I, aVL and V5–V6.
- 5 ECG of an 8-year-old girl shows normal P waves and QRS complexes but shows T-wave inversion in V1.

THEME 5 Treatment of heart failure

- A Nifedipine
- B Metolazone
- C Lidocaine
- D Sit upright, controlled oxygen if hypoxaemic, intravenous furosemide and sublingual glyceryl trinitrate
- E Oral isosorbide mononitrate
- F Digoxin
- G Oxygen, oral morphine, oral furosemide and intravenous glyceryl trinitrate
- H Intravenous adenosine
- I Spironolactone
- J Start cardiopulmonary resuscitation (CPR)
- K Oral furosemide

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man with heart failure requires rate control to treat coexisting atrial fibrillation.
- 2 A 65-year-old woman being treated with large doses of loop diuretic requires add-on therapy for oedema refractory to treatment.
- 3 A 69-year-old woman with asthma being treated with a loop diuretic, ACE inhibitor and long-acting nitrate is prescribed a drug to reduce long-term mortality.
- 4 A 70-year-old woman with a history of chronic heart failure presents with severe pulmonary oedema.
- 5 Treatment of mild symptoms of shortness of breath and ankle oedema in a 65-year-old man with left ventricular dysfunction caused by ischaemic heart disease. He is already taking an ACE inhibitor.

Answers · Themes 1-5

IRFAN EMQS

Theme 1 · Cardiac murmurs

- 1 **A Mitral stenosis** — Tapping apex, loud S1 and apical mid-diastolic rumble (left lateral, expiration) is mitral stenosis.
- 2 **G Aortic stenosis** — Heaving undisplaced apex, absent A2 and ejection murmur to the carotids indicate severe calcific aortic stenosis.
- 3 **K Tricuspid regurgitation** — Pansystolic murmur at the lower left sternal edge louder on inspiration with a pulsatile liver is tricuspid regurgitation.
- 4 **J Mitral regurgitation** — Displaced volume-loaded apex, soft S1 and apical pansystolic murmur to the axilla is chronic mitral regurgitation.
- 5 **L Ventricular septal defect** — Harsh pansystolic murmur with thrill at the lower left sternal edge, not varying with respiration and with a normal JVP, is a VSD.

Theme 2 · Jugular venous pressure

- 1 **J Superior vena cava obstruction** — A raised non-pulsatile JVP means obstruction above the RA: superior vena cava obstruction (lung cancer, lymphoma).
- 2 **E Tricuspid regurgitation** — Giant systolic v waves occur when the RV ejects back through an incompetent tricuspid valve: tricuspid regurgitation.
- 3 **A Tricuspid stenosis** — Large a waves with a slow y descent and ascites indicate tricuspid stenosis, usually rheumatic with mitral stenosis.
- 4 **B Complete heart block** — Cannon a waves (atrium contracting against a closed tricuspid valve) indicate AV dissociation, typically complete heart block.
- 5 **G Constrictive pericarditis** — A JVP rising on inspiration (Kussmaul sign) is classic of constrictive pericarditis.

Theme 3 · ECG abnormalities (Set 1)

- 1 **B Pulmonary embolus** — Sinus tachycardia is the commonest ECG finding in PE; S1Q3T3 reflects RV strain but is insensitive. Use Wells, D-dimer and CTPA.
- 2 **C Right bundle-branch block** — RSR' in V1 with T inversion V1–V3, broad S in I and V6 and QRS ≥ 120 ms is right bundle-branch block.
- 3 **I Hypokalaemia** — Hypokalaemia: long PR, ST depression, flat T waves and prominent U waves.

- 4 **M Mitral stenosis** — Broad bifid P mitrale in II with a deep negative terminal P in V1 indicates left atrial enlargement, classically mitral stenosis.
- 5 **F Inferior myocardial infarction** — ST elevation in II, III and aVF is inferior MI (usually RCA): primary PCI and check V4R for RV infarction.

Theme 4 · ECG abnormalities (Set 2)

- 1 **A Hyperkalaemia** — Tall tented T waves are the earliest sign of hyperkalaemia (ACEi + spironolactone); check K⁺ and give IV calcium if ECG changes.
- 2 **K Cardiac tamponade** — PEA after blunt chest trauma with distended neck veins, central trachea and equal breath sounds is tamponade: immediate pericardiocentesis.
- 3 **B Atrial fibrillation** — Irregularly irregular narrow complexes without P waves is atrial fibrillation, often triggered by acute illness such as pneumonia.
- 4 **E Left bundle-branch block** — Broad QRS with M-shaped R in V5–V6 and lateral T inversion is LBBB; cardiac sarcoid infiltrates the conducting system.
- 5 **G Normal ECG** — T-wave inversion in V1 (often V2–V3) is a normal juvenile pattern in children.

Theme 5 · Treatment of heart failure

- 1 **F Digoxin** — Digoxin slows AF rate without negative inotropy, suiting heart failure (with a beta-blocker). Avoid verapamil and diltiazem.
- 2 **B Metolazone** — Metolazone (thiazide-like) acts synergistically with loop diuretics in resistant oedema; monitor K⁺, Na⁺ and renal function closely.
- 3 **I Spironolactone** — Spironolactone reduces HFrEF mortality (RALES) added to an ACE inhibitor; a beta-blocker is avoided here because of asthma.
- 4 **D Sit upright, controlled oxygen if hypoxaemic, intravenous furosemide and sublingual glyceryl trinitrate** — Acute pulmonary oedema: sit up, oxygen if hypoxaemic, IV furosemide and nitrates if SBP > 110; routine opiates are no longer advised.
- 5 **K Oral furosemide** — Oral furosemide relieves congestion but not mortality; use the lowest effective dose while prognostic therapy is optimised.

SET 2 OF 2

THEME 6 Cardiovascular emergencies

- A Intravenous dobutamine
- B DC shock and adrenaline
- C Oxygen, subcutaneous low-molecular-weight heparin and intravenous fluids
- D Sublingual nifedipine
- E Atropine
- F Aspirin and heparin
- G DC shock and atropine
- H Intravenous labetalol
- I Emergency renal dialysis
- J Aspirin and streptokinase
- K Aspirin, a P2Y12 inhibitor and primary PCI

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 57-year-old businessman presents with a 4-h history of crushing chest pain. The ECG shows ST elevation in II, III and aVF.
- 2 A 65-year-old man presenting with chest pain becomes unresponsive. His ECG shows ventricular fibrillation.
- 3 A 40-year-old woman collapses after a flight with breathlessness and right-sided pleuritic chest pain.
- 4 A 45-year-old man with chronic glomerulonephritis presents with severe headache, papilloedema and bilateral retinal haemorrhages. His BP is 240/132 mmHg.
- 5 A 55-year-old man requires immediate pharmacological management for severe symptomatic sinus bradycardia.

Answers • Theme 6

IRFAN EMQS

Theme 6 • Cardiovascular emergencies

- 1 **K Aspirin, a P2Y12 inhibitor and primary PCI** — Inferior STEMI < 12 h: aspirin, P2Y12 loading and primary PCI within 120 min; fibrinolysis only if PCI is not available in time.
- 2 **B DC shock and adrenaline** — VF: immediate shock and CPR; adrenaline 1 mg and amiodarone 300 mg after the third shock.
- 3 **C Oxygen, subcutaneous low-molecular-weight heparin and intravenous fluids** — Collapse with breathlessness and

pleuritic pain after a flight is PE: oxygen, therapeutic anticoagulation and fluids; thrombolysis if unstable.

- 4 **H Intravenous labetalol** — Malignant hypertension (papilloedema, haemorrhages) from glomerulonephritis: IV labetalol in a monitored bed, lowering MAP $\leq$ 25%. Avoid sublingual nifedipine.
- 5 **E Atropine** — Symptomatic sinus bradycardia: IV atropine 500 μ g, repeated to 3 mg; then isoprenaline/adrenaline or transcutaneous pacing.

19 THEMES • 94 SCENARIOS • AFTER GET AHEAD! MEDICINE EMQS

Get Ahead EMQs

SET 1 OF 4

THEME 1 Management of hypertension

- A Admit to high-dependency unit for monitoring and intravenous antihypertensive medication
- B Admit to hospital for oral antihypertensive medication
- C Ambulatory blood pressure monitoring
- D Confirm over 4 weeks and treat if persistent
- E Confirm over 12 weeks and treat if persistent
- F Lifestyle advice only
- G No further assessment required
- H Start oral antihypertensive treatment as an outpatient
- I Refer for investigation of secondary hypertension
- J Provide lifestyle advice and reassess annually
- K Sublingual nifedipine

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 63-year-old man is found by his GP to have a blood pressure of 135/60 mmHg. He is otherwise fit and well and has no significant past medical history.
- 2 A 63-year-old man with newly diagnosed type 2 diabetes has a blood pressure of 155/89 mmHg at his first outpatient appointment. Fundoscopy shows arteriovenous nicking but no haemorrhages or papilloedema.
- 3 A 65-year-old woman has a blood pressure persistently above 160/90 mmHg over a 2-month assessment period with her GP, despite repeated lifestyle advice. She is otherwise fit and well and asymptomatic.

- 4** A 67-year-old man presents to the emergency department with headaches. His blood pressure is 220/120 mmHg. Fundoscopy shows some arteriovenous nipping but no haemorrhages, exudates or papilloedema, and he has no other symptoms or signs of acute organ damage.
- 5** A 62-year-old man is brought to the emergency department by his wife with headaches and blurred vision. He is orientated in time and person but not in place. His blood pressure is 240/140 mmHg and heart rate 84 beats/min. Fundoscopy shows bilateral papilloedema and flame haemorrhages. Neurological examination is otherwise normal, and an emergency head CT shows no neurological cause for his symptoms.

THEME 2 Chronic heart failure

- A** Angiotensin II receptor blockers
- B** Angiotensin-converting enzyme inhibitors
- C** Amlodipine
- D** Beta-blockers
- E** B-type natriuretic peptide (BNP or NT-proBNP)
- F** Digoxin
- G** Lipid profile
- H** Loop diuretics
- I** Verapamil
- J** Nicorandil
- K** Spironolactone
- L** Thiazide diuretics
- M** Thyroid function tests
- N** Transoesophageal echocardiography
- O** Transthoracic Doppler echocardiography
- P** Troponin I

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** An 80-year-old woman with ischaemic heart disease presents to the emergency department with a 3-month history of worsening breathlessness and swollen ankles. Her ECG shows no abnormality, but you suspect heart failure. Which simple blood test could effectively rule out heart failure?
- 2** An 80-year-old woman with suspected heart failure has a raised NT-proBNP. Which investigation should be requested next to confirm chronic heart failure?
- 3** An 80-year-old woman has newly confirmed heart failure with reduced ejection fraction and is starting a beta-blocker. Which class of drug acting on the renin-angiotensin system is recommended first line, reduces mortality and helps reverse left ventricular remodeling?
- 4** An 80-year-old woman admitted with chronic heart failure is still breathless with ankle oedema 2 days after admission. Which class of diuretic is the first-line symptomatic treatment of fluid overload in chronic heart failure?

- 5** An 80-year-old woman with heart failure with reduced ejection fraction remains breathless, especially when lying flat, 6 months after diagnosis. She takes enalapril, bisoprolol, aspirin and furosemide at optimised doses, and her potassium and renal function are normal. Which additional treatment should be prescribed?

THEME 3 Acute ECG interpretation

- A** 3rd-degree heart block
- B** Atrial fibrillation
- C** Atrial flutter
- D** Bifid P-waves
- E** Delta wave
- F** J-waves
- G** Sinus arrhythmia
- H** Sinus tachycardia
- I** Tented T-waves
- J** U-waves

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 34-year-old woman has been suffering from profuse diarrhoea and vomiting for 4 days. She is now complaining of extreme muscle weakness and muscle cramps.
- 2** A 55-year-old man known to be in end-stage renal failure presents to the emergency department feeling generally unwell and experiencing palpitations. He admits to missing his last haemodialysis appointment as he went away on holiday.
- 3** A 40-year-old woman presents with left-sided chest pain that is worse on breathing in. She is also experiencing shortness of breath on exertion and has coughed up a small amount of frank blood. She is otherwise well, with no significant past medical history. Her only regular medication is the oral contraceptive pill.
- 4** An 85-year-old man with dementia is brought to the emergency department by the ambulance after escaping from his care home during the night. He was found wandering on the local moors by a farmer in the early hours of the morning. He is shivering and confused. His core temperature is 31°C.
- 5** A 32-year-old man is brought to the emergency department after a road traffic collision. He has no obvious injuries but is anxious and in pain, with a regular pulse of 118 beats/min.

THEME 4 Complications of acute MI

- A First-degree heart block
- B Acute mitral valve regurgitation
- C Acute pericarditis
- D Atrial ectopic beats
- E Atrial fibrillation
- F Cardiac tamponade
- G Complete (third-degree) heart block
- H Dressler's syndrome
- I Pulmonary embolism
- J Subacute left ventricular failure

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old woman had an inferior myocardial infarction 2 days ago. She has become acutely short of breath and is coughing up pink frothy sputum. There is a new pansystolic murmur at the apex radiating to the axilla, and bilateral crackles are heard at the lung bases.
- 2 A 58-year-old man complains of central chest pain, fever and lethargy 4 weeks after suffering a myocardial infarction. The chest pain is worse on inspiration, is stabbing in nature, and is relieved by ibuprofen and sitting forward.
- 3 You are asked to review a 72-year-old woman who had a myocardial infarction 12 hours ago. Her heart rate is 40 beats/min and her blood pressure is 90/60 mmHg. The ECG shows abnormally wide QRS complexes occurring at a rate of 40 beats/min and P-waves occurring at a rate of 90 beats/min. There is no relationship between the P-waves and QRS complexes.
- 4 A 66-year-old man, known to have had an anterior myocardial infarction 2 days previously, collapses. On examination, you notice that he has a raised jugular venous pressure that rises with inspiration. Auscultation seems normal. His blood pressure is 80/50 mmHg and his heart rate is 100 beats/min. The ECG shows sinus tachycardia with small QRS complexes.
- 5 A 60-year-old man who had a myocardial infarction 4 weeks ago presents to his GP complaining of increasing shortness of breath since discharge. It is worse at night time on lying flat, and he occasionally wakes up acutely short of breath in the early hours of the morning.

THEME 5 Secondary hypertension

- A Acromegaly
- B Adult polycystic kidney disease
- C Aortic dissection
- D Berry aneurysm
- E Coarctation of the aorta
- F Conn's syndrome
- G Cushing's syndrome
- H Non-steroidal anti-inflammatory drug use
- I Oral contraception
- J Pheochromocytoma
- K Pre-eclampsia
- L Stress at work
- M White coat syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 30-year-old man presents to the emergency department with headache and muscle weakness. He also complains of increased thirst and urine output. Apart from a blood pressure of 190/95 mmHg, examination is normal. Serum potassium is 2.2 mmol/L, sodium 150 mmol/L, urea 4.3 mmol/L and creatinine 69 µmol/L.
- 2 A previously fit and well 35-year-old female doctor presents to her GP with a history of 'panic attacks', during which she feels extremely anxious and experiences palpitations, chest tightness and headaches. While in the surgery, she experiences a similar episode, becoming flushed and anxious. Her heart rate was 120 beats/min and her blood pressure reached 185/80 mmHg.
- 3 A 27-year-old woman with known Turner's syndrome presents to her GP with recurrent nose bleeds. On examination, her blood pressure is 190/95 mmHg. Auscultation reveals a mid-late systolic murmur best heard in between the shoulder blades.
- 4 A 45-year-old man was found to have a persistently raised blood pressure by his GP. On examination, a vague mass could be felt on the left side of the abdomen. He mentioned that his father died at 26 years of age following a 'brain haemorrhage'.
- 5 A 55-year-old woman is noted to have an elevated blood pressure on three separate occasions by her GP. She has a recent history of persisting acne and hirsutism, for which a dermatology referral has been made. Marked bruising is observed on examination.

Answers • Themes 1-5

Theme 1 • Management of hypertension

- 1 **J Provide lifestyle advice and reassess annually** — Clinic BP < 140/90 is not hypertension: lifestyle advice and periodic re-measurement.

- 2 **C Ambulatory blood pressure monitoring** — A clinic reading of 140/90–179/119 must be confirmed by ABPM (or HBPM) before diagnosis (NICE NG136); then treat promptly given his diabetes.
- 3 **H Start oral antihypertensive treatment as an outpatient** — Confirmed persistent hypertension despite lifestyle change

needs drug treatment started in primary care; admission is unnecessary.

- 4 **H Start oral antihypertensive treatment as an outpatient** — BP 220/120 without acute organ damage is hypertensive urgency; start oral treatment and review early; rapid lowering is dangerous.
- 5 **A Admit to high-dependency unit for monitoring and intravenous antihypertensive medication** — BP 240/140 with papilloedema, haemorrhages and confusion is a hypertensive emergency; HDU, arterial line, IV labetalol/nitroprusside.

Theme 2 · Chronic heart failure

- 1 **E B-type natriuretic peptide (BNP or NT-proBNP)** — A normal NT-proBNP (< 400 ng/L non-acute) makes heart failure very unlikely; above 2000 ng/L, echo within 2 weeks.
- 2 **O Transthoracic Doppler echocardiography** — Transthoracic Doppler echo confirms heart failure and its type (EF), valve disease and wall motion.
- 3 **B Angiotensin-converting enzyme inhibitors** — An ACE inhibitor with a beta-blocker is first line in HFrEF, reducing mortality and reversing remodelling; ARB if intolerant.
- 4 **H Loop diuretics** — Loop diuretics are first-line for fluid overload in heart failure; thiazides are add-on for resistance.
- 5 **K Spironolactone** — Still symptomatic on ACEi and beta-blocker: add spironolactone (RALES ~30% mortality reduction) and an SGLT2 inhibitor; check K⁺.

Theme 3 · Acute ECG interpretation

- 1 **J U-waves** — Diarrhoea and vomiting cause hypokalaemia: weakness, cramps and prominent U waves. Replace K⁺ and check magnesium.
- 2 **I Tented T-waves** — Missed dialysis causes hyperkalaemia, first seen as tall tented T waves; give IV calcium, insulin-glucose and urgent dialysis.
- 3 **H Sinus tachycardia** — Pleuritic pain, haemoptysis and breathlessness on the pill suggest PE, whose commonest ECG finding is sinus tachycardia; confirm with CTPA.
- 4 **F J-waves** — Hypothermia (31 °C) produces J (Osborn) waves at the QRS–ST junction; rewarm gently as the cold heart is prone to VF.

- 5 **H Sinus tachycardia** — Pain and anxiety after trauma cause sinus tachycardia; persistent tachycardia should prompt a search for occult bleeding.

Theme 4 · Complications of acute MI

- 1 **B Acute mitral valve regurgitation** — Pulmonary oedema with a new apical pansystolic murmur 2–7 days after inferior MI is acute MR from posteromedial papillary rupture; urgent echo and surgery.
- 2 **H Dressler's syndrome** — Fever and pleuritic pain relieved by sitting forward 4 weeks after MI is Dressler syndrome; aspirin/NSAID plus colchicine.
- 3 **G Complete (third-degree) heart block** — P waves at 90/min unrelated to broad QRS at 40/min is complete heart block; with hypotension give atropine and temporary pacing.
- 4 **F Cardiac tamponade** — Collapse 2–5 days after anterior MI with hypotension, raised JVP and small QRS complexes is tamponade from free-wall rupture.
- 5 **J Subacute left ventricular failure** — Orthopnoea and PND weeks after MI indicate left ventricular failure; echo, diuretics and prognostic HF therapy.

Theme 5 · Secondary hypertension

- 1 **F Conn's syndrome** — Hypertension with K⁺ 2.2, high-normal sodium and polyuria is Conn's syndrome; aldosterone:renin ratio, then adrenal CT and venous sampling.
- 2 **J Pheochromocytoma** — Paroxysmal anxiety, palpitations, headache and flushing with hypertension suggest pheochromocytoma; measure metanephrines, alpha-block first.
- 3 **E Coarctation of the aorta** — Turner syndrome with hypertension and an interscapular systolic murmur is coarctation; check radiofemoral delay.
- 4 **B Adult polycystic kidney disease** — Hypertension, flank mass and a family history of early brain haemorrhage (berry aneurysm) point to ADPKD; confirm by renal ultrasound.
- 5 **G Cushing's syndrome** — Hypertension with acne, hirsutism and bruising suggests Cushing's syndrome; screen with overnight dexamethasone suppression, then ACTH.

SET 2 OF 4

THEME 6 Heart sounds

- A** Austin Flint murmur
- B** Continuous murmur radiating to the back
- C** Ejection systolic murmur heard at the right second intercostal space only
- D** Ejection systolic murmur heard best in the right second intercostal space that radiates into the right carotid artery
- E** Graham Steell murmur
- F** Muffled heart sounds
- G** Mid-systolic click loudest at the apex
- H** Fixed splitting of the second heart sound
- I** Normal heart sounds
- J** Opening snap best heard at the apex
- K** Pansystolic murmur heard best at the apex with radiation into the axilla
- L** Physiological splitting of the second heart sound
- M** Reverse splitting of the second heart sound
- N** Summation gallop rhythm

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 67-year-old man presents with symptoms of acute left ventricular failure. On examination, you notice that his heart rate is 120 beats/min. On auscultation, there are more than two heart sounds present, but you cannot distinguish them from each other.
- 2** A 55-year-old man presents to the emergency department with a severe headache. He is known to have mitral valve regurgitation.
- 3** You are called to see an 87-year-old female inpatient who is complaining of chest pain. On examination, you notice a murmur. A brief look in her hospital notes confirms that she has known ischaemic heart disease, hypertension and aortic valve sclerosis.
- 4** A 65-year-old woman with known systemic lupus erythematosus presents with central chest pain that is very similar to the pain that she feels when suffering from pericarditis. Echocardiography demonstrates a pericardial effusion that is likely to be chronic. On auscultation, the heart sounds do not sound as they should.
- 5** A 45-year-old man presents with abdominal pain. He is known to have Marfan's syndrome with advanced aortic valve regurgitation.

THEME 7 Signs of cardiovascular disease

- A** First-degree heart block
- B** Third-degree heart block
- C** Acute aortic regurgitation
- D** Aortic stenosis
- E** Atrial fibrillation
- F** Atrial flutter
- G** Chronic aortic regurgitation
- H** Heterozygous familial hypercholesterolaemia
- I** Homozygous familial hypercholesterolaemia
- J** Mitral regurgitation
- K** Mobitz type II heart block
- L** Myocardial infarction
- M** Rheumatic fever
- N** Secondary hypercholesterolaemia
- O** Subacute bacterial endocarditis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** On examination of a 75-year-old man who presented with shortness of breath, mild chest pain and palpitations, you note that he has an irregularly irregular pulse.
- 2** You are asked to see a 37-year-old man who presented to the emergency department with chest pain. He is extremely anxious, as both his father and grandfather died suddenly in their 30s. On examination, you notice a pale ring around his cornea and yellow plaques just below his eye. When you mention these findings to the patient, he says that he has similar lesions on his elbows and within his Achilles' tendon.
- 3** On examination of a 50-year-old man who presented with fever, weight loss and lethargy, you notice small linear red lesions beneath his fingernails.
- 4** While examining a 62-year-old man who presented in a state of collapse, you note that his heart rate is only 40 beats/min. Additionally, there appears to be an abnormally large wave within the jugular venous pressure waveform.
- 5** While examining a 65-year-old man who is being treated for a suspected type A aortic dissection with sudden acute pulmonary oedema, you hear a short, soft early diastolic murmur. He is tachycardic and hypotensive, and his pulse pressure is not widened.

THEME 8 Side-effects of cardiovascular medication

- A** Amlodipine
- B** Amiodarone
- C** Atenolol
- D** Candesartan
- E** Dalteparin
- F** Digoxin
- G** Furosemide
- H** Glyceryl trinitrate
- I** Ramipril
- J** Sando-K
- K** Simvastatin
- L** Spironolactone
- M** Streptokinase
- N** Verapamil

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 64-year-old man develops a dry cough about 1 week after starting a new blood pressure medication.
- 2** A 58-year-old woman develops bilateral swelling of her ankles 2 weeks after her GP has prescribed an anti-hypertensive drug.
- 3** A 74-year-old man is being treated for left ventricular failure as an inpatient. He is noticed to have U-waves on a routine ECG.
- 4** A 56-year-old woman presents to her GP with profound muscular pain, weakness and fever 5 days after beginning a course of an unknown 'heart tablet'.
- 5** A 55-year-old man who is being treated for ST-elevation myocardial infarction in the emergency department resuscitation room suddenly becomes hypotensive.

THEME 9 Drugs used during cardiac arrest

- A** Adrenaline
- B** Amiodarone
- C** Atropine
- D** Calcium gluconate
- E** Digoxin
- F** Dopamine
- G** Magnesium sulfate
- H** Naloxone
- I** Salbutamol

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** This drug is used during cardiac arrest to increase coronary perfusion pressure.
- 2** This anti-arrhythmic drug is used in the standard management of ventricular fibrillation and pulseless ventricular tachycardia.
- 3** This drug is the first-line treatment of peri-arrest bradycardia with adverse features, given as 500 micrograms IV and repeated to a maximum of 3 mg.

- 4** This drug is given in cardiac arrest or peri-arrest instability caused by polymorphic ventricular tachycardia (torsades de pointes), or when hypomagnesaemia is suspected.
- 5** This drug should be considered for use in cardiac arrest secondary to hyperkalaemia.

THEME 10 Diagnosis of chest pain

- A** Acute myocardial infarction
- B** Aortic dissection
- C** Boerhaave's syndrome
- D** Decubitus angina
- E** Gastro-oesophageal reflux disease
- F** Herpes zoster
- G** Mitral valve prolapse
- H** Oesophageal spasm
- I** Pericarditis
- J** Pulmonary oedema
- K** Pulmonary embolism
- L** Stable angina
- M** Tietze's syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 60-year-old man with known hypertension presents to the emergency department with a 2-hour history of central chest pain and interscapular back pain. Over the next 30 minutes, he develops slurred speech and reduced power and sensation on his left side.
- 2** A 30-year-old woman with a known congenital heart defect presents to her GP with a 4-week history of left-sided chest pain and palpitations that are worse on exertion. The pain does not radiate, has no other associated symptoms and is not worse on inspiration. On examination, there is a mid-systolic click best heard at the apex on auscultation of the heart.
- 3** A 70-year-old man presents to his GP following an episode of central crushing chest pain that started 30 minutes after going to bed. The pain did not radiate and there were no associated symptoms. Following two puffs of sublingual glyceryl trinitrate spray, his chest pain resolved. The ECG shows no abnormalities. This is the third time in the last 2 weeks that these symptoms have occurred on going to bed.
- 4** A 36-year-old man with alcohol dependence presents to the emergency department with a 5-hour history of vomiting followed by chest pain. His heart sounds are normal. There is reduced air entry at the left lung base, which is also dull to percussion, and you feel crepitus (surgical emphysema) in the neck. The ECG shows sinus tachycardia.

- 5 A 70-year-old man presents to the emergency department with retrosternal chest pain that is crushing in nature. This has happened on several occasions previously, and usually occurs after drinking hot fluids. Despite there being no ECG changes, he is given glyceryl trinitrate, which partially relieves the pain. After 30 minutes, the pain relieves spontaneously.

Answers · Themes 6–10

GET AHEAD EMQS

Theme 6 · Heart sounds

- 1 **N Summation gallop rhythm** — In acute LVF at fast rates S3 and S4 fuse into one loud extra sound: a summation gallop.
- 2 **K Pansystolic murmur heard best at the apex with radiation into the axilla** — MR gives a pansystolic murmur at the apex radiating to the axilla; loudness does not reflect severity.
- 3 **C Ejection systolic murmur heard at the right second intercostal space only** — Aortic sclerosis gives an ejection murmur at the right 2nd space without carotid radiation, with a normal pulse and A2.
- 4 **F Muffled heart sounds** — A large chronic pericardial effusion muffles the heart sounds; look for tamponade signs (pulsus paradoxus, raised JVP).
- 5 **A Austin Flint murmur** — Severe AR can cause an Austin Flint murmur (apical mid-diastolic rumble from the jet hitting the mitral leaflet), without opening snap.

Theme 7 · Signs of cardiovascular disease

- 1 **E Atrial fibrillation** — An irregularly irregular pulse is characteristic of AF; confirm by ECG and assess CHA₂DS₂-VASc.
- 2 **H Heterozygous familial hypercholesterolaemia** — Arcus, xanthelasma and tendon xanthomas with premature family CAD is heterozygous familial hypercholesterolaemia (~1 in 250); cascade testing.
- 3 **O Subacute bacterial endocarditis** — Fever, weight loss and splinter haemorrhages suggest subacute bacterial endocarditis; 3 blood cultures and echo.
- 4 **B Third-degree heart block** — Bradycardia with intermittent cannon a waves and collapse is complete heart block (Stokes–Adams); permanent pacing.
- 5 **C Acute aortic regurgitation** — Acute AR (e.g. type A dissection) causes shock with a short soft diastolic murmur and no wide pulse pressure; needs emergency surgery.

Theme 8 · Side-effects of cardiovascular medication

- 1 **I Ramipril** — ACE inhibitors (ramipril) cause dry cough in ~10% via bradykinin; switch to an ARB.
- 2 **A Amlodipine** — Amlodipine causes dose-dependent ankle oedema from precapillary dilatation; it responds poorly to diuretics.

- 3 **G Furosemide** — Furosemide causes hypokalaemia, giving U waves, flat T waves and ST depression.
- 4 **K Simvastatin** — Statins (simvastatin) can cause myositis and rhabdomyolysis, especially with interacting drugs; stop it and check CK.
- 5 **M Streptokinase** — Streptokinase commonly causes hypotension and allergy (bacterial protein) and must not be reused because of antibodies.

Theme 9 · Drugs used during cardiac arrest

- 1 **A Adrenaline** — Adrenaline's alpha effect raises aortic diastolic and coronary perfusion pressure; 1 mg IV every 3–5 min.
- 2 **B Amiodarone** — Amiodarone 300 mg IV after the third shock (150 mg after the fifth) is the standard antiarrhythmic in VF/pVT.
- 3 **C Atropine** — Atropine 500 µg IV (to 3 mg) blocks vagal tone and is first-line for unstable bradycardia; no longer used in asystole/PEA.
- 4 **G Magnesium sulfate** — Magnesium sulfate 2 g IV suppresses torsades de pointes and corrects hypomagnesaemia.
- 5 **D Calcium gluconate** — In hyperkalaemic arrest IV calcium stabilises the myocardium; it does not lower K⁺ (insulin–glucose, dialysis do).

Theme 10 · Diagnosis of chest pain

- 1 **B Aortic dissection** — Tearing chest and interscapular pain then a stroke in a hypertensive man suggests dissection into the carotids; CT aortography, no thrombolysis.
- 2 **G Mitral valve prolapse** — Atypical chest pain and palpitations with an apical mid-systolic click is mitral valve prolapse; echo confirms.
- 3 **D Decubitus angina** — Angina on lying flat (decubitus angina) reflects increased venous return and implies severe CAD or LV dysfunction; refer for angiography.
- 4 **C Boerhaave's syndrome** — Chest pain and surgical emphysema after forceful vomiting in a drinker is Boerhaave syndrome; CT with water-soluble contrast and surgery.
- 5 **H Oesophageal spasm** — Crushing pain triggered by hot or cold drinks, partly relieved by GTN, with a normal ECG is diffuse oesophageal spasm.

SET 3 OF 4

THEME 11 Diseases of the myocardium and pericardium

- A Arrhythmogenic right ventricular cardiomyopathy
- B Atrial myxoma
- C Cardiac tamponade
- D Constrictive pericarditis
- E Dilated cardiomyopathy
- F Hypertrophic obstructive cardiomyopathy
- G Myocarditis
- H Non-ST-elevation myocardial infarction
- I Restrictive cardiomyopathy
- J Subacute bacterial endocarditis
- K Uraemic pericarditis
- L Viral pericarditis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 23-year-old amateur footballer collapses and dies while training, with no contact from other players. He had two previous episodes of syncope at the gym for which no cause was found, and a harsh ejection systolic murmur that became louder on standing had been noted at a pre-season medical.
- 2 A 46-year-old man presents to the GP with a history of shortness of breath that occurs on exertion and when lying down at night. On general inspection, he notices multiple spider naevi on his chest and face. On examination of his heart, the apex beat is displaced laterally and there is a pansystolic murmur that is difficult to identify since the heartbeat is irregularly irregular.
- 3 A 64-year-old woman is referred for echocardiography after a history of weight loss, low-grade fever, non-specific chest pain and palpitations. The echocardiogram shows a pedunculated mass in the left atrium attached to the interatrial septum.
- 4 A 45-year-old man presents to the emergency department with stabbing central chest pain that is worse when lying down at night. Potassium is 5.4 mmol/L, sodium 145 mmol/L, urea 15 mmol/L and creatinine 540 µmol/L.
- 5 A 46-year-old woman presents to the emergency department with a short history of worsening shortness of breath and a reduced exercise tolerance. She has been suffering with lethargy, muscle aches, fever and palpitations for 2 weeks. Routine blood tests show a raised white blood cell count and a raised C-reactive protein. A plasma troponin returns at 1.2 ng/mL.

THEME 12 Tachycardias and tachyarrhythmias

- A Atrioventricular node re-entry tachycardia
- B Chronic atrial fibrillation
- C Multifocal atrial tachycardia
- D Paroxysmal atrial fibrillation
- E Sick sinus syndrome
- F Sinus tachycardia
- G Torsades de pointes
- H Ventricular extrasystoles
- I Ventricular fibrillation
- J Wolff-Parkinson-White syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 76-year-old man with known ischaemic heart disease is referred for a 24-hour ECG tape after suffering occasional palpitations. The tape shows periods of sinus bradycardia followed by episodes of sinus tachycardia.
- 2 A 55-year-old man is noticed to have a heart rate of 110 beats/min while on the high-dependency unit following abdominal surgery. His other observations include blood pressure 146/92 mmHg, respiratory rate 14/min and oxygen saturations 99% on 2 L of oxygen via nasal cannulae. His epidural analgesia was stopped that morning, and he is complaining of abdominal pain with a severity of 8/10.
- 3 A 75-year-old woman who suffers occasional episodes of palpitations presents to the emergency department with a painful left leg. She was previously fit and well. The leg is pale, cold and slightly mottled at presentation. The ankle-brachial pulse index is 0.3 on the left and 0.9 on the right. An ECG performed by the nursing staff is normal.
- 4 A 55-year-old man is brought to the emergency department after a sudden collapse while gardening. He is unconscious and unresponsive, with no palpable pulse or respiratory effort. The monitor shows a disorganised, irregular, broad-complex rhythm with a fluctuating baseline.
- 5 A 36-year-old woman presents to the emergency department with chest pain and a sensation of feeling her heart pounding. Apart from a history of schizophrenia, she has no medical problems. A heart monitor is attached and shows a broad complex tachycardia with a rate of 210 beats/min on a variable axis.

THEME 13 Management of acute coronary syndromes

- A** Aspirin
- B** Beta-blockers
- C** Calcium-channel blockers
- D** Clopidogrel
- E** Cyclizine
- F** Emergency coronary artery bypass graft
- G** Emergency external pacing
- H** Loop diuretics
- I** Metoclopramide
- J** Past medical history of a haemorrhagic stroke 6 weeks ago
- K** Past medical history of ischaemic stroke 2 years ago
- L** Past medical history of peptic ulcer disease
- M** Primary percutaneous coronary intervention (angioplasty ± stenting)
- N** Previous thrombolysis with streptokinase
- O** ST depression of 2 mm in leads II, III and aVF
- P** Statins
- Q** ST elevation of 2 mm in leads II, III and aVF
- R** ST elevation of 2 mm in leads V1–V5
- S** Tented T waves

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 62-year-old man with diabetes presents to a district general hospital with sudden-onset, central, crushing chest pain that began at rest. The junior doctor on call records an ECG showing an acute inferior myocardial infarction that meets the criteria for reperfusion therapy. Which ECG change has he seen?
- 2** A 62-year-old diabetic man presents to a large teaching hospital with sudden-onset, central, crushing chest pain that commenced at rest. An ECG shows significant ST elevation in the inferior leads. Before the attending junior doctor can administer thrombolysis, the cardiology registrar suggests a preferable intervention.
- 3** A 62-year-old diabetic man presents to the district general hospital with sudden-onset, central, crushing chest pain that commenced at rest. An ECG shows significant ST elevation in the inferior leads, and the attending junior doctor considers thrombolysis. What would be considered as an absolute contraindication to thrombolysis?
- 4** A 62-year-old diabetic man presents to a large teaching hospital with sudden-onset, central, crushing chest pain that commenced at rest. He is complaining of a variety of symptoms, including pain, breathlessness, nausea and headache. Before considering management, which drug should ideally be avoided in this case?
- 5** A 62-year-old man with diabetes presents to a district general hospital with sudden-onset, central, crushing chest pain that began at rest. The junior doctor starts treatment according to an evidence-based hospital protocol. Which drug, given as early as possible as a 300 mg loading dose, reduced 5-week vascular mortality after acute myocardial infarction by about a quarter in the ISIS-2 trial?

THEME 14 Valvular heart disease

- A** Aortic regurgitation
- B** Aortic stenosis
- C** Atrial septal defect
- D** Bicuspid aortic valve
- E** Coarctation of the aorta
- F** Marfan's syndrome
- G** Mitral regurgitation
- H** Mitral stenosis
- I** Prosthetic heart valve
- J** Pulmonary stenosis
- K** Tetralogy of Fallot
- L** Tricuspid regurgitation
- M** Tricuspid stenosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 55-year-old man is admitted with tiredness and worsening shortness of breath. On examination, you note gynaecomastia and a midline sternotomy scar. He is subsequently found to have a normocytic anaemia with a number of red cell fragments on the blood film. Coombs' test is negative.
- 2** A 47-year-old woman has developed aortic stenosis in association with a degree of aortic regurgitation. She is told by her cardiologist that it is likely that she has a congenital abnormality of one of her valves.
- 3** A 58-year-old man is admitted following an episode of haematemesis. He is a heavy drinker and is already under investigation for heart failure. On examination, he is found to have a murmur that is best heard at the apex and radiates into his axilla.
- 4** A 35-year-old Indian woman who has known valvular heart disease as a result of rheumatic fever presents to the emergency department with chest pain. An ECG demonstrates bifid P-waves.
- 5** A 27-year-old woman with known primary pulmonary hypertension is admitted with a long history of abdominal pain. On examination, you find a murmur and peripheral oedema. Her liver is enlarged and tender to palpation.

THEME 15 Pathogens of infective endocarditis

- A** Candida albicans
- B** Coxsackie B virus
- C** Enterococcus faecalis
- D** Mycobacterium tuberculosis
- E** Neisseria meningitidis
- F** Staphylococcus aureus
- G** Streptococcus bovis
- H** Staphylococcus epidermidis
- I** Streptococcus pyogenes
- J** Viridans streptococcus
- K** Trypanosoma cruzi

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 32-year-old male intravenous drug user presents with a 4-week history of fever, lethargy and weight loss. On examination, you find that he has swollen ankles and a raised jugular venous pressure. On auscultation, you notice a pansystolic murmur that is best heard over the left sternal border.
- 2 A 45-year-old woman presents with a 3-week history of night sweats, weight loss and palpitations approximately 1 month after having a dental abscess removed. On auscultation, you notice a pansystolic murmur that is best heard at the apex. She had rheumatic fever as a child.
- 3 A 64-year-old man presents to the emergency department with fever. His temperature is 40.2 °C, blood pressure 100/48 mmHg and heart rate 114 beats/min. He had a metallic aortic valve replacement 1 month ago. Transoesophageal echocardiography shows vegetations at the sewing ring, and blood cultures grow Gram-positive cocci in clusters that are coagulase negative.
- 4 A 72-year-old man with known benign prostatic hyper trophy and a past history of rheumatic fever presents to the emergency department. He was last admitted 3 weeks ago with acute urinary retention, for which he was catheterized and then taught intermittent self-catheterization. He is lethargic, feverish and has lost a stone (6 kg) in weight since his discharge. On examination, you notice a diastolic murmur and signs of left ventricular failure that were not documented on his previous admission.
- 5 A 65-year-old woman, who was diagnosed with sigmoid carcinoma 5 years previously, presents with a 5-week history of malaise, fever, palpitations and weight loss. On examination, you notice a number of splinter haemorrhages under her fingernails. Auscultation reveals a diastolic murmur that is best heard over the second intercostal space on the left-hand side while in expiration. The murmur was not present at her last colonoscopy 2 years ago.

Answers · Themes 11-15

GET AHEAD EMQS

Theme 11 · Diseases of the myocardium and pericardium

- 1 **F Hypertrophic obstructive cardiomyopathy** — Sudden death in a young athlete with a murmur louder on standing is HOCM; screen first-degree relatives.
- 2 **E Dilated cardiomyopathy** — Heavy alcohol use with heart failure, functional MR and AF is alcoholic dilated cardiomyopathy; abstinence may reverse it.
- 3 **B Atrial myxoma** — A pedunculated LA mass on the septum with fever and weight loss is atrial myxoma; excise surgically.
- 4 **K Uraemic pericarditis** — Positional pericarditic pain in advanced renal failure is uraemic pericarditis: start or intensify dialysis.
- 5 **G Myocarditis** — Breathlessness after a febrile viral illness with raised CRP and troponin suggests myocarditis; cardiac MRI and avoid exercise 3–6 months.

Theme 12 · Tachycardias and tachyarrhythmias

- 1 **E Sick sinus syndrome** — Alternating sinus bradycardia and tachycardia in an older patient is sick sinus syndrome; pacemaker if symptomatic.
- 2 **F Sinus tachycardia** — Tachycardia with severe pain after epidural removal is sinus tachycardia from pain; give analgesia and exclude bleeding or sepsis.
- 3 **D Paroxysmal atrial fibrillation** — Embolic limb ischaemia with intermittent palpitations but a normal ECG suggests paroxysmal AF; ambulatory monitoring and anticoagulation.
- 4 **I Ventricular fibrillation** — Pulseless, unresponsive collapse with a chaotic, irregular broad-complex rhythm and a fluctuating baseline is ventricular fibrillation. Start CPR and defibrillate immediately.

- 5 **G Torsades de pointes** — Antipsychotics prolong QT and cause torsades (broad tachycardia with twisting axis): IV magnesium, stop drug, defibrillate if pulseless.

Theme 13 · Management of acute coronary syndromes

- 1 **Q ST elevation of 2 mm in leads II, III and aVF** — ST elevation ≥ 1 mm in two contiguous inferior leads (II, III, aVF) defines inferior STEMI, an indication for reperfusion.
- 2 **M Primary percutaneous coronary intervention (angioplasty $\pm$ stenting)** — Primary PCI within 120 min is preferred over fibrinolysis: better patency, fewer reinfarctions, strokes and deaths.
- 3 **J Past medical history of a haemorrhagic stroke 6 weeks ago** — Any previous intracranial haemorrhage is an absolute contraindication to thrombolysis (also ischaemic stroke < 6 months, active bleeding, dissection).
- 4 **E Cyclizine** — Cyclizine raises SVR and LV filling pressure in MI/heart failure; use metoclopramide as the antiemetic.
- 5 **A Aspirin** — Aspirin cut 5-week vascular mortality by ~23% in ISIS-2 (additive to streptokinase); give 300 mg chewed at once.

Theme 14 · Valvular heart disease

- 1 **I Prosthetic heart valve** — Sternotomy scar with Coombs-negative anaemia and schistocytes is mechanical haemolysis from a prosthetic valve; echo for paravalvular leak.
- 2 **D Bicuspid aortic valve** — Bicuspid aortic valve (1–2%) causes mixed AS/AR presenting in middle age; image the whole aorta.
- 3 **G Mitral regurgitation** — Apical pansystolic murmur to the axilla in a heavy drinker is functional MR from alcoholic dilated cardiomyopathy.

- 4 **H Mitral stenosis** — Bifid P waves (P mitrale) reflect LA enlargement, the hallmark of rheumatic mitral stenosis.
- 5 **L Tricuspid regurgitation** — Pulmonary hypertension causes functional TR with a pansystolic murmur, oedema and a tender congested liver.

Theme 15 · Pathogens of infective endocarditis

- 1 **F Staphylococcus aureus** — Tricuspid endocarditis in a drug injector is most often *S. aureus*; septic pulmonary emboli are common.
- 2 **J Viridans streptococcus** — Subacute IE on a rheumatic mitral valve after dental work is classically viridans streptococci.

SET 4 OF 4

THEME 16 Investigation of cardiovascular disease

- A 24-hour ECG tape
- B Ambulatory blood-pressure monitoring
- C Chest X-ray
- D CT pulmonary angiogram
- E Doppler ultrasound scan
- F Exercise (stress) ECG
- G Myocardial biopsy
- H Myocardial perfusion scan
- I Resting ECG
- J Serum brain-type natriuretic peptide (BNP)
- K Serum D-dimer levels
- L Serum lactate dehydrogenase
- M Serum myoglobin
- N Serum troponin
- O Thrombophilia screen
- P Transoesophageal echocardiography
- Q Transthoracic echocardiography
- R CT coronary angiography

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old man presents to his GP following several episodes of central chest pain that occurred while gardening. On each occasion, the pain was relieved by rest within 5 minutes. His resting ECG, full blood count and thyroid function tests are normal. Which investigation should be requested first to diagnose the cause?
- 2 A 67-year-old woman presents to the emergency department with a 12-hour history of central chest pain that commenced at rest and radiated to both arms. She is nauseated and appears pale and sweaty. The resting ECG shows an ST depression of 1 mm in leads II, III and aVF, associated with non-specific T-wave changes. A chest X-ray is normal.
- 3 A 40-year-old female inpatient develops a red, tender swelling of her left leg 10 days after a hysterectomy. The affected leg is painful and hard to touch, and the peripheral veins appear engorged. A full blood count is unremarkable and her observations are stable.

- 3 **H Staphylococcus epidermidis** — Early prosthetic valve IE with coagulase-negative staphylococci is *S. epidermidis* (biofilm on the sewing ring); surgery plus vancomycin, rifampicin, gentamicin.
- 4 **C Enterococcus faecalis** — *Enterococcus faecalis* IE follows genitourinary instrumentation in older men; treat with amoxicillin plus gentamicin or ceftriaxone.
- 5 **G Streptococcus bovis** — *S. bovis* (gallolyticus) IE is strongly linked to colorectal neoplasia; any such bacteraemia needs colonoscopy.

- 4 A 50-year-old man with a past history of rheumatic fever becomes acutely unwell 5 days after an elective tooth extraction. On examination, he has a temperature of 38.3°C and has a pansystolic murmur that was not documented on his previous admission. Blood cultures have been sent, but the results are not yet available.
- 5 A 56-year-old man with hypertension and ischaemic heart disease presents to his GP with almost daily episodes of palpitations. Each episode lasts approximately 1 hour and terminates spontaneously. A resting ECG performed in the practice shows no abnormalities. Routine blood tests taken a week previously were within normal limits.

THEME 17 Symptoms of cardiovascular disease

- A Acute angina
- B Acute mesenteric infarction
- C Acute myocardial infarction
- D Acute pulmonary oedema
- E Chronic left ventricular failure (NYHA class I)
- F Chronic left ventricular failure (NYHA class III)
- G Drug side-effect
- H Epilepsy
- I Hypertrophic cardiomyopathy
- J Infective endocarditis
- K Mesenteric ischaemia
- L Postural hypotension
- M Rheumatic fever
- N Stokes-Adams collapse
- O Vasovagal syncope

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old man wakes from sleep feeling extremely breathless and anxious. His breathlessness resolves after a few minutes of walking around the bedroom. During the day he is comfortable at rest but becomes breathless while getting dressed or walking a few metres around the house.
- 2 A 65-year-old man presents with a 6-hour history of shortness of breath and palpitations, and is coughing up pink frothy sputum. He refuses to be laid flat.

- 3** A 70-year-old man with known angina, hypertension and hypercholesterolaemia is experiencing severe episodic abdominal pains. These occur approximately 30 minutes after eating. A gastroscopy and colonoscopy show no abnormalities.
- 4** A 12-year-old girl is complaining of lethargy, fever and shortness of breath. These symptoms were initially associated with painful and swollen knee joints, which have now resolved to be replaced by a swollen elbow joint. Apart from a recent sore throat, she is otherwise fit and well.
- 5** A 45-year-old woman with known Addison's disease collapsed while getting out of bed in the morning. She denies any chest pain or palpitations. She was unresponsive for 40 seconds, during which her 'legs twitched a little'. She recovered fully after 2 minutes. This has happened on several previous occasions when standing up from a lying or sitting position.

THEME 18 Management of cardiovascular disease

- A** Adenosine
B Atenolol
C Digoxin
D Non-steroidal anti-inflammatory drugs
E Paracetamol
F Streptokinase
G Sublingual glyceryl trinitrate
H Synchronized DC cardioversion
I Tenecteplase alone
J Tenecteplase followed by intravenous heparin infusion
K Unsynchronized DC cardioversion
L Verapamil
M Warfarin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 67-year-old man presents to the emergency department with palpitations. He has known ischaemic heart disease and hypertension. On examination, he is pale and appears confused, and his hands are cool and clammy. His pulse is 160 beats/min and irregularly irregular, his blood pressure is 70/40 mmHg and his capillary refill time is 6 s.
- 2** A 45-year-old woman presents to her GP with a 2-day history of sharp central chest pain. The pain does not radiate and there is no associated shortness of breath. She only gains relief by sitting forward in bed at night. Examination is unremarkable, and an ECG shows widespread concave ST elevation with PR-segment depression. She has a past history of rheumatoid arthritis.
- 3** A 25-year-old asthmatic man presents to the emergency department with palpitations. Examination reveals a regular heart rate of 168 beats/min. The ECG shows a regular narrow complex tachycardia with no obvious P-waves. The Valsalva manoeuvre and carotid sinus massage have failed to slow the heart rate.

- 4** A 69-year-old man presents to a rural hospital with central chest pain radiating to his left arm that came on suddenly at rest 1 hour ago. He is nauseated, sweaty and short of breath. He had a heart attack 4 years ago, during which he received a 'clot-busting' drug that he was told never to have again. The ECG shows ST elevation of 4 mm in V2, V3 and V4. The nearest PCI-capable centre is more than 2 hours away and he has no contraindications to thrombolysis. Aspirin, clopidogrel, morphine and metoclopramide have been given so far.
- 5** A 57-year-old man with known end-stage renal failure awaiting renal transplantation collapses at home with chest pain. He is brought to the emergency department attached to a cardiac monitor. The monitor shows a regular broad complex tachycardia with a rate of 230 beats/min. On examination, the carotid pulse is not palpable.

THEME 19 Causes of collapse

- A** Acute myocardial infarction
B Cerebrovascular accident
C Dissecting aortic aneurysm
D Drug allergy
E First-dose hypotension
F Left ventricular failure
G Ruptured abdominal aortic aneurysm
H Stable angina
I Stokes-Adams attack
J Supraventricular tachycardia
K Vasovagal syncope
L Ventricular rupture
M Wolff-Parkinson-White syndrome

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 45-year-old woman collapses during a church service in summer. Her husband states that she turned pale and collapsed shortly after standing up. She did not jerk or lose urinary continence. She denies chest pain, shortness of breath and palpitations. She recovered after 30 seconds and now feels back to her usual self. All observations – including blood tests, ECG, and lying and standing blood pressure – are normal.
- 2** A 25-year-old man collapses suddenly while visiting his grandmother in hospital. There appears to be no pulse. The crash team is called, and manage to resuscitate him after identifying ventricular fibrillation on the cardiac monitor. The attending anaesthetist records a repeat ECG and notes that the QRS complexes are broad with slurred upstroke of the R wave.
- 3** A 75-year-old woman is brought to the emergency department after collapsing. On examination, her heart rate is 80 beats/min and irregular, with a blood pressure of 150/90 mmHg. She has difficulty in moving her left arm and leg.

- 4 A 67-year-old man presents to the emergency department following a collapse. He has a history of ischaemic heart disease and was diagnosed with hypertension last week, when he was started on anti-hypertensive treatment. He felt dizzy when he got out of his chair to go to the toilet and then fainted. He regained consciousness almost immediately and had fully recovered within 5 minutes.

Answers · Themes 16–19

GET AHEAD EMQS

Theme 16 · Investigation of cardiovascular disease

- 1 **R CT coronary angiography** — Typical stable angina: CT coronary angiography is first line (NICE, ESC 2024); exercise ECG is no longer used for diagnosis.
- 2 **N Serum troponin** — Rest pain with inferior ST depression is NSTEMI; serial high-sensitivity troponin (0/1 h or 0/3 h) separates NSTEMI from unstable angina.
- 3 **E Doppler ultrasound scan** — Swollen tender leg 10 days after pelvic surgery is high-probability DVT (Wells ≥ 2): compression ultrasound. D-dimer is raised post-op anyway.
- 4 **Q Transthoracic echocardiography** — Suspected native-valve IE: blood cultures then transthoracic echo first; TOE if TTE negative with ongoing suspicion, prosthetic valves or abscess.
- 5 **A 24-hour ECG tape** — Near-daily palpitations with a normal ECG: 24–48 h Holter monitoring; rarer episodes need an event or loop recorder.

Theme 17 · Symptoms of cardiovascular disease

- 1 **F Chronic left ventricular failure (NYHA class III)** — PND and breathlessness on less than ordinary activity but comfort at rest is chronic LVF, NYHA class III.
- 2 **D Acute pulmonary oedema** — Pink frothy sputum and inability to lie flat is acute pulmonary oedema: sit up, oxygen, IV furosemide, nitrates if BP allows.
- 3 **K Mesenteric ischaemia** — Post-prandial abdominal pain with normal endoscopy in an arteriopathy is chronic mesenteric ischaemia ('intestinal angina'); CT angiography.
- 4 **M Rheumatic fever** — Migratory large-joint arthritis, fever and carditis after a sore throat is acute rheumatic fever; ASO titre, penicillin, aspirin, prophylaxis.
- 5 **L Postural hypotension** — Recurrent collapse on standing in Addison's disease is postural hypotension; check lying/standing BP and fludrocortisone dose.

Theme 18 · Management of cardiovascular disease

- 1 **H Synchronized DC cardioversion** — Fast AF with shock is unstable: synchronised DC cardioversion (up to 3 attempts), then amiodarone if it fails.
- 2 **D Non-steroidal anti-inflammatory drugs** — Pleuritic pain eased sitting forward with saddle ST elevation and PR depression is pericarditis: NSAID with PPI plus colchicine.
- 3 **L Verapamil** — Regular narrow tachycardia after failed vagal manoeuvres in an asthmatic: IV verapamil, as adenosine may cause bronchospasm.
- 4 **J Tenecteplase followed by intravenous heparin infusion** — Anterior STEMI without timely PCI: fibrinolysis with tenecteplase (not repeat streptokinase) plus heparin, then transfer for angiography.
- 5 **K Unsynchronized DC cardioversion** — Pulseless broad-complex tachycardia is pulseless VT: unsynchronised shocks with CPR; correct likely hyperkalaemia.

Theme 19 · Causes of collapse

- 1 **K Vasovagal syncope** — Brief faint with pallor on prolonged standing in a warm place, rapid recovery and normal ECG is vasovagal syncope.
- 2 **M Wolff–Parkinson–White syndrome** — A slurred QRS upstroke (delta wave) with short PR is WPW; AF conducting down the pathway can cause VF and collapse. Ablate.
- 3 **B Cerebrovascular accident** — Collapse with AF and new left hemiparesis is a right-hemisphere stroke (cardioembolic): urgent CT before thrombolysis/thrombectomy.
- 4 **E First-dose hypotension** — Postural syncope soon after starting an antihypertensive (e.g. doxazosin) is first-dose hypotension; start low at bedtime.

2 THEMES • 6 SCENARIOS • OTHER COLLEGES & SOURCES

Other Sources EMQs

THEME 1 Valvular and related heart disease

- A Tricuspid stenosis
- B Tricuspid regurgitation
- C Mitral stenosis
- D Mitral regurgitation
- E Aortic regurgitation (insufficiency)
- F Aortic stenosis
- G Hypertrophic cardiomyopathy
- H Pulmonic stenosis
- I Pulmonic regurgitation (insufficiency)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A tall, thin 19-year-old woman who has had little previous health care complains mainly of reduced vision. She has a strong pulse, BP 180/70 mmHg and a high-pitched, blowing, early diastolic decrescendo murmur.
- 2 A 23-year-old graduate student has had extreme fatigue and a vague sense of feeling ill for a few weeks; he is slightly agitated. BP 110/70 mmHg, pulse 100/min, temperature 38.0 °C. The neck veins are distended with prominent v waves, and a holosystolic murmur at the left sternal border intensifies on inspiration.
- 3 An 18-year-old man is sent from his physical education class with dizziness and palpitations after exercise; his instructor thinks he is faking. Vital signs are normal. He has a rapidly rising carotid pulse, an S4 and a harsh systolic crescendo–decrescendo murmur beginning well after S1, best heard at the lower left sternal border.

THEME 2 Antiarrhythmic management

- A Amiodarone
- B Atropine
- C Digoxin
- D Diltiazem
- E Isoprenaline
- F Lidocaine
- G Metoprolol
- H Quinidine
- I Observation

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 72-year-old man has a 2-hour history of chest pain, acute ST elevation in leads II, III and aVF, and sinus bradycardia at 40/min. His BP is 80/40 mmHg.
- 2 A 58-year-old female smoker is admitted to the ICU with respiratory distress due to pneumonia. Her course is complicated by an anterior myocardial infarction, and she is on the highest beta-blocker dose her lungs tolerate. Serum potassium and magnesium are normal. She now has 10–12 ventricular premature beats per hour, occasional couplets and a few short runs of non-sustained ventricular tachycardia; her BP and oxygen saturation remain stable.
- 3 A 60-year-old man on his first day in the coronary care unit after a myocardial infarction develops an accelerated idioventricular rhythm at a rate of 80/min.

Answers • Themes 1–2

OTHER SOURCES EMQS

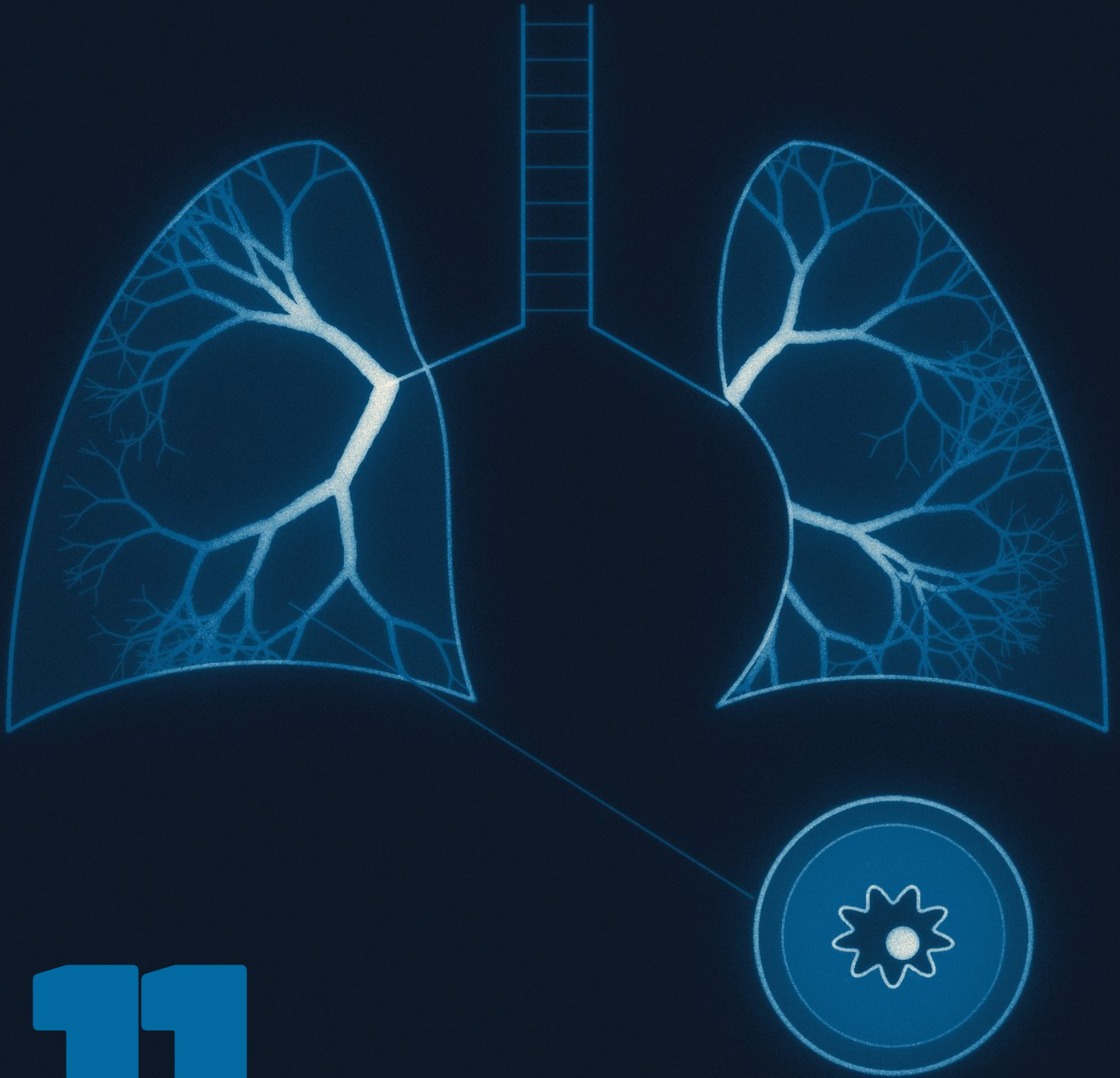
Theme 1 • Valvular and related heart disease

- 1 **E Aortic regurgitation (insufficiency)** — Tall thin woman with lens dislocation (Marfan), wide pulse pressure and early diastolic decrescendo murmur: aortic regurgitation from root dilatation.
- 2 **B Tricuspid regurgitation** — Fever with prominent v waves and a pansystolic murmur louder on inspiration is tricuspid regurgitation, likely right-sided endocarditis.
- 3 **G Hypertrophic cardiomyopathy** — Exertional dizziness with a brisk carotid upstroke, S4 and crescendo–decrescendo mur-

mur at the lower left sternal border is HCM; AS has a slow-rising pulse.

Theme 2 • Antiarrhythmic management

- 1 **B Atropine** — Symptomatic sinus bradycardia with hypotension in inferior STEMI: IV atropine 500 µg (to 3 mg) first; pacing or isoprenaline if it fails; perfuse.
- 2 **I Observation** — Asymptomatic ectopics and short NSVT with stable haemodynamics: observe, continue beta-blocker and correct K/Mg. Prophylactic antiarrhythmics do not help.
- 3 **I Observation** — Accelerated idioventricular rhythm (60–120/min) after reperfusion is benign and self-limiting: observe.



11

OBSTRUCTIVE AIRWAY DISEASE

Obstructive Airway Disease

Asthma and COPD side by side: diagnosis and spirometry, acute severe attacks and exacerbations, oxygen, ventilation and long-term management.

80

MCQs

0

EMQ SCENARIOS

7

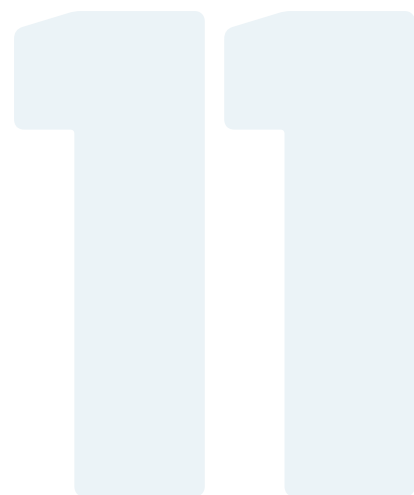
SOURCES

15

REPEATED IDEAS

IN THIS CHAPTER

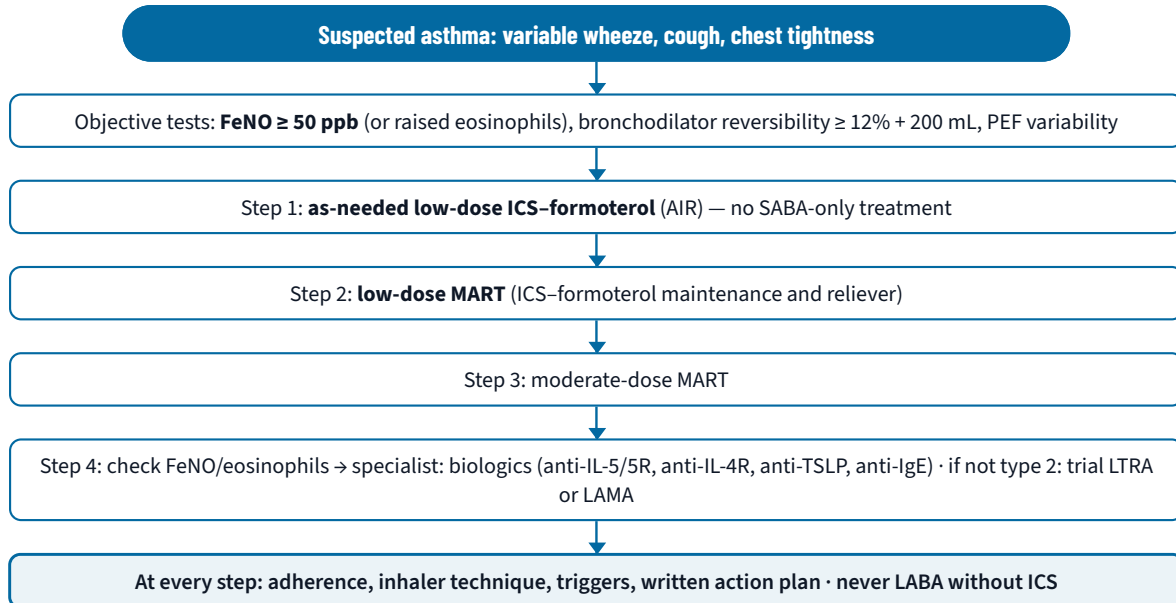
01	Management flowcharts	5 charts	page 192
02	Local Exams MCQs	34 Q	page 195
03	Davidson MCQs	9 Q	page 200
04	Harrison MCQs	12 Q	page 202
05	Crash Course MCQs	8 Q	page 204
06	PreTest MCQs	2 Q	page 205
07	Passmedicine MCQs	7 Q	page 206
08	PasTest MCQs	8 Q	page 207



Step by step

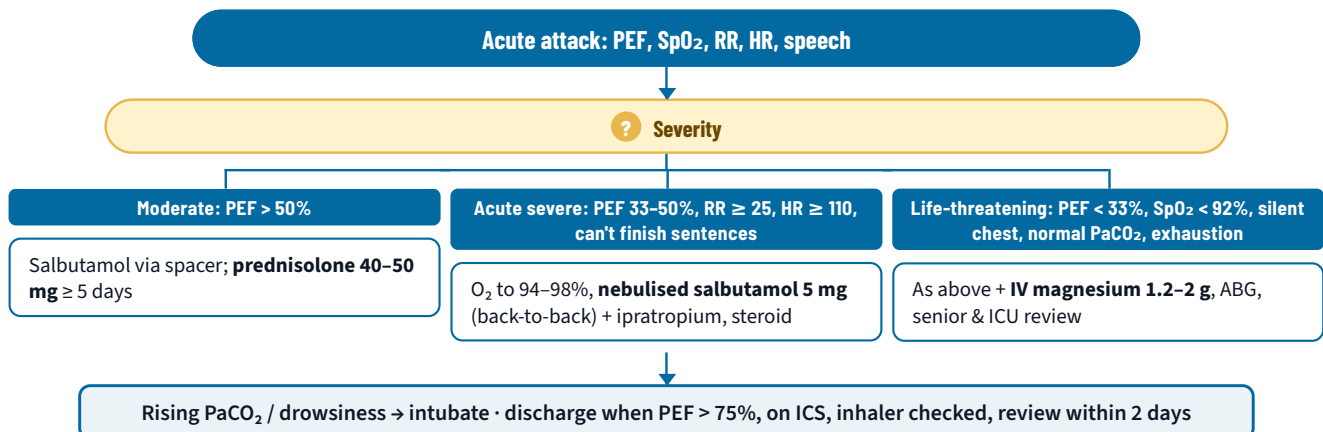
Asthma in adults: diagnosis and stepwise treatment

BTS/NICE/SIGN 2024 • GINA 2025



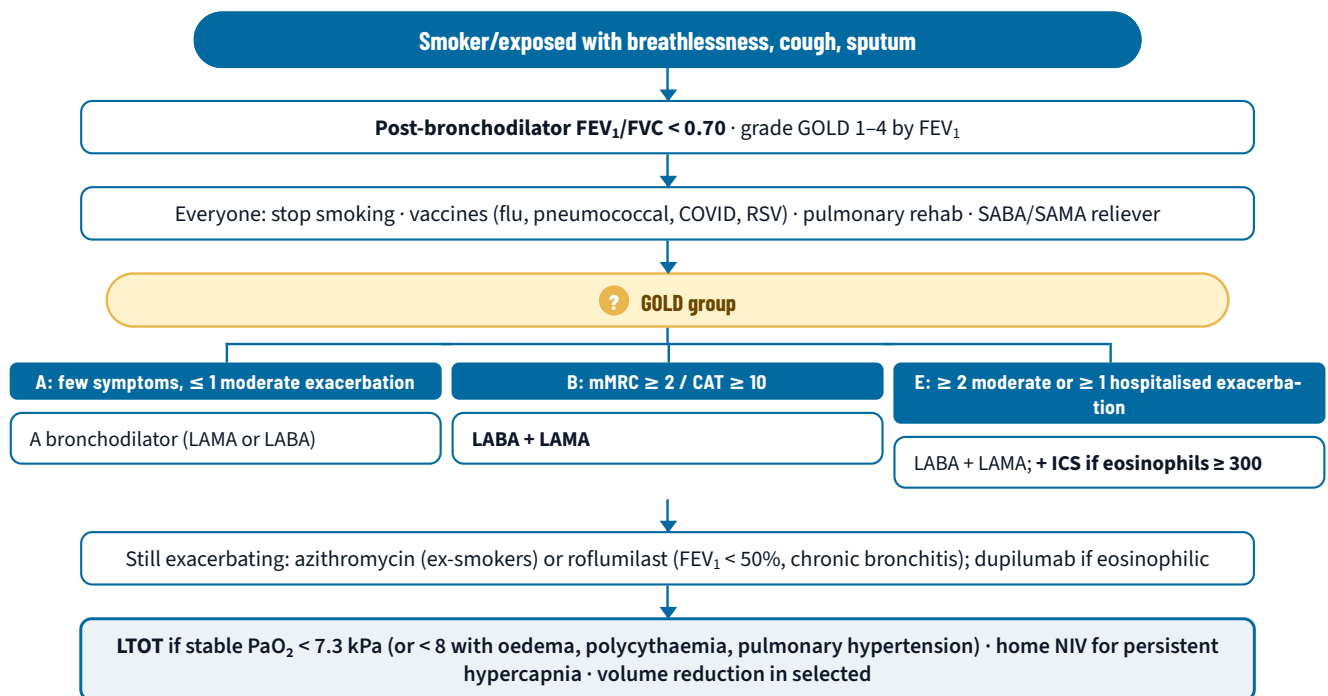
Acute asthma

BTS/SIGN • NICE



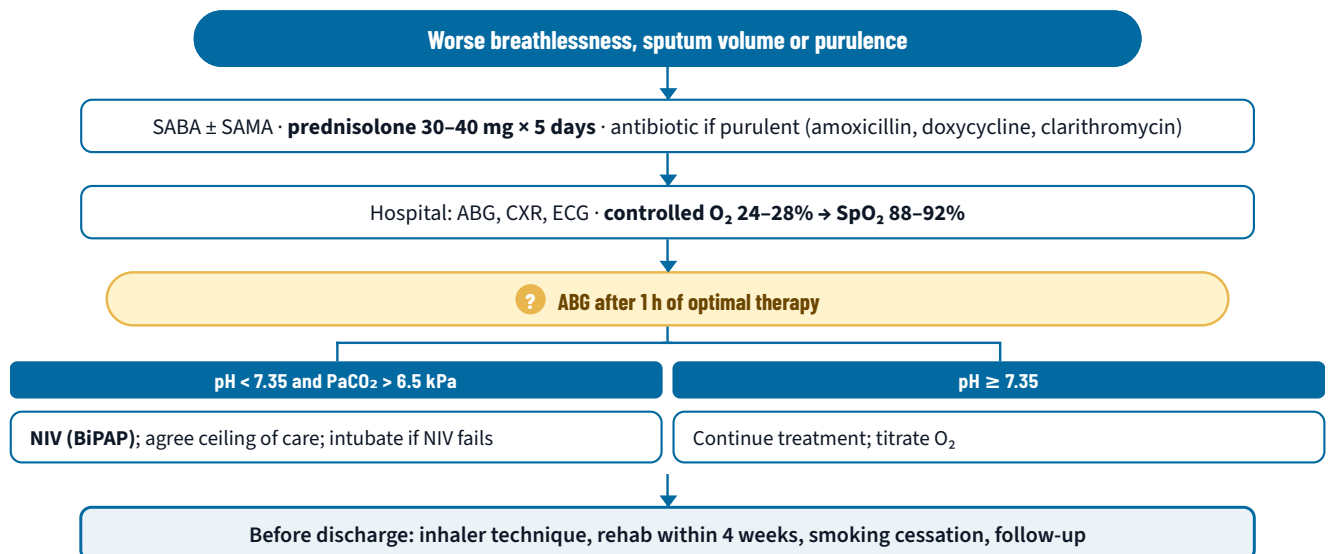
COPD: diagnosis and maintenance

GOLD 2025 · NICE NG115



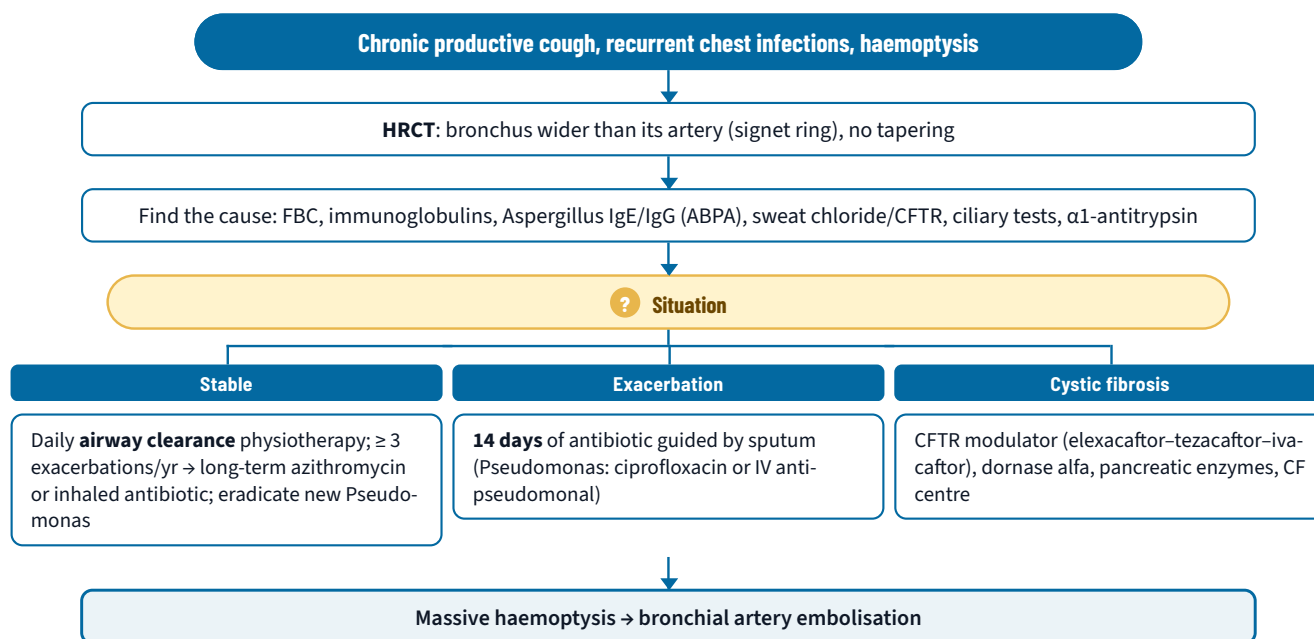
COPD exacerbation

GOLD 2025 · BTS NIV · NICE NG115



Bronchiectasis and cystic fibrosis

BTS 2019 · NICE NG78



Local Exams MCQs

SET 1 OF 2

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 22-year-old woman with asthma has had wheeze and breathlessness for 6 hours despite using her salbutamol inhaler several times. HR 128/min, RR 30/min, SpO₂ 89% on air; peak flow 40% of predicted. What is the most appropriate next step?

A High-dose inhaled corticosteroid
 B Nebulised salbutamol plus ipratropium and IV magnesium sulfate
 C Oral prednisolone only
 D Add a long-acting muscarinic antagonist
 E Discharge with increased short-acting beta-agonist use

- 2** A 35-year-old man with asthma has had worsening breathlessness and chest tightness for 1 day with night-time cough, and has used his reliever 8 times in 24 hours. HR 118/min, RR 32/min, BP 128/74 mmHg, SpO₂ 93% on air; there is diffuse wheeze with accessory muscle use. PEF is 45% of predicted and a chest X-ray shows no pneumonia. What is the next best step?

A High-dose ICS/LABA inhaler
 B Nebulised salbutamol every 20 minutes for 1 hour and oral prednisolone 40 mg
 C IV magnesium sulfate as first-line treatment
 D Add a LAMA inhaler (tiotropium)
 E Start oral montelukast

REPEATED 2* • ALSO ASKED IN
 Crash Course

- 3** A 68-year-old man with severe COPD has had 3 days of worsening breathlessness, cough with yellow sputum and wheeze. HR 108/min, RR 28/min, BP 138/84 mmHg, SpO₂ 87% on air. ABG on air: pH 7.32, PaCO₂ 7.7 kPa (58 mmHg), HCO₃⁻ 30 mmol/L, PaO₂ 7.7 kPa (58 mmHg). There is diffuse wheeze with prolonged expiration. What is the most appropriate next step?

A Systemic corticosteroid and nebulised salbutamol/ipratropium
 B Long-term oxygen therapy
 C Inhaled corticosteroid/LABA combination
 D Oral montelukast
 E Roflumilast

- 4** A 72-year-old man with advanced COPD is reviewed when clinically stable, 5 weeks after his last exacerbation. ABG on air: PaO₂ 7.5 kPa (56 mmHg), PaCO₂ 6.9 kPa (52 mmHg); SpO₂ 89%. He has mild ankle oedema and echocardiography shows pulmonary hypertension. What is the most appropriate next step?

A Oral acetazolamide
 B Long-term oxygen therapy
 C Increase inhaled bronchodilator therapy
 D Repeat ABG in 3 months
 E Nocturnal oxygen only

REPEATED 3* • ALSO ASKED IN
 39th batch final Paper 1 • Harrison

- 5** Breathlessness with wheeze can be caused by all of the following EXCEPT:

A Acute bronchitis
 B Pneumothorax
 C Asthma
 D Foreign-body inhalation

REPEATED 3* • ALSO ASKED IN
 End-block exam 2025 • Previous-years exam

- 6** Which of these measures increases survival in COPD?

A Stopping smoking and weight reduction
 B Stopping smoking and long-term oxygen therapy
 C Stopping smoking and regular combined LABA/ICS
 D Stopping smoking and intermittent short courses of steroids

REPEATED 3* • ALSO ASKED IN
 End-block exam 2025 • Log book 2025

- 7** A 65-year-old man has had cough and sputum for 4 years and has been diagnosed with COPD. Which test can differentiate chronic bronchitis from emphysema?

A Chest X-ray
 B Diffusing capacity for carbon monoxide (DLCO)
 C Peak expiratory flow rate
 D Vital capacity and FEV₁

- 8** Which of the following is a feature of life-threatening asthma?

A PEF 40% of predicted
 B SpO₂ 90%
 C Inability to complete sentences in one breath
 D Respiratory rate 22/min

- 9 Which of the following is NOT characteristic of asthma?
- A It can be seasonal or perennial
 - B The patient can be completely free of symptoms between episodes
 - C The disease can start at any age
 - D In adults it is more common and more severe in men than in women
 - E Attacks of cough and wheeze occur at night

REPEATED 3* · ALSO ASKED IN

End-block exam 2025 · Previous-years exam

- 10 A patient with long-standing chronic bronchitis presents with increased cough, sputum and severe breathlessness. ABG on air: PaO_2 8.2 kPa, PaCO_2 8.6 kPa, pH 7.20. Which of the following should be avoided?
- A Nebulised salbutamol
 - B Nebulised ipratropium bromide
 - C 35% oxygen
 - D Oral prednisolone
 - E Non-invasive ventilation

- 11 A patient with asthma presents with an acute severe attack. Which of the following is an indication for mechanical ventilation?
- A Unable to speak more than two words in a breath
 - B Exhaustion, confusion and drowsiness
 - C Peak expiratory flow 40% of predicted
 - D Respiratory rate 28/min

- 12 All of the following are true in COPD EXCEPT:

- A Radiographic signs correlate with the severity of airflow limitation
- B Measurement of lung volumes assesses hyperinflation
- C In severe COPD with large bullae, body plethysmography is preferred to estimate lung volumes
- D A low peak flow is consistent with COPD but is non-specific

REPEATED 2* · ALSO ASKED IN

End-block exam 2025

- 13 All of the following are true regarding bronchodilators in COPD EXCEPT:

- A The inhaled route is preferred to systemic bronchodilators
- B Short-acting bronchodilators are more appropriate for patients with moderate to severe disease
- C Breathlessness may improve significantly despite minimal change in FEV_1
- D Tiotropium is more appropriate than ipratropium for moderate to severe disease

- 14 A patient with asthma using a regular low-dose inhaled corticosteroid plus an as-needed short-acting beta₂-agonist wakes at night with cough and occasional wheeze three to four times a week. Inhaler technique and adherence are good. What is the best next step?

- A Maintenance oral prednisolone
- B Add a leukotriene receptor antagonist
- C Switch to low-dose ICS-formoterol as maintenance and reliever therapy (MART)
- D Add modified-release theophylline

REPEATED 2* · ALSO ASKED IN

Passmedicine

CASE · QUESTIONS 15-19

A 54-year-old smoker has cough, sputum and breathlessness. There is vesicular breathing with a prolonged expiratory phase, low-pitched expiratory wheezes and coarse crackles throughout the chest. Spirometry: FEV_1/VC 56%, rising to 68% after bronchodilator; post-bronchodilator FEV_1 62% of predicted.

- 15 What is the diagnosis?

- A Bronchial asthma
- B Extrinsic allergic alveolitis
- C Chronic obstructive pulmonary disease
- D Idiopathic pulmonary fibrosis

- 16 According to the spirometric classification of severity based on post-bronchodilator FEV_1 , he has:

- A Mild disease
- B Moderate disease
- C Severe disease
- D Very severe disease

- 17 What is the most important step in his management?

- A Short-acting bronchodilators
- B Long-acting bronchodilators
- C Ipratropium bromide
- D Smoking cessation

REPEATED 2* · ALSO ASKED IN

PreTest

- 18 Which of the following statements about the treatment of his disease is NOT true?

- A Oral steroids can be used as maintenance therapy
- B Roflumilast improves lung function and reduces moderate to severe exacerbations
- C Azithromycin 500 mg three times weekly can reduce exacerbations
- D Pneumococcal and annual influenza vaccination are indicated

- 19 Which of the following is NOT a complication of his disease?

- A Secondary polycythaemia
- B Finger clubbing
- C Pneumothorax
- D Right-sided heart failure

CASE · QUESTIONS 20–23

A 62-year-old man has had progressive breathlessness and a productive cough for 8 years, with thick whitish sputum that is worst in the morning; his symptoms have recently worsened. He has a 40 pack-year smoking history. He is cyanosed with a raised JVP and bilateral pitting ankle oedema, and coarse crackles and rhonchi are heard over both lungs. RR 22/min, BP 145/85 mmHg; PaO₂ 55 mmHg, PaCO₂ 66 mmHg, haemoglobin 18 g/dL.

- 20** What is the most likely diagnosis?
- A Bronchial asthma
 - B COPD with cor pulmonale
 - C Bronchiectasis
 - D Interstitial lung disease
 - E Idiopathic pulmonary fibrosis
- 21** What is the most important risk factor for this condition?
- A Cigarette smoking
 - B Occupational silica exposure
 - C Alpha-1 antitrypsin deficiency
 - D Viral infection
 - E Air pollution
- 22** The raised haemoglobin is most likely due to:
- A Bone marrow malignancy
 - B Infection
 - C Secondary polycythaemia due to chronic hypoxia
 - D Acute blood loss
 - E Iron deficiency anaemia

- 23** The raised JVP is most likely caused by:
- A Left-sided heart failure
 - B Right-sided heart failure
 - C Renal failure
 - D Liver failure
 - E Deep vein thrombosis

END-BLOCK EXAMS 2025

End-block & induction exams

- 24** A 16-year-old boy has a 6-month history of worsening breathlessness and a daily productive cough yielding about a cupful of sputum. He had severe measles in childhood and has smoked 10 cigarettes a day for 2 years. He looks unwell, with mild dyspnoea, early finger clubbing and bilateral coarse crackles over the lower zones; cardiac examination is normal. What is the most likely diagnosis?
- A Bronchiectasis
 - B Chronic bronchitis
 - C Pulmonary tuberculosis
 - D Asthma
 - E Interstitial lung disease

REPEATED 3* · ALSO ASKED IN
Log book 2025 · Crash Course

- 25** A 16-year-old boy has a daily productive cough yielding about a cupful of purulent sputum, bilateral coarse basal crackles and early finger clubbing. He had severe measles in early childhood and has no occupational exposures. Which underlying cause of his condition is most likely?
- A Post-infectious damage following severe measles
 - B Alpha-1 antitrypsin deficiency
 - C Primary pulmonary hypertension
 - D Occupational asbestos exposure
 - E Sarcoidosis

REPEATED 2* · ALSO ASKED IN
End-block exam 2025

Answers 1–25

- 1** **B Nebulised salbutamol plus ipratropium and IV magnesium sulfate** — SpO₂ < 92% is a life-threatening feature: oxygen, nebulised salbutamol + ipratropium, steroids and IV magnesium 1.2–2 g, with early ICU review.
- 2** **B Nebulised salbutamol every 20 minutes for 1 hour and oral prednisolone 40 mg** — **UPDATED** Acute severe asthma (PEF 33–50%, no life-threatening feature): repeated nebulised salbutamol, oxygen to 94–98% and prednisolone 40–50 mg.
- 3** **A Systemic corticosteroid and nebulised salbutamol/ipratropium** — COPD exacerbation with acidosis: controlled O₂ (88–92%), nebulised salbutamol/ipratropium, prednisolone, antibiotic; NIV if acidosis persists.
- 4** **B Long-term oxygen therapy** — LTOT (≥ 15 h/day) for stable PaO₂ < 7.3 kPa, or 7.3–8 with oedema, polycythaemia or pulmonary hypertension; it improves survival.
- 5** **B Pneumothorax** — Wheeze comes from narrowed airways (asthma, bronchitis, foreign body). Pneumothorax gives reduced breath sounds and hyper-resonance, not wheeze.
- 6** **B Stopping smoking and long-term oxygen therapy** — Smoking cessation and LTOT (for chronic hypoxaemia) are the classic interventions that prolong survival in COPD.
- 7** **B Diffusing capacity for carbon monoxide (DLCO)** — DLCO is low in emphysema (alveolar destruction) but near normal in chronic bronchitis; spirometry is obstructive in both.
- 8** **B SpO₂ 90%** — SpO₂ < 92% is life-threatening asthma (with PEF < 33%, silent chest, normal PaCO₂, exhaustion); PEF 33–50% is acute severe.
- 9** **D In adults it is more common and more severe in men than in women** — Childhood asthma is commoner in boys, but after puberty it is commoner and more severe in women.

- 10 **C 35% oxygen** — **UPDATED** In hypercapnic COPD with PaO₂ 8.2 kPa (SpO₂ ~90%), 35% oxygen would worsen CO₂ retention; use 24–28% Venturi to SpO₂ 88–92%.
- 11 **B Exhaustion, confusion and drowsiness** — Ventilate in asthma for exhaustion, drowsiness, confusion, arrest or rising PaCO₂ despite maximal therapy; involve ICU early.
- 12 **A Radiographic signs correlate with the severity of airflow limitation** — CXR signs of COPD correlate poorly with airflow limitation, which is graded by post-bronchodilator FEV₁ (GOLD 1–4).
- 13 **B Short-acting bronchodilators are more appropriate for patients with moderate to severe disease** — Long-acting bronchodilators (LAMA + LABA) are maintenance in moderate-severe COPD; short-acting agents are relievers only.
- 14 **C Switch to low-dose ICS–formoterol as maintenance and reliever therapy (MART)** — **UPDATED** Uncontrolled asthma on low-dose ICS + SABA: switch to low-dose ICS–formoterol MART (BTS/NICE/SIGN 2024, GINA). Never LABA without ICS.
- 15 **C Chronic obstructive pulmonary disease** — Post-bronchodilator FEV₁/FVC < 0.70 in a symptomatic smoker confirms COPD; partial reversibility does not exclude it.
- 16 **B Moderate disease** — **UPDATED** GOLD grades by post-bronchodilator FEV₁: 1 ≥ 80%, 2 50–79% (moderate), 3 30–49%, 4 < 30%. FEV₁ 62% is moderate.
- 17 **D Smoking cessation** — Smoking cessation is the only measure that slows FEV₁ decline in COPD and improves survival; offer varenicline/NRT with support.
- 18 **A Oral steroids can be used as maintenance therapy** — Maintenance oral steroids are not used in COPD (no benefit,

many harms); short courses only. Roflumilast and azithromycin reduce exacerbations.

- 19 **B Finger clubbing** — **UPDATED** COPD does not cause clubbing (look for cancer, bronchiectasis or fibrosis). Polycythaemia, cor pulmonale and pneumothorax are complications.
- 20 **B COPD with cor pulmonale** — Chronic bronchitis in a heavy smoker with hypoxaemia, hypercapnia, cyanosis, raised JVP and oedema is COPD complicated by cor pulmonale (right heart failure from hypoxic pulmonary vasoconstriction).
- 21 **A Cigarette smoking** — Tobacco smoking causes most COPD and stopping is the single intervention that slows FEV₁ decline. Alpha-1 antitrypsin deficiency explains early, basal emphysema in a minority.
- 22 **C Secondary polycythaemia due to chronic hypoxia** — Chronic hypoxaemia drives renal erythropoietin release, giving secondary polycythaemia; it raises viscosity and is reduced by long-term oxygen. Polycythaemia vera would have a low EPO.
- 23 **B Right-sided heart failure** — Hypoxic pulmonary vasoconstriction causes pulmonary hypertension and right ventricular failure (cor pulmonale), seen as a raised JVP, hepatomegaly and peripheral oedema.
- 24 **A Bronchiectasis** — Copious purulent sputum, coarse basal crackles and clubbing after severe childhood measles is post-infective bronchiectasis; HRCT confirms.
- 25 **A Post-infectious damage following severe measles** — **UPDATED** Severe childhood measles, pertussis or TB damage bronchial walls, causing post-infectious bronchiectasis.

SET 2 OF 2

- 26 Apart from post-infective damage (e.g. after measles or whooping cough), which of the following is a recognised cause of bronchiectasis in young patients?
- A Cystic fibrosis
B Pulmonary embolism
C Mitral stenosis
D Spontaneous pneumothorax
E Coarctation of the aorta
- 27 Which of the following complications is commonly associated with bronchiectasis?
- A Haemoptysis
B Spontaneous pneumothorax
C Cardiac tamponade
D Aortic dissection
E Pulmonary embolism

REPEATED 4x · ALSO ASKED IN
End-block exam 2025 x2 · Log book 2025

- 28 Which HRCT finding is most specific for the diagnosis of bronchiectasis?
- A “Signet ring” sign
B Honeycombing
C Ground-glass opacities
D Tree-in-bud pattern
E Consolidation

- 29 The daily production of a large volume of sputum in bronchiectasis is primarily due to:

- A Chronic infection and inflammation of the bronchial walls
B Excess mucus production secondary to heart failure
C Neoplastic secretion from lung tumours
D Allergic reactions causing mucus hypersecretion
E Pulmonary oedema fluid accumulation

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 30 All of the following are characteristic of asthma EXCEPT:
- A It is rare before 5 years of age
B It can be seasonal or perennial
C It can start at any age
D The patient can be completely free of symptoms between episodes
- 31 Cough productive of sputum on most days for at least 3 months in each of 2 consecutive years is the clinical definition of which condition?
- A Bronchial asthma
B COPD (chronic bronchitis)
C Pulmonary oedema
D Pneumonia
E Pulmonary tuberculosis

- 32** Regarding treatment of acute exacerbation of COPD, the following are true EXCEPT:
- A** When sputum is purulent, simple antibiotic regimens such as amoxicillin or a macrolide are advised
 - B** Oral prednisolone reduces symptoms but cannot improve lung function
 - C** Mechanical ventilation may be considered in those with a reversible cause for deterioration (e.g. pneumonia) or no prior history of respiratory failure
 - D** Nebulised short-acting beta2-agonists can be combined with an anticholinergic agent

LOG BOOK 2025

Log-book questions

- 33** A 66-year-old heavy smoker presents with chronic productive cough, progressive dyspnoea, pursed-lip breathing, and a barrel chest. What is the most likely diagnosis?
- A** Asthma
 - B** Interstitial lung disease
 - C** Chronic obstructive pulmonary disease (COPD)
 - D** Pulmonary fibrosis
 - E** Bronchiectasis

REPEATED 2* · ALSO ASKED IN
Crash Course

- 34** Which pulmonary function test finding is most consistent with COPD?
- A** Reduced FEV₁/FVC ratio (obstructive pattern)
 - B** Reduced total lung capacity (restrictive pattern)
 - C** Increased diffusing capacity for carbon monoxide
 - D** Normal spirometry
 - E** Isolated reduced FVC with normal FEV₁

REPEATED 3* · ALSO ASKED IN
Harrison ×2

Answers 26–34

LOCAL EXAMS MCQS

- 26** **A Cystic fibrosis** — **UPDATED** Cystic fibrosis (thick mucus from CFTR defect) causes bronchiectasis in the young; also consider ciliary dyskinesia, immunodeficiency, ABPA.
- 27** **A Haemoptysis** — Haemoptysis is common in bronchiectasis (enlarged bronchial arteries); massive bleeding may need bronchial artery embolisation.
- 28** **A “Signet ring” sign** — The signet-ring sign (bronchus wider than its artery) is the key HRCT sign of bronchiectasis; honey-combing means fibrotic ILD.
- 29** **A Chronic infection and inflammation of the bronchial walls** — Chronic infection and neutrophilic inflammation of dilated bronchi cause the large daily sputum volumes; airway clearance is central.
- 30** **A It is rare before 5 years of age** — Asthma often begins before age 5 (most persistent cases), can start at any age, and is characteristically variable.
- 31** **B COPD (chronic bronchitis)** — Chronic bronchitis is a productive cough on most days for ≥ 3 months in 2 consecutive years (MRC definition).
- 32** **B Oral prednisolone reduces symptoms but cannot improve lung function** — **UPDATED** Steroids in a COPD exacerbation do improve FEV₁ and shorten recovery, so that statement is false; antibiotics for purulent sputum.
- 33** **C Chronic obstructive pulmonary disease (COPD)** — Chronic productive cough, dyspnoea, pursed-lip breathing and barrel chest in a heavy smoker is COPD; confirm FEV₁/FVC < 0.70.
- 34** **A Reduced FEV₁/FVC ratio (obstructive pattern)** — COPD is obstructive: post-bronchodilator FEV₁/FVC < 0.70, with gas trapping (raised RV/TLC) and low DLCO in emphysema. A low TLC means restriction.

Davidson MCQs

- 1** A 53-year-old woman with a 30-year history of asthma has worsening control despite maximal inhaled therapy, a chronic cough productive of green sputum, breathlessness and wheeze. A CT scan of her chest is shown. What does it show?



- A** Bronchiectasis
B Chronic bronchitis and emphysema
C Cystic fibrosis
D Honeycombing and interstitial fibrosis
E Typical changes of asthma
- 2** A 37-year-old woman with asthma has difficult asthma control that worsens every April and May. Serum allergy testing shows: total IgE 1241 kU/L; specific IgE to dog 24 kU/L, cat 4 kU/L, tree pollen 0.2 kU/L, oilseed rape 34 kU/L and grass pollen 25 kU/L. What is the most likely explanation for her presentation?
- A** She has exercise-induced asthma, so IgE testing is irrelevant
B She has concealed exposure to animals, as dog and cat specific IgE are raised
C She is non-adherent to medication, as total IgE remains high
D She is sensitised to montelukast, and the IgE results represent cross-reactivity
E Her asthma is triggered by seasonal pollens (oilseed rape and grass)
- 3** A 45-year-old smoker with cough, wheeze and productive sputum undergoes pulmonary function testing. Pre-bronchodilator: FEV₁ 2.6 L, FVC 4.4 L, FEV₁/FVC 0.59. Post-bronchodilator: FEV₁ 3.2 L, FVC 4.4 L, FEV₁/FVC 0.73. What does the testing show?
- A** A mixture of obstruction and restriction
B A restrictive defect
C A reversible obstructive defect
D An irreversible obstructive defect
E Inhaled salbutamol has confounded testing

- 4** A 27-year-old woman with severe atopic asthma, who is 26 weeks pregnant, presents with worsening nocturnal cough and breathlessness. PEF is 240 L/min (best 450 L/min). Her medications are high-dose ICS/LABA, oral montelukast and an as-needed SABA. What should be the next step in her management?

A Low-molecular-weight heparin (LMWH)
B Oral amoxicillin
C Oral prednisolone
D Reduce high-dose ICS/LABA
E Stop montelukast

- 5** A 48-year-old man with asthma has increasingly frequent exacerbations, nocturnal cough and blood eosinophilia. He uses fluticasone-salmeterol twice daily (good adherence and technique) and salbutamol as required, and has just started a 5-day course of prednisolone for the current exacerbation. According to current BTS/NICE/SIGN guidance, what is the most appropriate change to his maintenance therapy?

A Add amoxicillin
B Add montelukast
C Switch to budesonide-formoterol as maintenance and reliever therapy (MART)
D Provide a home nebuliser
E Start omalizumab

- 6** A 43-year-old woman who smokes and works in a bakery presents with cough, breathlessness and wheeze, which deteriorate after 3 weeks back at work. Which test is most helpful to assess for occupational asthma?

A Histamine challenge test
B Peak expiratory flow rate diary
C Specific IgE to flour
D Spirometry with reversibility
E Sputum eosinophils

7 A 55-year-old taxi driver has smoked 30 cigarettes a day since the age of 15 and is increasingly breathless on exertion. SpO₂ is 98% on air. Examination shows tracheal tug, a reduced cricosternal distance, a barrel chest, reduced cardiac dullness and symmetrically reduced air entry. The chest X-ray shows hyperinflation and spirometry shows moderate airflow obstruction. He walks 300 m on an incremental shuttle walk test before becoming breathless, and his saturations are maintained. Which pathological change best explains why he is breathless on exertion?

- A Activation of central chemoreceptors
- B Exercise-induced bronchospasm
- C Loss of elastic recoil
- D Paradoxical diaphragm movement
- E Pulmonary hypertension

8 A 67-year-old woman, an ex-smoker, has progressive exertional breathlessness. Spirometry is normal (FEV₁/FVC 0.74), but gas transfer (TLCO) is reduced to 55% of predicted. A CT scan of her chest is shown. What is the most likely cause of her breathlessness?



- A Asthma
- B Cardiac failure
- C Emphysema
- D Interstitial lung disease
- E Pulmonary thromboembolism

9 A 65 year old man with known COPD and mild obstruction presents with increasing exertional breathlessness despite LAMA and SABA. He does not desaturate on walk test. What is a reasonable addition to escalate therapy?

- A Ambulatory oxygen
- B Inhaled corticosteroid (ICS)/LABA combination inhaler
- C LAMA/LABA combination inhaler
- D Oral prednisolone
- E Salbutamol nebuliser

Answers 1-9

DAVIDSON MCQS

- 1** **A Bronchiectasis** — Dilated thick-walled central upper-lobe bronchi (signet ring) in a long-standing asthmatic suggest ABPA with bronchiectasis; check IgE and Aspergillus serology.
- 2** **E Her asthma is triggered by seasonal pollens (oilseed rape and grass)** — **UPDATED** Worsening every April–May with sensitisation to oilseed rape and grass pollen is seasonal allergic asthma; optimise ICS before the season.
- 3** **C A reversible obstructive defect** — FEV₁ rises 600 mL (23%) and the ratio normalises: a reversible obstructive defect; ≥ 400 mL strongly favours asthma.
- 4** **C Oral prednisolone** — PEF 53% of best with nocturnal symptoms in pregnancy is an exacerbation: prednisolone 40–50 mg ≥ 5 days; continue all asthma drugs.
- 5** **C Switch to budesonide–formoterol as maintenance and reliever therapy (MART)** — **UPDATED** Uncontrolled on fixed ICS/

LABA + SABA: switch to ICS–formoterol MART; if still uncontrolled with eosinophilia, refer for a biologic.

- 6** **B Peak expiratory flow rate diary** — Serial PEF (≥ 4×/day for 3–4 weeks, at and away from work) is the first-line test for occupational asthma.
- 7** **C Loss of elastic recoil** — Emphysema destroys elastin; loss of elastic recoil collapses small airways in expiration, causing dynamic hyperinflation on exertion.
- 8** **C Emphysema** — **UPDATED** Upper-lobe low-attenuation areas without walls are centrilobular emphysema; it can cause breathlessness and low TLCO before spirometry changes.
- 9** **C LAMA/LABA combination inhaler** — Persistent breathlessness on a LAMA: escalate to LABA + LAMA (GOLD 2025). ICS only with exacerbations and eosinophils ≥ 300/μL.

Harrison MCQs

- 1 Which of the following is the major risk factor for asthma?
 - A Air pollution
 - B Atopy
 - C Diet
 - D Maternal cigarette smoking
 - E Upper respiratory viral infections
 - 2 A 24-year-old woman with eczema and allergic rhinitis has asthma. Which statement regarding her disease is true?
 - A Confirmation of the diagnosis requires methacholine challenge testing
 - B Mortality due to the disease has been increasing over the past decade
 - C The most common risk factor in individuals with the disorder is genetic predisposition
 - D The prevalence of the disorder has not changed in the past several decades
 - E The severity of the disease does not vary significantly within a given patient with the disease
 - 3 A 38-year-old woman dies of status asthmaticus. All of the following pathologic findings are likely EXCEPT:
 - A Infiltration of the airway mucosa with eosinophils and activated T lymphocytes
 - B Infiltration of the alveolar spaces with eosinophils and neutrophils
 - C Occlusion of the airway lumen by mucous plugs
 - D Thickening and oedema of the airway wall
 - E Thickening of the basement membrane of the airways with subepithelial collagen deposition
 - 4 Which of the following patients is appropriately diagnosed with asthma?
 - A A 24-year-old with cough and wheeze for 6 weeks after a viral respiratory infection
 - B A 26-year-old with cough and occasional wheeze when exercising in cold weather
 - C A 34-year-old with chronic cough whose FEV₁ increases from 1.68 L to 1.98 L (18%) after salbutamol
 - D A 44-year-old laboratory worker who handles mice and has cough and wheeze towards the end of each working week
 - E A 60-year-old heavy smoker with chronic breathlessness and airway hyperreactivity
 - 5 A 24-year-old woman with mild asthma uses her salbutamol inhaler 4–7 times a week and takes no other treatment. What do you advise?
 - A Switch to as-needed low-dose budesonide–formoterol
 - B Add inhaled salmeterol twice daily
 - C Add inhaled tiotropium
 - D Continue salbutamol alone as needed
 - E Add oral montelukast
- REPEATED 2* • ALSO ASKED IN**
Crash Course
- 6 A 28-year-old pregnant woman with mild persistent asthma is well controlled on an inhaled corticosteroid and salbutamol as needed. Which statement is correct?
 - A She should continue her current therapy and follow symptoms
 - B She should switch to inhaled tiotropium as needed
 - C She should switch from inhaled beclomethasone to salmeterol
 - D There is >70% chance her asthma will lessen in pregnancy
 - E There is >70% chance her asthma will worsen in pregnancy
 - 7 A 34-year-old man with severe persistent asthma expectorates brown mucus plugs and has marked blood eosinophilia. Which test is indicated next?
 - A Aspirin desensitization
 - B Chest CT
 - C Nasal nitric oxide testing
 - D Serum immunoglobulin (Ig) E level
 - E Sweat chloride testing
 - 8 All of the following are typically characterized as an obstructive lung disease EXCEPT:
 - A Asbestosis
 - B Asthma
 - C Bronchiectasis
 - D Chronic bronchitis
 - E Emphysema

- 9** A 69-year-old man, an ex-smoker with COPD (FEV₁ 45% of predicted), has had three exacerbations in the past year despite good adherence and inhaler technique on triple therapy (LAMA/LABA/ICS) and completion of pulmonary rehabilitation. Resting SpO₂ is 94% on air, PaCO₂ is normal and his ECG shows a normal QT interval. Which treatment is most likely to reduce the frequency of his exacerbations?
- A Azithromycin 250 mg three times weekly
 - B Continuous oxygen 2 L/min
 - C Nocturnal BiPAP therapy
 - D Montelukast 10 mg daily
 - E Theophylline 300 mg daily
- 10** All of the following are risk factors for COPD EXCEPT:
- A Airway hyperresponsiveness
 - B Coal dust exposure
 - C Passive cigarette smoke exposure
 - D Recurrent respiratory infections in adulthood
 - E Use of biomass fuels in poorly ventilated areas
- 11** A 56-year-old man has a severe COPD exacerbation with acute hypercapnic respiratory failure (pH 7.28, PaCO₂ 8.2 kPa) despite controlled oxygen and nebulised bronchodilators. Which therapy has been shown to produce the greatest reduction in mortality?
- A Inhaled bronchodilators
 - B IV glucocorticoids
 - C Broad-spectrum antibiotics
 - D Early intubation
 - E Noninvasive positive-pressure ventilation
- 12** A 62-year-old woman has focal bronchiectasis after Mycobacterium avium complex (MAC) infection, which has been treated. Her sputum cultures are now negative but she still produces sputum every day. What is the best treatment?
- A Consult thoracic surgery for lobectomy
 - B Continue MAC regimen
 - C Initiate airway clearance with oscillating positive expiratory pressure flutter valve
 - D Initiate dornase (DNase)
 - E Initiate prednisone 20 mg daily

Answers 1-12

- 1** **B Atopy** — Atopy (IgE sensitisation to common allergens) is the strongest identifiable risk factor for asthma.
- 2** **E The severity of the disease does not vary significantly within a given patient with the disease** — Asthma severity tends to stay constant within an individual (mild rarely becomes severe), though control fluctuates.
- 3** **B Infiltration of the alveolar spaces with eosinophils and neutrophils** — Fatal asthma shows mucus plugs, eosinophilic airway inflammation, oedema, smooth-muscle hypertrophy and basement membrane thickening; alveoli are spared.
- 4** **C A 34-year-old with chronic cough whose FEV₁ increases from 1.68 L to 1.98 L (18%) after salbutamol** — Asthma needs symptoms plus objective variability: FEV₁ rise ≥ 12% and ≥ 200 mL after bronchodilator (here 300 mL, 18%).
- 5** **A Switch to as-needed low-dose budesonide–formoterol** — **UPDATED** SABA-only treatment is no longer advised; mild asthma uses as-needed low-dose budesonide–formoterol (cuts severe exacerbations by ~two-thirds).
- 6** **A She should continue her current therapy and follow symptoms** — Asthma in pregnancy follows the rule of thirds; continue ICS and relievers, as uncontrolled asthma is the real fetal risk.
- 7** **D Serum immunoglobulin (Ig) E level** — Brown mucus plugs with eosinophilia in severe asthma suggest ABPA: total IgE (often > 1000) with Aspergillus-specific IgE first.
- 8** **A Asbestosis** — Asbestosis is interstitial fibrosis causing restriction; asthma, bronchiectasis, chronic bronchitis and emphysema are obstructive.
- 9** **A Azithromycin 250 mg three times weekly** — **UPDATED** Frequent exacerbator despite optimal inhalers: azithromycin 250 mg three times weekly (check ECG, LFTs, hearing). Roflumilast is the alternative.
- 10** **D Recurrent respiratory infections in adulthood** — **UPDATED** Smoking, occupational dust and biomass smoke cause COPD; adult respiratory infections trigger exacerbations but are not proven causes.
- 11** **E Noninvasive positive-pressure ventilation** — **UPDATED** In acute hypercapnic COPD failure, NIV roughly halves mortality and reduces intubation.
- 12** **C Initiate airway clearance with oscillating positive expiratory pressure flutter valve** — Persistent daily sputum in bronchiectasis needs daily airway clearance (e.g. oscillating PEP flutter device) taught by a physiotherapist.

8 QUESTIONS • AFTER CRASH COURSE SBAS & EMQS

Crash Course MCQs

- 1 A 30-year-old woman known to have asthma presents to the emergency department with a 2-day history of worsening breathlessness and chest tightness despite regular salbutamol inhalers. She has been suffering from an upper respiratory tract infection for the past few days. Which of the following findings on clinical examination would be an ominous sign for an impending respiratory failure in this patient?
 - A Tachycardia
 - B Tachypnoea
 - C Silent chest
 - D Increased wheezing
 - E All of the above
- 2 A 36-year-old woman with a history of moderate persistent asthma presents to the emergency department with progressive worsening of SOB, wheezing and cough over 3 days. She has been compliant with her maintenance asthma regimen, which consists of an inhaled corticosteroid and a leukotriene receptor antagonist for maintenance therapy and salbutamol as rescue therapy. Which one of the following is most consistent with a diagnosis of asthma?
 - A Reduced FEV₁ and reduced FEV₁:FVC ratio
 - B Reduced FEV₁ and normal FEV₁:FVC ratio
 - C Reduced FEV₁ and increased FEV₁:FVC ratio
 - D Reduced FVC and normal FEV₁:FVC ratio
 - E Reduced FVC and increased FEV₁:FVC ratio
- 3 A 60-year-old man with severe COPD is admitted with an acute exacerbation. What is the most appropriate initial investigation?
 - A Computed tomography (CT) chest scan
 - B Spirometry
 - C Arterial blood gas
 - D Peak flow
 - E Pulmonary function tests (PFTs)
- 4 A 68-year-old woman with known COPD is admitted with increasing breathlessness, wheeze and sputum production. Her respiratory rate is 26/min and SpO₂ is 84% on air; an arterial blood gas sample is being taken. What is the most appropriate immediate step in management?
 - A 60% oxygen
 - B 24% oxygen
 - C Oral steroids
 - D Noninvasive positive-pressure ventilation
 - E Antibiotics
- 5 A child is admitted with a cough productive of thick, foul-smelling green sputum. His mother has noticed episodes of him coughing up blood. On examination he has marked finger clubbing and coarse crackles at both lung bases. What is the most likely cause of his symptoms?
 - A Cystic fibrosis
 - B Inhaled foreign body
 - C Measles
 - D Klebsiella pneumonia
 - E Whooping cough
- 6 You suspect bronchiectasis in a young patient. Which of the following investigations is most likely to confirm the diagnosis?
 - A Sputum culture
 - B Sweat test
 - C Bronchoscopy
 - D Chest X-ray (CXR)
 - E High-resolution CT
- 7 A child recently diagnosed with cystic fibrosis is admitted following an exacerbation of his symptoms. What is the next most appropriate step in management?
 - A Prednisolone
 - B N-acetylcysteine
 - C Sodium reabsorption blockers
 - D Antibiotics
 - E Aminophylline
- 8 A teenager presents to the GP with symptoms suggestive of cystic fibrosis, and a sweat test is performed. Which result is diagnostic?
 - A A chloride value of 30–59 mmol/L
 - B A chloride value below 30 mmol/L
 - C A sodium value below 60 mmol/L
 - D A chloride value of 60 mmol/L or above
 - E A sodium value of 40–60 mmol/L

Answers 1-8

CRASH COURSE MCQS

- 1 **C Silent chest** — A silent chest means almost no airflow: a life-threatening feature signalling impending respiratory failure; urgent ICU review.
- 2 **A Reduced FEV₁ and reduced FEV₁:FVC ratio** — Asthma is obstructive: FEV₁ falls more than FVC (ratio < 0.70), with reversibility. Low FVC with normal ratio means restriction.
- 3 **C Arterial blood gas** — ABG is the key first test in a COPD admission: detects hypercapnia/acidosis and guides controlled oxygen and NIV.
- 4 **B 24% oxygen** — UPDATED Hypoxaemic COPD exacerbation: start controlled oxygen (24–28% Venturi) to SpO₂ 88–92% while the ABG is taken.
- 5 **A Cystic fibrosis** — Chronic foul purulent sputum, clubbing and bilateral crackles in a child suggests bronchiectasis from cystic fibrosis; sweat test.
- 6 **E High-resolution CT** — HRCT confirms bronchiectasis: bronchi wider than the artery (signet ring), no tapering, airways within 1 cm of pleura.
- 7 **D Antibiotics** — CF exacerbation: antibiotics guided by recent sputum (often IV ceftazidime + tobramycin for 14 days) plus physiotherapy.
- 8 **D A chloride value of 60 mmol/L or above** — UPDATED Sweat chloride ≥ 60 mmol/L (repeated) with a compatible picture is diagnostic of CF; 30–59 is intermediate.

2 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 28-year-old male with a long history of severe asthma presents to the emergency room with shortness of breath. He has previously required admission to the hospital and was once intubated for asthma. Which of the following findings on physical examination would predict a benign course?
 - A Silent chest
 - B Hypercapnia
 - C Thoracoabdominal paradox (paradoxical respiration)
 - D Pulsus paradoxus of 5 mm Hg
 - E Altered mental status
- 2 A 30-year-old athlete presents with intermittent wheezing that begins shortly after running. He smokes 20–40 cigarettes a day and has done so for 5 years. Which finding would be consistent with asthma?
 - A Hyperinflation present on chest x-ray
 - B Improvement in FEV₁ after bronchodilator
 - C Low oxygen saturation on finger oximetry
 - D Decreased FVC on PFT testing
 - E Dyspnoea on assuming a supine position

Answers 1-2

PRETEST MCQS

- 1 **D Pulsus paradoxus of 5 mm Hg** — Pulsus paradoxus up to 10 mmHg is normal, so 5 mmHg predicts a benign course; silent chest, hypercapnia and confusion are ominous.
- 2 **B Improvement in FEV₁ after bronchodilator** — Asthma's defining feature is reversible obstruction: FEV₁ rise ≥ 12% and ≥ 200 mL after bronchodilator.

7 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 28-year-old builder attends the respiratory clinic with a 6-month history of progressive exertional breathlessness and cough. His GP started a salbutamol inhaler, which he now uses several times a day. He has smoked since the age of 18 (10 pack-years) and reports occupational exposure to asbestos. Spirometry shows FEV_1/FVC 0.68; FEV_1 is 3.0 L before and 3.4 L after bronchodilator. What is the most appropriate intervention?
 - A Long-acting beta-agonist (LABA) and long-acting muscarinic antagonist (LAMA) inhaler
 - B Long-acting beta-agonist (LABA) inhaler
 - C Low-dose ICS–formoterol as maintenance and reliever therapy (MART)
 - D Omalizumab
 - E Oral glucocorticoids
- 2 A 30-year-old woman is admitted to the Emergency Department with an exacerbation of asthma. On arrival her peak flow is 30% of predicted, respiratory rate is 36/min and oxygen saturations are 98% on 100% high-flow oxygen. She is given back-to-back nebulisers, intravenous hydrocortisone and started on a magnesium infusion. Which one of the following would be the strongest indicator of a need for intubation and ventilation?
 - A PEF 35% of predicted
 - B pH 7.33
 - C Patient preference after informed consent
 - D Respiratory rate of 50/min
 - E Oxygen saturations of 95% on 100% high-flow oxygen
- 3 A 37-year-old woman with known asthma presents to the emergency department with severe shortness of breath. Her respiratory rate is 35/min, she is unable to complete a sentence and there is widespread wheeze. Initial ABG on air: pH 7.46, $PaCO_2$ 3.9 kPa, PaO_2 11.2 kPa, HCO_3^- 22 mmol/L. She is given oral prednisolone and back-to-back nebulised salbutamol and ipratropium. An hour later she is alert, but her chest is silent on auscultation. Repeat ABG: pH 7.35, $PaCO_2$ 5.7 kPa, PaO_2 11.0 kPa, HCO_3^- 24 mmol/L. Chest X-ray is unremarkable. What is the best next step in her management?
 - A Non-invasive ventilation
 - B Invasive mechanical ventilation
 - C Intravenous hydrocortisone
 - D Intravenous magnesium sulfate
 - E Omalizumab
- 4 A 65-year-old man attends the emergency department with a cough productive of green sputum. He has recently been diagnosed with COPD and is an active smoker. On examination, he is saturating at 94% on room air and is wheezy bilaterally. A chest X-ray does not reveal any consolidation. Blood tests: CRP 26 mg/L; Na^+ 133 mmol/L; K^+ 3.6 mmol/L; Bicarbonate 34 mmol/L; Urea 6 mmol/L; Creatinine 100 μ mol/L. What is the most likely underlying organism causing this clinical picture?
 - A Haemophilus influenzae
 - B Klebsiella pneumoniae
 - C Moraxella catarrhalis
 - D Staphylococcus aureus
 - E Streptococcus pneumoniae
- 5 A 58-year-old man with chronic obstructive pulmonary disease (COPD) presents with a 1-week history of increased breathlessness and reduced exercise tolerance. His COPD is usually managed with a salbutamol inhaler as required and a salmeterol inhaler. He has a chronic cough productive of clear sputum but denies any recent change in this. He appears breathless and his respiratory rate is 24 breaths per minute, but there is no use of accessory muscles at rest. All other clinical observations were normal. On auscultation of his chest you note expiratory wheeze throughout but good air entry and no other added sounds. Other than encouraging the use of his salbutamol inhaler, what is the most appropriate management?
 - A Give a course of oral prednisolone for 5 days
 - B Give a course of oral prednisolone for 7 days and oral amoxicillin for 5 days
 - C Give a course of oral prednisolone for 7 days and oral clarithromycin for 5 days
 - D Give a course of oral prednisolone for 5 days and oral amoxicillin for 5 days
 - E Give a course of oral prednisolone for 7 days and oral furosemide for 7 days

- 6** A 62-year-old woman with recently diagnosed chronic obstructive pulmonary disease (COPD) presents for review. Her FEV₁ is 65% of the predicted value. She has managed to give up smoking and was prescribed a salbutamol inhaler to use as required. Despite this, she is still symptomatic and complains of wheeze and shortness of breath. There is no history of asthma, eosinophilia or FEV₁ variation. What is the most appropriate next step?
- A Add an inhaled corticosteroid
 - B Add a combined long-acting beta2-agonist and long-acting muscarinic antagonist inhaler
 - C Refer for consideration of long-term oxygen therapy
 - D Add a long-acting beta2-agonist
 - E Add a combination long-acting beta2-agonist and corticosteroid inhaler
- 7** A 69-year-old man with COPD presents to the emergency department with breathlessness. Three days ago his GP started amoxicillin and prednisolone. In the department he has received back-to-back nebulised salbutamol and ipratropium, inspired oxygen has been titrated to 28% to achieve SpO₂ 88–92%, and because of a poor response an aminophylline infusion has been started. Thirty minutes later, his ABG on 28% oxygen shows pH 7.30, PaCO₂ 7.6 kPa and PaO₂ 8.1 kPa. What is the most appropriate next step in management?
- A Intravenous magnesium sulphate
 - B Intravenous hydrocortisone
 - C Decrease his oxygen to 24%
 - D Non-invasive ventilation
 - E Increase his oxygen to 35%

REPEATED 4* • ALSO ASKED IN
Formative exam • Harrison • Passmedicine

Answers 1–7

PASSMEDICINE MCQS

- 1** **C Low-dose ICS–formoterol as maintenance and reliever therapy (MART)** — **UPDATED** Obstruction with a 400 mL bronchodilator response in a young man is asthma; frequent symptoms need low-dose ICS–formoterol MART.
- 2** **B pH 7.33** — In acute asthma PaCO₂ should be low; acidosis (pH 7.33) means tiring with CO₂ retention — the strongest indication for intubation.
- 3** **D Intravenous magnesium sulfate** — **UPDATED** Silent chest and a 'normal' (rising) PaCO₂ are life-threatening; give IV magnesium 1.2–2 g, continue nebulisers and get ICU review.
- 4** **A Haemophilus influenzae** — Haemophilus influenzae is the commonest bacterial cause of infective COPD exacerbations (then pneumococcus, Moraxella).
- 5** **A Give a course of oral prednisolone for 5 days** — Non-infective COPD exacerbation (sputum unchanged): prednisolone 30 mg for 5 days and more bronchodilator; antibiotics only if purulent.
- 6** **B Add a combined long-acting beta2-agonist and long-acting muscarinic antagonist inhaler** — COPD still breathless on SABA without asthmatic features: add LABA + LAMA (NICE NG115, GOLD 2025).
- 7** **D Non-invasive ventilation** — **UPDATED** Respiratory acidosis (pH < 7.35, PaCO₂ > 6.5 kPa) persisting despite optimal therapy is the indication for NIV.

8 QUESTIONS • AFTER PASTEST

PasTest MCQs

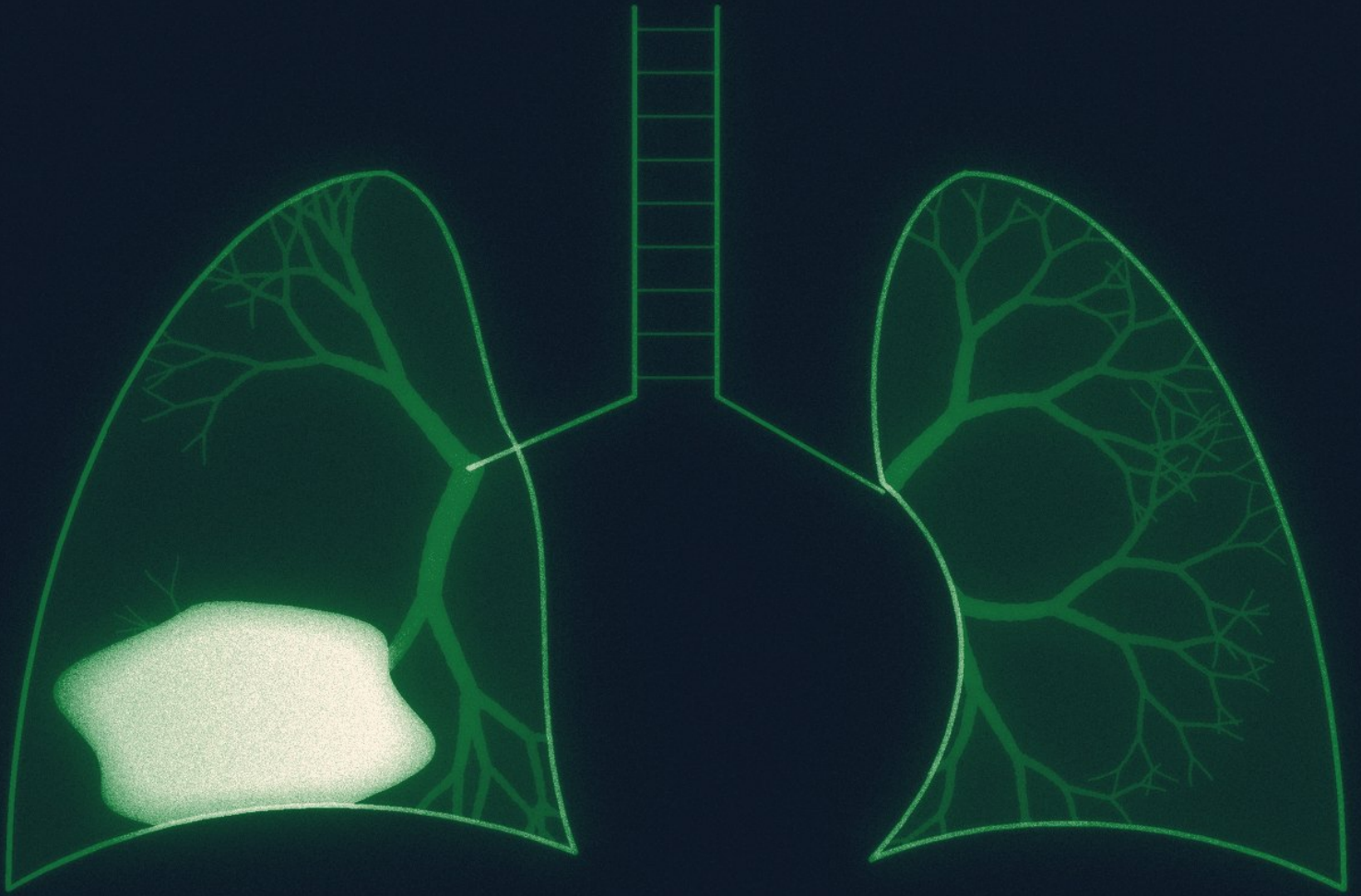
- 1** A 50-year-old woman has breathlessness on exertion that has been worsening for a year, with wheeze that is worse with exercise and during the summer, when she also has severe hay fever. She has never smoked. The chest X-ray shows mild hyperinflation but is otherwise unremarkable. Lung function: FEV₁ 1.1 L (predicted 1.7–3.0), FVC 2.7 L (2.0–3.5), FEV₁/FVC 41%, FRC 4.2 L (1.7–3.4), TLC 6.5 L (3.6–5.6), TLCO 5.86 mmol/min/kPa (5.7–9.5), KCO 1.63 mmol/min/kPa/L (predicted 1.66). What is the most likely diagnosis?
- A Asthma
 - B Bronchiectasis
 - C Chronic obstructive pulmonary disease (COPD)
 - D Interstitial lung disease
 - E Pulmonary haemorrhage
- 2** A 25-year-old man presents with breathlessness. Lung function tests show a PEF of 300 L/min and a reduced FEV₁/FVC ratio. He is otherwise healthy, smokes five cigarettes a day and keeps a pet cat. What is the most likely diagnosis?
- A Asthma
 - B Bronchial carcinoma
 - C Bronchiectasis
 - D Kyphoscoliosis
 - E Whooping cough

- 3** A 75-year-old man with a body mass index of 31 kg/m² has had worsening breathlessness and cough for about 1 week. He is normally short of breath and has had to give up walking to the shops, only 300 m away, and has a chronic cough that is worse in the morning. He smoked 20 cigarettes a day until this illness. ABG on air: pH 7.24, PaO₂ 6.2 kPa, PaCO₂ 9.8 kPa, HCO₃⁻ 33 mmol/L. What is the most likely diagnosis?
- A Acute asthma
 - B Acute exacerbation of chronic obstructive pulmonary disease
 - C Obstructive sleep apnoea
 - D Pulmonary embolus
 - E Pulmonary oedema
- 4** You review a 56-year-old man with COPD who is having an acute exacerbation. Which of the following features would suggest suitability for non-invasive ventilation with BiPAP?
- A Confusion
 - B Decreased conscious level
 - C Hypercapnia without profound hypoxia
 - D Concurrent pneumothorax
 - E Profound hypoxia
- 5** A 72-year-old man with severe COPD asks about home oxygen therapy. In which of the following settings have randomised controlled trials shown that long-term oxygen therapy (LTOT) reduces mortality?
- A Asthma
 - B Cor pulmonale caused by chronic airflow obstruction
 - C Cystic fibrosis
 - D Idiopathic pulmonary fibrosis
 - E Pulmonary sarcoidosis
- 6** A 66-year-old retired boilermaker attends the respiratory clinic. Spirometry shows post-salbutamol FEV₁ 1.81 L and FVC 2.7 L; transfer factor is 55% of predicted. What is the most likely diagnosis?
- A Asbestos-related pleural plaque disease
 - B Asthma
 - C Chronic obstructive pulmonary disease
 - D Pulmonary embolism
 - E Pulmonary fibrosis
- 7** A 40-year-old man presents with a long history of productive cough and breathlessness. He also complained of halitosis and reported having exacerbations that consisted of productive sputum, chest pain and occasional haemoptysis. Examination in the clinic reveals bilateral inspiratory crackles. Which of following treatments is the most likely to decrease the frequency of his exacerbations?
- A Cyclical antibiotic therapy
 - B Inhaled corticosteroids
 - C Nebulised bronchodilators
 - D Postural drainage
 - E Surgical resection
- 8** A 69-year-old woman with rheumatoid arthritis is referred with chronic cough productive of purulent sputum and recurrent chest infections. On examination she had coarse crepitations at both the bases. What is the most likely diagnosis?
- A Bronchiectasis
 - B Carcinoma of the lung
 - C Chronic bronchitis
 - D Diffuse interstitial lung fibrosis
 - E Tuberculosis

Answers 1-8

PASTEST MCQS

- 1** **A Asthma** — **UPDATED** Obstruction with hyperinflation but normal TLCO/KCO in a never-smoker with seasonal wheeze is asthma; emphysema lowers gas transfer.
- 2** **A Asthma** — Obstruction with a low peak flow and cat exposure in a 25-year-old is asthma; confirm with FeNO or reversibility.
- 3** **B Acute exacerbation of chronic obstructive pulmonary disease** — A chronic smoker with ABG showing acidosis, hypercapnia and raised bicarbonate has an acute-on-chronic type 2 failure from a COPD exacerbation.
- 4** **C Hypercapnia without profound hypoxia** — **UPDATED** NIV suits a conscious COPD patient with persistent hypercapnic acidosis; confusion, profound hypoxia, undrained pneumothorax and secretions favour intubation.
- 5** **B Cor pulmonale caused by chronic airflow obstruction** — LTOT ≥ 15 h/day improved survival in hypoxaemic COPD with cor pulmonale (MRC, NOTT); no survival benefit shown in CF, IPF or sarcoid.
- 6** **C Chronic obstructive pulmonary disease** — Post-bronchodilator FEV₁/FVC 0.67 with low transfer factor is COPD with emphysema; asbestosis would be restrictive.
- 7** **D Postural drainage** — Bronchiectasis exacerbations are reduced by daily airway clearance (postural drainage); add long-term azithromycin if ≥ 3 exacerbations/year.
- 8** **A Bronchiectasis** — Chronic purulent sputum and coarse basal crackles in rheumatoid arthritis is bronchiectasis; RA-ILD gives a dry cough and fine crackles.



12



RESPIRATORY INFECTION & INFLAMMATION

Respiratory Infection & Inflammation

Pneumonia, tuberculosis, bronchiectasis, lung abscess and granulomatous disease: organisms, severity assessment and treatment.

79

MCQs

0

EMQ SCENARIOS

6

SOURCES

17

REPEATED IDEAS

IN THIS CHAPTER

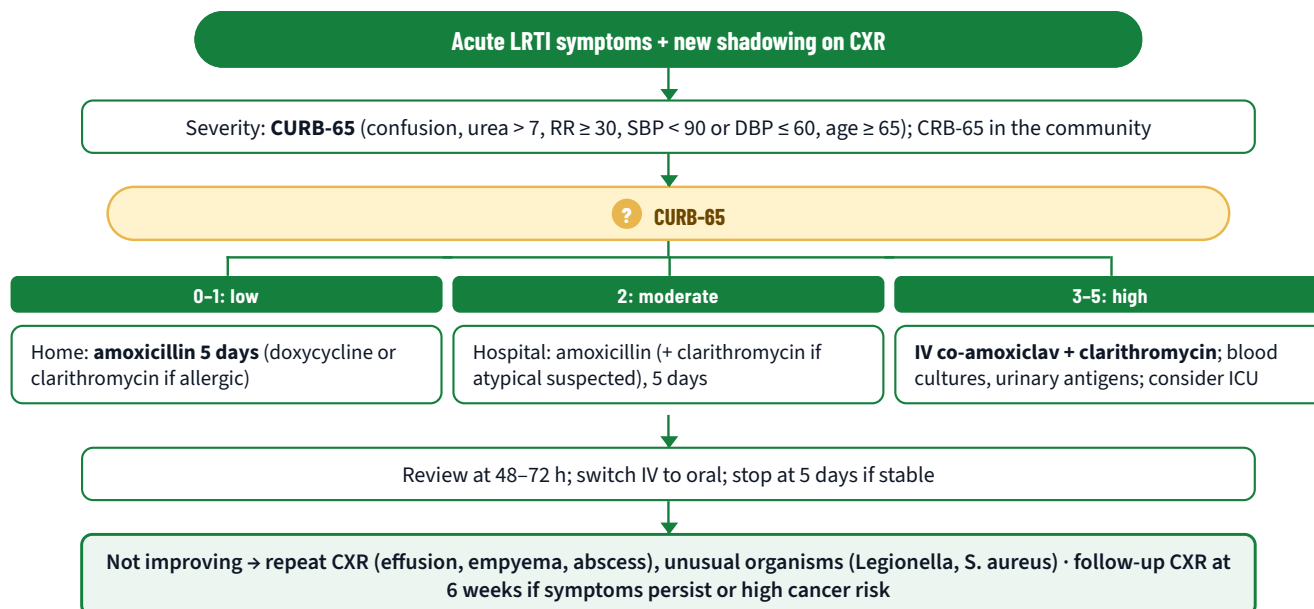
01	Management flowcharts	3 charts	page 212
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Step by step

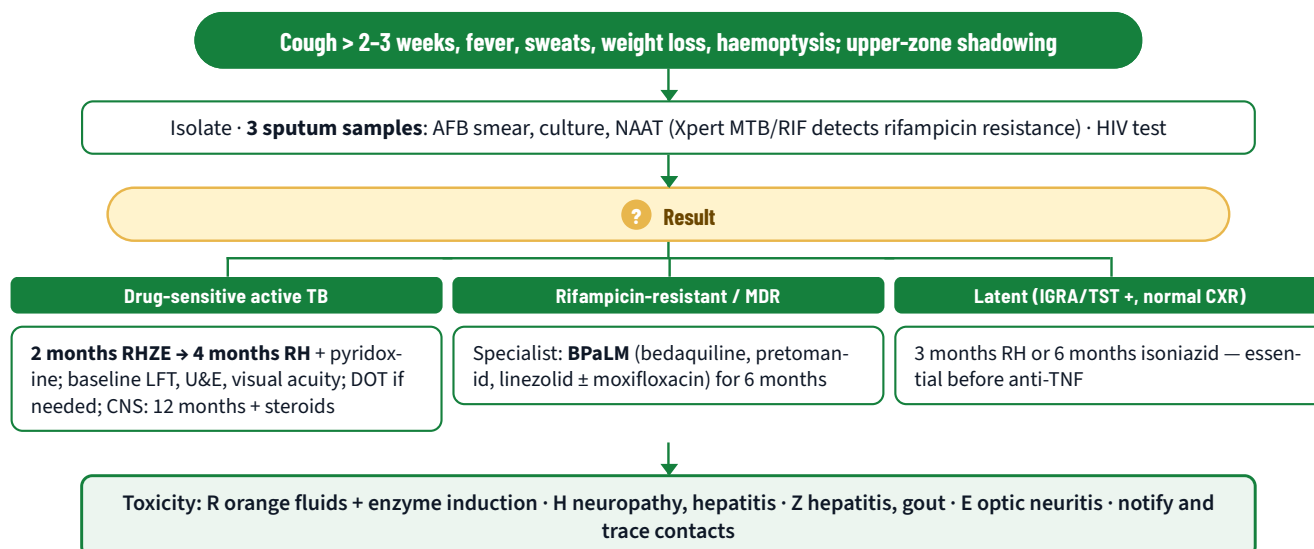
Community-acquired pneumonia

NICE • BTS



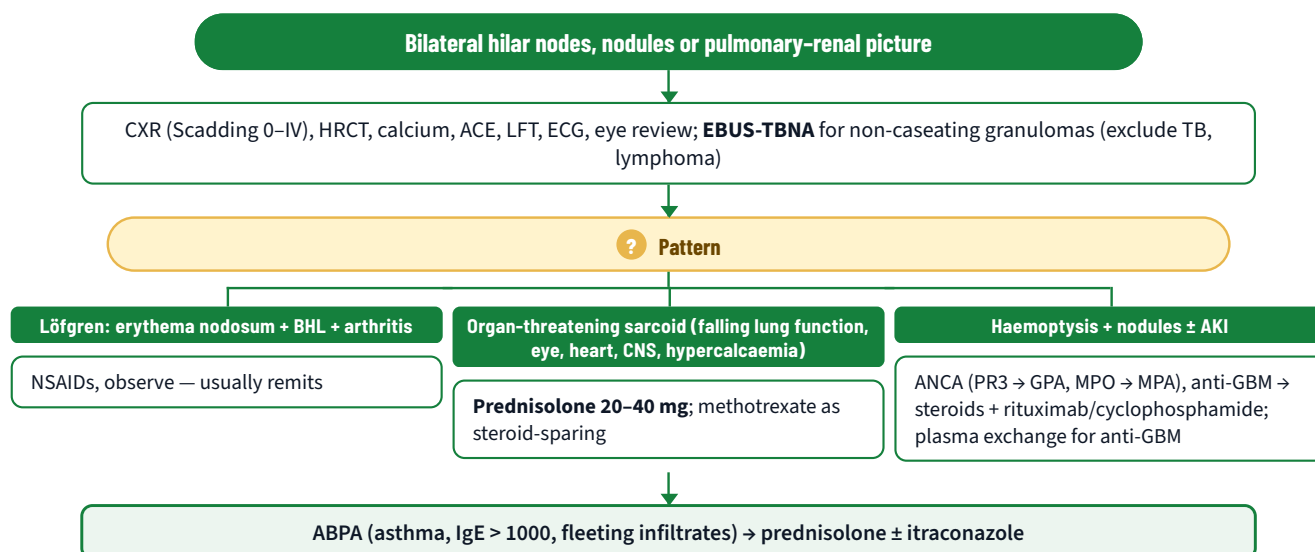
Tuberculosis

NICE NG33 • WHO 2022



Granulomatous and vasculitic lung disease

ERS 2021 SARCOIDOSIS · EULAR 2022 VASCULITIS



Local Exams MCQs

SET 1 OF 2

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

CASE • QUESTIONS 1–4

A 25-year-old woman with diabetes has had cough, fever, night sweats and weight loss for 4 months. She has finger clubbing and crackles over both apices; the chest X-ray shows ill-defined opacities in both upper zones.

- 1 What is the most important investigation?
 - A Full blood count and ESR
 - B C-reactive protein
 - C Bronchoscopy
 - D Sputum microscopy and culture for acid-fast bacilli
 - E Tuberculin skin test
- 2 Three weeks after starting treatment her family notices yellow discoloration of her sclera and skin. What is the most likely cause?
 - A Viral hepatitis
 - B Acute cholecystitis
 - C Drug-induced hepatitis
 - D Biliary cirrhosis
 - E Common bile duct stone
- 3 Risk factors for tuberculosis are grouped into patient-related factors and associated diseases. Which of the following is an associated disease rather than a patient-related factor?
 - A Overcrowding
 - B Smoking
 - C Diabetes mellitus
 - D Close contact with a smear-positive case
 - E Chest X-ray evidence of self-healed tuberculosis
- 4 Which of the following is NOT an indication to admit her to hospital?
 - A She is not adherent to her drugs
 - B She develops severe gastrointestinal upset from the drugs
 - C The physician is unsure of the diagnosis
 - D Drug resistance
 - E Diabetes mellitus

REPEATED 4* • ALSO ASKED IN
39th batch end-block (respiratory) ×2 • Crash Course

REPEATED 2* • ALSO ASKED IN
39th batch end-block (respiratory)

REPEATED 2* • ALSO ASKED IN
39th batch end-block (respiratory)

- 5 Which of the following is NOT a risk factor for drug-resistant tuberculosis?

- A Poor-quality drugs
- B Drug shortages
- C Diabetes mellitus
- D Previous treatment failure
- E Poor health-care provision

REPEATED 3* • ALSO ASKED IN
39th batch end-block (respiratory) • Previous-years exam

- 6 A 76-year-old man has fever, productive cough and worsening breathlessness for 2 days and is confused. T 39.1°C, BP 92/58 mmHg, RR 32/min, HR 122/min, SpO₂ 88% on air. Serum urea 12.1 mmol/L (BUN 34 mg/dL), WBC $18 \times 10^9/L$, lactate 3.1 mmol/L; the chest X-ray shows bilateral multilobar infiltrates. What is the most appropriate disposition?
 - A Discharge home on oral antibiotics
 - B Admit to the general medical ward
 - C Outpatient treatment with close follow-up
 - D Admit to the respiratory/intensive care unit
 - E Treat in the emergency department and observe for 6 hours
- 7 A 39-year-old man started standard four-drug treatment (rifampicin, isoniazid, pyrazinamide, ethambutol) for pulmonary tuberculosis 2 weeks ago. He now has abdominal pain, fatigue, dark urine and mild jaundice, and an acutely painful, swollen first metatarsophalangeal joint. AST 220 U/L, ALT 260 U/L, ALP normal, bilirubin 53 µmol/L (3.1 mg/dL), serum urate 0.62 mmol/L (raised). Which drug is most likely responsible for both problems?
 - A Ethambutol
 - B Isoniazid
 - C Rifampicin
 - D Pyrazinamide
 - E Streptomycin
- 8 A 52-year-old non-smoker has a long history of daily cough with purulent sputum, worse in the morning, occasional blood-streaked sputum and recurrent pneumonia. There are coarse inspiratory crackles at the bases and spirometry shows obstruction. Which is the best test to confirm bronchiectasis?
 - A Chest X-ray
 - B High-resolution CT chest
 - C Sputum culture
 - D Bronchoscopy
 - E MRI chest

REPEATED 5* • ALSO ASKED IN
End-block exam 2025 • Log book 2025 • Davidson • Harrison

- 9** A 45-year-old woman has community-acquired pneumonia with fever, cough and pleuritic chest pain. Which of the following is an indication for ICU admission?
- A CURB-65 score of 2
 - B Alkalosis
 - C Haemophilus influenzae as the causative organism
 - D Progressive hypercapnia
- 10** A 35-year-old man has had a flu-like illness for 5 days. Which finding supports a diagnosis of primary tuberculosis?
- A Erythema nodosum
 - B Erythema marginatum
 - C Hepatomegaly
 - D Finger clubbing
- 11** A 64-year-old man who drinks heavily has had high fever, pleuritic chest pain and cough with foul-smelling sputum for 15 days. He has finger clubbing, and the chest X-ray shows a right lower-lobe infiltrate with an air–fluid-filled cavity. What is the most likely diagnosis?
- A Pulmonary tuberculosis
 - B Acute exacerbation of COPD
 - C Lung abscess
 - D Pulmonary embolism
- 12** Which of the following is NOT a feature of primary tuberculosis?
- A Erythema nodosum
 - B Erythema marginatum
 - C Dactylitis
 - D Phlyctenular conjunctivitis
- REPEATED 2* · ALSO ASKED IN**
Previous-years exam
- 13** A patient with acute lobar pneumonia has herpes labialis (cold sores). Which organism is the most likely cause of the pneumonia?
- A Streptococcus pneumoniae
 - B Mycoplasma pneumoniae
 - C Staphylococcus epidermidis
 - D Haemophilus influenzae
- 14** Which of the following is NOT useful for diagnosing active tuberculosis in a high-prevalence country?
- A Chest X-ray
 - B Sputum culture
 - C Interferon-gamma release assay (IGRA)
 - D Rapid molecular test on sputum (Xpert MTB/RIF)
- 15** A patient with community-acquired pneumonia has a CURB-65 score of 2 and is allergic to penicillin. Which drug is suitable?
- A Meropenem
 - B Doxycycline
 - C Gentamicin
 - D Ceftriaxone
- 16** Brain abscess can complicate community-acquired pneumonia caused by:
- A Streptococcus pneumoniae
 - B Staphylococcus aureus
 - C Mycoplasma pneumoniae
 - D Legionella pneumophila
 - E Pseudomonas aeruginosa
- REPEATED 2* · ALSO ASKED IN**
Formative exam
- 17** Which of the following is NOT a patient-related risk factor for tuberculosis?
- A Smoking
 - B Age between 20 and 25 years
 - C Close contact with smear-positive pulmonary TB
 - D Overcrowding
 - E Chest X-ray evidence of self-healed TB
- REPEATED 3* · ALSO ASKED IN**
Formative exam · Previous-years exam
- 18** A tuberculous granuloma contains all of the following cells EXCEPT:
- A Lymphocytes
 - B Neutrophils
 - C Langhans giant cells
 - D Epithelioid cells
- REPEATED 2* · ALSO ASKED IN**
End-block exam 2025
- 19** A 22-year-old man has fever, night sweats, weight loss and cough productive of cupfuls of blood. What is the most likely diagnosis?
- A Pneumonia
 - B Haemophilia
 - C Pulmonary tuberculosis
 - D Granulomatosis with polyangiitis (Wegener)
- REPEATED 3* · ALSO ASKED IN**
Previous-years exam · Crash Course
- 20** Which is the important diagnostic test for pulmonary tuberculosis?
- A Tuberculin skin test
 - B Sputum for AFB smear and culture
 - C IGRA
 - D Chest X-ray
- REPEATED 2* · ALSO ASKED IN**
Previous-years exam

CASE · QUESTIONS 21–23

A 76-year-old man has fever, malaise, breathlessness and cough with green sputum. He is tachypnoeic with reduced expansion, reduced air entry, dullness and bronchial breathing in the left lower zone.

- 21** What is the most likely diagnosis?
- A Acute lobar pneumonia
 - B Bronchopneumonia
 - C COPD
 - D Acute bronchitis
 - E Asthma
- 22** What is the most likely causative organism?
- A Staphylococcus aureus
 - B Haemophilus influenzae
 - C Streptococcus pneumoniae
 - D Klebsiella pneumoniae
 - E Mycoplasma pneumoniae

REPEATED 2* · ALSO ASKED IN
Log book 2025

- 23** He is assessed as having low-severity pneumonia suitable for oral treatment. What is the appropriate antibiotic?
- A Ceftriaxone
 - B Metronidazole
 - C Amoxicillin
 - D Erythromycin

REPEATED 2* · ALSO ASKED IN
Crash Course

- 24** A 26-year-old man has had breathlessness and a productive cough for 3 months. He has finger clubbing and a 6-year history of infertility. What is the most likely cause?
- A Sarcoidosis
 - B Congenital heart disease
 - C Bronchiectasis
 - D Chronic bronchitis

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

- 25** A 24-year-old man has fever and cough productive of sputum after a recent influenza-like illness. He is unwell but alert and orientated; BP 80/60 mmHg, RR 35/min, urea 55 mg/dL (9.2 mmol/L). Chest X-ray shows multilobar consolidation with several thin-walled pneumatoceles. What is the most likely causative organism?
- A Staphylococcus aureus
 - B Streptococcus pneumoniae
 - C Klebsiella pneumoniae
 - D Legionella pneumophila
 - E Mycoplasma pneumoniae

Answers 1–25**LOCAL EXAMS MCQS**

- 1** **D Sputum microscopy and culture for acid-fast bacilli** — **UPDATED** Months of cough, fever, sweats and weight loss with upper-zone shadowing is TB: 2–3 sputum samples for AFB smear, culture and Xpert MTB/RIF.
- 2** **C Drug-induced hepatitis** — Jaundice weeks into RIPE therapy is drug-induced hepatitis; stop if ALT > 5× normal, then reintroduce drugs one at a time.
- 3** **C Diabetes mellitus** — **UPDATED** Diabetes is an associated disease raising TB risk (like HIV, steroids, CKD, silicosis); contact, overcrowding and smoking are patient factors.
- 4** **E Diabetes mellitus** — Most pulmonary TB is treated as outpatients; admit for uncertain diagnosis, intolerance, poor adherence, severe illness or MDR-TB. Diabetes alone is not a reason.
- 5** **C Diabetes mellitus** — Drug resistance comes from inadequate treatment and non-adherence (previous treatment is the strongest predictor); diabetes is not a cause.
- 6** **D Admit to the respiratory/intensive care unit** — CURB-65 = 5 with multilobar disease and high lactate is high-severity pneumonia: early ICU review, IV co-amoxiclav plus clarithromycin.
- 7** **D Pyrazinamide** — **UPDATED** Pyrazinamide causes both hepatitis and hyperuricaemia/gout; isoniazid causes neuropathy, rifampicin cholestasis, ethambutol optic neuritis.
- 8** **B High-resolution CT chest** — High-resolution CT confirms bronchiectasis (signet ring, no tapering); then test for a cause (immunoglobulins, ABPA, CF, cilia).
- 9** **D Progressive hypercapnia** — Rising PaCO₂ in pneumonia means the patient is tiring: an indication for ICU assessment. CURB-65 of 2 is ward care.
- 10** **A Erythema nodosum** — Primary TB may cause hypersensitivity: erythema nodosum and phlyctenular conjunctivitis, with a Ghon focus and hilar nodes.
- 11** **C Lung abscess** — A heavy drinker with foul sputum and a right lower-lobe cavity with air–fluid level has an aspiration lung abscess; prolonged antibiotics with anaerobic cover.
- 12** **B Erythema marginatum** — Primary TB can cause erythema nodosum, phlyctenular conjunctivitis and dactylitis; erythema marginatum belongs to rheumatic fever.
- 13** **A Streptococcus pneumoniae** — **UPDATED** S. pneumoniae is the commonest cause of lobar pneumonia; the fever often reactivates herpes labialis (cold sores).

- 14** **C Interferon-gamma release assay (IGRA)** — **UPDATED** IGRA cannot separate latent from active TB and is often positive in high-prevalence countries; sputum smear, culture and Xpert diagnose active disease.
- 15** **B Doxycycline** — CURB-65 = 2 with penicillin allergy: doxycycline or clarithromycin (amoxicillin otherwise).
- 16** **B Staphylococcus aureus** — *S. aureus* pneumonia is necrotising and bacteraemic, seeding metastatic abscesses including brain abscess.
- 17** **B Age between 20 and 25 years** — TB risk is highest in young children and the elderly; age 20–25 is not a risk factor.
- 18** **B Neutrophils** — TB granulomas contain epithelioid macrophages, Langhans giant cells and lymphocytes with caseation; neutrophils are not characteristic.
- 19** **C Pulmonary tuberculosis** — Fever, sweats, weight loss and haemoptysis in a young adult is pulmonary TB until proven otherwise; send sputum for AFB.
- 20** **B Sputum for AFB smear and culture** — Sputum AFB smear and culture ($\pm$ molecular test) is the key test for pulmonary TB; culture is the gold standard and gives sensitivities.
- 21** **A Acute lobar pneumonia** — Dullness, bronchial breathing and reduced expansion in one zone with fever and purulent sputum are the signs of lobar consolidation.
- 22** **C Streptococcus pneumoniae** — *S. pneumoniae* is the commonest identified cause of community-acquired pneumonia and classic lobar consolidation.
- 23** **C Amoxicillin** — Low-severity pneumonia (CURB-65 0–1): oral amoxicillin 500 mg tds for 5 days; doxycycline or clarithromycin if allergic.
- 24** **C Bronchiectasis** — Chronic productive cough, clubbing and male infertility suggest bronchiectasis from CF, primary ciliary dyskinesia or Young syndrome.
- 25** **A Staphylococcus aureus** — Necrotising pneumonia with pneumatoceles after influenza is typical of *S. aureus* (including PVL strains). *Pneumococcus* is commoner overall but rarely cavitates.

SET 2 OF 2

CASE · QUESTIONS 26–27

A 24-year-old man has fever and cough productive of sputum after a recent influenza-like illness. He is unwell but alert and orientated; BP 80/60 mmHg, RR 35/min, urea 55 mg/dL (9.2 mmol/L). Chest X-ray shows multilobar consolidation with several thin-walled pneumatoceles.

- 26** What is his CURB-65 score?
- A 1
B 2
C 3
D 4
E 5
- 27** What is the best initial treatment?
- A Vancomycin plus ceftriaxone
B Ceftriaxone plus azithromycin
C Levofloxacin alone
D Flucloxacillin alone
E Doxycycline
- 28** A patient is suspected of having pulmonary tuberculosis. How is the diagnosis confirmed?
- A Sputum acid-fast bacilli smear, culture and GeneXpert (NAAT)
B Blood culture
C Chest X-ray
D Tuberculin skin test
E Interferon-gamma release assay
- 29** Which antituberculous drug causes peripheral neuropathy and requires pyridoxine (vitamin B6) supplementation?
- A Isoniazid
B Rifampicin
C Ethambutol
D Streptomycin
E Pyrazinamide
- FORMATIVE & PREVIOUS YEARS**
College formative and previous-years exams
- 30** A 40-year-old man with early right lower-lobe pneumonia has arterial blood gases on air: PaO₂ 8.1 kPa (61 mmHg), PaCO₂ 4.2 kPa (32 mmHg). Why is he hypoxaemic without hypercapnia?
- A Increased ventilation of healthy alveoli removes CO₂ but cannot add extra oxygen to blood leaving the consolidated lung
B Supplemental oxygen suppresses carbon dioxide production
C The infection is too mild to affect gas exchange
D Fever reduces carbon dioxide production
E Consolidation increases the diffusing capacity for carbon dioxide
- 31** Which of the following is NOT a measure for the prevention and control of tuberculosis?
- A Chemoprophylaxis (preventive therapy) for latent infection
B BCG vaccination
C Isolation of smear-positive patients
D Pneumococcal vaccination
E Directly observed therapy

- 32** Regarding the adverse effects of antituberculous drugs, which statement is true?
- A** Asymptomatic rises in transaminases are common but serious liver damage is rare
 - B** Pyrazinamide commonly causes peripheral neuropathy
 - C** Ethambutol causes red-orange discoloration of urine
 - D** Streptomycin is safe in pregnancy

- 33** Which of the following is least useful in assessing the prognosis of community-acquired pneumonia?
- A** A very high C-reactive protein
 - B** Blood urea 30 mg/dL (above 7 mmol/L)
 - C** Age over 75 years
 - D** Hypotension

REPEATED 2* · ALSO ASKED IN
PasTest

- 34** A 34-year-old woman who initially complained of nasal obstruction develops cough, haemoptysis and pleuritic chest pain. Her chest X-ray shows multiple nodular masses. What is the most likely diagnosis?
- A** Granulomatosis with polyangiitis (Wegener)
 - B** Eosinophilic granulomatosis with polyangiitis (Churg–Strauss)
 - C** Goodpasture syndrome (anti-GBM disease)
 - D** Bronchogenic carcinoma

REPEATED 2* · ALSO ASKED IN
Previous-years exam

- 35** Which of the following findings best suggests active pulmonary tuberculosis?
- A** ESR 65 mm/h
 - B** Extensive fibrosis of the lungs
 - C** Positive tuberculin skin test
 - D** Cavitation on chest X-ray
 - E** Calcified hilar lymph nodes

LOG BOOK 2025

Log-book questions

- 36** A 30-year-old woman has high continuous fever, rigors, breathlessness and a cough productive of rusty sputum. Pneumococcal lobar pneumonia is suspected. If her sputum is sent for Gram stain, which finding is expected?
- A** Gram-negative rods
 - B** Gram-positive, lancet-shaped diplococci
 - C** Gram-positive cocci in clusters
 - D** Acid-fast bacilli
 - E** No organisms seen
- 37** What is the typical chest X-ray finding in lobar pneumonia?
- A** Diffuse interstitial infiltrates
 - B** Hyperlucent lung fields
 - C** Lobar consolidation
 - D** Cavitary lesions
 - E** Miliary nodules

Answers 26–37

LOCAL EXAMS MCQs

- 26** **C** 3 — Urea >7 mmol/L (1), RR ≥30 (1) and SBP <90 mmHg (1) score; he is not confused and is under 65. CURB-65 = 3, high severity: admit and consider critical care.
- 27** **A** Vancomycin plus ceftriaxone — Severe CAP needs a beta-lactam plus macrolide, but pneumatoceles after influenza suggest *S. aureus*, so add anti-MRSA cover (vancomycin or linezolid) until cultures return.
- 28** **A** Sputum acid-fast bacilli smear, culture and GeneXpert (NAAT) — Sputum for AFB smear, culture and a rapid NAAT (GeneXpert, which also detects rifampicin resistance) confirms active TB. Skin tests and IGRAs show infection only.
- 29** **A** Isoniazid — Isoniazid depletes pyridoxine and causes peripheral neuropathy, so pyridoxine is co-prescribed. Ethambutol causes optic neuritis; pyrazinamide causes gout.
- 30** **A** Increased ventilation of healthy alveoli removes CO₂ but cannot add extra oxygen to blood leaving the consolidated lung — **UPDATED** Overbreathing healthy alveoli removes CO₂ but cannot add oxygen to saturated blood, so shunt through consolidation causes type 1 failure.
- 31** **D** Pneumococcal vaccination — **UPDATED** TB control: case-finding, DOT, isolation, contact tracing with preventive therapy and BCG. Pneumococcal vaccine has no effect on TB.
- 32** **A** Asymptomatic rises in transaminases are common but serious liver damage is rare — **UPDATED** Mild transaminase rises are common early in TB treatment but serious hepatitis is rare; isoniazid neuropathy, ethambutol optic neuritis.
- 33** **A** A very high C-reactive protein — CURB-65 uses confusion, urea > 7 mmol/L, RR ≥ 30, low BP and age ≥ 65; CRP is useful for diagnosis and monitoring but is not a severity criterion.
- 34** **A** Granulomatosis with polyangiitis (Wegener) — Nasal disease then haemoptysis with multiple cavitating nodules is granulomatosis with polyangiitis (c-ANCA/PR3).
- 35** **D** Cavitation on chest X-ray — **UPDATED** Cavitation indicates active, usually smear-positive, post-primary TB; calcified nodes and fibrosis are old disease.
- 36** **B** Gram-positive, lancet-shaped diplococci — *S. pneumoniae* is a Gram-positive lancet-shaped diplococcus; clusters mean staphylococci.
- 37** **C** Lobar consolidation — Lobar pneumonia gives homogeneous consolidation bounded by fissures, with air bronchograms and a silhouette sign.

17 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1** A 45-year-old man returns from a holiday in Spain with a week of fever, left-sided pleuritic chest pain, cough and confusion. His chest X-ray is shown. What abnormality is seen?



- A Extensive left-sided consolidation
- B Large left-sided pleural effusion
- C Left-sided bronchial carcinoma
- D Left-sided empyema
- E Left-sided pulmonary infarct

- 2** A 45-year-old man returns from a hotel holiday in Spain with a week of fever, left-sided pleuritic chest pain, cough and new confusion (abbreviated mental test score 6/10). T 39.4°C, RR 26/min, BP 118/74 mmHg. Serum urea 9.8 mmol/L, sodium 128 mmol/L. The chest X-ray shows extensive left-sided consolidation. What is his CURB-65 score?

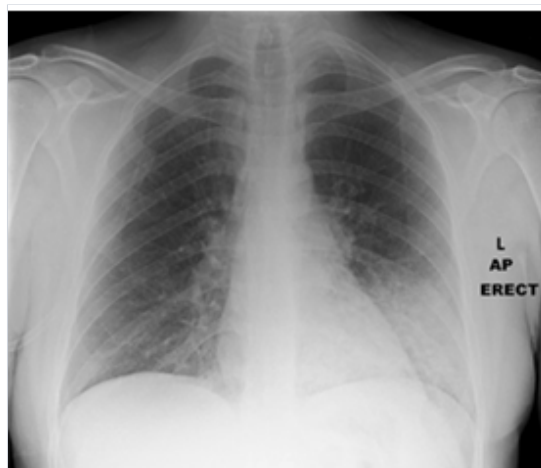
- A 0
- B 1
- C 2
- D 3
- E 4

- 3** A 45-year-old man returns from a hotel holiday in Spain with a week of fever, left-sided pleuritic chest pain, dry cough, diarrhoea and new confusion. Sodium 126 mmol/L, ALT 140 U/L, CK raised; the chest X-ray shows extensive left-sided consolidation. Which is the most likely causative organism?

- A Chlamydia pneumoniae
- B Haemophilus influenzae
- C Legionella pneumophila
- D Mycoplasma pneumoniae
- E Streptococcus pneumoniae

REPEATED 3x • ALSO ASKED IN
Crash Course x2

- 4** A 38-year-old man presents with cough productive of blood-streaked sputum, fever and left-sided pleuritic chest pain. He also has cold sores. His chest X-ray is shown. What does it show?



- A Left-sided apical pneumothorax
- B Left-sided basal consolidation
- C Left-sided collapse
- D Left-sided pleural effusion
- E Left-sided reticulonodular opacification

- 5** A 49-year-old man returns from a holiday in Portugal, where he stayed in a hotel and was in contact with a nephew who had influenza. He has fever, dry cough, breathlessness and confusion; his wife reports that at one point he did not recognise her. The chest X-ray shows bilateral infiltrates. What is the most useful diagnostic test?

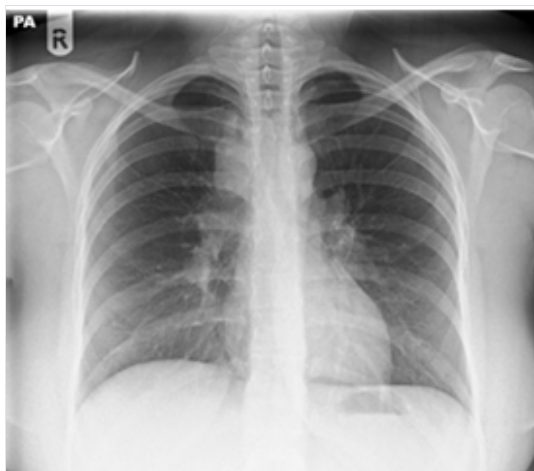
- A Differential cell count in bronchial lavage
- B High-resolution CT chest (HRCT)
- C Nucleic acid amplification testing for respiratory pathogens
- D Paired serology for mycoplasma and legionella
- E Sputum cytology

- 6** A 35-year-old woman with asthma attends clinic with deteriorating control. For 48 hours she has had fever of 38.6°C, a cough productive of green sputum and right-sided pleuritic chest pain. Her CRP is 180 mg/L with a neutrophilia; blood eosinophils are normal. Her chest X-ray is shown. What is the most likely cause of the deterioration?



- A** Bacterial infection
B Fungal infection
C Mycobacterial infection
D Parasitic infection
E Viral infection
- 7** A 32 year old man presents with a 5-day history of left-sided pleuritic chest pain, fever and cough productive of rusty sputum. He also has nasolabial cold sores. Which organism is likely responsible?
- A** *Aspergillus fumigatus*
B Herpes simplex virus (HSV)
C *Mycobacterium tuberculosis*
D *Pneumocystis jirovecii*
E *Streptococcus pneumoniae*
- 8** A 53 year old businessman returns from the Middle East with fever, chills, cough and shortness of breath. His CXR shows diffuse infiltrates. Which statement is most accurate?
- A** *Burkholderia pseudomallei* needs to be covered
B He should be isolated and tested for CPE
C He should be isolated and tested for MERS
D He should be isolated and tested for SARS
E Local antibiotic protocol for CAP should be followed
- 9** A 73 year old man in hospital for 3 days post-hip surgery becomes acutely confused with a new right-sided infiltrate on CXR. What is the appropriate approach?
- A** A CTPA should be ordered
B Blood cultures and watch-and-wait
C Local antibiotic guidelines for CAP
D Local antibiotic policy for HAP
E Local antibiotic policy for VAP
- 10** A 72-year-old man admitted 8 days ago with an exacerbation of COPD was initially improving but now spikes a fever, with green blood-streaked sputum and dense left-sided consolidation on the chest X-ray. Late-onset hospital-acquired pneumonia is most often caused by which of the following?
- A** *Acinetobacter*
B *Chlamydia*
C *Haemophilus*
D *Legionella*
E *Streptococcus*
- 11** The mortality from hospital-acquired pneumonia is approximately which of the following?
- A** 10%
B 20%
C 30%
D 40%
E 50%
- 12** A 34 year old woman who is an intravenous drug-user presents with high fever, pleuritic chest pain, cough productive of foul sputum and a fluctuant groin swelling. CXR shows multiple nodules and CT shows basal nodules, some cavitating. What is the likely explanation?
- A** Aspiration pneumonia
B Infective endocarditis
C Metastatic cancer
D Pulmonary thromboembolism
E Tuberculosis
- 13** A 54-year-old man is due to start anti-TNF monoclonal antibody therapy for Crohn's disease. He had BCG vaccination in childhood and his chest X-ray shows a minor abnormality. Which statement about the interferon-gamma release assay (IGRA) is true?
- A** A positive result should prompt antituberculous chemotherapy
B It is more specific than tuberculin skin testing
C It is now the first-line test for active TB
D It is only positive with systemic mycobacterial infection
E It measures the release of interferon-alpha from sensitised T cells
- 14** A 64-year-old woman with back pain and weight loss has a fluctuant, fluid-filled flank mass, from which pus is aspirated. What is the most important additional test on the pus?
- A** Bronchoscopy
B Echocardiogram to exclude septic embolus
C Flow cytometry
D Fluid biochemistry
E Mycobacterial testing

- 15 A 28-year-old woman develops painful red nodules on both shins, ankle arthralgia and fever, and her chest X-ray (shown) is performed. At 4-week review she feels well apart from mild lethargy. What is the most likely cause?



- A HIV
- B Lymphoma
- C Pneumonia
- D Sarcoidosis
- E Tuberculosis

- 16 A 45 year old non-smoking man with fever, weight loss, and breathlessness develops stridor on admission; CXR shows multiple bilateral nodules. His history includes sinusitis and conductive deafness. What is the most likely diagnosis?

- A Acute eosinophilic pneumonia
- B Bronchial carcinoma
- C Chronic eosinophilic pneumonia
- D Eosinophilic granulomatosis with polyangiitis
- E Granulomatosis with polyangiitis

- 17 Which of the following associations regarding lung disease is correct?

- A Anthrax and inadequately pasteurised milk
- B Chlamydia psittaci and hide factory workers
- C Coxiella burnetii and sewage workers
- D Francisella tularensis and muskrat contact
- E Leptospiral pneumonia and welding

Answers 1-17

DAVIDSON MCQS

- 1 **A Extensive left-sided consolidation** — Extensive left airspace opacity with air bronchograms and preserved volume is consolidation; hotel travel and confusion suggest Legionella.
- 2 **C 2** — **UPDATED** Confusion (1) + urea > 7 (1) = CURB-65 2: moderate severity, hospital treatment with amoxicillin plus clarithromycin; send Legionella antigen.
- 3 **C Legionella pneumophila** — **UPDATED** Hotel water systems, confusion, diarrhoea, hyponatraemia and deranged LFTs suggest Legionella; urinary antigen, treat with a macrolide or quinolone.
- 4 **B Left-sided basal consolidation** — Patchy left basal opacity obscuring the hemidiaphragm is left lower-lobe consolidation; with cold sores, typical pneumococcal pneumonia.
- 5 **C Nucleic acid amplification testing for respiratory pathogens** — Hotel stay plus a sick contact: multiplex PCR on sputum/swab rapidly detects Legionella, influenza and other pathogens.
- 6 **A Bacterial infection** — **UPDATED** Acute fever, purulent sputum, pleuritic pain, high CRP and lobar consolidation is bacterial pneumonia triggering the asthma attack; give antibiotics.
- 7 **E Streptococcus pneumoniae** — Fever, pleuritic pain and rusty sputum with cold sores is pneumococcal lobar pneumonia; the fever reactivates herpes simplex.
- 8 **C He should be isolated and tested for MERS** — Pneumonia within 14 days of travel to the Arabian Peninsula meets the possible MERS case definition: isolate and send MERS-CoV PCR.
- 9 **D Local antibiotic policy for HAP** — Pneumonia ≥ 48 h after admission is hospital-acquired (Gram-negatives, *S. aureus*): treat per local HAP policy, not CAP protocol.
- 10 **A Acinetobacter** — Late-onset HAP (≥ 5 days) is caused by resistant Gram-negatives (*Pseudomonas*, *Klebsiella*, *Acinetobacter*) and MRSA.
- 11 **C 30%** — Hospital-acquired pneumonia carries ~30% mortality (older, sicker patients, resistant organisms).
- 12 **B Infective endocarditis** — Fever with multiple cavitating peripheral nodules in a drug injector is septic emboli from tricuspid endocarditis (*S. aureus*); blood cultures and echo.
- 13 **B It is more specific than tuberculin skin testing** — IGRA uses TB-specific antigens absent from BCG, so it is more specific than the TST; a positive result means treat latent TB once active TB is excluded.
- 14 **E Mycobacterial testing** — A cold psoas abscess with back pain and weight loss suggests spinal TB (Pott disease): send pus for AFB, culture and molecular testing.
- 15 **D Sarcoidosis** — Bilateral hilar nodes with erythema nodosum, ankle arthritis and fever is Löfgren syndrome (acute sarcoidosis), which usually remits.
- 16 **E Granulomatosis with polyangiitis** — Sinusitis, conductive deafness, subglottic stenosis and multiple lung nodules are GPA; check c-ANCA (PR3), urine and biopsy.
- 17 **D Francisella tularensis and muskrat contact** — Tularaemia (*Francisella*) follows contact with rabbits or muskrats. Psittacosis = birds, Q fever = sheep/cattle, leptospirosis = rat urine.

3 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 A 38-year-old woman who recently started smoking develops a 5-day febrile illness with rapidly progressive respiratory failure and diffuse bilateral infiltrates. Bronchoalveolar lavage shows 40% eosinophils, blood cultures and respiratory viral PCR are negative, and she has been on broad-spectrum antibiotics for 48 hours without improvement. What is the best treatment now?
 - A Consult thoracic surgery for lung biopsy
 - B Continue IV antibiotics while awaiting cultures
 - C Initiate methylprednisolone 60 mg IV every 6 hours
 - D Initiate oseltamivir 75 mg BID
 - E Initiate TMP-SMX plus prednisone
- 2 Which finding is expected on bronchoalveolar lavage in diffuse alveolar haemorrhage?
 - A Atypical hyperplastic type II pneumocytes
 - B Ferruginous bodies
 - C Hemosiderin-laden macrophages
 - D Lymphocytosis with elevated CD4:CD8 ratio
 - E Milky appearance with foamy macrophages
- 3 A 52-year-old woman has progressive breathlessness; CT shows extensive calcified mediastinal soft tissue encasing and narrowing the central airways and pulmonary vessels (fibrosing mediastinitis). Which statement is true?
 - A A urine Histoplasma antigen test will be positive
 - B The most common cause is histoplasmosis or TB
 - C Patient should be referred for surgical resection
 - D Steroids will improve her condition
 - E Antifungals will improve her condition

Answers 1-3

HARRISON MCQS

- 1 **C Initiate methylprednisolone 60 mg IV every 6 hours** — Acute eosinophilic pneumonia after starting smoking (> 25% BAL eosinophils) responds dramatically to high-dose IV methylprednisolone.
- 2 **C Hemosiderin-laden macrophages** — Diffuse alveolar haemorrhage gives progressively bloodier BAL with hemosiderin-laden macrophages after 48–72 h.
- 3 **B The most common cause is histoplasmosis or TB** — Fibrosing mediastinitis follows burnt-out histoplasmosis or TB; drugs do not reverse it, and stenoses are stented.

9 QUESTIONS • AFTER CRASH COURSE SBAS & EMQS

Crash Course MCQs

- 1 A middle-aged woman is admitted to hospital with a cough, sputum production, fever and dyspnoea. A diagnosis of pneumonia is made. Which of the following organisms is most likely to account for her symptoms?
 - A Streptococcus pneumoniae
 - B Haemophilus influenzae
 - C Legionella pneumophila
 - D Staphylococcus aureus
 - E Aspergillus fumigatus
- 2 An elderly woman is admitted to hospital with pneumonia. The following are all risk factors used in predicting mortality in community-acquired pneumonia EXCEPT?
 - A Urea greater than 7 mmol/L
 - B Age of 65 years
 - C Respiratory rate of 20 breaths/min or more
 - D Systolic blood pressure less than 90 mmHg
 - E Confusion

- 3** A middle-aged man presents with a cough, sputum production, fever and dyspnoea. Red-coloured papules are noted on the back of his hands. Routine blood investigations demonstrate that he is anaemic. Which of the following organisms is most likely to account for his condition?
- A Chlamydia pneumoniae
 - B Haemophilus influenzae
 - C Staphylococcus aureus
 - D Mycoplasma pneumoniae
 - E Legionella pneumophila
- 4** A 42-year-old man presents to the GP with symptoms suggestive of pneumonia. He also complains of generalized muscular pains. He has many pets at home and mentions he recently bought two parrots. Which of the following organisms would account for his condition?
- A Chlamydia psittaci
 - B Haemophilus influenzae
 - C Staphylococcus aureus
 - D Mycoplasma pneumoniae
 - E Legionella pneumophila
- REPEATED 2* · ALSO ASKED IN**
Passmedicine
- 5** A 23-year-old known intravenous drug abuser is admitted with a fever, productive cough and dyspnoea. Which of the following organisms is most likely to account for his condition?
- A Chlamydia pneumoniae
 - B Haemophilus influenzae
 - C Staphylococcus aureus
 - D Mycoplasma pneumoniae
 - E Legionella pneumophila
- 6** You suspect pneumonia in a 35-year-old woman. Which of the following investigations is most likely to confirm the diagnosis?
- A Blood culture
 - B Sputum culture
 - C White cell count
 - D CXR
 - E Arterial blood gas
- 7** A man has recently been diagnosed with pulmonary tuberculosis. Which cells engulf the inhaled bacilli and go on to form the granulomatous lesions?
- A Neutrophils
 - B Macrophages
 - C Langhans giant cells
 - D Epithelioid cells
 - E T lymphocytes
- 8** A middle-aged man is diagnosed with drug-sensitive pulmonary tuberculosis. He has no known drug allergies. Which drug is least likely to be included in his initial treatment?
- A Ethambutol
 - B Pyrazinamide
 - C Streptomycin
 - D Rifampicin
 - E Isoniazid
- 9** A 55-year-old man has recently been diagnosed with TB. He has been started on quadruple therapy and complains of a colour change in his urine. What drug is most likely to be responsible for this?
- A Rifampicin
 - B Isoniazid
 - C Ethambutol
 - D Pyrazinamide
 - E Streptomycin

Answers 1-9

- 1** **A Streptococcus pneumoniae** — S. pneumoniae is the commonest cause of community-acquired pneumonia at all ages.
- 2** **C Respiratory rate of 20 breaths/min or more** — CURB-65: confusion, urea > 7, RR ≥ 30, low BP, age ≥ 65. A respiratory rate of 20 is not a criterion.
- 3** **D Mycoplasma pneumoniae** — Mycoplasma pneumonia with erythema multiforme and cold-agglutinin haemolytic anaemia; treat with a macrolide or doxycycline.
- 4** **A Chlamydia psittaci** — Pneumonia with myalgia after acquiring parrots is psittacosis (Chlamydia psittaci); treat with doxycycline.
- 5** **C Staphylococcus aureus** — Injecting drug users get S. aureus pneumonia (haematogenous or cavitating); blood cultures, echo and MRSA cover.
- 6** **D CXR** — Pneumonia is defined by new radiographic shadowing, so the chest X-ray confirms it; cultures identify an organism in a minority.
- 7** **B Macrophages** — Alveolar macrophages ingest M. tuberculosis and, activated by T-cell IFN-γ, become epithelioid and Langhans giant cells forming granulomas.
- 8** **C Streptomycin** — Standard TB therapy: 2 months RIPE then 4 months RI (with pyridoxine); streptomycin is no longer first line.
- 9** **A Rifampicin** — Rifampicin turns urine, sweat and tears orange-red and is a potent enzyme inducer (pill, warfarin, steroids).

3 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 63-year-old man presents to the emergency department with haemoptysis. He has antiphospholipid syndrome and takes warfarin. Chest auscultation is normal, but there is nasal crusting and reduced sensation to both ankles. The chest X-ray shows cavitating nodules in both mid-zones. Hb 138 g/L, urea 15.6 mmol/L, creatinine 240 µmol/L, CRP 145 mg/L, INR 1.4. What is the most likely diagnosis?
 - A Staphylococcus aureus pneumonia
 - B Anti-glomerular basement membrane disease
 - C Granulomatosis with polyangiitis
 - D Microscopic polyangiitis
 - E Pulmonary embolism
- 2 A 28-year-old man presents with a dry cough, progressive breathlessness and recurrent chest infections. His GP recently started a salbutamol inhaler, with little effect. There is diffuse wheeze. Investigations show blood eosinophilia and total IgE 2300 kU/L, and the chest X-ray shows bilateral pulmonary infiltrates. What is the treatment of choice?
 - A Intravenous immunoglobulin (IVIG)
 - B Low dose inhaled corticosteroids (Low-dose ICS inhaler)
 - C Nebulised antibiotics
 - D Oral glucocorticoids
 - E Rituximab
- 3 A middle-aged woman has progressive breathlessness. Her chest X-ray shows bilateral hilar lymphadenopathy. Calcium 2.7 mmol/L; angiotensin-converting enzyme 200 µg/L (normal below 40). Given the likely diagnosis, which feature is associated with a good prognosis?
 - A Lupus pernio
 - B Hypercalcaemia
 - C Age > 40
 - D Erythema nodosum
 - E HLA B13 subtype

Answers 1-3

PASSMEDICINE MCQS

- 1 **C Granulomatosis with polyangiitis** — UPDATED Nasal crusting, cavitating nodules, AKI and mononeuritis multiplex form the ENT–lung–kidney pattern of GPA; check PR3-ANCA.
- 2 **D Oral glucocorticoids** — Asthma with eosinophilia, IgE > 1000 kU/L and infiltrates is ABPA; oral prednisolone first line, itraconazole as steroid-sparing.
- 3 **D Erythema nodosum** — Löfgren syndrome (erythema nodosum with BHL) has the best prognosis in sarcoidosis, usually remitting within 2 years.

10 QUESTIONS • AFTER PASTEST

PasTest MCQs

- 1** A 20-year-old man with epilepsy presents to the Emergency Department with a three-day history of cough and increasing shortness of breath. He has foul-smelling sputum. He had a generalised tonic-clonic seizure one week ago. On examination his temperature is 40 °C, he has right-sided crepitations and his chest X-ray shows a right middle lobe infiltration. Which one of the following options is the most likely cause?

 - A Mycoplasma pneumonia
 - B Chemical pneumonitis
 - C Pneumonia due to Gram-negative aerobes
 - D Pneumonia due to Gram-positive aerobes
 - E Pneumonia due to anaerobes
- 2** A 47-year-old homeless man with alcohol misuse is admitted with increasing breathlessness, sweats and purulent sputum. A few days ago he was brought to the emergency department unconscious but self-discharged when he woke. He is febrile, with dullness and bronchial breathing in the right mid-zone, and the chest X-ray shows right middle-lobe consolidation. What is the most likely cause?

 - A Aspiration
 - B Chlamydia infection
 - C Legionella infection
 - D Mycoplasma infection
 - E Tuberculosis
- 3** An 81-year-old woman who saw her GP with influenza-like symptoms 2 weeks ago now presents with increasing cough, purulent sputum and haemoptysis, and has become acutely confused. She is agitated with a respiratory rate of 35/min and is hypoxic; WBC $20 \times 10^9/L$ (mainly neutrophils), creatinine 250 $\mu\text{mol/L}$. The chest X-ray shows patchy consolidation with necrosis and empyema. What diagnosis best fits this picture?

 - A Chlamydia psittaci pneumonia
 - B Haemophilus influenzae pneumonia
 - C Mycoplasma pneumonia
 - D Staphylococcus aureus pneumonia
 - E Streptococcus pneumoniae pneumonia
- 4** A 26-year-old office secretary who smokes 10–15 cigarettes a day presents with a couple of episodes of haemoptysis. She has felt tired recently and has passed cola-coloured urine. A full blood count shows anaemia and the chest X-ray shows blotchy shadowing over both lung fields. She is ANCA-negative. What is the most likely diagnosis?

 - A Bronchogenic carcinoma
 - B Goodpasture syndrome (anti-GBM disease)
 - C Granulomatosis with polyangiitis
 - D Pulmonary tuberculosis
 - E Sarcoidosis
- 5** A 25-year-old man with epilepsy aspirated gastric contents during a seizure (Mendelson syndrome). What is the most likely clinical sign?

 - A Bradypnoea
 - B Bronchospasm
 - C Laryngospasm
 - D Paradoxical breathing
 - E Tenderness to percussion
- 6** A 35-year-old woman being treated for asthma has worsening cough and wheeze with recurrent fleeting pulmonary infiltrates. She coughs up brownish plugs, and her chest X-ray shows pulmonary infiltration with some lobar consolidation. Which diagnosis best fits this picture?

 - A Allergic bronchopulmonary aspergillosis
 - B Aspergilloma
 - C Asthma
 - D Bronchiectasis
 - E Tuberculosis
- 7** A 22-year-old student attends clinic 6 weeks after admission with pneumonia. Which of the following complications of pneumonia is a chronic rather than an acute complication of the infection?

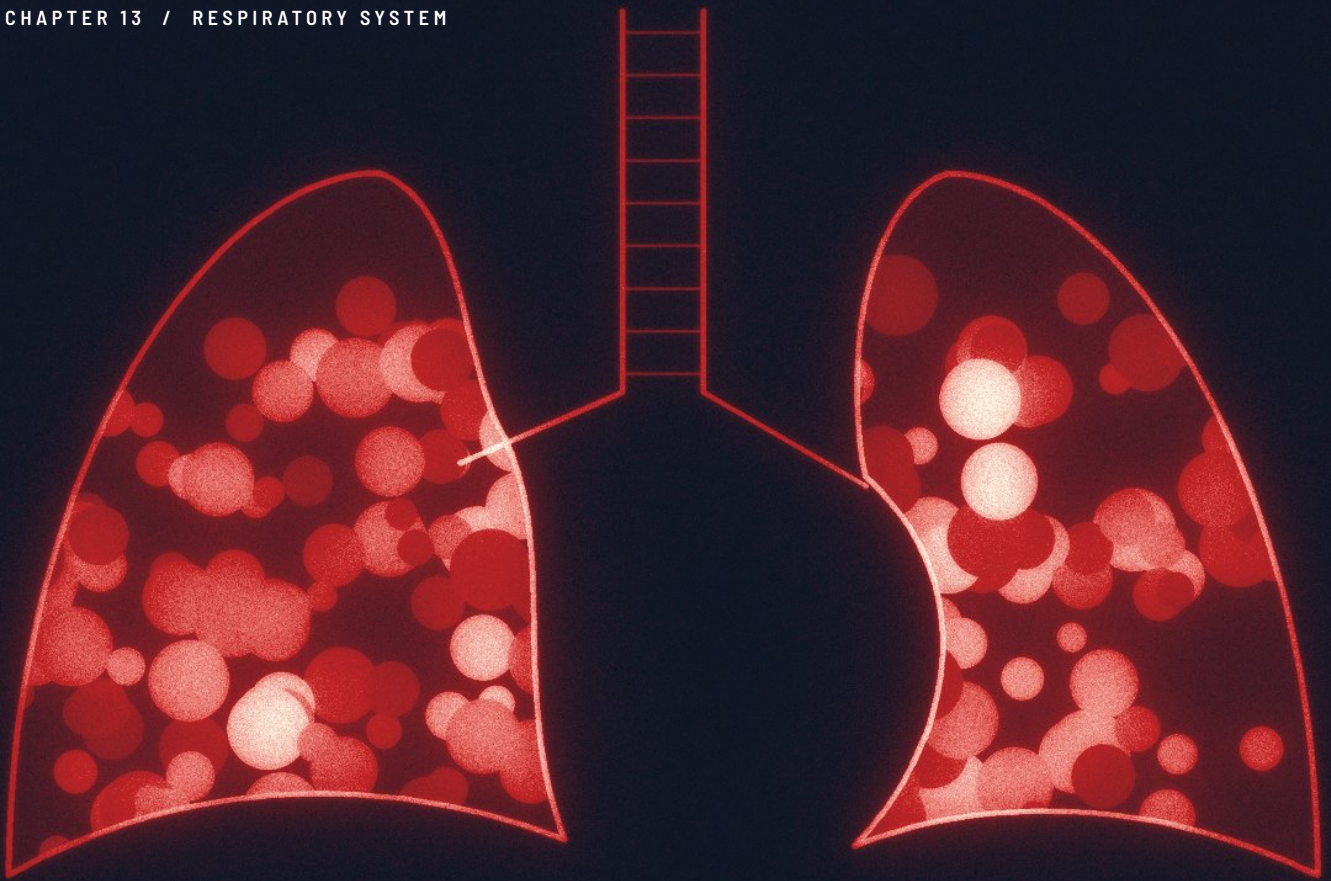
 - A Bronchiectasis
 - B Bronchopleural fistula
 - C Empyema
 - D Lung abscess
 - E Organising pneumonia

- 8 A 19-year-old army cadet is admitted to hospital with cough, fever and malaise. On examination there is fever and coarse crepitations at the bases of the lungs. Cold agglutinins are positive. A chest X-ray shows bibasal shadowing. What is the most likely diagnosis?
- A Klebsiella pneumonia
 - B Legionella pneumonia
 - C Mycoplasma pneumonia
 - D Q fever
 - E Viral pneumonia
- 9 A 62-year-old woman is brought to the emergency department with suspected pneumonia. Temperature 38.3°C, respiratory rate 27/min, BP 100/70 mmHg, and she is confused. Serum urea is 5.2 mmol/L. According to the CURB-65 criteria, which of these findings is associated with a worse prognosis?
- A Age 62
 - B Confusion
 - C Respiratory rate 27/min
 - D Systolic blood pressure 100 mmHg
 - E Urea 5.2 mmol/l
- 10 A 32-year-old man sees his GP with cough and fever, and tuberculosis is suspected. He comes from an area with high rates of drug resistance. Which blood test is most important before starting treatment?
- A Clotting screen
 - B Haemoglobin
 - C Liver function testing
 - D Platelet count
 - E Serum calcium

Answers 1-10

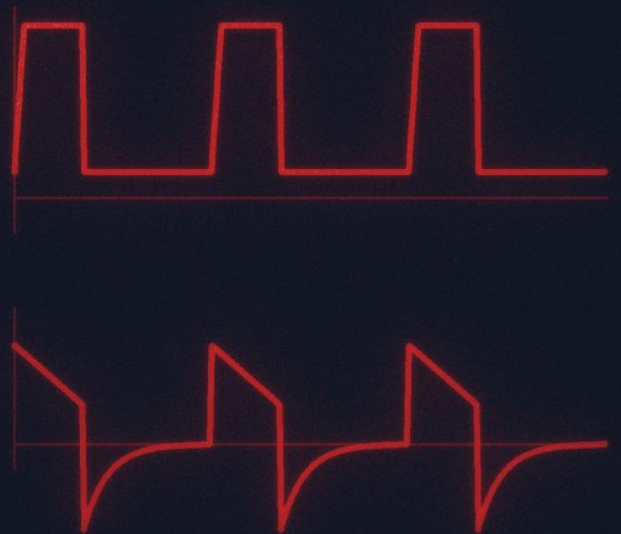
PASTEST MCQS

- 1 **E Pneumonia due to anaerobes** — Aspiration during a seizure with foul sputum days later is anaerobic pneumonia; treat with anaerobic cover (co-amoxiclav).
- 2 **A Aspiration** — **UPDATED** Recent alcohol-related unconsciousness causes aspiration pneumonia in a dependent segment (right middle/lower lobe); cover anaerobes.
- 3 **D Staphylococcus aureus pneumonia** — **UPDATED** Necrotising pneumonia with cavities and empyema after influenza is *S. aureus*; add flucloxacillin (or MRSA cover).
- 4 **B Goodpasture syndrome (anti-GBM disease)** — Haemoptysis, alveolar shadowing and nephritis with negative ANCA suggest anti-GBM (Goodpasture) disease; smoking predisposes to the bleeding.
- 5 **B Bronchospasm** — Gastric acid aspiration (Mendelson syndrome) causes chemical pneumonitis within hours with bronchospasm, tachypnoea and hypoxaemia.
- 6 **A Allergic bronchopulmonary aspergillosis** — Poor asthma control with brown mucus plugs and fleeting infiltrates is ABPA (IgE > 1000, *Aspergillus* IgE/IgG).
- 7 **A Bronchiectasis** — **UPDATED** Bronchiectasis is a chronic structural sequela of severe pneumonia; empyema, abscess and fistula are acute complications.
- 8 **C Mycoplasma pneumonia** — Atypical pneumonia in a barracks with cold agglutinins and an X-ray worse than the signs is *Mycoplasma*; treat with a macrolide.
- 9 **B Confusion** — Only new confusion meets a CURB-65 criterion here (urea 5.2, RR 27, BP 100/70 and age 62 all score zero).
- 10 **C Liver function testing** — First-line TB drugs are hepatotoxic, so baseline LFTs are essential; also renal function, HIV/hepatitis serology and visual acuity.



13

RESPIRATORY FAILURE & ARDS



Respiratory Failure & ARDS

Type 1 and type 2 respiratory failure, blood gases, oxygen and non-invasive ventilation, and the acute respiratory distress syndrome.

22

MCQs

0

EMQ SCENARIOS

5

SOURCES

1

REPEATED IDEAS

IN THIS CHAPTER

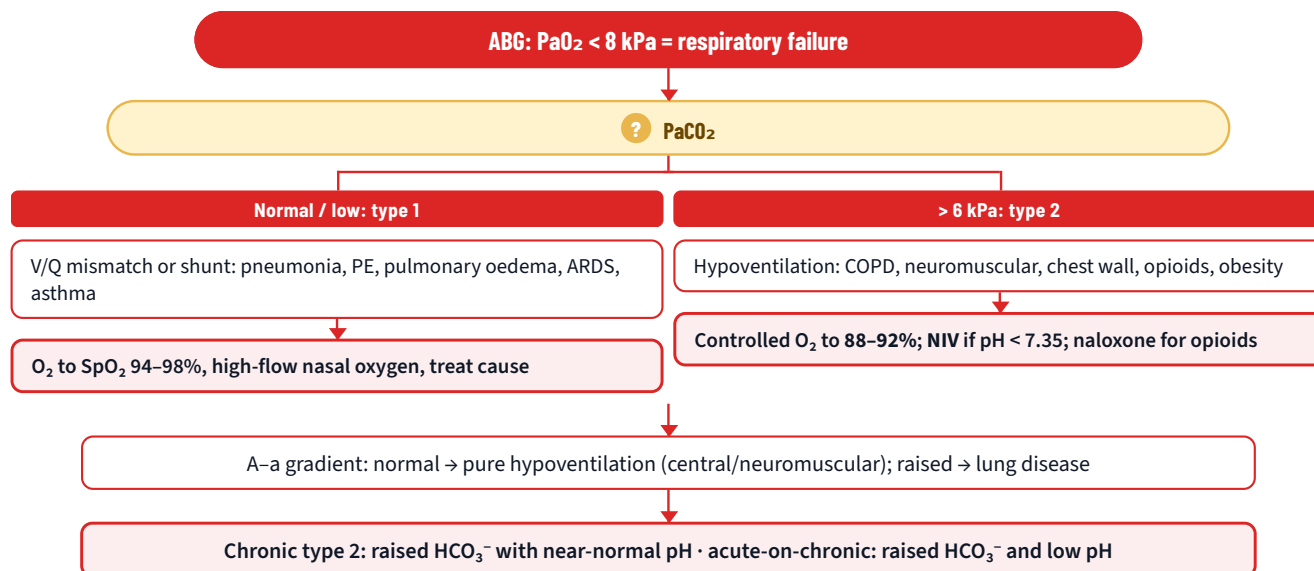
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13

Step by step

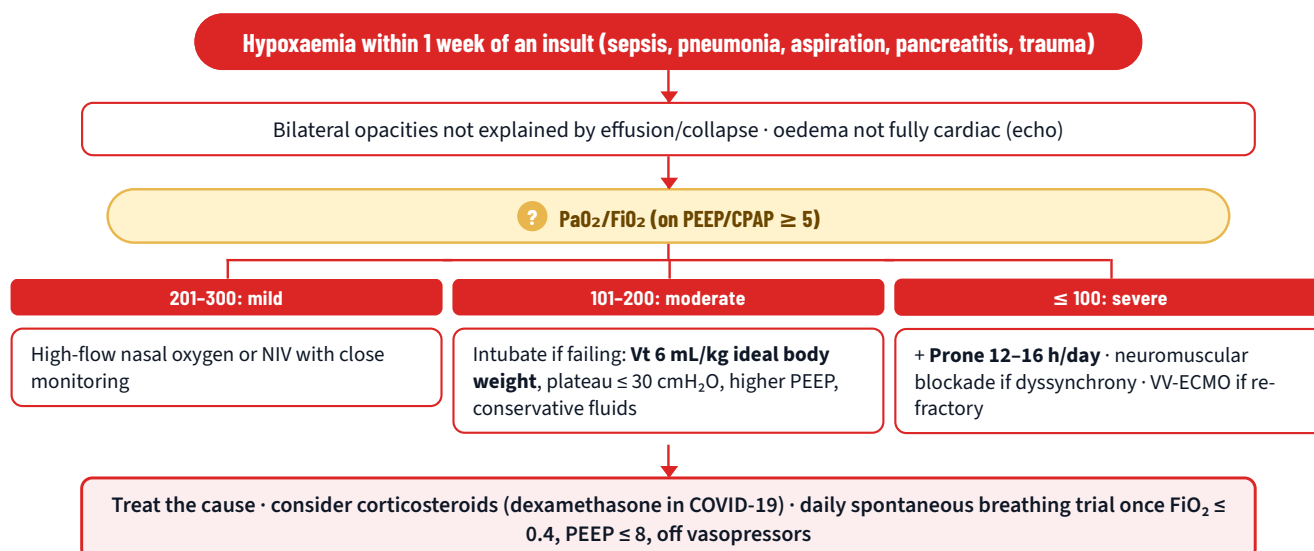
Respiratory failure: reading the ABG

BTS OXYGEN 2017



Acute respiratory distress syndrome

BERLIN + 2023 GLOBAL DEFINITION · ESICM 2023 · ATS 2024



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1 A patient with COVID-19 presents with severe breathlessness and cyanosis. Which of the following supports a diagnosis of acute respiratory distress syndrome?
 - A Chronic bilateral pulmonary infiltrates
 - B Alveoli filled with protein-rich fluid
 - C Normal PaO₂ with low PaCO₂
 - D A pulmonary artery wedge pressure of 12 mmHg excludes the diagnosis
- 2 Which of the following typically does NOT cause type 1 respiratory failure?
 - A Asthma
 - B Emphysema
 - C Chronic bronchitis
 - D Pulmonary embolism

REPEATED 2* • ALSO ASKED IN
Previous-years exam

- 3 A 50-year-old woman who had a viral chest infection 4 days ago develops severe breathlessness; she is cyanosed with SpO₂ 82% and bilateral crackles, and echocardiography is normal. What is the most likely cause of her breathlessness?
 - A Acute pulmonary oedema
 - B Bronchial asthma
 - C Severe anaemia
 - D Acute respiratory distress syndrome

CASE • QUESTIONS 4–5

A 29-year-old woman with COVID-19 is confused, tachypnoeic and breathless: pulse 130/min, BP 100/60 mmHg, RR 38/min, T 38.6°C, SpO₂ 67% on air rising to 86% on oxygen. WBC $13 \times 10^9/L$, urea 10 mmol/L (60 mg/dL), creatinine 124 µmol/L (1.4 mg/dL), D-dimer 0.6 mg/L. ECG: sinus tachycardia; chest X-ray: diffuse bilateral infiltrates with a normal heart size.

- 4 What is the diagnosis?
 - A Pulmonary embolism
 - B Acute pneumonia
 - C Acute respiratory distress syndrome
 - D Disseminated intravascular coagulation
 - E Lung abscess

- 5 Which of the following drugs is the most appropriate next step?
 - A Remdesivir
 - B Aciclovir
 - C Ceftriaxone
 - D Doxycycline
 - E Nebulised salbutamol

- 6 Which of the following is NOT a complication of acute respiratory distress syndrome?
 - A Respiratory failure
 - B Permanent pulmonary fibrosis
 - C Multi-organ failure
 - D Arrhythmia
 - E Increased lung compliance

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

CASE • QUESTIONS 7–8

A man with an exacerbation of COPD has shallow breathing at 10 breaths/min. Arterial blood gases: pH 7.31, PaCO₂ 70 mmHg (9.3 kPa), HCO₃⁻ 33 mmol/L (reference 22–26).

- 7 What is the acid–base disturbance?
 - A Respiratory acidosis
 - B Respiratory alkalosis
 - C Mixed metabolic and respiratory acidosis
 - D Metabolic acidosis
 - E Metabolic alkalosis
- 8 What does the raised bicarbonate indicate?
 - A No compensation
 - B Renal (metabolic) compensation
 - C Lactic acidosis
 - D Hyperventilation
 - E Laboratory error

Answers 1–8

- 1 **B Alveoli filled with protein-rich fluid** — ARDS is diffuse alveolar damage with protein-rich alveolar fluid and hyaline membranes, causing refractory hypoxaemia.
- 2 **C Chronic bronchitis** — Type 1 failure (V/Q mismatch): asthma, emphysema, PE. Chronic bronchitis ('blue bloater') typically causes type 2 failure.

- 3** **D Acute respiratory distress syndrome** — Severe hypoxaemia with bilateral crackles within a week of a viral infection and a normal echo fits ARDS (Berlin definition).
- 4** **C Acute respiratory distress syndrome** — Refractory hypoxaemia with bilateral infiltrates and normal heart size in COVID-19 is ARDS; dexamethasone and critical-care support.
- 5** **A Remdesivir** — Remdesivir is the only listed drug with evidence in hospitalised COVID-19 on oxygen; dexamethasone 6 mg is the key treatment.
- 6** **E Increased lung compliance** — ARDS reduces (not increases) lung compliance; complications include barotrauma, VAP, multi-organ failure and fibrosis.
- 7** **A Respiratory acidosis** — Low pH with high PaCO₂ is respiratory acidosis from hypoventilation (type 2 failure). The raised bicarbonate matches chronic compensation, so it is not a separate metabolic disorder.
- 8** **B Renal (metabolic) compensation** — Over days the kidneys retain bicarbonate (about 3.5 mmol/L per 10 mmHg rise in PaCO₂), so the high HCO₃⁻ shows chronic CO₂ retention with renal compensation.

2 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1** An 83-year-old woman was a passenger in a car that hit a lamppost. She complained of pain in her right hip and ribs and became drowsy after paramedics gave 2 mg of morphine. X-rays show a pelvic fracture and a single right rib fracture. Initially drowsy but responsive, she is now unresponsive, with no improvement after naloxone 400 micrograms IV. SpO₂ is 87% on 2 L/min oxygen via nasal cannulae; she is afebrile, BP 110/66 mmHg, pulse 65/min. There are no new findings on examination, and an urgent CT brain shows only small-vessel disease. Arterial blood gas: pH 7.22 (H⁺ 60 nmol/L), PaO₂ 8.7 kPa (65 mmHg), PaCO₂ 10 kPa (75 mmHg), HCO₃⁻ 26 mmol/L. What is the most likely cause of her deteriorating conscious level?
- A** Cholesterol embolism – ventilation/perfusion (V/Q) mismatch
- B** Chronic obstructive pulmonary disease (COPD) with oxygen toxicity – loss of hypoxic drive
- C** Flail segment due to rib fracture – loss of elastic recoil
- D** Opiate toxicity – suppression of the respiratory centre
- E** Undetected fracture of C3 – diaphragmatic failure
- 2** In which of the following conditions might acute type I respiratory failure be expected?
- A** Flail chest injury
- B** Lobar collapse
- C** Myasthenic crisis
- D** Obstructive sleep apnoea (OSA)
- E** Opioid toxicity

Answers 1-2

DAVIDSON MCQs

- 1** **E Undetected fracture of C3 – diaphragmatic failure** — **UPDATED** Acute type 2 failure with relative bradycardia after trauma suggests a high cervical cord injury (C3–C5) paralysing the diaphragm; immobilise and image.
- 2** **B Lobar collapse** — **UPDATED** Lobar collapse causes a shunt with normal/low PaCO₂ (type 1). Flail chest, OSA, myasthenia and opioids cause hypoventilation (type 2).

9 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 A 78-year-old woman has multilobar pneumonia requiring intubation. Arterial blood gas on a high inspired oxygen concentration shows PaO_2 7.2 kPa (54 mmHg). What is the main cause of the hypoxaemia?
 - A Hypoventilation alone
 - B Hypoventilation and ventilation-perfusion mismatch
 - C Shunt
 - D Ventilation-perfusion mismatch
 - E None
- 2 A 68-year-old man with multilobar pneumonia is in septic shock and is intubated. Chest X-ray shows new bilateral infiltrates, echocardiography shows normal left ventricular function, and his $\text{PaO}_2/\text{FiO}_2$ ratio is 90 mmHg on a PEEP of 10 cmH₂O. Which diagnosis fits best?
 - A Acute interstitial pneumonia
 - B Mild ARDS
 - C Moderate ARDS
 - D Multilobar community-acquired pneumonia alone
 - E Severe ARDS
- 3 A patient with ARDS (chest X-ray shown) is ventilated with a tidal volume of 6 mL/kg ideal body weight. The pressure-volume curve shows a lower inflection point at 12 cmH₂O. Current PEEP is 8 cmH₂O, plateau pressure 26 cmH₂O and SaO_2 86% on FiO_2 0.6. What is the best next step?
 
 - A Administer paralytic agent
 - B Decrease tidal volume to 350 mL
 - C Increase FiO_2 to 0.8
 - D Increase PEEP to 12 cmH₂O
 - E Increase tidal volume to 560 mL
- 4 In which of the following situations is non-invasive ventilation generally NOT recommended, because of a high failure rate and the risk of delaying intubation?
 - A Acute exacerbation of COPD with pH 7.28
 - B Severe ARDS with a $\text{PaO}_2/\text{FiO}_2$ ratio of 80 mmHg
 - C Obesity hypoventilation syndrome with acute hypercapnic failure
 - D Cardiogenic pulmonary oedema
 - E Post-extubation respiratory failure in COPD
- 5 A 62-year-old man intubated for pneumonia is being considered for weaning. Which of the following is a contraindication to a spontaneous breathing trial?
 - A Improving chest infiltrates
 - B FiO_2 of 0.45
 - C PEEP of 8 cmH₂O
 - D Ongoing need for vasopressor support
 - E Breathing 8 breaths/min above the set ventilator rate
- 6 A lung biopsy is taken 4 days after the onset of severe ARDS. Which histological finding is expected?
 - A Desquamative interstitial pneumonia
 - B Eosinophil-rich infiltrates
 - C Extensive fibrosis with bullae
 - D Homogeneous neutrophil infiltrates
 - E Diffuse alveolar damage with hyaline membranes
- 7 A 48-year-old patient with ARDS weighs 90 kg but has an ideal body weight of 59 kg. What is the most appropriate initial tidal volume?
 - A 236 mL
 - B 354 mL
 - C 472 mL
 - D 540 mL
 - E 590 mL
- 8 A 25-year-old man is brought in unresponsive. Arterial blood gas on room air: pH 7.16, PaCO_2 9.3 kPa (70 mmHg), PaO_2 7.3 kPa (55 mmHg). Which of the following is the most likely diagnosis?
 - A Asthma
 - B Narcotic overdose
 - C Pneumococcal pneumonia
 - D Pneumocystis pneumonia
 - E Pulmonary embolism

- 9 A patient with mild amyotrophic lateral sclerosis (motor neurone disease) has a PaCO_2 of ≥ 6 kPa (45 mmHg). Which accompanying symptom is an indication for non-invasive ventilation?
- A Orthopnoea
 - B Poor sleep quality
 - C Morning headaches
 - D Breathlessness on activities of daily living
 - E All of the above

Answers 1-9

HARRISON MCQS

- 1 **C Shunt** — Exudate-filled alveoli are perfused but not ventilated (true shunt), so hypoxaemia barely improves with high FiO_2 ; PEEP recruits them.
- 2 **E Severe ARDS** — **UPDATED** $\text{PaO}_2/\text{FiO}_2 \leq 100$ mmHg on PEEP ≥ 5 with new bilateral infiltrates is severe ARDS (Berlin); prone ventilation 12–16 h/day.
- 3 **D Increase PEEP to 12 cmH₂O** — **UPDATED** With plateau 26 cmH₂O there is room to raise PEEP to 12 to recruit alveoli while keeping plateau ≤ 30 ; tidal volume is already 6 mL/kg.
- 4 **B Severe ARDS with a $\text{PaO}_2/\text{FiO}_2$ ratio of 80 mmHg** — **UPDATED** NIV fails often in severe ARDS and delays intubation; it works in hypercapnic COPD and cardiogenic pulmonary oedema.
- 5 **D Ongoing need for vasopressor support** — **UPDATED** A spontaneous breathing trial needs haemodynamic stability; ongoing vasopressor dependence is a contraindication.
- 6 **E Diffuse alveolar damage with hyaline membranes** — **UPDATED** Early (exudative) ARDS shows diffuse alveolar damage with protein-rich oedema and hyaline membranes.
- 7 **B 354 mL** — **UPDATED** Lung-protective ventilation uses 6 mL/kg of ideal body weight: $6 \times 59 \approx 354$ mL, with plateau pressure ≤ 30 cmH₂O.
- 8 **B Narcotic overdose** — Acute respiratory acidosis with a normal A–a gradient (~ 7 mmHg) means pure hypoventilation, e.g. opioid overdose: give naloxone.
- 9 **E All of the above** — **UPDATED** In MND, NIV is offered when symptoms of hypoventilation (orthopnoea, poor sleep, morning headache, breathlessness) accompany hypercapnia.

2 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 30-year-old man is admitted after a motorcycle accident with a fracture of the right femur, which is managed with traction. Three days later he becomes confused and tachypnoeic, and a petechial rash is noted over the chest. The lungs are clear on auscultation. Arterial blood gases: PaO_2 6.7 kPa (50 mmHg), PaCO_2 3.7 kPa (28 mmHg), pH 7.49. What is the most likely diagnosis?
- A Unilateral pulmonary oedema
 - B Haematoma of the chest
 - C Fat embolism
 - D Pulmonary embolism
 - E Early *Staphylococcus aureus* pneumonia
- 2 A 70-year-old patient with chronic obstructive lung disease requires 2 L/min of nasal O_2 to treat his hypoxia, which is sometimes associated with angina. The patient develops pleuritic chest pain, fever, and purulent sputum. While using his oxygen at an increased flow of 5 L/min he becomes stuporous and develops a respiratory acidosis with CO_2 retention and worsening hypoxia. What would be the most appropriate next step in the management of this patient?
- A Stop oxygen
 - B Begin medroxyprogesterone
 - C Intubate and begin mechanical ventilation
 - D Observe for 24 hours before changing therapy
 - E Begin sodium bicarbonate

Answers 1-2

PRETEST MCQS

- 1 **C Fat embolism** — Hypoxaemia, confusion and petechiae 24–72 h after a long-bone fracture is fat embolism syndrome; supportive care, early fixation prevents it.
- 2 **C Intubate and begin mechanical ventilation** — High-flow oxygen caused CO_2 narcosis; stuporous with acidosis and worsening hypoxia, he cannot protect his airway: intubate and ventilate.

1 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 Which one of the following is not part of the diagnostic criteria of acute respiratory distress syndrome (ARDS)?
- A Bilateral infiltrates on chest X-ray
 - B Non-cardiogenic oedema
 - C $\text{PaO}_2/\text{FiO}_2 < 200$ mmHg
 - D Respiratory rate $> 24/\text{min}$
 - E Acute onset

Answer 1

PASSMEDICINE MCQs

- 1 **D Respiratory rate $> 24/\text{min}$** — Berlin ARDS criteria: onset < 1 week, bilateral opacities, not cardiac, $\text{PaO}_2/\text{FiO}_2 \leq 300$ on PEEP ≥ 5 . Respiratory rate is not a criterion.



14

PLEURAL DISEASE

Pleural Disease

Pleural effusion, empyema, pneumothorax and haemothorax: Light criteria, aspiration and drainage, and mesothelioma.

42

MCQs

0

EMQ SCENARIOS

7

SOURCES

5

REPEATED IDEAS

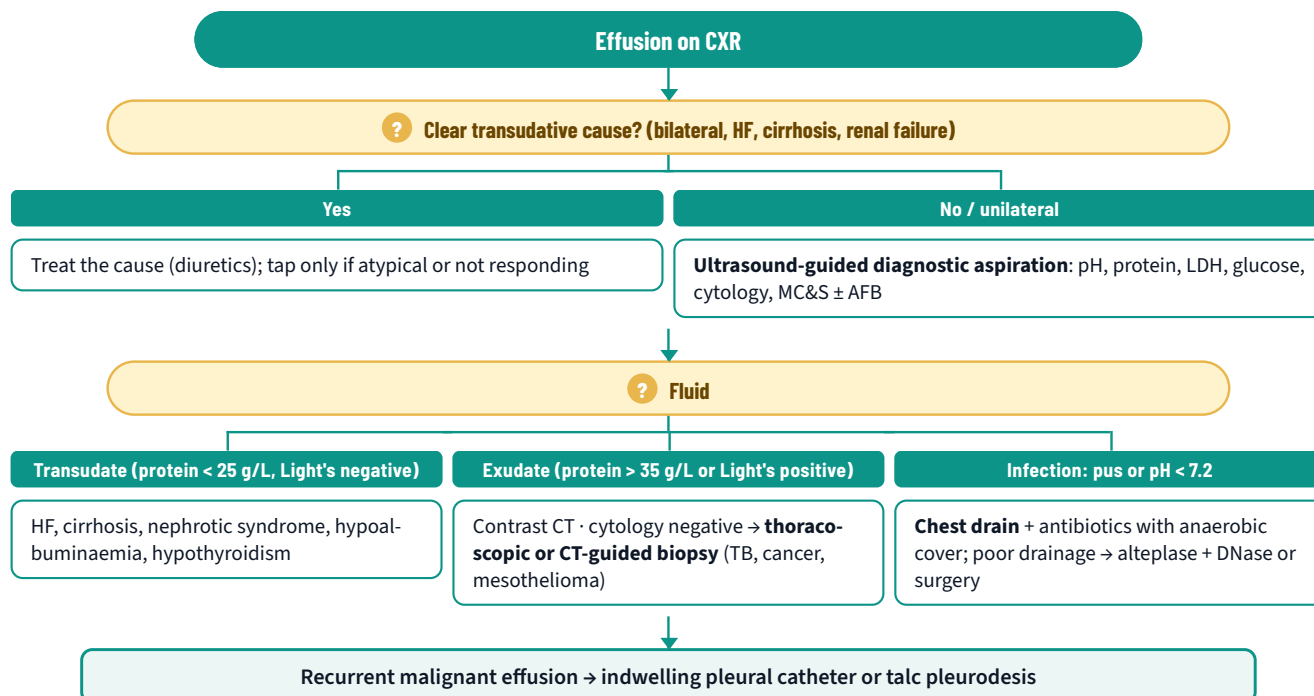
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Step by step

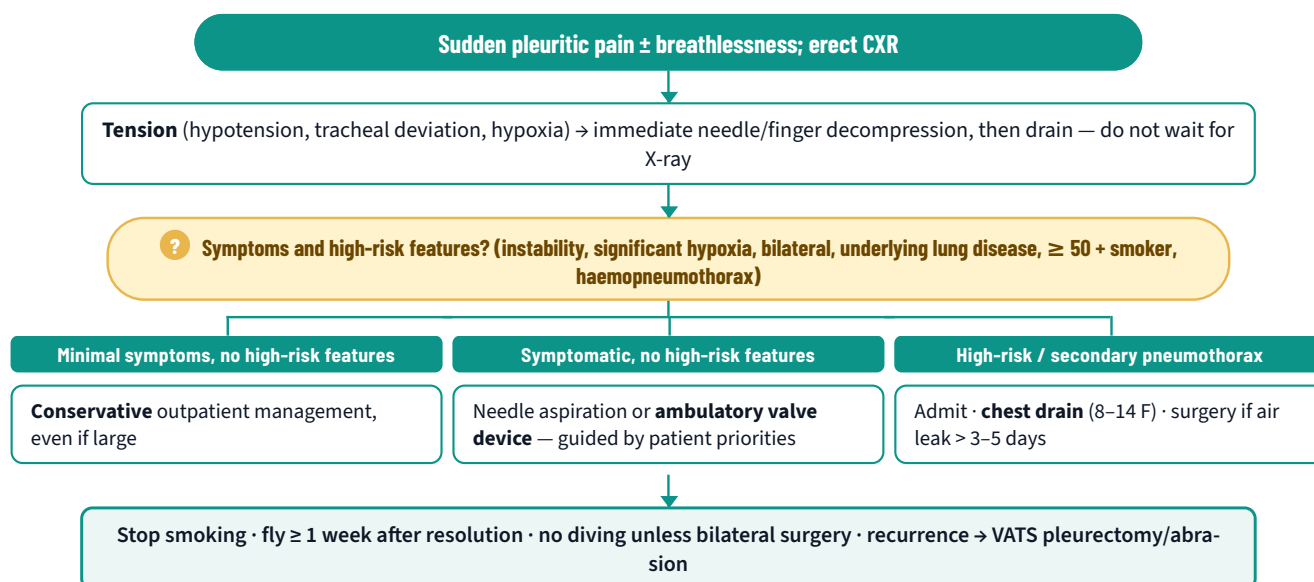
Pleural effusion

BTS PLEURAL 2023



Pneumothorax

BTS PLEURAL 2023



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** Which of the following CANNOT cause an exudative pleural effusion?
- A Parapneumonic effusion
 - B Malignancy
 - C Nephrotic syndrome
 - D Pulmonary infarction
- 2** A patient who had lobar pneumonia 4 weeks ago now has high fever, rigors, pleuritic chest pain, cough with sputum and breathlessness. Which complication has developed?
- A Parapneumonic pleural effusion
 - B Lung fibrosis
 - C Bronchopneumonia
 - D Empyema
- 3** A 25-year-old man who has recently had a chest infection has a pleural aspiration. The fluid is turbid and purulent-looking: protein 48 g/L, pH 7.1, glucose 1.8 mmol/L and a heavy neutrophil count. What is the most likely diagnosis?
- A Tuberculous pleurisy
 - B Simple (uncomplicated) parapneumonic effusion
 - C Empyema
 - D Malignant effusion
- 4** Common causes of empyema include all of the following EXCEPT:
- A Pneumonia
 - B Tuberculosis
 - C Infected haemothorax
 - D Rheumatic fever

REPEATED 4x • ALSO ASKED IN

End-block exam 2025 • Previous-years exam • Crash Course

REPEATED 2x • ALSO ASKED IN

Formative exam

REPEATED 2x • ALSO ASKED IN

Previous-years exam

- 5** A 56-year-old woman is breathless after a road traffic accident, with multiple bruises on the trunk. She is tachypnoeic with diminished air entry and stony dullness on the left. Her chest X-ray is shown. What is the likely diagnosis?



- A Pneumothorax
 - B Haemothorax
 - C Pleural effusion
 - D Lung collapse
- 6** A 56-year-old woman has a traumatic left haemothorax after a road traffic accident. What is the appropriate management now?
- A Urgent laparotomy
 - B Needle aspiration
 - C Right-sided chest tube
 - D Left-sided chest tube

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

CASE • QUESTIONS 7–10

A 45-year-old man is brought to the emergency department after a car accident. There is bruising over the right chest with paradoxical movement of a segment of the right chest wall, reduced expansion and diminished breath sounds on the right. He is conscious but in respiratory distress; BP 95/60 mmHg, SpO₂ 86% on air.

- 7** What is the most likely diagnosis?
- A Tension pneumothorax
 - B Flail chest
 - C Massive haemothorax
 - D Pulmonary contusion
 - E Lung collapse

- 8 Which of the following is the main reason his lung function is compromised?
- A Hyperventilation due to pain
 - B Underlying pulmonary contusion
 - C Mediastinal shift
 - D Air trapping during expiration
 - E Diaphragmatic rupture
- 9 Chest X-ray and CT show a massive haemothorax. What is the next immediate step?
- A Urgent thoracotomy
 - B Endotracheal intubation and positive-pressure ventilation
 - C Large-bore chest drain insertion
 - D Needle decompression in the second intercostal space
 - E Observation and repeat chest X-ray in 6 hours
- 10 His condition improves, but he remains hypoxic with persistent paradoxical chest movement despite oxygen and analgesia. What is the next definitive step?
- A Analgesia and oxygen only
 - B Intubation with mechanical ventilation
 - C Surgical fixation of the ribs
 - D Repeat chest X-ray after 24 hours
 - E External strapping of the chest wall

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 11 A 21-year-old high-jumper presents with acute-onset breathlessness and right-sided pleuritic chest pain. Examination reveals increased percussion resonance and reduced expansion on the right side. What is the most likely diagnosis?
- A Cardiac tamponade
 - B Aortic dissection
 - C Pneumothorax
 - D Pulmonary embolus

REPEATED 2* · ALSO ASKED IN
Crash Course

Answers 1-11

LOCAL EXAMS MCQs

- 1 **C Nephrotic syndrome** — Nephrotic syndrome causes a transudate (low oncotic pressure); pneumonia, malignancy and infarction cause exudates.
- 2 **D Empyema** — Recurrent high fever with rigors and pleuritic pain weeks after pneumonia is empyema; aspirate (pH < 7.2, pus) and drain.
- 3 **C Empyema** — **UPDATED** Purulent fluid with pH < 7.2, glucose < 3.3 mmol/L and neutrophils is empyema: chest drain plus antibiotics.
- 4 **D Rheumatic fever** — Empyema follows pneumonia, TB, infected haemothorax, surgery, oesophageal rupture or subphrenic abscess; rheumatic fever is not a cause.
- 5 **B Haemothorax** — Stony dullness and a dense homogeneous opacity after blunt chest trauma is a haemothorax.
- 6 **D Left-sided chest tube** — Traumatic haemothorax: left chest drain in the safe triangle; > 1500 mL initially or ~200 mL/h ongoing indicates thoracotomy.
- 7 **B Flail chest** — A segment moving paradoxically (in on inspiration) means two or more ribs broken in two or more places: a flail chest. Tension pneumothorax would shift the trachea and cause severe shock.
- 8 **B Underlying pulmonary contusion** — Hypoxaemia in flail chest comes mainly from the contused lung beneath the segment (alveolar haemorrhage and oedema), worsened by pain-limited breathing.
- 9 **C Large-bore chest drain insertion** — Resuscitate and insert a large-bore chest drain. Thoracotomy is needed if >1500 mL drains at once or >200 mL/h continues for 2–4 hours; needle decompression is for tension pneumothorax.
- 10 **B Intubation with mechanical ventilation** — Respiratory failure despite oxygen and analgesia needs intubation and positive-pressure ventilation, which splints the segment internally; rib fixation suits selected patients.
- 11 **C Pneumothorax** — **UPDATED** Sudden pleuritic pain in a tall young athlete with hyper-resonance and reduced expansion is primary spontaneous pneumothorax.

11 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

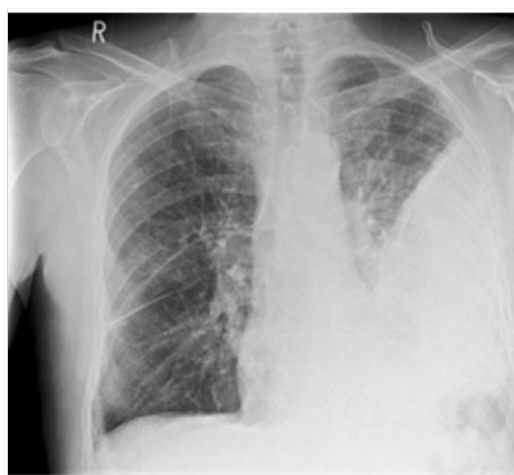
Davidson MCQs

- 1** A 64-year-old man presents with high fever, left-sided pleuritic chest pain and breathlessness; his chest X-ray is shown. Pleural fluid is turbid with pH 7.0, glucose 1.1 mmol/L and LDH 1800 U/L, but cultures are negative. He is treated for empyema with a chest drain and antibiotics. Two weeks later he re-presents with a right-sided pleural effusion, pleuritic pain, fever and a symmetrical small-joint polyarthritis. What is the most likely diagnosis?



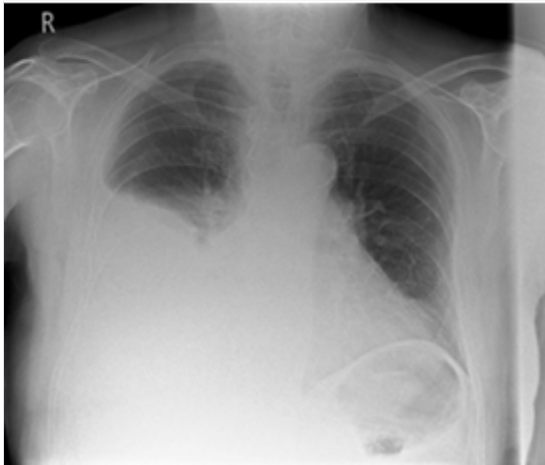
- A Empyema
- B Lung cancer
- C Lymphoma
- D Mesothelioma
- E Rheumatoid arthritis

- 2** A 52-year-old man presents with a 1-week history of fever, left-sided pleuritic chest pain and increasing shortness of breath. He has a history of alcohol dependency, alcoholic liver disease and chronic pancreatitis. He has smoked 10 cigarettes per day for 30 years and still drinks 28 units a week. In his 20s he worked as a labourer on building sites where asbestos was removed. His chest X-ray is shown. What is the most likely cause of the abnormality?



- A Bronchial carcinoma
- B Empyema
- C Hepatic hydrothorax
- D Mesothelioma
- E Pleural effusion secondary to pancreatitis

- 3** A 71 year old man presents with increasing breathlessness with no cough or systemic symptoms. He has atrial fibrillation and a recent echocardiogram with preserved LV function. What does his CXR show?



- A Right middle and lower lobe collapse
- B Right-sided bronchial carcinoma
- C Right-sided consolidation
- D Right-sided mesothelioma
- E Right-sided pleural effusion

- 4** A 71-year-old man with atrial fibrillation on apixaban presents with increasing breathlessness but no cough or systemic symptoms. A recent echocardiogram showed preserved left ventricular function. His chest X-ray shows a moderate right-sided pleural effusion. What is the best next step in his management?

- A Bedside thoracic ultrasound and pleural aspiration
- B Diuretic therapy
- C Repeat echocardiogram
- D Stop apixaban
- E Thoracic ultrasound scan-guided intercostal chest drain

- 5** A 35 year old man has been unwell for a week with fever, productive cough, rhinitis and aching muscles. He now presents with left-sided pleuritic chest pain. His CXR is shown. What is shown on the CXR?

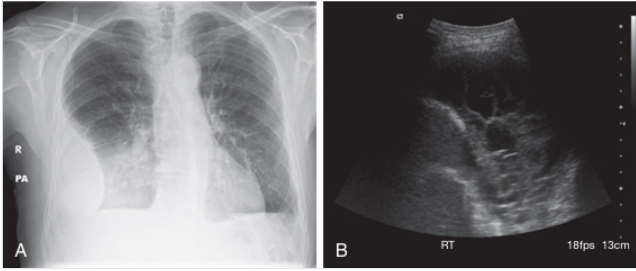


- A Bilateral hilar lymphadenopathy
- B Collapse/consolidation of the lingula
- C Diffuse right-sided infiltrates
- D Left hilar consolidation
- E Left-sided pneumothorax

- 6** A 29-year-old man presents with progressive shortness of breath over a week, new central chest discomfort and a productive cough. He is a heavy cannabis smoker with no significant past medical history. His chest X-ray (shown) shows large lucent upper-zone areas without lung markings, and it is unclear whether these are bullae or a pneumothorax. What is the next step in his management?



- A Admit for observation
- B CT scan
- C Human immunodeficiency virus (HIV) test
- D Intercostal chest drain
- E Therapeutic aspiration

- 7** A 75 year old man with asbestos exposure presents with a large right-sided pleural effusion. Pleural aspiration is exudative but cytology is negative and CT shows circumferential pleural thickening. Which test is most likely to give a diagnosis?
- Abrams needle biopsy
 - Bronchoscopy
 - Echocardiogram
 - Repeat pleural aspiration
 - Thoracoscopy
- 8** Which of the following is a cause of a transudative pleural effusion?
- Drug-induced pleural effusion
 - Hypothyroidism
 - Pulmonary infarction
 - Rheumatoid arthritis
 - Tuberculosis
- 9** Which one of the following statements is true regarding Light's criteria?
- Pleural fluid:serum protein ratio > 0.6
 - Pleural fluid LDH $>$ two-thirds of the upper limit of normal serum LDH
 - Pleural fluid LDH $<$ two-thirds of the upper limit of normal serum LDH
 - Pleural fluid LDH:serum LDH ratio > 0.5
 - Pleural fluid:serum protein ratio < 0.5
- 10** A 64 year old man presents with fever, sweats, right-sided pleuritic chest pain and breathlessness. He is an ex-smoker with a history of alcohol excess. Pleural fluid aspiration under ultrasound guidance is performed. Which value might be expected?
- 
- Pleural fluid glucose 5.6 mmol/L (101 mg/dL)
 - Pleural fluid LDH 150 U/L
 - Pleural fluid pH 6.9 (H+ 126 nmol/L)
 - Pleural fluid protein 27 g/L
 - Pleural fluid triglycerides 5.4 mmol/L (478 mg/dL)
- 11** A 34 year old man presents with acute-onset shortness of breath and left-sided pleuritic chest pain. Examination reveals O₂ saturations of 94% on room air and a hyper-resonant left chest. CXR shows a large left-sided pneumothorax. Therapeutic aspiration of 2.5 L of air is performed with no radiographic improvement. What should be the next step?
- Admit for observation and oxygen
 - Bronchoscopy
 - Cardiothoracic surgery
 - Continue therapeutic aspiration
 - Intercostal chest drain

Answers 1-11

- 1** **E Rheumatoid arthritis** — Rheumatoid effusions are exudates with very low glucose and pH, mimicking empyema; new polyarthritis reveals the diagnosis.
- 2** **B Empyema** — **UPDATED** A large left effusion with a week of fever and pleuritic pain in a heavy drinker is most likely empyema; aspirate urgently.
- 3** **E Right-sided pleural effusion** — Homogeneous lower-zone opacity with a meniscus and lost costophrenic angle is a pleural effusion; unilateral with normal LV needs investigation.
- 4** **A Bedside thoracic ultrasound and pleural aspiration** — **UPDATED** Unilateral effusion with normal LV function: bedside ultrasound and diagnostic aspiration; no need to stop apixaban (BTS 2023).
- 5** **E Left-sided pneumothorax** — **UPDATED** A visible pleural edge with no markings beyond and a hyperlucent hemithorax is a left pneumothorax.
- 6** **B CT scan** — **UPDATED** In a cannabis smoker, a large bulla can mimic pneumothorax; CT distinguishes them before any drain (risk of bronchopleural fistula).
- 7** **E Thoracoscopy** — Asbestos, cytology-negative exudate and circumferential thickening suggest mesothelioma; thoroscopic biopsy (yield $> 90\%$).
- 8** **B Hypothyroidism** — Hypothyroidism can cause a transudate; infarction, RA, TB and drugs cause exudates.
- 9** **B Pleural fluid LDH $>$ two-thirds of the upper limit of normal serum LDH** — Light's exudate criteria: protein ratio > 0.5 , LDH ratio > 0.6 , or fluid LDH $> 2/3$ of the upper limit of serum LDH.
- 10** **C Pleural fluid pH 6.9 (H+ 126 nmol/L)** — A septated effusion with fever means complicated parapneumonic effusion/empyema: pH < 7.2 (here 6.9) with low glucose and high LDH.
- 11** **E Intercostal chest drain** — **UPDATED** If needle aspiration fails to re-expand the lung, insert a chest drain (or ambulatory device); stop aspiration at 2.5 L.

5 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 A 53-year-old man with pneumonia is found to have gross pus in the pleural space (empyema), and a chest tube is placed along with intravenous antibiotics. Which additional treatment improves pleural drainage and reduces the need for surgery?

A Immediate decortication
 B Intrapleural alteplase alone
 C Intrapleural alteplase plus DNase
 D Intrapleural DNase alone
 E Intrapleural streptokinase

- 2 A 44-year-old man on mechanical ventilation suddenly becomes hypotensive. The peak airway pressure has risen to 55 cmH₂O and breath sounds are absent on the right. What is the next step?

A IV fluid bolus
 B Disconnect ventilator to allow exhalation
 C Needle decompression of the right chest
 D Sedate patient for synchrony
 E Suction for mucus plug

- 3 A 62-year-old man with pneumonia has a pleural effusion. Which finding does NOT indicate a complicated parapneumonic effusion?

A Loculated fluid on ultrasound
 B Pleural fluid pH <7.20
 C Pleural fluid glucose <3.3 mmol/L (60 mg/dL)
 D Positive pleural fluid Gram stain or culture
 E Reaccumulation of fluid after the initial thoracentesis

- 4 A patient has a moderate right-sided pleural effusion. The pleural fluid analysis is shown. Which of the following is NOT a likely cause of this effusion?

Appearance	Serosanguineous
pH	7.48
Protein	5.8 g/dL (serum protein 7.2 g/dL)
LDH	285 IU/L (serum LDH 320 IU/L)
Glucose	66 mg/dL
WBC	3800/μL
RBC	24,000/μL
PMNs	10%
Lymphocytes	80%
Mesothelial cells	10%
Cytology	Lymphocytosis with chronic inflammation and no malignant cells or organisms identified

A Cirrhosis
 B Lung cancer
 C Mesothelioma
 D Pulmonary embolism
 E Tuberculosis

- 5 A 28-year-old tall, thin man who smokes has a small right-sided primary spontaneous pneumothorax. Which statement is true?

A Smoking does not affect his risk of recurrence
 B If it recurs, VATS pleurectomy or pleural abrasion reduces the recurrence rate to about 1–5%
 C Most patients need tube thoracostomy for the pneumothorax to resolve
 D The recurrence rate after a first episode is less than 5%
 E He can fly safely 24 hours after the pneumothorax resolves on X-ray

Answers 1-5

HARRISON MCQs

- 1 **C Intrapleural alteplase plus DNase** — In poorly draining pleural infection, intrapleural alteplase plus DNase improves drainage and reduces surgery (MIST-2).
- 2 **C Needle decompression of the right chest** — Sudden hypotension, high airway pressure and absent breath sounds in a ventilated patient is tension pneumothorax: needle decompression.
- 3 **E Reaccumulation of fluid after the initial thoracentesis** — **UPDATED** Complicated parapneumonic effusion: loculation,

pus, positive Gram stain/culture, pH < 7.2 or glucose < 3.3 mmol/L. Reaccumulation alone is not one.

- 4 **A Cirrhosis** — **UPDATED** The fluid is a lymphocytic exudate (protein ratio 0.8, LDH ratio 0.89); cirrhosis causes a transudate, so it is not the cause.
- 5 **B If it recurs, VATS pleurectomy or pleural abrasion reduces the recurrence rate to about 1–5%** — **UPDATED** Recurrent primary pneumothorax is treated by VATS pleurectomy/abrasion (recurrence 1–5%); stopping smoking cuts recurrence.

5 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1 With regard to pleural fluid, which of the following is most important in determining whether an effusion is a transudate or exudate?
 - A Bacterial count
 - B Amylase levels
 - C Malignant cells
 - D Glucose levels
 - E Protein levels
- 2 You are called to review a 70-year-old man admitted with pneumococcal pneumonia who has become more unwell, with swinging fevers, worsening breathlessness and pleuritic chest pain. His inflammatory markers are rising and a repeat chest X-ray shows a new right-sided pleural effusion. What is the most appropriate next diagnostic investigation?
 - A CT scan
 - B Bronchoscopy
 - C Ultrasound-guided diagnostic pleural tap
 - D PFTs
 - E Recheck the sputum for resistant organisms
- 3 A 76-year-old man with COPD is admitted to the emergency department with acute-on-chronic breathlessness and has a chest drain inserted for a pneumothorax. Which of the following is true of chest drain management?
 - A He should have his drain clamped if he has to travel to the radiology department without a medical escort
 - B The drain should only be removed when it has stopped bubbling
 - C The drain may be clamped if it only bubbles on coughing
 - D The drain should only be removed when it has stopped swinging
 - E Pain relief is usually not necessary in the first few days after insertion
- 4 A 76-year-old man is admitted to the intensive care unit for intravenous potassium replacement. A central venous line is inserted via the subclavian vein, and during the procedure he becomes acutely breathless with falling oxygen saturation. What is the most likely diagnosis?
 - A The guidewire has entered the right ventricle
 - B His chronic obstructive airways disease has been exacerbated
 - C A pneumothorax has been created
 - D A myocardial infarction has been induced
 - E Air embolism has occurred

REPEATED 2* • ALSO ASKED IN
PasTest

- 5 A 58-year-old man with longstanding emphysema presents to the emergency department with sudden breathlessness and left-sided pleuritic chest pain. He is hypoxic, tachypnoeic, hypotensive and tachycardic. His trachea is deviated to the right and breath sounds are markedly reduced on the left. What is the next step in management?
 - A Chest X-ray
 - B Insert a large-bore cannula into the left second intercostal space
 - C Insert a left-sided chest drain with suction
 - D 1 L saline bolus
 - E Oxygen via nasal cannulae

Answers 1-5

CRASH COURSE MCQs

- 1 **E Protein levels** — Protein separates transudate from exudate (ratio > 0.5, or > 35 g/L); Light's criteria are used when protein is 25–35 g/L.
- 2 **C Ultrasound-guided diagnostic pleural tap** — A new effusion with ongoing sepsis during pneumonia: ultrasound-guided diagnostic tap; pH < 7.2 or pus means a drain.
- 3 **B The drain should only be removed when it has stopped bubbling** — Remove a pneumothorax drain only when bubbling has stopped and the lung is re-expanded; never clamp a bubbling drain.
- 4 **C A pneumothorax has been created** — **UPDATED** Sudden breathlessness and desaturation during subclavian line insertion is iatrogenic pneumothorax; urgent CXR.
- 5 **B Insert a large-bore cannula into the left second intercostal space** — Hypotension and tracheal deviation away from a silent hemithorax is tension pneumothorax: immediate large-bore cannula decompression.

3 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A patient with low-grade fever and weight loss has poor excursion on the right side of the chest with decreased fremitus, flatness to percussion, and decreased breath sounds all on the right. The trachea is deviated to the left. Which of the following is the most likely diagnosis?
 - A Pneumothorax
 - B Pleural effusion
 - C Consolidated pneumonia
 - D Atelectasis
 - E Chronic obstructive lung disease
- 2 A 64-year-old woman is found to have a right-sided pleural effusion on chest X-ray. Pleural fluid analysis shows a pleural fluid/serum protein ratio of 0.38, a pleural fluid lactate dehydrogenase (LDH) of 110 IU/L (serum upper limit of normal 190 IU/L) and a pleural fluid/serum LDH ratio of 0.46. Which disorder is most likely?
 - A Bronchogenic carcinoma
 - B Congestive heart failure
 - C Pulmonary embolism
 - D Sarcoidosis
 - E Systemic lupus erythematosus
- 3 A 50-year-old male with emphysema develops the sudden onset of shortness of breath and left-sided pleuritic chest pain. Pneumothorax is suspected. Which of the following physical examination findings would be consistent with the diagnosis?
 - A Localized wheezes at the left base
 - B Hyper-resonance, decreased breath sounds in the left chest with trachea deviation to right
 - C Increased tactile fremitus on the left side
 - D Decreased breath sounds on the left side with deviation of the trachea to the left
 - E Rales at the left base

Answers 1-3

PRETEST MCQS

- 1 **B Pleural effusion** — Stony dullness, reduced breath sounds and fremitus with the trachea pushed away is a large effusion (TB or malignancy here).
- 2 **B Congestive heart failure** — **UPDATED** All Light's criteria negative (protein 0.38, LDH 0.46, LDH < 127): a transudate, most commonly heart failure.
- 3 **B Hyper-resonance, decreased breath sounds in the left chest with trachea deviation to right** — **UPDATED** Pneumothorax: hyper-resonance and reduced breath sounds on the affected side, with a large one pushing the trachea away.

4 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 29-year-old man is admitted with sudden onset dyspnoea and pleuritic chest pain. He is a smoker but has no history of respiratory disease. He considers himself healthy and regularly plays rugby. On admission he has a chest x-ray that shows a pneumothorax with a 3cm rim of air. Aspiration is successful and he is discharged. A follow-up chest x-ray two weeks later shows a complete resolution. What is the single most important piece of advice to reduce his risk of further pneumothoraces?
 - A Avoid flying for 12 months
 - B Avoid contact sports for 12 months
 - C Stop smoking
 - D Arrange a course of respiratory physiotherapy
 - E Seek prompt medical advice for potential respiratory infections
- 2 A 72-year-old woman is referred to ambulatory care with a 3-month history of increasing breathlessness. She has no significant respiratory history. On examination her observations are normal, with reduced air entry at the right base. Chest X-ray shows a large right-sided pleural effusion. Pleural aspiration results: pleural fluid protein 34 g/L (serum protein 61 g/L); white cell count 2500 cells/ μ L, predominantly lymphocytes; no organisms on microscopy; no growth on culture after 48 hours. What is the most likely cause of this effusion?
 - A Congestive cardiac failure
 - B Hypoalbuminaemia
 - C Lung cancer
 - D Nephrotic syndrome
 - E Parapneumonic effusion
- 3 A 35-year-old woman presents to the emergency department with sudden right-sided chest pain and shortness of breath that began at the start of her menstrual period. A similar but milder episode occurred during her last period and resolved. The pain now radiates to her shoulder. She has two children, the second delivered by emergency caesarean section. What is the most likely cause of her symptoms?
 - A Meigs' syndrome
 - B Takotsubo cardiomyopathy
 - C Catamenial pneumothorax
 - D Pleural effusion
 - E Hiatus hernia
- 4 A 32-year-old man with no significant past medical history presents to the emergency department with sudden-onset left-sided chest pain and breathlessness. There is no trauma or recent infection. Respiratory rate 21/min, SpO₂ 96% on air, BP 124/78 mmHg, heart rate 88/min, afebrile. The trachea is central; the left side of the chest is hyper-resonant with reduced breath sounds. Chest X-ray shows a 2.5 cm apical rim of air. His pain and breathlessness are troubling him and he wants the air treated, but he strongly wishes to avoid hospital admission. What is the most appropriate initial management?
 - A Ambulatory device
 - B Chest drain
 - C Discharge and follow-up in 1-2 weeks
 - D Inpatient monitoring
 - E Oxygen therapy

Answers 1-4

PASSMEDICINE MCQS

- 1 **C Stop smoking** — Smoking (including cannabis) is the main modifiable risk for pneumothorax recurrence: advise stopping.
- 2 **C Lung cancer** — **UPDATED** A lymphocytic sterile exudate with a 3-month history in an older woman suggests malignancy (lung cancer); cytology and CT.
- 3 **C Catamenial pneumothorax** — Recurrent right pneumothorax within 72 h of menstruation is catamenial pneumothorax from thoracic endometriosis.
- 4 **A Ambulatory device** — **UPDATED** Symptomatic primary pneumothorax without high-risk features, wishing to avoid admission: ambulatory valve device (BTS 2023).

3 QUESTIONS • AFTER PASTEST

PasTest MCQs

- 1 A 70-year-old man with moderate chronic obstructive pulmonary disease (COPD) attends the Emergency Department with increasing dyspnoea. He has been found to have a 3-cm pneumothorax. What would be the optimal management for this patient?
 - A Chest drain insertion if needle aspiration fails
 - B Chest drain insertion initially
 - C Conservative management, with observation and repeat chest X-ray after 4 hours
 - D Conservative management, with observation and repeat chest X-ray after 12 hours
 - E Needle aspiration
- 2 A 43-year-old woman has a pleural effusion together with arthralgia and a photosensitive rash that has worsened after sun exposure over the past few months. What is the most likely diagnosis?
 - A Actinomycosis
 - B Adenocarcinoma of the lung
 - C Progressive massive fibrosis
 - D Granulomatosis with polyangiitis
 - E Systemic lupus erythematosus
- 3 After a tennis match, a thin 20-year-old woman develops sudden left-sided pleuritic chest pain and breathlessness. A few minutes later she becomes cyanosed and hypotensive; her trachea is deviated to the right and the left hemithorax is hyper-resonant with absent breath sounds. What is the diagnosis?
 - A Acute pancreatitis
 - B Ectopic pregnancy
 - C Myocardial infarction
 - D Pulmonary embolism
 - E Tension pneumothorax

Answers 1-3

PASTEST MCQS

- 1 **B Chest drain insertion initially** — UPDATED Secondary pneumothorax in COPD is high-risk; symptomatic or > 2 cm needs admission and a chest drain.
- 2 **E Systemic lupus erythematosus** — Photosensitive rash, arthralgia and pleural effusion suggest SLE serositis; check ANA, anti-dsDNA and complement.
- 3 **E Tension pneumothorax** — UPDATED Pneumothorax with cyanosis, hypotension and tracheal deviation away is tension pneumothorax: needle decompression then drain.



15

INTERSTITIAL LUNG DISEASE

Interstitial Lung Disease

Idiopathic pulmonary fibrosis, sarcoidosis, hypersensitivity pneumonitis, occupational and drug-induced lung disease.

60

MCQs

0

EMQ SCENARIOS

7

SOURCES

8

REPEATED IDEAS

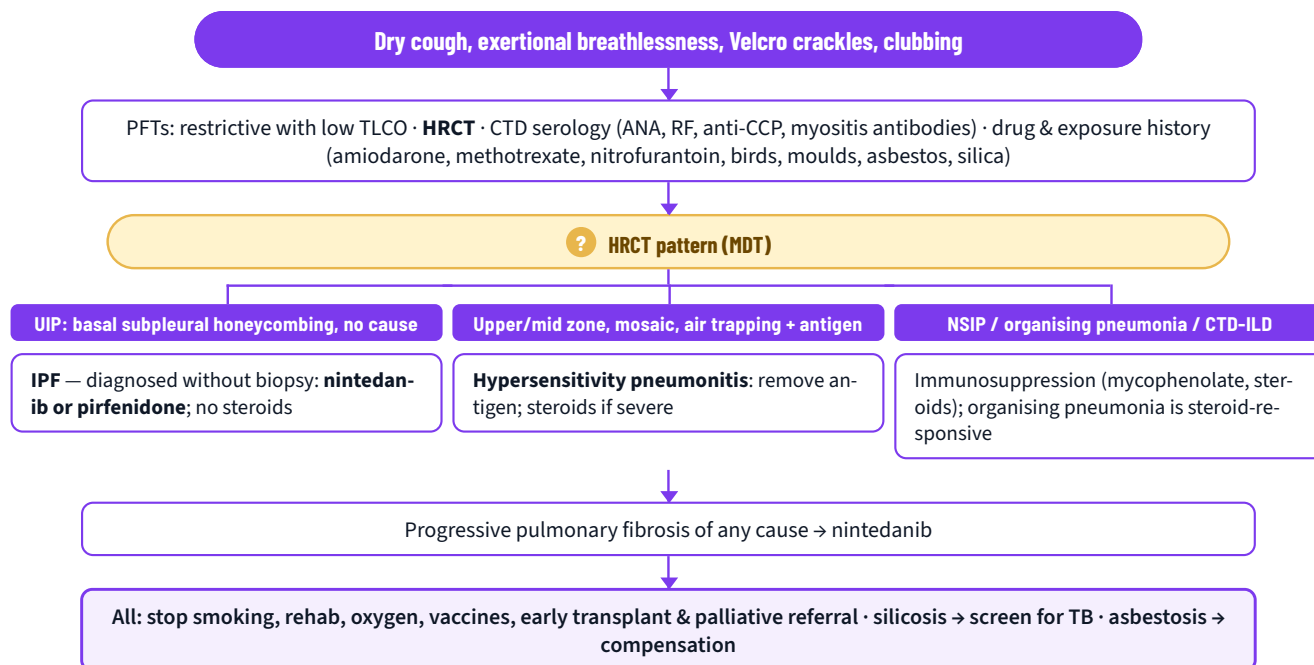
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Step by step

Interstitial lung disease

ATS/ERS/JRS/ALAT 2022 • NICE TA



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

CASE • QUESTIONS 1–5

A 56-year-old woman has had a persistent dry cough and increasing breathlessness for 6 months. She is thin with finger clubbing, breathless at rest (RR 24/min), with bibasal end-inspiratory crackles; SpO_2 is 88% on air and cardiovascular examination is normal.

- 1 What is the most likely diagnosis?
 - A Chronic bronchitis
 - B Asthma
 - C Idiopathic pulmonary fibrosis
 - D Bronchiectasis
 - E Emphysema
- 2 Which pulmonary function result is most consistent with her disease?
 - A Increased FEV_1
 - B Markedly increased FVC
 - C Normal or increased FEV_1/FVC ratio
 - D Increased total lung capacity
 - E Normal FVC
- 3 What is the most characteristic radiological finding?
 - A Hyperinflated chest
 - B Honeycomb fibrosis
 - C Bronchial wall thickening
 - D Patchy opacities
 - E Pleural effusion
- 4 Which arterial blood gas finding is consistent with her disease?
 - A Raised PaO_2
 - B Raised $PaCO_2$
 - C Low PaO_2 with low $PaCO_2$
 - D Raised serum bicarbonate
 - E Low PaO_2 with high $PaCO_2$
- 5 What is the main pharmacological treatment?
 - A Inhaled corticosteroids
 - B Oral steroids
 - C Antifibrotic drugs (pirfenidone or nintedanib)
 - D Long-acting muscarinic antagonist
 - E Beta-agonists

REPEATED 4* • ALSO ASKED IN

39th batch end-block (respiratory) • Previous-years exam • Crash Course

- 6 A 58-year-old man has progressive exertional breathlessness and a dry cough for 8 months. There are basal "Velcro" crackles and finger clubbing. PFTs show a restrictive pattern (FVC 62%) with DLCO 45%; HRCT shows basal subpleural reticulation with honeycombing and traction bronchiectasis. What is the most likely diagnosis?
 - A Pulmonary embolism
 - B Asthma
 - C Idiopathic pulmonary fibrosis
 - D Heart failure with pulmonary oedema
 - E COPD

REPEATED 2* • ALSO ASKED IN

Davidson

- 7 Which of the following best describes fibrosing alveolitis (idiopathic pulmonary fibrosis)?
 - A Orthopnoea and bibasal crackles
 - B Cough and occasional haemoptysis
 - C Productive cough and wheeze
 - D Finger clubbing and "Velcro" crackles

REPEATED 2* • ALSO ASKED IN

End-block exam 2025

- 8 Which of the following is NOT typically associated with idiopathic pulmonary fibrosis?
 - A Finger clubbing
 - B Pulmonary hypertension
 - C Traction bronchiectasis
 - D Median survival of 3–5 years after diagnosis
 - E Common age of onset 30–40 years
- 9 A woman who had radiotherapy for breast cancer 2 months ago develops a cough and breathlessness. What is the treatment of choice?
 - A Corticosteroids
 - B Oral bronchodilators
 - C Montelukast
 - D Tiotropium

- 10 A 50-year-old woman with rheumatoid arthritis, recently started on methotrexate and naproxen, becomes progressively breathless. Spirometry shows an FEV_1/FVC ratio of 0.50 with no significant bronchodilator reversibility. What is the most likely cause of her breathlessness?
 - A Bronchiolitis obliterans
 - B Idiopathic pulmonary fibrosis
 - C Methotrexate pneumonitis
 - D Severe anaemia of chronic disease

REPEATED 2* • ALSO ASKED IN

Passmedicine

- 11** A 60-year-old woman has progressive exertional breathlessness and dry cough for 2 months, with finger clubbing, restricted expansion and bilateral basal fine end-inspiratory crackles. Chest X-ray and CT show bilateral reticulonodular shadowing. What is the next step in management?
- A Bronchoscopy
 - B Start bronchodilators
 - C Start prednisolone 60 mg/day
 - D Pulmonary function tests

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 12** Which of the following is not typically associated with idiopathic pulmonary fibrosis?
- A Tachypnoea
 - B Fine inspiratory crackles
 - C Tremor or hyperreflexia
 - D Central cyanosis in advanced disease
 - E Finger clubbing

Answers 1-12

LOCAL EXAMS MCQS

- 1** **C Idiopathic pulmonary fibrosis** — Dry cough, progressive dyspnoea, clubbing, Velcro crackles and hypoxaemia in an older adult is IPF; HRCT for UIP pattern.
- 2** **C Normal or increased FEV₁/FVC ratio** — IPF is restrictive: TLC, FVC and FEV₁ fall together, so FEV₁/FVC is normal or high (> 0.8); TLC falls early.
- 3** **B Honeycomb fibrosis** — The IPF hallmark is a UIP pattern: basal subpleural reticulation with honeycombing and traction bronchiectasis.
- 4** **C Low PaO₂ with low PaCO₂** — **UPDATED** IPF causes V/Q mismatch with hyperventilation: low PaO₂ with low PaCO₂ (type 1 failure); hypercapnia is pre-terminal.
- 5** **C Antifibrotic drugs (pirfenidone or nintedanib)** — Anti-fibrotics (pirfenidone, nintedanib) halve FVC decline in IPF; prednisolone ± azathioprine increased mortality (PANTHER).
- 6** **C Idiopathic pulmonary fibrosis** — Velcro crackles, clubbing, restriction with low DLCO and a UIP pattern on HRCT is IPF once exposures and CTD are excluded.
- 7** **D Finger clubbing and "Velcro" crackles** — Fibrosing alveolitis (IPF) gives fine late-inspiratory Velcro crackles and clubbing in up to half of patients.
- 8** **E Common age of onset 30–40 years** — IPF is a disease of older adults (60s–70s), rare before 50; onset at 30–40 should prompt a search for another cause.
- 9** **A Corticosteroids** — Radiation pneumonitis 1–3 months after chest radiotherapy is treated with oral corticosteroids once infection is excluded.
- 10** **A Bronchiolitis obliterans** — **UPDATED** Fixed obstruction (FEV₁/FVC 0.50, no reversibility) in RA suggests obliterative bronchiolitis; methotrexate pneumonitis is restrictive.
- 11** **D Pulmonary function tests** — With CT evidence of fibrotic ILD, full PFTs (spirometry, volumes, TLC) confirm restriction and set a baseline before MDT review.
- 12** **C Tremor or hyperreflexia** — **UPDATED** IPF gives tachypnoea, crackles, clubbing and cyanosis; tremor and hyperreflexia are unrelated neurological signs.

11 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

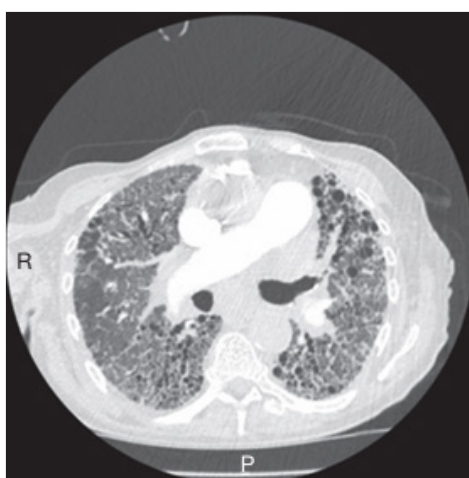
- 1** A 72 year old woman with a 20-year history of rheumatoid arthritis is found on CXR and CT to have multiple smooth nodules that have been present for at least 5 years. What is the likely cause?
- A Bronchiectasis
 - B Metastatic cancer
 - C Pulmonary embolism
 - D Pulmonary fibrosis
 - E Rheumatoid nodules
- 2** A 55 year old geologist with cough, breathlessness, and arthralgia since home renovation (with damp and dust exposure) and beryllium handling at work presents with bilateral hilar lymphadenopathy on CXR. What is the most likely diagnosis?
- A Berylliosis
 - B Dysfunctional breathing
 - C Hypersensitivity pneumonitis
 - D Sarcoidosis
 - E Silicosis

- 3 A 69 year old woman undergoing a CT coronary angiogram is incidentally found to have diffuse ground glass and centrilobular nodules. She has a pet parrot. What is the likely diagnosis?

A Aspirin sensitivity
 B Breathing artefact
 C Hypersensitivity pneumonitis
 D Idiopathic pulmonary fibrosis
 E Sarcoidosis

REPEATED 2x • ALSO ASKED IN
 Harrison

- 4 A 73-year-old woman has a 1-year history of progressive exertional breathlessness and a dry cough. She has finger clubbing and bibasal crackles. Her CT chest is shown. What is the most likely cause of the CT appearance?



- A Bronchiectasis
 B Emphysema
 C Idiopathic pulmonary fibrosis
 D Lymphangitis carcinomatosa
 E Scarring related to pulmonary tuberculosis
- 5 A 71-year-old man presents with increasing breathlessness. He has no relevant occupational or drug exposures and no features of connective tissue disease. HRCT shows bilateral subpleural, basal-predominant reticulation with honeycombing and traction bronchiectasis. Which statement about confirming the diagnosis is correct?
- A A typical CT appearance is sufficient for diagnosis
 B Blind transbronchial needle aspiration would likely give the diagnosis
 C Bronchoscopy with cytopathological analysis of washings would help
 D Endobronchial ultrasound of his subcarinal node would exclude more diffuse disease
 E Transbronchial lung biopsy would give a tissue diagnosis

- 6 A 62-year-old ex-smoker presents with breathlessness on exertion. Pulmonary function tests: FEV₁ 3.2 L (98% predicted), FVC 3.8 L (97% predicted), FEV₁/FVC 0.84, TLC 96% predicted, TLCO 44% predicted, KCO 56% predicted. How should these results be interpreted?

A Gas transfer reduction indicates emphysema
 B Gas transfer reduction indicates hypersensitivity pneumonitis
 C No evidence of airways obstruction
 D There is a restrictive picture
 E There is an obstructive picture

- 7 Respiratory bronchiolitis–interstitial lung disease (RBILD) is more common in which group?

A Non-smokers
 B Patients > 65 years of age
 C Patients with connective tissue diseases
 D Smokers
 E Women

- 8 A 55 year old man with a 2-year history of increasing breathlessness and dry cough, with restrictive PFTs and HRCT showing bilateral hilar lymphadenopathy, bronchovascular nodules and extensive fibrosis is evaluated. What is the most likely diagnosis?

A Sarcoidosis stage 0
 B Sarcoidosis stage I
 C Sarcoidosis stage II
 D Sarcoidosis stage III
 E Sarcoidosis stage IV

- 9 A 26 year old non-smoking woman presents for the second time with pneumothorax. CT identifies multiple lung cysts and an upper pole abnormality of the right kidney. What diagnosis should be considered?

A Alveolar microlithiasis
 B Alveolar proteinosis
 C Lymphangioleiomyomatosis
 D Lymphocytic interstitial pneumonia
 E Pulmonary Langerhans cell histiocytosis (histiocytosis X)

- 10 Which of the following statements is true regarding progressive massive fibrosis (PMF)?

A Characterised by small radiographic nodules
 B Chyloptysis is associated
 C Finger clubbing and basal crackles are characteristic
 D It has no impact on lung function
 E It may progress even after exposure ceases

- 11 A 41 year old stonemason with shortness of breath is found to have silicosis on imaging. Silicosis results from the inhalation of which substance?

A Coal
 B Cotton
 C Quartz
 D Silicone
 E Tin

Answers 1-11

DAVIDSON MCQs

- 1 **E Rheumatoid nodules** — Multiple smooth nodules stable for 5 years in long-standing seropositive RA are rheumatoid (necrobiotic) nodules.
- 2 **A Berylliosis** — Sarcoid-like granulomatous disease with BHL after occupational exposure is berylliosis; beryllium lymphocyte proliferation test confirms.
- 3 **C Hypersensitivity pneumonitis** — Diffuse ground-glass with centrilobular nodules and a pet parrot is hypersensitivity pneumonitis (bird fancier's lung); avoid the antigen.
- 4 **C Idiopathic pulmonary fibrosis** — **UPDATED** Subpleural reticulation with honeycombing and traction bronchiectasis on CT is a UIP pattern: IPF.
- 5 **A A typical CT appearance is sufficient for diagnosis** — **UPDATED** A typical UIP pattern on HRCT in the right context is sufficient to diagnose IPF after MDT discussion (ATS/ERS 2022); no biopsy needed.
- 6 **C No evidence of airways obstruction** — **UPDATED** FEV₁/FVC 0.84 with normal volumes excludes obstruction and restriction; an isolated low TLC needs Hb, HRCT and echo.
- 7 **D Smokers** — **UPDATED** RB-ILD is a smoking-related ILD of current heavy smokers; smoking cessation is the key treatment.
- 8 **E Sarcoidosis stage IV** — Scadding stage IV sarcoidosis is established fibrosis, even if hilar nodes remain; stage I (BHL alone) usually resolves.
- 9 **C Lymphangioleiomyomatosis** — Recurrent pneumothorax with diffuse thin-walled cysts and a renal angiomyolipoma in a young woman is LAM; VEGF-D, sirolimus.
- 10 **E It may progress even after exposure ceases** — Progressive massive fibrosis (complicated CWP/silicosis) forms upper-zone conglomerate masses and can progress after exposure stops.
- 11 **C Quartz** — Silicosis is caused by respirable crystalline silica (quartz) in stonemasonry, quarrying and sandblasting.

9 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 A 56-year-old man has subacute fever, cough and breathlessness with patchy alveolar infiltrates that do not respond to antibiotics. Lung biopsy shows plugs of granulation tissue filling the distal airways and alveolar ducts (organising pneumonia pattern). What is the best therapy?
 - A Azathioprine 100 mg daily
 - B Nintedanib 150 mg twice daily
 - C Pirfenidone 2403 mg daily
 - D Prednisone 1 mg/kg daily
 - E Referral for transplantation
- 2 Which of the following pulmonary function test results is consistent with Idiopathic pulmonary fibrosis?
 - A Increased TLC, decreased vital capacity, decreased FEV₁/FVC ratio
 - B Decreased TLC, decreased VC, decreased RV, increased FEV₁/FVC ratio, normal MIP
 - C Decreased TLC, increased RV, normal FEV₁/FVC ratio, decreased MIP
 - D Normal TLC, normal RV, normal FEV₁/FVC ratio, normal MIP
 - E Normal TLC, decreased VC, increased RV, decreased FEV₁/FVC ratio, normal MIP
- 3 A 65-year-old man has progressive dyspnoea with bibasilar crackles, and interstitial lung disease is suspected. Which test is most likely to determine the aetiology?
 - A Bronchoscopy with transbronchial lung biopsy
 - B CT pulmonary angiography
 - C Echocardiography
 - D High-resolution CT scan of the chest
 - E Nuclear medicine stress test
- 4 A 34-year-old woman who works in a pet shop has had subacute breathlessness and a dry cough for 2 months. HRCT shows diffuse ground-glass opacity with poorly defined centrilobular nodules, and transbronchial biopsy shows loosely formed non-caseating granulomas with a lymphocytic interstitial infiltrate. What is the most appropriate management?
 - A Amphotericin
 - B Doxycycline
 - C Glucocorticoids
 - D Glucocorticoids plus azathioprine
 - E Glucocorticoids plus removal of antigen
- 5 Chronic silicosis is related to increased risk of which conditions?
 - A Infection with invasive *Aspergillus*
 - B Infection with *Mycobacterium tuberculosis*
 - C Lung cancer
 - D Rheumatoid arthritis
 - E All of the above

- 6** All of the following occupational lung diseases are correctly matched EXCEPT:
- A** Berylliosis—High-technology electronics
 - B** Byssinosis—Cotton milling
 - C** Farmer's lung—Moldy hay
 - D** Progressive massive fibrosis—Shipyard workers
 - E** Metal fume fever—Welding
- 7** A 53-year-old farmer develops sudden fever, chills and breathlessness a few hours after stacking mouldy hay. Which organism is most likely responsible?
- A** *Nocardia asteroides*
 - B** *Histoplasma capsulatum*
 - C** *Cryptococcus neoformans*
 - D** *Actinomyces* species
 - E** *Aspergillus fumigatus*
- 8** A 62-year-old man has exertional dyspnoea and bibasal crackles. His HRCT (shown) suggests idiopathic pulmonary fibrosis. What would lung histopathology most likely show?
- 
- A** Desquamative interstitial pneumonia
 - B** Diffuse alveolar damage
 - C** Loosely organized granulomatous inflammation
 - D** Nonspecific interstitial pneumonia
 - E** Usual interstitial pneumonia
- 9** What is the recommended treatment for idiopathic pulmonary fibrosis?
- A** Azathioprine plus prednisone
 - B** Cyclophosphamide 100 mg daily
 - C** Nintedanib 150 mg twice a day
 - D** Prednisone 60 mg daily
 - E** No therapy is effective for IPF

Answers 1-9

- 1** **D Prednisone 1 mg/kg daily** — Organising pneumonia unresponsive to antibiotics responds rapidly to prednisolone ~0.75–1 mg/kg tapered over 6–12 months.
- 2** **B Decreased TLC, decreased VC, decreased RV, increased FEV₁/FVC ratio, normal MIP** — **UPDATED** IPF: all lung volumes fall (TLC, VC, RV) with a high FEV₁/FVC and normal muscle strength (MIP).
- 3** **D High-resolution CT scan of the chest** — HRCT pattern and distribution often give the specific ILD diagnosis without tissue and guide any biopsy.
- 4** **E Glucocorticoids plus removal of antigen** — **UPDATED** Hypersensitivity pneumonitis from pet-shop birds: remove the antigen, with oral steroids for significant disease.
- 5** **E All of the above** — Silicosis increases risk of TB/NTM, fungal infection, lung cancer and autoimmune disease (RA, systemic sclerosis).
- 6** **D Progressive massive fibrosis—Shipyard workers** — PMF follows coal or silica exposure; shipyard workers get asbestos-related disease.
- 7** **D Actinomyces species** — Fever and breathlessness hours after mouldy hay is farmer's lung, from thermophilic actinomycetes (*Saccharopolyspora*).
- 8** **E Usual interstitial pneumonia** — The histological pattern of IPF is usual interstitial pneumonia: patchy, temporally heterogeneous fibrosis with fibroblastic foci.
- 9** **C Nintedanib 150 mg twice a day** — Nintedanib 150 mg twice daily (or pirfenidone) slows FVC decline in IPF; prednisolone/azathioprine is harmful.

9 QUESTIONS • AFTER CRASH COURSE SBAS & EMQS

Crash Course MCQs

- 1 A 35-year-old farmer develops cough, wheeze, breathlessness and fever a few hours after working in a barn with stored hay. He has smoked 10 cigarettes a day for 6 months and runs marathons. His grandfather had asthma as a child, and he has lived in central London for the past year. What is the most likely cause of his condition?
 - A Asthma
 - B COPD
 - C Occupation
 - D Exercise
 - E Air pollution
- 2 An Afro-Caribbean man presents with increasing dyspnoea. A chest X-ray shows bilateral hilar lymphadenopathy and his serum calcium is raised. Which extrapulmonary organ is most commonly involved clinically in this condition?
 - A Heart
 - B Central nervous system
 - C Kidneys
 - D Parotid glands
 - E Eyes
- 3 You suspect sarcoidosis in an Afro-Caribbean man who presents with a cough and evidence of lymphadenopathy. Which of the following investigations is likely to be most useful in confirming a diagnosis?
 - A CXR
 - B Lung function tests
 - C CT chest
 - D Serum angiotensin-converting enzyme (ACE)
 - E Lymph node biopsy
- 4 Extrinsic allergic alveolitis is often named after the exposure that causes it. Malt worker's lung results from which activity?
 - A Forking mouldy hay
 - B Handling pigeons
 - C Turning germinating barley
 - D Turning mushroom compost
 - E Contaminated humidifier systems in air conditioners
- 5 A shipbuilding yard worker presents with breathlessness, and CT shows asbestos-related changes. He asks whether he can claim UK Industrial Injuries Disablement Benefit. Which of the following asbestos-related conditions would NOT entitle him to a claim?
 - A Pleural plaques
 - B Asbestosis
 - C Mesothelioma
 - D Bilateral diffuse pleural thickening
 - E Primary lung cancer with asbestosis
- 6 A 35-year-old man completed chemotherapy and mediastinal radiotherapy for Hodgkin lymphoma 6 weeks ago. He now has a non-productive cough. His chest X-ray shows paramediastinal upper-zone shadowing with a straight lateral margin, and his oxygen saturation is 95% on air. What is the most likely diagnosis?
 - A Bleomycin toxicity
 - B Pneumocystis carinii pneumonia
 - C Pulmonary lymphoma
 - D Radiation pneumonitis
 - E TB
- 7 A 56-year-old female is seen in the respiratory clinic with a 1-year history of breathlessness. Physical examination shows tachypnoea, widespread crackles and finger clubbing. Spirometry shows a low vital capacity and total lung capacity associated with a relatively well-preserved peak flow rate. Arterial blood gases reveal hypoxaemia. What is the most likely diagnosis?
 - A Emphysema
 - B Bronchial asthma
 - C Pulmonary fibrosis
 - D Bilateral pleural effusions
 - E Bronchitis

REPEATED 2x • ALSO ASKED IN
PasTest

- 8 A 65-year-old retired rock miner sees his GP with worsening breathlessness, decreased exercise tolerance, dry cough and weight loss. He reports no past medical history. On examination he appears slightly tachypnoeic at rest. Fine inspiratory crackles are heard in the upper zones. Subsequent chest radiograph reveals multiple round opacities in the upper lobes with calcified hilar lymphadenopathy. What is the most likely diagnosis?
 - A Asbestosis
 - B Silicosis
 - C Cystic fibrosis
 - D Aspergillosis
 - E TB

REPEATED 2x • ALSO ASKED IN
PasTest

- 9 A 60-year-old nonsmoker presents to his GP with worsening breathlessness for several months, mainly on exertion. He has also developed a nonproductive cough and nonpainful deformities of his fingernails. He worked in the building industry for 30 years before his retirement this year. On examination, there are bibasal fine end-inspiratory crackles and evidence of finger clubbing. A subsequent plain chest radiograph shows evidence of lower lobe fibrotic changes. What is the most likely cause in this patient's condition?

- A Smoking
- B Silicosis
- C Asbestosis
- D Chronic bronchitis
- E Hypersensitivity pneumonitis

REPEATED 2x • ALSO ASKED IN
PasTest

Answers 1-9

- 1 **C Occupation** — Symptoms 4–8 h after mouldy hay in a barn are acute farmer's lung, so his occupation is the cause.
- 2 **E Eyes** — **UPDATED** Eye involvement (uveitis) is the commonest clinical extrapulmonary site in sarcoidosis (25–50%); screen every patient.
- 3 **E Lymph node biopsy** — Sarcoidosis is confirmed by non-caseating granulomas, usually from EBUS-guided lymph node sampling; ACE is non-specific.
- 4 **C Turning germinating barley** — **UPDATED** Malt worker's lung is EAA from *Aspergillus clavatus* released when turning germinating barley.
- 5 **A Pleural plaques** — **UPDATED** Pleural plaques are benign markers of exposure and not compensable; asbestosis, diffuse thickening, mesothelioma and lung cancer are.
- 6 **D Radiation pneumonitis** — **UPDATED** Opacities with a sharp straight edge following the radiation field 6 weeks after mediastinal radiotherapy are radiation pneumonitis.
- 7 **C Pulmonary fibrosis** — Low VC and TLC with preserved peak flow, clubbing and crackles is a restrictive defect from pulmonary fibrosis.
- 8 **B Silicosis** — Rock mining with upper-zone nodules and egg-shell-calcified hilar nodes is silicosis; screen for TB.
- 9 **C Asbestosis** — Decades in building work with lower-lobe fibrosis, crackles and clubbing is asbestosis.

1 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 34-year-old Black woman has cough, dyspnoea and fatigue. Examination shows hepatomegaly and a palpable spleen tip, with no palpable peripheral lymph nodes. Her chest X-ray is shown. What is the best approach to establishing the diagnosis?



- A Open lung biopsy
- B Liver biopsy
- C Bronchoscopy with EBUS-guided needle aspiration of hilar nodes and transbronchial lung biopsy
- D Mediastinoscopy
- E Serum angiotensin-converting enzyme (ACE) level

Answer 1

PRETEST MCQs

- 1 **C** Bronchoscopy with EBUS-guided needle aspiration of hilar nodes and transbronchial lung biopsy — UPDATED BHL with hepatosplenomegaly in a young Black woman suggests

sarcoidosis; EBUS-TBNA plus transbronchial biopsy confirms granulomas.

10 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1** A 66-year-old man presents to the respiratory clinic with worsening shortness of breath. He is otherwise well and is a retired dockyard worker. He is a lifelong non-smoker. On examination he has fine crackles at both lung bases. A chest X-ray shows fine reticulonodular shadowing at both lung bases and several pleural plaques. He undergoes pulmonary function testing. Which of these pulmonary function tests fits with the likely diagnosis?

 - A Transfer factor for CO (TLCO) 45% of predicted
 - B Total lung capacity (TLC) 120% of predicted
 - C A concave flow-volume loop
 - D FEV₁/FVC ratio 70%
 - E Transfer coefficient (KCO) 110% of predicted
- 2** A 67-year-old man has had progressively worsening exertional dyspnoea and a dry cough for 9 months. He stopped smoking 20 cigarettes/day 30 years ago. His oxygen saturation is 97% on air and respiratory rate 14/min; there are fine bibasal crackles and finger clubbing. B-type natriuretic peptide is 88 pg/mL (normal <100); ECG shows sinus rhythm at 72/min. Spirometry: FEV₁ 1.57 L (50% predicted), FVC 1.63 L (39% predicted), FEV₁/FVC 96%. Which investigation is most likely to confirm the diagnosis?

 - A Acute vasodilator testing
 - B Echocardiogram
 - C Flow-volume loop
 - D High-resolution CT scan
 - E Ventilation/perfusion scan
- 3** A 62-year-old man is referred to the respiratory clinic with exertional shortness of breath. He has a past medical history of rheumatoid arthritis and recurrent urinary tract infections and takes methotrexate, folic acid and nitrofurantoin as prophylaxis against infection. He is a pigeon fancier and has worked as a roofer for 40 years. On examination, chest auscultation reveals crackles at the lung apices. The jugular venous pulse is not raised and there is no peripheral oedema. A chest radiograph demonstrates reticular shadowing of the lung apices. Given the likely diagnosis, what is the probable cause?

 - A Asbestosis
 - B Extrinsic allergic alveolitis
 - C Methotrexate treatment
 - D Nitrofurantoin
 - E Rheumatoid arthritis
- 4** A 55-year-old man is referred to the chest clinic. He first presented to his GP 6 weeks ago with breathlessness, a dry cough, fever and malaise, and several courses of antibiotics have not helped. There is no haemoptysis. He is previously fit, has never smoked and works as an accountant. There are inspiratory crackles on chest examination but otherwise the examination is normal. Blood tests show a leukocytosis and raised ESR and CRP, and a chest X-ray shows patchy bilateral opacities. Pulmonary function tests show an FEV₁/FVC ratio of 0.82 with a transfer factor of 67% predicted. What is the most likely diagnosis?

 - A Cryptogenic organising pneumonia
 - B Hypersensitivity pneumonitis
 - C Community acquired pneumonia
 - D Lung cancer
 - E Pulmonary haemorrhage
- 5** A 39-year-old woman presents to the GP with a painful rash affecting both shins. Multiple raised erythematous nodules are noted bilaterally and the patient reports they have been present for 3 weeks and appear to be worsening. Of note, she also reports a persisting cough for the preceding 6 weeks and a chest x-ray reveals hilar lymphadenopathy. Which of the following is most likely to be abnormal?

 - A C-ANCA
 - B Creatinine kinase
 - C DsDNA
 - D P-ANCA
 - E Serum ACE level
- 6** A 54-year-old man presents to the emergency department with bilateral ankle swelling and a rash on his shins. He has no past medical history. On examination, there is synovitis of the bilateral ankles and a few erythematous tender lumps on his anterior tibiae. A chest x-ray demonstrates bilateral hilar lymphadenopathy. Given the likely diagnosis, what is the most likely outcome?

 - A Progressive disability and death despite treatment
 - B Relapsing remitting course despite treatment
 - C Resolution with need for maintenance therapy
 - D Resolution with treatment but without need for maintenance therapy
 - E Spontaneous resolution

- 7** A 34-year-old Afro-Caribbean man presents to his GP with a 4-week history of fever, joint pains, lymphadenopathy and a tender red rash on his legs. Bloods show mild inflammatory markers; chest x-ray shows bilateral hilar lymphadenopathy. What is the most appropriate management?
- A Azathioprine
 - B Cyclophosphamide
 - C Methotrexate
 - D Monitoring
 - E Prednisolone
- 8** A 58-year-old man originally from India presents with progressive shortness of breath. History includes ankylosing spondylitis and atrial flutter for which he takes naproxen, amiodarone, and omeprazole. He previously worked in stone and coal mines. Examination reveals fine basal crepitations bilaterally. What is the most likely underlying cause?
- A Amiodarone
 - B Ankylosing spondylitis
 - C Coal worker's pneumoconiosis
 - D Silicosis
 - E Tuberculosis
- 9** A 68-year-old male patient presents with a 6-month history of shortness of breath worse on exertion and a dry cough. Examination shows fine crackles at both lung bases. You suspect idiopathic pulmonary fibrosis and organise investigations. Which of the following investigation findings would support a diagnosis of idiopathic pulmonary fibrosis?
- A Reticular changes on CT imaging that is worse at the bases
 - B Obstructive picture on spirometry
 - C Extensive ground glass opacities on CT imaging
 - D Increased transfer factor on spirometry
 - E A lymphocytosis on bronchoalveolar lavage
- 10** Which one of the following is responsible for farmer's lung?
- A Aspergillus clavatus
 - B Saccharopolyspora rectivirgula
 - C Candida albicans
 - D Mycobacterium avium
 - E Avian proteins

Answers 1-10

PASSMEDICINE MCQs

- 1** **A Transfer factor for CO (TLCO) 45% of predicted** — UPDATED Asbestosis is a fibrotic restriction with reduced TLCO (e.g. 45%); a raised KCO suggests extrapulmonary restriction.
- 2** **D High-resolution CT scan** — Restriction (FVC 39%, ratio 96%) with clubbing and normal BNP suggests fibrotic ILD; HRCT confirms.
- 3** **B Extrinsic allergic alveolitis** — Upper-zone fibrosis in a pigeon keeper is chronic extrinsic allergic alveolitis; asbestos and drugs cause lower-zone disease.
- 4** **A Cryptogenic organising pneumonia** — UPDATED Subacute flu-like illness with migratory opacities unresponsive to antibiotics is cryptogenic organising pneumonia; responds to steroids.
- 5** **E Serum ACE level** — Erythema nodosum with BHL suggests sarcoidosis; serum ACE is raised in ~60%.
- 6** **E Spontaneous resolution** — UPDATED Löfgren syndrome (erythema nodosum, ankle arthritis, BHL) remits spontaneously in > 90% within 1–2 years.
- 7** **D Monitoring** — Löfgren syndrome needs only monitoring and NSAIDs; steroids are for organ-threatening sarcoidosis.
- 8** **A Amiodarone** — Amiodarone causes lower-zone pneumonitis/fibrosis; silicosis, CWP, TB and ankylosing spondylitis cause upper-zone disease.
- 9** **A Reticular changes on CT imaging that is worse at the bases** — Basal-predominant subpleural reticulation on CT supports IPF; IPF reduces (not increases) TLCO.
- 10** **B Saccharopolyspora rectivirgula** — Farmer's lung is caused by Saccharopolyspora rectivirgula in mouldy hay (type III/IV hypersensitivity).

8 QUESTIONS • AFTER PASTEST

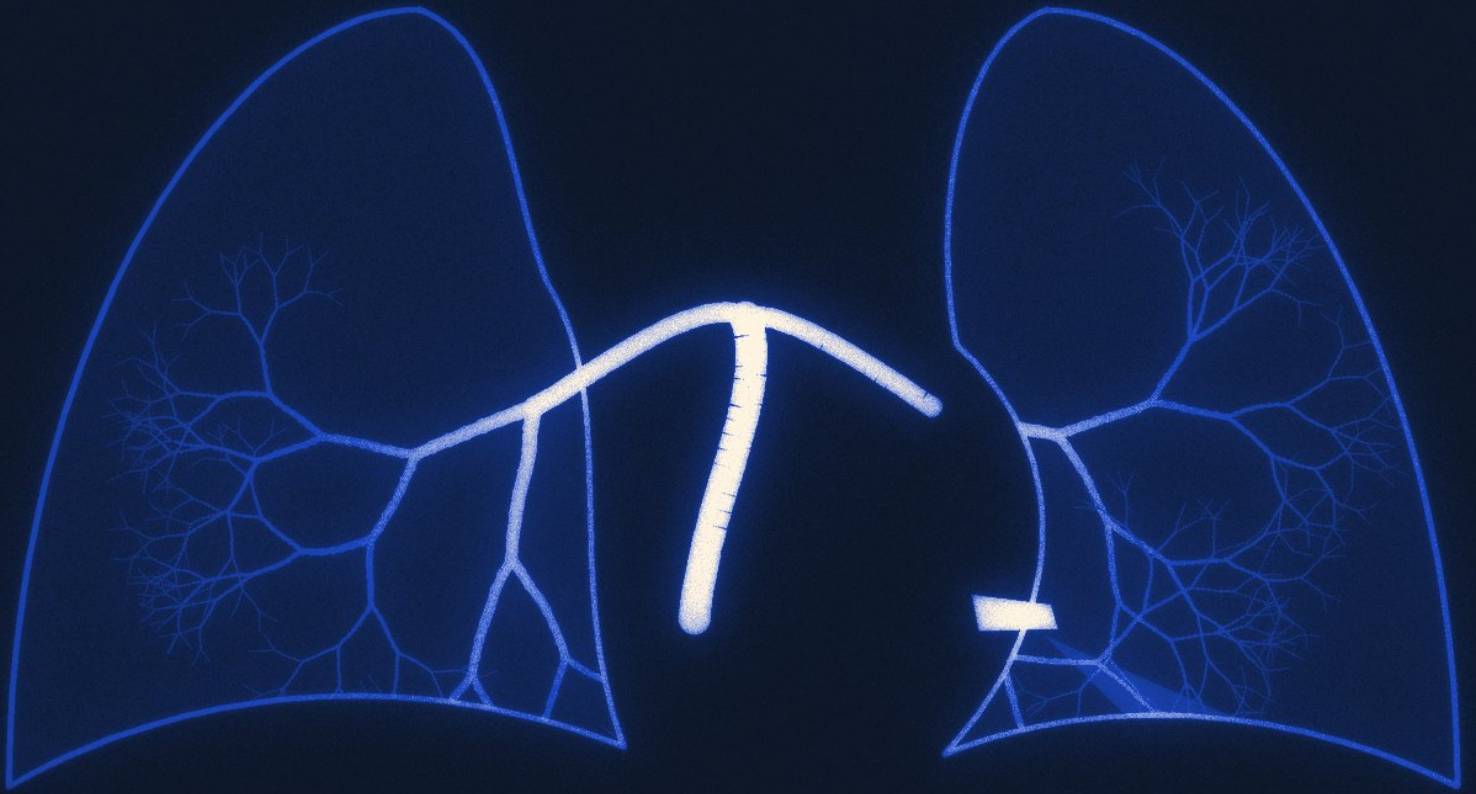
PasTest MCQs

- 1 A young adult was referred because of cough and shortness of breath. Extrinsic allergic alveolitis was diagnosed. Apart from reduction of exposure to the allergen, which other therapy is most likely to be successful?
 - A Antibiotics
 - B Corticosteroids
 - C Desensitisation
 - D Immunoglobulins
 - E Non-steroidal anti-inflammatory drugs
- 2 A 48-year-old man who has kept parrots for many years has had slowly progressive breathlessness and a dry cough for 2 years. There are fine crackles in the upper and mid zones. You suspect chronic extrinsic allergic alveolitis (hypersensitivity pneumonitis) due to parrot antigen. Which finding would be most consistent with this diagnosis?
 - A Bilateral upper lobe fibrosis
 - B Chlamydia antibody positivity
 - C Elevated IgE
 - D Obstructive picture on lung function
- 3 A 60-year-old man with a 30 pack-year smoking history has had worsening shortness of breath for 6 months. He works as a baker and keeps racing pigeons. He is clubbed, his saturation is 91% on air and there are widespread fine inspiratory crepitations. The chest X-ray shows reticulonodular shadowing, and CT shows mainly subpleural, basal-predominant reticulation with some honeycombing, without mosaic attenuation or air trapping. Avian precipitins are negative. What is the diagnosis?
 - A Hypersensitivity pneumonitis
 - B Langerhans cell histiocytosis
 - C Occupational asthma
 - D Pulmonary sarcoidosis
 - E Usual interstitial pneumonitis
- 4 A 30-year-old woman who has recently moved from Nigeria to the United Kingdom sees her GP with breathlessness and a dry cough. She has well-controlled type 2 diabetes, does not smoke or use illicit drugs, and has no exposure to pets or birds. Slit-lamp examination confirms left anterior uveitis. Serum ACE is normal and ANCA is negative. Her chest X-ray shows bilateral hilar lymphadenopathy. What is the most likely diagnosis?
 - A Histoplasmosis
 - B Hypersensitivity pneumonitis
 - C Non-Hodgkin lymphoma
 - D Pulmonary tuberculosis
 - E Sarcoidosis
- 5 A 55-year-old woman with systemic sclerosis has been referred with breathlessness and a dry cough. On examination, fine crackles are heard at the lung bases. What is the most likely cause of her symptoms?
 - A Chest infection
 - B Interstitial lung disease
 - C Late asthma
 - D Pulmonary embolism
 - E Tuberculosis
- 6 A 67-year-old retired boilermaker is referred to the respiratory clinic with shortness of breath, a dry cough and weight loss. Examination shows inspiratory crackles in both mid zones and finger clubbing. Which single investigation would be most useful to confirm the likely diagnosis?
 - A Bronchoalveolar lavage
 - B Carbon monoxide transfer factor
 - C High-resolution computed tomography
 - D Serum angiotensin-converting enzyme
 - E Transbronchial biopsy
- 7 A 70-year-old woman with rheumatoid arthritis started methotrexate 2 months ago. Over the last 3 weeks she has developed progressive breathlessness, a dry cough and a low-grade fever. There are inspiratory crackles throughout both lung fields and her oxygen saturation is 89% on air. Haemoglobin is mildly reduced and serum creatinine is raised; the chest X-ray shows new bilateral diffuse interstitial infiltrates. What is the most likely diagnosis?
 - A Rheumatoid interstitial lung disease (pulmonary fibrosis)
 - B Community-acquired pneumonia
 - C Methotrexate pneumonitis
 - D Pulmonary embolus
 - E Congestive heart failure
- 8 A 65-year-old man has proximal muscle weakness and breathlessness. Investigations show a creatine kinase of 890 U/L and an ESR of 75 mm/hour. Which of the following is the most likely underlying cause of his interstitial lung disease?
 - A Ankylosing spondylitis
 - B Idiopathic pulmonary fibrosis
 - C Polymyositis
 - D Rheumatoid arthritis
 - E Systemic lupus erythematosus

Answers 1-8

PASTEST MCQS

- 1 **B Corticosteroids** — UPDATED EAA: avoid the antigen; corticosteroids speed recovery in severe or progressive disease. It is immune-mediated, not infective.
- 2 **A Bilateral upper lobe fibrosis** — UPDATED Chronic hypersensitivity pneumonitis causes upper/mid-zone fibrosis with restriction and low TLCO; avian precipitins support it.
- 3 **E Usual interstitial pneumonitis** — UPDATED Clubbing, crackles and basal subpleural honeycombing with negative precipitins is UIP/IPF; HP is upper/mid-zone with air trapping.
- 4 **E Sarcoidosis** — Uveitis with BHL in a young West African woman is sarcoidosis; a normal ACE does not exclude it.
- 5 **B Interstitial lung disease** — UPDATED ILD (usually NSIP) affects most patients with systemic sclerosis, especially anti-Scl-70, and is the leading cause of death; PAH is the other lung complication.
- 6 **C High-resolution computed tomography** — In a retired boilermaker with clubbing and crackles, HRCT best confirms asbestosis (basal reticulation with pleural plaques) and screens for cancer.
- 7 **C Methotrexate pneumonitis** — UPDATED Fever, dry cough, hypoxaemia and new diffuse infiltrates within months of methotrexate is methotrexate pneumonitis: stop it, exclude infection, give steroids.
- 8 **C Polymyositis** — Proximal weakness with high CK and ESR is polymyositis, complicated by ILD in up to 40% (anti-Jo-1); screen for malignancy.



16

PULMONARY VASCULAR DISEASE

Pulmonary Vascular Disease

Pulmonary embolism — probability, imaging and treatment — and pulmonary hypertension.

14

MCQs

0

EMQ SCENARIOS

6

SOURCES

3

REPEATED IDEAS

IN THIS CHAPTER

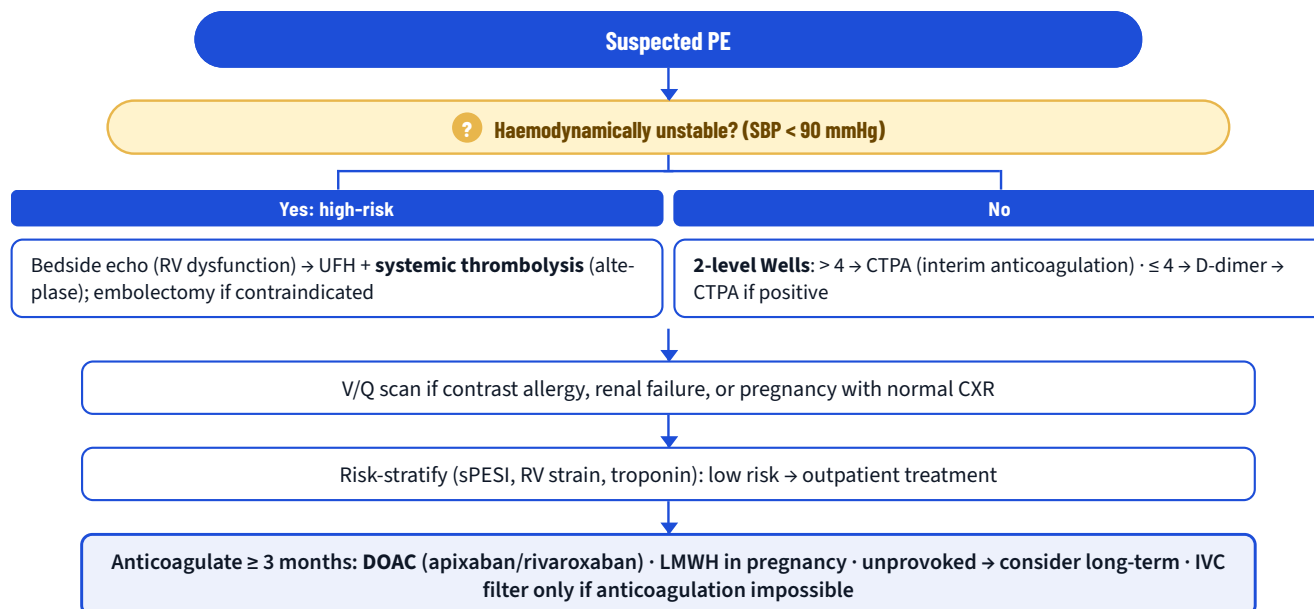
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16

Step by step

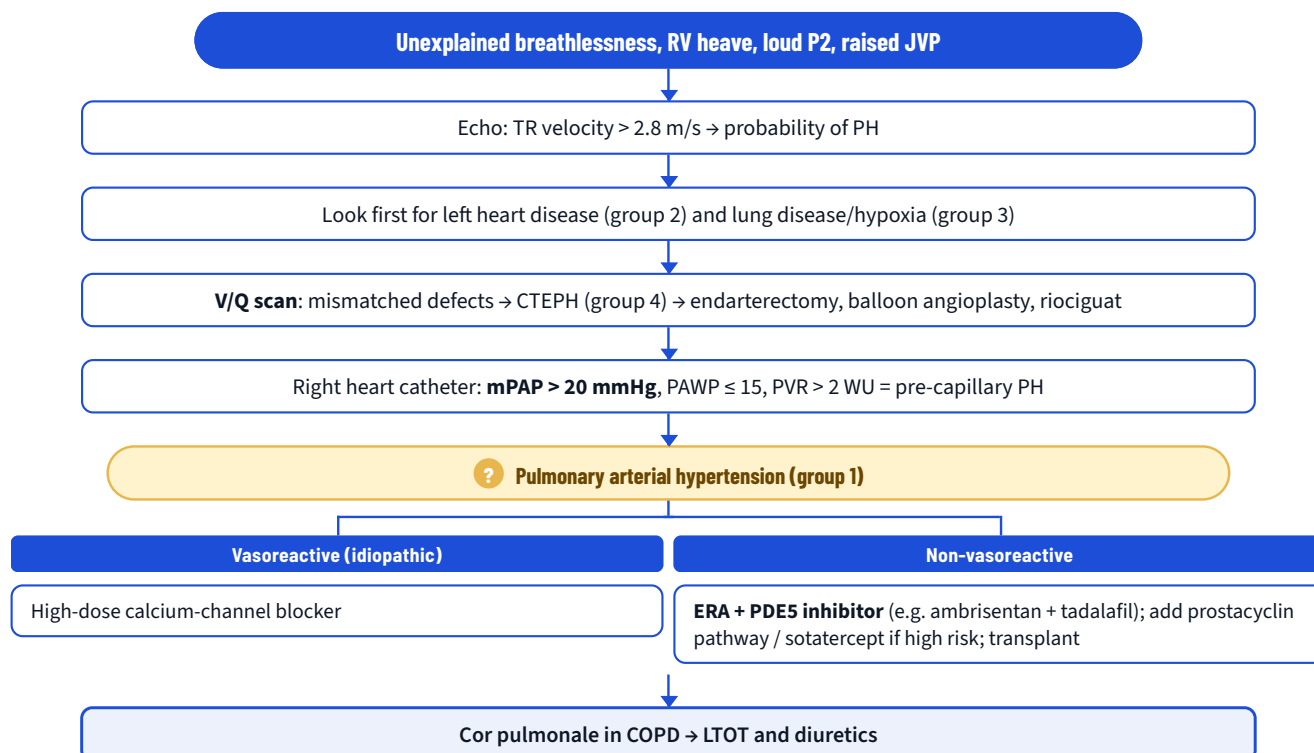
Suspected pulmonary embolism

NICE NG158 • ESC 2019 PE



Pulmonary hypertension

ESC/ERS 2022 PH



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 34-year-old woman on the combined oral contraceptive pill has sudden breathlessness and right-sided pleuritic chest pain for 2 hours, without cough or fever. HR 122, RR 30, BP 118/74, SpO₂ 89% on air, T 37.1°C. The lungs are clear; there is mild left calf tenderness without swelling. ECG shows sinus tachycardia and the chest X-ray is normal. What is the most appropriate next diagnostic step?

- A D-dimer
- B CT pulmonary angiography
- C Lower-limb Doppler ultrasound only
- D Non-contrast CT chest
- E Bronchodilator therapy and reassess

REPEATED 5* • ALSO ASKED IN
Log book 2025 ×2 • Davidson • Crash Course

- 2** A woman who is 7 months pregnant develops sudden breathlessness at rest, haemoptysis and cyanosis with a raised JVP. The ECG shows sinus tachycardia with partial right bundle branch block. What is the most likely diagnosis?

- A Pneumothorax
- B Pulmonary hypertension
- C Pulmonary embolism
- D Pulmonary oedema

REPEATED 3* • ALSO ASKED IN
Previous-years exam ×2

- 3** In acute massive pulmonary embolism all of the following are true EXCEPT:

- A Pleuritic chest pain and haemoptysis
- B Central cyanosis
- C Hypoxia and hypocapnia
- D Raised JVP

LOG BOOK 2025

Log-book questions

- 4** In a patient with COPD, the presence of distended neck veins and bilateral leg oedema is most suggestive of:

- A Left-sided heart failure
- B Cor pulmonale
- C Constrictive pericarditis
- D Renal failure
- E Superior vena cava obstruction

- 5** On precordial examination in a patient with cor pulmonale, which finding is most likely?

- A Displaced apical impulse
- B Parasternal heave
- C Loud S3 gallop
- D Friction rub
- E Continuous murmur

Answers 1–5

- 1** **B CT pulmonary angiography** — Two-level Wells ≥ 4.5 (PE likely): go straight to CTPA with interim anticoagulation; D-dimer only rules out unlikely PE.
- 2** **C Pulmonary embolism** — Sudden breathlessness, haemoptysis, cyanosis and raised JVP in pregnancy is PE with RV strain; treat with LMWH (no DOAC/warfarin).
- 3** **A Pleuritic chest pain and haemoptysis** — **UPDATED** Massive PE causes collapse, hypoxaemia and RV failure; pleuritic pain

and haemoptysis come from infarction by smaller distal emboli.

- 4** **B Cor pulmonale** — Raised JVP and leg oedema in COPD are cor pulmonale (hypoxic pulmonary hypertension); LTOT improves survival.
- 5** **B Parasternal heave** — **UPDATED** RV hypertrophy in cor pulmonale gives a left parasternal heave, loud P2 and functional TR.

3 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

Davidson MCQs

- 1 A 46 year old man has recently fractured his leg, which is in a plaster cast. He suddenly becomes very breathless, unwell and collapses. The attending doctor suspects a pulmonary embolus from a deep vein thrombosis. The doctor performs an ECG. What is the most common ECG change in patients with pulmonary embolism?
 - A Anterior T-wave inversion
 - B Atrial fibrillation
 - C 'S1Q3T3'
 - D Sinus tachycardia
 - E ST elevation
- 2 A 26-year-old woman with idiopathic (primary) pulmonary arterial hypertension is evaluated for therapy. Which of the following therapies may be indicated?
 - A Bosentan
 - B Cyclizine
 - C Etanercept
 - D Infliximab
 - E Isosorbide mononitrate
- 3 A 24-year-old woman presents with worsening breathlessness and palpitations. Examination shows an elevated JVP, and the chest X-ray shows prominent proximal pulmonary arteries with a paucity of peripheral vascular markings. What is the most appropriate next investigation?
 - A D-dimer
 - B Dysfunctional breathing studies and Nijmegen questionnaire
 - C HRCT
 - D Spirometry
 - E Transthoracic echocardiography

Answers 1-3

DAVIDSON MCQS

- 1 **D Sinus tachycardia** — The commonest ECG finding in PE is sinus tachycardia; S1Q3T3 and RBBB are uncommon.
- 2 **A Bosentan** — Bosentan (endothelin receptor antagonist) treats PAH; others are PDE-5 inhibitors, riociguat and prostacyclin agents. Monitor LFTs.
- 3 **E Transthoracic echocardiography** — Raised JVP with pruned peripheral vessels suggests pulmonary hypertension; transthoracic echo is the first test, then RHC.

2 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 Which of the following sets of pulmonary function test results is most consistent with heritable (familial) pulmonary arterial hypertension?
 - A Increased TLC, decreased vital capacity, decreased FEV₁/FVC ratio
 - B Decreased TLC, decreased VC, decreased RV, increased FEV₁/FVC ratio, normal MIP
 - C Decreased TLC, increased RV, normal FEV₁/FVC ratio, decreased MIP
 - D Normal TLC, normal RV, normal FEV₁/FVC ratio, normal MIP
 - E None of the above
- 2 A 47-year-old woman with idiopathic pulmonary arterial hypertension and right heart failure despite maximal medical therapy is referred for transplantation. Which of the following statements is true?
 - A She will require heart-lung transplantation
 - B Patients with idiopathic PAH have worse 5-year survival after transplantation than all other recipient groups
 - C Single-lung transplantation is the preferred procedure
 - D Her right ventricular function will recover after lung transplantation
 - E She is at high risk of recurrent PAH in the transplanted lungs

Answers 1-2

HARRISON MCQS

- 1 **D** Normal TLC, normal RV, normal FEV₁/FVC ratio, normal MIP — **UPDATED** PAH is vascular, so volumes, spirometry and muscle strength are normal; the typical abnormality is an isolated low DLCO.
- 2 **D** Her right ventricular function will recover after lung transplantation — RV dysfunction in PAH recovers after bilateral lung transplantation as afterload falls; heart–lung only for complex disease.

2 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 40-year-old woman has had increasing fatigue and breathlessness for several years. Chest X-ray shows right ventricular enlargement and enlarged central pulmonary arteries. Pulmonary embolism has been excluded by CT pulmonary angiography, and other causes of pulmonary hypertension have been ruled out. Right heart catheterisation shows a pulmonary artery pressure of 75/30 mmHg. What is the best next step in management?
 - A Acute drug testing with short-acting pulmonary vasodilators
 - B High-dose nifedipine
 - C Intravenous prostacyclin
 - D Lung transplantation
 - E Empiric trial of sildenafil
- 2 A 35-year-old woman complains of slowly progressive breathlessness. She has no cough, sputum, pleuritic chest pain or history of thrombophlebitis, but has taken appetite suppressants at various times. Examination reveals a raised JVP, a palpable right ventricular heave and a loud P2. The chest X-ray shows clear lung fields, oxygen saturation is 94% and the ECG shows right axis deviation. A perfusion lung scan is normal, with no segmental defects. What is the most likely diagnosis?
 - A Pulmonary arterial hypertension
 - B Recurrent pulmonary emboli
 - C Right-to-left cardiac shunt
 - D Interstitial lung disease
 - E Left ventricular diastolic dysfunction

REPEATED 2x • ALSO ASKED IN
PasTest

Answers 1-2

PRETEST MCQS

- 1 **A** Acute drug testing with short-acting pulmonary vasodilators — In idiopathic PAH, perform acute vasoreactivity testing at RHC; the ~10% responders get high-dose CCBs, others combination PAH therapy.
- 2 **A** Pulmonary arterial hypertension — **UPDATED** Pulmonary hypertension with clear lungs, normal perfusion scan and prior appetite suppressants is drug-induced PAH; confirm by RHC.

1 QUESTIONS • AFTER PASTEST

PasTest MCQs

- 1 A 50-year-old woman presents with breathlessness that has increased over several months. She has peripheral oedema and clinical evidence of right-sided heart failure without signs of left ventricular failure. She is a non-smoker with no chronic cough or sputum, and her chest X-ray lung fields and spirometry are normal. What is the most likely cause of her cor pulmonale?
- A Asthma
 - B Bronchiectasis
 - C Carcinoma of the lung
 - D Recurrent pneumonias
 - E Recurrent small pulmonary emboli

Answer 1

PASTEST MCQs

- 1 **E Recurrent small pulmonary emboli** — UPDATED Cor pulmonale with normal lung fields and spirometry suggests

CTEPH from recurrent small emboli; V/Q scan, endarterectomy may cure.

1 QUESTIONS • OTHER COLLEGES & SOURCES

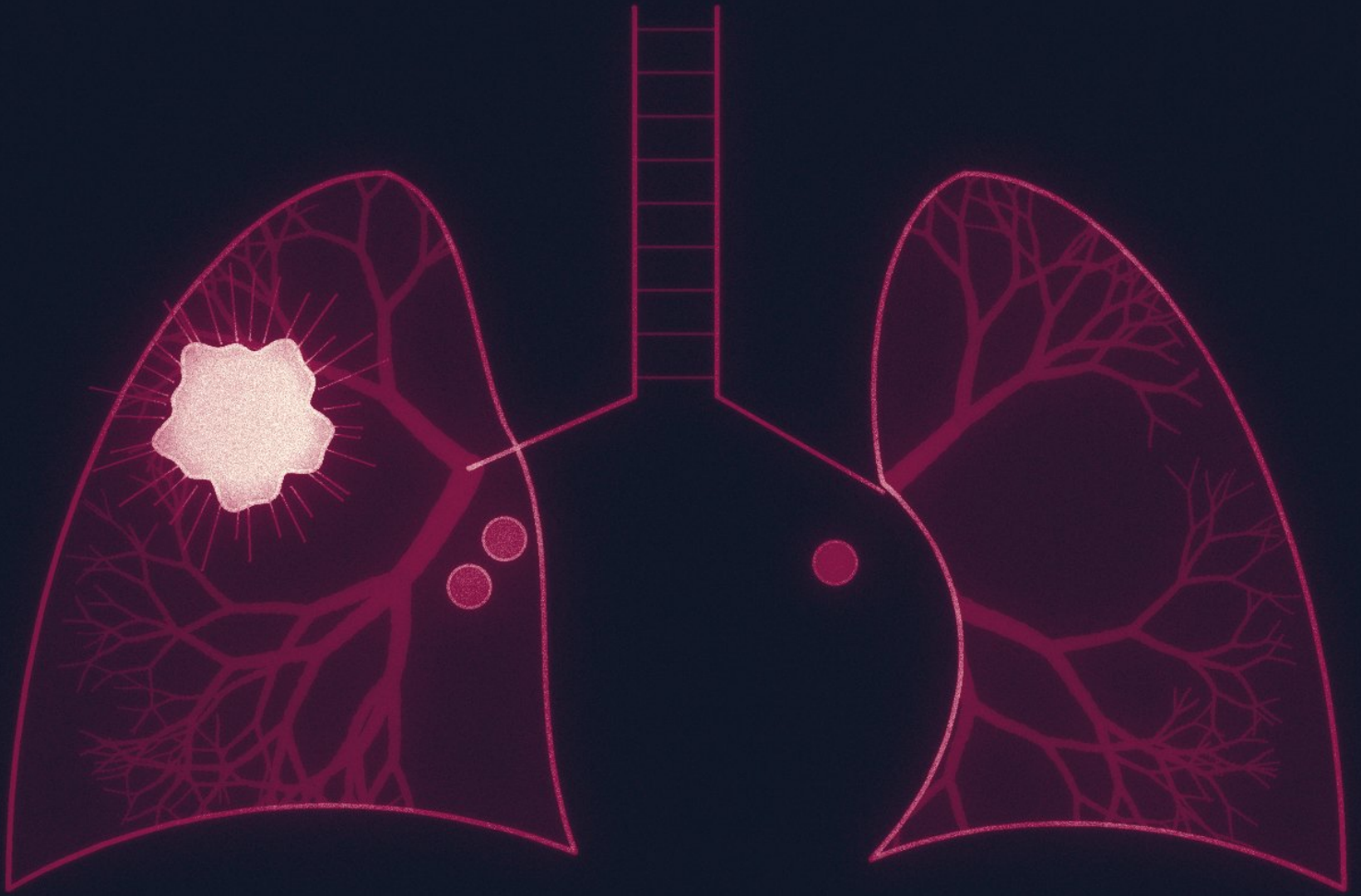
Other Sources MCQs

- 1 A 66-year-old woman presents to the emergency department with a 2-day history of progressive shortness of breath and sharp right-sided chest pain that is worse on deep inspiration. She also has mild pain in the right leg. Cardiovascular and respiratory examinations are unremarkable; heart rate is 96/min and respiratory rate 12/min. She denies weight loss or long-haul flights but had a nasal polypectomy 3 weeks ago. What is the most likely diagnosis?
- A Muscular strain
 - B Heart failure
 - C Pneumothorax
 - D Angina
 - E Pulmonary embolism

Answer 1

OTHER SOURCES MCQs

- 1 **E Pulmonary embolism** — UPDATED Pleuritic pain and breathlessness with calf pain 3 weeks after surgery is PE: Wells score, CTPA and anticoagulate meanwhile.



17

LUNG CANCER

Lung Cancer

Presentation, paraneoplastic syndromes, diagnosis, staging and treatment of lung cancer and other thoracic tumours.

53

MCQS

0

EMQ SCENARIOS

7

SOURCES

3

REPEATED IDEAS

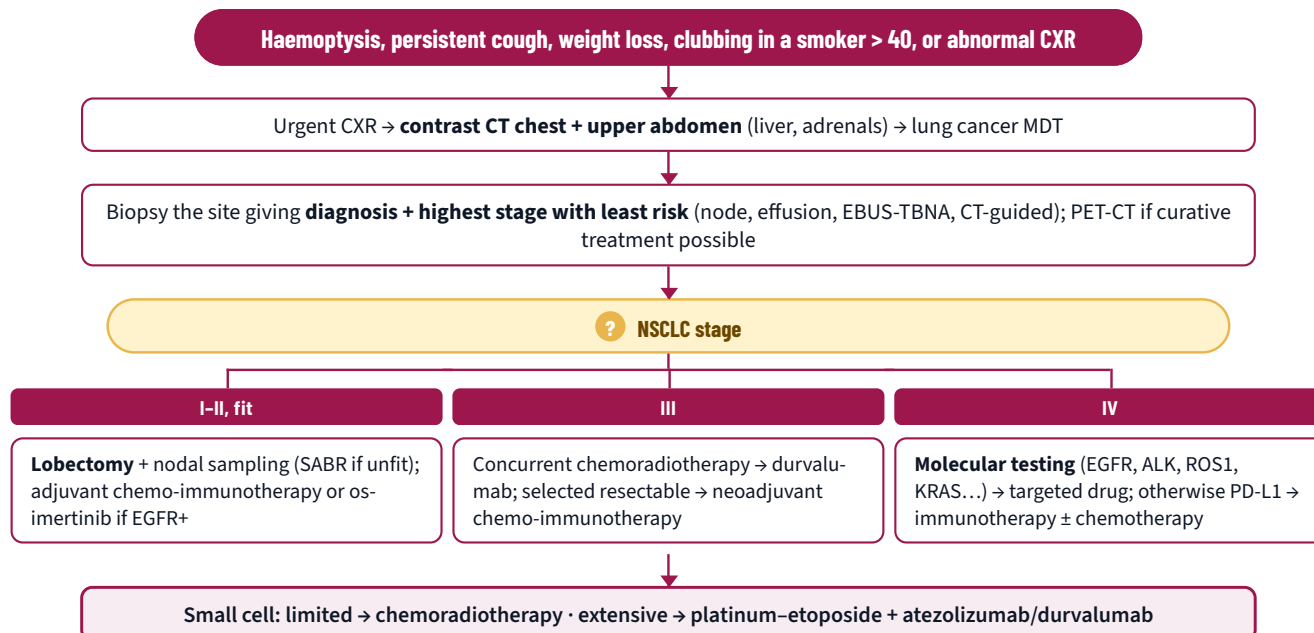
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01	Management flowcharts	3 charts	page 276
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Step by step

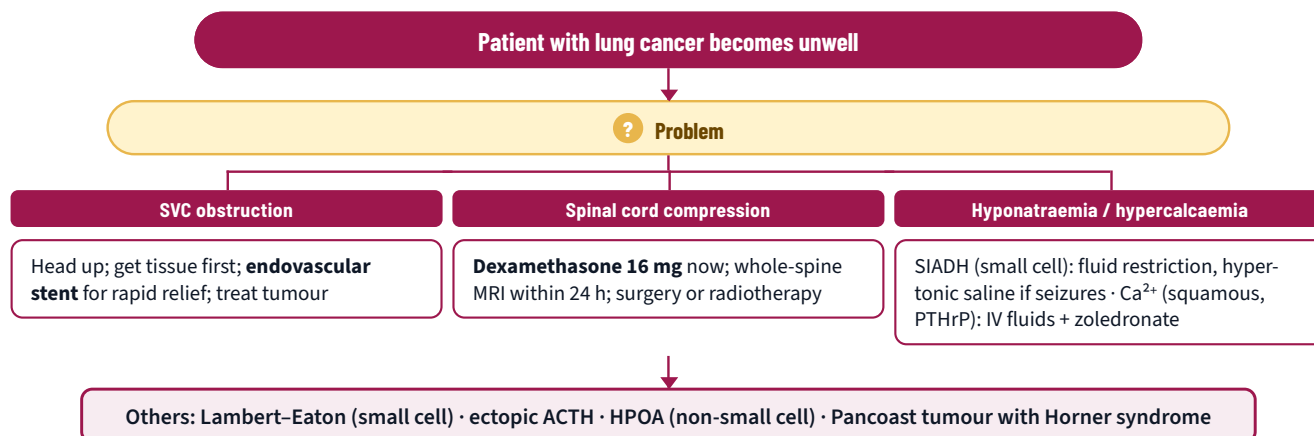
Suspected lung cancer

NICE NG122 (2024)



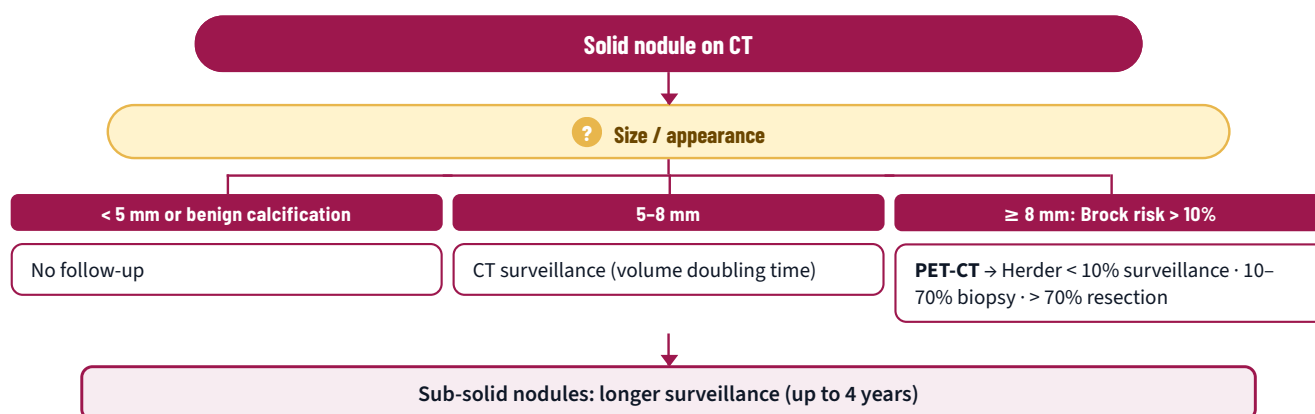
Lung cancer emergencies and syndromes

NICE NG234 · NICE NG122



Incidental pulmonary nodule

BTS PULMONARY NODULES



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1** A 70-year-old man has cough, haemoptysis and breathlessness. He has finger clubbing and localised wheeze over the right lower lobe; CT shows a rounded mass in the right lower zone. What is the most appropriate next step?

A Sputum for acid-fast bacilli
B Bronchoscopy and biopsy
C Peak expiratory flow rate
D CT pulmonary angiography

- 2** A 78-year-old man has recurrent blood-streaked sputum with cough and breathlessness. He is anaemic with finger clubbing, and a localised wheeze is heard over the right middle zone. Six weeks later he is more breathless; there is stony dullness from the right base up to the mid zone, with absent breath sounds and reduced vocal resonance over this area, and the trachea is central. What is the most likely cause?

A Pleural effusion
B Lung collapse
C Acute lobar pneumonia
D Lung fibrosis
E Emphysema

- 3** A 73-year-old heavy smoker with bronchial carcinoma develops nausea, vomiting, irritability and a seizure. Hb 9.8 g/dL; serum sodium 104 mmol/L (135–145). Which complication has developed?

A Syndrome of inappropriate ADH secretion
B Horner's syndrome
C Encephalopathy
D Cerebellar degeneration
E Carcinoid syndrome

REPEATED 2* • ALSO ASKED IN
Formative exam

- 4** Each of the following is a risk factor for lung cancer EXCEPT:

A Asbestos
B Radon
C Passive smoking
D Lead poisoning

- 5** A breathless patient with lung cancer has bilateral hilar enlargement and a pleural effusion on chest X-ray. What is the next step?

A CT of chest, abdomen and head for staging
B Pulmonary function tests
C Pleural fluid aspiration for cytology
D Palliative radiotherapy because the patient is not a candidate for curative resection

- 6** Which of the following is a cutaneous manifestation of bronchogenic carcinoma?

A Dermatomyositis
B Dermatitis herpetiformis
C Psoriasis
D Cushing's syndrome

CASE • QUESTIONS 7–8

A 70-year-old heavy smoker has haemoptysis, finger clubbing and has lost 2 stones (13 kg) in weight over the last month. There is reduced air entry and wheeze in the right middle zone, and the chest X-ray shows a homogeneous opacity in the middle of the right lung.

- 7** What is the most likely diagnosis?

A Bronchogenic carcinoma
B Tuberculosis
C Lobar pneumonia
D Idiopathic pulmonary fibrosis
E Bronchiectasis

REPEATED 4* • ALSO ASKED IN

39th batch end-block (respiratory) • Crash Course • PasTest

- 8** Some weeks later he develops a plethoric (moon) face, weight gain and striae. What is the likely cause?

A Gynaecomastia
B Cushing's syndrome
C Obesity
D Hypothyroidism
E SIADH

- 9** A smoker with a lung mass develops Cushingoid features. Which test best distinguishes pituitary ACTH (Cushing disease) from ectopic ACTH secretion?

A Low-dose dexamethasone suppression test
B Serum cortisol
C High-dose dexamethasone suppression test
D Short Synacthen test

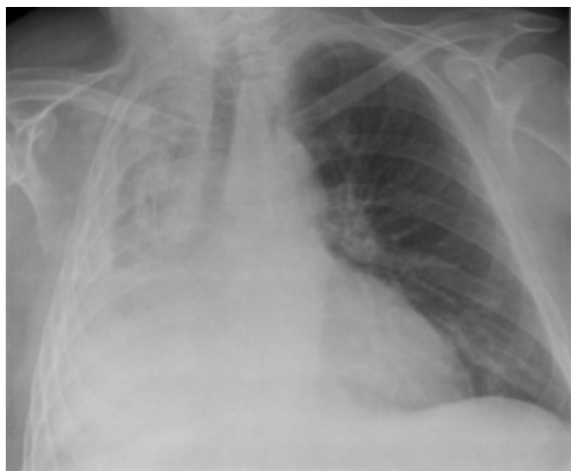
- 10** A 70-year-old heavy smoker has haemoptysis, finger clubbing and has lost 2 stones (13 kg) in weight over the last month. There is reduced air entry and wheeze in the right middle zone, and the chest X-ray shows a homogeneous opacity in the middle of the right lung. Later he has fatigue and concentrated urine; serum sodium 123 mmol/L and potassium 4.0 mmol/L. What is the likely diagnosis?

A Diabetes insipidus
B SIADH
C Diuretic effect
D Adrenal insufficiency

REPEATED 2* • ALSO ASKED IN

39th batch end-block (respiratory)

- 11 A man with a central bronchial carcinoma develops increasing breathlessness, with reduced breath sounds and dullness to percussion over the right chest. His chest X-ray is shown. What is the diagnosis?



- A Lung collapse due to bronchial obstruction
- B Secondary pneumonia
- C Pleural effusion
- D Empyema

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

CASE · QUESTIONS 12–15

An elderly man who is a heavy smoker has weight loss, haemoptysis, fatigue, constipation and thirst; he has diabetes mellitus. Chest X-ray shows a right hilar mass, and PET-CT shows no distant metastasis.

- 12 What is the most likely type of lung cancer?
- A Adenocarcinoma
 - B Small cell carcinoma
 - C Large cell carcinoma
 - D Carcinoid tumour
 - E Squamous cell carcinoma

- 13 What is the most likely cause of his hypercalcaemia?
- A Bone metastases
 - B Paraneoplastic secretion of PTH-related peptide
 - C Primary hyperparathyroidism
 - D Vitamin D toxicity
 - E Milk-alkali syndrome

- 14 Lung function tests show good pulmonary reserve. What is the most appropriate management?
- A Chemotherapy
 - B Radiotherapy
 - C Surgical resection
 - D Immunotherapy
 - E Best supportive care

- 15 Lung function tests show good pulmonary reserve. One year later he develops contralateral chest pain; a pleural effusion is found and aspiration confirms malignant cells. What is the best treatment to prevent recurrence of the effusion?
- A Pleurodesis
 - B Systemic chemotherapy
 - C Repeated therapeutic aspiration
 - D Diuretics
 - E Radiotherapy to the chest wall

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 16 Which of the following is the strongest determinant of prognosis in lung cancer?
- A Age
 - B Sex
 - C Stage of disease at diagnosis
 - D Tumour laterality (left or right lung)
 - E Number of pack-years smoked
- 17 Causes of dyspnoea in bronchial carcinoma may be the following EXCEPT:
- A Phrenic nerve palsy
 - B Bronchial obstruction
 - C Pleural effusion
 - D Recurrent laryngeal nerve paralysis

Answers 1–17

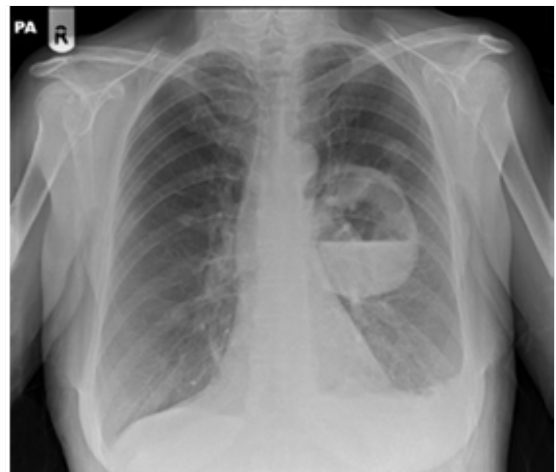
- 1 **B Bronchoscopy and biopsy** — Haemoptysis, clubbing and localised wheeze with a mass in a 70-year-old is bronchial carcinoma: bronchoscopy and biopsy ($\pm$ EBUS).
- 2 **A Pleural effusion** — **UPDATED** Stony dullness with absent breath sounds and a central trachea is a pleural effusion, likely malignant; aspirate for cytology.
- 3 **A Syndrome of inappropriate ADH secretion** — Na^+ 104 with seizure in lung cancer is SIADH (small cell); give hypertonic saline boluses, raising $\text{Na}^+ \leq 8\text{--}10$ mmol/L in 24 h.
- 4 **D Lead poisoning** — Lung cancer risks: smoking, radon, asbestos, metals (chromium, nickel, arsenic), pollution, radiotherapy. Lead is not a carcinogen here.
- 5 **C Pleural fluid aspiration for cytology** — Effusion in suspected lung cancer: ultrasound-guided aspiration for cytology diagnoses and stages (M1a) and relieves breathlessness.
- 6 **A Dermatomyositis** — Dermatomyositis is a paraneoplastic skin manifestation of lung cancer; screen adults presenting with it.
- 7 **A Bronchogenic carcinoma** — Haemoptysis, weight loss, clubbing and localised wheeze with an opacity in a heavy smoker is bronchogenic carcinoma; urgent CT and tissue.
- 8 **B Cushing's syndrome** — Cushingoid features in a smoker with lung cancer are ectopic ACTH (small cell or carcinoid); often hypokalaemic alkalosis dominates.

- 9 **C High-dose dexamethasone suppression test** — High-dose dexamethasone suppresses pituitary (Cushing disease) but not usually ectopic ACTH; petrosal sinus sampling is more reliable.
- 10 **B SIADH** — Hyponatraemia with normal K^+ and concentrated urine in lung cancer is SIADH; fluid restriction, treat tumour, tolvaptan if refractory.
- 11 **A Lung collapse due to bronchial obstruction** — A central tumour occluding a bronchus causes collapse: opacification with volume loss and the mediastinum pulled towards it.
- 12 **E Squamous cell carcinoma** — A central (hilar) tumour in a smoker with hypercalcaemia (thirst, constipation) is squamous cell carcinoma, which secretes PTH-related peptide. Small cell causes SIADH and ectopic ACTH instead.
- 13 **B Paraneoplastic secretion of PTH-related peptide** — With no metastases on PET, hypercalcaemia is humoral: PTHrP from the squamous tumour, with a suppressed PTH. Treat with IV saline then a bisphosphonate.
- 14 **C Surgical resection** — Non-metastatic non-small cell cancer in a patient fit for surgery (adequate FEV_1 /TLCO) is resected, usually by lobectomy with nodal sampling.
- 15 **A Pleurodesis** — A symptomatic recurrent malignant effusion is drained and then controlled by talc pleurodesis (or an in-dwelling pleural catheter if the lung is trapped). Repeated aspiration only buys days.
- 16 **C Stage of disease at diagnosis** — **UPDATED** TNM stage at diagnosis is the strongest prognostic factor (stage IA ~70–90% vs stage IV < 10% 5-year survival).
- 17 **D Recurrent laryngeal nerve paralysis** — **UPDATED** Recurrent laryngeal nerve palsy causes hoarseness and bovine cough, not breathlessness.

20 QUESTIONS • AFTER DAVIDSON'S PRINCIPLES & PRACTICE OF MEDICINE SELF-ASSESSMENT

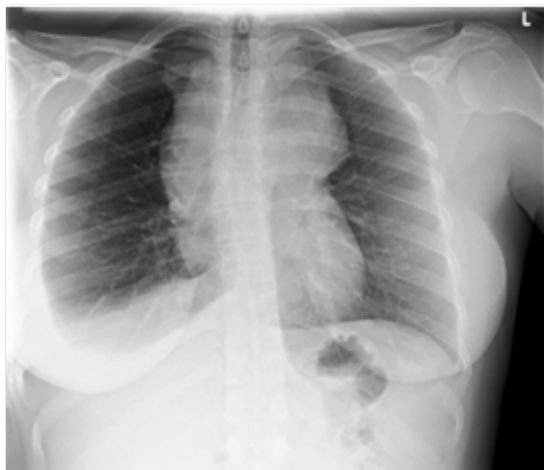
Davidson MCQs

- 1 A 46-year-old woman was diagnosed with adenocarcinoma of the lung at bronchoscopy 1 week ago. She presents to the emergency department acutely short of breath with a non-productive cough and a central chest ache that is worse on inspiration. She is afebrile; SpO_2 91% on 40% oxygen, respiratory rate 30/min, BP 100/65 mmHg, pulse 110/min. Expansion is reduced on the right, with dullness to percussion throughout the right side and reduced breath sounds; the trachea is deviated to the right and the apex beat is not palpable. What is the most likely diagnosis?
- A Collapse of the right lung
B Pericardial effusion
C Right-sided pleural effusion
D Right-sided pneumonia
E Right-sided pneumothorax
- 2 A 52-year-old man presents with 4 days of haemoptysis. He has lost weight over the last 2 months without night sweats. He has smoked 30 cigarettes per day for 40 years and worked in construction, where he may have been exposed to asbestos. He returned from a holiday in Guyana 1 week ago. His chest X-ray is shown. What is the most likely cause of the X-ray findings?



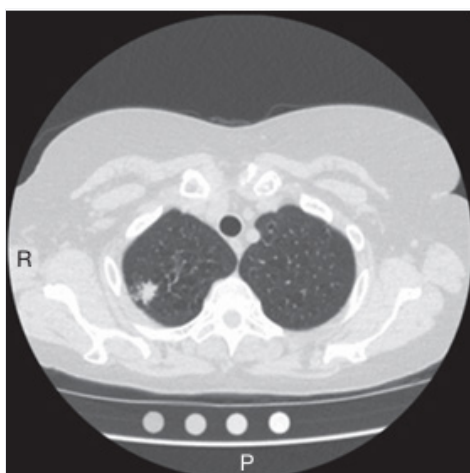
- A Bronchial carcinoma
B Granulomatous polyangiitis
C Lung abscess
D Pulmonary and pleural tuberculosis (TB)
E Pulmonary infarct

- 3** A 26-year-old woman presents with 24 hours of central chest pain and a month of progressive shortness of breath. Pulse 105/min, BP 105/65 mmHg, SpO₂ 96% on air, respiratory rate 16/min; she is afebrile. The chest is clear and the heart sounds are normal without murmurs. ECG shows sinus tachycardia only. Haemoglobin 110 g/L, WCC $9 \times 10^9/L$, platelets $340 \times 10^9/L$, urea and electrolytes normal, CRP 67 mg/L. Her chest X-ray (shown) shows a widened superior mediastinum. What should the next investigation be?



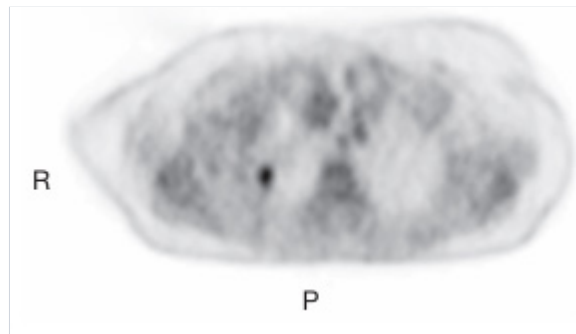
- A Bronchoscopy
- B CT chest, abdomen, pelvis
- C D-dimer
- D Echocardiogram
- E Positron emission tomography (PET) scan

- 4** A 63-year-old woman had a CT pulmonary angiogram for left-sided pleuritic chest pain. There was no pulmonary embolism, but a 15-mm spiculated nodule at the apex of the right lung was noted (shown). Her pain resolved within 24 hours and she is otherwise asymptomatic; she has smoked 10 cigarettes per day for 40 years. What is the best next step?



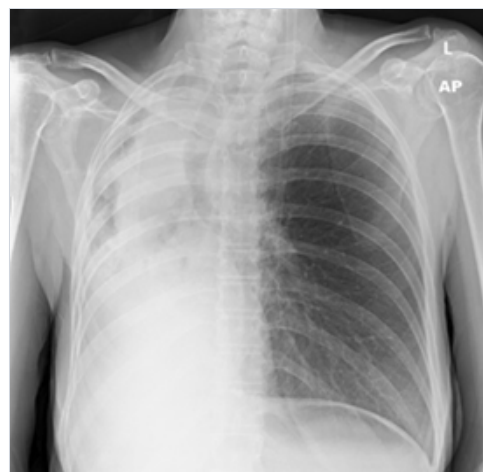
- A Bronchoscopy
- B Commence antibiotics
- C CT-guided biopsy
- D Interferon-gamma release assay
- E PET scan

- 5** A 63-year-old woman had a pulmonary nodule identified incidentally on a CT pulmonary angiogram. She has a 20 pack-year smoking history but no past medical history, is asymptomatic with performance status 0 and has normal pulmonary function. PET-CT (shown) shows FDG avidity in the right upper lobe nodule with no nodal or distant uptake. What should be the next step?



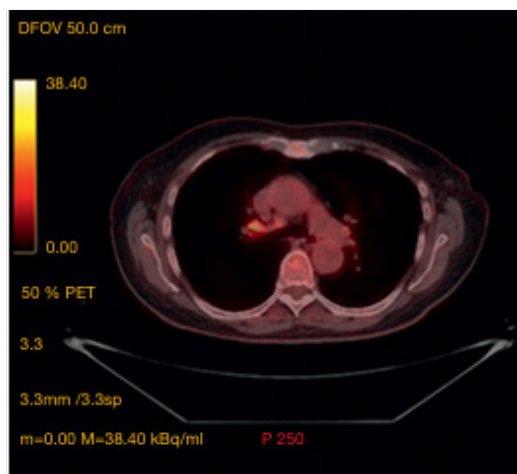
- A Antituberculous chemotherapy
- B Further follow-up scan in 3 months
- C Palliative chemotherapy
- D Palliative radiotherapy
- E Right upper lobectomy

- 6** A 51-year-old woman presents to the emergency department with sudden-onset breathlessness and right-sided pleuritic chest pain. She is a non-smoker with no significant past medical history. She needs 40% oxygen to maintain saturations of 92%; she is afebrile, BP 101/62 mmHg, pulse 112/min, respiratory rate 30/min. What does her chest X-ray (shown) demonstrate?



- A Dextrocardia
- B Left-sided tension pneumothorax
- C Right-sided lung collapse
- D Right-sided pleural effusion
- E Right-sided pneumothorax

- 7** A 54-year-old woman attends the rapid-access lung cancer clinic with haemoptysis and an abnormal chest X-ray. CT shows a large cavitating right upper lobe mass and bronchoscopic biopsy confirms squamous cell carcinoma. What does her PET-CT scan (shown) demonstrate?



- A** Uptake confined to the right upper lobe lesion
B Uptake in the right upper lobe lesion and a spinal metastasis
C Uptake in the right upper lobe lesion and indeterminate uptake in pulmonary vessels
D Uptake in the right upper lobe lesion and right hilar lymph node
E Uptake in the right upper lobe lesion, a spinal metastasis and a sternal metastasis
- 8** A 59 year old smoker presents with a new cough. CT reveals a 4-cm peripheral tumour in the left lower lobe with an ipsilateral enlarged hilar lymph node. Due to emphysema, CT-guided biopsy is high risk. PET shows high FDG avidity in the tumour and indeterminate hilar uptake. What should be the next investigation?
- A** CT-guided biopsy
B Endobronchial ultrasound-guided fine-needle aspiration (EBUS-FNA)
C Flexible bronchoscopy
D Repeat CT scanning
E Thoracoscopy
- 9** An 84-year-old retired shipyard insulation worker who smokes a pipe has a 3-month history of a nagging left-sided chest ache, worst in the axilla and radiating to the spine. Examination shows dullness from the left base to the mid zone and erythema ab igne over the left chest wall. What is the most likely cause of his chest pain?
- A** Chronic thromboembolic disease
B Herpes zoster
C Mesothelioma
D Pneumonia
E Tietze's syndrome
- 10** A 67-year-old man undergoing CT colonography is found to have a 6-mm nodule in the right lower lobe. Which feature increases the risk of malignancy in a pulmonary nodule?
- A** A smooth margin
B Central deposition of calcification
C Lack of smoking history
D Size < 4 mm
E Upper lobe distribution
- 11** A 65-year-old woman with rheumatoid arthritis has CT for suspected interstitial lung disease, which shows mild basal interlobular septal thickening and a 1.5-cm spiculated right upper lobe nodule. What is true regarding PET scanning for pulmonary nodules?
- A** It always detects bronchoalveolar carcinoma
B Inflammatory nodules are reliably excluded
C It has replaced tissue biopsy
D It is not useful in nodules < 1 cm in diameter
E It offers no more information than a standard CT
- 12** A 63 year old female non-smoker with a 5-cm lesion in the right mid-zone undergoes bronchoscopy with tumour biopsy. Which is the most common histological type of lung cancer?
- A** Adenocarcinoma
B Large cell carcinoma
C Mesothelioma
D Small cell carcinoma
E Squamous carcinoma
- 13** A 73-year-old woman with metastatic non-small cell lung cancer develops increasingly painful wrists. What is the most likely cause?
- A** Bone metastases
B Finger clubbing
C Horner's syndrome
D Hypercalcaemia
E Hypertrophic pulmonary osteoarthropathy (HPOA)
- 14** A 56-year-old smoker with a new cough has a CT showing a 4-cm peripheral left lung mass with an enlarged ipsilateral hilar lymph node and a pleural effusion; pleural fluid cytology is positive for malignant cells. What is the correct stage?
- A** I
B II
C III
D IV
E Staging not yet complete
- 15** A 36-year-old non-smoker has isolated haemoptysis. CT shows collapse of the left upper lobe with a vascular tumour at the left upper lobe orifice. What is the most likely tumour type?
- A** Adenocarcinoma
B Breast
C Carcinoid
D Small cell
E Squamous

- 16** A 64 year old woman with breast cancer metastatic to liver, lungs and bone develops increasing breathlessness despite antibiotics. CT shows diffuse interstitial thickening radiating from the hila. What is the likely diagnosis?
- A Drug-induced pneumonitis
 - B Lymphangitis carcinomatosa
 - C Pneumocystis pneumonia
 - D Pulmonary oedema
 - E Venous thromboembolism
- 17** A 28-year-old man has a large anterior mediastinal mass with cervical and axillary lymphadenopathy. What is the most likely diagnosis?
- A Bronchogenic cyst
 - B Lymphoma
 - C Neurogenic tumour
 - D Oesophageal tumour
 - E Retrosternal goitre
- 18** A 72 year old man with a history of benign asbestos pleurisy presents with a right-sided pleural effusion, pleural plaques and mediastinal pleural thickening. What is the most likely reason for his presentation?
- A Asbestos pleural plaques
 - B Asbestosis
 - C Benign asbestos pleurisy
 - D Diffuse pleural thickening
 - E Mesothelioma
- 19** A 72-year-old smoker has a 5-cm peripheral left lung mass, left hilar lymphadenopathy, an indeterminate liver lesion and a palpable, hard 2-cm left supraclavicular lymph node. What should be the next diagnostic test?
- A CT-guided biopsy
 - B Endobronchial ultrasound
 - C Flexible bronchoscopy
 - D Liver biopsy
 - E Supraclavicular lymph node biopsy
- 20** A 55-year-old woman with a left apical lung cancer and suspected mediastinal node involvement undergoes mediastinoscopy under general anaesthesia, followed by radiotherapy. She has been hoarse since waking from the procedure, but laryngoscopy shows normal vocal cord movement. What is the most likely cause?
- A Chronic laryngitis
 - B Endotracheal intubation during mediastinoscopy
 - C Laryngeal tuberculosis
 - D Left recurrent laryngeal nerve involvement by the tumour
 - E Psychogenic aphonia

Answers 1-20

DAVIDSON MCQs

- 1** **A Collapse of the right lung** — Whole-hemithorax dullness with the trachea pulled towards it is lung collapse from an obstructing tumour; an effusion pushes it away.
- 2** **A Bronchial carcinoma** — A thick-walled cavitating mass in a smoker with haemoptysis and weight loss is most likely cavitating squamous carcinoma.
- 3** **B CT chest, abdomen, pelvis** — **UPDATED** A widened mediastinum with inflammation in a young woman suggests lymphoma; contrast CT chest, abdomen, pelvis to stage and find biopsy sites.
- 4** **E PET scan** — **UPDATED** A spiculated solid upper-lobe nodule ≥ 8 mm in a smoker (Brock risk $> 10\%$) needs PET-CT next (BTS).
- 5** **E Right upper lobectomy** — **UPDATED** An FDG-avid nodule without nodal or distant uptake in a fit patient is presumed stage I: lobectomy (SABR if unfit).
- 6** **C Right-sided lung collapse** — Right hemithorax opacification with volume loss and the mediastinum pulled right is right lung collapse; bronchoscopy for the cause (e.g. carcinoid).
- 7** **D Uptake in the right upper lobe lesion and right hilar lymph node** — PET-CT shows uptake in the right upper lobe primary and right hilar node (N1); sample PET-positive nodes (EBUS) before deciding.
- 8** **B Endobronchial ultrasound-guided fine-needle aspiration (EBUS-FNA)** — EBUS-TBNA of PET-avid hilar nodes gives diagnosis and stage in one low-risk procedure (NICE NG122).
- 9** **C Mesothelioma** — **UPDATED** Persistent dull chest wall pain with effusion decades after asbestos exposure is mesothelioma; CT and pleural biopsy.
- 10** **E Upper lobe distribution** — Upper-lobe location raises nodule malignancy risk (Brock), as do age, female sex, emphysema, size, spiculation and part-solid nature.
- 11** **D It is not useful in nodules < 1 cm in diameter** — PET-CT is unreliable for nodules < 8 –10 mm, misses low-avidity tumours (carcinoid, AIS) and is falsely positive in inflammation.
- 12** **A Adenocarcinoma** — Adenocarcinoma is the commonest lung cancer (40–50%), especially in never-smokers and women; test for EGFR, ALK, ROS1 and PD-L1.
- 13** **E Hypertrophic pulmonary osteoarthropathy (HPOA)** — **UPDATED** Painful wrists/ankles with clubbing in NSCLC is hypertrophic pulmonary osteoarthropathy (periostitis); NSAIDs, bisphosphonates, treat tumour.
- 14** **D IV** — A malignant pleural effusion is M1a, so the cancer is stage IVA regardless of T and N; treatment is systemic.
- 15** **C Carcinoid** — A vascular endobronchial tumour causing collapse in a young non-smoker is a bronchial carcinoid; resection has an excellent prognosis.

- 16 **B Lymphangitis carcinomatosa** — Beaded septal and bronchovascular thickening radiating from the hila in metastatic breast cancer is lymphangitis carcinomatosa.
- 17 **B Lymphoma** — Anterior mediastinal mass with cervical and axillary nodes in a young adult is lymphoma (the 4 Ts: thymoma, teratoma, thyroid, lymphoma).
- 18 **E Mesothelioma** — A unilateral effusion with mediastinal pleural thickening after asbestos exposure is highly suspicious of mesothelioma; pleural biopsy.
- 19 **E Supraclavicular lymph node biopsy** — **UPDATED** Biopsy the palpable supraclavicular node (N3); it gives histology, molecular markers and the highest stage with least risk (NICE NG122).
- 20 **B Endotracheal intubation during mediastinoscopy** — **UPDATED** Hoarseness right after anaesthesia with normal cord movement is intubation-related laryngeal trauma; a nerve palsy would paralyse the cord.

5 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

- 1 A woman with systemic sclerosis is found to have a 3-cm lung lesion that is FDG-avid on PET-CT. Which statement is most accurate?
- A Additional diagnostic studies are indicated
 - B The findings on PET/CT make infection very likely
 - C The findings on PET/CT make infection very unlikely
 - D The findings on PET/CT make malignancy very likely
 - E The findings on PET/CT make malignancy very unlikely
- 2 A 75-year-old man with previous asbestos exposure has a pleural effusion suspicious for mesothelioma. Which statement about this condition is true?
- A Cigarette smoking increases the likelihood of developing the condition
 - B Death is usually due to widespread distant metastases
 - C Exposure can be brief (1–2 years) and the latency may be 40 years or more
 - D Repeated pleural fluid cytology usually gives a definitive diagnosis
 - E Combined surgical resection and chemotherapy significantly improves survival
- 3 A 64-year-old man has facial swelling and distended neck veins, and superior vena cava obstruction is suspected. Which CT finding is most likely?
- A A central mass obstructing right main bronchus
 - B An apical mass invading the chest wall
 - C A large pericardial effusion
 - D Massive right pleural effusion
 - E Enlarged mediastinal lymph nodes compressing the SVC
- 4 A 64-year-old man has facial swelling and distended neck veins from superior vena cava obstruction caused by enlarged mediastinal lymph nodes. He has no stridor or confusion, and no tissue diagnosis has yet been made. Initial management includes all of the following EXCEPT:
- A Intravenous furosemide for diuresis
 - B Elevation of the head of the bed
 - C Emergency radiotherapy
 - D Low-sodium diet
 - E Oxygen
- 5 A 58-year-old woman with metastatic breast cancer develops acute paraplegia from suspected metastatic spinal cord compression. What is the first step in management?
- A Dexamethasone 10 mg IV
 - B Neurosurgery consult for decompression
 - C Radiation oncology consult
 - D MRI of the brain
 - E MRI entire spinal cord

Answers 1–5

HARRISON MCQs

- 1 **A Additional diagnostic studies are indicated** — FDG uptake also occurs in infection and inflammation, so an avid lesion still needs tissue diagnosis.
- 2 **C Exposure can be brief (1–2 years) and the latency may be 40 years or more** — Mesothelioma can follow brief asbestos exposure with a 30–40+ year latency; smoking does not raise its risk; cytology is often negative.
- 3 **E Enlarged mediastinal lymph nodes compressing the SVC** — SVC obstruction is usually malignant (lung cancer, lymphoma) from mediastinal nodes compressing the SVC; stent for rapid relief.
- 4 **C Emergency radiotherapy** — **UPDATED** In SVC obstruction without stridor or cerebral oedema, get tissue first; emergency radiotherapy may obscure the diagnosis.
- 5 **A Dexamethasone 10 mg IV** — Suspected MSCC with deficit: give high-dose dexamethasone immediately, then whole-spine MRI within 24 h and definitive treatment.

6 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1 A 65-year-old woman presents with breathlessness and haemoptysis. She is diagnosed with squamous cell carcinoma (SCC) of the lung. She has been a smoker for as long as she can remember. She has lived in central London for almost 30 years now and she worked in a shipbuilding yard for most of her working life. Both her father and mother died of lung cancer. Which of the following is the most likely cause of her cancer?
 - A Smoking
 - B Gender
 - C Family history
 - D Occupation
 - E City living
- 2 A middle-aged man is diagnosed with lung cancer. He worked in a shipbuilding yard for most of his life. He is distraught with the diagnosis as he says he has never smoked in his life. Which of the following types of lung cancer is he most likely to suffer from?
 - A Squamous
 - B Large cell
 - C Adenocarcinoma
 - D Alveolar cell
 - E Small cell
- 3 A 53-year-old woman is diagnosed with lung cancer. Her main symptoms on presentation included breathlessness and haemoptysis. She complained of stretch marks on her abdomen and that her skin bruised easily. On examination you note purple abdominal striae. Which of the following types of lung cancer is she most likely to suffer from?
 - A Squamous
 - B Large cell
 - C Adenocarcinoma
 - D Alveolar cell
 - E Small cell
- 4 A 63-year-old man is diagnosed with lung cancer. Which of the following extrapulmonary manifestations is he most likely to suffer from?
 - A Clubbing
 - B Dermatomyositis
 - C Lambert–Eaton myasthenic syndrome (LEMS)
 - D Hypercalcaemia
 - E Thrombophlebitis migrans
- 5 A 66-year-old woman with history of weight loss presents to the respiratory clinic with an 8-week history of nonproductive cough and chronic clearing of throat. She has a 52-pack year smoking history and quit 25 years ago. She was subsequently diagnosed with carcinoma of the lung. Which of the following is NOT a complication of carcinoma of the lung?
 - A Adrenal metastases
 - B Hypertrophic pulmonary osteoarthropathy
 - C Hypocalcaemia
 - D Hyponatraemia
 - E Ptosis
- 6 A 65-year-old factory worker presents with a dry cough, right-sided chest pain and finger clubbing. He has lost 4 kg in weight in the last 2 weeks and has a 30-year smoking history. The right lower chest is dull to percussion with reduced expansion, and a chest X-ray shows a right peripheral lung mass with a small pleural effusion. What is the most likely diagnosis?
 - A Asbestosis
 - B Non-small cell lung cancer
 - C Mesothelioma
 - D Coal worker's pneumoconiosis
 - E Small cell lung cancer

Answers 1–6

CRASH COURSE MCQs

- 1 **A Smoking** — Smoking causes ~85% of lung cancer, especially squamous and small cell; asbestos multiplies its effect.
- 2 **C Adenocarcinoma** — Adenocarcinoma is the commonest lung cancer in never-smokers; asbestos raises risk of all types.
- 3 **E Small cell** — Cushing's (striae, bruising) in lung cancer means ectopic ACTH from small cell carcinoma (or carcinoid).
- 4 **A Clubbing** — Clubbing is the commonest extrapulmonary sign of lung cancer (up to a third); new clubbing in > 40s needs an urgent CXR.
- 5 **C Hypocalcaemia** — Lung cancer causes hypercalcaemia (PTHrP, bone metastases), not hypocalcaemia; SIADH, HPOA, adrenal metastases and Horner's are recognised.
- 6 **B Non-small cell lung cancer** — **UPDATED** A peripheral mass with clubbing, weight loss and effusion in a smoker is non-small cell lung cancer; clubbing is uncommon in small cell.

1 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 50-year-old man with long-standing COPD develops insidious aching in the distal limbs, particularly both wrists, and has lost 4.5 kg (10 lb). The skin over the wrists is warm and red, and there is bilateral finger clubbing. Plain X-rays of the wrists show periosteal thickening, reported as possible osteomyelitis. What is the most appropriate next step?
- A Start ciprofloxacin
 - B Obtain chest x-ray
 - C Aspirate both wrists
 - D Begin methotrexate therapy
 - E Obtain erythrocyte sedimentation rate

Answer 1

PRETEST MCQS

- 1 **B Obtain chest x-ray** — Clubbing with painful symmetrical periostitis is HPOA, usually from lung cancer; COPD does not cause clubbing, so get a chest X-ray.

2 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 42-year-old man has had a grade III chondrosarcoma of the right maxilla resected. He now has breathlessness and a dry cough, and a chest X-ray suggests lung metastases. Which radiological feature of lung metastases is characteristic of this tumour?
- A Consolidation
 - B Cavitation
 - C Haemorrhage
 - D Calcification
 - E Miliary pattern
- 2 A 66-year-old man with COPD is referred by his GP to the rapid-access chest clinic after several courses of antibiotics and steroids for an exacerbation that has not fully resolved. His chest X-ray shows a suspicious 3-cm mass at the right hilum with background changes of COPD. Blood tests: Hb 142 g/L, platelets $323 \times 10^9/L$, WBC $9.1 \times 10^9/L$, Na 128 mmol/L, K 3.9 mmol/L, urea 6.4 mmol/L, creatinine 99 $\mu\text{mol/L}$. There is no clubbing, and a fixed monophonic wheeze is heard in the right lung. What is the most likely underlying cause?
- A Tuberculosis
 - B Sarcoidosis
 - C Squamous cell lung cancer
 - D Small cell lung cancer
 - E Lymphoma

Answers 1-2

PASSMEDICINE MCQS

- 1 **D Calcification** — **UPDATED** Calcified lung metastases are characteristic of bone/cartilage sarcomas (osteoid-, chondrosarcoma) and thyroid or mucinous cancers.
- 2 **D Small cell lung cancer** — A central mass with fixed monophonic wheeze and hyponatraemia in a smoker suggests small cell lung cancer (SIADH).

2 QUESTIONS • AFTER PASTEST

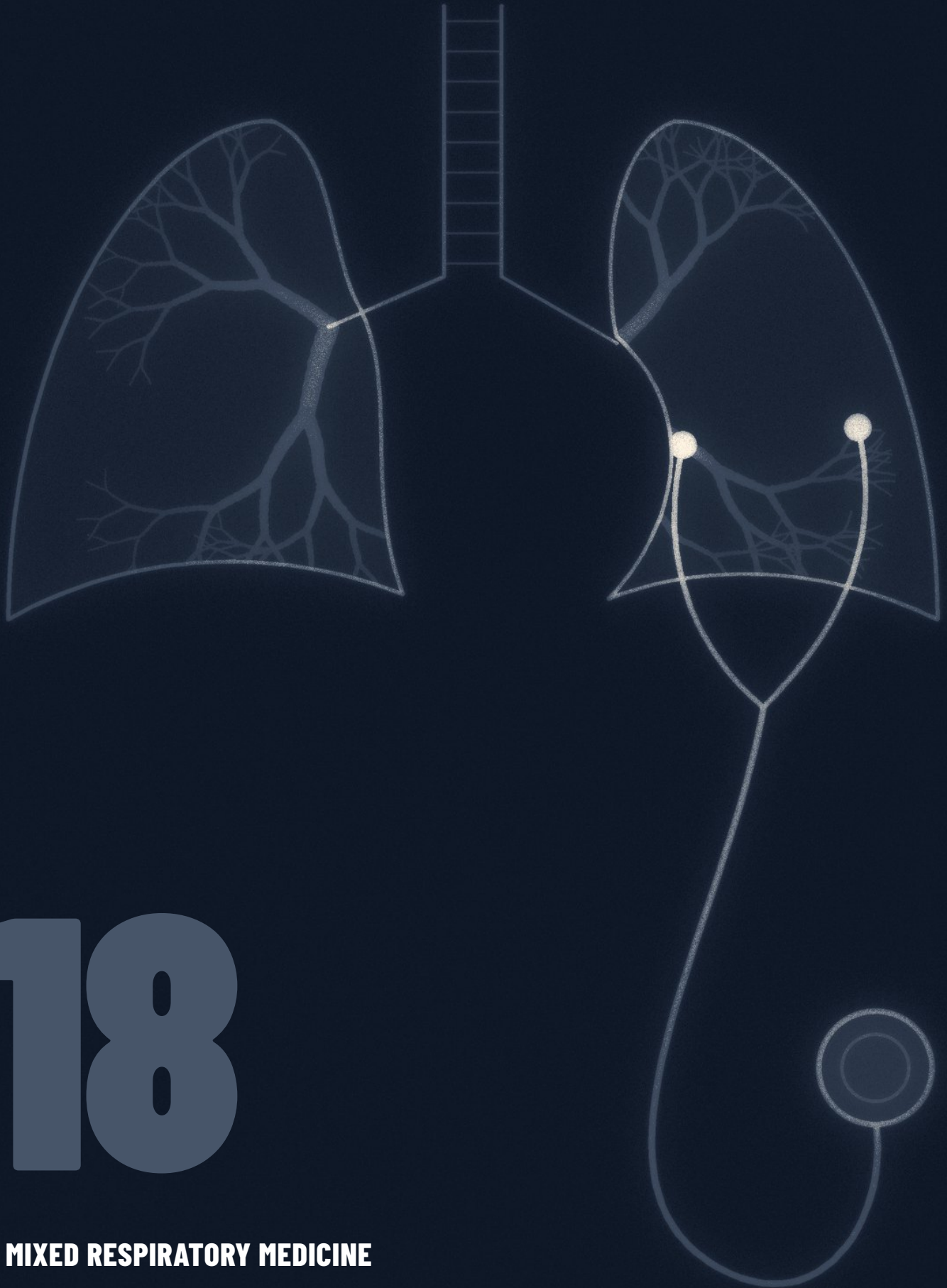
PasTest MCQs

- 1** A 74-year-old man who has smoked 40 cigarettes per day for 30 years presents with increasing shortness of breath. He takes diclofenac and paracetamol for joint pain. His chest X-ray shows a spiculated left hilar mass with dense consolidation of the left lower lobe beyond it, and air bronchograms indicate that the lobar bronchus is still patent. Which finding would be expected on examination over the left lower lobe?
- A** Decreased breath sounds
 - B** Hyper-resonance
 - C** Inspiratory crackles
 - D** Polyphonic wheeze
 - E** Whispering pectoriloquy
- 2** A 68-year-old man who smokes 40 cigarettes per day has a proven diagnosis of bronchial carcinoma, and radical radiotherapy is being considered. Which of the following is the most significant contraindication to radical radiotherapy?
- A** Adenocarcinoma
 - B** Forced expiratory volume in 1 s (FEV₁) < 60%
 - C** Malignant pleural effusion
 - D** Superior vena caval obstruction
 - E** Tumour adjacent to the hilum

Answers 1-2

PASTEST MCQs

- 1** **E Whispering pectoriloquy** — UPDATED Consolidation with a patent bronchus gives bronchial breathing, increased vocal resonance and whispering pectoriloquy.
- 2** **C Malignant pleural effusion** — A malignant pleural effusion (M1a, stage IV) cannot be included in a radical radiotherapy field, so radical treatment is inappropriate.



18

MIXED RESPIRATORY MEDICINE

Mixed Respiratory Medicine

Integrated respiratory questions across the whole system for final revision.

90

MCQS

0

EMQ SCENARIOS

7

SOURCES

4

REPEATED IDEAS

IN THIS CHAPTER

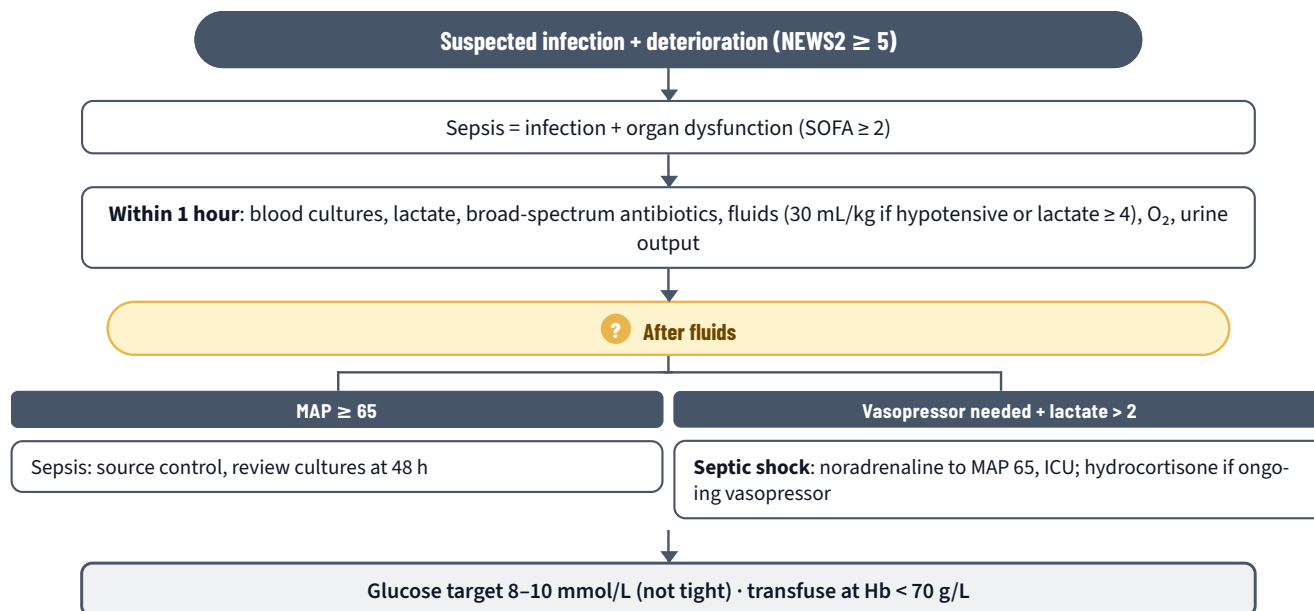
01	Management flowcharts	3 charts	page 290
02	Local Exams MCQs	4 Q	page 292
03	Davidson MCQs	19 Q	page 293
04	Harrison MCQs	37 Q	page 296
05	Crash Course MCQs	1 Q	page 301
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07	Passmedicine MCQs	10 Q	page 304
08	PasTest MCQs	10 Q	page 306

18

Step by step

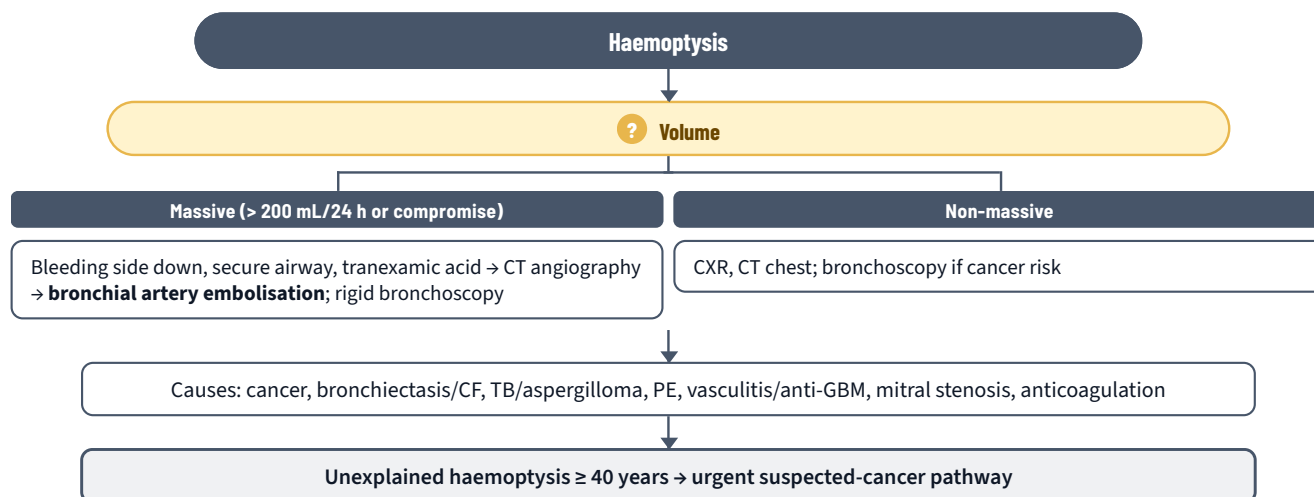
Sepsis

SURVIVING SEPSIS 2021 • NICE NG51



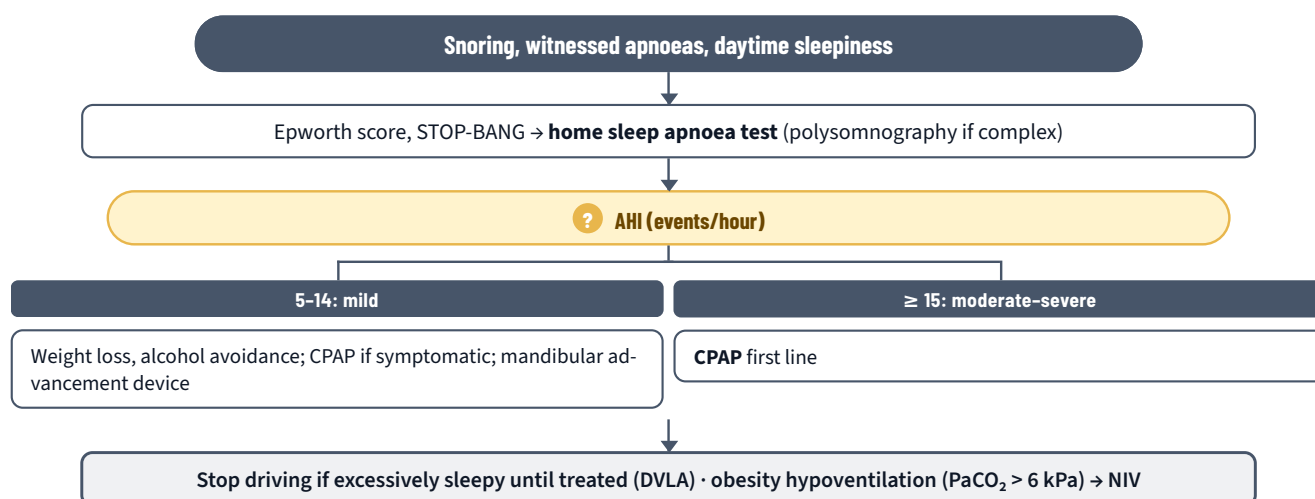
Haemoptysis

NICE NG12 • BTS



Obstructive sleep apnoea

NICE NG202



Local Exams MCQs

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

- 1 After an adrenalectomy a patient becomes breathless and cyanosed. His chest X-ray is shown. What is the cause?



- A Hospital-acquired pneumonia
- B Atelectasis (lung collapse)
- C Pleural effusion
- D Empyema

REPEATED 2* • ALSO ASKED IN
Passmedicine

FORMATIVE & PREVIOUS YEARS

College formative and previous-years exams

- 2 A 28-year-old man of Chinese origin on treatment for pulmonary tuberculosis telephones 2 weeks into therapy complaining of painful eyes and blurred vision. What is the likely cause?
- A Drug resistance to ethambutol and pyrazinamide
 - B Ethambutol
 - C Immune reconstitution
 - D Intercurrent viral infection
 - E Non-tuberculous mycobacterium
- 3 A 58-year-old woman with a 28-year history of rheumatoid arthritis presents with progressive shortness of breath. All of the following are recognised pulmonary complications of rheumatoid arthritis EXCEPT:
- A Pleural effusion
 - B Interstitial lung disease
 - C Pulmonary nodules
 - D Bronchiolitis obliterans
 - E Apical (upper-lobe) fibrobullous disease

LOG BOOK 2025

Log-book questions

- 4 What is the immediate treatment for a haemodynamically unstable patient with suspected pulmonary embolism?
- A Intravenous antibiotics
 - B Thrombolytic therapy
 - C Oral beta-blockers
 - D High-flow oxygen alone
 - E Urgent coronary angioplasty

Answers 1–4

LOCAL EXAMS MCQs

- 1 **B** Atelectasis (lung collapse) — Dense hemithorax opacity with volume loss after upper abdominal surgery is collapse from retained secretions; analgesia, physiotherapy.
- 2 **B** Ethambutol — **UPDATED** Painful eyes and blurred vision on TB treatment is ethambutol optic neuritis: stop it and get urgent ophthalmology review.
- 3 **E** Apical (upper-lobe) fibrobullous disease — **UPDATED** RA causes effusions, ILD, nodules, bronchiolitis obliterans and bronchiectasis; apical fibrobullous disease belongs to ankylosing spondylitis.
- 4 **B** Thrombolytic therapy — Haemodynamically unstable (high-risk) PE: systemic thrombolysis (alteplase) after starting heparin; embolectomy if contraindicated.

Davidson MCQs

- 1 A 53-year-old man has had a non-productive cough for 3 months. His spirometry is normal and he already takes omeprazole for reflux. Which statement about his management is true?
 - A A CXR should be performed
 - B A trial of an inhaled corticosteroid (ICS) is warranted
 - C He should see an ENT surgeon
 - D His cough has become a habit and he must get used to it
 - E His reflux medication should be increased
- 2 A 24-year-old man with cystic fibrosis asks about ivacaftor. Ivacaftor corrects the function of which CFTR defect?
 - A $\Delta F508$
 - B G542X
 - C G551D
 - D R117H
 - E W1282X
- 3 A 56-year-old non-smoking man with metastatic adenocarcinoma of the left lung (EGFR mutation positive) is evaluated. What is the treatment of choice?
 - A Adjuvant chemotherapy and lobectomy
 - B Best supportive care
 - C Cisplatin and pemetrexed
 - D EGFR tyrosine kinase inhibitor (e.g. erlotinib or osimertinib)
 - E Radiotherapy
- 4 A 24-year-old patient with cystic fibrosis fails to recover from an exacerbation, and sputum cultures now isolate *Pseudomonas aeruginosa*. Which statement regarding *P. aeruginosa* in cystic fibrosis is true?
 - A Intravenous antibiotic therapy is rarely required
 - B It is a benign coloniser of the bronchiectatic airways
 - C It is one of the earliest bacteria isolated in sputum from CF patients
 - D Nebulised azithromycin 3 times a week suppresses infection
 - E Nebulised tobramycin is an effective treatment in chronic colonisation
- 5 A 61 year old woman on IV daptomycin for staphylococcal discitis, previously improving, spikes a fever after 2 weeks with rising WCC (including eosinophilia) and new bilateral upper zone consolidation on CXR. What should be the treatment?
 - A Antibiotics to cover HAP
 - B Antituberculosis chemotherapy
 - C Continue current therapy
 - D IV furosemide
 - E Stop daptomycin and give prednisolone
- 6 A 70 year old woman with a 6-week history of cough, low-grade fever, and malaise, on nitrofurantoin for UTIs, has bi-basal infiltrates on CXR. What is the most likely diagnosis?
 - A Atypical pneumonia
 - B Cryptogenic organising pneumonia
 - C Idiopathic pulmonary fibrosis
 - D Nitrofurantoin lung disease
 - E Non-specific interstitial pneumonia (NSIP)
- 7 A 75-year-old woman has a 10-year history of daily non-productive cough. She has back pain from a vertebral fracture, and the chest X-ray shows a large hiatus hernia and kyphosis. What is the most likely cause of her cough?
 - A Asthma
 - B Gastro-oesophageal reflux
 - C Hypercalcaemia
 - D Lung cancer
 - E Tuberculosis
- 8 A 42-year-old non-smoking woman has a 4-month history of breathlessness occurring only in the daytime, which began after an upper respiratory tract infection with nasal blockage. Her D-dimer was positive but CTPA is normal, and spirometry is normal. What should be the next investigation?
 - A Arterial blood gas
 - B Bronchoscopy
 - C Dysfunctional breathing studies and a Nijmegen questionnaire
 - D Echocardiogram
 - E Lung volume and gas transfer studies
- 9 A 64-year-old woman with COPD presents with sudden right-sided pleuritic chest pain and breathlessness after returning from a 3-week holiday. SpO₂ is 84% on room air; there is reduced air entry and mild bilateral ankle oedema, and the D-dimer is elevated. What is the most likely cause of her chest pain?
 - A Bronchospasm
 - B Malignant pleural disease
 - C Pneumonia
 - D Pneumothorax
 - E Pulmonary thromboembolism

- 10** A 45-year-old non-smoking man with finger clubbing, noted since his teens and also present in his father, presents with a sprained wrist. He has no other symptoms. What is the next step the family physician should take?
- Check bloods (including LFTs, thyroid function tests and ESR)
 - CXR to exclude cancer
 - Reassurance
 - Referral to respiratory clinic
 - Sweat test to exclude cystic fibrosis
- 11** A patient with known bronchiectasis and a long-standing right apical cavity from old tuberculosis now coughs up 50 mL of fresh red blood. The chest X-ray shows a rounded mass partially filling the cavity. What is the likely cause of his haemoptysis?
- Carcinoma
 - Mycetoma
 - Pulmonary infarction
 - Suppurative pneumonia
 - Tuberculosis
- 12** A 32-year-old patient with cystic fibrosis presents with massive haemoptysis (250 mL of fresh red blood) but is haemodynamically stable. What is the most useful investigation?
- Bronchial artery angiography
 - Bronchoscopy
 - Coagulation studies
 - CTPA
 - Sputum culture
- 13** Acute coryza is most commonly caused by which of the following?
- Bordetella pertussis
 - Haemophilus influenzae
 - Mycoplasma pneumoniae
 - Rhinovirus
 - Streptococcus pneumoniae
- 14** A 28-year-old woman recovering from ACL repair in an above-knee cast has near-syncope on standing, with BP 80/45 mmHg and sinus tachycardia. The chest X-ray is clear and the D-dimer is elevated, and she then has a cardiac arrest. What should the immediate management include?
- Apixaban
 - Intravenous heparin infusion
 - LMWH
 - Thrombolysis
 - Warfarin
- 15** A 45-year-old woman has a cough every May that resolves by autumn, with nasal discharge and watering eyes. Examination and spirometry are normal. What is the likely diagnosis?
- Allergic asthma
 - Allergic rhinitis
 - Bordetella pertussis
 - Perennial rhinitis
 - Viral upper respiratory tract infection
- 16** Which of the following statements is true regarding breathing during sleep?
- Abnormal ventilatory drive is present in obstructive sleep apnoea
 - During sleep muscle tone increases
 - Forty per cent of middle-aged women snore
 - Hypoventilation accompanies normal sleep
 - Palatoglossus and genioglossus contract actively during expiration
- 17** A 55-year-old man with excessive daytime sleepiness, an Epworth Sleepiness Scale score of 18 and a BMI of 36 kg/m² undergoes a sleep study. What is the likely result?
- 5 apnoea/hypopneas per hour
 - 10 apnoea/hypopneas per hour
 - 20 apnoea/hypopneas per hour
 - Central sleep apnoea
 - Narcolepsy
- 18** A 72-year-old man with ischaemic heart disease, hypertension and childhood poliomyelitis has increasing breathlessness and nocturnal symptoms. His chest X-ray is under-inspired, a sleep study shows significant nocturnal hypoventilation, and arterial blood gases show decompensated type 2 respiratory failure. What is the likely diagnosis?
- Bronchial carcinoma
 - Eventration of the diaphragm
 - Hernia through foramen of Bochdalek
 - Idiopathic diaphragmatic paralysis
 - Post-polio syndrome
- 19** Which of these associations between pathophysiological features and conditions is correct?
- Diaphragmatic rupture and ankylosing spondylitis
 - Eventration of diaphragm and foramen of Bochdalek
 - Pectus carinatum and poliomyelitis
 - Pectus excavatum and asthma
 - Thoracic kyphoscoliosis and tuberculosis

Answers 1-19

DAVIDSON MCQS

- 1** **A** A CXR should be performed — Chronic cough (> 8 weeks) always needs a chest X-ray and spirometry to exclude serious disease before treating common causes.
- 2** **C** G551D — Ivacaftor is a CFTR potentiator for gating mutations, classically G551D; F508del needs a corrector plus ivacaftor.

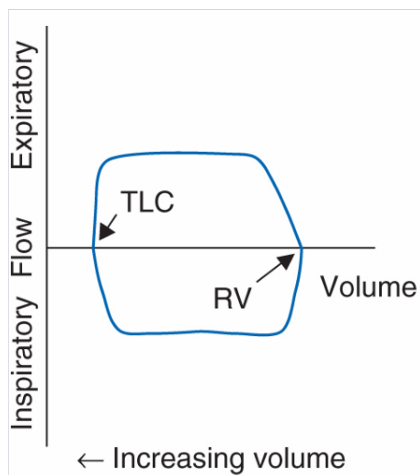
- 3 **D** EGFR tyrosine kinase inhibitor (e.g. erlotinib or osimertinib) — **UPDATED** Metastatic EGFR-mutant adenocarcinoma is treated first line with an oral EGFR TKI (osimertinib preferred).
- 4 **E** Nebulised tobramycin is an effective treatment in chronic colonisation — **UPDATED** Chronic *Pseudomonas* in CF accelerates decline; long-term nebulised tobramycin (or colistin, aztreonam) reduces exacerbations.
- 5 **E** Stop daptomycin and give prednisolone — **UPDATED** Fever, eosinophilia and new infiltrates after 2 weeks of daptomycin is drug-induced eosinophilic pneumonia: stop it and give steroids.
- 6 **D** Nitrofurantoin lung disease — **UPDATED** Chronic cough, fever and bibasal infiltrates on long-term nitrofurantoin is nitrofurantoin lung; stopping the drug is diagnostic and therapeutic.
- 7 **B** Gastro-oesophageal reflux — Chronic dry cough with a large hiatus hernia and kyphosis is most likely GORD; trial a PPI.
- 8 **C** Dysfunctional breathing studies and a Nijmegen questionnaire — Daytime-only breathlessness after a viral illness with normal imaging and spirometry suggests dysfunctional breathing; Nijmegen questionnaire.
- 9 **E** Pulmonary thromboembolism — Sudden pleuritic pain with hypoxaemia after travel and a raised D-dimer in COPD is most likely PE; CTPA (CXR to exclude pneumothorax).
- 10 **C** Reassurance — Lifelong familial clubbing without symptoms is hereditary clubbing; reassure. New clubbing needs investigation.
- 11 **B** Mycetoma — **UPDATED** A mass within an old TB cavity with an air crescent is an aspergilloma (mycetoma), which causes haemoptysis.
- 12 **A** Bronchial artery angiography — Massive haemoptysis in CF comes from bronchial arteries: bronchial artery angiography localises and embolises the bleeding.
- 13 **D** Rhinovirus — The common cold (acute coryza) is caused mainly by rhinoviruses; symptomatic treatment only.
- 14 **D** Thrombolysis — Cardiac arrest from suspected massive PE: immediate thrombolysis during CPR, continuing CPR for 60–90 min.
- 15 **B** Allergic rhinitis — Seasonal cough with rhinorrhoea and watery eyes every May with normal spirometry is seasonal allergic rhinitis; antihistamine and nasal steroid.
- 16 **D** Hypoventilation accompanies normal sleep — Normal sleep reduces ventilatory drive and muscle tone, so mild hypoventilation (small PaCO_2 rise) is normal.
- 17 **C** 20 apnoea/hypopneas per hour — Epworth 18 in an obese man suggests significant OSA; AHI 20/h is moderate (mild 5–14, moderate 15–29, severe ≥ 30).
- 18 **E** Post-polio syndrome — New respiratory muscle weakness decades after polio is post-polio syndrome, causing nocturnal then daytime hypercapnia; nocturnal NIV.
- 19 **E** Thoracic kyphoscoliosis and tuberculosis — Spinal TB (Pott disease) causes angular thoracic kyphosis/kyphoscoliosis and can lead to restrictive ventilatory failure.

37 QUESTIONS • AFTER HARRISON'S SELF-ASSESSMENT & BOARD REVIEW

Harrison MCQs

SET 1 OF 2

- 1 At what lung volume does the outward recoil of the chest wall equal the inward elastic recoil of the lung?
- Expiratory reserve volume
 - Functional residual capacity
 - Residual volume
 - Tidal volume
 - Total lung capacity
- 2 A 65-year-old man has stridor-like, low-pitched wheeze heard over the central airway. His flow-volume loop is shown. What is the most likely cause of his symptoms?



- Aspirated foreign body
 - Chronic obstructive pulmonary disease
 - Idiopathic pulmonary fibrosis
 - Subglottic stenosis
 - Unilateral vocal cord paralysis
- 3 A 22-year-old pregnant woman presents with acute dyspnoea. Arterial blood gas shows a raised A-a gradient and the chest X-ray is normal. What is the next best step in diagnosis and management?
- Initiate therapy with amoxicillin for acute bronchitis
 - Perform a computed tomography (CT) pulmonary angiogram
 - Perform an echocardiogram
 - Reassure the patient that dyspnoea is normal
 - Treat with clonazepam for a panic attack
- 4 Which of the following pulmonary function test results is consistent with myasthenia gravis?
- Increased TLC, decreased vital capacity, decreased FEV₁/FVC ratio
 - Decreased TLC, decreased VC, decreased RV, increased FEV₁/FVC ratio, normal MIP
 - Decreased TLC, increased RV, normal FEV₁/FVC ratio, decreased MIP
 - Normal TLC, normal RV, normal FEV₁/FVC ratio, normal MIP
 - None
- 5 A 36-year-old woman rescued from a house fire remains obtunded, with seizures, hypotension and a bitter almond odour on her breath. Which test will confirm the diagnosis?
- Ammonia concentration
 - Blood lactate level
 - Carboxyhaemoglobin level
 - Red blood cell cyanide concentration
 - Methaemoglobin level
- 6 A 28-year-old has recurrent sinus and lung infections and cystic fibrosis is suspected. Which finding best confirms the diagnosis?
- DNA analysis showing one F508del allele
 - Decreased baseline nasal potential difference
 - Persistent Pseudomonas in sputum cultures
 - Sweat chloride >40 mmol/L
 - Sweat chloride >60 mmol/L
- 7 A 42-year-old man has progressive dyspnoea and weight loss. Bronchoalveolar lavage returns milky fluid containing PAS-positive material. What is the diagnosis?
- Cryptogenic organising pneumonia
 - Desquamative interstitial pneumonia
 - Nocardiosis
 - Pneumocystis pneumonia
 - Pulmonary alveolar proteinosis
- 8 A patient has symptomatic pulmonary alveolar proteinosis with hypoxaemia. What is the most appropriate treatment?
- Doxycycline
 - Prednisolone
 - Prednisolone and cyclophosphamide
 - Co-trimoxazole
 - Whole-lung lavage

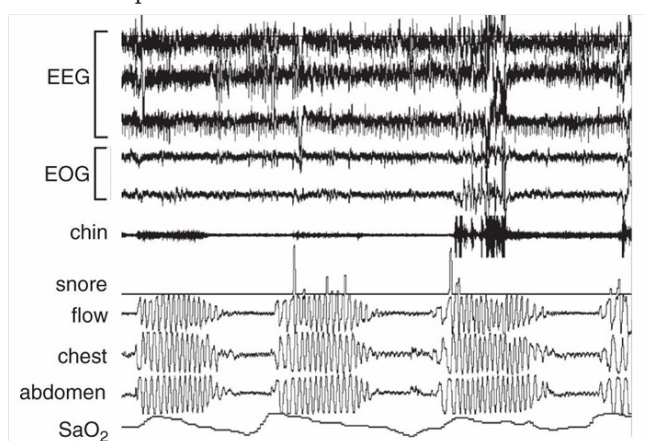
- 9 A 52-year-old man has obesity hypoventilation syndrome. Which statement is true?
- A Mutation of the PHOX2B gene is associated with this condition
 - B CPAP alone is always adequate therapy
 - C Intubation is mandatory for acute hypercapnia
 - D Coexistent obstructive sleep apnoea occurs in only 25% of cases
 - E Weight loss will lead to improvements in PaCO₂ over time

- 10 A 52-year-old man with loud snoring, witnessed apnoeas, daytime sleepiness and a high BMI is thought to have obstructive sleep apnoea. He has no significant cardiopulmonary or neuromuscular disease and no symptoms of another sleep disorder. What is the most appropriate next step?
- A Attended in-laboratory polysomnography
 - B Home sleep apnoea test
 - C Overnight oximetry alone
 - D Tonsillectomy referral
 - E Empirical CPAP without testing

REPEATED 2* · ALSO ASKED IN
PreTest

- 11 A 48-year-old man with moderate obstructive sleep apnoea refuses CPAP. Which statement about untreated obstructive sleep apnoea is correct?
- A It has no significant effect on mortality
 - B It is not associated with depression
 - C It increases the risk of motor vehicle accidents
 - D It has no effect on blood pressure
 - E Its complications are prevented by sleeping with extra pillows

- 12 The polysomnogram shown demonstrates a waxing and waning (crescendo–decrescendo) breathing pattern with central apnoeas. Which patient is most likely to have this pattern?



- A A 6-year-old with enlarged tonsils and adenoids
- B A 36-year-old on methadone maintenance
- C A 48-year-old obese man with loud snoring
- D A 52-year-old menopausal woman with frequent awakenings
- E A 68-year-old with atrial fibrillation and ischaemic cardiomyopathy (EF 15%), raised JVP and oedema

- 13 A 48-year-old man has symptomatic obstructive sleep apnoea with an apnoea–hypopnoea index of 42/hour. What is the most appropriate initial therapy?
- A Continuous positive airway pressure (CPAP)
 - B Mandibular advancement device
 - C Oxygen therapy
 - D Uvulopalatopharyngoplasty
 - E Weight loss alone

REPEATED 3* · ALSO ASKED IN
Crash Course · PreTest

- 14 A 25-year-old with cystic fibrosis is referred for lung transplantation. What is the main impediment to long-term survival after transplantation?
- A Bronchiolitis obliterans syndrome
 - B CMV infection
 - C Chronic kidney disease
 - D Primary graft dysfunction
 - E Posttransplantation lymphoproliferative disorder

- 15 A 42-year-old man in haemorrhagic shock has a lactate of 4.8 mmol/L. Which factor is least important in determining oxygen delivery to his tissues?
- A Cardiac output
 - B Haemoglobin concentration
 - C PaO₂
 - D SaO₂
 - E All contribute equally

- 16 A 67-year-old woman is intubated for pneumonia on a background of COPD. All of the following are measures that prevent ICU complications EXCEPT:
- A Enoxaparin prophylaxis
 - B Omeprazole 20 mg daily
 - C Aggressive (tight) blood glucose control
 - D Early mobilisation
 - E Standard central line care bundle

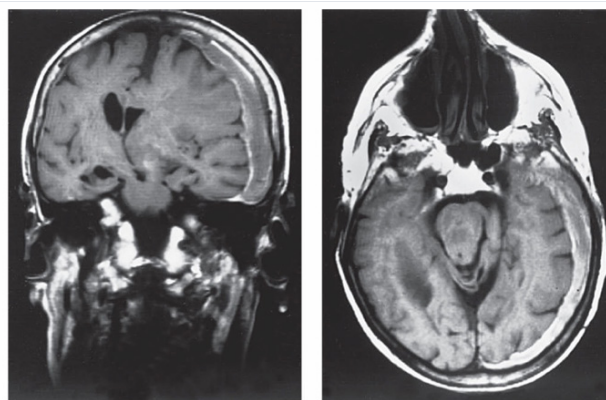
All of the following statements regarding shock pathophysiology are true EXCEPT:

- A Hypotension inhibits the vasomotor centre, reducing adrenergic output
- B Metabolic derangements raise extracellular osmolarity, pulling fluid from cells
- C Mitochondrial dysfunction leads to lactate accumulation
- D Proinflammatory mediators contribute to organ failure
- E Severe stress increases cortisol, promoting gluconeogenesis and lipolysis

A patient with a C2–3 spinal cord transection is in shock. Which haemodynamic profile is expected?

- A Low CO, high SVR, low RA pressure, low PCWP, low SvO₂
- B Low CO, high SVR, high RA pressure, high PCWP, low SvO₂
- C High CO, low SVR, low RA pressure, low PCWP, high SvO₂
- D Low CO, normal SVR, high RA pressure, low PCWP, low SvO₂
- E Low CO, low SVR, low RA pressure, low PCWP, low SvO₂

- 19** All of the following statements about the epidemiology of sepsis are true EXCEPT:
- A Gram-positive bacteria are the most commonly isolated causative organisms
 - B In septic shock, blood cultures are positive in 40%-70%
 - C Most sepsis occurs in patients with significant comorbidities
 - D Respiratory infections are the most common cause of sepsis
 - E Incidence of severe sepsis is ~3 per 1000 and rising
- 20** A 62-year-old man with right lower lobe pneumonia is confused, with RR 28/min, HR 118/min and BP 84/50 mmHg, which rises to 112/70 mmHg after 2 L of IV crystalloid without vasopressors. ABG on air: pH 7.46, PaCO₂ 4.3 kPa (32 mmHg), PaO₂ 10.1 kPa (76 mmHg); lactate 1.6 mmol/L. How should his condition be classified?
- A Refractory septic shock
 - B Septic shock
 - C Sepsis
 - D Bacteraemia
 - E Systemic inflammatory response syndrome without infection
- 21** A 68-year-old man with previous splenectomy is in septic shock with no clear source. Local rates of cephalosporin-resistant *Streptococcus pneumoniae* exceed 1%. Which empirical regimen is most appropriate?
- A Cefotaxime
 - B Cefotaxime plus vancomycin
 - C Levofloxacin
 - D Levofloxacin plus vancomycin
 - E Vancomycin alone
- 22** All of the following therapies initially looked promising but have not improved mortality in sepsis EXCEPT:
- A Activated protein C (drotrecogin alfa)
 - B Red cell transfusion only when haemoglobin falls below 70 g/L
 - C Intensive insulin therapy for tight glucose control
 - D ACTH stimulation testing to select patients for steroid therapy
 - E Tissue factor pathway inhibitor
- 23** Which of the following conditions can produce true coma, rather than being a state that mimics coma?
- A Akinetic mutism
 - B Catatonia
 - C Locked-in state
 - D Metabolic encephalopathy
 - E Persistent vegetative state
- 24** A 74-year-old man has a large subdural haematoma with uncal herniation (MRI shown). What is the cause of his coma?



- A** Compression of the anterior and posterior cerebral arteries
- B** Compression of the midbrain against the opposite tentorial edge
- C** Compression of the opposite cerebral peduncle
- D** Downward displacement of the cerebellar tonsils into the foramen magnum
- E** Entrapment of the ventricles causing hydrocephalus
- 25** A 44-year-old man is found comatose, flushed, with 5 mm pupils, muscle spasms and rapid breathing. What is the most likely cause of his coma?
- A Concussion
 - B Cortical vein thrombosis
 - C Drug intoxication
 - D Systemic infection
 - E Meningitis

Answers 1-25

HARRISON MCQS

- 1** **B** Functional residual capacity — At FRC the inward recoil of the lung equals the outward recoil of the chest wall; FRC rises in emphysema and falls in fibrosis or obesity.
- 2** **D** Subglottic stenosis — Flattening of both inspiratory and expiratory limbs of the flow-volume loop is a fixed central obstruction such as subglottic stenosis.
- 3** **B** Perform a computed tomography (CT) pulmonary angiogram — Acute dyspnoea with a raised A-a gradient in pregnancy needs imaging for PE (CTPA or V/Q); D-dimer is unhelpful in pregnancy.
- 4** **C** Decreased TLC, increased RV, normal FEV₁/FVC ratio, decreased MIP — Myasthenia: respiratory muscle weakness gives low TLC, raised RV, normal FEV₁/FVC and low MIP.
- 5** **D** Red blood cell cyanide concentration — Coma, seizures, shock and bitter almond breath after a house fire suggest cyanide; red cell cyanide confirms, but give hydroxocobalamin without waiting.
- 6** **E** Sweat chloride >60 mmol/L — Sweat chloride ≥ 60 mmol/L on two occasions with a compatible picture confirms cystic fibrosis.

- 7 E** Pulmonary alveolar proteinosis — Milky BAL with PAS-positive material is pulmonary alveolar proteinosis (usually autoimmune anti-GM-CSF); CT shows crazy paving.
- 8 E** Whole-lung lavage — Symptomatic PAP is treated by whole-lung lavage; inhaled GM-CSF is a newer option. Steroids do not help.
- 9 E** Weight loss will lead to improvements in PaCO₂ over time — Obesity hypoventilation (daytime PaCO₂ > 6 kPa, BMI ≥ 30) improves with weight loss; CPAP or bilevel NIV at night.
- 10 B** Home sleep apnoea test — **UPDATED** Uncomplicated adult with high probability of OSA: a home sleep apnoea test is the first step; full polysomnography if complex or inconclusive.
- 11 C** It increases the risk of motor vehicle accidents — **UPDATED** Untreated OSA roughly doubles to triples the risk of road accidents; advise not to drive while sleepy.
- 12 E** A 68-year-old with atrial fibrillation and ischaemic cardiomyopathy (EF 15%), raised JVP and oedema — Crescendo-decrescendo breathing with central apnoeas is Cheyne–Stokes respiration, typical of advanced heart failure.
- 13 A** Continuous positive airway pressure (CPAP) — AHI ≥ 30/h is severe OSA: CPAP is first line and improves sleepiness and BP. Oral devices suit mild disease; weight loss helps but is not enough alone.
- 14 A** Bronchiolitis obliterans syndrome — Bronchiolitis obliterans syndrome (chronic allograft dysfunction) is the main barrier to long-term survival after lung transplant.
- 15 C** PaO₂ — O₂ delivery = cardiac output × (1.34 × Hb × SaO₂ + small dissolved fraction), so PaO₂ matters least once saturated.
- 16 C** Aggressive (tight) blood glucose control — Tight glucose control (4.4–6.1 mmol/L) increased mortality (NICE-SUGAR); target ~8–10 mmol/L.
- 17 A** Hypotension inhibits the vasomotor centre, reducing adrenergic output — Hypotension reduces baroreceptor firing, which increases (not inhibits) sympathetic output in shock.
- 18 E** Low CO, low SVR, low RA pressure, low PCWP, low SvO₂ — **UPDATED** High cervical cord transection: neurogenic shock with low SVR, low filling pressures, bradycardia, low CO and low SvO₂.
- 19 A** Gram-positive bacteria are the most commonly isolated causative organisms — Gram-negative organisms remain the commonest isolates in sepsis (~60%); respiratory infection is the commonest source.
- 20 C** Sepsis — **UPDATED** Pneumonia with confusion and hypotension responding to fluids is sepsis (Sepsis-3); septic shock needs vasopressors and lactate > 2.
- 21 B** Cefotaxime plus vancomycin — Asplenic septic shock: cefotaxime/ceftriaxone for encapsulated organisms plus vancomycin where resistant pneumococci are prevalent.
- 22 B** Red cell transfusion only when haemoglobin falls below 70 g/L — A restrictive transfusion threshold (Hb 70 g/L) is safe in sepsis (TRISS); activated protein C and tight glucose control failed.
- 23 D** Metabolic encephalopathy — **UPDATED** Diffuse metabolic encephalopathy causes true coma; vegetative state, akinetic mutism, catatonia and locked-in syndrome mimic it.
- 24 B** Compression of the midbrain against the opposite tentorial edge — In uncal herniation, coma results from the midbrain being pushed against the opposite tentorial edge, distorting the reticular activating system.
- 25 C** Drug intoxication — Coma without focal signs, with flushing, reactive pupils, twitching and hyperventilation suggests drug intoxication; toxicology screen.

SET 2 OF 2

- 26** In which patient is intraparenchymal intracranial pressure monitoring preferred over a ventriculostomy?
- A A 24-year-old with traumatic brain injury and GCS 6
 B A 35-year-old with fulminant hepatic failure and INR 3.5
 C A 48-year-old man with subarachnoid haemorrhage and hydrocephalus
 D A 68-year-old with a large haemorrhagic stroke and midline shift
 E All of the above
- 27** A 56-year-old woman with subarachnoid haemorrhage has an external ventricular drain in place, and her intracranial pressure rises to 45 mmHg. What is the next management step?
- A Hypertonic saline 23.4% bolus
 B Dexamethasone 4 mg every 6 hours
 C No action, as cerebral perfusion pressure is adequate
 D Open the ventriculostomy to drain CSF to an ICP below 20–25 mmHg
 E Hyperventilate to a PaCO₂ of 4 kPa (30 mmHg)
- 28** A 67-year-old man has been in the ICU with sepsis for 3 weeks and now cannot be weaned from the ventilator. He has symmetrical weakness of all limbs, worse distally, with absent tendon reflexes and reduced distal sensation; nerve conduction studies show an axonal sensorimotor polyneuropathy. What is the most likely cause?
- A Critical illness myopathy
 B Critical illness polyneuropathy
 C Fibroproliferative ARDS changes
 D Guillain–Barré syndrome
 E ICU-associated delirium
- 29** A 56-year-old woman with a subarachnoid haemorrhage improves initially, but on day 7 has an abrupt neurological decline. CT shows no new blood. What is the most likely cause?
- A Cerebral oedema
 B Hydrocephalus
 C Hyponatraemia
 D Aneurysm rerupture
 E Vasospasm

- 30** A 56-year-old woman develops delayed neurological deterioration on day 7 after a coiled aneurysmal subarachnoid haemorrhage. CT shows no new blood, and CT angiography confirms severe cerebral vasospasm. She is already taking nimodipine. What treatment do you recommend now?
- A Neurosurgical clipping of the aneurysm
 - B Free water restriction
 - C Ventriculostomy placement
 - D Hyperventilation and mannitol
 - E IV noradrenaline to raise blood pressure and referral for balloon angioplasty
- 31** A 21-year-old with acute lymphoblastic leukaemia and a white cell count of $156 \times 10^9/L$ starts induction chemotherapy. Which is NOT a feature of tumour lysis syndrome?
- A Acute kidney injury
 - B Hypercalcaemia
 - C Hyperkalaemia
 - D Hyperphosphataemia
 - E Hyperuricaemia
- 32** All of the following are important in preventing tumour lysis syndrome EXCEPT:
- A Allopurinol prophylaxis
 - B Adequate IV hydration (about 3 L/m² per day)
 - C Rasburicase for patients at high risk
 - D Frequent serum biochemistry (every 4–6 hours) in high-risk patients
 - E Prophylactic haemodialysis before chemotherapy
- 33** In which of the following patients presenting to the emergency department with acute dyspnoea would a positive D-dimer result prompt further testing for pulmonary embolism?
- A A 24-year-old woman who is 32 weeks pregnant
 - B A 48-year-old man with no past medical history who has calf pain after prolonged air travel and a normal alveolar-arterial oxygen gradient
 - C A 56-year-old woman receiving chemotherapy for breast cancer
 - D A 62-year-old man who had hip replacement surgery 4 weeks ago
 - E A 72-year-old man who had an acute myocardial infarction 2 weeks ago
- 34** A 42-year-old woman presents with acute breathlessness 2 days after developing calf pain and swelling following a 9-hour car journey. BP 98/60 mmHg, HR 114/min, RR 28/min, SpO₂ 92% on room air, weight 89 kg; the lungs are clear and the right leg is swollen and tender. ABG: pH 7.52, PaCO₂ 3.3 kPa (25 mmHg), PaO₂ 9.1 kPa (68 mmHg). Kidney and liver function are normal, and CT confirms pulmonary embolism. All of the following agents can be used alone as initial therapy EXCEPT:
- A Enoxaparin 1 mg/kg subcutaneously (SC) twice daily
 - B Fondaparinux 7.5 mg SC once daily
 - C Tinzaparin 175 units/kg SC once daily
 - D Unfractionated heparin IV adjusted to maintain activated partial thromboplastin time (aPTT) two to three times the upper limit of normal
 - E Warfarin 7.5 mg orally once daily to maintain international normalized ratio at 2–3
- 35** Which of the following statements regarding pulmonary embolism is true?
- A Airway resistance usually decreases in the setting of an acute pulmonary embolism
 - B Almost all patients with pulmonary embolism have evidence of deep vein thrombosis at the time
 - C Alveolar hyperventilation is a typical physiologic abnormality in the presence of pulmonary embolism
 - D Hypotension in the setting of acute pulmonary embolism is often due to acute left ventricular systolic dysfunction
 - E The size of the arterial-alveolar gradient is invariably correlated with the size of the pulmonary embolism
- 36** A 57-year-old man with hypertension is admitted with pulmonary embolism. BP 132/62 mmHg, HR 85/min, SpO₂ 95% on room air; echocardiography shows normal right heart size and function and troponin is undetectable. Doppler shows extensive femoral–popliteal DVT. He is started on LMWH with warfarin and asks whether he should have an inferior vena cava (IVC) filter. What is the most appropriate answer?
- A “No, you would not benefit from an IVC filter at this point.”
 - B “No, there are currently no indications for IVC filter placement in the setting of deep vein thrombus or pulmonary embolism.”
 - C “We will need to go back to review your images and see how large your pulmonary embolus clot burden is before deciding whether an IVC filter is appropriate for you or not.”
 - D “Yes, given the presence of residual lower extremity thrombus, we will schedule you for IVC filter insertion.”
 - E “Yes, given your age and gender, we will schedule you for an IVC filter insertion.”

- 37** A 45-year-old woman with morbid obesity (weight 140 kg), hypertension and diabetes is on day 4 after open gastric bypass surgery. She develops sudden breathlessness, tachycardia, hypoxia and a sense of impending doom. CT pulmonary angiography shows a saddle pulmonary embolism. Her blood pressure is 70/45 mmHg and urine output is poor. After intravenous 0.9% saline and dobutamine, her blood pressure improves only modestly. What is the most appropriate next step?
- A Subcutaneous enoxaparin
 - B Inferior vena cava filter
 - C Surgical pulmonary embolectomy
 - D Systemic alteplase (tPA)
 - E Systemic urokinase

Answers 26–37

HARRISON MCQs

- 26** **B** A 35-year-old with fulminant hepatic failure and INR 3.5 — With coagulopathy (fulminant hepatic failure) an intraparenchymal ICP monitor is preferred; an EVD carries more bleeding risk.
- 27** **D** Open the ventriculostomy to drain CSF to an ICP below 20–25 mmHg — With an EVD in place and ICP 45 mmHg, drain CSF first (target < 20–25 mmHg), then osmotherapy if needed.
- 28** **B** Critical illness polyneuropathy — **UPDATED** Axonal sensorimotor polyneuropathy after weeks of sepsis causing failure to wean is critical illness polyneuropathy.
- 29** **E** Vasospasm — Neurological decline on day 7 after SAH without new blood is vasospasm (delayed cerebral ischaemia).
- 30** **E** IV noradrenaline to raise blood pressure and referral for balloon angioplasty — Symptomatic vasospasm after a secured aneurysm: euvolaemia and induced hypertension (noradrenaline) plus angioplasty.
- 31** **B** Hypercalcaemia — Tumour lysis causes high K⁺, phosphate and urate with hypocalcaemia (calcium bound by phosphate), not hypercalcaemia.
- 32** **E** Prophylactic haemodialysis before chemotherapy — **UPDATED** TLS prevention: hydration, allopurinol or rasburicase and monitoring; dialysis is for established TLS, not prophylaxis.
- 33** **B** A 48-year-old man with no past medical history who has calf pain after prolonged air travel and a normal alveolar–arterial oxygen gradient — A positive D-dimer is meaningful only in low/intermediate-probability outpatients without pregnancy, cancer or recent surgery.
- 34** **E** Warfarin 7.5 mg orally once daily to maintain international normalized ratio at 2–3 — Warfarin cannot be used alone initially (slow onset, early prothrombotic effect); LMWH, fondaparinux, UFH or apixaban/rivaroxaban can.
- 35** **C** Alveolar hyperventilation is a typical physiologic abnormality in the presence of pulmonary embolism — PE typically causes alveolar hyperventilation with hypocapnia; hypotension results from RV failure.
- 36** **A** “No, you would not benefit from an IVC filter at this point.” — An IVC filter is only for contraindication to or failure of anticoagulation; it does not improve survival.
- 37** **C** Surgical pulmonary embolectomy — Massive PE with shock 4 days after major surgery (thrombolysis contraindicated): surgical or catheter embolectomy.

1 QUESTIONS • AFTER CRASH COURSE SBAS & EMQs

Crash Course MCQs

- 1** An overweight middle-aged man sees his GP after his wife complains about his loud snoring. He drinks large amounts of alcohol in the evenings before bed. Which of the following symptoms is he most likely to report?
- A Morning headaches
 - B Nocturnal choking
 - C Reduced libido
 - D Daytime sleepiness
 - E Personality changes

Answer 1

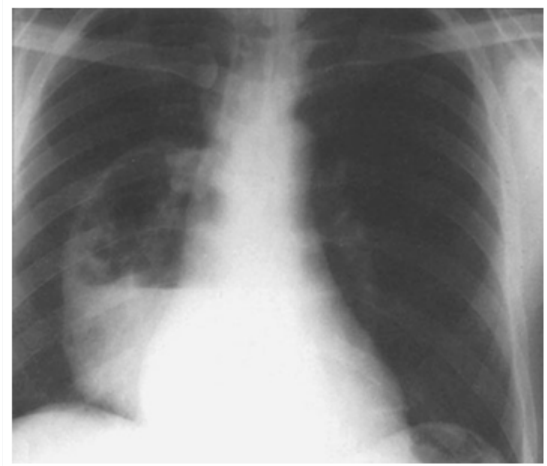
- 1 **D** Daytime sleepiness — Daytime sleepiness is the cardinal OSA symptom; obesity and evening alcohol worsen airway collapse.

9 QUESTIONS • AFTER PRETEST MEDICINE

PreTest MCQs

- 1 A 30-year-old man with tetraplegia and a long-term indwelling urinary catheter presents with fever, breathlessness and cough. He has had recurrent urinary tract infections for years and takes long-term prophylactic therapy to suppress them. There is mild bilateral wheeze and a diffuse erythematous rash. Chest X-ray shows diffuse alveolar infiltrates. WBC $13.5 \times 10^9/L$ (neutrophils 50%, lymphocytes 30%, eosinophils 20%). What is the most likely diagnosis?
- A Sepsis with ARDS secondary to urinary tract infection
 B Healthcare-associated pneumonia
 C A drug reaction to one of his medications
 D An acute exacerbation of COPD
 E Lymphocytic interstitial pneumonitis
- 2 A 68-year-old woman with asthma attends for a routine review. She has occasional palpitations and tremor, but her breathlessness is well controlled. She was admitted with mild heart failure 2 months ago and has occasional postprandial acid reflux. Her medications are lisinopril, digoxin, furosemide, an as-required short-acting inhaled beta-agonist and theophylline, and she takes an over-the-counter tablet (name unknown) for reflux. Her heart rate is 112/min, heart sounds are normal and there is a mild tremor of the outstretched hands. What is the best next step?
- A Chest X-ray to exclude heart failure
 B Serum theophylline level
 C Spirometry before and after bronchodilator
 D Lorazepam 0.5 mg orally three times daily
 E Stop the beta-agonist and substitute inhaled ipratropium

- 3 A 40-year-old man with alcohol dependence develops cough and fever. His chest X-ray (shown) demonstrates a cavity with an air–fluid level in the superior segment of the right lower lobe. What is the most likely causative agent?



- A Streptococcus pneumoniae
 B Haemophilus influenzae
 C Legionella pneumophila
 D Anaerobes
 E Mycoplasma pneumoniae
- 4 A 25-year-old man is referred for evaluation of infertility. He has had a productive cough and recurrent chest infections all his life, together with chronic abdominal pain, diarrhoea and difficulty gaining weight. He also has diabetes mellitus. His chest X-ray suggests bronchiectasis. What is the most likely diagnosis?
- A COPD
 B Primary ciliary dyskinesia
 C Cystic fibrosis
 D Common variable immunodeficiency
 E Asthma

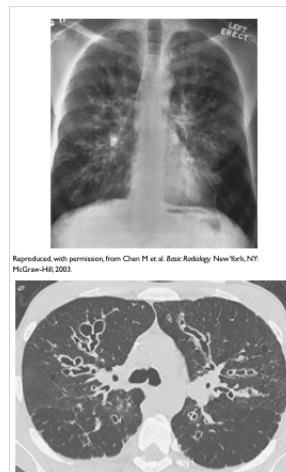
5 A previously well 40-year-old man presents with a 3-day history of fever and shaking chills, including a 15-minute rigor. He also has a non-productive cough, anorexia and right-sided pleuritic chest pain, and has been breathless for 12 hours. Chest X-ray shows right middle lobe consolidation, and the full blood count shows neutrophilia with many band forms. Which of the following statements about this patient's pneumonia is correct?

- A If the sputum Gram stain shows many squamous epithelial cells and culture grows mixed flora, the patient probably has a viral infection
- B If the Gram stain shows numerous Gram-positive diplococci, numerous white cells and few epithelial cells, *Streptococcus pneumoniae* is the most likely pathogen
- C Although *S. pneumoniae* is the most likely cause, the diagnosis would be unlikely if blood cultures are negative
- D The absence of rigors would rule out pneumococcal pneumonia
- E A negative urinary pneumococcal antigen test excludes pneumococcal pneumonia

6 A 60-year-old man becomes short of breath 2 days after cholecystectomy. He has no fever, chills, sputum or pleuritic chest pain. Temperature 37.2 °C, pulse 75/min, respiratory rate 20/min, BP 120/70 mmHg. There are diminished breath sounds and dullness at the left base, and the trachea is shifted to the left. Chest X-ray shows a retrocardiac opacity obscuring the left hemidiaphragm. What is the most likely underlying problem?

- A An acute inflammatory process (pneumonia)
- B A left lower lobe mass
- C Loss of volume of the left lower lobe (postoperative atelectasis)
- D Acute bronchospasm caused by surgery
- E Acute pneumothorax

7 A 55-year-old woman with long-standing chronic lung disease and recurrent episodes of acute bronchitis reports that her sputum, now produced daily and thick, has increased markedly over several months. There are flecks of blood in the sputum and she has lost 3.6 kg (8 lb). She has no fever or chills, and sputum cultures have not grown specific pathogens. Her chest X-ray and CT are shown. What is the most likely cause of her symptoms?



- A Pulmonary tuberculosis
- B Exacerbation of chronic lung disease
- C Bronchiectasis
- D Anaerobic lung abscess
- E Carcinoma of the lung

8 A 20-year-old firefighter attends the emergency department with headache and dizziness after helping to put out a garage fire. He is not breathless, and arterial blood gas shows a normal PaO₂. What is the best first step in management?

- A Give high-flow 100% oxygen via a non-rebreather mask
- B Obtain a chest X-ray
- C Reassure him, as a normal PaO₂ excludes significant poisoning
- D Obtain a CT scan of the head
- E Evaluate for anaemia

9 An 18-year-old man's family have noticed blue discoloration of his nose, ears and fingers over the past few weeks. Which of the following statements is true?

- A This patient has peripheral cyanosis
- B Total haemoglobin is important in determining whether a patient will have cyanosis
- C Central cyanosis with hypoxia suggests methaemoglobinemia or sulphaemoglobinemia
- D The presence of clubbing in a cyanotic patient confirms an acute cause of the cyanosis
- E Large amounts of methaemoglobin are required to produce cyanosis

Answers 1–9

PRETEST MCQs

- 1 **C** A drug reaction to one of his medications — Fever, wheeze, rash, infiltrates and 20% eosinophils on long-term UTI suppression is a drug (nitrofurantoin) reaction; stop it.
- 2 **B** Serum theophylline level — Tachycardia and tremor on theophylline (narrow range 10–20 mg/L, raised by cimetidine) suggest toxicity: check the level.
- 3 **D** Anaerobes — A cavity with air–fluid level in the superior segment of the right lower lobe in an alcoholic is an aspiration abscess from anaerobes.
- 4 **C** Cystic fibrosis — **UPDATED** Bronchiectasis with pancreatic insufficiency, diabetes and male infertility (absent vas) is cystic fibrosis; sweat test and genotyping.
- 5 **B** If the Gram stain shows numerous Gram-positive diplococci, numerous white cells and few epithelial cells, *Streptococcus pneumoniae* is the most likely pathogen — **UPDATED** A good sputum sample with Gram-positive diplococci and many neutrophils strongly supports *S. pneumoniae*.
- 6 **C** Loss of volume of the left lower lobe (postoperative atelectasis) — Tracheal shift towards a retrocardiac opacity after upper abdominal surgery is left lower lobe collapse (atelectasis); physiotherapy.
- 7 **C** Bronchiectasis — **UPDATED** Dilated thick-walled bronchi larger than their arteries (signet ring) with daily sputum is bronchiectasis; investigate the cause.
- 8 **A** Give high-flow 100% oxygen via a non-rebreather mask — **UPDATED** CO poisoning after smoke exposure (normal PaO₂): give 100% oxygen via non-rebreather immediately; measure COHb by co-oximetry.
- 9 **B** Total haemoglobin is important in determining whether a patient will have cyanosis — Cyanosis depends on total Hb (needs ~40–50 g/L deoxy-Hb): easy in polycythaemia, absent in severe anaemia.

10 QUESTIONS • AFTER PASSMEDICINE

Passmedicine MCQs

- 1 A 62-year-old man presents with breathlessness, wheeze and rhinorrhoea. On examination there is bilateral expiratory wheeze. Blood results: Hb 120 g/L, platelets 228×10^9 /L, WBC 14.2×10^9 /L, eosinophils 2.2×10^9 /L, Na⁺ 138 mmol/L, K⁺ 4.1 mmol/L, urea 5.8 mmol/L, creatinine 62 µmol/L, CRP 8 mg/L. p-ANCA is positive. Which of the following drugs should be avoided?
 - A Corticosteroids
 - B Ciclosporin
 - C Ipratropium
 - D Montelukast
 - E Salbutamol
- 2 A 23-year-old woman is started on varenicline to help her stop smoking. Which one of the following adverse effects is most likely to occur?
 - A Vivid dreams
 - B Nausea
 - C Constipation
 - D Insomnia
 - E Drug-induced lupus
- 3 A 47-year-old man is reviewed in the smoking cessation clinic. Which one of the following conditions would contraindicate the prescription of bupropion?
 - A History of supraventricular tachycardia
 - B Previous episodes of acute pancreatitis
 - C Epilepsy
 - D Depression
 - E Hypertension
- 4 A 77-year-old man with type 2 diabetes mellitus is admitted with worsening breathlessness. His GP started amoxicillin 5 days ago. BP 88/60 mmHg, respiratory rate 36/min. Chest X-ray shows left lower lobe consolidation. Arterial blood gas on air: pH 7.37, PaCO₂ 5.5 kPa, PaO₂ 9.1 kPa. What is the most suitable antibiotic therapy?
 - A Oral amoxicillin + erythromycin
 - B Intravenous ceftriaxone
 - C Intravenous co-amoxiclav + clarithromycin
 - D Intravenous vancomycin + clarithromycin
 - E Oral co-amoxiclav + erythromycin
- 5 A 22-year-old with cystic fibrosis is assessed for lung transplantation after multiple chest infections and a poor response to chest physiotherapy, nutritional supplementation and a trial of lumacaftor/ivacaftor. He meets the criteria and is placed on the transplant waiting list. Chronic infection with which organism would be a contraindication to transplantation?
 - A *Aspergillus fumigatus*
 - B *Burkholderia cepacia*
 - C *Haemophilus influenzae*
 - D *Pseudomonas aeruginosa*
 - E *Scedosporium apiospermum*

- 6** A 64-year-old woman with asthma is referred with worsening breathlessness over the past few months. Her asthma was diagnosed in her teens; she uses an inhaled corticosteroid and rarely needs her salbutamol. She has no wheeze or nocturnal symptoms, her inhaler technique is excellent and she is adherent. She has no other medical problems and does not smoke. She is comfortable at rest, SpO₂ is 96% on air, BMI is 30 kg/m² and the chest is clear. Chest X-ray shows a normal heart size and clear lung fields. Lung function: FEV₁ 3.30 L (96.1%), FVC 3.92 L (88.6%), FEV₁/FVC 0.84, TLC 77.4%, KCO 101.9%. What is the likely cause of her breathlessness?
- Obesity
 - Acute exacerbation of asthma
 - Pulmonary haemorrhage
 - Pulmonary fibrosis
 - Polycythaemia
- 7** A 76-year-old woman with alcohol dependence is in hospital receiving antibiotics for a urinary tract infection. Over the last few days she has developed a cough productive of thick, blood-stained, jelly-like sputum and feels very nauseous. Chest X-ray shows bilateral cavitating upper-lobe opacities. What is the most likely causative organism?
- Klebsiella pneumoniae
 - Pseudomonas aeruginosa
 - Mycoplasma pneumoniae
 - Legionella pneumophila
 - Staphylococcus aureus
- 8** A 54-year-old man who lives alone and works at a bird sanctuary is admitted with a 10-day history of worsening breathlessness, cough and fever. He is started on intravenous co-amoxiclav, but after 72 hours there is no clinical improvement. PCR of a respiratory sample is positive for Chlamydia psittaci. What is the most appropriate action?
- Continue co-amoxiclav
 - Switch to co-trimoxazole
 - Switch to doxycycline
 - Switch to levofloxacin
 - Switch to piperacillin-tazobactam
- 9** A 29-year-old man on an expedition on Mount Everest develops a headache at about 5000 m, followed by poor coordination and slurred speech. What is the most appropriate treatment?
- Descent and acetazolamide
 - Descent and dexamethasone
 - Descent and mannitol
 - Descent and nifedipine
 - Descent and oxygen
- REPEATED 2x · ALSO ASKED IN**
Passmedicine
- 10** A 34-year-old man climbing Mount Kilimanjaro has had nausea and headache for 2 days. At 4500 m he develops breathlessness and a cough with pink frothy sputum, and there are bibasal crackles. Other than descent, what is the most appropriate treatment?
- Nifedipine
 - Furosemide
 - Mannitol
 - Hydralazine
 - Third-generation cephalosporin

Answers 1-10

- 1** **D** Montelukast — **UPDATED** Late asthma with eosinophilia and positive p-ANCA suggests EGPA; montelukast may unmask it and is best avoided.
- 2** **B** Nausea — **UPDATED** Nausea is the commonest varenicline adverse effect (~a third); vivid dreams and insomnia also occur.
- 3** **C** Epilepsy — Bupropion lowers the seizure threshold and is contraindicated in epilepsy.
- 4** **C** Intravenous co-amoxiclav + clarithromycin — CURB-65 ≥ 3 pneumonia failing amoxicillin: IV co-amoxiclav plus clarithromycin.
- 5** **B** Burkholderia cepacia — **UPDATED** Chronic Burkholderia cepacia (especially cenocepacia) infection worsens transplant survival and is usually a contraindication.
- 6** **A** Obesity — No obstruction, clear CXR and a slightly low TLCO with normal KCO (small alveolar volume) point to obesity as the cause of breathlessness.
- 7** **A** Klebsiella pneumoniae — **UPDATED** Upper-lobe cavitating pneumonia in an alcohol-dependent inpatient is classically Klebsiella pneumoniae ('redcurrant jelly' sputum).
- 8** **C** Switch to doxycycline — **UPDATED** C. psittaci is intracellular, so beta-lactams fail: switch to doxycycline 100 mg bd for 10–14 days (macrolide if pregnant).
- 9** **B** Descent and dexamethasone — Ataxia and slurred speech after ascent is high-altitude cerebral oedema: descend immediately and give dexamethasone (and oxygen).
- 10** **A** Nifedipine — **UPDATED** Pink frothy sputum and crackles at altitude is HAPE: descent, oxygen and nifedipine to lower pulmonary artery pressure.

10 QUESTIONS • AFTER PASTEST

PasTest MCQs

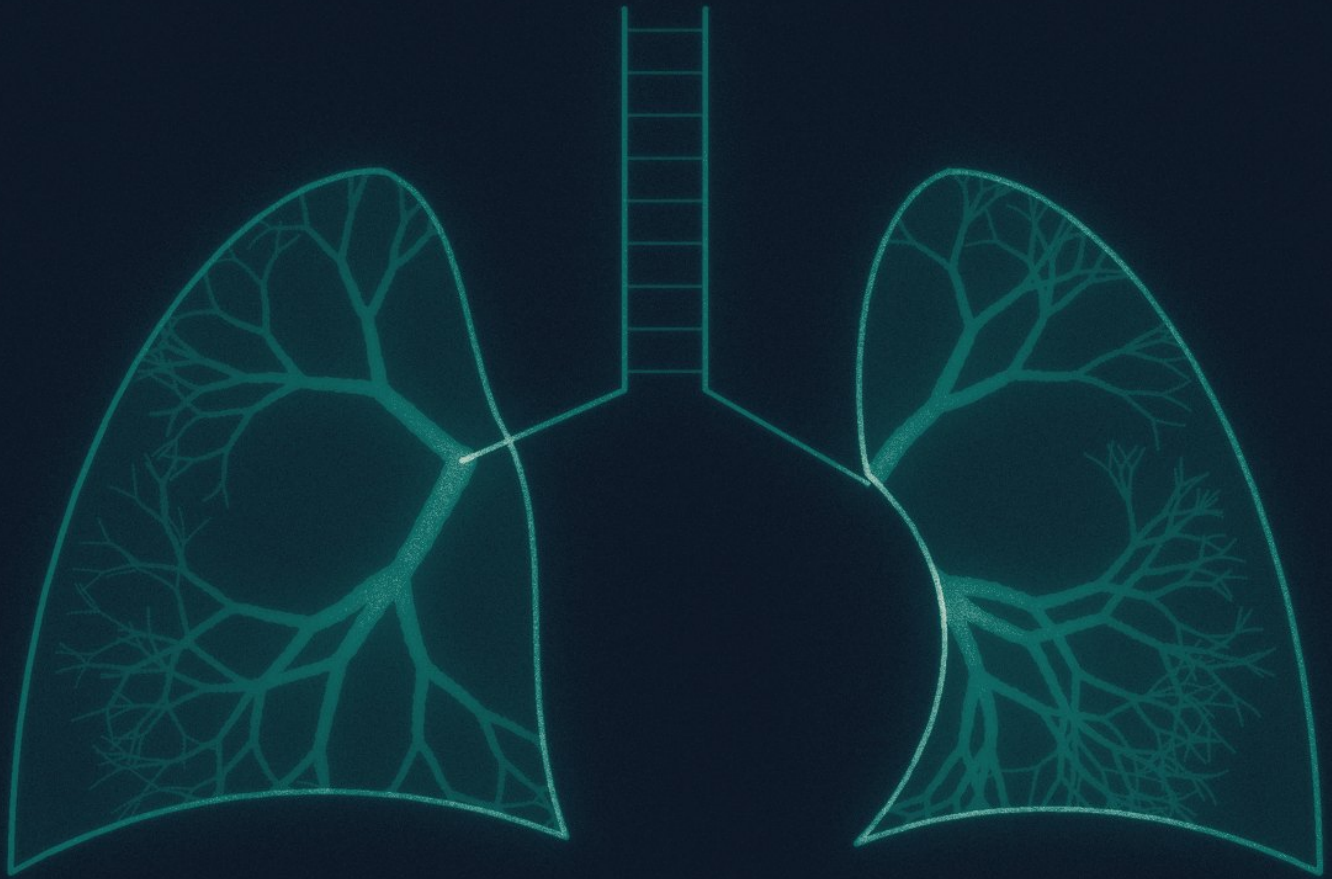
- 1 An 18-year-old woman has an acute pulmonary embolism at 9 weeks of pregnancy. There are no obvious risk factors. What is the most appropriate treatment for the remainder of her pregnancy?
 - A Aspirin
 - B Intravenous unfractionated heparin
 - C Subcutaneous low-molecular-weight heparin
 - D Subcutaneous unfractionated heparin
 - E Warfarin
- 2 A 51-year-old woman with asthma and severe rheumatoid arthritis presents with 3 days of cough, wheeze and fever following a cold. She has taken naproxen for several years without any respiratory symptoms. Chest X-ray is clear, peak flow is 380 L/min (predicted 510 L/min) and CRP is 49 mg/L. She has already received nebulised salbutamol in the clinic with only partial improvement. Which single treatment is most appropriate now?
 - A Oral amoxicillin
 - B Nebulised ipratropium bromide
 - C Oral prednisolone
 - D Stop naproxen
 - E Oral theophylline
- 3 A 26-year-old man presents with fever, headache and a non-productive cough. He has mildly raised liver enzymes, raised lactate dehydrogenase (LDH) and anaemia and he is positive for cold agglutinins. What is the most appropriate treatment?
 - A Ampicillin
 - B Clindamycin
 - C Erythromycin
 - D Imipenem
 - E Piperacillin
- 4 A 46-year-old woman is admitted with acute breathlessness. Arterial blood gas on air: PaO₂ 7.4 kPa, PaCO₂ 3.8 kPa, pH 7.45. Peak expiratory flow is 250 L/min (about 50% of predicted) and the chest X-ray is normal. What is the most appropriate initial management step?
 - A Heparin infusion
 - B Intravenous aminophylline
 - C Intubate and ventilate
 - D Ipratropium nebuliser
 - E Nebulised salbutamol
- 5 A 59-year-old man is reviewed in clinic 3 months after coronary artery bypass grafting. He is breathless on exertion and particularly when lying flat in bed. The operation went well and he has no angina. His medications are ramipril 10 mg, bisoprolol 5 mg and atorvastatin 20 mg. He looks well; BP 123/72 mmHg, pulse 70/min and regular, BMI 31 kg/m², and the chest is clear. Hb 131 g/L, WCC 5.6×10^9 /L, platelets 178×10^9 /L, Na⁺ 138 mmol/L, K⁺ 4.8 mmol/L, creatinine 110 µmol/L. Vital capacity is 88% of predicted sitting and falls to 55% of predicted lying down. What is the most likely cause of his breathlessness?
 - A Diaphragmatic palsy
 - B Heart failure
 - C Obesity
 - D Pulmonary fibrosis
 - E Renal artery stenosis
- 6 An anxious 22-year-old woman has had mild exertional breathlessness and intermittent double vision over the last few weeks. Chest X-ray, lung function tests and ECG are all normal. What is the most likely diagnosis?
 - A Eaton–Lambert syndrome
 - B Myasthenia gravis
 - C Somatisation disorder
 - D Transient ischaemic attack
 - E Unstable angina
- 7 A 37-year-old man with increasing breathlessness is found to have unilateral emphysema. Which of the following is most likely to have predisposed him to this?
 - A Allergic bronchopulmonary aspergillosis
 - B Bronchial adenoma
 - C Childhood bronchiolitis
 - D Crohn's disease
 - E Hypogammaglobulinaemia
- 8 In which of the following respiratory diseases is finger clubbing NOT a feature?
 - A Asbestosis
 - B Bronchial carcinoma
 - C Chronic bronchitis
 - D Idiopathic pulmonary fibrosis
 - E Lung abscess
- 9 Which one of the following features is seen least frequently in patients with obstructive sleep apnoea syndrome?
 - A Daytime sleepiness
 - B Female gender
 - C Hypertension
 - D Large neck size
 - E Snoring

- 10 A 34-year-old woman who does not smoke is found to have normal lung volumes but a reduced transfer factor for carbon monoxide (TLCO) of 45% of predicted. What is the most likely cause?
- A Acute asthma
 - B Emphysema
 - C Goodpasture syndrome
 - D Polycythaemia
 - E Pulmonary arteriovenous malformation

Answers 1-10

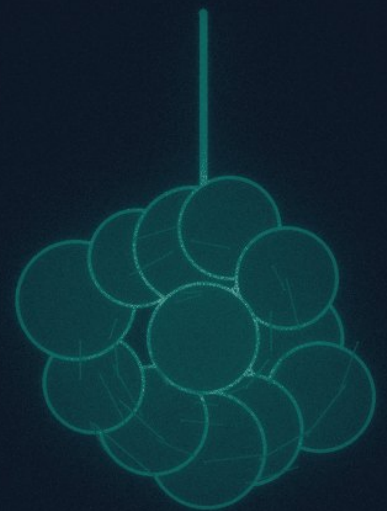
PASTEST MCQS

- 1 **C** Subcutaneous low-molecular-weight heparin — PE in pregnancy: weight-adjusted LMWH for the rest of pregnancy and ≥ 6 weeks postpartum; warfarin and DOACs are avoided.
- 2 **C** Oral prednisolone — **UPDATED** Moderate viral-triggered asthma exacerbation with partial SABA response: oral prednisolone 40–50 mg for ≥ 5 days.
- 3 **C** Erythromycin — Mycoplasma (cold agglutinins, haemolysis) lacks a cell wall: use a macrolide (erythromycin) or doxycycline, not beta-lactams.
- 4 **E** Nebulised salbutamol — Acute severe asthma with $\text{PaO}_2 < 8$ kPa: oxygen to 94–98% and high-dose nebulised salbutamol first.
- 5 **A** Diaphragmatic palsy — A fall in VC > 20 –25% from upright to supine indicates diaphragm weakness — here phrenic palsy after cardiac surgery.
- 6 **B** Myasthenia gravis — Fluctuating diplopia with breathlessness in a young woman suggests myasthenia gravis; antibodies, EMG and CT for thymoma.
- 7 **C** Childhood bronchiolitis — Unilateral hyperlucent emphysematous lung is Swyer–James (MacLeod) syndrome after childhood obliterative bronchiolitis.
- 8 **C** Chronic bronchitis — Clubbing is not a feature of chronic bronchitis/COPD; it occurs in cancer, fibrosis, asbestosis and chronic suppurative lung disease.
- 9 **B** Female gender — OSA is 2–3 times commoner in men, so female gender is the least frequent feature listed.
- 10 **E** Pulmonary arteriovenous malformation — Isolated low TLCO with normal volumes suggests pulmonary vascular disease, e.g. arteriovenous malformation (HHT); bubble echo or CTPA.



19

RESPIRATORY EMQS



Respiratory EMQs

Extended matching questions on the respiratory system: breathlessness, haemoptysis, clubbing, chest X-rays, blood gases and pleural fluid.

0

MCQS

214

EMQ SCENARIOS

4

SOURCES

6

REPEATED IDEAS

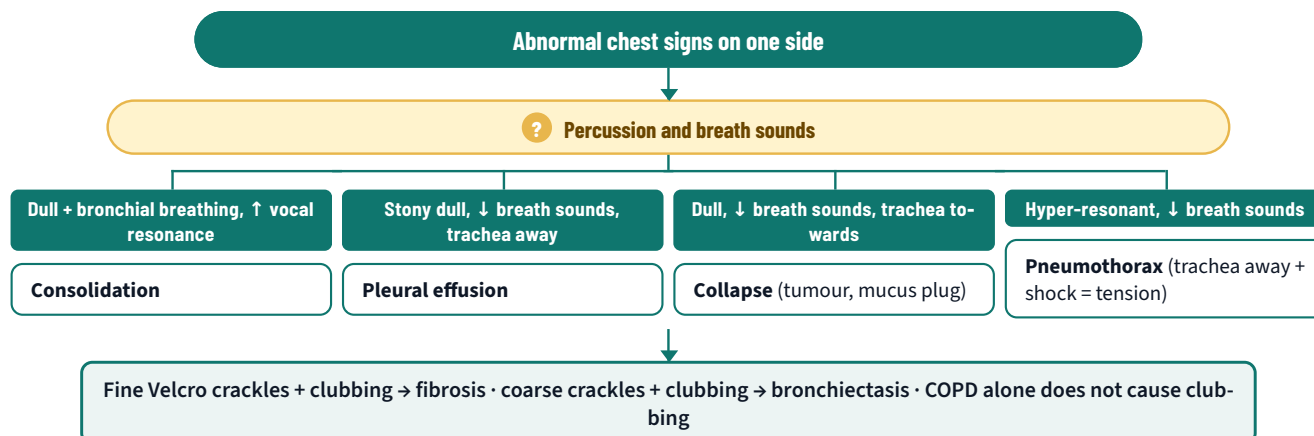
IN THIS CHAPTER

01	Management flowcharts	3 charts	page 310
02	Local Exams EMQs	90 Q	page 311
03	Crash Course EMQs	38 Q	page 320
04	Irfan EMQs	26 Q	page 324
05	Get Ahead EMQs	60 Q	page 327

Step by step

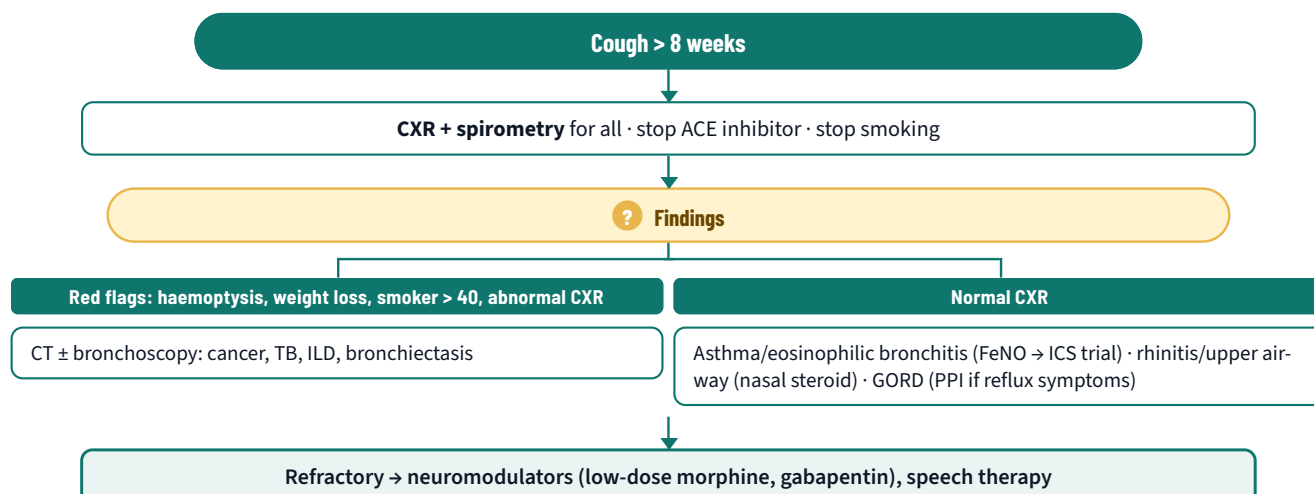
Chest examination decoder

CLINICAL EXAMINATION



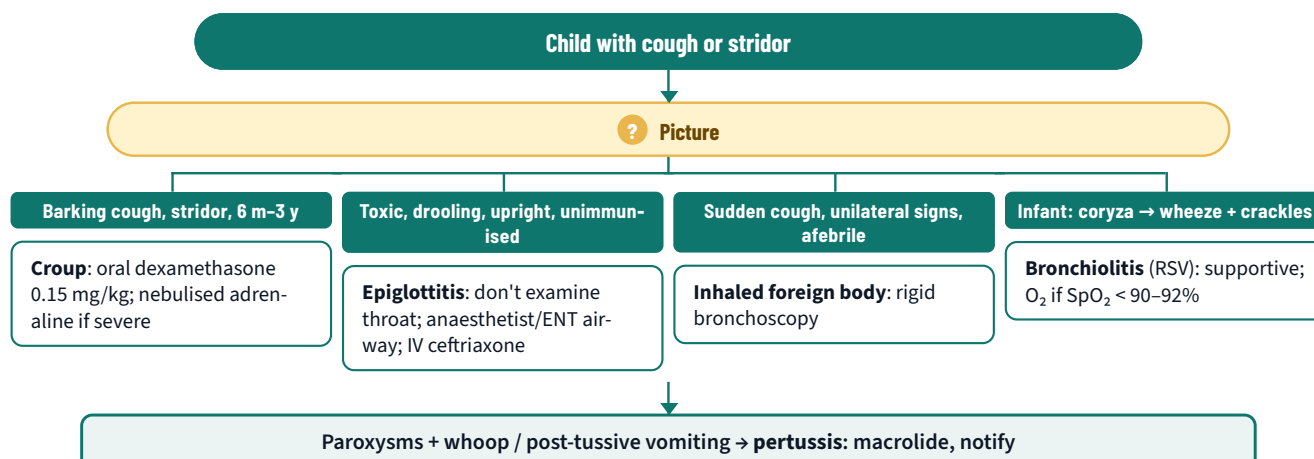
Chronic cough

ERS 2020 · BTS



The child with cough or stridor

NICE NG9 · RCPCH



Local Exams EMQs

SET 1 OF 4

39 BLOCKS QUESTIONS

End-block papers of the 39th batch (2024–2025)

THEME 1 Presentations of breathlessness

- A Bronchial asthma
- B Bronchiectasis
- C Lobar pneumonia
- D Mitral regurgitation
- E Cardiac tamponade
- F Aortic stenosis
- G Empyema
- H Bronchial carcinoma
- I Pleural effusion
- J Right-sided heart failure
- K Hypertrophic obstructive cardiomyopathy
- L Dilated cardiomyopathy
- M Pulmonary oedema
- N Emphysema
- O Pneumothorax
- P Chronic constrictive pericarditis
- Q Chronic bronchitis
- R Pulmonary embolism
- S Bronchopneumonia
- T Extrinsic allergic alveolitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 21-year-old man has had productive cough and breathlessness for 2 years; he had severe measles in childhood. There is finger clubbing, cyanosis and bilateral coarse crackles.
- 2 A 63-year-old man presents with massive haemoptysis after 3 months of breathlessness, cough and blood-streaked sputum. He is anaemic, with gynaecomastia and localised wheeze over the right middle zone.
- 3 A 66-year-old man has sudden breathlessness with pink frothy sputum; he is cold, clammy and cyanosed, pulse 116/min regular, BP 90/55 mmHg, with bilateral fine inspiratory crackles.
- 4 A 70-year-old woman has sudden breathlessness, fever, rigors, rusty sputum and left-sided pleuritic pain; there is reduced expansion, dullness and bronchial breathing over the left lower zone.
- 5 A 33-year-old farmer has recurrent dry cough and breathlessness several hours after starting work; he is febrile with diffuse end-inspiratory crackles, and the chest X-ray shows mid-zone mottling.
- 6 An 18-year-old student has had fever, night sweats and cough for 3 months and increasing breathlessness for 2 weeks; there is reduced expansion with stony dullness and absent breath sounds over the right middle and lower zones.

- 7 A 24-year-old man has sudden left-sided pleuritic chest pain and breathlessness; he is not cyanosed, pulse 96/min, SpO₂ 96%, with a resonant percussion note and absent breath sounds on the left.
- 8 A postoperative patient has sudden chest discomfort and breathlessness; he is cyanosed with a raised JVP, pulse 112/min small volume, BP 90/60 mmHg and SpO₂ 86%, and heart and chest examination are otherwise unremarkable.
- 9 A 62-year-old man with an MI 5 days ago suddenly becomes breathless; pulse 118/min, BP 85/50 mmHg with pulsus paradoxus, a raised JVP and soft heart sounds.
- 10 A 25-year-old man has sudden chest discomfort, breathlessness and syncope while playing football; the pulse is small-volume, the apex forceful with a fourth heart sound, and there is an apical pansystolic murmur and a mid-systolic murmur at the base.
- 11 A 16-year-old girl has recurrent episodes of dry cough and breathlessness; there is expiratory wheeze throughout the chest and the chest X-ray is normal.

REPEATED 5× · ALSO ASKED IN
Formative exam ×3 · Irfan

THEME 2 Finger clubbing

- A Lung cancer
- B Infective endocarditis
- C Idiopathic pulmonary fibrosis
- D Lung abscess
- E Cyanotic congenital heart disease
- F Bronchiectasis
- G Sarcoidosis
- H Atrial myxoma
- I Pleural effusion
- J Pulmonary tuberculosis
- K Thyrotoxicosis (thyroid acropachy)
- L Empyema

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 20-year-old woman has syncope, palpitations and breathlessness; there is finger clubbing and echo shows a large mass arising from the left atrium and obstructing the mitral valve.
- 2 A 40-year-old woman has fever, fatigue and breathlessness; there is finger clubbing and a new mitral regurgitant murmur.
- 3 A 30-year-old woman has high remittent fever, rigors, cough, sputum and pleuritic chest pain; there is finger clubbing with stony dullness and absent air entry over the right lower zone.

- 4 A 36-year-old man has a chronic cough with yellow sputum, sometimes blood-stained; there is clubbing of fingers and toes, weight loss and bibasal coarse crackles, and he has 6 years of infertility.
- 5 A 55-year-old woman has progressive cough and breathlessness, with finger clubbing, restricted expansion and bilateral fine inspiratory crackles.
- 6 A 35-year-old woman with finger clubbing has tremor, palpitations, weight loss, exophthalmos and pretibial myxoedema.

THEME 3 Treatment of respiratory emergencies and diseases

- A Mechanical ventilation
- B High-dose IV antibiotics
- C Alteplase
- D Beclometasone inhaler
- E Pirfenidone
- F IV steroid plus IV antihistamine
- G Chest tube
- H Iloprost inhaler
- I IV furosemide
- J Bosentan
- K Intramuscular adrenaline

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 34-year-old man has severe breathlessness after a car accident; there is central cyanosis, RR 28/min, pulse 124/min small volume, BP 80/50 mmHg, chest expansion 2 cm, a resonant percussion note throughout and absent air entry and breath sounds on the right.
- 2 A patient with an acute asthma attack has central cyanosis. He is given IV hydrocortisone, nebulised salbutamol plus ipratropium and 2 g of IV magnesium sulfate; a few hours later the blood gases show PaO₂ 7.4 kPa (54 mmHg) and PaCO₂ 5.8 kPa (44 mmHg).

- 3 A 65-year-old bed-bound patient suddenly becomes severely breathless; there is cyanosis, a raised JVP, pulse 118/min small volume, BP 80/50 mmHg, a right ventricular gallop and a loud P2.
- 4 A 60-year-old woman has had a dry cough and breathlessness for 2 months; there is finger clubbing, central cyanosis, reduced expansion and bilateral fine inspiratory crackles.
- 5 A 65-year-old woman takes a new drug prescribed for knee pain and develops generalised itching, severe breathlessness, swelling of the face, eyes, lips and tongue, and widespread wheeze.

THEME 4 Cough and breathlessness

- A Empyema
- B Bronchial carcinoma
- C COPD
- D Bronchiectasis
- E ACE inhibitor
- F Bronchial asthma
- G Pulmonary fibrosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old man started treatment for hypertension 3 months ago and has had a dry cough ever since; his chest examination and chest X-ray are normal.
- 2 A 70-year-old woman treated for left lower lobe pneumonia is still febrile with rigors after a week of antibiotics; there is stony dullness and absent breath sounds at the left base, and pleural aspiration yields turbid fluid with pH 7.0.
- 3 An older man with 40 pack-years of smoking has breathlessness for 4 years, worse over the last 3; there is reduced chest movement with coarse crackles and wheeze.

Answers · Themes 1–4

LOCAL EXAMS EMQS

Theme 1 · Presentations of breathlessness

- 1 **B Bronchiectasis** — Chronic productive cough after childhood measles with clubbing and coarse crackles is bronchiectasis; HRCT confirms.
- 2 **H Bronchial carcinoma** — Haemoptysis, anaemia and a fixed localised wheeze with gynecomastia (ectopic hCG) is bronchial carcinoma.
- 3 **M Pulmonary oedema** — Sudden pink frothy sputum, bilateral crackles, cold peripheries and hypotension is cardiogenic pulmonary oedema with shock.
- 4 **C Lobar pneumonia** — Fever, rusty sputum and pleuritic pain with dullness and bronchial breathing is lobar pneumonia (pneumococcal).

- 5 **T Extrinsic allergic alveolitis** — Fever, cough and breathlessness 4–8 h after work in a farmer is acute extrinsic allergic alveolitis (farmer's lung).
- 6 **I Pleural effusion** — Stony dullness with absent breath sounds is a pleural effusion; with fever and sweats in a young person, likely tuberculous.
- 7 **O Pneumothorax** — Sudden pleuritic pain with hyper-resonance and absent breath sounds in a young man is primary spontaneous pneumothorax.
- 8 **R Pulmonary embolism** — Post-operative hypoxaemia, raised JVP and hypotension with a clear chest is massive PE: systemic thrombolysis.
- 9 **E Cardiac tamponade** — Hypotension, pulsus paradoxus, raised JVP and soft heart sounds days after MI is tamponade (free-wall rupture).

- 10 **K Hypertrophic obstructive cardiomyopathy** — Exertional syncope with S4, basal ejection murmur and apical MR murmur in a young athlete is HOCM.
- 11 **A Bronchial asthma** — Recurrent cough and breathlessness with widespread wheeze and a normal CXR is asthma; confirm variable obstruction.

Theme 2 · Finger clubbing

- 1 **H Atrial myxoma** — A left atrial mass obstructing the mitral valve with syncope and clubbing is atrial myxoma; excise surgically.
- 2 **B Infective endocarditis** — Fever with a new regurgitant murmur and clubbing suggests subacute infective endocarditis; blood cultures and echo.
- 3 **L Empyema** — Swinging fever, pleuritic pain and stony dullness with clubbing is empyema; aspirate (pus, pH < 7.2) and drain.
- 4 **F Bronchiectasis** — Chronic purulent sputum with clubbing, crackles and infertility is bronchiectasis (CF or ciliary dyskinesia).
- 5 **C Idiopathic pulmonary fibrosis** — Progressive dry cough with clubbing and bilateral Velcro crackles is idiopathic pulmonary fibrosis.
- 6 **K Thyrotoxicosis (thyroid acropachy)** — Clubbing (thyroid acropachy) with exophthalmos and pretibial myxoedema is specific to Graves' disease.

SET 2 OF 4

THEME 5 Investigation of respiratory and cardiovascular presentations

- A Sputum for acid-fast bacilli and culture
- B Full blood count and ESR
- C CT pulmonary angiography
- D Bronchoscopy and biopsy
- E Spirometry
- F Cardiac enzymes
- G Lipid profile
- H Urine metanephrine and normetanephrine
- I Tuberculin skin test
- J Sputum Gram stain

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 58-year-old heavy smoker with chronic cough and sputum recovered poorly from a recent pneumonia; his sputum is increasing and occasionally bloody, and the chest X-ray shows a right hilar opacity.
- 2 A 45-year-old woman has had a chronic cough with white-grey sputum for 2 months with low-grade fever, sweats and marked weight loss; there are crackles at the right apex and bilateral apical opacities on chest X-ray.
- 3 A 64-year-old man with hypertension and diabetes has severe retrosternal chest pain for 6 hours radiating to the left shoulder, with nausea and vomiting; the ECG shows ST depression in V4–V6.

Theme 3 · Treatment of respiratory emergencies and diseases

- 1 **G Chest tube** — Resonant silent hemithorax with hypotension after trauma is tension pneumothorax: decompress, then chest tube.
- 2 **A Mechanical ventilation** — PaO₂ < 8 kPa with a normal PaCO₂ despite maximal asthma therapy means tiring; ICU for intubation and ventilation.
- 3 **C Alteplase** — Sudden breathlessness with RV strain and SBP < 90 in a bed-bound patient is massive PE: alteplase.
- 4 **E Pirfenidone** — Dry cough, clubbing and fine crackles is IPF; pirfenidone or nintedanib slows FVC decline.
- 5 **K Intramuscular adrenaline** — Itch, facial/tongue swelling and wheeze after a new drug is anaphylaxis: IM adrenaline 0.5 mg, repeat after 5 min if needed.

Theme 4 · Cough and breathlessness

- 1 **E ACE inhibitor** — Dry cough after starting antihypertensive treatment with a normal CXR is ACE-inhibitor cough; switch to an ARB.
- 2 **A Empyema** — Persistent fever with a turbid effusion of pH 7.0 after pneumonia is empyema: chest drain plus antibiotics.
- 3 **C COPD** — Progressive breathlessness in a 40 pack-year smoker with reduced chest expansion, wheeze and coarse crackles is COPD. Diagnosis is confirmed by post-bronchodilator FEV₁/FVC below 0.7.

- 4 A 16-year-old student has had a night-time cough for 2 months, since moving to a new house; amoxicillin did not help. He does not smoke, and one of his brothers has hay fever.
- 5 A 38-year-old man has recurrent episodes of headache, palpitations and vomiting; he is pale and sweaty, with BP 190/110 mmHg sitting and 150/90 mmHg standing.

REPEATED 3× · ALSO ASKED IN
Formative exam ×2

THEME 6 Treatment of cardiorespiratory presentations

- A Rosuvastatin
- B Heparin
- C Lisinopril
- D Prednisolone
- E IV furosemide
- F Short-acting beta2-agonist inhaler as needed
- G Chest tube
- H Isosorbide mononitrate
- I Pirfenidone
- J Steroid inhaler

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 67-year-old man with diabetes, hypertension and ischaemic heart disease is woken at 4 am by severe breathlessness relieved by sitting up, with cough and white frothy sputum; pulse 126/min, BP 120/70 mmHg.

- 2** A 31-year-old woman 4 days after cholecystectomy has sudden severe breathlessness; pulse 112/min regular, BP 95/60 mmHg and a raised JVP.
- 3** A 62-year-old woman has progressive exertional breathlessness and dry cough for months; there is finger clubbing and chest expansion of 2 cm. FVC 1.65 L (predicted 2.75 L), FEV₁ 1.45 L (predicted 2.3 L).
- 4** A middle-aged woman is severely breathless after an accident; she is cyanosed, pulse 120/min, BP 90/55 mmHg, with absent air entry on the left and the mediastinum shifted to the right.
- 5** A 65-year-old woman attends her GP for annual review of heart failure with reduced ejection fraction; BP 165/95 mmHg on furosemide and aspirin. Which medication should be added?

REPEATED 4* · ALSO ASKED IN
Formative exam ×3

THEME 7 Management of COPD

- A** Inhaled salbutamol
B 60% oxygen, nebulised salbutamol, oral prednisolone, oral amoxicillin
C Intravenous salbutamol
D Long-term oxygen therapy
E Lung transplantation
F 28% oxygen, nebulised salbutamol plus ipratropium, oral prednisolone, oral amoxicillin
G 28% oxygen, inhaled bronchodilators, intravenous amoxicillin
H Oral aminophylline
I ICU admission and intubation
J 100% oxygen and amoxicillin
K Inhaled long-acting bronchodilators (LABA plus LAMA)
L Non-invasive positive pressure ventilation (BiPAP)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A previously healthy 65-year-old smoker with early COPD is breathless on exertion.
- 2** A 65-year-old woman with long-standing COPD presents with breathlessness and cough productive of coloured sputum.
- 3** A 70-year-old man with an acute severe exacerbation of COPD does not respond to oxygen, nebulised bronchodilators and oral steroids; on 28% oxygen his ABG shows pH 7.28, PO₂ 8 kPa and PCO₂ 9.4 kPa.

- 4** A 65-year-old non-smoker with advanced COPD on maximal inhaled therapy is still breathless; his PaO₂ when clinically stable, measured twice 3 weeks apart, is 6.5 kPa.
- 5** A 55-year-old patient with COPD needs regular add-on therapy because bronchodilators do not control his symptoms.

REPEATED 2* · ALSO ASKED IN
Irfan

THEME 8 Haemoptysis

- A** Tuberculosis
B Haemothorax
C Bronchogenic carcinoma
D Polyarteritis nodosa
E Granulomatosis with polyangiitis (Wegener)
F Bronchiectasis
G Goodpasture's syndrome (anti-GBM disease)
H Haemophilia
I Pulmonary oedema
J Pneumonia
K Pulmonary embolus
L Eosinophilic granulomatosis with polyangiitis (Churg–Strauss)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 65-year-old smoker has breathlessness, a gallop rhythm and pink frothy sputum.
- 2** A 24-year-old man with cough and intermittent haemoptysis develops haematuria a few weeks later; renal biopsy shows crescentic glomerulonephritis with linear deposition on immunofluorescence.
- 3** A 34-year-old woman with nasal obstruction develops cough, haemoptysis and pleuritic chest pain; her chest radiograph shows multiple nodular masses.
- 4** A 22-year-old man has fever, night sweats, weight loss and cough productive of cupfuls of blood; Ziehl–Neelsen stain is positive for acid-fast bacilli.
- 5** A 35-year-old businessman returning from a trip abroad collapses at the airport with haemoptysis and pleuritic chest pain; he has a sinus tachycardia and right axis deviation on ECG.

REPEATED 3* · ALSO ASKED IN
Formative exam · Irfan

Answers · Themes 5–8

LOCAL EXAMS EMQS

Theme 5 · Investigation of respiratory and cardiovascular presentations

- 1** **D** **Bronchoscopy and biopsy** — A persistent hilar opacity with bloody sputum in a heavy smoker suggests central carcinoma: CT, then bronchoscopy and biopsy.

- 2** **A** **Sputum for acid-fast bacilli and culture** — Chronic cough, sweats, weight loss and apical opacities is TB: sputum for AFB smear, culture and NAAT.

- 3 **F Cardiac enzymes** — Ischaemic pain with ST depression: high-sensitivity troponin to separate NSTEMI from unstable angina.
- 4 **E Spirometry** — Nocturnal cough after moving house in an atopic non-smoker suggests cough-variant asthma: spirometry with reversibility ($\pm$ FeNO).
- 5 **H Urine metanephrine and normetanephrine** — Paroxysmal headache, palpitations and sweating with hypertension and postural drop suggest pheochromocytoma: metanephrines.

Theme 6 • Treatment of cardiorespiratory presentations

- 1 **E IV furosemide** — Nocturnal breathlessness with frothy sputum in IHD is acute pulmonary oedema: IV furosemide (plus nitrates if BP allows).
- 2 **B Heparin** — Post-operative PE with SBP > 90 is not high-risk: anticoagulate with heparin; thrombolysis only for shock.
- 3 **I Pirfenidone** — Restriction (FVC 60%, ratio ~0.88) with clubbing is IPF; pirfenidone is recommended at FVC 50–80%.
- 4 **G Chest tube** — Absent air entry with the mediastinum shifted away and hypotension after trauma is tension pneumothorax: decompress and drain.
- 5 **C Lisinopril** — HFrEF on furosemide and aspirin with BP 165/95: add an ACE inhibitor (lisinopril) for survival and BP.

Theme 7 • Management of COPD

- 1 **A Inhaled salbutamol** — New COPD with exertional breathlessness: start an as-needed short-acting bronchodilator (salbutamol), stop smoking, rehab.

SET 3 OF 4

39TH BATCH FINAL PAPERS

Final exam Paper 1 & Paper 2 of the 39th batch (recalled)

THEME 9 Management of cough

- A Inhaled corticosteroid
- B Short-acting beta2-agonist
- C Long-acting beta2-agonist
- D Proton-pump inhibitor
- E Antihistamine plus decongestant
- F Antituberculous therapy
- G Stop the current medication
- H Reduce exposure
- I Loop diuretic
- J Anticoagulant
- K Antibiotics
- L Antifibrotic therapy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old diabetic smoker has chronic productive cough and progressive breathlessness; spirometry shows persistent airflow limitation. He takes empagliflozin and metformin.
- 2 A 22-year-old woman has a chronic dry cough that is worse at night with occasional wheeze; the chest X-ray is normal and spirometry shows reversible obstruction.

- 2 **F 28% oxygen, nebulised salbutamol plus ipratropium, oral prednisolone, oral amoxicillin** — Infective COPD exacerbation: 28% oxygen, nebulised salbutamol + ipratropium, prednisolone and oral amoxicillin.
- 3 **L Non-invasive positive pressure ventilation (BiPAP)** — COPD exacerbation with persistent acidosis (pH 7.28, PaCO₂ 9.4) despite treatment: non-invasive ventilation (BiPAP).
- 4 **D Long-term oxygen therapy** — Stable PaO₂ 6.5 kPa twice in a non-smoker with COPD: long-term oxygen therapy ($\geq$ 15 h/day) improves survival.
- 5 **K Inhaled long-acting bronchodilators (LABA plus LAMA)** — COPD still symptomatic on short-acting bronchodilators: add a LABA + LAMA inhaler.

Theme 8 • Haemoptysis

- 1 **I Pulmonary oedema** — Pink frothy sputum with a gallop rhythm is acute cardiogenic pulmonary oedema.
- 2 **G Goodpasture's syndrome (anti-GBM disease)** — Haemoptysis then crescentic GN with linear IgG on the GBM is anti-GBM (Goodpasture's) disease.
- 3 **E Granulomatosis with polyangiitis (Wegener)** — Nasal obstruction then haemoptysis with multiple nodules is granulomatosis with polyangiitis (c-ANCA).
- 4 **A Tuberculosis** — Fever, sweats, weight loss and haemoptysis with AFB on ZN stain is pulmonary TB.
- 5 **K Pulmonary embolus** — Collapse with haemoptysis and pleuritic pain after a long flight, with RV strain on ECG, is PE.

- 3 A 45-year-old man taking steroids for glomerulonephritis has a chronic cough that is worse after meals and on lying flat; chest examination is normal.
- 4 A 52-year-old has a chronic cough with a sensation of postnasal drip and frequent throat clearing, especially at night.
- 5 A 60-year-old hypertensive man taking ramipril has had a dry cough for 4 weeks.
- 6 A 40-year-old woman has a chronic cough with night sweats, weight loss and haemoptysis.
- 7 A 65-year-old man has a dry cough and exertional breathlessness with fine inspiratory crackles and finger clubbing.
- 8 A 58-year-old man who works in a cement factory has a chronic cough; the chest X-ray shows diffuse nodular opacities.
- 9 A 50-year-old woman has acute fever, cough with purulent sputum, pleuritic chest pain and bronchial breathing.
- 10 A 72-year-old woman coughs when lying flat and has frothy sputum, bilateral basal crackles and ankle swelling.

FORMATIVE & PREVIOUS YEARS

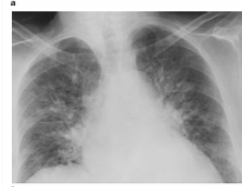
THEME 10 Pleural fluid analysis

- A** Unilateral, turbid: WBC 90,000/ μ L (95% neutrophils); protein 45 g/L (serum 52 g/L); LDH 255 U/L (serum 290 U/L); pH 6.84; glucose 1.1 mmol/L; Gram stain and culture pending
- B** Right-sided, straw-coloured: WBC 150/ μ L (20% neutrophils, 35% lymphocytes, 45% mesothelial cells); protein 14 g/L (serum 54 g/L); LDH 66 U/L (serum 175 U/L); pH 7.42; glucose 5.6 mmol/L
- C** Bilateral, slightly turbid: WBC 980/ μ L (10% neutrophils, 30% lymphocytes, 60% mesothelial cells); protein 39 g/L (serum 38 g/L); LDH 225 U/L (serum 240 U/L); pH 7.00; glucose 0.3 mmol/L
- D** Bilateral, straw-coloured: WBC 4200/ μ L (100% lymphocytes); protein 30 g/L (serum 50 g/L); LDH 560 U/L (serum 450 U/L); pH 7.27; glucose 4.3 mmol/L
- E** Right-sided, bloody: WBC 1200/ μ L (15% neutrophils, 5% lymphocytes, 80% 'reactive' mesothelial cells); RBC 130,000/ μ L; protein 42 g/L (serum 46 g/L); LDH 560 U/L (serum 226 U/L); pH 7.30; glucose 6.7 mmol/L
- F** Left-sided, turbid, food particles seen: WBC 54,000/ μ L (92% neutrophils, 8% lymphocytes); protein 52 g/L (serum 52 g/L); LDH 400 U/L (serum 200 U/L); pH 6.00; glucose 2.2 mmol/L
- G** Left-sided, straw-coloured: WBC 2000/ μ L (80% neutrophils, 10% lymphocytes, 10% mesothelial cells); protein 20 g/L (serum 48 g/L); LDH 158 U/L (serum 220 U/L); pH 7.40; Gram stain negative; amylase 32,000 U/L

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man complains of orthopnoea and paroxysmal nocturnal dyspnoea. There is distension of the neck veins and bilateral basal crackles. A chest X-ray shows cardiomegaly and bilateral pleural effusions, the right larger than the left.
- 2 A 52-year-old man with alcohol dependence develops left-sided chest pain after repeated bouts of vomiting. He is sweaty, temperature 38.6 °C, pulse 126/min, BP 84/52 mmHg, with crackles and moderate dullness at the left base, a clear right lung and subcutaneous emphysema in the left supraclavicular fossa.
- 3 A 72-year-old woman is admitted from a nursing home with fever and cough, and a chest X-ray shows right lower lobe pneumonia with a moderate pleural effusion. Despite vancomycin and levofloxacin she remains febrile, her breathlessness worsens and a repeat chest X-ray shows the effusion enlarging.
- 4 A 52-year-old woman is admitted with abdominal pain and hypertriglyceridaemia. Her amylase is elevated and she is treated for acute pancreatitis with IV fluids and opioid analgesia. Over the next few days she becomes increasingly breathless and dullness develops at the left base.
- 5 A 68-year-old retired construction worker has right-sided chest pain, breathlessness and a dry cough, with marked weight loss and anorexia. A chest X-ray shows a right pleural effusion with pleural thickening.

THEME 11 Chest X-ray interpretation (images a-d shown)



- A** Chest X-ray a (shown)
B Chest X-ray b (shown)
C Chest X-ray c (shown)
D Chest X-ray d (shown)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 60-year-old man develops fever, chills and a productive cough in hospital after surgery. There is increased tactile fremitus in the right mid-zone. Sputum Gram stain shows many neutrophils, few squamous cells and Gram-positive cocci in clusters.
- 2 A 45-year-old man with known coronary artery disease develops breathlessness and wakes at night gasping for breath. There is dullness to percussion at the right base.
- 3 An 85-year-old man who recently arrived from Vietnam has had cough and night sweats for more than a year. There are bilateral upper-zone crackles.
- 4 A 50-year-old woman has long-standing, poorly controlled hypertension. The apex beat is displaced to the sixth intercostal space.

THEME 12 Arterial blood gases

- A pH 7.50, PaO₂ 10.0 kPa (75 mmHg), PaCO₂ 3.7 kPa (28 mmHg)
- B pH 7.14, PaO₂ 10.4 kPa (78 mmHg), PaCO₂ 12.7 kPa (95 mmHg)
- C pH 7.06, PaO₂ 4.8 kPa (36 mmHg), PaCO₂ 12.7 kPa (95 mmHg)
- D pH 7.06, PaO₂ 14.4 kPa (108 mmHg), PaCO₂ 1.7 kPa (13 mmHg)
- E pH 7.37, PaO₂ 6.4 kPa (48 mmHg), PaCO₂ 7.2 kPa (54 mmHg)

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 30-year-old obese woman who is a bus driver develops sudden left-sided pleuritic chest pain, haemoptysis and breathlessness.

- 2 A 60-year-old heavy smoker has severe chronic bronchitis, peripheral oedema and cyanosis.
- 3 A 22-year-old man who injects drugs is brought to the emergency department by friends who cannot wake him.
- 4 A 62-year-old man with chronic bronchitis develops chest pain and is given high-flow oxygen by face mask in the ambulance. He is lethargic on arrival in the emergency department.
- 5 A 20-year-old man with diabetes mellitus attends the emergency department with diffuse abdominal pain, rapid breathing and fever.

Answers · Themes 9–12

Theme 9 · Management of cough

- 1 **C Long-acting beta2-agonist** — COPD: stop smoking, then a long-acting bronchodilator (LABA, with a LAMA); his diabetes drugs do not cause cough.
- 2 **A Inhaled corticosteroid** — Nocturnal cough with reversible obstruction is asthma; treatment is based on inhaled corticosteroid (ICS–formoterol per GINA).
- 3 **D Proton-pump inhibitor** — Cough after meals and when lying flat suggests reflux, made worse by steroids: give a trial of a proton-pump inhibitor.
- 4 **E Antihistamine plus decongestant** — Upper-airway cough syndrome (postnasal drip) is treated with an antihistamine–decongestant or intranasal steroid.
- 5 **G Stop the current medication** — ACE-inhibitor cough (bradykinin) settles after stopping ramipril; switch to an ARB.
- 6 **F Antituberculous therapy** — Night sweats, weight loss and haemoptysis suggest pulmonary TB: confirm on sputum, then 6 months of RIPE therapy.
- 7 **L Antifibrotic therapy** — Dry cough, Velcro crackles and clubbing suggest IPF; after HRCT confirmation, give an antifibrotic (pirfenidone or nintedanib).
- 8 **H Reduce exposure** — Dust-related pneumoconiosis (e.g. silicosis) has no specific drug treatment; remove him from exposure.
- 9 **K Antibiotics** — Lobar pneumonia: assess CURB-65 and start antibiotics (amoxicillin for low severity).
- 10 **I Loop diuretic** — Orthopnoea, frothy sputum, basal crackles and oedema mean heart failure: give a loop diuretic.

Theme 10 · Pleural fluid analysis

- 1 **B Right-sided, straw-coloured: WBC 150/μL (20% neutrophils, 35% lymphocytes, 45% mesothelial cells); protein 14 g/L (serum 54 g/L); LDH 66 U/L (serum 175 U/L); pH 7.42; glucose 5.6 mmol/L** — Heart failure gives a transudate (protein and LDH ratios below Light's thresholds, few cells), often bilateral; treat with diuretics.

- 2 **F Left-sided, turbid, food particles seen: WBC 54,000/μL (92% neutrophils, 8% lymphocytes); protein 52 g/L (serum 52 g/L); LDH 400 U/L (serum 200 U/L); pH 6.00; glucose 2.2 mmol/L** — Pain and surgical emphysema after vomiting is Boerhaave rupture: left effusion with pH ~6, food particles and high amylase.
- 3 **A Unilateral, turbid: WBC 90,000/μL (95% neutrophils); protein 45 g/L (serum 52 g/L); LDH 255 U/L (serum 290 U/L); pH 6.84; glucose 1.1 mmol/L; Gram stain and culture pending** — Turbid neutrophilic fluid with pH < 7.2 and very low glucose not responding to antibiotics is empyema: chest drain.
- 4 **G Left-sided, straw-coloured: WBC 2000/μL (80% neutrophils, 10% lymphocytes, 10% mesothelial cells); protein 20 g/L (serum 48 g/L); LDH 158 U/L (serum 220 U/L); pH 7.40; Gram stain negative; amylase 32,000 U/L** — Pancreatitis causes a left-sided exudate with very high pleural amylase.
- 5 **E Right-sided, bloody: WBC 1200/μL (15% neutrophils, 5% lymphocytes, 80% 'reactive' mesothelial cells); RBC 130,000/μL; protein 42 g/L (serum 46 g/L); LDH 560 U/L (serum 226 U/L); pH 7.30; glucose 6.7 mmol/L** — A bloody high-LDH exudate with atypical mesothelial cells after asbestos exposure suggests mesothelioma; pleural biopsy.

Theme 11 · Chest X-ray interpretation (images a–d shown)

- 1 **A Chest X-ray a (shown)** — Gram-positive cocci in clusters in hospital pneumonia mean *S. aureus*: patchy consolidation with cavities and pneumatoceles (film a); cover MRSA.
- 2 **B Chest X-ray b (shown)** — PND with a right effusion in IHD is LVF: film b shows cardiomegaly, upper-lobe diversion and interstitial oedema.
- 3 **C Chest X-ray c (shown)** — Chronic cough and sweats in a migrant with upper-zone signs is post-primary TB: film c shows bilateral upper-lobe fibronodular shadowing.
- 4 **D Chest X-ray d (shown)** — Long-standing hypertension with a displaced apex gives the rounded enlarged LV contour of film d; ECG and echo confirm LVH.

Theme 12 · Arterial blood gases

- 1 **A** pH 7.50, PaO₂ 10.0 kPa (75 mmHg), PaCO₂ 3.7 kPa (28 mmHg) — PE causes hypoxaemia with hyperventilation: respiratory alkalosis (pH 7.50, PaCO₂ 3.7) with low PaO₂.
- 2 **E** pH 7.37, PaO₂ 6.4 kPa (48 mmHg), PaCO₂ 7.2 kPa (54 mmHg) — Chronic hypercapnic COPD with cor pulmonale: low PaO₂, high PaCO₂, near-normal pH from renal compensation.
- 3 **C** pH 7.06, PaO₂ 4.8 kPa (36 mmHg), PaCO₂ 12.7 kPa (95 mmHg) — Opioid overdose causes acute hypoventilation: pH 7.06, PaCO₂ 12.7, PaO₂ 4.8 on air. Ventilate and give naloxone.

SET 4 OF 4

THEME 13 Diagnosis of chest disease

- A Small cell carcinoma of the lung
- B Adenocarcinoma in situ (formerly bronchoalveolar carcinoma)
- C Silicosis
- D Pneumonia
- E Cystic fibrosis
- F Hodgkin lymphoma
- G Asbestosis
- H Hypersensitivity pneumonitis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 20-year-old man, a non-smoker, has a chronic cough and recurrent bronchitis with thick green sputum. He has also been treated for abdominal cramps and malabsorption.
- 2 A 40-year-old construction worker has had increasing breathlessness and cough for many years. There are bilateral inspiratory crackles, and the chest X-ray shows eggshell calcification of the hilar nodes and small nodules in both upper zones.
- 3 A 42-year-old man has fever, night sweats and pruritus. There is a 2-cm firm, rubbery supraclavicular lymph node and bilateral axillary nodes, and the chest X-ray shows mediastinal lymphadenopathy.

THEME 14 Diagnosis of lung disease

- A Tuberculosis
- B Primary lung tumour
- C Pulmonary embolism
- D Metastatic lung cancer
- E Asbestosis
- F Histoplasmosis
- G Idiopathic pulmonary fibrosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 32-year-old man born in India has a cough with yellow, blood-streaked sputum, night sweats and 4.5 kg weight loss. There is dullness to percussion above both clavicles, and the chest X-ray shows bilateral upper-lobe infiltrates with cavitation.

- 4 **B** pH 7.14, PaO₂ 10.4 kPa (78 mmHg), PaCO₂ 12.7 kPa (95 mmHg) — High-flow oxygen in a CO₂ retainer causes acute-on-chronic acidosis with high PaCO₂ but adequate PaO₂; target 88–92%.
- 5 **D** pH 7.06, PaO₂ 14.4 kPa (108 mmHg), PaCO₂ 1.7 kPa (13 mmHg) — DKA: severe metabolic acidosis with Kussmaul breathing, so PaCO₂ is very low (1.7 kPa) and PaO₂ high-normal.

- 2 A 55-year-old woman who smokes heavily has a cough with small amounts of bright red blood, loss of appetite and 5.5 kg weight loss. The chest X-ray shows a solitary 3-cm pulmonary nodule with spiculated (shaggy) margins.
- 3 A 65-year-old man retiring from work as a plumber has a dry cough and breathlessness on walking. There are bilateral basal crackles and finger clubbing; the chest X-ray shows bilateral lower-zone linear shadowing and CT shows calcified pleural plaques.

THEME 15 Treatment of asthma

- A Low-dose oral theophylline
- B Oral prednisolone
- C Regular inhaled beclometasone
- D Inhaled ipratropium bromide
- E Inhaled salbutamol as needed
- F Nebulised adrenaline
- G Salbutamol metered-dose inhaler via a spacer
- H Home oxygen
- I Inhaled salmeterol added to the inhaled corticosteroid
- J Oral sodium cromoglicate

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 7-year-old girl has slight wheeze and breathlessness despite using her salbutamol metered-dose inhaler as needed; she cannot coordinate pressing the canister with breathing in.
- 2 A 22-year-old student with asthma well controlled on regular low-dose inhaled beclometasone gets mild wheeze during football and asks what to take before exercise.
- 3 A 17-year-old with asthma treated with salbutamol alone uses it most days and wakes with wheeze once a week.
- 4 A 33-year-old with asthma remains uncontrolled despite high-dose inhaled corticosteroid, a long-acting beta2-agonist and slow-release theophylline, with good adherence and technique.

- 5 A 25-year-old woman needs add-on therapy because regular inhaled beclometasone plus salbutamol as needed does not control her symptoms.

REPEATED 2* · ALSO ASKED IN
Irfan

THEME 16 Best investigation in respiratory presentations

- A Chest X-ray
- B High-resolution CT (HRCT) of the chest
- C Spirometry and full lung function tests
- D Arterial blood gas (ABG)
- E Bronchoscopy
- F Ventilation-perfusion (V/Q) scan
- G CT pulmonary angiography (CTPA)
- H Flow-volume loop
- I Sputum culture and cytology
- J ECG
- K Contrast-enhanced CT of the chest and upper abdomen

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 67-year-old lifelong smoker has progressive breathlessness; there are bilateral crackles that clear with coughing, scattered rhonchi and ankle oedema.
- 2 A 50-year-old woman has had a dry cough and exertional breathlessness for several months, with bibasal fine crackles and finger clubbing.
- 3 A 24-year-old man with asthma develops acute breathlessness after taking propranolol; he is tachypnoeic, hypoxaemic and wheezy.
- 4 A tall, slim 20-year-old man has sudden right-sided pleuritic chest pain with reduced breath sounds on the right.
- 5 A 35-year-old woman has fever, cough, pleuritic chest pain, confusion and hypotension, with bronchial breathing over the left lower zone.
- 6 A 70-year-old heavy smoker has cough, sputum, haemoptysis, hoarseness and hypercalcaemia.
- 7 A 72-year-old woman who has never smoked has had a daily cough with copious green sputum, sometimes blood-streaked, since whooping cough in childhood; there are coarse crackles and digital clubbing.
- 8 A 38-year-old woman with hypertriglyceridaemia and severe acute pancreatitis becomes increasingly breathless and hypoxaemic, with bilateral infiltrates on chest X-ray.
- 9 A 55-year-old man has chronic breathlessness and inspiratory stridor with a large thyroid goitre.
- 10 A 68-year-old woman develops sudden chest tightness, breathlessness and hypotension after a hip replacement, with few findings on auscultation.

Answers · Themes 13–16

Theme 13 · Diagnosis of chest disease

- 1 **E Cystic fibrosis** — Recurrent suppurative chest infections with malabsorption in a young non-smoker is cystic fibrosis; sweat test.
- 2 **C Silicosis** — Upper-zone nodules with eggshell hilar node calcification after silica exposure is silicosis (raises TB risk).
- 3 **F Hodgkin lymphoma** — Rubbery painless lymphadenopathy with B symptoms, pruritus and mediastinal nodes suggests Hodgkin lymphoma. Excision biopsy of a node (Reed–Sternberg cells) and PET-CT staging are required.

Theme 14 · Diagnosis of lung disease

- 1 **A Tuberculosis** — Bilateral upper-lobe cavitation with sweats, weight loss and haemoptysis in a migrant is post-primary TB; isolate and send sputum.
- 2 **B Primary lung tumour** — A solitary spiculated nodule with haemoptysis and weight loss in a heavy smoker is primary lung cancer until proven otherwise.
- 3 **E Asbestosis** — Lower-zone fibrosis with clubbing and calcified pleural plaques in a plumber is asbestosis.

Theme 15 · Treatment of asthma

- 1 **G Salbutamol metered-dose inhaler via a spacer** — A child who cannot coordinate an MDI should use it through a spacer before treatment is stepped up.

- 2 **E Inhaled salbutamol as needed** — Exercise-induced symptoms in otherwise controlled asthma: salbutamol shortly before exercise.
- 3 **C Regular inhaled beclometasone** — Reliever use most days or night waking means uncontrolled asthma: add a regular inhaled corticosteroid (or ICS–formoterol).
- 4 **B Oral prednisolone** — Uncontrolled on high-dose ICS + LABA + theophylline: lowest-dose maintenance prednisolone and specialist referral for biologics.
- 5 **I Inhaled salmeterol added to the inhaled corticosteroid** — Uncontrolled on low-dose ICS: add a LABA (salmeterol) in a combination inhaler (or switch to MART).

Theme 16 · Best investigation in respiratory presentations

- 1 **C Spirometry and full lung function tests** — Chronic breathlessness in a smoker with secretions and oedema suggests COPD: spirometry ($FEV_1/FVC < 0.7$) with full lung function tests.
- 2 **B High-resolution CT (HRCT) of the chest** — Dry cough, Velcro crackles and clubbing suggest ILD (IPF); HRCT is the key investigation.
- 3 **D Arterial blood gas (ABG)** — Bronchospasm after propranolol with $SpO_2 < 92\%$: ABG to detect a normal/rising $PaCO_2$ (life-threatening asthma).
- 4 **A Chest X-ray** — Suspected primary spontaneous pneumothorax is confirmed with a standard erect chest X-ray, which

also shows its size. CT is reserved for uncertain or complex cases.

- 5 **A Chest X-ray** — A chest X-ray confirms pneumonia; confusion and hypotension give CURB-65 ≥ 2 , so admit for IV antibiotics.
- 6 **K Contrast-enhanced CT of the chest and upper abdomen** — Haemoptysis, hoarseness and hypercalcaemia in a smoker suggest advanced lung cancer: contrast CT chest and upper abdomen first.
- 7 **B High-resolution CT (HRCT) of the chest** — Lifelong purulent sputum after whooping cough with clubbing is bronchiectasis; HRCT confirms.

- 8 **D Arterial blood gas (ABG)** — ARDS complicating pancreatitis is graded by $\text{PaO}_2/\text{FiO}_2$ on PEEP, so an ABG is needed.
- 9 **H Flow-volume loop** — Stridor from a goitre is extrathoracic obstruction: a flattened inspiratory limb on the flow-volume loop.
- 10 **G CT pulmonary angiography (CTPA)** — Sudden breathlessness and hypotension after hip replacement with a clear chest is PE: CTPA (bedside echo if unstable).

8 THEMES • 38 SCENARIOS • AFTER CRASH COURSE SBAS & EMQS

Crash Course EMQs

SET 1 OF 2

THEME 1 Causes of pneumonia

- A *Mycoplasma pneumoniae*
- B *Chlamydia psittaci*
- C *Staphylococcus aureus*
- D *Legionella pneumophila*
- E *Haemophilus influenzae*
- F Aspiration pneumonia
- G *Pseudomonas aeruginosa*
- H *Pneumocystis jirovecii*
- I *Streptococcus pneumoniae*
- J *Mycobacterium avium* complex

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 32-year-old man with known cystic fibrosis presents with shortness of breath and feels generally unwell.
- 2 A Gram-negative coccobacillus that is a common cause of infective exacerbations and pneumonia in patients with COPD.
- 3 A middle-aged man presents with a cough and generalised muscle pains. He is a keen collector of rare birds.
- 4 A man who injects drugs presents with shortness of breath and fever.
- 5 A middle-aged woman who has recently returned from a hotel stay in Las Vegas presents with a non-productive cough, fever and diarrhoea.

THEME 2 Respiratory drugs

- A Ethambutol
- B Pyrazinamide
- C Isoniazid
- D Rifampicin
- E Metronidazole
- F Erythromycin
- G Clarithromycin
- H Flucloxacillin
- I Tobramycin
- J Ticarcillin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A drug associated with induction of hepatic enzymes.
- 2 A drug associated with blurred vision and red-green colour blindness.
- 3 A man who injects drugs is diagnosed with pneumonia with patchy consolidation on his chest X-ray; blood cultures grow meticillin-sensitive *Staphylococcus aureus*.
- 4 A first-line antituberculous drug that causes hepatitis and can precipitate gout.
- 5 A middle-aged woman with a recent stroke and an unsafe swallow aspirates and develops pneumonia with foul-smelling sputum. A drug with anaerobic cover is to be added to amoxicillin.

THEME 3 Complications of lung cancer

- A Lambert–Eaton myasthenic syndrome (LEMS)
- B Dermatomyositis
- C Herpes zoster
- D Acanthosis nigricans
- E Hypertrophic pulmonary osteoarthropathy
- F Horner syndrome
- G Pancoast tumour
- H Cushing syndrome
- I Thrombophlebitis migrans
- J Clubbing

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 75-year-old man recently diagnosed with lung cancer complains of painful, swollen wrists and ankles.
- 2 A middle-aged man presents with haemoptysis and pain in the shoulder radiating down the inner arm.
- 3 A 65-year-old woman presents with breathlessness, haemoptysis and velvety, pigmented thickening of the skin in her axillae.
- 4 A middle-aged woman with known lung cancer presents with drooping of the right upper eyelid, a constricted right pupil and apparent sinking of the right eye.
- 5 A patient recently diagnosed with small cell lung cancer has proximal lower-limb weakness and difficulty climbing stairs. Tendon reflexes are absent but re-appear after sustained muscle contraction; there is no rash.

THEME 4 Diseases of the respiratory tract

- A Bronchiectasis
- B Sinusitis
- C Rhinitis
- D Pharyngitis
- E Laryngotracheobronchitis
- F Influenza
- G Emphysema
- H Acute bronchitis
- I Asthma
- J Obstructive sleep apnoea

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Permanent enlargement of the airspaces distal to the terminal bronchioles, with destruction of their walls.

- 2 A 23-year-old man presents in winter with sudden-onset fever, dry cough and muscle aches.
- 3 A condition associated with a hoarse voice, barking cough and stridor.
- 4 An 18-year-old woman with cystic fibrosis has a persistent cough productive of yellow–green sputum.
- 5 A previously well middle-aged man develops a cough productive of sputum, mild chest tightness and wheeze a few days after a cold. He is afebrile, his symptoms resolve over 2–3 weeks and his chest X-ray is normal.

THEME 5 Investigation findings in respiratory disease

- A COPD
- B Obstructive sleep apnoea
- C Sarcoidosis
- D Granulomatosis with polyangiitis (Wegener)
- E Bronchiectasis
- F Cystic fibrosis
- G Extrinsic allergic alveolitis
- H Asthma
- I Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- J Pneumonia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old woman has progressive breathlessness, finger clubbing and fine bibasal inspiratory crackles. HRCT shows subpleural, basal reticulation with honeycombing.
- 2 A 55-year-old farmer presenting with a cough and SOB. His CXR demonstrates fluffy nodular shadowing.
- 3 A lifelong smoker has an FEV₁/FVC ratio of 62%, with an increase in FEV₁ of only 60 mL (4%) after salbutamol.
- 4 A 23-year-old woman presents with fever, cough and shortness of breath. Her white cell count is $23 \times 10^9/L$ and a chest X-ray shows patchy consolidation.
- 5 A 45-year-old man presents with haemoptysis. His chest X-ray shows cavitating lesions and blood tests show a positive c-ANCA (anti-PR3) antibody.

Answers • Themes 1–5

Theme 1 • Causes of pneumonia

- 1 **G Pseudomonas aeruginosa** — Adults with CF are chronically colonised by *Pseudomonas aeruginosa*; exacerbations need two IV antipseudomonal agents.

- 2 **E Haemophilus influenzae** — *Haemophilus influenzae*, a Gram-negative coccobacillus, is the commonest bacterium in COPD exacerbations.

- 3 **B Chlamydia psittaci** — Atypical pneumonia with myalgia in a bird keeper suggests psittacosis caused by *Chlamydia psittaci*. Serology confirms it and doxycycline is the treatment.
- 4 **C Staphylococcus aureus** — Drug injectors with fever and breathlessness have *S. aureus* (tricuspid IE, septic emboli); blood cultures and echo.
- 5 **D Legionella pneumophila** — Dry cough, fever and diarrhoea after a hotel stay is *Legionella*; check sodium, LFTs and urinary antigen.

Theme 2 · Respiratory drugs

- 1 **D Rifampicin** — Rifampicin is a potent hepatic enzyme inducer (reduces warfarin, pill, steroids) and turns secretions orange.
- 2 **A Ethambutol** — Ethambutol causes optic neuritis with red-green colour blindness; check vision before and during treatment.
- 3 **H Flucloxacillin** — MSSA pneumonia in a drug injector: high-dose IV flucloxacillin (vancomycin if MRSA).
- 4 **B Pyrazinamide** — Pyrazinamide is hepatotoxic and reduces renal urate excretion, causing hyperuricaemia and gout. Isoniazid is also hepatotoxic but causes peripheral neuropathy rather than gout.
- 5 **E Metronidazole** — Metronidazole adds anaerobic cover to amoxicillin in aspiration pneumonia with foul sputum.

Theme 3 · Complications of lung cancer

- 1 **E Hypertrophic pulmonary osteoarthropathy** — Painful swollen wrists and ankles in lung cancer is hypertrophic pulmonary osteoarthropathy (periostitis, usually with clubbing).
- 2 **G Pancoast tumour** — Shoulder pain down the inner arm with haemoptysis is a Pancoast tumour invading the brachial plexus (C8–T1).
- 3 **D Acanthosis nigricans** — Sudden flexural velvety pigmentation in an older adult is acanthosis nigricans, a paraneoplastic marker.
- 4 **F Horner syndrome** — Ptosis, miosis and apparent enophthalmos is Horner syndrome from an apical tumour.

SET 2 OF 2

THEME 6 Lung diseases

- A Asbestosis
- B Asthma
- C Bird fancier's lung
- D COPD
- E Cor pulmonale
- F Pleural effusion
- G Pneumoconiosis
- H Pulmonary embolus
- I Rheumatoid lung
- J Pneumothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old woman has tar-stained fingers, central cyanosis and widespread wheeze.

- 5 **A Lambert-Eaton myasthenic syndrome (LEMS)** — Proximal weakness with reflexes returning after contraction in small cell cancer is Lambert-Eaton syndrome (anti-VGCC).

Theme 4 · Diseases of the respiratory tract

- 1 **G Emphysema** — Permanent enlargement of airspaces distal to the terminal bronchioles with wall destruction defines emphysema.
- 2 **F Influenza** — Abrupt winter fever, myalgia and dry cough is influenza; oseltamivir for at-risk patients within 48 h.
- 3 **E Laryngotracheobronchitis** — Hoarseness, barking cough and stridor is laryngotracheobronchitis (croup); oral dexamethasone.
- 4 **A Bronchiectasis** — Chronic purulent sputum in cystic fibrosis reflects bronchiectasis; physiotherapy and culture-guided antibiotics.
- 5 **H Acute bronchitis** — Self-limiting productive cough with transient wheeze after a cold and a normal CXR is acute bronchitis; no antibiotics.

Theme 5 · Investigation findings in respiratory disease

- 1 **I Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)** — Basal subpleural reticulation with honeycombing is a UIP pattern, diagnostic of IPF in the right setting.
- 2 **G Extrinsic allergic alveolitis** — Cough and breathlessness with fluffy nodular shadowing in a farmer is extrinsic allergic alveolitis (farmer's lung).
- 3 **A COPD** — FEV₁/FVC 62% with only 60 mL (4%) reversibility in a smoker indicates COPD.
- 4 **J Pneumonia** — Acute fever, cough and a marked neutrophil leucocytosis with new consolidation indicate bacterial pneumonia. CURB-65 guides severity and management.
- 5 **D Granulomatosis with polyangiitis (Wegener)** — Cavitating nodules with haemoptysis and PR3-ANCA are granulomatosis with polyangiitis; steroids plus rituximab or cyclophosphamide.

- 2 A 30-year-old man involved in a road traffic accident has bruising over the left chest and is short of breath. The left side is hyper-resonant to percussion with reduced breath sounds; the trachea is central and he is haemodynamically stable.
- 3 A 55-year-old smoker with weight loss and a cough has stony dullness to percussion, reduced vocal resonance and absent breath sounds at the right base. The underlying cause is thought to be a bronchial carcinoma.
- 4 A 24-year-old male master's student is admitted acutely short of breath for the past day. He had a similar episode during his first year at university. He returned from a summer tour of Australasia 1 week ago. His oxygen saturation is 93% on air and he is haemodynamically stable. There are reduced breath sounds on the left, with asymmetrical chest expansion and hyper-resonance. A medical student tells you that 'the lungs look different' on the chest X-ray.

- 5** A 73-year-old woman who worked for many years in an asbestos textile mill is admitted breathless with mild haemoptysis. She has had three similar admissions in the last 5 months. She is thin, with finger clubbing, an increased anteroposterior chest diameter, scattered wheeze and fine bibasal crackles (right more than left). Arterial blood gas on 28% oxygen: PaO₂ 13.6 kPa, PaCO₂ 5.6 kPa. Chest X-ray: borderline cardiomegaly, blunted right costophrenic angle, chronic changes throughout and a bulky left hilum.

THEME 7 Causes of shortness of breath

- A Pulmonary embolism
- B Aortic stenosis
- C Congestive cardiac failure
- D Asthma
- E Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- F Pleural effusion
- G Bronchial carcinoma
- H COPD
- I Right middle lobe pneumonia
- J Pneumothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A 55-year-old baker is admitted with increasing shortness of breath over the past 6 weeks, with exertional chest tightness and two episodes of syncope while at work. He had asthma as a child and has smoked 20 pack-years. He has a slow-rising pulse. Admission blood tests, including D-dimer, are unremarkable. Chest X-ray: no active disease. Peak expiratory flow rate 450 L/min.
- 2** A 32-year-old woman has recently returned from a 2-month sabbatical in Thailand. She has been short of breath for the past week. There is reduced air entry in the right middle and lower zones and no calf swelling. Oxygen saturation 92%, D-dimer 0.45 mg/L (normal <0.5), white cell count $13.6 \times 10^9/L$. Chest X-ray: indistinct right heart border.

- 3** A 61-year-old retired plumber is admitted with increasing shortness of breath over the past 2 months, associated with mild pleuritic chest pain. He smokes 20 cigarettes a day. There is stony dullness, reduced breath sounds and reduced vocal resonance at the left base. Chest X-ray: a homogeneous opacity with a meniscus obscuring the left hemidiaphragm.

THEME 8 Respiratory infections

- A Cytomegalovirus
- B Staphylococcus aureus
- C Pseudomonas aeruginosa
- D Escherichia coli
- E Legionella pneumophila
- F Streptococcus pneumoniae
- G Candida albicans
- H Mycobacterium tuberculosis
- I Coxiella burnetii
- J Mycoplasma pneumoniae

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1** A patient has been ventilated on the intensive care unit for a week after a complicated laparotomy and now has new lung infiltrates and mucopurulent secretions from the endotracheal tube.
- 2** A young, previously fit man presents acutely unwell with pneumonia and his chest radiograph shows numerous small abscesses.
- 3** A middle-aged man is admitted to intensive care with severe acute pneumonia. He is the second intensive care admission and the third case of pneumonia among workers at the same factory in town.
- 4** A 71-year-old woman develops a chest infection 3 days after an uncomplicated resection of a Dukes A colonic carcinoma. Blood cultures and a catheter urine specimen both grow the same lactose-fermenting Gram-negative bacillus.
- 5** A 36-year-old woman has a 3-day history of fever, cough and pleuritic pain. She has coughed up a bit of blood in some purulent sputum. She has a pleural rub and her chest radiograph shows segmental consolidation in the left upper lobe.

Answers · Themes 6–8

Theme 6 · Lung diseases

- 1** **D COPD** — Heavy smoking with central cyanosis and diffuse wheeze indicates COPD (the chronic bronchitic 'blue bloater' pattern with hypoxaemia). Spirometry with FEV₁/FVC below 0.7 confirms it.
- 2** **J Pneumothorax** — Hyper-resonance and reduced breath sounds after blunt trauma with a central trachea and stable BP is a simple pneumothorax; chest drain.

- 3** **F Pleural effusion** — Stony dullness with absent breath sounds in a smoker with weight loss is a (likely malignant) pleural effusion; aspirate for cytology.
- 4** **J Pneumothorax** — A recurrent episode with unilateral reduced breath sounds and an asymmetric CXR is recurrent primary pneumothorax; refer for surgery.
- 5** **A Asbestosis** — Clubbing, basal crackles and pleural disease after asbestos textile work is asbestosis; the bulky hilum needs CT for cancer.

Theme 7 • Causes of shortness of breath

- B Aortic stenosis** — Exertional dyspnoea, angina and syncope with a slow-rising pulse is severe aortic stenosis; normal PEF and D-dimer argue against asthma and PE.
- I Right middle lobe pneumonia** — Loss of the right heart border with neutrophilia is right middle lobe pneumonia; a normal D-dimer makes PE unlikely.
- F Pleural effusion** — Stony dullness with a meniscus obscuring the hemidiaphragm is a pleural effusion; tap for cytology (cancer or mesothelioma).

Theme 8 • Respiratory infections

- C Pseudomonas aeruginosa** — Late ventilator-associated pneumonia (> 5 days) is commonly Pseudomonas; anti-pseudomonal therapy per local policy.

- B Staphylococcus aureus** — Pneumonia with multiple small abscesses in a fit young man is *S. aureus* (possibly PVL); anti-staphylococcal plus anti-toxin agents.
- E Legionella pneumophila** — A cluster of severe pneumonia among workers at one site suggests Legionella from contaminated water; notify.
- D Escherichia coli** — A lactose-fermenting Gram-negative bacillus in blood and urine after bowel surgery is *E. coli*.
- F Streptococcus pneumoniae** — Fever, pleuritic pain, rusty sputum and segmental consolidation in a healthy adult is pneumococcal pneumonia.

5 THEMES • 26 SCENARIOS • IRFAN EMQ COLLECTION

Irfan EMQs

SET 1 OF 2

THEME 1 Causes of pneumonia

- A Legionella pneumophila
- B Aspergillus fumigatus
- C Pneumocystis jirovecii (formerly *P. carinii*)
- D Streptococcus pneumoniae
- E Cytomegalovirus (CMV)
- F Chlamydia psittaci
- G Coxiella burnetii
- H Mycoplasma pneumoniae
- I Escherichia coli
- J Chlamydia pneumoniae
- K Moraxella catarrhalis
- L Varicella zoster virus
- M Staphylococcus aureus
- N Pseudomonas aeruginosa

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- An 80-year-old man presents with bilateral cavitating bronchopneumonia after an influenza infection.
- A 24-year-old student presents with severe headache, fever, dry cough and arthralgia. He has recently bought several parrots and was previously fit and well.
- A 40-year-old man with untreated HIV infection presents with fever, dry cough, weight loss and exertional breathlessness with desaturation on exertion.
- A 75-year-old man presents with headache, dry cough, anaemia and a skin rash. Blood tests detect cold agglutinins.
- A 25-year-old air-conditioning technician, who suffered from flu-like symptoms a week ago, has developed a dry cough. His chest radiograph shows multilobar shadowing. Blood tests show hyponatraemia and lymphopenia. Urinalysis reveals haematuria.

THEME 2 Chest radiograph pathology

- A Mitral stenosis
- B Bronchiectasis
- C Right upper lobe collapse
- D Kartagener's syndrome
- E Aortic stenosis
- F Previous varicella pneumonitis
- G Asbestosis
- H Right lower lobe collapse
- I Granulomatosis with polyangiitis (Wegener's)
- J Caplan's syndrome
- K Post-fracture fat embolism
- L Right ventricular failure
- M Left ventricular failure

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- The chest X-ray of a former coal miner with rheumatoid arthritis shows multiple bilateral well-defined nodules 0.5–5 cm in diameter.
- The chest X-ray of a retired shipyard worker shows calcified plaques along both hemidiaphragms and reticular shadowing in both lower zones.
- The chest X-ray shows a triangular opacity behind the right side of the heart with loss of the outline of the right hemidiaphragm; the right heart border remains visible.
- The chest X-ray shows numerous tiny calcified nodules, each less than 5 mm, scattered through both lungs in an adult who had severe chickenpox in adulthood.

- 5 The chest X-ray of a man with chronic productive cough, recurrent sinusitis and infertility shows the cardiac apex and aortic knuckle on the right with tramline and ring shadows in both lower zones.

THEME 3 Chest radiograph pathology (clinical cases)

- A Pneumonia
- B Silicosis
- C Bronchiectasis
- D Aspergillosis
- E Mesothelioma
- F Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- G Extrinsic allergic alveolitis
- H Sarcoidosis
- I Emphysema

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 28-year-old African-Caribbean man presents with dry cough and progressive shortness of breath. His chest radiograph shows bilateral hilar lymphadenopathy.
- 2 The chest X-ray of a 13-year-old boy with cystic fibrosis shows tramline and ring shadows.
- 3 A 65-year-old dockyard worker presents with weight loss and shortness of breath. He is clubbed and cachectic. His chest radiograph shows pleural calcification and a lobulated pleural mass.
- 4 A 40-year-old woman presents with gross clubbing and progressive shortness of breath. Examination reveals fine end-inspiratory crackles. Her chest radiograph shows a ground-glass appearance of the lung.
- 5 A 65-year-old smoker has breathlessness. On the chest X-ray eight anterior ribs are visible above the diaphragm in the mid-clavicular line on each side, and the diaphragms are flat.

THEME 4 Emergency management of respiratory distress

- A Intubation and mechanical ventilation
- B Left-sided needle decompression
- C Chest X-ray
- D Non-invasive positive pressure ventilation (NIV)
- E 28% oxygen, nebulised salbutamol, oral prednisolone
- F Emergency tracheostomy
- G 28% oxygen, nebulised salbutamol, intravenous hydrocortisone
- H Right-sided needle decompression
- I 100% oxygen, intramuscular adrenaline
- J Intercostal chest drain
- K 100% oxygen, intravenous hydrocortisone
- L 28% oxygen, nebulised salbutamol, intramuscular adrenaline

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 65-year-old man with long-standing COPD presents with severe breathlessness. He has been treated with controlled oxygen, nebulised bronchodilators and steroids. An hour later, on 28% oxygen, his ABG shows PaO₂ 7.5 kPa, PaCO₂ 9.0 kPa and pH 7.26.
- 2 A 17-year-old woman presents with wheeze and marked perioral swelling after eating a peanut snack: PaO₂ 7.0 kPa on 28% oxygen, PaCO₂ 4.1 kPa.
- 3 A 14-year-old with asthma has an acute attack and is tiring despite nebulised bronchodilators and steroids: PaO₂ 10.0 kPa on 28% oxygen, PaCO₂ 8.0 kPa.
- 4 A 28-year-old man involved in a road traffic accident presents with severe respiratory distress. Examination reveals decreased expansion on the right side of the chest with mediastinal shift to the left.
- 5 A young man presents with sudden breathlessness. Examination reveals reduced expansion on the right; he is haemodynamically stable with SaO₂ 95%.
- 6 A 30-year-old man with a stab wound to the right chest is breathless but haemodynamically stable; there is dullness and absent breath sounds at the right base, and the chest X-ray shows a large right haemothorax.

Answers · Themes 1-4

Theme 1 · Causes of pneumonia

- 1 **M** *Staphylococcus aureus* — Cavitating bronchopneumonia after influenza is classically *S. aureus*; flucloxacillin (or MRSA cover).
- 2 **F** *Chlamydia psittaci* — Headache, fever, dry cough and arthralgia after buying parrots is psittacosis (*Chlamydia psittaci*); doxycycline.
- 3 **C** *Pneumocystis jirovecii* (formerly *P. carinii*) — Dry cough and exertional desaturation in untreated HIV is *Pneumocystis* pneumonia; high-dose co-trimoxazole ± steroids.

- 4 **H** *Mycoplasma pneumoniae* — Dry cough, headache, rash and cold-agglutinin anaemia is *Mycoplasma pneumoniae*; macrolide or doxycycline.
- 5 **A** *Legionella pneumophila* — Multilobar pneumonia with hyponatraemia, lymphopenia and haematuria in an air-conditioning technician is *Legionella*; urinary antigen.

Theme 2 · Chest radiograph pathology

- 1 **J** Caplan's syndrome — Multiple rheumatoid nodules in a dust-exposed worker with RA is Caplan's syndrome.
- 2 **G** Asbestosis — Calcified diaphragmatic plaques with lower-zone reticulation in a shipyard worker is asbestosis.

- 3 **H Right lower lobe collapse** — A triangular retrocardiac opacity obscuring the right hemidiaphragm (heart border preserved) is right lower lobe collapse.
- 4 **F Previous varicella pneumonitis** — Healed varicella pneumonia leaves multiple widespread punctate (1–3 mm) calcified nodules. Other causes of this pattern include healed miliary tuberculosis or histoplasmosis.
- 5 **D Kartagener's syndrome** — Dextrocardia with bronchiectasis, sinusitis and infertility is Kartagener's syndrome (primary ciliary dyskinesia).

Theme 3 • Chest radiograph pathology (clinical cases)

- 1 **H Sarcoidosis** — BHL with dry cough in a young African-Caribbean adult is pulmonary sarcoidosis.
- 2 **C Bronchiectasis** — Tramline and ring shadows are the plain-film signs of bronchiectasis, universal in CF.
- 3 **E Mesothelioma** — A lobulated pleural mass with pleural calcification in a dockyard worker is mesothelioma; pleural biopsy.
- 4 **F Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)** — Gross clubbing with fine end-inspiratory crackles and ground-glass/reticular shadowing is IPF.

SET 2 OF 2

THEME 5 Treatment of respiratory infections

- A Intravenous benzylpenicillin
B Oral flucloxacillin
C Oral tetracycline
D Oral ciprofloxacin
E Intravenous ceftazidime
F Oral isoniazid plus rifampicin
G Intravenous flucloxacillin
H Oral amoxicillin
I High-dose zidovudine plus pyrazinamide
J Intravenous amoxicillin
K Intravenous co-trimoxazole
L Intravenous teicoplanin
M Intravenous cefuroxime plus erythromycin

- 5 **I Emphysema** — More than six anterior ribs above flat diaphragms indicate hyperinflation, as in emphysema.

Theme 4 • Emergency management of respiratory distress

- 1 **D Non-invasive positive pressure ventilation (NIV)** — Persistent hypercapnic acidosis (pH 7.26, PaCO₂ 9.0) despite treatment in COPD: non-invasive ventilation.
- 2 **I 100% oxygen, intramuscular adrenaline** — Wheeze and angio-oedema after peanuts is anaphylaxis: IM adrenaline 0.5 mg and high-flow oxygen.
- 3 **A Intubation and mechanical ventilation** — A raised PaCO₂ in acute asthma is near-fatal asthma, reflecting exhaustion; the patient needs ICU for intubation and mechanical ventilation. NIV is not recommended in acute asthma.
- 4 **H Right-sided needle decompression** — Respiratory distress after trauma with mediastinal shift away from the right is right tension pneumothorax: needle decompression.
- 5 **C Chest X-ray** — A stable suspected spontaneous pneumothorax needs a chest X-ray to confirm and size it before deciding management.
- 6 **J Intercoastal chest drain** — Large traumatic haemothorax: large-bore chest drain (> 1500 mL initially or 200 mL/h means thoracotomy).

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 Standard therapy for community-acquired pneumococcal pneumonia not requiring hospital admission.
- 2 A 35-year-old inpatient, admitted 10 days ago for another condition, develops severe pneumonia.
- 3 A 40-year-old builder presents with a severe community-acquired pneumonia. Atypical pathogens are suspected.
- 4 A 22-year-old man with HIV infection presents with *Pneumocystis jirovecii* pneumonia.
- 5 A 19-year-old man contracts pneumonia with symptoms of headache, fever and dry cough. Serology shows evidence of chlamydia infection.

Answers • Theme 5

IRFAN EMQS

Theme 5 • Treatment of respiratory infections

- 1 **H Oral amoxicillin** — Low-severity CAP managed at home: oral amoxicillin (macrolide or doxycycline if allergic).
- 2 **E Intravenous ceftazidime** — Late-onset severe hospital-acquired pneumonia needs antipseudomonal cover, e.g. IV ceftazidime or piperacillin-tazobactam.
- 3 **M Intravenous cefuroxime plus erythromycin** — Severe CAP: IV beta-lactam plus macrolide (e.g. cefuroxime plus erythromycin) to cover atypicals.
- 4 **K Intravenous co-trimoxazole** — *Pneumocystis pneumonia*: high-dose co-trimoxazole (IV if moderate-severe) with steroids if PaO₂ < 9.3 kPa.
- 5 **C Oral tetracycline** — Chlamydial pneumonia is intracellular: treat with a tetracycline (doxycycline) or macrolide, not beta-lactams.

12 THEMES • 60 SCENARIOS • AFTER GET AHEAD! MEDICINE EMQs

Get Ahead EMQs

SET 1 OF 3

THEME 1 Diagnosis of respiratory conditions (1)

- A Asbestos-related lung disease
- B Asthma
- C Bronchiectasis
- D Chronic obstructive pulmonary disease
- E Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- F Cystic fibrosis
- G Extrinsic allergic alveolitis
- H Mesothelioma
- I Non-small cell lung cancer
- J Open pneumothorax
- K Sarcoidosis
- L Secondary lung metastases
- M Obstructive sleep apnoea
- N Small cell lung cancer
- O Tuberculosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 68-year-old retired accountant who smokes has a 3-week history of small-volume haemoptysis. He has noticed that he is passing less urine than normal, although he is otherwise well. His serum sodium is 122 mmol/L with an inappropriately concentrated urine. On examination, a monophonic wheeze is heard over the left upper zone.
- 2 A 12-year-old boy is brought to the GP by his mother. Over the last couple of years he has had repeated chest infections, and his mother is worried that he is not putting on weight. On examination, there are coarse crackles throughout both lung fields.
- 3 A 58-year-old builder complains of left-sided chest discomfort that has been progressing over the last fortnight. He is otherwise well. On examination, the left lower zone is dull to percussion. A pleural tap yields blood-stained fluid.
- 4 A 53-year-old shopfitter is admitted for an elective inguinal hernia repair. A routine chest X-ray shows irregular white plaques over both lungs that cross the lobar boundaries. He feels well and auscultation is normal.
- 5 A 16-year-old boy presents to his GP with a 6-month history of night cough and intermittent wheeze. He has previously been well and has never had a chest infection. Examination of the chest is unremarkable.

THEME 2 Causes of shortness of breath

- A Acute coryza
- B Asbestosis
- C Aspergillosis
- D Bronchiectasis
- E Chronic obstructive pulmonary disease
- F Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- G Extrinsic allergic alveolitis
- H Pneumoconiosis
- I Sarcoidosis
- J Tuberculosis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 54-year-old man presents to the GP with worsening shortness of breath and a cough productive of bloody sputum. He has lost 6 kg in weight over the last month and has night sweats. A chest X-ray shows consolidation at the left apex.
- 2 A 42-year-old woman presents to the GP with a 6-month history of worsening shortness of breath associated with a dry cough. A chest X-ray shows bilateral hilar lymphadenopathy.
- 3 A 56-year-old man presents to the GP with a 6-month history of worsening shortness of breath. Examination reveals bilateral end-inspiratory crackles and clubbing. A chest X-ray shows reticulonodular shadowing, especially in the lower zones.
- 4 A 47-year-old man who has kept racing pigeons for many years presents to the GP with a 4-month history of worsening shortness of breath. Examination reveals bilateral end-inspiratory crackles without clubbing. A chest X-ray shows reticulonodular shadowing, especially in the upper zones.
- 5 A 38-year-old man presents with an episode of massive haemoptysis. He says that he has had two similar episodes in the last month. He often feels short of breath, but has recently finished a course of treatment for pulmonary tuberculosis. Examination is unremarkable. A chest X-ray shows a cavitating lesion with an opaque mass within it.

THEME 3 Diagnosis of respiratory conditions (2)

- A Acute coryza
- B Asthma
- C Bronchiectasis
- D Chronic obstructive pulmonary disease
- E Closed non-tension pneumothorax
- F Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)
- G Cystic fibrosis
- H Extrinsic allergic alveolitis
- I Open non-tension pneumothorax
- J Pneumoconiosis
- K Pneumonia
- L Pulmonary embolism
- M Obstructive sleep apnoea
- N Tension pneumothorax

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 67-year-old man has a 3-day history of worsening shortness of breath. He can normally walk 100 m but now can barely get to the front door. Although he says he normally coughs up small amounts of clear sputum, he is now expectorating large amounts of greeny-yellow sputum. Auscultation reveals widespread wheeze.
- 2 A 32-year-old man is stabbed in the right side of his chest. He struggles to the hospital. On arrival, he is in some respiratory distress. On examination, there is decreased air entry and hyper-resonance to percussion on the right side. The trachea is not deviated. The doctor notes the presence of a 'sucking' wound in the right chest.
- 3 A 26-year-old man presents with a 2-day history of feeling generally unwell and shortness of breath. Today, he started coughing up thick green sputum that had flecks of dry blood in. He complains of a sharp pain over the right side of his chest, which is worse on breathing in. Auscultation reveals bronchial breathing at the right base.
- 4 A 48-year-old woman presents with sudden-onset shortness of breath and a sharp pain on the right side of her chest. She was previously well. On examination, there is no wheeze or crackles and no areas of hyper-resonance.
- 5 A 21-year-old woman is involved in a road traffic accident. When the paramedics arrive, she has a heart rate of 122 beats/min, a respiratory rate of 32/min and a blood pressure of 86/42 mmHg. On examination, there are absent breath sounds and hyper-resonance to percussion on the left side. The trachea is deviated to the right.

THEME 4 Investigation of respiratory disease

- A Bronchoscopy and biopsy
- B Chest X-ray
- C CT pulmonary angiography
- D D-dimers
- E Lung function tests
- F Kveim-Siltzbach test
- G Mantoux test
- H No investigation required
- I Pleural biopsy
- J Sputum culture
- K Sweat test
- L Ventilation-perfusion scan

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 12-year-old boy is brought to the GP by his mother. Over the last couple of years he has had repeated chest infections, and his mother is worried that he is not putting on weight. On examination, there are coarse crackles throughout both lung fields and his fingers are clubbed.
- 2 A 67-year-old man complains of a long history of shortness of breath on exertion and a productive cough. He has been coughing up sputum for the last 4 months, but has had symptoms on and off for a few years. He attributed his symptoms to smoking, but he stopped last year.
- 3 A 59-year-old man presents to the GP with worsening shortness of breath and a cough productive of bloody sputum. He has lost a stone in weight over the last month and has been having night sweats. A chest X-ray shows scarring at the apices.
- 4 A 67-year-old man presents with a 3-week history of a cough productive of blood-stained sputum. He has lost 2 stone (12 kg) in weight over the last 2 months. Auscultation of the lungs is unremarkable, and a chest X-ray shows a 2 cm density in the left lower lobe.
- 5 A 58-year-old man who worked in shipbuilding complains of left-sided chest discomfort that has been progressing over the last fortnight. He is otherwise well. On examination, the left lower zone is dull to percussion, and a chest X-ray confirms a left pleural effusion. A diagnostic pleural tap yields blood-stained fluid, but cytology is non-diagnostic.

THEME 5 Investigation of acute dyspnoea

- A Bronchoscopy and biopsy
- B Chest X-ray
- C CT pulmonary angiography
- D D-dimers
- E Lung function tests
- F Kveim-Siltzbach test
- G Mantoux test
- H No investigation required
- I Pleural biopsy
- J Sputum culture
- K Sweat test
- L Ventilation-perfusion scan

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 42-year-old woman presents with sudden-onset shortness of breath and sharp right-sided chest pain. She is obviously distressed. Auscultation of the chest is normal and there is no hyper-resonance to percussion.
- 2 A 42-year-old woman presents with sudden-onset shortness of breath and sharp right-sided chest pain. She is obviously distressed. Auscultation of the chest is normal and there is no hyper-resonance to percussion. A chest X-ray shows clear lung fields. She had an anaphylactic reaction to iodinated contrast medium in the past.
- 3 A 42-year-old woman presents with sudden-onset shortness of breath and sharp right-sided chest pain. She is obviously distressed. Auscultation of the chest is normal and there is no hyper-resonance to percussion. A chest X-ray shows occasional patchy shadowing.
- 4 A 42-year-old woman presents with sudden-onset shortness of breath and sharp right-sided chest pain. She is obviously distressed. She is coughing up large amounts of green sputum. There are crackles at the right lower zone, and a chest X-ray confirms right lower lobe consolidation.
- 5 A 42-year-old woman presents with sudden-onset shortness of breath and sharp right-sided chest pain. She is obviously distressed. There are absent breath sounds and hyper-resonance to percussion over the right side, and the trachea is deviated to the left.

Answers · Themes 1-5

GET AHEAD EMQS

Theme 1 · Diagnosis of respiratory conditions (1)

- 1 **N Small cell lung cancer** — Haemoptysis with a monophonic wheeze and SIADH suggests small cell lung cancer.
- 2 **F Cystic fibrosis** — Recurrent chest infections, failure to thrive and coarse crackles in a child suggest cystic fibrosis; sweat test.
- 3 **H Mesothelioma** — Progressive chest-wall pain with a blood-stained effusion in a builder is mesothelioma; pleural biopsy.
- 4 **A Asbestos-related lung disease** — Holly-leaf plaques crossing lobar boundaries in an asymptomatic tradesman are calcified pleural plaques (asbestos exposure); benign.
- 5 **B Asthma** — Six months of nocturnal cough and intermittent wheeze in a well teenager is asthma; confirm objectively.

Theme 2 · Causes of shortness of breath

- 1 **J Tuberculosis** — Haemoptysis, weight loss and sweats with apical consolidation is post-primary TB; isolate and send 3 sputum samples.
- 2 **I Sarcoidosis** — BHL with dry cough and breathlessness is most often sarcoidosis.
- 3 **F Idiopathic pulmonary fibrosis (cryptogenic fibrosing alveolitis)** — Clubbing, Velcro crackles and lower-zone reticulation is IPF; HRCT UIP pattern, antifibrotics.
- 4 **G Extrinsic allergic alveolitis** — Upper-zone reticulation in a pigeon keeper without clubbing is chronic extrinsic allergic alveolitis; avoid antigen.

- 5 **C Aspergillosis** — A mass within a cavity with recurrent massive haemoptysis is an aspergilloma; Aspergillus IgG, embolise or resect.

Theme 3 · Diagnosis of respiratory conditions (2)

- 1 **D Chronic obstructive pulmonary disease** — Increased purulent sputum and breathlessness on a background of COPD is an infective exacerbation: bronchodilators, prednisolone, antibiotic.
- 2 **I Open non-tension pneumothorax** — A sucking chest wound with central trachea is an open (non-tension) pneumothorax: three-sided dressing, then drain away from the wound.
- 3 **K Pneumonia** — Fever, pleuritic pain and bronchial breathing at one base is pneumonia; assess with CURB-65.
- 4 **L Pulmonary embolism** — Sudden pleuritic pain and breathlessness with a normal chest examination is PE until proven otherwise.
- 5 **N Tension pneumothorax** — Hyper-resonance, absent breath sounds, tracheal deviation away and shock is tension pneumothorax: decompress without waiting for X-ray.

Theme 4 · Investigation of respiratory disease

- 1 **K Sweat test** — Suspected cystic fibrosis is diagnosed by the sweat test (chloride > 60 mmol/L) with CFTR genotyping.
- 2 **E Lung function tests** — Suspected COPD is confirmed by lung function tests (post-bronchodilator FEV₁/FVC < 0.7).
- 3 **J Sputum culture** — Suspected reactivation TB: three sputum samples for AFB smear, culture (with sensitivities) and NAAT.

- 4 **A Bronchoscopy and biopsy** — Haemoptysis, weight loss and a lung mass: staging CT then bronchoscopy and biopsy.
- 5 **I Pleural biopsy** — Blood-stained effusion with negative cytology after asbestos exposure: pleural biopsy (image-guided or thorascopic).

Theme 5 - Investigation of acute dyspnoea

- 1 **B Chest X-ray** — Acute pleuritic pain and breathlessness: chest X-ray first to exclude pneumothorax, consolidation and effusion.

- 2 **L Ventilation-perfusion scan** — High-probability PE with contrast allergy and a clear CXR: V/Q scan.
- 3 **C CT pulmonary angiography** — Suspected PE with an abnormal CXR: CTPA, as V/Q would be indeterminate.
- 4 **J Sputum culture** — Confirmed lobar consolidation explains the pain: sputum (and blood) culture to identify the organism.
- 5 **H No investigation required** — Tension pneumothorax is a clinical diagnosis: no investigation, immediate needle decompression.

SET 2 OF 3

THEME 6 Management of respiratory disease (1)

- A Amoxicillin
- B Cannula into second intercostal space, midclavicular line
- C Cannula into sixth intercostal space, midclavicular line
- D Chemotherapy
- E Co-trimoxazole
- F Erythromycin
- G Inhaled anticholinergic
- H Inhaled β_2 -agonist
- I Low-molecular-weight heparin
- J Amoxicillin plus metronidazole
- K No management required
- L Radiotherapy
- M Rifampicin, isoniazid, pyrazinamide and ethambutol
- N Steroids
- O Warfarin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 47-year-old man presents with a 2-day history of feeling generally unwell and shortness of breath. Today, he started coughing up thick green sputum. Auscultation reveals bronchial breathing at the right base. He is allergic to penicillin.
- 2 A 67-year-old woman with viral gastroenteritis aspirates some of her vomit. A few days later she develops a cough productive of foul-smelling sputum. A chest X-ray confirms right middle lobe consolidation.
- 3 A 38-year-old woman with HIV infection has a 2-month history of worsening shortness of breath and a dry cough. On examination, her oxygen saturation is 89% and falls further on exertion. A chest X-ray shows bilateral perihilar interstitial shadowing.
- 4 A 49-year-old man presents to the GP with worsening shortness of breath and a cough productive of bloody sputum. He has lost 6 kg in weight over the last month. A chest X-ray shows consolidation at both apices, and sputum smear shows acid-fast bacilli.
- 5 A 68-year-old man has a 3-week history of haemoptysis and has lost 6 kg in weight. A biopsy taken at bronchoscopy confirms non-small cell lung cancer. Staging shows locally advanced disease that is not surgically resectable, with no distant metastases.

THEME 7 Management of respiratory disease (2)

- A Amoxicillin
- B Cannula into second intercostal space, midclavicular line
- C Cannula into sixth intercostal space, midclavicular line
- D Chemotherapy
- E Co-trimoxazole
- F Erythromycin
- G Inhaled anticholinergic
- H Inhaled β_2 -agonist alone, as needed
- I Low-molecular-weight heparin
- J Metronidazole
- K No management required
- L Radiotherapy
- M Rifampicin and isoniazid
- N Low-dose inhaled corticosteroid with as-needed inhaled β_2 -agonist
- O Warfarin

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 38-year-old woman presents with sudden-onset shortness of breath and chest pain. Examination is unremarkable. A chest X-ray is clear, but the diagnosis is eventually confirmed on a ventilation-perfusion scan.
- 2 A 21-year-old woman is involved in a road traffic accident. When the paramedics arrive, she has a heart rate of 122 beats/min, a respiratory rate of 32/min and a blood pressure of 86/42 mmHg. On examination, breath sounds are absent and there is hyper-resonance to percussion on the left side. The trachea is deviated to the right.
- 3 A 68-year-old man has a 3-week history of haemoptysis. He has lost a stone (6 kg) in weight over this period of time. A biopsy taken at bronchoscopy confirms a small cell lung tumour. A CT scan shows that the disease is not surgically resectable.
- 4 An 11-year-old boy presents to his GP with a 6-month history of night cough and intermittent wheeze. He has previously been well. On examination, he is small for his age and has mild eczema on his upper limbs.
- 5 A 53-year-old man is admitted after breaking his left arm. The X-ray of the arm includes part of the lung fields, which show holly-leaf-shaped white lesions that cross the lobar boundaries. He feels well and auscultation is normal.

THEME 8 Airway management

- A Cricothyroidotomy
- B Endotracheal tube
- C Head tilt and chin lift
- D Laryngeal mask airway
- E Nasopharyngeal airway
- F Oropharyngeal airway
- G Suction
- H Tracheostomy

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 18-year-old man is post-ictal following an epileptic seizure. He is making snoring noises when breathing. His Glasgow Coma Scale score is 6 and he has no gag reflex.
- 2 An 18-year-old man is post-ictal following an epileptic seizure. He is making snoring noises when breathing. His Glasgow Coma Scale score is 12, his jaw is clenched and his gag reflex is intact.
- 3 A 65-year-old man collapsed unconscious in a restaurant toilet after complaining of chest pain. An ambulance has been called, but has yet to arrive.
- 4 A 30-year-old woman has significant facial and upper airway burns. An attempt at bag-and-mask ventilation and intubation has failed due to laryngeal oedema.
- 5 A 67-year-old woman is in ventricular fibrillation. An anaesthetist has yet to arrive.

THEME 9 Pathogens in pneumonia

- A Chlamydia pneumoniae
- B Chlamydia psittaci
- C Haemophilus influenzae
- D Klebsiella pneumoniae
- E Legionella pneumophila
- F Mycobacterium tuberculosis
- G Mycoplasma pneumoniae
- H Pneumocystis jirovecii
- I Pseudomonas aeruginosa
- J Staphylococcus aureus
- K Streptococcus pneumoniae

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 26-year-old man has a 2-day history of malaise, fever and shortness of breath associated with pain over the right side of his chest. He has developed a cough and is expectorating rusty sputum. A chest X-ray confirms consolidation of the right lower lobe.
- 2 A 67-year-old woman who had influenza a week ago presents with a 5-day history of malaise and a cough productive of yellow sputum. Examination is unremarkable. A chest X-ray shows two cavities with air-fluid levels.

- 3 A 42-year-old woman with HIV infection has a 2-month history of worsening shortness of breath and a dry cough. On examination, her oxygen saturation is 89% and falls further on exertion. A chest X-ray shows bilateral perihilar interstitial shadowing.
- 4 A 53-year-old man is brought to the emergency department after a fit. His wife says that he has been complaining of malaise, muscle aches and a dry cough for the last 7 days since they returned from holiday. Blood tests reveal a sodium of 124 mmol/L and a potassium of 4.2 mmol/L.
- 5 A 62-year-old man who keeps parrots presents with a 2-week history of shortness of breath, a dry cough and widespread joint aches. He has felt feverish and unwell over this time. Examination reveals an enlarged liver and spleen. A chest X-ray shows patchy consolidation of both lower lobes.

THEME 10 Management of asthma

- A Inhaled high-dose corticosteroid
- B Inhaled low-dose corticosteroid
- C Inhaled short-acting β_2 -agonist
- D Intravenous magnesium sulphate
- E Nebulised short-acting β_2 -agonist
- F Oral amoxicillin
- G No treatment required
- H Oral leukotriene receptor antagonist
- I Oral corticosteroid
- J Oxygen via face mask and inhaled short-acting β_2 -agonist
- K Oxygen via face mask, inhaled short-acting β_2 -agonist and corticosteroid
- L Oxygen via face mask and nebulised short-acting β_2 -agonist
- M Oxygen via face mask, nebulised short-acting β_2 -agonist and corticosteroid

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 An 8-year-old boy presents to his GP with a 6-month history of night cough and intermittent wheeze. He has previously been well. On examination, he is small for his age and has mild eczema on his upper limbs.
- 2 A 16-year-old boy attends the GP with poorly controlled chronic asthma. He is currently being managed with an inhaled long-acting β_2 -agonist and high-dose inhaled steroids. His parents want to know if there is anything else that he can be given.
- 3 A 14-year-old girl attends the GP with poorly controlled chronic asthma despite good adherence and inhaler technique. She is taking an inhaled long-acting β_2 -agonist, a high-dose inhaled corticosteroid and montelukast. Her parents want to know if anything else can be done.
- 4 A 23-year-old man with known asthma is brought to the emergency department with shortness of breath. He has widespread wheeze, is too breathless to talk, and has a heart rate of 115 beats/min and a respiratory rate of 36/min.

- 5** A 21-year-old woman with known asthma is brought to the emergency department with shortness of breath. She can barely talk, has a silent chest and a respiratory rate of 36/min. Despite some improvement with oxygen, nebulised salbutamol and steroids, her peak flow is still 30% of her best.

Answers · Themes 6–10

GET AHEAD EMQS

Theme 6 · Management of respiratory disease (1)

- 1** **F Erythromycin** — CAP in penicillin allergy: erythromycin/clarithromycin or doxycycline.
- 2** **J Amoxicillin plus metronidazole** — Aspiration pneumonia with foul sputum: amoxicillin plus metronidazole (or co-amoxiclav) for anaerobic cover.
- 3** **E Co-trimoxazole** — Subacute cough with exertional desaturation and perihilar shadowing in HIV is PCP: co-trimoxazole.
- 4** **M Rifampicin, isoniazid, pyrazinamide and ethambutol** — Smear-positive pulmonary TB: rifampicin, isoniazid, pyrazinamide and ethambutol for 2 months, then RI for 4 months, with pyridoxine.
- 5** **L Radiotherapy** — Unresectable stage III NSCLC: radical (chemo)radiotherapy followed by immunotherapy; radiotherapy also palliates haemoptysis.

Theme 7 · Management of respiratory disease (2)

- 1** **I Low-molecular-weight heparin** — Confirmed PE: immediate anticoagulation; LMWH here (DOAC preferred by NICE where suitable). Warfarin alone is too slow.
- 2** **B Cannula into second intercostal space, midclavicular line** — Tension pneumothorax: immediate needle decompression in the 2nd intercostal space, midclavicular line (or 4th/5th anterior axillary), then drain.
- 3** **D Chemotherapy** — Unresectable small cell lung cancer: platinum–etoposide chemotherapy (± immunotherapy), the tumour being highly chemosensitive.
- 4** **N Low-dose inhaled corticosteroid with as-needed inhaled β_2 -agonist** — Atopic child with 6 months of night cough and wheeze: paediatric low-dose ICS plus as-needed SABA.
- 5** **K No management required** — Leaf-shaped calcified plaques crossing lobar boundaries in a well man are benign pleural plaques: no management required.

Theme 8 · Airway management

- 1** **F Oropharyngeal airway** — Snoring with GCS 6 and no gag reflex: insert an oropharyngeal (Guedel) airway; intubate if consciousness does not improve.
- 2** **E Nasopharyngeal airway** — Semi-conscious with clenched jaw and intact gag: a nasopharyngeal airway is tolerated; an oral airway would provoke gagging.

- 3** **C Head tilt and chin lift** — Out-of-hospital collapse without equipment: open the airway with head tilt and chin lift, then check breathing and start CPR.
- 4** **A Cricothyroidotomy** — Failed intubation and bag-mask ventilation from laryngeal oedema ('can't intubate, can't oxygenate'): cricothyroidotomy.
- 5** **D Laryngeal mask airway** — During arrest, non-intubators use a supraglottic airway (laryngeal mask/i-gel) with minimal interruption to compressions.

Theme 9 · Pathogens in pneumonia

- 1** **K Streptococcus pneumoniae** — Fever, pleuritic pain and rusty sputum with lobar consolidation in a young adult is pneumococcal pneumonia.
- 2** **J Staphylococcus aureus** — Cavitating pneumonia with abscesses after influenza is characteristic of *S. aureus*.
- 3** **H Pneumocystis jirovecii** — Dry cough with exertional desaturation and perihilar shadowing in HIV is *Pneumocystis jirovecii* pneumonia.
- 4** **E Legionella pneumophila** — Pneumonia after travel with myalgia, dry cough and hyponatraemia causing a seizure suggests *Legionella*; urinary antigen.
- 5** **B Chlamydia psittaci** — Atypical pneumonia with hepatosplenomegaly in a parrot owner is psittacosis (*C. psittaci*); doxycycline.

Theme 10 · Management of asthma

- 1** **B Inhaled low-dose corticosteroid** — Atopic child with nocturnal cough and wheeze: start paediatric low-dose ICS plus as-needed SABA.
- 2** **H Oral leukotriene receptor antagonist** — Uncontrolled on high-dose ICS + LABA (after checking adherence and technique): add montelukast; refer if still poor.
- 3** **I Oral corticosteroid** — Uncontrolled on high-dose ICS, LABA and montelukast: lowest-dose oral steroid under specialist care; consider biologics.
- 4** **M Oxygen via face mask, nebulised short-acting β_2 -agonist and corticosteroid** — Too breathless to talk, HR 115, RR 36 is acute severe asthma: oxygen to 94–98%, nebulised salbutamol (+ ipratropium) and steroids.
- 5** **D Intravenous magnesium sulphate** — Silent chest with PEF 30% despite initial treatment is life-threatening: IV magnesium 1.2–2 g and early ICU referral.

SET 3 OF 3

THEME 11 Respiratory diseases of childhood

- A Aspiration of vomit
- B Asthma
- C Bronchiolitis
- D Croup
- E Cystic fibrosis
- F Epiglottitis
- G Infectious mononucleosis
- H Inhalation of foreign body
- I Pertussis

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 5-year-old unimmunized boy presents to the GP with a chronic cough. His mother reports that he has had this cough for 3 months. It started with a normal 'cold and cough', but he has not managed to shake it off. For the first 4 weeks, the cough would occur in bursts and he would get very distressed, occasionally going blue and vomiting.
- 2 A 2-year-old boy presents with a non-productive cough that started suddenly yesterday afternoon. He is otherwise well and has a temperature of 36.5°C. On examination, there are reduced breath sounds on the right side.
- 3 An 18-month-old girl presents with a non-productive cough and stridor. The cough is described as 'barking' in nature. She has a temperature of 38°C and is restless. She does not appear systemically unwell and is not cyanotic, but there is a moderate stridor on exertion.
- 4 A 7-year-old boy presents to the emergency department. His mother says that he has become very unwell very quickly. It started with a high fever and a sore throat. He is now very quiet, and is drooling while sitting upright on the edge of the bed. You examine him briefly, and he appears shocked and tachypnoeic and has a very faint stridor. He has a past medical history of asthma and has never been immunized.

- 5 A 6-month-old baby presents with a cough. The illness began 4 days ago with a runny nose and a mild fever. Over the last 24 hours, he has developed a high-pitched cough associated with a wheeze. On examination, crepitations are heard throughout the lung fields.

THEME 12 Scoring the severity of pneumonia (CURB-65)

- A CURB-65 score 0
- B CURB-65 score 1
- C CURB-65 score 2
- D CURB-65 score 3
- E CURB-65 score 4
- F CURB-65 score 5
- G Not a pneumonia

For each scenario, choose the single most appropriate option. Each option may be used once, more than once or not at all.

- 1 A 55-year-old man presents with a cough productive of purulent sputum. A chest X-ray shows left basal consolidation. His blood pressure is 118/78 mmHg, respiratory rate 18/min and urea 5.4 mmol/L. He scores 10/10 on the Abbreviated Mental Test Score (AMTS).
- 2 A 69-year-old man presents with a cough productive of purulent sputum. A chest X-ray shows left basal consolidation. His blood pressure is 86/62 mmHg, respiratory rate 34/min and urea 9.6 mmol/L. He scores 6/10 on the AMTS.
- 3 A 71-year-old man presents with a cough productive of purulent sputum. A chest X-ray shows no consolidation. His blood pressure is 96/68 mmHg, respiratory rate 28/min and urea 6.6 mmol/L. He scores 9/10 on the AMTS.
- 4 A 71-year-old man presents with a cough productive of purulent sputum. A chest X-ray shows left basal consolidation. His blood pressure is 96/68 mmHg, respiratory rate 32/min and urea 8.3 mmol/L. He scores 9/10 on the AMTS.
- 5 A 42-year-old man presents with a cough productive of purulent sputum. A chest X-ray shows left basal consolidation. His blood pressure is 106/78 mmHg, respiratory rate 24/min and urea 7.3 mmol/L. He scores 10/10 on the AMTS.

Answers · Themes 11-12

GET AHEAD EMQS

Theme 11 · Respiratory diseases of childhood

- 1 **I Pertussis** — Paroxysmal coughing with cyanosis and post-tussive vomiting for months in an unimmunised child is pertussis; macrolide and notify.
- 2 **H Inhalation of foreign body** — Sudden cough in a well afebrile toddler with unilateral reduced breath sounds is an inhaled foreign body; rigid bronchoscopy.

- 3 **D Croup** — Barking cough and stridor with mild fever in a toddler is croup (parainfluenza); oral dexamethasone 0.15 mg/kg.
- 4 **F Epiglottitis** — Rapid high fever, drooling, upright posture and soft stridor in an unimmunised child is epiglottitis (Hib); secure the airway, don't examine the throat.

- 5 **C** **Bronchiolitis** — Coryza progressing to cough, wheeze and widespread crackles in an infant is bronchiolitis (RSV); supportive care.

Theme 12 · Scoring the severity of pneumonia (CURB-65)

- 1 **A** **CURB-65 score 0** — Under 65, alert, urea ≤ 7 , RR < 30 and normal BP: CURB-65 = 0, home treatment with amoxicillin.
- 2 **F** **CURB-65 score 5** — Confusion, urea > 7 , RR ≥ 30 , SBP < 90 and age ≥ 65 : CURB-65 = 5, high severity.

- 3 **G** **Not a pneumonia** — Pneumonia needs new radiological shadowing; with a clear CXR this is a lower respiratory tract infection, not pneumonia.
- 4 **D** **CURB-65 score 3** — Urea > 7 , RR ≥ 30 and age ≥ 65 score (BP and AMTS 9 do not): CURB-65 = 3, high severity.
- 5 **B** **CURB-65 score 1** — Only urea > 7 scores: CURB-65 = 1, low severity, usually home treatment.

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Practise like the exam. Learn from every answer.

- 1,512 questions across 19 chapters, numbered by source
- Management flowcharts updated to current guidelines in every chapter
- Two-to-three-line explanations · 297 answers corrected
- Repeated questions merged with a note of where they were asked

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